

SEPARATION PROCESS PRINCIPLES

Second Edition



J. D. SEADER / ERNEST J. HENLEY

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SECOND EDITION

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Preface to the Second Edition

NEW TO THIS EDITION

“Time and tide wait for no man” and most certainly not for engineering textbooks. The seven years since publication of the first edition of “Separation Process Principles” have witnessed: (1) advances in the fundamentals of mass, heat, and momentum transport and wide availability of computer programs to facilitate the application of complex transport mathematical models; (2) changes in the practice of chemical engineering design; and (3) restructuring of the chemical engineering curriculum. In response to what we have noted and what has been pointed out in strong reviews solicited by the publishers, we have included the following revisions and additions to this second edition:

- A new section on dimensions and units to facilitate the use of the SI, AE, and CGS systems, which permeate applications to separation processes.
- The addition to each chapter of a list of instructional objectives.
- Increased emphasis on the many ways used to express the composition of chemical mixtures.
- New material on the thermodynamics of difficult mixtures, including electrolytes, polymer solutions, and mixtures of light gases and polar organic compounds.
- Tables of typical diffusivity values.
- Table of formulae and meanings of dimensionless groups.
- A subsection on the recent theoretical analogy of Churchill and Zajic.
- New sections on hybrid systems and membrane cascades.
- Discussions of the fourth generation of random packings and high-capacity trays.
- A brief discussion of the rate-based multicell model.
- New section on optimal control as a third mode of operation for batch distillation.
- New discussion on concentration polarization and fouling.
- New sections on ultrafiltration and microfiltration.
- New subsection on Continuous, Countercurrent Adsorption Systems.
- Revision of the subsection on the McCabe–Thiele Method for Bulk Separation by adsorption.
- New subsection on Simulated (and True) Moving Bed Systems for Adsorption.

The following three chapters were not in the first edition of the book, but were available in hard copy, as supplemental chapters, to instructors. They are now included in the second edition:

Chapter 16 on Leaching and Washing, with an added subsection on the espresso machine.

Chapter 17 on Crystallization, Desublimation, and Evaporation.

Chapter 18 on Drying of Solids, including Psychrometry.

In the first edition, each topic was illustrated by at least one detailed example and was accompanied by at least three homework exercises. This continues to be true for most of the added topics and chapters. There are now 214 examples and 649 homework exercises. In addition, 839 references are cited.

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Nomenclature

Latin Capital and Lowercase Letters

A	constant in equations of state; constant in Margules equation; area for mass transfer; area for heat transfer; area; coefficient in Freundlich equation; absorption factor = L/KV ; total area of a tray; frequency factor	C_L	constant in (6-132) and Table 6.8
A_a	active area of a sieve tray	C_V	constant in (6-133) and Table 6.8
A_b	active bubbling area of a tray	C_h	packing constant in Table 6.8
A_d	downcomer cross-sectional area of a tray	C_o	orifice coefficient
A_{da}	area for liquid flow under downcomer	C_P, C_p	specific heat at constant pressure; packing constant in Table 6.8
A_h	hole area of a sieve tray	$C_{P_V}^o$	ideal gas heat capacity at constant pressure
A_{ij}	binary interaction parameter in van Laar equation	c	molar concentration; constant in the BET equation; speed of light
$\bar{A}_{ij}$	binary interaction parameter in Margules two-constant equation	c^*	liquid concentration in equilibrium with gas at its bulk partial pressure
A_j, B_j, C_j, D_j	material-balance parameters defined by (10-7) to (10-11)	c'	concentration in liquid adjacent to a membrane surface
A_M	membrane surface area	c_m	metastable limiting solubility of crystals
A_p	pre-exponential (frequency) factor	c_s	humid heat; normal solubility of crystals
A_w	specific surface area of a particle	c_t	total molar concentration
a	activity; constants in the ideal-gas heat capacity equation; constant in equations of state; interfacial area per unit volume; surface area; characteristic dimension of a solid particle; equivalents exchanged in ion exchange; interfacial area per stage	Δc_{limit}	limiting supersaturation
$\bar{a}$	interfacial area per unit volume of equivalent clear liquid on a tray	$D, \mathcal{D}$	diffusivity; distillate flow rate; amount of distillate; desorbent (purge) flow rate; discrepancy functions in inside-out method of Chapter 10.
a_h	specific hydraulic area of packing	D_B	bubble diameter
a_{mk}	group interaction parameter in UNIFAC method	D_E	eddy diffusion coefficient in (6-36)
a_v	surface area per unit volume	D_e, D_{eff}	effective diffusivity [see (3-49)]
B	constant in equations of state, bottoms flow rate; number of binary azeotropes	D_H	diameter of perforation of a sieve tray
B^0	rate of nucleation per unit volume of solution	D_i	impeller diameter
b	molar availability function = $h - T_0s$; constant in equations of state; component flow rate in bottoms; surface perimeter	D_{ij}	mutual diffusion coefficient of i in j
C	general composition variable such as concentration, mass fraction, mole fraction, or volume fraction; number of components; constant; capacity parameter in (6-40); constant in tray liquid holdup expression given by (6-50); rate of production of crystals	D_K	Knudsen diffusivity
C_1	constant in (6-126)	D_L	longitudinal eddy diffusivity
C_2	constant in (6-127)	$\bar{D}_N$	arithmetic-mean diameter
C_D	drag coefficient	D_o	diffusion constant in (3-57)
C_F	entrainment flooding factor in Figure 6.24 and (6-42)	D_P, D_p	effective packing diameter; particle diameter
		$\bar{D}_p$	average of apertures of two successive screen sizes
		D_s	surface diffusivity
		$\bar{D}_S$	surface (Sauter) mean diameter
		D_T	tower or vessel diameter
		$\bar{D}_V$	volume-mean diameter
		$\bar{D}_W$	mass-mean diameter
		d	component flow rate in distillate
		d_e	equivalent drop diameter; pore diameter
		d_H	hydraulic diameter = $4r_H$
		d_m	molecule diameter
		d_p	droplet or particle diameter; pore diameter
		d_{vs}	Sauter mean diameter defined by (8-35)

E	activation energy; dimensionless concentration change defined in (3-80); extraction factor defined in (4-24); amount or flow rate of extract; turbulent diffusion coefficient; voltage; wave energy; evaporation rate		partial molar enthalpy
E^0	standard electrical potential	H	Henry's law coefficient defined by (6-121)
E_b	radiant energy emitted by a black body	H_j	residual of energy balance equation (10-5)
E_D	activation energy of diffusion in a polymer		heat of adsorption
$E_{i,j}$	residual of equilibrium equation (10-2)		heat of condensation
E_{MD}	fractional Murphree dispersed-phase efficiency		heat of crystallization
E_{MV}	fractional Murphree vapor efficiency		heat of dilution
E_{OV}	fractional Murphree vapor point efficiency		integral heat of solution at saturation
E_o	fractional overall stage (tray) efficiency		heat of solution at infinite dilution
E_p	activation energy		molar enthalpy of vaporization
	radiant energy of a given wavelength emitted by a black body	H_G	height of a transfer unit for the gas
$E\{t\}dt$	fraction of effluent with a residence time between t and $t + dt$	H_i	distance of impeller above tank bottom
	number of independent equations in Gibbs phase rule	H_L	height of a transfer unit for the liquid
ΔE^{vap}	molar internal energy of vaporization	H_{OG}	height of an overall transfer unit based on the gas phase =
e	entrainment rate; heat transfer rate across a phase boundary	H_{OL}	height of an overall transfer unit based on the liquid phase =
F	Faraday's constant = 96,490 coulomb/g-equivalent; feed flow rate; force; F -factor defined below (6-67)		humidity
F_b	buoyancy force		molal humidity
F_d	drag force		percentage humidity
F_F	foaming factor in (6-42)		relative humidity
F_g	gravitational force		saturation humidity
F_{HA}	hole-area factor in (6-42)		saturation humidity at temperature T_w
F_{LV}, F_{LG}	kinetic-energy ratio defined in Figure 6.24	HETP	height equivalent to a theoretical plate
F_P	Packing factor in Table 6.8	HETS	height equivalent to a theoretical stage (same as HETP)
F_{ST}	surface tension factor in (6-42)	HTU	height of a transfer unit
F_V	solids volumetric velocity in volume per unit cross-sectional area per unit time	h	molar enthalpy; heat-transfer coefficient; specific enthalpy; liquid molar enthalpy; height of a channel; height; Planck's constant =
$F\{t\}$	fraction of eddies with a contact time less than t	h_d	dry tray pressure drop as head of liquid
	number of degrees of freedom	h_{da}	head loss for liquid flow under downcomer
f	pure-component fugacity; Fanning friction factor; function; component flow rate in feed; residual	h_{dc}	clear liquid head in downcomer
f_f	fraction of flooding velocity	h_{df}	height of froth in downcomer
f_i	fugacity of component i in a mixture	h_f	height of froth on tray
f_v	volume shape factor	h_l	equivalent head of clear liquid on tray
	partial fugacity	h_L	specific liquid holdup in a packed column
f_w	function of the acentric factor in the S-R-K and P-R equations	h_t	total tray pressure drop as head of liquid
G	Gibbs free energy; mass velocity; volumetric holdup on a tray; rate of growth of crystal size	h_w	weir height
G_{ij}	binary interaction parameter in NRTL equation		pressure drop due to surface tension as head of liquid
g	molar Gibbs free energy; acceleration due to gravity	I	electrical current
g_c	universal	i	current density
g_{ij}	energy of interaction in NRTL equation	J_i	molar flux of i by ordinary molecular diffusion relative to the molar-average velocity of the mixture
H	Henry's law coefficient defined in Table 2.3; Henry's law constant defined in (3-50); height or length of vessel; molar enthalpy	j_D	Chilton-Colburn j -factor for mass transfer $\equiv$
		j_H	Chilton-Colburn j -factor for heat transfer $\equiv$
		j_M	Chilton-Colburn j -factor for momentum transfer
		j_i	mass flux of i by ordinary molecular diffusion relative to the mass-average velocity of the mixture.

K	equilibrium ratio for vapor-liquid equilibria; equilibrium partition coefficient in (3-53) and for a component distributed between a fluid and a membrane; overall mass-transfer coefficient; adsorption equilibrium constant	LES	length of equilibrium (spent) section of adsorption bed
K	overall mass-transfer coefficient for UM diffusion	LUB	length of unused bed in adsorption
K_a	chemical equilibrium constant based on activities	L_w	weir length
K_c	solubility product; overall mass-transfer coefficient for crystallization	l	constant in UNIQUAC and UNIFAC equations; component flow rate in liquid; length
K_D	equilibrium ratio for liquid-liquid equilibria	l_{ij}	binary interaction parameter
	equilibrium ratio in mole- or mass-ratio compositions for liquid-liquid equilibria	l_M	membrane thickness
K_G	overall mass-transfer coefficient based on the gas phase with a partial pressure driving force	I_T	packed height
K_{ij}	molar selectivity coefficient in ion exchange	M	molecular weight; mixing-point amount or flow rate, molar liquid holdup
K_L	overall mass-transfer coefficient based on the liquid phase with a concentration driving force	M_i	moles of i in batch still
K_s	capacity parameter defined by (6-53)	$M_{i,j}$	residual of component material-balance equation (10-1)
K_W	wall factor given by (6-111)	M_T	mass of crystals per unit volume of magma
K_X	overall mass-transfer coefficient based on the liquid phase with a mole ratio driving force	M_t	total mass
K_x	overall mass-transfer coefficient based on the liquid phase with a mole-fraction driving force	m	slope of equilibrium curve; mass flow rate; mass
K_Y	overall mass-transfer coefficient based on the gas phase with a mole ratio driving force	m_c	mass of crystals per unit volume of mother liquor
K_y	overall mass-transfer coefficient based on the gas phase with a mole-fraction driving force		molality of i in solution
K_r	restrictive factor for diffusion in a pore	m_p	mass of adsorbent or particle
k	thermal conductivity; mass-transfer coefficient in the absence of the bulk-flow effect	m_s	mass of solid on a dry basis; solids flow rate
k	mass-transfer coefficient that takes into account the bulk-flow effect as in (3-229) and (3-230)	m_v	mass evaporated; rate of evaporation
k_c	mass-transfer coefficient based on a concentration, c , driving force; thermal conductivity of crystal layer	m_x	tangent to the vapor-liquid equilibrium line in the region of liquid-film mole fractions as in Figure 3.22
k_{ij}	binary interaction parameter	m_y	tangent to the vapor-liquid equilibrium line in the region of gas-film mole fractions as in Figure 3.22
k_i	mass-transfer coefficient for integration into crystal lattice	MTZ	length of mass-transfer zone in adsorption bed
k_N	constant	N	number of phases; number of moles; molar flux = n/A ; number of equilibrium (theoretical, perfect) stages; rate of rotation; number of transfer units; cumulative number of crystals of size, L , and smaller; number of stable nodes; molar flow rate
	constant	N_A	number of additional variables; Avogadro's number molecules/mol
k_p	mass-transfer coefficient for the gas phase based on a partial pressure, p , driving force	N_a	number of actual trays
k_x	mass-transfer coefficient for the liquid phase based on a mole-fraction driving force	N_{Bi}	Biot number for heat transfer Biot number for mass transfer
k_y	mass-transfer coefficient for the gas phase based on a mole-fraction driving force	N_D	number of degrees of freedom
	liquid molar flow rate in stripping section	N_E	number of independent equations
L	liquid; length; height; liquid flow rate; underflow flow rate; crystal size	N_{Eo}	Eotvos number defined by (8-49)
L	solute-free liquid molar flow rate; liquid molar flow rate in an intermediate section of a column	N_{Fo}	Fourier number for heat transfer = $\alpha t/a^2$ = dimensionless time Fourier number for mass transfer = Dt/a^2 = dimensionless time
L_B	length of adsorption bed	N_{Fr}	Froude number = inertial force/gravitational force
L_e	entry length	N_G	number of gas-phase transfer units defined in Table 6.7
L_{pd}	predominant crystal size	N_L	number of liquid-phase transfer units defined in Table 6.7
L_S	liquid molar flow rate of sidestream	N_{Le}	Lewis number = N_{Sc}/N_{Pr}

N_{Lu}	Luikov number = $1/N_{Le}$		vapor pressure in a pore
N_{min}	minimum number of stages for specified split	P_0	adsorbate vapor pressure at test conditions
N_{Nu}	Nusselt number = dh/k = temperature gradient at wall or interface/temperature gradient across fluid (d = characteristic length)	p	partial pressure
		p	partial pressure in equilibrium with liquid at its bulk concentration
N_{OG}	number of overall gas-phase transfer units defined in Table 6.7	p_j, q_j	material-balance parameters for Thomas algorithm in Chapter 10
N_{OL}	number of overall liquid-phase transfer units defined in Table 6.7	Q	rate of heat transfer; volume of liquid; volumetric flow rate
N_{Pe}	Peclet number for heat transfer = $N_{Re}N_{Pr}$ = convective transport to molecular transfer	Q_C	rate of heat transfer from condenser
	Peclet number for mass transfer = = convective transport to molecular transfer	Q_L	volumetric liquid flow rate
		Q_{ML}	volumetric flow rate of mother liquor
		Q_R	rate of heat transfer to reboiler
N_{Po}	Power number defined in (8-21)	Q_k	area parameter for functional group k in UNIFAC method
N_{Pr}	Prandtl number = momentum diffusivity/thermal diffusivity	q	relative surface area of a molecule in UNIFAC and UNIFAC equations; heat flux; loading or concentration of adsorbate on adsorbent; feed condition in distillation defined as the ratio of increase in liquid molar flow rate across feed stage to molar feed rate
N_R	number of redundant equations		volume-average adsorbate loading defined for a spherical particle by (15-103)
N_{Re}	Reynolds number inertial force/viscous force (d = characteristic length)	q^e	surface excess in liquid adsorption
NRX	number of reactions	q_L	liquid flow rate across a tray
N_{Sc}	Schmidt number momentum diffusivity/mass diffusivity	R	universal gas constant: 1.987 cal/mol K or Btu/lbmol 8315 J/kmol K or Pa m ³ /kmol K 82.06 atm cm ³ /mol K 0.7302 atm ft ³ /lbmol R 10.73 psia ft ³ /lbmol R;
N_{Sh}	Sherwood number concentration gradient at wall or interface/concentration gradient across fluid (d = characteristic length)		molecule radius; amount or flow rate of raffinate; ratio of solvent to insoluble solids; reflux ratio; drying-rate flux; inverted binary mass-transfer coefficients defined by (12-31) and (12-32)
N_{St}	Stanton number for heat transfer = h/GC_P Stanton number for mass transfer	R	drying-rate per unit mass of bone-dry solid
NTU	number of transfer units	R_c	drying-rate flux in the constant-rate period
N_T	total number of crystals per unit volume of mother liquor; number of transfer units for heat transfer	R_f	drying-rate flux in the falling-rate period
N_i	number of equilibrium (theoretical) stages	R_k	volume parameter for functional group k in UNIFAC method
N_V	number of variables	R_L	liquid-phase withdrawal factor in (10-80)
N_{We}	Weber number defined by (8-37)	R_{min}	minimum reflux ratio for specified split
	number of moles	R_p	particle radius
n	molar flow rate; moles; constant in Freundlich equation; number of pores per cross-sectional area of membrane; number of crystals per unit size per unit volume	R_V	vapor-rate withdrawal factor in (10-81)
n_c	number of crystals per unit volume of mother liquor	r	relative number of segments per molecule in UNIFAC and UNIFAC equations; radius; ratio of permeate to feed pressure for a membrane; distance in direction of diffusion; reaction rate; fraction of a stream exiting a stage that is removed as a sidestream; molar rate of mass transfer per unit volume of packed bed
n^0	initial value for number of crystals per unit size per unit volume	r_c	radius at reaction interface
n_+, n_-	valences of cation and anion, respectively	r_H	hydraulic radius = flow cross section/wetted perimeter
P	pressure; power; electrical power	r_p	pore radius
P, P	difference points	r_s	radius at surface of particle
	parachor; number of phases in Gibbs phase rule	r_w	radius at tube wall
P_c	critical pressure		
P_M	permeability		
	permeance		
P_r	reduced pressure, P/P_c		
	vapor pressure		

S	solid; rate of entropy; total entropy; solubility equal to H in (3-50); cross-sectional area for flow; solvent flow rate; mass of adsorbent; stripping factor = KV/L ; surface area; inert solid flow rate; flow rate of crystals; supersaturation; belt speed; number of saddles	u_s	superficial velocity
S_{ij}	separation factor in ion exchange	u_V	gas velocity
S_g	surface area per unit volume of a porous particle	u_0	characteristic rise velocity of a droplet
S_x	residual of liquid-phase mole-fraction summation equation (10-3)	V	vapor; volume; vapor flow rate; overflow flow rate
S_y	residual of vapor-phase mole-fraction summation equation (10-4)		vapor molar flow rate in an intermediate section of a column; solute-free molar vapor rate
s	molar entropy; fractional rate of surface renewal; relative supersaturation	V_B	boilup ratio
s_p	particle external surface area	V_H	holdup as a fraction of dryer volume
SF	split fraction defined by (1-2)	V_{LH}	volumetric liquid holdup
SP	separation power or relative split ratio defined by (1-4); salt passage defined by (14-70)	V_{ML}	volume of mother liquor in magma
SR	split ratio defined by (1-3)	V_p	pore volume per unit mass of particle
T	temperature	V_V	volume of a vessel
T_c	critical temperature		vapor molar flow rate in stripping section
T_g	glass-transition temperature for a polymer		number of variables in Gibbs phase rule
T_{ij}	binary interaction parameter in UNIQUAC and UNIFAC equations	v	molar volume; velocity; component flow rate in vapor; volume of gas adsorbed
T_m	melting temperature for a polymer		average molecule velocity
T_0	datum temperature for enthalpy; reference temperature; infinite source or sink temperature	v_i	species velocity relative to stationary coordinates
T_r	reduced temperature = T/T_c		species diffusion velocity relative to the molar average velocity of the mixture
T_s	source or sink temperature	v_c	critical molar volume
T_v	moisture evaporation temperature	v_H	humid volume
t	time; residence time	v_M	molar average velocity of a mixture
	average residence time	v_p	particle volume
t_b	time to breakthrough in adsorption	v_r	reduced molar volume,
t_c	contact time in the penetration theory	v_s	molar volume of crystals
t_E	elution time in chromatography	v_0	superficial velocity
t_F	feed pulse time in chromatography		summation of atomic and structural diffusion volumes in (3-36)
t_L	contact time of liquid in penetration theory; residence time of crystals to reach size L	W	rate of work; width of film; bottoms flow rate; amount of adsorbate; washing factor in leaching = S/RF_A ; baffle width; moles of liquid in a batch still; moisture content on a wet basis; vapor sidestream molar flow rate; weir length
t_{res}	residence time	W_{min}	minimum work of separation
U	superficial velocity; overall heat-transfer coefficient; liquid sidestream molar flow rate; reciprocal of extraction factor	WES	weight of equilibrium (spent) section of adsorption bed
U_a	superficial vapor velocity based on tray active bubbling area	WUB	weight of unused adsorption bed
U_f	flooding velocity	W_s	rate of shaft work
u	velocity; interstitial velocity	w	mass fraction; width of a channel; weighting function in (10-90)
	bulk-average velocity; flow-average velocity	X	mole or mass ratio; mass ratio of soluble material to solvent in underflow; moisture content on a dry basis; general variable; parameter in (9-34)
	relative or slip velocity	X	equilibrium moisture on a dry basis
u_{all}	allowable velocity	X_B	bound moisture content on a dry basis
u_c	velocity of concentration wave in adsorption	X_c	critical free moisture content on a dry basis
u_{ij}	energy of interaction in UNIQUAC equation	X_T	total moisture content on a dry basis
u_L	superficial liquid velocity	X_i	mass of solute per volume of solid
u_{mf}	minimum fluidization velocity	X_m	mole fraction of functional group m in UNIFAC method
u_o	hole velocity for sieve tray; superficial gas velocity in a packed column		

x	mole fraction in liquid phase; mole fraction in any phase; distance; mass fraction in raffinate; mass fraction in underflow; mass fraction of particles	y	vector of mole fractions in vapor phase
x	normalized mole fraction = 1	Z	compressibility factor = Pv/RT ; total mass; height
x	vector of mole fractions in liquid phase	Z_f	froth height on a tray
x_n	fraction of crystals of size smaller than L	Z_L	length of liquid flow path across a tray
Y	mole or mass ratio; mass ratio of soluble material to solvent in overflow; pressure-drop factor for packed columns defined by (6-102); concentration of solute in solvent; parameter in (9-34)		lattice coordination number in UNIQUAC and UNIFAC equations
y	mole fraction in vapor phase; distance; mass fraction in extract; mass fraction in overflow	z	mole fraction in any phase; overall mole fraction in combined phases; distance; overall mole fraction in feed; dimensionless crystal size; length of liquid flow path across tray
		z	vector of mole fractions in overall mixture

Greek Letters

thermal diffusivity, ; relative volatility; surface area per adsorbed molecule

ideal separation factor for a membrane

relative volatility of component i with respect to component j for vapor-liquid equilibria; parameter in NRTL equation

energy-balance parameters defined by (10-23) to (10-26)

relative selectivity of component i with respect to component j for liquid-liquid equilibria

film flow rate/unit width of film; thermodynamic function defined by (12-37)

residual activity coefficient of functional group k in UNIFAC equation

specific heat ratio; activity coefficient

change (final – initial)

solubility parameter; film thickness; velocity boundary layer thickness; thickness of the laminar sublayer in the Prandtl analogy

concentration boundary layer thickness

Kronecker delta

exponent parameter in (3-40); fractional porosity; allowable error; tolerance in (10-31)

bed porosity (external void fraction)

eddy diffusivity for diffusion (mass transfer)

eddy diffusivity for heat transfer

eddy diffusivity for momentum transfer

particle porosity (internal void fraction)

Murphree vapor-phase plate efficiency in (10-73)

area fraction in UNIQUAC and UNIFAC equations; dimensionless concentration change defined in (3-80); correction factor in Edmister group method; cut equal to permeate flow rate to feed flow rate for a membrane; contact angle; fractional coverage in Langmuir equation; solids residence time in a dryer; root of the Underwood equation, (9-28)

average liquid residence time on a tray

Maxwell-Stefan mass-transfer coefficient in a binary mixture

binary interaction parameter in Wilson equation

mV/L ; radiation wavelength

limiting ionic conductances of cation and anion, respectively

energy of interaction in Wilson equation

chemical potential or partial molar Gibbs free energy; viscosity

momentum diffusivity (kinematic viscosity), ; wave frequency; stoichiometric coefficient

number of functional groups of kind k in molecule i in UNIFAC method

fractional current efficiency; dimensionless distance in adsorption defined by (15-115); dimensionless warped time in (11-2)

osmotic pressure; product of ionic concentrations

mass density

bulk density

crystal density

particle density

true (crystalline) solid density

surface tension; interfacial tension; Stefan-Boltzmann constant = $5.671 \times 10^{-8} \text{ W/m}^2 \text{ K}^4$

interfacial tension

interfacial tension between crystal and solution

tortuosity; shear stress; dimensionless time in adsorption defined by (15-116); retention time of mother liquor in crystallizer; convergence criterion in (10-32)

binary interaction parameter in NRTL equation

shear stress at wall

v number of ions per molecule

v volume fraction; parameter in Underwood equations (9-24) and (9-25)

local volume fraction in the Wilson equation

probability function in the surface renewal theory

pure-species fugacity coefficient; association factor in the Wilke-Chang equation; recovery factor in absorption and stripping; volume fraction; concentration ratio defined by (15-125)

partial fugacity coefficient
 froth density
 effective relative density of froth defined by (6-48)
 particle sphericity
 segment fraction in UNIQUAC equation; V/F in flash calculations; E/F in liquid-liquid equilibria calculations for single-stage extraction; sphericity defined before Example 15.7

dry-packing resistance coefficient given by (6-113)
 fractional entrainment; loading ratio defined by (15-126); sphericity
 acentric factor defined by (2-45); segment fraction in UNIFAC method

Subscripts

A solute

<i>a</i> , <i>ads</i>	adsorption	LM	log mean of two values, A and B = $(A - B)/\ln(A/B)$
avg	average	LP	low pressure
<i>B</i>	bottoms	<i>M</i>	mass transfer; mixing-point condition; mixture
<i>b</i>	bulk conditions; buoyancy	<i>m</i>	mixture; maximum
bubble	bubble-point condition	max	maximum
<i>C</i>	condenser; carrier; continuous phase	min	minimum
<i>c</i>	critical; convection; constant-rate period	<i>N</i>	stage
cum	cumulative	<i>n</i>	stage
<i>D</i>	distillate, dispersed phase; displacement	<i>O</i>	overall
<i>d</i>	drag; desorption	<i>o</i> , <i>0</i>	reference condition; initial condition
<i>d</i> , <i>db</i>	dry bulb	<i>out</i>	leaving
des	desorption	<i>OV</i>	overhead vapor
dew	dew-point condition	<i>P</i>	permeate
<i>ds</i>	dry solid	<i>R</i>	reboiler; rectification section; retentate
<i>E</i>	enriching (absorption) section	<i>r</i>	reduced; reference component; radiation
<i>e</i>	effective; element	res	residence time
eff	effective	<i>S</i>	solid; stripping section; sidestream; solvent; stage; salt
<i>F</i>	feed	<i>s</i>	source or sink; surface condition; solute; saturation
<i>f</i>	flooding; feed; falling-rate period	<i>T</i>	total
<i>G</i>	gas phase	<i>t</i>	turbulent contribution
GM	geometric mean of two values, A and B = square root of A times B	<i>V</i>	vapor
<i>g</i>	gravity	<i>W</i>	batch still
<i>gi</i>	gas in	<i>w</i>	wet solid-gas interface
<i>go</i>	gas out	<i>w</i> , <i>wb</i>	wet bulb
<i>H</i> , <i>h</i>	heat transfer	<i>ws</i>	wet solid
<i>I</i> , <i>I</i>	interface condition	<i>X</i>	exhausting (stripping) section
<i>i</i>	particular species or component	<i>x</i> , <i>y</i> , <i>z</i>	directions
<i>in</i>	entering		at the edge of the laminar sublayer
<i>irr</i>	irreversible	0	surroundings; initial
<i>j</i>	stage number		infinite dilution; pinch-point zone
<i>k</i>	particular separator; key component		
<i>L</i>	liquid phase; leaching stage		

Superscripts

E excess; extract phase

<i>F</i>	feed	<i>o</i>	pure species; standard state; reference condition
<i>ID</i>	ideal mixture	<i>p</i>	particular phase
(<i>k</i>)	iteration index	<i>R</i>	raffinate phase
<i>LF</i>	liquid feed		

s saturation condition
 VF vapor feed
 $-$ partial quantity; average value
infinite dilution

(1), (2) denotes which liquid phase
I, II denotes which liquid phase
at equilibrium

Abbreviations

Angstrom	m	in.	inch
ARD	asymmetric rotating disk contactor	J	joule
atm	atmosphere	K	degrees Kelvin
avg	average	kg	kilogram
BET	Brunauer-Emmett-Teller	kmol	kilogram-mole
BP	bubble-point method	L	liter; low boiler
B-W-R	Benedict-Webb-Rubin equation of state	LHS	left-hand side of an equation
bar	0.9869 atmosphere or 100 kPa	LK	light-key component
barrer	membrane permeability unit, 1 barrer = $10^{-10} \text{ cm}^3 (\text{STP}) \text{ cm}/(\text{cm}^2 \text{ s cm Hg})$	LLK	lighter than light key component
bbf	barrel	L-K-P	Lee-Kessler-Plöcker equation of state
Btu	British thermal unit	LM	log mean
C_i	paraffin with i carbon atoms	LW	lost work
C_i^-	olefin with i carbon atoms	lb	pound
C-S	Chao-Seader equation	lb _f	pound-force
C	degrees Celsius, K-273.2	lb _m	pound-mass
cal	calorie	lbmol	pound-mole
cfh	cubic feet per hour	ln	logarithm to the base e
cfm	cubic feet per minute	log	logarithm to the base 10
cfs	cubic feet per second	M	molar
cm	centimeter	MSMPR	mixed-suspension, mixed-product removal
cmHg	pressure in centimeters head of mercury	MSC	molecular-sieve carbon
cP	centipoise	MSA	mass separating agent
cw	cooling water	MW	megawatts
EMD	equimolar counter diffusion	m	meter
EOS	equation of state	meq	milliequivalents
ESA	energy separating agent	mg	milligram
ESS	error sum of squares	min	minute
eq	equivalents	mm	millimeter
F	degrees Fahrenheit, R 459.7	mmHg	pressure in mm head of mercury
FUG	Fenske-Underwood-Gilliland	mmol	millimole (0.001 mole)
ft	feet	mol	gram-mole
GLC-EOS	group-contribution equation of state	mole	gram-mole
GP	gas permeation	N	newton; normal
g	gram	NLE	nonlinear equation
gmol	gram-mole	NRTL	nonrandom, two-liquid theory
gpd	gallons per day	nbp	normal boiling point
gph	gallons per hour	ODE	ordinary differential equation
gpm	gallons per minute	PDE	partial differential equation
gps	gallons per second	POD	Podbielniak extractor
H	high boiler	P-R	Peng-Robinson equation of state
HHK	heavier than heavy key component	ppm	parts per million (usually by weight)
HK	heavy-key component	PSA	pressure-swing adsorption
hp	horsepower	psi	pounds force per square inch
h	hour	psia	pounds force per square inch absolute
I	intermediate boiler	PV	pervaporation

RDC	rotating-disk contactor	scfh	standard cubic feet per hour
RHS	right-hand side of an equation	scfm	standard cubic feet per minute
R-K	Redlich-Kwong equation of state	stm	steam
R-K-S	Redlich-Kwong-Soave equation of state (same as S-R-K)	TSA	temperature-swing adsorption
RO	reverse osmosis	UMD	unimolecular diffusion
RTL	raining-bucket contactor	UNIFAC	UNIQUAC functional group activity coefficients
R	degrees Rankine	UNIQUAC	universal quasi-chemical theory
SC	simultaneous-correction method	VOC	volatile organic compound
SG	silica gel	VPE	vibrating-plate extractor
S.G.	specific gravity	vs	versus
SR	stiffness ratio; sum-rates method	VSA	vacuum-swing adsorption
S-R-K	Soave-Redlich-Kwong equation of state	wt	weight
STP	standard conditions of temperature and pressure (usually 1 atm and either 0C or 60F)	y	year
s	second	yr	year
scf	standard cubic feet	μm	micron = micrometer
scfd	standard cubic feet per day		

Mathematical Symbols

d	differential	$\ln$	natural logarithm
e	exponential function	$\log$	logarithm to the base 10
$\text{erf}(x)$	error function of		partial differential
$\text{erfc}(x)$	complementary error function of $x = 1 - \text{erf}(x)$	$\{ \}$	delimiters for a function
$\exp$	exponential function		delimiters for absolute value
f	function		sum
i	imaginary part of a complex value		product; $\pi = 3.1416$

Dimensions and Units

Chemical engineers must be proficient in the use of three systems of units: (1) the International System of Units, *SI System* (Système Internationale d'Unités), which was established in 1960 by the 11th General Conference on Weights and Measures and has been widely adopted; (2) the *AE* (American Engineering) *System*, which is based largely upon an English system of units adopted when the Magna Carta was signed in 1215 and is the preferred system in the United States; and (3) the *CGS* (centimeter-gram-second) *System*, which was devised in 1790 by the National Assembly of France, and served as the basis for the development of the SI System. A useful index to units and systems of units is given on the website at <http://www.sizes.com/units/index.htm>

Engineers must deal with *dimensions* and *units* to express the dimensions in terms of numerical *values*. Thus, for 10 gallons of gasoline, the dimension is volume, the unit is gallons, and the value is 10. As detailed in NIST (National Institute of Standards and Technology) Special Publication 811 (1995 edition), which is available at the website <http://physics.nist.gov/cuu/pdf/sp811.pdf>, units are *base* or *derived*.

BASE UNITS

The base units are those that cannot be subdivided, are independent, and are accurately defined. The base units are for dimensions of length, mass, time, temperature, molar amount, electrical current, and luminous intensity, all of which can be measured independently. Derived units are expressed in terms of base units or other derived units and include dimensions of volume, velocity, density, force, and energy. In this book we deal with the first five of the base dimensions. For these, the base units are:

Base Dimension	SI Unit	AE Unit	CGS Unit
Length	meter, m	foot, ft	centimeter, cm
Mass	kilogram, kg	pound, lb _m	gram, g
Time	second, s	hour, h	second, s
Temperature	kelvin, K	Fahrenheit, F	Celsius, C
Molar amount	gram-mole, mol	pound-mole, lbmol	gram-mole, mol

DERIVED UNITS

Many derived dimensions and units are used in chemical engineering. Several are listed in the following table:

Derived Dimension	SI Unit	AE Unit	CGS Unit
Area = Length ²	m ²	ft ²	cm ²
Volume = Length ³	m ³	ft ³	cm ³
Mass flow rate = Mass/Time	kg/s	lb _m /h	g/s
Molar flow rate = Molar amount/Time	mol/s	lbmol/h	mol/s
Velocity = Length/Time	m/s	ft/h	cm/s
Acceleration = Velocity/Time	m/s ²	ft/h ²	cm/s ²
Force = Mass Acceleration	newton, N = 1 kg m/s ²	lb _f	dyne = 1 g cm/s ²

(Continued)

Derived Dimension	SI Unit	AE Unit	CGS Unit
Pressure = Force/Area	pascal, Pa = 1 N/m ² = 1 kg/m s ²	lb _f /in ²	atm
Energy = Force Length	joule, J = 1 N m = 1 kg m ² /s ²	ft lb _f , Btu	erg = 1 dyne cm = 1 g cm ² /s ² , cal
Power = Energy/Time = Work/Time	Watt, W = 1 J/s = 1 N m/s = 1 kg m ² /s ³	hp	erg/s
Density = Mass/Volume	kg/m ³	lb _m /ft ³	g/cm ³

OTHER UNITS ACCEPTABLE FOR USE WITH THE SI SYSTEM

A major advantage of the SI System is the consistency of the derived units with the base units. However, some acceptable deviations from this consistency and some other acceptable base units are given in the following table:

Dimension	Base or Derived SI Unit	Acceptable SI Unit
Time	s	minute (min), hour (h), day (d), year (y)
Volume	m ³	liter (L) = 10 ⁻³ m ³
Mass	kg	metric ton or tonne (t) = 10 ³ kg
Pressure	Pa	bar = 10 ⁵ Pa

PREFIXES

Also acceptable for use with the SI System are decimal multiples and submultiples of SI units formed by prefixes. The following table lists the more commonly used prefixes:

Prefix	Factor	Symbol
giga	10 ⁹	G
mega	10 ⁶	M
kilo	10 ³	k
deci	10 ⁻¹	d
centi	10 ⁻²	c
milli	10 ⁻³	m
micro	10 ⁻⁶	μ
nano	10 ⁻⁹	n
pico	10 ⁻¹²	p

USING THE AE SYSTEM OF UNITS

The AE System is more difficult to use than the SI System because of the units used with force, energy, and power. In the AE System, the force unit is the pound-force, lb_f, which is defined to be numerically equal to the pound-mass, lb_m, at sea-level of the Earth. Accordingly, Newton's second law of motion is written,

where F = force in lb_f, m = mass in lb_m, g = acceleration due to gravity in ft/s², and to complete the definition, $g_c = 32.174 \text{ lb}_m \text{ ft/lb}_f \text{ s}^2$, where 32.174 ft/s² is the acceleration due to gravity at sea-level of the Earth. The constant, g_c , is not used with the SI System or the CGS System because the former does not define a kg_f and the CGS System does not use a g_f.

Thus, when using AE units in an equation that includes force and mass, incorporate g_c to adjust the units.

Example

A 5,000-pound-mass weight, m , is held at a height, h , of 4,000 feet above sea-level. Calculate its potential energy above sea-level, $P.E. = mgh$, using each of the three systems of units. Factors for converting units are given on the inside back cover of this book.

SI System:

CGS System:

AE System:

However, the accepted unit of energy for the AE System is $\text{ft} \cdot \text{lb}_f$, which is obtained by dividing by g_c . Therefore, $P.E. = 643.5/32.174 = 20.00 \text{ ft} \cdot \text{lb}_f$

Another difficulty with the AE System is the differentiation between energy as work and energy as heat. As seen in the preceding table, the work unit is $\text{ft} \cdot \text{lb}_f$, while the heat unit is Btu. A similar situation exists in the CGS System with corresponding units of erg and calorie (cal). In older textbooks, the conversion factor between work and heat is often incorporated into an equation with the symbol J , called Joule's constant or the mechanical equivalent of heat, where,

Thus, in the previous example, the heat equivalents are

AE System:

CGS System:

In the SI System, the prefix M, mega, stands for million. However, in the natural gas and petroleum industries of the United States, when using the AE System, M stands for thousand and MM stands for million. Thus, MBtu stands for thousands of Btu, while MM Btu stands for millions of Btu.

It should be noted that the common pressure and power units in use for the AE System are not consistent with the base units. Thus, for pressure, pounds per square inch, psi or lb_f/in^2 , is used rather than lb_f/ft^2 . For power, hp is used instead of $\text{ft} \cdot \text{lb}_f/\text{h}$, where, the conversion factor is

CONVERSION FACTORS

Physical constants may be found on the inside front cover of this book. Conversion factors are given on the inside back cover. These factors permit direct conversion of AE and CGS values to SI values. The following is an example of such a conversion together with the reverse conversion.

Example

Convert 50 psia (lb_f/in^2 absolute) to kPa:
The conversion factor for lb_f/in^2 to Pa is 6895, which results in
 $50(6895) = 345000 \text{ Pa}$ or 345 kPa

Convert 250 kPa to atm:
 $250 \text{ kPa} = 250000 \text{ Pa}$. The conversion factor for atm to Pa is
. Therefore, dividing by the conversion factor,
atm

Three of the units [gallons (gal), calories (cal), and British thermal unit (Btu)] in the list of conversion factors have two or more definitions. The gallons unit cited here is the U.S. gallon, which is 83.3% of the Imperial gallon. The cal and Btu units used here are international (IT). Also in common use are the thermochemical cal and Btu, which are 99.964% of the international cal and Btu.

FORMAT FOR EXERCISES IN THIS BOOK

In numerical exercises throughout this book, the system of units to be used to solve the problem is stated. Then when given values are substituted into equations, units are not appended to the values. Instead, the conversion of a given value to units in the above tables of base and derived units is done prior to substitution in the equation or carried out directly in the equation as in the following example.

Example

Using conversion factors on the inside back cover of this book, calculate a Reynolds number, Re , given $D = 4.0 \text{ ft}$, $v = .5 \text{ ft/s}$, $\rho = 62.4 \text{ lb}_m/\text{ft}^3$, and $\mu = 2.0 \text{ cP}$ (i.e., centipoise).

Using the SI System (kg-m-s),

Using the CGS System (g-cm-s),

Using the AE System ($\text{lb}_m\text{-ft-h}$)

Convert the viscosity of 0.02 g/cm s to $\text{lb}_m/\text{ft h}$:

**SEPARATION
PROCESS
PRINCIPLES**

Part 1

Fundamental Concepts

In the first five chapters, fundamental concepts are presented that apply to processes for the separation of chemical mixtures. Emphasis is on industrial processes, but many of the concepts apply to small-scale separations as well. In Chapter 1, the role of separation operations in chemical processes is illustrated. Five general separation techniques are enumerated, each being driven by energy and/or the addition of mass to alter properties important to separation. For each technique, equipment types are briefly described. Various ways of specifying separation operations are discussed, including component recovery and product purity, and the use of these specifications in making mass balances is illustrated. The selection of feasible equipment for a particular separation problem is briefly covered.

The degree to which a separation can be achieved depends on differing rates of mass transfer of the individual components of the mixture, with limits dictated by thermodynamic phase equilibrium. Chapter 2 is a review of thermodynamics applicable to separation operations, particularly those involving fluid phases. Chapter 3 is an extensive discussion of mass transfer of individual components in binary mixtures under

stagnant, laminar-flow, and turbulent-flow conditions, by analogy to conductive and convective heat transfer wherever possible.

Many separation operations are designed on the basis of the limit of attaining thermodynamic phase equilibrium. Chapter 4 covers mass-balance calculations for phase equilibrium in a single contacting stage that may include vapor, liquid, and/or solid phases. Often the degree of separation can be greatly improved by using multiple contacting stages, with each stage approaching equilibrium, in a cascade and/or by using a sequence of two or more different types of separation methods in a hybrid system. These are of great importance to industrial separation processes and are briefly described in Chapter 5, before proceeding to subsequent chapters in this book, each focusing on detailed descriptions and calculations for a particular separation operation. Included in Chapter 5 is a detailed discussion of degrees-of-freedom analysis, which determines the number of allowable specifications for cascades and hybrid systems. This type of analysis is used throughout this book, and is widely used in process simulators such as ASPEN PLUS, CHEMCAD, and HYSYS.

Chapter 1

Separation Processes

The separation of chemical mixtures into their constituents has been practiced, as an art, for millennia. Early civilizations developed techniques to (1) extract metals from ores, perfumes from flowers, dyes from plants, and potash from the ashes of burnt plants, (2) evaporate sea water to obtain salt, (3) refine rock asphalt, and (4) distill liquors. The human body could not function for long if it had no kidney, a membrane that selectively removes water and waste products of metabolism from blood.

Separations, including enrichment, concentration, purification, refining, and isolation, are important to chemists and chemical engineers. The former use *analytical separation methods*, such as chromatography, to determine compositions of complex mixtures quantitatively. Chemists also use small-scale *preparative separation techniques*, often similar to analytical separation methods, to recover and purify chemicals. Chemical engineers are more concerned with the manufacture of chemicals using economical, large-scale separation methods, which may differ considerably from laboratory techniques. For example, in a laboratory, chemists separate and analyze light-hydrocarbon mixtures by gas-liquid chromatography, while in a large manufacturing plant a chemical engineer uses distillation to separate the same hydrocarbon mixtures.

This book presents the principles of large-scale component separation operations, with emphasis on methods applied by chemical engineers to produce useful chemical products economically. Included are treatments of classical separation methods, such as distillation, absorption, liquid-liquid extraction, leaching, drying, and crystallization, as well as newer methods, such as adsorption and membrane separation. Separation operations for gas, liquid, and solid phases are covered. Using the principles of separation operations, chemical engineers can successfully develop, design, and operate industrial processes.

Increasingly, chemical engineers are being called upon to deal with industrial separation problems on a smaller scale, e.g., manufacture of specialty chemicals by batch processing, recovery of biological solutes, crystal growth of semiconductors, recovery of valuable chemicals from wastes, and the development of products (such as the artificial kidney) that involve the separation of chemical mixtures. Many of the separation principles for these smaller-scale problems are covered in this book and illustrated in examples and homework exercises.

1.0 INSTRUCTIONAL OBJECTIVES

After completing this chapter, you should be able to:

- Explain the role of separation operations in an industrial chemical process.
- Explain what constitutes the separation of a chemical mixture and enumerate the five general separation techniques.
- Explain the use of an energy-separating agent (ESA) and/or a mass-separating agent (MSA) in a separation operation.
- Explain how separations are made by phase creation or phase addition and list the many separation operations that use these two techniques.
- Explain how separations are made by introducing selective barriers and list several separation operations that utilize membranes.
- Explain how separations are made by introducing solid agents and list the three major separation operations that utilize this technique.
- Explain the use of external fields to separate chemical mixtures.
- Calculate component material balances around a separation operation based on specifications of component recovery (split ratios or split fractions) and/or product purity.

- Use the concepts of key components and separation power to measure the degree of separation between two key components.
- Make a selection of feasible separation operations based on factors involving the feed, products, property differences among chemical components, and characteristics of different separation operations.

1.1 INDUSTRIAL CHEMICAL PROCESSES

The chemical industry manufactures products that differ in chemical content from process feeds, which can be (1) naturally occurring raw materials, (2) plant or animal matter, (3) chemical intermediates, (4) chemicals of commerce, or (5) waste products. Especially common are oil refineries [1], which, as indicated in Figure 1.1, produce a variety of useful products. The relative amounts of these products produced from, say, 150,000 bbl/day of crude oil depend on the constituents of the crude oil and the types of refinery processes. Processes include distillation to separate crude oil into various boiling-point fractions or cuts, alkylation to combine small hydrocarbon molecules into larger molecules, catalytic reforming to change the structure of medium-size hydrocarbon molecules, fluid catalytic cracking to break apart large hydrocarbon molecules, hydrocracking to break apart even larger molecules, and other processes to convert the crude-oil residue to coke and lighter fractions.

A chemical process is conducted in either a *batchwise*, *continuous*, or *semicontinuous* manner. The operations may be classified as *key operations*, which are unique to chemical engineering because they involve changes in chemical composition, or *auxiliary operations*, which are necessary to the success of the key operations but may be designed by mechanical engineers as well because the auxiliary operations do not involve changes in chemical composition. The key operations are (1) chemical reactions and (2) separation of chemical mixtures. The auxiliary operations include phase separation, heat addition or removal (to change temperature or phase condition), shaft-work addition or removal (to

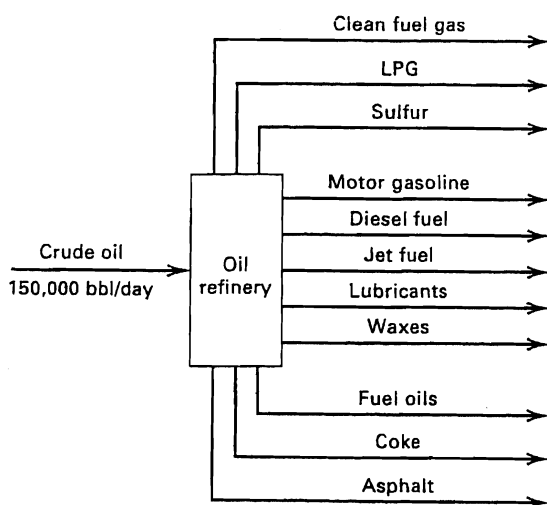


Figure 1.1 Refinery for converting crude oil into a variety of marketable products.

change pressure), mixing or dividing of streams or batches of material, solids agglomeration, size reduction of solids, and separation of solids by size.

The key operations for the separation of chemical mixtures into new mixtures and/or essentially pure components are of central importance. Most of the equipment in the average chemical plant is there to purify raw materials, intermediates, and products by the separation techniques described briefly in this chapter and discussed in detail in subsequent chapters.

Block-flow diagrams are used to represent chemical processes. They indicate, by square or rectangular blocks, chemical reaction and separation steps and, by connecting lines, the major process streams that flow from one processing step to another. Considerably more detail is shown in *process-flow diagrams*, which also include auxiliary operations and utilize symbols that depict more realistically the type of equipment employed. The block-flow diagram of a continuous process for manufacturing hydrogen chloride gas from evaporated chlorine and electrolytic hydrogen [2] is shown in Figure 1.2. The heart of the process is a chemical reactor, where the high-temperature gas-phase combustion reaction, $\text{H}_2 + \text{Cl}_2 \rightarrow 2\text{HCl}$, occurs. The only auxiliary equipment required consists of pumps and compressors to deliver feeds to the reactor and product to storage, and a heat exchanger to cool the product. For this process, no separation operations are necessary because complete conversion of chlorine occurs in the reactor. A slight excess of hydrogen is used, and the product, consisting of 99% HCl and small amounts of H_2 , N_2 , H_2O , CO , and CO_2 , requires no purification. Such simple commercial processes that require no separation of chemical species are very rare.

Some industrial chemical processes involve no chemical reactions, but only operations for separating chemicals and phases, together with auxiliary equipment. A block-flow diagram for such a process is shown in Figure 1.3, where wet natural gas is continuously separated into six light-paraffin

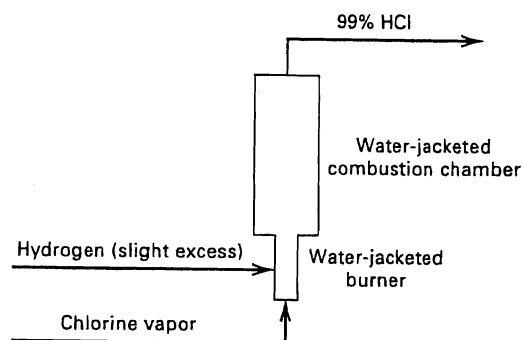


Figure 1.2 Synthetic process for anhydrous HCl production.

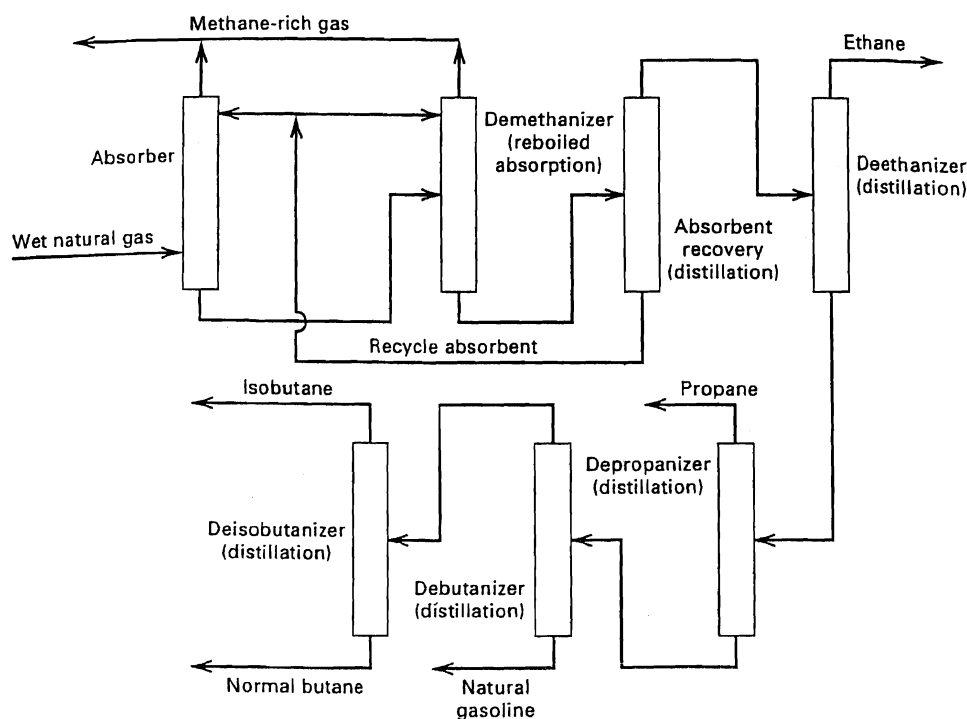


Figure 1.3 Process for recovery of light hydrocarbons from casinghead gas.

hydrocarbon components and mixtures by a train of separators [3]. A train or sequence of separators is used because it is often impossible to produce more than two products with a single piece of separation equipment.

Many industrial chemical processes involve at least one chemical reactor accompanied by one or more separation trains. An example is the continuous, direct hydration of ethylene to ethyl alcohol [4]. The heart of the process is a reactor packed with solid-catalyst particles, operating at 572 K and 6.72 MPa (570°F and 975 psia), in which the hydration reaction, $C_2H_4 + H_2O \rightarrow C_2H_5OH$, takes place. Because of thermodynamic equilibrium limitations, the conversion of ethylene is only 5% per pass through the reactor. The unreacted ethylene is recovered in a separation step and recycled back to the reactor. By this recycle technique, which is common to many industrial processes, essentially complete conversion of the ethylene fed to the process is achieved. If pure ethylene were available as a feedstock and no side reactions

occurred, the relatively simple process in Figure 1.4 could be constructed, in which two by-products (light ends and waste water) are also produced. This process uses a reactor, a partial condenser for ethylene recovery, and distillation to produce aqueous ethyl alcohol of near-azeotropic composition (93 wt%). Unfortunately, a number of factors frequently combine to increase the complexity of the process, particularly with respect to separation-equipment requirements. These factors include impurities in the ethylene feed, and side reactions involving both ethylene and feed impurities such as propylene. Consequently, the separation system must also deal with diethyl ether, isopropyl alcohol, acetaldehyde, and other chemicals. The resulting industrial process, shown in Figure 1.5, is much more complicated. After the hydration reaction, a partial condenser and high-pressure water absorber recover unreacted ethylene for recycling. The pressure of the liquid from the bottom of the absorber is reduced, causing partial vaporization. Vapor is separated from the

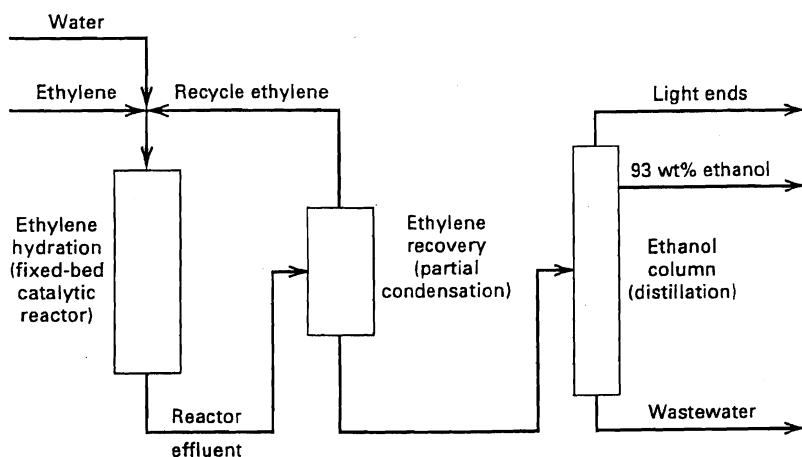


Figure 1.4 Hypothetical process for hydration of ethylene to ethanol.

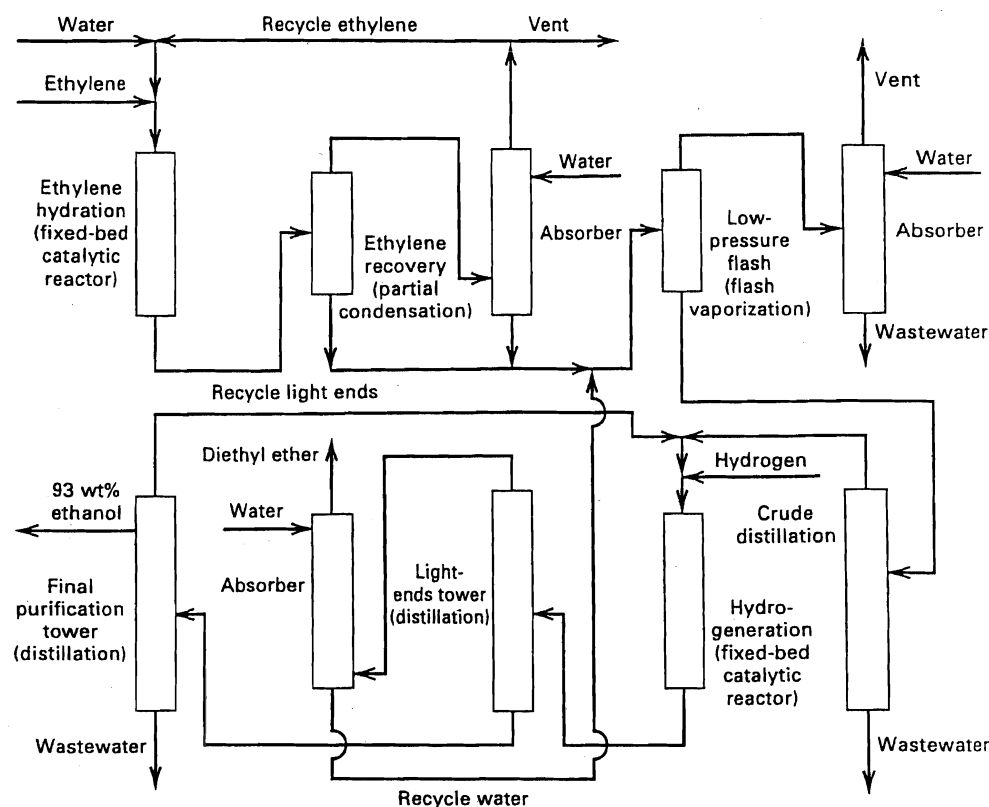


Figure 1.5 Industrial process for hydration of ethylene to ethanol.

remaining liquid in the low-pressure flash drum. Vapor from the low-pressure flash is scrubbed with water in an absorber to remove alcohol and prevent its loss to the vent gas. Crude, concentrated ethanol containing diethyl ether and acetaldehyde is distilled overhead in the crude-distillation (stripper) column and then catalytically hydrogenated in the vapor phase to convert acetaldehyde to ethanol. Diethyl ether is removed by distillation in the light-ends tower and scrubbed with water in an absorption tower. The final product is prepared by distillation in the final-purification tower, where 93 wt% aqueous ethanol product is withdrawn several trays below the top tray, light ends are concentrated in the so-called pasteurization-tray section above the product-withdrawal tray and recycled to the catalytic-hydrogenation reactor, and wastewater is removed from the bottom of the tower. Besides the separation equipment shown, additional separation steps may be necessary to concentrate the ethylene feed to the process and remove impurities that poison the catalysts. In the development of a new process from the laboratory stage through the pilot-plant stage, experience shows that more separation steps than originally anticipated are usually needed.

The above examples serve to illustrate the importance of separation operations in industrial chemical processes. Such operations are employed not only to separate a feed mixture into other mixtures and relatively pure components, to recover solvents for recycle, and to remove wastes, but also, when used in conjunction with chemical reactors, to purify reactor feeds, recover reactants from reactor effluents for recycle, recover by-products, and recover and purify products

to meet required specifications. Sometimes a separation operation, such as absorption of SO_2 by limestone slurry, may be accompanied by a chemical reaction that serves to facilitate the separation. In this book, emphasis is on separation operations that do not rely on concurrent chemical reactions; however, reactive distillation is discussed in Chapter 11.

Chemical engineers also design products. A significant product that involves the separation of chemicals is the espresso machine for making a cup of coffee that is superior to that made in a filter-drip machine. The goal in coffee making is to leach from the coffee beans the best oils, leaving behind ingredients responsible for acidity and bitterness in the cup of coffee. The espresso machine accomplishes this by conducting the leaching operation rapidly in 20–30 seconds with water at high temperature and pressure. If the operation is carefully controlled, the resulting cup of espresso, if immediately consumed, has: (1) a topping of creamy foam that traps the extracted chemicals, (2) a fullness of body due to emulsification, and (3) a richness of aroma. Typically, 25% of the coffee bean is extracted and the espresso contains less caffeine than filtered coffee. Cussler and Moggridge [13] and Seider, Seader, and Lewin [14] discuss many other examples of products designed by chemical engineers, some of which involve the separation of chemicals.

1.2 MECHANISM OF SEPARATION

Mixing of chemicals is a spontaneous, natural process that is accompanied by an increase in entropy or randomness. The inverse process, the separation of that mixture into its

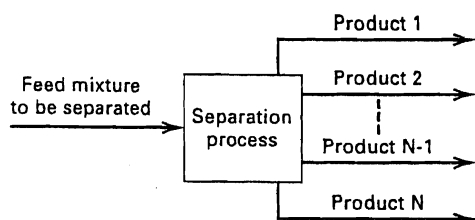


Figure 1.6 General separation process.

constituent chemical species, is not a spontaneous process; it requires an expenditure of energy. A mixture to be separated usually originates as a single, homogeneous phase (solid, liquid, or gas). If it exists as two or more immiscible phases, it is often best to first use a mechanical means based on gravity, centrifugal force, pressure reduction, or an electric and/or magnetic field to separate the phases. Then, appropriate separation techniques are applied to each phase.

A schematic diagram of a general separation process is shown in Figure 1.6. The feed mixture can be vapor, liquid, or solid, while the two or more products may differ in composition from each other and the feed, and may differ in phase state from each other and/or from the feed. The separation is accomplished by forcing the different chemical species in the feed into different spatial locations by any of five general separation techniques, or combinations thereof, as shown in Figure 1.7. The most common industrial technique, Figure 1.7a, involves the creation of a second phase (vapor, liquid, or solid) that is immiscible with the feed phase. The creation is accomplished by energy (heat and/or shaft-work) transfer to or from the process or by pressure

reduction. A second technique, Figure 1.7b, is to introduce the second phase into the system in the form of a solvent that selectively dissolves some of the species in the feed. Less common, but of growing importance, is the use of a barrier, Figure 1.7c, which restricts and/or enhances the movement of certain chemical species with respect to other species. Also of growing importance are techniques that involve the addition of solid particles, Figure 1.7d, which act directly or as inert carriers for other substances so as to cause separation. Finally, external fields, Figure 1.7e, of various types are sometimes applied for specialized separations.

For all the techniques of Figure 1.7, separations are achieved by enhancing the rate of mass transfer by diffusion of certain species relative to mass transfer of all species by bulk movement within a particular phase. The driving force and direction of mass transfer by diffusion is governed by thermodynamics, with the usual limitations of equilibrium. Thus, both transport and thermodynamic considerations are crucial in separation operations. The rate of separation is governed by *mass transfer*, while the extent of separation is limited by *thermodynamic equilibrium*. These two topics are treated in Chapters 2, 3, and 4. Fluid mechanics and heat transfer also play important roles, and applicable principles are included in appropriate chapters, particularly with respect to phase separation, phase change, pressure drop, temperature change, and entrainment.

The extent of separation achieved between or among the product phases for each of the chemical species present in the feed depends on the exploitation of differences in molecular, thermodynamic, and transport properties of the

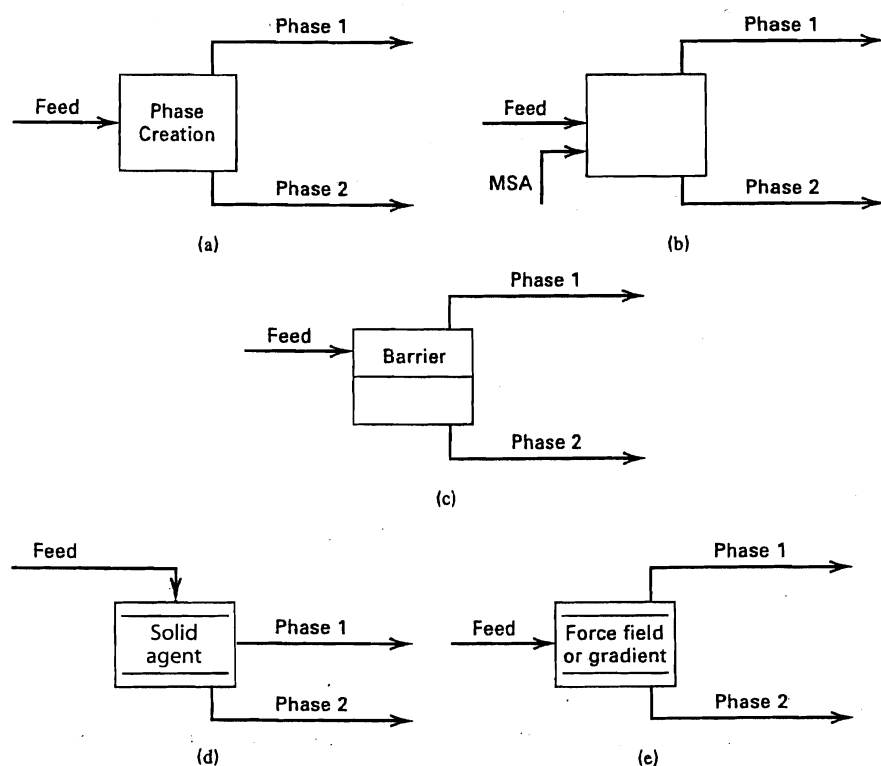


Figure 1.7 General separation techniques: (a) separation by phase creation; (b) separation by phase addition; (c) separation by barrier; (d) separation by solid agent; (e) separation by force field or gradient.

species in the different phases present. Some properties of importance are:

1. Molecular properties

Molecular weight	Polarizability
van der Waals volume	Dielectric constant
van der Waals area	Electric charge
Molecular shape (acentric factor)	Radius of gyration
Dipole moment	

2. Thermodynamic and transport properties

Vapor pressure	Adsorptivity
Solubility	Diffusivity

Values of these properties for many substances are available in handbooks, specialized reference books, and journals. Many of these properties can also be estimated using computer-aided, process-simulation programs. When they are not available, these properties must be estimated or determined experimentally if a successful application of the appropriate separation operation(s) is to be achieved.

1.3 SEPARATION BY PHASE ADDITION OR CREATION

If the feed mixture is a homogeneous, single-phase solution (gas, liquid, or solid), a second immiscible phase must often be developed or added before separation of chemical species can be achieved. This second phase is created by an *energy-separating agent* (ESA) and/or added as a *mass-separating agent* (MSA). Application of an ESA involves heat transfer and/or transfer of shaft work to or from the mixture to be separated. Alternatively, vapor may be created from a liquid phase by reducing the pressure. An MSA may be partially immiscible with one or more of the species in the mixture. In this case, the MSA frequently remains the constituent of highest concentration in the added phase. Alternatively, the MSA may be completely miscible with a liquid mixture to be separated, but may selectively alter the partitioning of species between liquid and vapor phases. This facilitates a more complete separation when used in conjunction with an ESA, as in extractive distillation.

Although separations that use an ESA are generally preferred, an MSA can make possible a separation that is not feasible with an ESA. Disadvantages of the use of an MSA are: (1) need for an additional separator to recover the MSA for recycle, (2) need for MSA makeup, (3) possible contamination of the product with the MSA, and (4) more difficult design procedures.

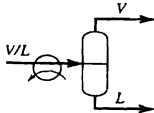
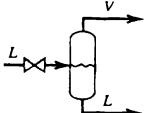
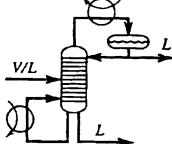
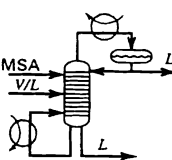
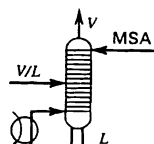
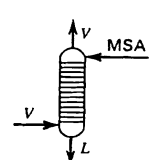
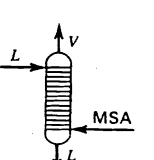
When two immiscible fluid phases are contacted, intimate mixing of the two phases is important in enhancing mass-transfer rates so that the thermodynamic-maximum degree-of-partitioning of species can be approached more rapidly. After phase contact, the separation operation is completed by employing gravity and/or an enhanced technique, such as centrifugal force, to disengage the two phases. Table 1.1 is a compilation of the more common industrial-

separation operations based on interphase mass transfer between two phases, one of which is created by an ESA or added as an MSA. Graphic symbols that are suitable for process-flow diagrams are included in the table. Vapor and liquid and/or solid phases are designated by V, L, and S, respectively. Design procedures have become fairly routine for the operations prefixed by an asterisk (*) in the first column of Table 1.1. Such procedures have been incorporated as mathematical models into widely used commercial computer-aided, chemical-process simulation and design (CAPD) programs for continuous, steady-state operations and are treated in considerable detail in subsequent chapters of this book. Batchwise modes of these operations are also treated in this book when appropriate.

When the feed mixture includes species that differ widely in their tendency to vaporize and condense, *partial condensation* or *partial vaporization*, Separation Operation (1) in Table 1.1 may be adequate to achieve the desired separation or recovery of a particular component. A vapor feed is partially condensed by removing heat, and a liquid feed is partially vaporized by adding heat. Alternatively, partial vaporization can be caused by *flash vaporization*, Operation (2), by reducing the pressure of the feed with a valve. In both of these operations, after partitioning of species by interphase mass transfer has occurred, the resulting vapor phase is enriched with respect to the species that are most volatile (most easily vaporized), while the liquid phase is enriched with respect to the least volatile species. After this single contact, the two phases, which, except near the critical region, are of considerably different density, are separated by gravity.

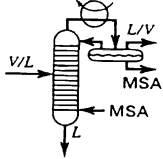
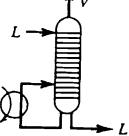
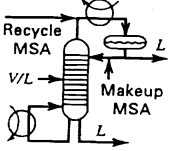
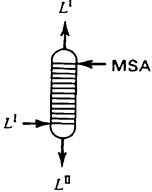
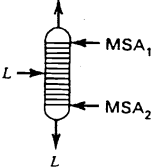
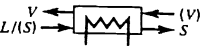
Often, the degree of species separation achieved by a single, partial vaporization or partial condensation step is inadequate because the volatility differences among species in the feed are not sufficiently large. In that case, it may still be possible to achieve a desired separation of the species in the feed mixture, without introducing an MSA, by employing *distillation*, Operation (3) in Table 1.1, the most widely utilized industrial separation method. Distillation involves multiple contacts between countercurrently flowing liquid and vapor phases. Each contact consists of mixing the two phases to promote rapid partitioning of species by mass transfer, followed by phase separation. The contacts are often made on horizontal trays (referred to as *stages*) arranged in a vertical column as shown in the symbol for distillation in Table 1.1. Vapor, while flowing up the column, is increasingly enriched with respect to the more volatile species. Correspondingly, liquid flowing down the column is increasingly enriched with respect to the less-volatile species. Feed to the distillation column enters on a tray somewhere between the top and bottom trays, and often near the middle of the column. The portion of the column above the feed entry is called the *enriching* or *rectification section*, and that below is the *stripping section*. Feed vapor starts up the column; feed liquid starts down. Liquid is required for making contacts with vapor above the feed tray, and vapor is required for making contacts with liquid below the feed tray.

Table 1.1 Separation Operations Based on Phase Creation or Addition

Separation Operation	Symbol ^a	Initial or Feed Phase	Created or Added Phase	Separating Agent(s)	Industrial Example ^b
Partial condensation or vaporization* (1)		Vapor and/or liquid	Liquid or vapor	Heat transfer (ESA)	Recovery of H ₂ and N ₂ from ammonia by partial condensation and high-pressure phase separation (Vol. 2, pp. 494–496)
Flash vaporization* (2)		Liquid	Vapor	Pressure reduction	Recovery of water from sea water (Vol. 24, pp. 343–348)
Distillation* (3)		Vapor and/or liquid	Vapor and liquid	Heat transfer (ESA) and sometimes work transfer	Purification of styrene (Vol. 21, pp. 785–786)
Extractive distillation* (4)		Vapor and/or liquid	Vapor and liquid	Liquid solvent (MSA) and heat transfer (ESA)	Separation of acetone and methanol (Suppl. Vol., pp. 153–155)
Reboiled absorption* (5)		Vapor and/or liquid	Vapor and liquid	Liquid absorbent (MSA) and heat transfer (ESA)	Removal of ethane and lower molecular weight hydrocarbons for LPG production (Vol. 14, pp. 384–385)
Absorption* (6)		Vapor	Liquid	Liquid absorbent (MSA)	Separation of carbon dioxide from combustion products by absorption with aqueous solutions of an ethanolamine (Vol. 4, pp. 730–735)
Stripping* (7)		Liquid	Vapor	Stripping vapor (MSA)	Stream stripping of naphtha, kerosene, and gas oil side cuts from crude distillation unit to remove light ends (Vol. 17, pp. 199–201)

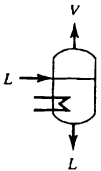
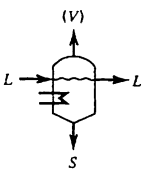
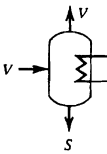
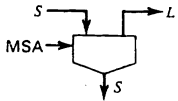
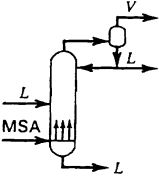
(Continued)

Table 1.1 (Continued)

Separation Operation	Symbol ^a	Initial or Feed Phase	Created or Added Phase	Separating Agent(s)	Industrial Example ^b
Refluxed stripping (steam distillation)* (8)		Vapor and/or liquid	Vapor and liquid	Stripping vapor (MSA) and heat transfer (ESA)	Separation of products from delayed coking (Vol. 17, pp. 210–215)
Reboiled stripping* (9)		Liquid	Vapor	Heat transfer (ESA)	Recovery of amine absorbent (Vol. 17, pp. 229–232)
Azeotropic distillation* (10)		Vapor and/or liquid	Vapor and liquid	Liquid entrainer (MSA) and heat transfer (ESA)	Separation of acetic acid from water using <i>n</i> -butyl acetate as an entrainer to form an azeotrope with water (Vol. 3, pp. 365–368)
Liquid–liquid extraction* (11)		Liquid	Liquid	Liquid solvent (MSA)	Recovery of aromatics (Vol. 9, pp. 707–709)
Liquid–liquid extraction (two-solvent)* (12)		Liquid	Liquid	Two liquid solvents (MSA ₁ and MSA ₂)	Use of propane and cresylic acid as solvents to separate paraffins from aromatics and naphthenes (Vol. 17, pp. 223–224)
Drying (13)		Liquid and often solid	Vapor	Gas (MSA) and/or heat transfer (ESA)	Removal of water from polyvinylchloride with hot air in a fluid-bed dryer (Vol. 23, pp. 901–904)

(Continued)

Table 1.1 (Continued)

Separation Operation	Symbol ^a	Initial or Feed Phase	Created or Added Phase	Separating Agent(s)	Industrial Example ^b
Evaporation (14)		Liquid	Vapor	Heat transfer (ESA)	Evaporation of water from a solution of urea and water (Vol. 23, pp. 555–558)
Crystallization (15)		Liquid	Solid (and vapor)	Heat transfer (ESA)	Crystallization of <i>p</i> -xylene from a mixture with <i>m</i> -xylene (Vol. 24, pp. 718–723)
Desublimation (16)		Vapor	Solid	Heat transfer (ESA)	Recovery of phthalic anhydride from noncondensable gas (Vol. 17, pp. 741–742)
Leaching (liquid–solid extraction) (17)		Solid	Liquid	Liquid solvent	Extraction of sucrose from sugar beets with hot water (Vol. 21, pp. 907–908)
Foam fractionation (18)		Liquid	Gas	Gas bubbles (MSA)	Recovery of detergents from waste solutions (Vol. 10, pp. 544–545)

*Design procedures are fairly well accepted.

^aTrays are shown for columns, but alternatively packing can be used. Multiple feeds and side streams are often used and may be added to the symbol.

^bCitations refer to volume and page(s) of *Kirk-Othmer Encyclopedia of Chemical Technology*, 3rd ed., John Wiley and Sons, New York (1978–1984).

Often, vapor from the top of the column is condensed in a condenser by cooling water or a refrigerant to provide contacting liquid, called *reflux*. Similarly, liquid at the bottom of the column passes through a reboiler, where it is heated by condensing steam or some other heating medium to provide contacting vapor, called *boilup*.

When volatility differences between species to be separated are so small as to necessitate more than about 100 trays in a distillation operation, *extractive distillation*, Operation (4), is often considered. Here, an MSA, acting as a solvent, is used to increase volatility differences between selected species of the feed, thereby reducing the number of required trays. Generally, the MSA, which must be completely miscible with the liquid phase throughout the column, is less volatile than any of the species in the feed mixture and is introduced to a stage near the top of the column. Reflux to the top tray is utilized to minimize MSA content in the top product. A subsequent separation operation, usually distillation, is used to recover the MSA for recycling back to the extractive distillation column.

If condensation of vapor leaving the top of a distillation column is not easily accomplished by heat transfer to cooling water or a refrigerant, a liquid MSA called an *absorbent* may be introduced to the top tray in place of reflux. The resulting separation operation is called *reboiled absorption*, (5). If the feed is all vapor and the stripping section of the column is not needed to achieve the desired separation, the operation is referred to as *absorption*, (6). This operation may not require an ESA and is frequently conducted at ambient temperature and high pressure. Constituents of the vapor feed dissolve in the absorbent to varying extents depending on their solubilities. Vaporization of a small fraction of the absorbent also generally occurs.

The inverse of absorption is *stripping*, Operation (7) in Table 1.1. Here, a liquid mixture is separated, generally at elevated temperature and ambient pressure, by contacting liquid feed with a stripping agent. This MSA eliminates the need to reboil the liquid at the bottom of the column, which may be important if the liquid is not thermally stable. If contacting trays are also needed above the feed tray in order to achieve the desired separation, a *refluxed stripper*, (8), may be employed. If the bottoms product from a stripper is thermally stable, it may be reboiled without using an MSA. In that case, the column is called a *reboiled stripper*, (9). Additional separation operations are required to recover, for recycling, MSAs used in absorption and stripping operations.

The formation of minimum-boiling azeotropic mixtures makes *azeotropic distillation*, (10), another useful tool where separation by distillation is not feasible. In the example cited in Table 1.1, the MSA, *n*-butyl acetate, which forms a heterogeneous (i.e., two liquid phases present), minimum-boiling azeotrope with water, is used as an entrainer to facilitate the separation of acetic acid from water. The azeotrope is taken overhead and then condensed and separated into acetate and water layers. The MSA is recirculated, and the distillate water layer and bottoms acetic acid are removed as products.

Liquid-liquid extraction, (11) and (12), using one or two solvents, respectively, is widely used when distillation is impractical, especially when the mixture to be separated is temperature-sensitive and/or more than about 100 distillation stages would be required. When one solvent is used, it selectively dissolves only one or a fraction of the components in the feed mixture. In a two-solvent extraction system, each solvent has its own specific selectivity for dissolving the components of the feed mixture. Thus, if a feed mixture consists of species A and B, solvent C might preferentially dissolve A, while solvent D dissolves B. As with extractive distillation, additional separation operations are generally required to recover, for recycling, solvent from streams leaving the extraction operation.

A variation of liquid-liquid extraction is *supercritical-fluid extraction*, where the extraction temperature and pressure are slightly above the critical point of the solvent. In this region, solute solubility in the supercritical fluid changes drastically with small changes in temperature and pressure. Following extraction, the pressure of the extract can be reduced to release the solvent, which is then recycled. For the processing of foodstuffs, the supercritical fluid is an inert substance such as CO₂, which will not contaminate the product.

Since many chemicals are processed wet but sold as dry solids, one of the more common manufacturing steps is *drying*, (13), which involves removal of a liquid from a solid by vaporization of the liquid. Although the only basic requirement in drying is that the vapor pressure of the liquid to be evaporated be higher than its partial pressure in the gas stream, the design and operation of dryers represents a complex problem in heat transfer, fluid mechanics, and mass transfer. In addition to the effect of such external conditions as temperature, humidity, air flow rate, and degree of solid subdivision on drying rate; the effect of internal conditions of liquid and vapor diffusion, capillary flow, equilibrium moisture content, and heat sensitivity in the solid must be considered. Although drying is a multiphase mass-transfer process, equipment-design procedures differ from those of any of the other processes discussed in this chapter because the thermodynamic concepts of equilibrium are difficult to apply to typical drying situations, where the concentration of vapor in the gas is so far from saturation, and concentration gradients in the solid are such that mass-transfer driving forces are undefined. Also, heat transfer rather than mass transfer may well be the limiting rate process. Therefore, the typical dryer design procedure is for the process engineer to send a few tons of representative, wet sample material for pilot-plant tests by one or two reliable dryer manufacturers and to purchase the equipment that produces a satisfactorily dried product at the lowest cost. The types of commercial dryers are discussed in detail in *Perry's Chemical Engineers' Handbook* [5] and Chapter 18.

Evaporation, Operation (14) in Table 1.1, is generally defined as the transfer of volatile components of a liquid into a gas by volatilization caused by heat transfer. Humidification and evaporation are synonymous in the scientific sense; however, *humidification* or *dehumidification* implies that

one is intentionally adding vapor to or removing vapor from a gas. Major applications of evaporation are humidification, conditioning of air, cooling of water, and the concentration of aqueous solutions.

Crystallization, (15), is carried out in many organic, and almost all inorganic, chemical manufacturing plants where the desired product is a finely divided solid. Since crystallization is essentially a purification step, the conditions in the crystallizer must be such that impurities do not precipitate with the desired product. In *solution crystallization*, the mixture, which includes a solvent, is cooled and/or the solvent is evaporated to cause crystallization. In *melt crystallization*, two or more soluble species, in the absence of a solvent, are separated by partial freezing. A particularly versatile melt crystallization technique is *zone melting* or *refining*, which relies on selective distribution of impurity solutes between a liquid and a solid phase to achieve a separation. Many metals are refined by this technique, which, in its simplest form, involves moving a molten zone slowly through an ingot by moving the heater or drawing the ingot past the heater. The manufacture of single crystals has been a vital development in the semiconductor industry in recent years. Typically single crystals of very high purity silicon are produced worldwide by the Czochralski technique, wherein a single crystal is pulled from a melt. Typical crystal dimensions, after shaping into a uniform rod with diamond grinding machines, are 150-mm diameter $\times$ 1-m long, from which wafers of 675-micron thickness are sawed.

Sublimation is the transfer of a substance from the solid to the gaseous state without formation of an intermediate liquid phase, usually at a relatively high vacuum. Major applications have been in the removal of a volatile component from an essentially nonvolatile one. Examples are separation of sulfur from impurities, purification of benzoic acid, and freeze-drying of foods. The reverse process, *desublimation*, (16), is also practiced, for example, in the recovery of phthalic anhydride from gaseous reactor effluent. The most common application of sublimation in everyday life occurs in the use of Dry Ice as a refrigerant for storing ice cream, vegetables, and other perishables. The sublimed gas, unlike water, does not puddle and spoil the frozen materials.

Liquid-solid extraction, often referred to as *leaching*, (17), is widely used in the metallurgical, natural product, and food industries under batch, semicontinuous, or continuous operating conditions. The major problem in leaching is to promote diffusion of the solute out of the solid and into the liquid solvent. The most effective way of doing this is to reduce the dimensions of the solid to the smallest feasible particle size. For large-scale applications, in the metallurgical industries in particular, large, open tanks are used in countercurrent operation. The major difference between solid-liquid and liquid-liquid systems centers about the difficulty of transporting the solid, or the solid slurry, from stage to stage. For this reason, the solid may be left in the same tank, with only the liquid transferred from tank to tank. In the pharmaceutical, food, and natural-product industries, countercurrent solid transport is provided by complicated

mechanical devices. A supercritical fluid is sometimes used as the solvent in leaching.

In adsorptive-bubble separation methods, surface-active material collects at solution interfaces, establishing a concentration gradient between a solute in the bulk and in the surface layer. If the (very thin) surface layer can be collected, partial solute removal from the solution will have been achieved. The major application of this phenomenon is in ore flotation processes, where solid particles migrate to and attach themselves to rising gas bubbles and literally float out of the solution. This is essentially a three-phase system. *Foam fractionation*, (18), a two-phase adsorptive-bubble separation method, is a process where natural or chelate-induced surface activity causes a solute to migrate to rising bubbles and is, thus, removed as a foam. This method is not covered in this book.

Each equipment symbol shown in Table 1.1 corresponds to the simplest configuration for the operation represented. More complex versions are possible and frequently desirable. For example, a more complex version of the reboiled absorber, Separation Operation (5) in Table 1.1, is shown in Figure 1.8. This reboiled absorber has two feeds, an intercooler, a side stream, and both an interreboiler and a bottoms reboiler. Acceptable design procedures must handle such complex situations. It is also possible to conduct chemical reactions simultaneously with separation operations in a single column. Siirola [6] describes the evolution of an advanced commercial process for producing methyl acetate by the esterification of methanol and acetic acid. The process is conducted in a single column in an integrated process that involves three reaction zones and three separation zones.

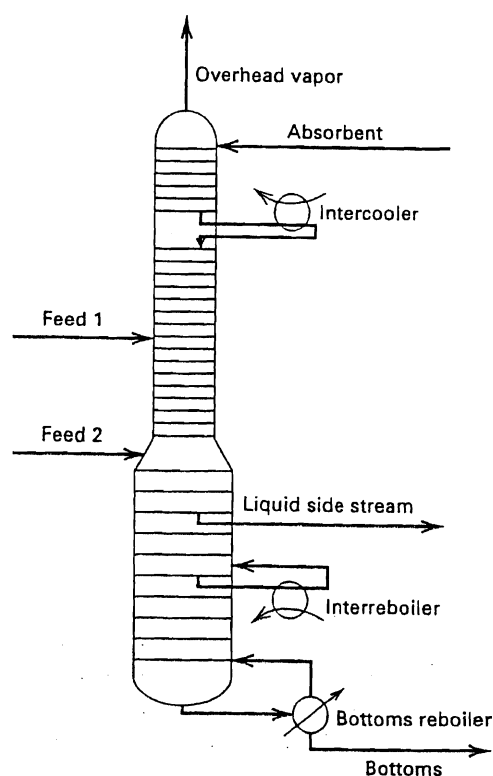


Figure 1.8 Complex reboiled absorber.

1.4 SEPARATION BY BARRIER

The use of microporous and nonporous membranes as semi-permeable barriers for difficult and highly selective separations is rapidly gaining adherents. Membranes are fabricated from natural fibers, synthetic polymers, ceramics, or metals, but may also consist of liquid films. Solid membranes are fabricated into flat sheets, tubes, hollow fibers, or spiral-wound sheets, which are incorporated into commercial modules or cartridges, generally available only in certain sizes. For microporous membranes, separation is effected by differing rates of diffusion through the pores; while for nonporous membranes, separation occurs because of differences in both solubility in the membrane and rate of diffusion through the membrane. The most complex and selective membranes are found in the trillions of cells in the human body.

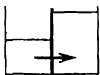
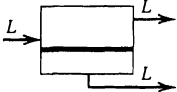
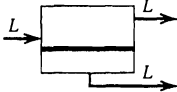
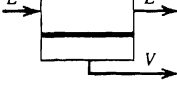
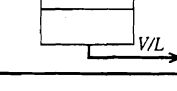
Table 1.2 lists the more widely used membrane separation operations. *Osmosis*, Operation (1) in Table 1.2, involves transfer, by a concentration gradient, of a solvent through a membrane into a mixture of solute and solvent. The membrane is almost nonpermeable to the solute. In *reverse osmosis*, (2), transport of solvent in the opposite direction is effected by imposing a pressure, higher than the

osmotic pressure, on the feed side. Using a nonporous membrane, reverse osmosis successfully desalts water. *Dialysis*, (3), is the transport, by a concentration gradient, of small solute molecules, sometimes called crystalloids, through a porous membrane. The molecules unable to pass through the membrane are small, insoluble, nondiffusible particles, sometimes referred to as colloids.

Microporous membranes can be used in a manner similar to reverse osmosis to selectively allow small solute molecules and/or solvents to pass through the membrane and to prevent large dissolved molecules and suspended solids from passing through. *Microfiltration*, (4), refers to the retention of molecules typically in the size range from 0.02 to 10 μm . *Ultrafiltration*, (5), refers to the range from 1 to 20 nm. To retain molecules down to 0.1 nm, reverse osmosis, sometimes called *hyperfiltration*, can be used.

Although reverse osmosis can be used to separate organic and aqueous-organic liquid mixtures, high pressures are required. Alternatively, *pervaporation*, (6), in which the species being absorbed by and transported through the nonporous membrane are evaporated, can be used. This method, which uses much lower pressures than reverse osmosis, but

Table 1.2 Separation Operations Based on a Barrier

Separation Operation	Symbol ^a	Initial or Feed Phase	Separating Agent	Industrial Example ^b
Osmosis (1)		Liquid	Nonporous membrane	—
Reverse osmosis* (2)		Liquid	Nonporous membrane with pressure gradient	Desalinization of sea water (Vol. 24, pp. 349–353)
Dialysis (3)		Liquid	Porous membrane with pressure gradient	Recovery of caustic from hemicellulose (Vol. 7, p. 572)
Microfiltration (4)		Liquid	Microporous membrane with pressure gradient	Removal of bacteria from drinking water (Vol. 15, p. 115)
Ultrafiltration (5)		Liquid	Microporous membrane with pressure gradient	Separation of whey from cheese (Vol. 15, pp. 562–564)
Pervaporation* (6)		Liquid	Nonporous membrane with pressure gradient	Separation of azeotropic mixtures (Vol. 15, pp. 116–117)
Gas permeation* (7)		Vapor	Nonporous membrane with pressure gradient	Hydrogen enrichment (Vol. 20, pp. 709–710)
Liquid membrane (8)		Vapor and/or liquid	Liquid membrane with pressure gradient	Removal of hydrogen sulfide (Vol. 15, p. 119)

*Design procedures are fairly well accepted.

^aSingle units are shown. Multiple units can be cascaded.

^bCitations refer to volume and page(s) of *Kirk-Othmer Encyclopedia of Chemical Technology*, 3rd ed., John Wiley and Sons, New York (1978–1984).

where the heat of vaporization must be supplied, is used to separate azeotropic mixtures.

The separation of gas mixtures by selective *gas permeation*, (7), through membranes, using pressure as the driving force, is a relatively simple process, first used in the 1940s with porous fluorocarbon barriers to separate $^{235}\text{UF}_6$ and $^{238}\text{UF}_6$ at great expense because it required enormous amounts of electric power. More recently, nonporous polymer membranes are used commercially to enrich gas mixtures containing hydrogen, recover hydrocarbons from gas streams, and produce nitrogen-enriched and oxygen-enriched air.

Liquid membranes, (8), of only a few molecules in thickness can be formed from surfactant-containing mixtures that locate at the interface between two fluid phases. With such a membrane, aromatic hydrocarbons can be separated from paraffinic hydrocarbons. Alternatively, the membrane can be formed by imbibing the micropores with liquids that are doped with additives to facilitate transport of certain solutes, such as CO_2 and H_2S .

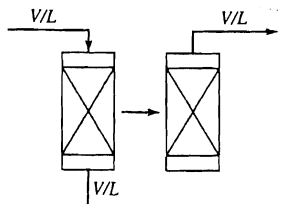
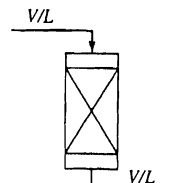
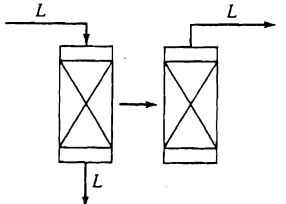
1.5 SEPARATION BY SOLID AGENT

Separation operations that use solid mass-separating agents are listed in Table 1.3. The solid, usually in the form of a granular material or packing, acts as an inert support for a thin layer of adsorbent or enters directly into the separation operation by selective adsorption of, or chemical reaction with, certain species in the feed mixture. Adsorption is confined to the sur-

face of the solid adsorbent, unlike absorption, which occurs throughout the bulk of the absorbent. In all cases, the active separating agent eventually becomes saturated with solute and must be regenerated or replaced periodically. Such separations are often conducted batchwise or semicontinuously. However, equipment is available to simulate continuous operation.

Adsorption, Separation Operation (1) in Table 1.3, is used to remove components present in low concentrations in non-adsorbing solvents or gases and to separate the components in gas or liquid mixtures by selective adsorption on solids, followed by desorption to regenerate the adsorbents, which include activated carbon, aluminum oxide, silica gel, and synthetic sodium or calcium aluminosilicate zeolite adsorbents (molecular sieves). The sieves differ from the other adsorbents in that they are crystalline and have pore openings of fixed dimensions, making them very selective. A simple adsorption device consists of a cylindrical vessel packed with a bed of solid adsorbent particles through which the gas or liquid flows. Regeneration of the adsorbent is conducted periodically, so two or more vessels are used, one vessel desorbing while the other(s) adsorb(s). If the vessel is arranged vertically, it is usually advantageous to employ downward flow of a gas. With upward flow, jiggling of the bed can cause particle attrition and a resulting increase in pressure drop and loss of material. However, for liquid flow, better distribution is achieved by upward flow. Regeneration is accomplished by one of four methods: (1) vaporizing the adsorbate with a hot purge gas (*thermal-swing adsorption*), (2) reducing the

Table 1.3 Separation Operations Based on a Solid Agent

Separation Operation	Symbol ^a	Initial or Feed Phase	Separating Agent	Industrial Example ^b
Adsorption* (1)		Vapor or liquid	Solid adsorbent	Purification of <i>p</i> -xylene (Vol. 24, pp. 723–725)
Chromatography* (2)		Vapor or liquid	Solid adsorbent or liquid adsorbent on a solid support	Separation of xylene isomers and ethylbenzene (Vol. 24, pp. 726–727)
Ion exchange* (3)		Liquid	Resin with ion-active sites	Demineralization of water (Vol. 13, pp. 700–701)

*Design procedures are fairly well accepted.

^aSingle units are shown. Multiple units can be cascaded.

^bCitations refer to volume and page(s) of *Kirk-Othmer Encyclopedia of Chemical Technology*, 3rd ed., John Wiley and Sons, New York (1978–1984).

pressure to vaporize the adsorbate (*pressure-swing adsorption*), (3) inert purge stripping without change in temperature or pressure, and (4) displacement desorption by a fluid containing a more strongly adsorbed species.

Chromatography, Separation Operation (2) in Table 1.3, is a method for separating the components of a feed gas or liquid mixture by passing the feed through a bed of packing. The feed may be volatilized into a carrier gas, and the bed may be a solid adsorbent (gas-solid chromatography) or a solid-inert support that is coated with a very viscous liquid that acts as an absorbent (gas-liquid chromatography). Because of selective adsorption on the solid adsorbent surface or absorption into liquid absorbent, followed by desorption, different components of the feed mixture move through the bed at different rates, thus effecting the separation. In *affinity chromatography*, a macromolecule (called a ligate) is selectively adsorbed by a ligand (e.g., an ammonia molecule in a coordination compound) that is covalently bonded to a solid-support particle. Ligand-ligate pairs include inhibitors–enzymes, antigens–antibodies, and antibodies–proteins. Chromatography in its various forms is finding use in bioseparations.

Ion exchange, (3), resembles adsorption in that solid particles are used and regeneration is necessary. However, a chemical reaction is involved. In water softening, a typical ion-exchange application, an organic or inorganic polymer in its sodium form removes calcium ions by exchanging calcium for sodium. After prolonged use, the (spent) polymer, which becomes saturated with calcium, is regenerated by contact with a concentrated salt solution.

1.6 SEPARATION BY EXTERNAL FIELD OR GRADIENT

External fields can be used to take advantage of differing degrees of response of molecules and ions to forces and gradients. Table 1.4 lists common techniques, with combinations of these techniques with each other and with previously described separation methods also being possible.

Centrifugation, Operation (1) in Table 1.4, establishes a pressure field that separates fluid mixtures according to molecular weight. This technique is used to separate $^{235}\text{UF}_6$ from $^{238}\text{UF}_6$, and large polymer molecules according to molecular weight.

If a rather large temperature gradient is applied to a homogeneous solution, concentration gradients can be established and *thermal diffusion*, (2), is induced. It has been used to enhance the separation of uranium isotopes in gas permeation processes.

Natural water contains 0.000149 atom fraction of deuterium. When water is decomposed by *electrolysis*, (3), into hydrogen at the cathode and oxygen at the anode, the deuterium concentration in the hydrogen produced is lower than that in the water. Until 1953, this process was the only commercial source of heavy water (D_2O). In *electrodialysis*, (4), cation- and anion-permeable membranes carry a fixed charge, preventing the migration of species of like charge. This operation can be used to desalinate (remove salts from) sea water. A somewhat related process is *electrophoresis*, (5), which exploits the different migration velocities of charged colloidal or suspended species in an electric field. Positively charged species, such as dyes, hydroxide sols, and colloids, migrate to the cathode; while most small, suspended, negatively charged particles are attracted to the anode. By changing the solvent from an acidic to a basic condition, migration direction can sometimes be changed, particularly for proteins. Electrophoresis is a highly versatile method for separating biochemicals.

Another separation technique for biochemicals and difficult-to-separate heterogeneous mixtures of micromolecular and colloidal materials is *field-flow fractionation*, (6). For the mixture to be separated, an electrical field, magnetic field, or thermal gradient is established in a direction perpendicular to a laminar-flow field. Components of the mixture are driven to different locations in the stream; thus, they travel in the flow direction at different velocities, so a separation is achieved.

Table 1.4 Separation Operations by Applied Field or Gradient

Separation Operation	Initial or Feed Phase	Force Field or Gradient	Industrial Example ^a
Centrifugation (1)	Vapor	Centrifugal force field	Separation of uranium isotopes (Vol. 23, pp. 531–532)
Thermal diffusion (2)	Vapor or liquid	Thermal gradient	Separation of chlorine isotopes (Vol. 7, p. 684)
Electrolysis (3)	Liquid	Electrical force field	Concentration of heavy water (Vol. 7, p. 550)
Electrodialysis (4)	Liquid	Electrical force field and membrane	Desalination of sea water (Vol. 24, pp. 353–359)
Electrophoresis (5)	Liquid	Electrical force field	Recovery of hemicelluloses (Vol. 4, p. 551)
Field-flow fractionation (6)	Liquid	Laminar flow in force field	

^aCitations refer to volume and page(s) of *Kirk–Othmer Encyclopedia of Chemical Technology*, 3rd ed., John Wiley and Sons, New York (1978–1984).

1.7 COMPONENT RECOVERIES AND PRODUCT PURITIES

Separation operations are subject to the conservation of mass. Accordingly, if no chemical reactions occur and the process operates in a continuous, steady-state fashion, then for each component, i , in a mixture of C components, the molar (or mass) flow rate in the feed, $n_i^{(F)}$, is equal to the sum of the product molar (or mass) flow rates, $n_i^{(p)}$, for that component in the N product phases, p . Thus, referring to Figure 1.6,

$$n_i^{(F)} = \sum_{p=1}^N n_i^{(p)} = n_i^{(1)} + n_i^{(2)} + \cdots + n_i^{(N-1)} + n_i^{(N)} \quad (1-1)$$

To solve (1-1) for values of $n_i^{(p)}$, from specified values of $n_i^{(F)}$, we need an additional $N - 1$ independent expressions involving $n_i^{(p)}$. This gives a total of NC equations in NC unknowns. For example, if a feed mixture containing C components is separated into N product phases, $C(N - 1)$ additional expressions are needed. General forms of these expressions, which deal with the extent of separation, are considered in this and the next section. If more than one stream is fed to the separation process, $n_i^{(F)}$ is the summation for all feeds.

Equipment for separating components of a mixture is designed and operated to meet desired or required specifications, which are typically given as *component recoveries* and/or *product purities*. In Figure 1.9, the block-flow diagram for a hydrocarbon separation system, the feed is the bottoms product from a reboiled absorber used to deethanize (i.e., remove ethane and components of smaller molecular weight) a mixture of refinery gases and liquids. The separation process of choice in this example is a sequence of three multistage distillation columns. The composition of the feed to the process is included in Figure 1.9, where components

are rank-listed by decreasing volatility, and hydrocarbons heavier (i.e., of greater molecular weight) than normal pentane and in the hexane (C_6)-to-undecane (C_{11}) range are lumped together in a so-called C_6^+ fraction. The three distillation columns of Figure 1.9 separate the deethanized feed into four products: a C_5^+ -rich bottoms, a C_3 -rich distillate, an iC_4 -rich distillate, and an nC_4 -rich bottoms. For each column, each component in the feed is partitioned between the overhead and the bottoms, according to a unique *split fraction* or *split ratio* that depends on (1) the component thermodynamic and transport properties in the vapor and liquid phases, (2) the number of contacting stages, and (3) the relative vapor and liquid flows through the column. The *split fraction*, SF, for component i in separator k is the fraction of that component found in the first product:

$$SF_{i,k} = \frac{n_{i,k}^{(1)}}{n_{i,k}^{(F)}} \quad (1-2)$$

where $n^{(1)}$ and $n^{(F)}$ refer to component molar flow rates in the first product and the feed, respectively. Alternatively, a *split ratio*, SR, between two products, may be defined as

$$SR_{i,k} = \frac{n_{i,k}^{(1)}}{n_{i,k}^{(2)}} = \frac{SF_{i,k}}{(1 - SF_{i,k})} \quad (1-3)$$

where $n^{(2)}$ refers to a component molar flow rate in the second product. Alternatively, SF and SR can be defined in terms of component mass flow rates.

If the process shown in Figure 1.9 is part of an operating plant with the measured material balance of Table 1.5, the split fractions and split ratios in Table 1.6 are determined from (1-2) and (1-3). In Table 1.5, it is seen that only two of the four products are relatively pure: C_3 overhead from the second column and iC_4 overhead from the third column. The molar purity of C_3 in the C_3 overhead is (54.80/56.00) or

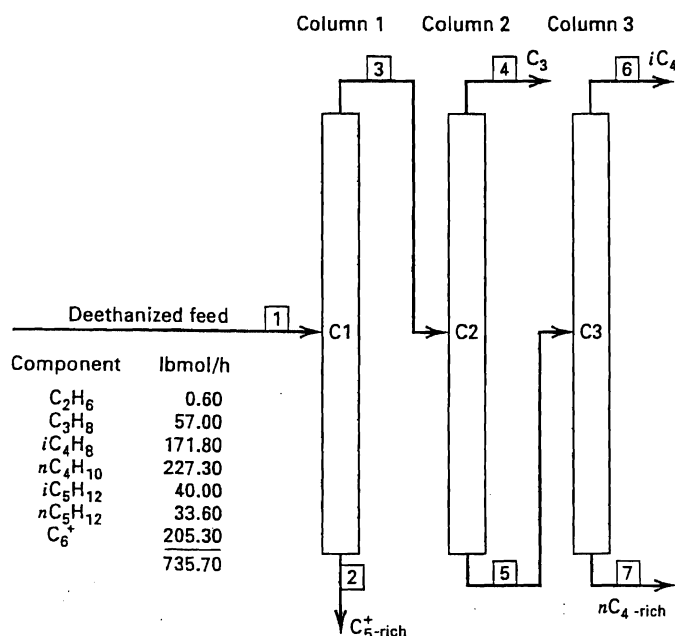


Figure 1.9 Hydrocarbon recovery process.

Table 1.5 Operating Material Balance for Hydrocarbon Recovery Process

Component	lbmol/h in Stream						
	1 Feed to C1	2 C ₅ ⁺ -rich	3 Feed to C2	4 C ₃	5 Feed to C3	6 iC ₄	7 nC ₄ -rich
C ₂ H ₆	0.60	0.00	0.60	0.60	0.00	0.00	0.00
C ₃ H ₈	57.00	0.00	57.00	54.80	2.20	2.20	0.00
iC ₄ H ₁₀	171.80	0.10	171.70	0.60	171.10	162.50	8.60
nC ₄ H ₁₀	227.30	0.70	226.60	0.00	226.60	10.80	215.80
iC ₅ H ₁₂	40.00	11.90	28.10	0.00	28.10	0.00	28.10
nC ₅ H ₁₂	33.60	16.10	17.50	0.00	17.50	0.00	17.50
C ₆ ⁺	205.30	205.30	0.00	0.00	0.00	0.00	0.00
Total	735.60	234.10	501.50	56.00	445.50	175.50	270.00

Table 1.6 Split Fractions and Split Ratios for Hydrocarbon Recovery Process

Component	Column 1		Column 2		Column 3		Overall Percent Recovery
	SF	SR	SF	SR	SF	SR	
C ₂ H ₆	1.00	Large	1.00	Large	—	—	100
C ₃ H ₈	1.00	Large	0.9614	24.91	1.00	Large	96.14
iC ₄ H ₁₀	0.9994	1,717	0.0035	0.0035	0.9497	18.90	94.59
nC ₄ H ₁₀	0.9969	323.7	0.00	0.00	0.0477	0.0501	94.94
iC ₅ H ₁₂	0.7025	2.361	0.00	0.00	0.00	0.00	29.75
nC ₅ H ₁₂	0.5208	1.087	0.00	0.00	0.00	0.00	47.92
C ₆ ⁺	0.00	Small	—	—	—	—	100

97.86%, while the iC₄ overhead purity is (162.50/175.50) or 92.59% iC₄. The nC₄-rich bottoms from Column 3 has an nC₄ purity of only (215.80/270.00) or 79.93%. Each of the three columns is designed to make a split between two adjacent components (called the *key components*) in the list of components ordered in decreasing volatility. As seen by the horizontal dividing lines in Table 1.6, the three key splits are nC₄H₁₀/iC₅H₁₂, C₃H₈/iC₄H₁₀, and iC₄H₁₀/nC₄H₁₀ for Columns 1, 2, and 3, respectively. From the split fractions listed in Table 1.6, it is seen that all splits are relatively sharp (SF > 0.95 for the light key and SF < 0.05 for the heavy key), except for column 1, where the split ratio for the heavy key (iC₅H₁₂) is not sharp at all, and ultimately causes the nC₄-rich bottoms to be relatively impure in nC₄, even though the split between the two key components in the third column is relatively sharp.

In Table 1.6, for each column, values of SF and SR decrease in the order of the ranked component list. It is also noted in Table 1.6 that SF may be a better quantitative measure of degree of separation than SR because SF is bounded between 0 and 1, while SR can range from 0 to a very large value.

Two other common measures of extent of separation can be applied to each column, or to the separation system as a whole. One measure is the *percent recovery* in a designated system product of each component in the feed to the system.

These values, as computed from the data of Table 1.5, are listed in the last column of Table 1.6. As shown, the component recoveries are all relatively high (>95%) except for the two pentane isomers. The other measure of extent of separation is *product purity*. These purities for the main component were computed for all except the C₅⁺-rich product, which is [(11.90 + 16.10 + 205.30)/234.10] or 99.66% pure with respect to the pentanes and heavier. Such a product is a *multi-component product*. One of the most common multicomponent products is gasoline.

Product impurity levels and a designation of the impurities are included in product specifications for chemicals of commerce. The product purity with respect to each component in each of the three final products for the hydrocarbon recovery process, as computed from the process operating data of Table 1.5, is given in Table 1.7, where the values are also extremely important because maximum allowable percentages of impurities are compared to the product specifications. The C₅⁺ fraction is not included because it is an intermediate that is sent to an isomerization process. From the comparison in Table 1.7, it is seen that two products easily meet their specifications, while the iC₄ product barely meets its specification. If the process is equipped with effective controllers, it might be possible to reduce the energy input to the process and still meet C₃ and nC₄-rich product

Table 1.7 Comparison of Measured Product Purities with Specifications

Component	mol% in Product					
	Propane		Isobutane		Normal Butane	
	Data	Spec	Data	Spec	Data	Spec
C ₂ H ₆	1.07	5 max	0	—	0	—
C ₃ H ₈	97.86	93 min	1.25	3 max	0	1 max
iC ₄ H ₁₀	1.07	2 min	92.60	92 min	{ 83.11	{ 80 min
nC ₄ H ₁₀	0	—	6.15	7 max		
C ₅ ⁺	0	—	0	—	16.89	20 max
Total	100.00		100.00		100.00	

specifications. Although the product purities in Table 1.7 are given in mol%, this designation is usually restricted to gas mixtures for which purities in vol% are equivalent to mol%. For liquid mixtures, purities are often specified in wt%. To meet environmental regulations, maximum amounts of impurities in gas, liquid, and solids streams are typically specified in *ppm* (parts per million) or *ppb* (parts per billion), usually by volume (same as moles) for gases and by weight (mass) for liquids and solids. For aqueous solutions, especially those containing acids and bases, common designations for composition are *molarity* or molar concentration (moles of solute per liter of solution, M), *normality* (number of equivalent weights of solute per liter of solution, N), and *molality* (moles of solute per kilogram of solvent). For some chemical products, an attribute, such as color, may be used in place of a purity in terms of composition.

1.8 SEPARATION POWER

Some separations in Table 1.1 are often inadequate for making a sharp split between two key components of a feed mixture, and can only effect the desired recovery of a single key component. Examples are Operations 1, 2, 6, 7, 8, 9, 11, 13, 14, 15, 16, and 17 in Table 1.1. For these, either a single separation stage is utilized as in Operations 1, 2, 13, 14, 15, 16, and 17 or the feed enters at one end (not near the middle) of a multistage separator as in Separation Operations 6, 7, 8, 9, and 11. The split ratio, SR, split fraction, SF, recovery, or purity that can be achieved for the key component depends on a number of factors. For the simplest case of a single separation stage, the factors that influence SR and SF values include: (1) the relative molar amounts of the two phases leaving the separator and (2) thermodynamic, mass transport, and other properties of the key components. For multistage separators, an additional factor must be added, namely, (3) the number of stages and their configuration. The quantitative relationships involving these factors are unique to each type of separator. Therefore, detailed discussion of these relationships is deferred to subsequent chapters, where individual

separation operations are discussed in detail. A general but brief discussion of some of the important property factors is given in the next section.

When multistage separators are utilized and the feed mixture enters somewhere near the middle of the separator, such that the separator consists of two sections of stages, one on either side of the feed stage, it is often possible to achieve a relatively sharp separation between two key components. One section acts to remove one key component, while the other section acts to remove the other key component. Examples are Separation Operations 3, 4, 5, 10, and 12 in Table 1.1. For these operations, a convenient measure of the relative degree of separation between two components, *i* and *j*, is the *separation power* (also referred to as the relative split ratio and the separation factor), SP, of the separation equipment, defined in terms of the component splits achieved, as measured by the compositions of the two products, (1) and (2):

$$SP_{i,j} = \frac{C_i^{(1)}/C_i^{(2)}}{C_j^{(1)}/C_j^{(2)}} \quad (1-4)$$

where *C* is some measure of composition such as mole fraction, mass fraction, or concentration in moles or mass per unit volume. Most commonly, mole fractions or concentrations are used, but in any case, the separation power is readily converted to the following forms in terms of split fractions or split ratios:

$$SP_{i,j} = \frac{SR_i}{SR_j} \quad (1-5)$$

$$SP_{i,j} = \frac{SF_i/SF_j}{(1 - SF_i)/(1 - SF_j)} \quad (1-6)$$

Achievable values of SP depend on the number of stages and the relative thermodynamic and mass transport properties of components *i* and *j*. In general, when applied to the two key components, components *i* and *j* and products 1 and 2 are selected so that $SP_{i,j} > 1.0$. Then, a large value corresponds to a relatively high degree of separation or high separation

Table 1.8 Main Separation Factors for Hydrocarbon Recovery Process

Key-Component Split	Column	Separation Factor, SP
$n\text{C}_4\text{H}_{10}/i\text{C}_5\text{H}_{12}$	C1	137.1
$\text{C}_3\text{H}_8/i\text{C}_4\text{H}_{10}$	C2	7103
$i\text{C}_4\text{H}_{10}/n\text{C}_4\text{H}_{10}$	C3	377.6

power; a small value larger than but close to 1.0 corresponds to a low degree of separation power. For example, if $\text{SP} = 10,000$ and $\text{SR}_i = 1/\text{SR}_j$, then, from (1-5), $\text{SR}_i = 100$ and $\text{SR}_j = 0.01$, corresponding to a sharp separation. However, if $\text{SP} = 9$ and $\text{SR}_i = 1/\text{SR}_j$, then $\text{SR}_i = 3$ and $\text{SR}_j = \frac{1}{3}$, corresponding to a nonsharp separation.

For the hydrocarbon recovery process of Figure 1.9, the values of SP in Table 1.8 are computed from the data in Tables 1.5 or 1.6 for the main split in each of the three separators. The separation factor in Column C1 is relatively small because the split for the heavy key, $i\text{C}_5\text{H}_{12}$, is not sharp. The largest separation factor occurs in column C2, where the separation is relatively easy because of the fairly large volatility difference between the two keys. Much more difficult is the butane-isomer split in Column C3, where only a moderately sharp split is achieved.

When applying the conservation of mass principle to separation operations using (1-1), component specifications in terms of component recoveries are easily applied, while those in terms of split ratios and, particularly, purities are more difficult, as shown in the following example.

EXAMPLE 1.1

A feed, F , of 100 kmol/h of air containing 21 mol% O_2 and 79 mol% N_2 is to be partially separated by a membrane unit according to each of the following four sets of specifications. For each case, compute the amounts, in kmol/h, and compositions, in mol%, of the two products (retentate, R , and permeate, P). The membrane is more permeable to the oxygen.

Case 1: 50% recovery of O_2 to the permeate and 87.5% recovery of N_2 to the retentate.

Case 2: 50% recovery of O_2 to the permeate and 50 mol% purity of O_2 in the permeate.

Case 3: 85 mol% purity of N_2 in the retentate and 50 mol% purity of O_2 in the permeate.

Case 4: 85 mol% purity of N_2 in the retentate and a split ratio of O_2 in the permeate to the retentate equal to 1.1.

SOLUTION

The feed is

$$n_{\text{O}_2}^{(F)} = 0.21(100) = 21 \text{ kmol/h}$$

$$n_{\text{N}_2}^{(F)} = 0.79(100) = 79 \text{ kmol/h}$$

Case 1: This is the simplest case to calculate because two recoveries are given:

$$n_{\text{O}_2}^{(P)} = 0.50(21) = 10.5 \text{ kmol/h}$$

$$n_{\text{N}_2}^{(R)} = 0.875(79) = 69.1 \text{ kmol/h}$$

$$n_{\text{O}_2}^{(R)} = 21 - 10.5 = 10.5 \text{ kmol/h}$$

$$n_{\text{N}_2}^{(P)} = 79 - 69.1 = 9.9 \text{ kmol/h}$$

Case 2: With the recovery for O_2 given, calculate its distribution into the two products:

$$n_{\text{O}_2}^{(P)} = 0.50(21) = 10.5 \text{ kmol/h}$$

$$n_{\text{O}_2}^{(R)} = 21 - 10.5 = 10.5 \text{ kmol/h}$$

Using the purity of O_2 in the permeate, the total permeate is

$$n^{(P)} = 10.5/0.5 = 21 \text{ kmol/h}$$

By a total permeate material balance,

$$n_{\text{N}_2}^{(P)} = 21 - 10.5 = 10.5 \text{ kmol/h}$$

By an overall N_2 material balance,

$$n_{\text{N}_2}^{(R)} = 79 - 10.5 = 68.5 \text{ kmol/h}$$

Case 3: With two purities given, write two simultaneous material-balance equations, one for each component, in terms of the total retentate and total permeate.

For nitrogen, with a fractional purity of $1.00 - 0.50 = 0.50$ in the permeate,

$$n_{\text{N}_2} = 0.85n^{(R)} + 0.50n^{(P)} = 79 \text{ kmol/h} \quad (1)$$

For oxygen, with a fractional purity of $1.00 - 0.85 = 0.15$ in the retentate,

$$n_{\text{O}_2} = 0.50n^{(P)} + 0.15n^{(R)} = 21 \text{ kmol/h} \quad (2)$$

Solving (1) and (2) simultaneously for the total products gives

$$n^{(P)} = 17.1 \text{ kmol/h} \quad n^{(R)} = 82.9 \text{ kmol/h}$$

Therefore, the component flow rates are

$$n_{\text{N}_2}^{(R)} = 0.85(82.9) = 70.5 \text{ kmol/h}$$

$$n_{\text{O}_2}^{(R)} = 82.9 - 70.5 = 12.4 \text{ kmol/h}$$

$$n_{\text{O}_2}^{(P)} = 0.50(17.1) = 8.6 \text{ kmol/h}$$

$$n_{\text{N}_2}^{(P)} = 17.1 - 8.6 = 8.5 \text{ kmol/h}$$

Case 4: First compute the O_2 flow rates using the split ratio and an overall O_2 material balance,

$$\frac{n_{\text{O}_2}^{(P)}}{n_{\text{O}_2}^{(R)}} = 1.1 \quad 21 = n_{\text{O}_2}^{(P)} + n_{\text{O}_2}^{(R)}$$

Solving these two equations simultaneously gives

$$n_{\text{O}_2}^{(R)} = 10 \text{ kmol/h} \quad n_{\text{O}_2}^{(P)} = 21 - 10 = 11 \text{ kmol/h}$$

Since the retentate contains 85 mol% N_2 and, therefore, 15 mol% O_2 , the flow rates for the N_2 are

$$n_{\text{N}_2}^{(R)} = \frac{85}{15}(10) = 56.7 \text{ kmol/h} \quad n_{\text{N}_2}^{(P)} = 79 - 56.7 = 22.3 \text{ kmol/h}$$

1.9 SELECTION OF FEASIBLE SEPARATION PROCESSES

Selection of a best separation process must be made from among a number of feasible candidates. When the feed mixture is to be separated into more than two products, a combination of two or more operations may be best. Even when only two products are to be produced, a hybrid process of two or more different types of operations may be most economical. Only an introduction to the selection of a separation process is given here. A detailed treatment is given in Chapter 7 of Seider, Seader, and Lewin [14].

Important factors in the selection of feasible separation operations are listed in Table 1.9. These factors have to do with feed and product conditions, property differences that can be exploited, and certain characteristics of the candidate separation operations. The most important feed conditions are composition and flow rate, because the other conditions (temperature, pressure, and phase condition) can be altered by pumps, compressors, and heat exchangers to fit a particular candidate separation operation. In general, however, the vaporization of a liquid feed that has a high heat of vaporization, the condensation of a vapor feed with a refrigerant, and/or the compression of a vapor feed can add significantly to the cost. Some separation operations, such as those based on the use of barriers or solid agents, perform best on feeds that are dilute in the species to be recovered. The most important product conditions are the required purities because, again, the other conditions listed can be altered by energy transfer after the separation is achieved.

Table 1.9 Factors That Influence the Selection of Feasible Separation Operations

A. Feed conditions	
1.	Composition, particularly concentration of species to be recovered or separated
2.	Flow rate
3.	Temperature
4.	Pressure
5.	Phase state (solid, liquid, and/or gas)
B. Product conditions	
1.	Required purities
2.	Temperatures
3.	Pressures
4.	Phase states
C. Property differences that may be exploited	
1.	Molecular
2.	Thermodynamic
3.	Transport
D. Characteristics of separation operation	
1.	Ease of scale-up
2.	Ease of staging
3.	Temperature, pressure, and phase-state requirements
4.	Physical size limitations
5.	Energy requirements

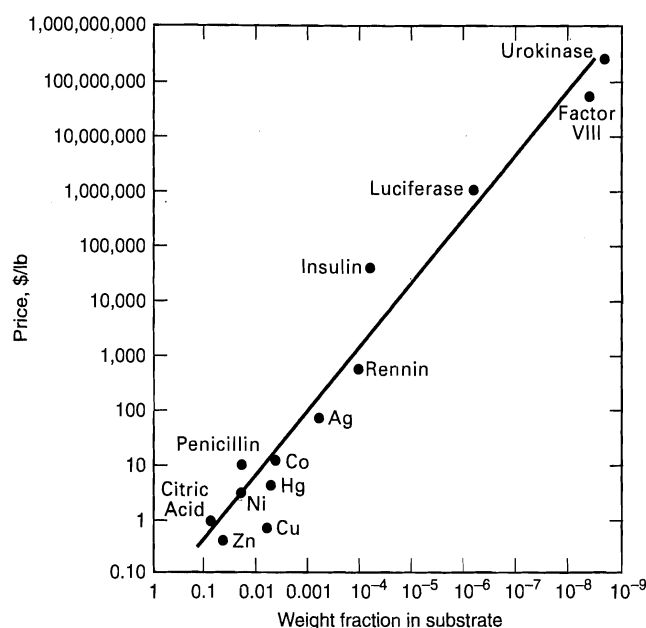


Figure 1.10 Effect of concentration of product in feed material on price [9].

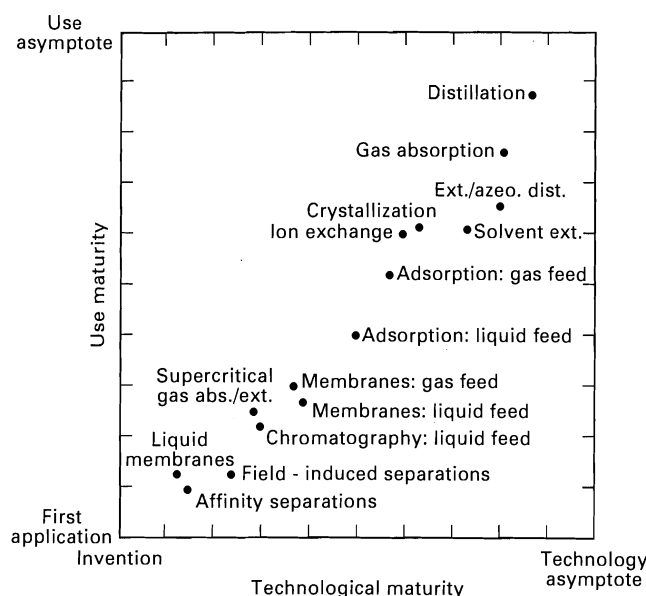
In general, as demonstrated by Sherwood, Pigford, and Wilke [7] and updated recently, using additional data for biological materials from Dwyer [8], by Keller [9], the cost of recovering and purifying a chemical contained in a mixture can depend strongly on the concentration of that chemical in the feed mixture. Keller's correlation is given in Figure 1.10, where it is seen that the more dilute the chemical is in the mixture, the higher is its sales price.

When very pure products are required, either large differences in certain properties must exist or significant numbers of stages must be provided. It is important to consider both molecular and bulk thermodynamic and transport properties, some of which are listed near the end of Section 1.2. Data and estimation methods for many bulk properties are given by Poling, Prausnitz, and O'Connell [10] and for both molecular and bulk properties by Daubert and Danner [11].

Some separation operations are well understood and can be readily designed from a mathematical model and/or scaled up to a commercial size from laboratory data. The results of a survey by Keller [9], shown in Figure 1.11, show that the degree to which a separation operation is technologically mature correlates well with its commercial use. Operations based on a barrier are more expensive to stage than those based on use of a solid agent or the creation or addition of a second phase. Some separation equipment is limited to a maximum size. For capacities requiring a larger size, parallel units must be provided. The choice of single or parallel units must be given careful consideration. Except for size constraints or fabrication problems, the capacity of a single unit can be doubled for an additional investment cost of only about 50%. If two parallel units are installed, the additional investment is 100%. Table 1.10 is a list of the more common separation operations ranked according to ease of scale-up. Those operations ranked near the top are frequently designed without the need for any laboratory data or

Table 1.10 Ease of Scale-up of the Most Common Separation Operations

Operation in Decreasing Ease of Scale-up	Ease of Staging	Need for Parallel Units
Distillation	Easy	No need
Absorption	Easy	No need
Extractive and azeotropic distillation	Easy	No need
Liquid-liquid extraction	Easy	Sometimes
Membranes	Repressurization required between stages	Almost always
Adsorption	Easy	Only for regeneration cycle
Crystallization	Not easy	Sometimes
Drying	Not convenient	Sometimes

**Figure 1.11** Technological and use maturities of separation processes [9].

pilot-plant tests. Operations near the middle usually require laboratory data, while operations near the bottom require pilot-plant tests on actual feed mixtures. Also included in the table is an indication of the ease of providing multiple stages and to what extent parallel units may be required to handle high capacities. A detailed discussion of the selection of alternative techniques for the separation of components from both homogeneous and heterogeneous phases, with many examples, is given by Woods [12]. Ultimately, the process having the lowest operating, maintenance, and capital costs is selected.

EXAMPLE 1.2

Propylene and propane are among the light hydrocarbons produced by thermal and catalytic cracking of heavy petroleum fractions. Propane is valuable as a fuel by itself and in liquefied natural gas (LPG), and as a feedstock for producing propylene and ethylene. Propylene is used to make acrylonitrile monomer for synthetic rubber, isopropyl alcohol, cumene, propylene oxide, and polypropylene.

Although propylene and propane have close boiling points, they are traditionally separated by distillation. Representative conditions are shown in Figure 1.12, where it is seen that a large number of stages is needed and the reflux and boilup flow rates compared to the feed flow rate are also large. Accordingly, considerable attention has been given to the possible replacement of distillation with a more economical and less energy-intensive separation operation. Based on the factors in Table 1.9, the characteristics in Table 1.10, and the list of species properties that might be exploited, given at the end of Section 1.2, propose some feasible alternatives to distillation to produce products from the feed in Figure 1.12.

SOLUTION

First, note that the component feed and product flow rates in Figure 1.12 satisfy (1-1), the conservation of mass. Table 1.11 compares properties of the two species, taken mainly from Daubert and Danner [11], where it is seen that the only listed property that might be exploited is the dipole moment. Because of the asymmetric location of the double bond in propylene, its dipole moment is significantly greater than that of propane, making propylene a polar compound, although weakly so (some define a polar compound as one with a dipole moment greater than 1 debye). Separation operations that can exploit this difference are:

1. Extractive distillation with a polar solvent such as furfural or an aliphatic nitrile that will reduce the volatility of propylene (Ref.: U.S. Patent 2,588,056, March 4, 1952).

Table 1.11 Comparison of Properties for Example 1.2

Property	Propylene	Propane
Molecular weight	42.081	44.096
van der Waals volume, m^3/kmol	0.03408	0.03757
van der Waals area, $\text{m}^2/\text{kmol} \times 10^{-8}$	5.060	5.590
Acentric factor	0.142	0.152
Dipole moment, debyes	0.4	0.0
Radius of gyration, $\text{m} \times 10^{10}$	2.254	2.431
Normal melting point, K	87.9	85.5
Normal boiling point, K	225.4	231.1
Critical temperature, K	364.8	369.8
Critical pressure, MPa	4.61	4.25

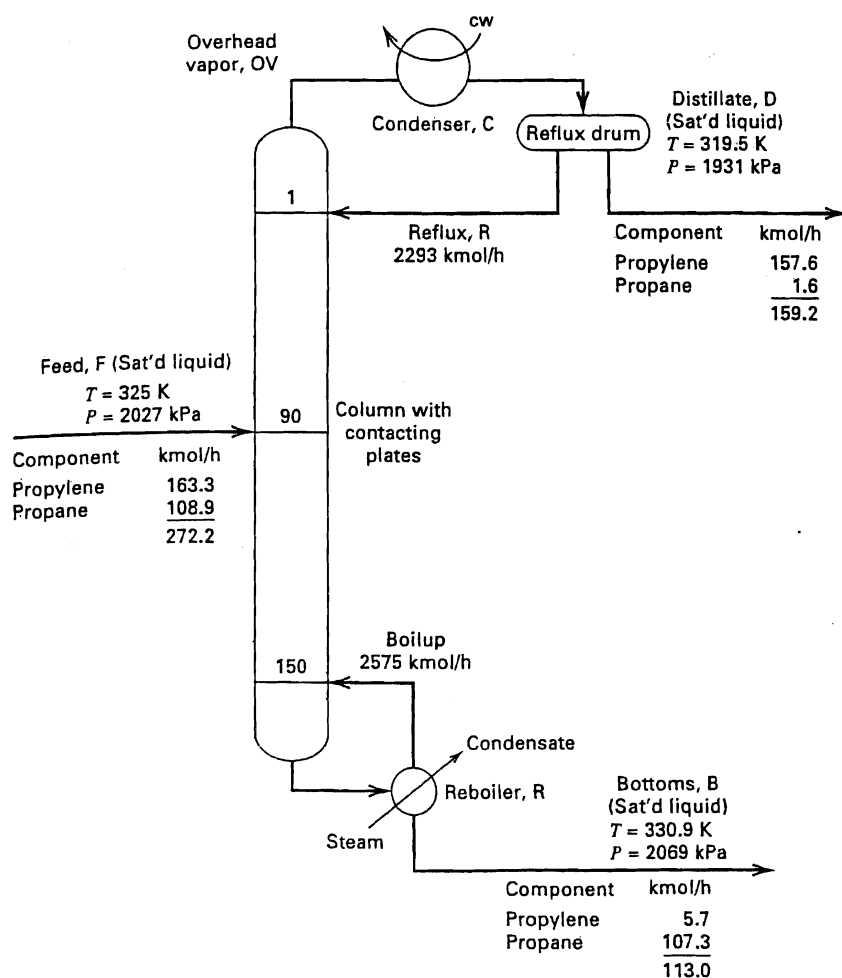


Figure 1.12 Distillation of a propylene-propane mixture.

2. Adsorption with silica gel or a zeolite that will selectively adsorb propylene [Ref.: *J. Am. Chem. Soc.*, **72**, 1153–1157 (1950)].

3. Facilitated transport membranes using impregnated silver nitrate to carry propylene selectively through the membrane [Ref.: *Recent Developments in Separation Science*, Vol. IX, 173–195 (1986)].

SUMMARY

1. Almost all industrial chemical processes include equipment for separating chemicals contained in the process feed(s) and/or produced in reactors within the process.

2. More than 25 different separation operations are commercially important.

3. The extent of separation achievable by a particular separation operation depends on exploitation of the differences in certain properties of the species.

4. The more widely used separation operations involve transfer of species between two phases, one of which is created by energy transfer or the reduction of pressure, or by introduction as a MSA.

5. Less commonly used separation operations are based on the use of a barrier, a solid agent, or a force field to cause species being separated to diffuse at different rates and/or to be selectively absorbed or adsorbed.

6. Separation operations are subject to the conservation of mass. The degree of separation of a component in a separator is indicated by a split fraction, SF, given by (1-2), and/or by a split ratio, SR, given by (1-3).

7. For a sequence, system, or train of separators, overall component recoveries and product purities are of prime importance and are related by material balances to individual SF and/or SR values for the separators in the system.

8. Some separation operations, such as absorption, are capable of only a specified degree of separation for a single species. Other separation operations, such as distillation, can effect a sharp split between two so-called key components.

9. The degree of separation between two key components by a particular separation operation can be indicated by a separation power (separation factor), SP, given by (1-4) and related to SF and SR values by (1-5) and (1-6).

10. For given feed(s) and product specifications, the best separation process must frequently be selected from among a number of feasible candidates. The choice may depend on factors listed in Table 1.9. The cost of recovering and purifying a chemical depends on its concentration in the feed mixture. The extent of industrial use of a separation operation depends on the technological maturity of the operation.

REFERENCES

1. Kirk-Othmer *Encyclopedia of Chemical Technology*, 3rd ed., John Wiley and Sons, New York, Vol. 17, pp. 183–256 (1982).
2. MAUDE, A.H., *Trans. AIChE*, **38**, 865–882 (1942).
3. CONSIDINE, D.M., Ed., *Chemical and Process Technology Encyclopedia*, McGraw-Hill, New York, pp. 760–763 (1974).
4. CARLE, T.C., and D.M. STEWART, *Chem. Ind. (London)*, May 12, 1962, 830–839.
5. PERRY, R.H., and D.W. GREEN, Eds., *Perry's Chemical Engineers' Handbook*, 7th ed., McGraw-Hill, New York (1997).
6. SIROLA, J.J., *AIChE Symp. Ser.*, **91**(304), 222–233 (1995).
7. SHERWOOD, T.K., R.L. PIGFORD, and C.R. WILKE, *Mass Transfer*, McGraw-Hill, New York (1975).
8. DWYER, J.L., *Biotechnology*, **1**, 957 (Nov. 1984).
9. KELLER, G.E., II, *AIChE Monogr. Ser.*, **83**(17) (1987).
10. POLING, B.E., J.M. PRAUSNITZ, and J.P. O'CONNELL, *The Properties of Gases and Liquids*, 5th ed., McGraw-Hill, New York (2001).
11. DAUBERT, T.E., and R.P. DANNER, *Physical and Thermodynamic Properties of Pure Chemicals—Data Compilation*, DIPPR, AIChE, Hemisphere, New York (1989).
12. WOODS, D.R., *Process Design and Engineering Practice*, Prentice-Hall, Englewood Cliffs, NJ (1995).
13. CUSSLER, E.L., and G.D. MOGGIDGE, *Chemical Product Design*, Cambridge University Press, Cambridge, UK (2001).
14. SEIDER, W.D., J.D. SEADER, and D.R. LEWIN, *Product & Process Design Principles*, John Wiley & Sons, New York (2004).

EXERCISES

Section 1.1

1.1 The book, *Chemical Process Industries*, 4th edition, by R. Norris Shreve and J. A. Brink, Jr. (McGraw-Hill, New York, 1984), contains process descriptions, process flow diagrams, and technical data for processes used commercially in 38 chemical industries. For each of the following processes, draw a block-flow diagram of just the reaction and separation steps and describe the process in terms of just those steps, giving careful attention to the particular chemicals being formed in the reactor and separated in each of the separation operations:

- (a) Coal chemicals, pp. 72–74
- (b) Natural gas purification, pp. 84–86
- (c) Acetylene, pp. 115–117
- (d) Magnesium compounds, pp. 174–177
- (e) Chlorine and caustic soda, pp. 214–219
- (f) Potassium chloride, pp. 269–270
- (g) Ammonia, pp. 278–282
- (h) Sulfuric acid, pp. 299–310
- (i) Fluorocarbons, pp. 321–323
- (j) Uranium, pp. 338–340
- (k) Titanium dioxide, pp. 388–390
- (l) Cottonseed oil, pp. 468–471
- (m) Glycerin, pp. 502–503
- (n) Industrial alcohol, pp. 530–534
- (o) Polyethylene, pp. 587–588
- (p) Formaldehyde, pp. 596–598
- (q) Styrene, pp. 630–635
- (r) Natural-gas liquids, pp. 660–661

Section 1.2

1.2 Explain in detail, using thermodynamic principles, why the mixing of pure chemicals to form a homogeneous mixture is a so-called spontaneous process, while the separation of that mixture into its pure (or nearly pure) species is not.

1.3 Explain in detail, using the first and second laws of thermodynamics, why the separation of a mixture into essentially pure species or other mixtures of differing compositions requires the transfer of energy to the mixture or a degradation of its energy.

Section 1.3

1.4 Compare the advantages and disadvantages of making separations using an ESA versus using an MSA.

1.5 Every other year, the magazine *Hydrocarbon Processing* publishes a petroleum-refining handbook, which gives process-flow diagrams and data for more than 75 commercial processes. For each of the following processes in the November 1990 handbook, list the separation operations of the type given in Table 1.1 and indicate what chemical(s) is(are) being separated:

- (a) Hydrotreating (Chevron), p. 114
- (b) Ethers (Phillips), p. 128
- (c) Alkylation (Exxon), p. 130
- (d) Treating of BTX cut (GKT), p. 136

1.6 Every other year, the magazine *Hydrocarbon Processing* publishes a petrochemical handbook, which gives process-flow diagrams and data for more than 50 commercial processes. For each of the following processes in the March 1991 handbook, list the separation operations of the type given in Table 1.1 and indicate what chemical(s) is(are) being separated:

- (a) Linear alkylbenzene (UOP), p. 130
- (b) Methyl amines (Acid-Amine Technologies), p. 133
- (c) Butene-2 (Phillips), p. 144
- (d) Caprolactam (SNIA), p. 150
- (e) Ethylene glycols (Scientific Design), p. 156
- (f) Styrene (Monsanto), p. 188

Section 1.4

1.7 Explain why osmosis is not used as a separation operation.

1.8 The osmotic pressure, π , of sea water is given approximately by the expression $\pi = RTc/M$, where c is the concentration of the dissolved salts (solutes) in g/cm^3 and M is the average molecular weight of the solutes as ions. If pure water is to be recovered from sea water at 298 K and containing 0.035 g of salts/ cm^3 of sea water and $M = 31.5$, what is the minimum required pressure difference across the membrane in kPa?

1.9 It has been shown that a liquid membrane of aqueous ferrous ethylenediaminetetraacetic acid, maintained between two sets of microporous, hydrophobic, hollow fibers that are packed in a

permeator cell, can selectively and continuously remove sulfur dioxide and nitrogen oxides from the flue gas of power-generating plants. Prepare a detailed drawing of a possible device to carry out such a separation. Show all locations of inlet and outlet streams, the arrangement of the hollow fibers, and a method for handling the membrane liquid. Should the membrane liquid be left in the cell or circulated? Would a sweep fluid be needed to remove the oxides?

Section 1.5

1.10 Explain the differences, if any, between adsorption and gas-solid chromatography.

1.11 In gas-liquid chromatography, is it essential that the gas flow through the packed tube in plug flow? Discuss in detail.

Section 1.6

1.12 In electrophoresis, explain why most small suspended particles are negatively charged.

1.13 In field-flow fractionation, could a turbulent-flow field be used? Why or why not?

Section 1.7

1.14 The feed to Column C3 of the distillation sequence in Figure 1.9 is given in Table 1.5. However, the separation is to be altered so as to produce a distillate that is 95 mol% pure isobutane with a recovery of isobutane in the distillate (SF) of 96%. Because of the relatively sharp separation in Column C3 between iC_4 and nC_4 , assume all propane goes to the distillate and all C_5 s goes to the bottoms.

(a) Compute the flow rates in lbmol/h of each component in each of the two products leaving Column C3.

(b) What is the percent purity of the normal butane in the bottoms product?

(c) If the purity of the isobutane in the distillate is fixed at 95%, what percent recovery of isobutane in the distillate will maximize the percent purity of normal butane in the bottoms product?

1.15 Five hundred kmol/h of a liquid mixture of light alcohols containing, by moles, 40% methanol (M), 35% ethanol (E), 15% isopropanol (IP), and 10% normal propanol (NP) is distilled in a sequence of two distillation columns. The distillate from the first column is 98% pure M with a 96% recovery of M. The distillate from the second column is 92% pure E with a 95% recovery of E from the process feed. Assume no propanols in the distillate from Column C1, no M in the bottoms from Column C2, and no normal propanol in the distillate from Column C2.

(a) By material balances, assuming negligible propanols in the distillate from the first column, compute the flow rates in kmol/h of each component in each feed, distillate, and bottoms. Draw a labeled block-flow diagram like Figure 1.9. Include the results of the material balances in a table like Table 1.5 and place the table below your block-flow diagram.

(b) Compute the mole-percent purity of the propanol mixture leaving as bottoms from the second column in the sequence.

(c) If the recovery of ethanol is fixed at 95%, what is the maximum purity that can be achieved for the ethanol in the distillate from the second column?

(d) If instead, the purity of the ethanol is fixed at 92%, what is the maximum recovery of ethanol (based on the process feed) that can be achieved?

1.16 A mixture of ethanol and benzene is separated in a network of distillation and membrane separation steps. In one intermediate step, a near-azeotropic liquid mixture of 8,000 kg/h of 23 wt% ethanol in benzene is fed to a pervaporation membrane consisting of a thin ionomeric film of perfluorosulfonic acid polymer cast on a porous Teflon support. The membrane is selective for ethanol such that the vapor permeate contains 60 wt% ethanol, while the non-permeate liquid contains 90 wt% benzene.

(a) Draw a flow diagram of the pervaporation step using the appropriate symbol from Table 1.2 and include on the diagram all of the given information.

(b) Compute the component flow rates in kg/h in the feed stream and in the two product streams and enter these results on the diagram.

(c) What separation operation could be used to further purify the vapor permeate?

Section 1.8

1.17 The Prism gas permeation process developed by the Monsanto Company is highly selective for hydrogen when using hollow-fiber membranes of materials such as silicone-coated polysulphone. In a typical application, a gas at 16.7 MPa and 40°C, containing the following components in kmol/h: 42.4 H_2 , 7.0 CH_4 , and 0.5 N_2 , is separated into a nonpermeate gas at 16.2 MPa and a permeate gas at 4.56 MPa.

(a) If the membrane is nonpermeable to nitrogen, the Prism membrane separation index, on a mole basis (SP) for hydrogen relative to methane is 34.13, and the split fraction (SF) for hydrogen to the permeate gas is 0.6038, calculate the kmol/h of each component and the total flow in kmol/h of both the nonpermeate gas and the permeate gas.

(b) Compute the percent purity of the hydrogen in the permeate gas.

(c) Using an average heat capacity ratio, γ , of 1.4, estimate the outlet temperatures of the two exiting gas streams by assuming the ideal gas law and reversible expansions for each gas and no heat transfer between the two exiting gas streams.

(d) Draw a process-flow diagram of the membrane process and indicate on the diagram for each stream the pressure, temperature, and component flow rates.

1.18 Nitrogen gas can be injected into oil wells to increase the recovery of crude oil (enhanced oil recovery). Usually, natural gas is produced with the oil and it is desirable to recover the nitrogen from the gas for reinjection into the well. Furthermore, the natural gas must not contain more than 3 mol% nitrogen if the natural gas is to be put into a pipeline. A total of 170,000 SCFH (based on 60°F and 14.7 psia) of natural gas containing 18% N_2 , 75% CH_4 , and 7% C_2H_6 at 100°F and 800 psia is to be processed for N_2 removal. A two-step separation process has been proposed consisting of (1) membrane separation with a nonporous glassy polyimide membrane, followed by (2) pressure-swing adsorption using molecular sieves to which the permeate gas is fed. The membrane separator is highly selective for N_2 ($SP_{N_2,CH_4} = 16$) and completely impermeable to ethane. The pressure-swing adsorption step selectively adsorbs methane, giving 97% pure methane product in the adsorbate, with an 85% recovery of CH_4 fed to the adsorber. The non-permeate (retentate) gas from the membrane step and adsorbate from the pressure-swing adsorption step are combined to give a methane stream that contains 3.0% N_2 . The pressure drop across the membrane is 760 psia. The permeate at 20°F is compressed in

two stages to 275 psia and cooled to 100°F before entering the adsorption step. The adsorbate gas, which exits the adsorber during regeneration at 100°F and 15 psia, is compressed in three stages to 800 psia and cooled to 100°F before being combined with nonpermeate gas to give the final pipeline natural gas.

(a) Draw a process-flow diagram of the separation process using appropriate symbols from Tables 1.2 and 1.3. Include the gas compressors and heat exchangers. Label the diagram with all of the data given above, and number all process streams.

(b) Compute by material balances, using the data above, the component flow rates of N_2 , CH_4 , and C_2H_6 in lbmol/h for all process streams entering and exiting the two separation operations. Place the results in a material-balance table similar to Table 1.5.

Section 1.9

1.19 A mixture of ethylbenzene (EB) and the three isomers (ortho, meta, and para) of xylene is widely available in petroleum refineries.

(a) Based on differences in normal boiling points, verify that the separation between *meta*-xylene (MX) and *para*-xylene (PX) by distillation is far more difficult than the separations between EB and PX, and MX and *ortho*-xylene (OX).

(b) Prepare a list of properties for MX and PX similar to Table 1.11. From that list, which property differences might be the best ones to exploit to separate a mixture of these two xylenes?

(c) Explain why melt crystallization and adsorption are used commercially to separate MX and PX.

1.20 When a mixture of ethanol and water is distilled at ambient pressure, the products are a distillate of ethanol and water of near-azeotrope composition (89.4 mol% ethanol) and a bottoms product of nearly pure water. Based on differences in certain properties of ethanol and water, explain how the following separation operations might be able to recover almost pure ethanol from the distillate:

- Extractive distillation
- Azeotropic distillation
- Liquid–liquid extraction
- Crystallization
- Pervaporation membrane
- Adsorption

1.21 A stream containing 7,000 kmol/h of water and 3,000 parts per million (ppm) by weight of ammonia at 350 K and 1 bar is to be processed to remove 90% of the ammonia. What type of separation operation would you use? If it involves a mass-separating agent, propose one.

1.22 A light-hydrocarbon feed stream contains 45.4 kmol/h of propane, 136.1 kmol/h of isobutane, 226.8 kmol/h of *n*-butane, 181.4 kmol/h of isopentane, and 317.4 kmol/h of *n*-pentane. This stream is to be separated in a sequence of three distillation columns, similar to that in Figure 1.9, into four products: (1) propane-rich,

(2) isobutane-rich, (3) *n*-butane-rich, and (4) combined pentanes-rich. However, the distillate from the first column is to be the propane-rich product; the distillate from Column 2 is to be the isobutane-rich product; and the distillate from Column 3 is to be the *n*-butane-rich product, with the combined pentanes being the bottoms from Column 3. The recovery of each main component in each product is to be 98%. For example, 98% of the propane in the feed stream is to appear in the propane-rich product, and 98% of the combined pentanes in the feed stream is to appear in the bottoms product from Column 3.

(a) Draw a process-flow diagram, similar to Figure 1.9.

(b) Complete a material balance for each column and summarize the results in a table similar to Table 1.5. To complete the material balance, you will have to make some assumptions about the flow rates of: (1) isobutane in the distillates for Columns 1 and 3 and (2) *n*-butane in the distillates for Columns 1 and 2, consistent with the specified recoveries. Assume that propane will not be found in the distillate from Column 3 and pentanes will not be found in the distillate from Column 2.

(c) Calculate the mol% purities of each of the products and summarize your results in a table similar to Table 1.7, but without the specifications, which are not given here.

1.23 The need to remove organic pollutants from wastewater is common to many industrial processes. Separation methods that may be considered are: (1) adsorption, (2) distillation, (3) liquid–liquid extraction, (4) membrane separation, (5) stripping with air, and (6) stripping with steam. Discuss the advantages and disadvantages of each method for this application. Be sure to consider the fate of the organic material.

1.24 Many waste gas streams in processing plants contain volatile organic compounds (VOCs), which must be removed. Recovery of the VOCs may be accomplished by several separation methods, including: (1) absorption, (2) adsorption, (3) condensation, (4) freezing, and (5) membrane separation. Discuss the advantages and disadvantages of each method, paying particular attention to the fate of the VOC. For the case of a stream containing 3 mol% acetone in air, draw a flow diagram for a process based on absorption. Choose a reasonable absorbent and include in your process a means to recover the acetone and recycle the absorbent.

1.25 Describe three methods suitable for the separation of air into nitrogen and oxygen.

1.26 What separation methods can be used to separate azeotropic mixtures of water and an organic chemical such as ethanol?

1.27 An aqueous stream contains 5% by weight of magnesium sulfate. Devise a process, complete with a process-flow diagram, for the production of nearly pure magnesium sulfate from this stream.

1.28 Explain why the separation of a stream containing 10 wt% acetic acid in water might be more economical by liquid–liquid extraction with ethyl acetate than by distillation.

Chapter 2

Thermodynamics of Separation Operations

Thermodynamic properties and equations play a major role in separation operations, particularly with respect to energy requirements, phase equilibria, and sizing equipment. This chapter discusses applied thermodynamics for separation processes. Equations for energy balances, entropy and availability balances, and for determining phase densities and phase compositions at equilibrium are developed. These involve thermodynamic properties, including specific volume

or density, enthalpy, entropy, availability, and fugacities and activities together with their coefficients, all as functions of temperature, pressure, and phase composition. Methods for estimating properties for ideal and nonideal mixtures are summarized. However, this chapter is not a substitute for any of the excellent textbooks on chemical engineering thermodynamics. Furthermore, emphasis here is on fluid phases, with little consideration of solid phases.

2.0 INSTRUCTIONAL OBJECTIVES

After completing this chapter, you should be able to:

- Make energy, entropy, and availability balances around a separation process using the first and second laws of thermodynamics.
- Calculate lost work and second-law efficiency of a separation process.
- Explain the concept of phase equilibria in terms of Gibbs free energy, chemical potential, fugacity, fugacity coefficients, activity, and activity coefficients.
- Understand the concept and usefulness of the equilibrium ratio (K -value) for problems involving liquid and/or vapor phases at equilibrium.
- Derive expressions for K -values in terms of fugacity coefficients and activity coefficients.
- Write vapor–liquid K -value expressions for Raoult's law (ideal), a modified Raoult's law, and Henry's law.
- Calculate density, enthalpy, and entropy of ideal mixtures.
- Utilize graphical correlations to obtain thermodynamic properties of ideal and near-ideal mixtures.
- Use nomographs to estimate vapor–liquid K -values of nonideal hydrocarbon and light–gas mixtures.
- Explain how computer programs use equations of state (e.g., Soave-Redlich-Kwong or Peng-Robinson) to compute thermodynamic properties of vapor and liquid mixtures, including K -values.
- Explain how computer programs use liquid-phase activity-coefficient correlations (e.g., Wilson, NRTL, UNIQUAC, or UNIFAC) to compute thermodynamic properties, including K -values, for nonideal vapor and liquid mixtures at equilibrium.

2.1 ENERGY, ENTROPY, AND AVAILABILITY BALANCES

Most industrial separation operations utilize large quantities of energy in the form of heat and/or shaft work. A study by Mix et al. [1] reports that two quads ($1 \text{ quad} = 10^{15} \text{ Btu}$) of energy were consumed by distillation separations in petroleum, chemical, and natural-gas processing plants in the United States in 1976. This amount of energy was 2.7% of the total U.S. energy consumption of 74.5 quads and is equivalent to the energy obtained from approximately 1 million bbl of crude oil per day over a one-year period. This amount of oil can be compared to 13 million bbl/day, the

average amount of crude oil processed by petroleum refineries in the United States in early 1991. At a crude oil price of approximately \$40/bbl, the energy consumption by distillation in the United States is approximately \$20 trillion per year. Thus, it is of considerable interest to know the extent of energy consumption in a separation process, and to what degree energy requirements might be reduced. Such energy estimates can be made by applying the first and second laws of thermodynamics.

Consider the continuous, steady-state, flow system for a general separation process in Figure 2.1. One or more feed streams flowing into the system are separated into two or

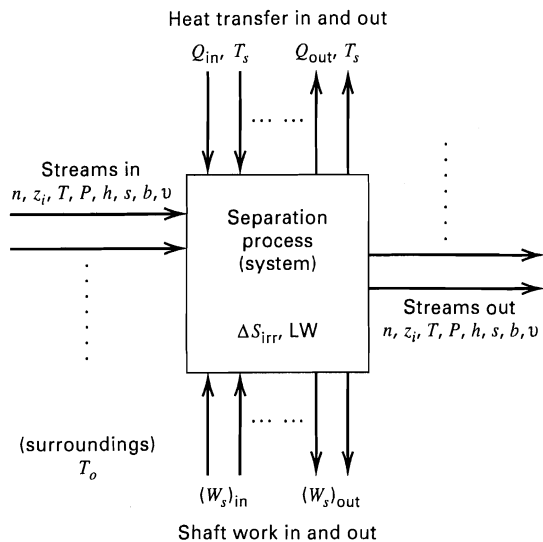


Figure 2.1 General separation system.

more product streams that flow out of the system. For all these streams, we denote the molar flow rates by n , the component mole fractions by z_i , the temperature by T , the pressure by P , the molar enthalpies by h , the molar entropies by s , and the molar availabilities by b . If chemical reactions occur in the process, enthalpies and entropies are referred to the elements, as discussed by Felder and Rousseau [2]; otherwise they can be referred to the compounds. Heat flows in

or out of the system are denoted by Q , and shaft work crossing the boundary of the system is denoted by W_s . At steady state, if kinetic, potential, and surface energy changes are neglected, the first law of thermodynamics (also referred to as the conservation of energy or the energy balance), states that the sum of all forms of energy flowing into the system equals the sum of the energy flows leaving the system:

$$\begin{aligned} & (\text{stream enthalpy flows} + \text{heat transfer} \\ & \quad + \text{shaft work})_{\text{leaving system}} - (\text{stream enthalpy flows} \\ & \quad + \text{heat transfer} + \text{shaft work})_{\text{entering system}} \\ & = 0 \end{aligned}$$

In terms of symbols, the energy balance is given by Eq. (1) in Table 2.1, where all flow rate, heat transfer, and shaft work terms are positive. Molar enthalpies may be positive or negative depending on the reference state.

All separation processes must satisfy the energy balance. Inefficient separation processes require large transfers of heat and/or shaft work both into and out of the process; efficient processes require smaller levels of heat transfer and/or shaft work. The first law of thermodynamics provides no information on energy efficiency, but the second law of thermodynamics (also referred to as the entropy balance), given by Eq. (2) in Table 2.1, does. In words, the steady-state entropy balance is

$$\begin{aligned} & (\text{Stream entropy flows} \\ & \quad + \text{entropy flows by heat transfer})_{\text{leaving system}} \\ & \quad - (\text{stream entropy flows} \\ & \quad + \text{entropy flows by heat transfer})_{\text{entering system}} \\ & = \text{production of entropy by the process} \end{aligned}$$

In the entropy balance equation, the heat sources and sinks in Figure 2.1 are at absolute temperatures T_s . For example, if condensing steam at 150°C supplies heat, Q , to the reboiler of a distillation column, $T_s = 150 + 273 = 423\text{ K}$. If cooling water at an average temperature of 30°C removes heat, Q , in a condenser, $T_s = 30 + 273 = 303\text{ K}$. Unlike the energy balance, which states that energy is conserved, the entropy balance predicts the production of entropy, ΔS_{irr} , which is the irreversible increase in the entropy of the universe. This term, which must be a positive quantity, is a quantitative measure of the thermodynamic inefficiency of a process. In the limit, as a reversible process is approached, ΔS_{irr} tends to zero. Note that the entropy balance contains no terms related to shaft work.

Although ΔS_{irr} is a measure of energy inefficiency, it is difficult to relate to this measure because it does not have the units of energy/time (power). A more useful measure or process inefficiency can be derived by combining (1) and (2) in Table 2.1 to obtain a combined statement of the first and second laws of thermodynamics, which is given as (3) in Table 2.1. To perform this derivation, it is first necessary to define an infinite source of or sink for heat transfer at the absolute temperature, $T_s = T_0$, of the surroundings. This temperature is typically about 300 K and represents the largest

Table 2.1 Universal Thermodynamic Laws for a Continuous, Steady-State, Flow System

Energy balance:

$$(1) \sum_{\text{out of system}} (nh + Q + W_s) - \sum_{\text{in to system}} (nh + Q + W_s) = 0$$

Entropy balance:

$$(2) \sum_{\text{out of system}} \left(ns + \frac{Q}{T_s} \right) - \sum_{\text{in to system}} \left(ns + \frac{Q}{T_s} \right) = \Delta S_{\text{irr}}$$

Availability balance:

$$(3) \sum_{\text{in to system}} \left[nb + Q \left(1 - \frac{T_0}{T_s} \right) + W_s \right] - \sum_{\text{out of system}} \left[nb + Q \left(1 - \frac{T_0}{T_s} \right) + W_s \right] = LW$$

Minimum work of separation:

$$(4) W_{\text{min}} = \sum_{\text{out of system}} nb - \sum_{\text{in to system}} nb$$

Second-law efficiency:

$$(5) \eta = \frac{W_{\text{min}}}{LW + W_{\text{min}}}$$

where $b = h - T_0 s = \text{availability function}$

$LW = T_0 \Delta S_{\text{irr}} = \text{lost work}$

source of coolant associated with the processing plant being analyzed. This might be the average temperature of cooling water, air, or a nearby river, lake, or ocean. Heat transfer associated with this surrounding coolant and transferred from (or to) the process is termed Q_0 . Thus, in both (1) and (2) in Table 2.1, the Q and Q/T_s terms include contributions from Q_0 and Q_0/T_0 , respectively.

The derivation of (3) in Table 2.1 can be made, as shown by de Nevers and Seader [3], by combining (1) and (2) to eliminate Q_0 . The resulting equation is referred to as an *availability* (or *exergy*) balance, where the term availability means "available for complete conversion to shaft work." The stream availability function, b , as defined by

$$b = h - T_0s \quad (2-1)$$

is a measure of the maximum amount of stream energy that can be converted into shaft work if the stream is taken to the reference state. It is similar to Gibbs free energy, $g = h - Ts$, but differs in that the infinite surroundings temperature, T_0 , appears in the equation instead of the stream temperature, T . Terms in (3) in Table 2.1 containing Q are multiplied by $(1 - T_0/T_s)$, which, as shown in Figure 2.2, is the reversible Carnot heat-engine cycle efficiency, representing the maximum amount of shaft work that can be produced from Q at T_s , where the residual amount of energy ($Q - W_s$) is transferred as heat to a sink at T_0 . Shaft work, W_s , remains at its full value in (3). Thus, although Q and W_s have the same thermodynamic worth in (1) of Table 2.1, heat transfer has less worth in (3). This is because shaft work can be converted completely to heat (by friction), but heat cannot be converted completely to shaft work unless the heat is available at an infinite temperature.

Availability, like entropy, is not conserved in a real, irreversible process. The total availability (i.e., ability to produce shaft work) passing into a system is always greater than the total availability leaving a process. Thus (3) in Table 2.1 is written with the "into system" terms first. The difference

is the *lost work*, LW , which is also called the loss of availability (or exergy), and is defined by

$$LW = T_0 \Delta S_{irr} \quad (2-2)$$

Lost work is always a positive quantity. The greater its value, the greater is the energy inefficiency. In the lower limit, as a reversible process is approached, lost work tends to zero. The lost work has the same units as energy, thus making it easy to attach significance to its numerical value. In words, the steady-state availability balance is

$$\begin{aligned} &(\text{Stream availability flows} + \text{availability of heat} \\ &+ \text{shaft work})_{\text{entering system}} - (\text{stream availability flows} \\ &+ \text{availability of heat} + \text{shaft work})_{\text{leaving system}} \\ &= \text{loss of availability (lost work)} \end{aligned}$$

For any separation process, lost work can be computed from (3) in Table 2.1. Its magnitude depends on the extent of process irreversibilities, which include fluid friction, heat transfer due to finite temperature-driving forces, mass transfer due to finite concentration or activity-driving forces, chemical reactions proceeding at finite displacements from chemical equilibrium, mixing of streams at differing conditions of temperature, pressure, and/or composition, and so on. Thus, to reduce the lost work, driving forces for momentum transfer, heat transfer, mass transfer, and chemical reaction must be reduced. Practical limits to this reduction exist because, as driving forces are decreased, equipment sizes increase, tending to infinitely large sizes as driving forces approach zero.

For a separation process that occurs without chemical reaction, the summation of the stream availability functions leaving the process is usually greater than the same summation for streams entering the process. In the limit for a reversible process ($LW = 0$), (3) of Table 2.1 reduces to (4), where $W_{\min}$ is the minimum shaft work required to conduct the separation and is equivalent to the difference in the heat transfer and shaft work terms in (3). This minimum work is independent of the nature (or path) of the separation process. The work of separation for an actual irreversible process is always greater than the minimum value computed from (4).

From (3) of Table 2.1, it is seen that as a separation process becomes more irreversible, and thus more energy inefficient, the increasing LW causes the required equivalent work of separation to increase by the same amount. Thus, the equivalent work of separation for an irreversible process is given by the sum of lost work and minimum work of separation. The *second-law efficiency*, therefore, can be defined as

$$\begin{aligned} &(\text{fractional second-law efficiency}) \\ &= \left(\frac{\text{minimum work of separation}}{\text{equivalent actual work of separation}} \right) \end{aligned}$$

In terms of symbols, the efficiency is given by (5) in Table 2.1.

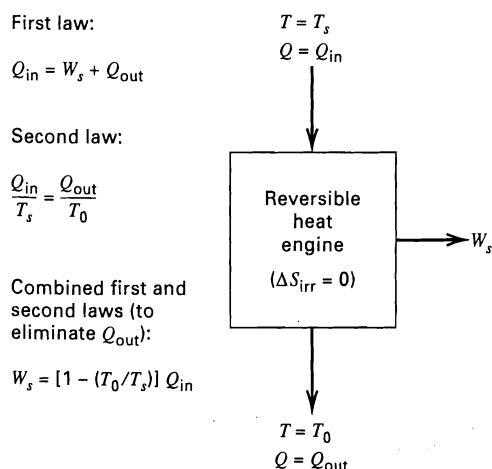


Figure 2.2 Carnot heat engine cycle for converting heat to shaft work.

EXAMPLE 2.1

For the propylene–propane separation of Figure 1.12, using the following thermodynamic properties for certain streams, as estimated from the Soave–Redlich–Kwong equation of state discussed in Section 2.5, and the relations given in Table 2.1, compute in SI units:

- The condenser duty, Q_C
- The reboiler duty, Q_R
- The irreversible entropy production, assuming 303 K for the condenser cooling-water sink and 378 K for the reboiler steam source
- The lost work, assuming $T_0 = 303$ K
- The minimum work of separation
- The second-law efficiency

Stream	Phase Condition	Enthalpy (h), kJ/kmol	Entropy (s), kJ/kmol-K
Feed (F)	Liquid	13,338	-4.1683
Overhead vapor (OV)	Vapor	24,400	24.2609
Distillate (D) and reflux (R)	Liquid	12,243	-13.8068
Bottoms (B)	Liquid	14,687	-2.3886

SOLUTION

Place the condenser (C) cooling water and the reboiler (R) steam outside the distillation system. Thus, Q_C and Q_R cross the boundary of the system. The following calculations are made using the stream flow rates in Figure 1.12 and properties above.

- Compute condenser duty from an energy balance around the condenser. From (1), Table 2.1, noting that the overhead-vapor molar flow rate is given by $n_{OV} = n_R + n_D$ and $h_R = h_D$, the condenser duty is

$$\begin{aligned} Q_C &= n_{OV}(h_{OV} - h_R) \\ &= (2,293 + 159.2)(24,400 - 12,243) \\ &= 29,811,000 \text{ kJ/h} \end{aligned}$$

- Compute reboiler duty from an energy balance around the entire distillation operation. (An energy balance around the reboiler cannot be made because data are not given for the boilup rate.) From (1), Table 2.1,

$$\begin{aligned} Q_R &= n_D h_D + n_B h_B + Q_C - n_F h_F \\ &= 159.2(12,243) + 113(14,687) \\ &\quad + 29,811,000 - 272.2(13,338) \\ &= 29,789,000 \text{ kJ/h} \end{aligned}$$

- Compute the production of entropy from an entropy balance around the entire distillation system. From Eq. (2), Table 2.1,

$$\begin{aligned} \Delta S_{\text{irr}} &= n_D s_D + n_B s_B + Q_C/T_C - n_F s_F - Q_R/T_R \\ &= 159.2(-13.8068) + 113(-2.3886) + 29,811,000/303 \\ &\quad - 272.2(-4.1683) - 29,789,000/378 \\ &= 18,246 \text{ kJ/h-K} \end{aligned}$$

- Compute lost work from its definition at the bottom of Table 2.1:

$$\begin{aligned} LW &= T_0 \Delta S_{\text{irr}} \\ &= 303(18,246) = 5,529,000 \text{ kJ/h} \end{aligned}$$

Alternatively, compute lost work from an availability balance around the entire distillation system. From (3), Table 2.1, where the availability function, b , is defined near the bottom of Table 2.1,

$$\begin{aligned} LW &= n_F b_F + Q_R(1 - T_0/T_R) \\ &\quad - n_D b_D - n_B b_B - Q_C(1 - T_0/T_C) \\ &= 272.2[13,338 - (303)(-4.1683)] \\ &\quad + 29,789,000(1 - 303/378) \\ &\quad - 159.2[12,243 - (303)(-13.8068)] \\ &\quad - 113[14,687 - (303)(-2.3886)] \\ &\quad - 29,811,000(1 - 303/303) \\ &= 5,529,000 \text{ kJ/h} \quad (\text{same result}) \end{aligned}$$

- Compute the minimum work of separation for the entire distillation system. From (4), Table 2.1,

$$\begin{aligned} W_{\text{min}} &= n_D b_D + n_B b_B - n_F b_F \\ &= 159.2[12,243 - (303)(-13.8068)] \\ &\quad + 113[14,687 - (303)(-2.3886)] \\ &\quad - 272.2[13,338 - (303)(-4.1683)] \\ &= 382,100 \text{ kJ/h} \end{aligned}$$

- Compute the second-law efficiency for the entire distillation system. From (5), Table 2.1,

$$\begin{aligned} \eta &= \frac{W_{\text{min}}}{LW + W_{\text{min}}} \\ &= \frac{382,100}{5,529,000 + 382,100} \\ &= 0.0646 \quad \text{or} \quad 6.46\% \end{aligned}$$

This low second-law efficiency is typical of a difficult distillation separation, which in this case requires 150 theoretical stages with a reflux ratio of almost 15 times the distillate rate.

2.2 PHASE EQUILIBRIA

Analysis of separations equipment frequently involves the assumption of phase equilibria as expressed in terms of Gibbs free energy, chemical potentials, fugacities, or activities. For each phase in a multiphase, multicomponent system, the total Gibbs free energy is

$$G = G(T, P, N_1, N_2, \dots, N_C)$$

where N_i = moles of species i . At equilibrium, the total G for all phases is a minimum, and methods for determining this minimum are referred to as *free-energy minimization techniques*. Gibbs free energy is also the starting point for the derivation of commonly used equations for expressing phase equilibria. From classical thermodynamics, the total differential of G is given by

$$dG = -SdT + VdP + \sum_{i=1}^C \mu_i dN_i \quad (2-3)$$

where μ_i is the chemical potential or partial molar Gibbs free energy of species i . When (2-3) is applied to a closed system consisting of two or more phases in equilibrium at uniform temperature and pressure, where each phase is an open system capable of mass transfer with another phase, then

$$dG_{\text{system}} = \sum_{p=1}^N \left[\sum_{i=1}^C \mu_i^{(p)} dN_i^{(p)} \right]_{P,T} \quad (2-4)$$

where the superscript (p) refers to each of N phases in equilibrium. Conservation of moles of each species, in the absence of chemical reaction, requires that

$$dN_i^{(1)} = - \sum_{p=2}^N dN_i^{(p)} \quad (2-5)$$

which, upon substitution into (2-4), gives

$$\sum_{p=2}^N \left[\sum_{i=1}^C (\mu_i^{(p)} - \mu_i^{(1)}) dN_i^{(p)} \right] = 0 \quad (2-6)$$

With $dN_i^{(1)}$ eliminated in (2-6), each $dN_i^{(p)}$ term can be varied independently of any other $dN_i^{(p)}$ term. But this requires that each coefficient of $dN_i^{(p)}$ in (2-6) be zero. Therefore,

$$\mu_i^{(1)} = \mu_i^{(2)} = \mu_i^{(3)} = \dots = \mu_i^{(N)} \quad (2-7)$$

Thus, the chemical potential of a particular species in a multicomponent system is identical in all phases at physical equilibrium.

Fugacities and Activity Coefficients

Chemical potential cannot be expressed as an absolute quantity, and the numerical values of chemical potential are difficult to relate to more easily understood physical quantities. Furthermore, the chemical potential approaches an infinite negative value as pressure approaches zero. For these reasons, the chemical potential is not favored for phase equilibria calculations. Instead, fugacity, invented by G. N. Lewis in 1901, is employed as a surrogate.

The partial fugacity of species i in a mixture is like a pseudo-pressure, defined in terms of the chemical potential by

$$\bar{f}_i = C \exp \left(\frac{\mu_i}{RT} \right) \quad (2-8)$$

where C is a temperature-dependent constant. Regardless of the value of C , it is shown by Prausnitz, Lichtenthaler, and Azevedo [4] that (2-7) can be replaced with

$$\bar{f}_i^{(1)} = \bar{f}_i^{(2)} = \bar{f}_i^{(3)} = \dots = \bar{f}_i^{(N)} \quad (2-9)$$

Thus, at equilibrium, a given species has the same partial fugacity in each existing phase. This equality, together with

equality of phase temperatures and pressures,

$$T^{(1)} = T^{(2)} = T^{(3)} = \dots = T^{(N)} \quad (2-10)$$

and

$$P^{(1)} = P^{(2)} = P^{(3)} = \dots = P^{(N)} \quad (2-11)$$

constitutes the required conditions for phase equilibria. For a pure component, the partial fugacity, $\bar{f}_i$, becomes the pure-component fugacity, f_i . For a pure, ideal gas, fugacity is equal to the pressure, and for a component in an ideal-gas mixture, the partial fugacity is equal to its partial pressure, $p_i = y_i P$. Because of the close relationship between fugacity and pressure, it is convenient to define their ratio for a pure substance as

$$\phi_i = \frac{f_i}{P} \quad (2-12)$$

where ϕ_i is the pure-species fugacity coefficient, which has a value of 1.0 for an ideal gas. For a mixture, partial fugacity coefficients are defined by

$$\bar{\phi}_{iV} \equiv \frac{\bar{f}_{iV}}{y_i P} \quad (2-13)$$

$$\bar{\phi}_{iL} \equiv \frac{\bar{f}_{iL}}{x_i P} \quad (2-14)$$

such that as ideal-gas behavior is approached, $\bar{\phi}_{iV} \rightarrow 1.0$ and $\bar{\phi}_{iL} \rightarrow P_i^s / P$, where P_i^s = vapor (saturation) pressure.

At a given temperature, the ratio of the partial fugacity of a component to its fugacity in some defined standard state is termed the *activity*. If the standard state is selected as the pure species at the same pressure and phase condition as the mixture, then

$$a_i \equiv \frac{\bar{f}_i}{f_i^o} \quad (2-15)$$

Since at phase equilibrium, the value of f_i^o is the same for each phase, substitution of (2-15) into (2-9) gives another alternative condition for phase equilibria,

$$a_i^{(1)} = a_i^{(2)} = a_i^{(3)} = \dots = a_i^{(N)} \quad (2-16)$$

For an ideal solution, $a_{iV} = y_i$ and $a_{iL} = x_i$.

To represent departure of activities from mole fractions when solutions are nonideal, *activity coefficients* based on concentrations in mole fractions are defined by

$$\gamma_{iV} \equiv \frac{a_{iV}}{y_i} \quad (2-17)$$

$$\gamma_{iL} \equiv \frac{a_{iL}}{x_i} \quad (2-18)$$

For ideal solutions, $\gamma_{iV} = 1.0$ and $\gamma_{iL} = 1.0$.

For convenient reference, thermodynamic quantities that are useful in phase equilibria calculations are summarized in Table 2.2.

Table 2.2 Thermodynamic Quantities for Phase Equilibria

Thermodynamic Quantity	Definition	Physical Significance	Limiting Value for Ideal Gas and Ideal Solution
Chemical potential	$\mu_i \equiv \left(\frac{\partial G}{\partial N_i} \right)_{P,T,N_j}$	Partial molar free energy, $\bar{g}_i$	$\mu_i = \bar{g}_i$
Partial fugacity	$\bar{f}_i \equiv C \exp \left(\frac{\mu_i}{RT} \right)$	Thermodynamic pressure	$\bar{f}_{iV} = y_i P$ $\bar{f}_{iL} = x_i P_i^s$
Fugacity coefficient of a pure species	$\phi_i \equiv \frac{f_i}{P}$	Deviation to fugacity due to pressure	$\phi_{iV} = 1.0$ $\phi_{iL} = \frac{P_i^s}{P}$
Partial fugacity coefficient of a species in a mixture	$\bar{\phi}_{iV} \equiv \frac{\bar{f}_{iV}}{y_i P}$ $\bar{\phi}_{iL} \equiv \frac{\bar{f}_{iL}}{x_i P}$	Deviations to fugacity due to pressure and composition	$\bar{\phi}_{iV} = 1.0$ $\bar{\phi}_{iL} = \frac{P_i^s}{P}$
Activity	$a_i \equiv \frac{\bar{f}_i}{f_i^o}$	Relative thermodynamic pressure	$a_{iV} = y_i$ $a_{iL} = x_i$
Activity coefficient	$\gamma_{iV} \equiv \frac{a_{iV}}{y_i}$ $\gamma_{iL} \equiv \frac{a_{iL}}{x_i}$	Deviation to fugacity due to composition	$\gamma_{iV} = 1.0$ $\gamma_{iL} = 1.0$

K-Values

A *phase-equilibrium ratio* is the ratio of mole fractions of a species present in two phases at equilibrium. For the vapor–liquid case, the constant is referred to as the *K-value* (vapor–liquid equilibrium ratio or *K-factor*):

$$K_i \equiv \frac{y_i}{x_i} \quad (2-19)$$

For the liquid–liquid case, the ratio is referred to as the distribution coefficient or liquid–liquid equilibrium ratio:

$$K_{D_i} \equiv \frac{x_i^{(1)}}{x_i^{(2)}} \quad (2-20)$$

For equilibrium-stage calculations involving the separation of two or more components, separation factors, like (1-4), are defined by forming ratios of equilibrium ratios. For the vapor–liquid case, *relative volatility* is defined by

$$\alpha_{ij} \equiv \frac{K_i}{K_j} \quad (2-21)$$

For the liquid–liquid case, the *relative selectivity* is

$$\beta_{ij} \equiv \frac{K_{D_i}}{K_{D_j}} \quad (2-22)$$

Equilibrium ratios can be expressed by the quantities in Table 2.2 in a variety of rigorous formulations. However, the only ones of practical interest are developed as follows. For vapor–liquid equilibrium, (2-9) becomes, for each component,

$$\bar{f}_{iV} = \bar{f}_{iL}$$

To form an equilibrium ratio, these partial fugacities are commonly replaced by expressions involving mole fractions as derived from the definitions in Table 2.2:

$$\bar{f}_{iL} = \gamma_{iL} x_i f_{iL}^o \quad (2-23)$$

or

$$\bar{f}_{iL} = \bar{\phi}_{iL} x_i P \quad (2-24)$$

and

$$\bar{f}_{iV} = \bar{\phi}_{iV} y_i P \quad (2-25)$$

If (2-24) and (2-25) are used with (2-19), a so-called *equation-of-state form* of the *K-value* is obtained:

$$K_i = \frac{\bar{\phi}_{iL}}{\bar{\phi}_{iV}} \quad (2-26)$$

This expression has received considerable attention, with applications of importance being the Starling modification of the Benedict, Webb, and Rubin (B–W–R–S) equation of state [5], the Soave modification of the Redlich–Kwong (S–R–K or R–K–S) equation of state [6], the Peng–Robinson (P–R) equation of state [7], and the Plöcker et al. modification of the Lee–Kesler (L–K–P) equation of state [8].

If (2-23) and (2-25) are used, a so-called *activity coefficient form* of the *K-value* is obtained:

$$K_i = \frac{\gamma_{iL} f_{iL}^o}{\bar{\phi}_{iV} P} = \frac{\gamma_{iL} \phi_{iL}}{\bar{\phi}_{iV}} \quad (2-27)$$

Table 2.3 Useful Expressions for Estimating K -Values for Vapor–Liquid Equilibria ($K_i \equiv y_i/x_i$)

	Equation	Recommended Application
Rigorous forms:		
(1) Equation-of-state	$K_i = \frac{\bar{\phi}_{iL}}{\bar{\phi}_{iV}}$	Hydrocarbon and light gas mixtures from cryogenic temperatures to the critical region
(2) Activity coefficient	$K_i = \frac{\gamma_{iL} \bar{\phi}_{iL}}{\bar{\phi}_{iV}}$	All mixtures from ambient to near-critical temperature
Approximate forms:		
(3) Raoult's law (ideal)	$K_i = \frac{P_i^s}{P}$	Ideal solutions at near-ambient pressure
(4) Modified Raoult's law	$K_i = \frac{\gamma_{iL} P_i^s}{P}$	Nonideal liquid solutions at near-ambient pressure
(5) Poynting correction	$K_i = \gamma_{iL} \bar{\phi}_{iV}^s \left(\frac{P_i^s}{P} \right) \exp \left(\frac{1}{RT} \int_{P_i^s}^P v_{iL} dP \right)$	Nonideal liquid solutions at moderate pressure and below the critical temperature
(6) Henry's law	$K_i = \frac{H_i}{P}$	Low-to-moderate pressures for species at supercritical temperature

Since 1960, (2-27) has received considerable attention with applications to important industrial systems presented by Chao and Seader (C–S) [9], with a modification by Grayson and Streed [10].

Table 2.3 is a summary of useful formulations for estimating K -values for vapor–liquid equilibrium. Included are the two rigorous expressions given by (2-26) and (2-27), from which the other approximate formulations are derived. The so-called Raoult's law or ideal K -value is obtained from (2-27) by substituting from Table 2.2, for an ideal gas and ideal gas and liquid solutions, $\gamma_{iL} = 1.0$, $\bar{\phi}_{iL} = P_i^s/P$, and $\bar{\phi}_{iV} = 1.0$. The modified Raoult's law relaxes the assumption of an ideal liquid solution by including the liquid-phase activity coefficient. The Poynting-correction form for moderate pressures is obtained by approximating the pure-component liquid fugacity coefficient in (2-27) by the expression

$$\bar{\phi}_{iL} = \bar{\phi}_{iV}^s \frac{P_i^s}{P} \exp \left(\frac{1}{RT} \int_{P_i^s}^P v_{iL} dP \right) \quad (2-28)$$

where the exponential term is the Poynting factor or correction. If the liquid molar volume is reasonably constant over the pressure range, the integral in (2-28) becomes $v_{iL}(P - P_i^s)$. For a light gas species, whose critical temperature is less than the system temperature, the Henry's law form for the K -value is convenient provided that a value of H_i , the empirical Henry's law coefficient, is available. This constant for a particular species, i , depends on liquid-phase composition, temperature, and pressure. As pointed out in other chapters, other forms of Henry's law are used besides the one in Table 2.3. Included in Table 2.3 are recommendations for the application of each of the vapor–liquid K -value expressions.

Regardless of which thermodynamic formulation is used for estimating K -values, the accuracy depends on the particular correlations used for the thermodynamic properties

required (i.e., vapor pressure, activity coefficient, and fugacity coefficients). For practical applications, the choice of K -value formulation is a compromise among considerations of accuracy, complexity, convenience, and past experience.

For liquid–liquid equilibria, (2-9) becomes

$$\bar{f}_{iL}^{(1)} = \bar{f}_{iL}^{(2)} \quad (2-29)$$

where superscripts (1) and (2) refer to the two immiscible liquid phases. A rigorous formulation for the distribution coefficient is obtained by combining (2-23) with (2-20) to obtain an expression involving only activity coefficients:

$$K_{Di} = \frac{x_i^{(1)}}{x_i^{(2)}} = \frac{\gamma_{iL}^{(2)} \bar{f}_{iL}^{o(2)}}{\gamma_{iL}^{(1)} \bar{f}_{iL}^{o(1)}} = \frac{\gamma_{iL}^{(2)}}{\gamma_{iL}^{(1)}} \quad (2-30)$$

For vapor–solid equilibria, a useful formulation can be derived if the solid phase consists of just one of the components of the vapor phase. In that case, the combination of (2-9) and (2-25) gives

$$f_{iS} = \bar{\phi}_{iV} y_i P \quad (2-31)$$

At low pressure, $\bar{\phi}_{iV} = 1.0$ and the solid fugacity can be approximated by the vapor pressure of the solid to give for the vapor-phase mole fraction of the component forming the solid phase:

$$y_i = \frac{(P_i^s)_{\text{solid}}}{P} \quad (2-32)$$

For liquid–solid equilibria, a similar useful formulation can be derived if again the solid phase is a pure component. Then the combination of (2-9) and (2-23) gives

$$f_{iS} = \gamma_{iL} x_i \bar{f}_{iL}^o \quad (2-33)$$

At low pressure, the solid fugacity can be approximated by vapor pressure to give, for the component in the solid phase,

$$x_i = \frac{(P_i^s)_{\text{solid}}}{\gamma_{iL}(P_i^s)_{\text{liquid}}} \quad (2-34)$$

EXAMPLE 2.2

Estimate the K -values of a vapor-liquid mixture of water and methane at 2 atm total pressure for temperatures of 20 and 80°C.

SOLUTION

At the conditions of temperature and pressure, water will exist mainly in the liquid phase and will follow Raoult's law, as given in Table 2.3. Because methane has a critical temperature of -82.5°C , well below the temperatures of interest, it will exist mainly in the vapor phase and follow Henry's law, in the form given in Table 2.3. From *Perry's Chemical Engineers' Handbook*, 6th ed., pp. 3-237 and 3-103, the following vapor pressure data for water and Henry's law coefficients for CH_4 are obtained:

$T, ^\circ\text{C}$	P^s for H_2O , atm	H for CH_4 , atm
20	0.02307	3.76×10^4
80	0.4673	6.82×10^4

K -values for water and methane are estimated from (3) and (6), respectively, in Table 2.3, using $P = 2$ atm, with the following results:

$T, ^\circ\text{C}$	$K_{\text{H}_2\text{O}}$	K_{CH_4}
20	0.01154	18,800
80	0.2337	34,100

The above K -values confirm the assumptions of the phase distribution of the two species. The K -values for H_2O are low, but increase rapidly with increasing temperature. The K -values for methane are extremely high and do not change rapidly with temperature for this example.

2.3 IDEAL-GAS, IDEAL-LIQUID-SOLUTION MODEL

Design procedures for separation equipment require numerical values for phase enthalpies, entropies, densities, and phase-equilibrium ratios. Classical thermodynamics provides a means for obtaining these quantities in a consistent manner from P - v - T relationships, which are usually referred to as *equation-of-state models*. The simplest model applies when both liquid and vapor phases are ideal solutions (all activity coefficients equal 1.0) and the vapor is an ideal gas. Then the thermodynamic properties can be computed from unary constants for each of the species in the mixture in a relatively straightforward manner using the equations given in Table 2.4. In general, these ideal

equations apply only at near-ambient pressure, up to about 50 psia (345 kPa), for mixtures of isomers or components of similar molecular structure.

For the vapor, the molar volume and mass density are computed from (1), the ideal-gas law in Table 2.4, which involves the molecular weight, M , of the mixture and the universal gas constant, R . For a mixture, the ideal-gas law assumes that both Dalton's law of additive partial pressures and Amagat's law of additive pure-species volumes apply.

The molar vapor enthalpy is computed from (2) by integrating, for each species, an equation in temperature for the zero-pressure heat capacity at constant pressure, C_{Pv}^o , starting from a reference (datum) temperature, T_0 , to the temperature of interest, and then summing the resulting species vapor enthalpies on a mole-fraction basis. Typically, T_0 is taken as 0 K or 25°C . Although the reference pressure is zero, pressure has no effect on the enthalpy of an ideal gas. A common empirical representation of the effect of

Table 2.4 Thermodynamic Properties for Ideal Mixtures

Ideal gas and ideal-gas solution:

$$(1) \quad v_v = \frac{V}{\sum_{i=1}^C N_i} = \frac{M}{\rho_v} = \frac{RT}{P}, \quad M = \sum_{i=1}^C y_i M_i$$

$$(2) \quad h_v = \sum_{i=1}^C y_i \int_{T_0}^T (C_{Pv}^o)_{iV} dT = \sum_{i=1}^C y_i h_{iv}^o$$

$$(3) \quad s_v = \sum_{i=1}^C y_i \int_{T_0}^T \frac{(C_{Pv}^o)_{iV}}{T} dT - R \ln \left(\frac{P}{P_0} \right) - R \sum_{i=1}^C y_i \ln y_i$$

where the first term is s_v^o

Ideal-liquid solution:

$$(4) \quad v_L = \frac{V}{\sum_{i=1}^C N_i} = \frac{M}{\rho_L} = \sum_{i=1}^C x_i v_{iL}, \quad M = \sum_{i=1}^C x_i M_i$$

$$(5) \quad h_L = \sum_{i=1}^C x_i (h_{iV}^o - \Delta H_i^{\text{vap}})$$

$$(6) \quad s_L = \sum_{i=1}^C x_i \left[\int_{T_0}^T \frac{(C_{Pv}^o)_{iV}}{T} dT - \frac{\Delta H_i^{\text{vap}}}{T} \right] - R \ln \left(\frac{P}{P_0} \right) - R \sum_{i=1}^C x_i \ln x_i$$

Vapor-liquid equilibria:

$$(7) \quad K_i = \frac{P_i^s}{P}$$

Reference conditions (datum): h , ideal gas at T_0 and zero pressure; s , ideal gas at T_0 and 1 atm pressure.

Refer to elements if chemical reactions occur; otherwise refer to components.

temperature on the zero-pressure vapor heat capacity of a pure component is the following fourth-degree polynomial equation:

$$C_{P_v}^o = [a_0 + a_1T + a_2T^2 + a_3T^3 + a_4T^4]R \quad (2-35)$$

where the constants a_k depend on the species. Values of the constants for hundreds of compounds, with T in K, are tabulated by Poling, Prausnitz, and O'Connell [11]. Because $C_P = dh/dT$, (2-35) can be integrated for each species to give the ideal-gas species molar enthalpy:

$$h_v^o = \int_{T_0}^T C_{P_v}^o dT = \sum_{k=1}^5 \frac{a_{k-1}(T^k - T_0^k)R}{k} \quad (2-36)$$

The vapor molar entropy is computed from (3) in Table 2.4 by integrating $C_{P_v}^o/T$ from T_0 to T for each species, summing on a mole-fraction basis, adding a term for the effect of pressure referenced to a datum pressure, P_0 , which is generally taken to be 1 atm (101.3 kPa), and adding a term for the entropy change of mixing. Unlike the ideal vapor enthalpy, the ideal vapor entropy includes terms for the effects of pressure and mixing. The reference pressure is not taken to be zero, because the entropy is infinity at zero pressure. If (2-35) is used for the heat capacity,

$$\int_{T_0}^T \left(\frac{C_{P_v}^o}{T} \right) dT = \left[a_0 \ln \left(\frac{T}{T_0} \right) + \sum_{k=1}^4 \frac{a_k(T^k - T_0^k)}{k} \right] R \quad (2-37)$$

The liquid molar volume and mass density are computed from the pure species molar volumes using (4) in Table 2.4 and the assumption of additive volumes (not densities). The effect of temperature on pure-component liquid density from the freezing point to the critical region at saturation pressure is correlated well by the empirical two-constant equation of Rackett [12]:

$$\rho_L = AB^{-(1-T/T_c)^{2/7}} \quad (2-38)$$

where values of the constants A , B , and the critical temperature, T_c , are tabulated for approximately 700 organic compounds by Yaws et al. [13].

The vapor pressure of a pure liquid species is well represented over a wide range of temperature from below the normal boiling point to the critical region by an empirical extended Antoine equation:

$$\ln P^s = k_1 + k_2/(k_3 + T) + k_4T + k_5 \ln T + k_6T^{k_7} \quad (2-39)$$

where the constants k_k depend on the species. Values of the constants for hundreds of compounds are built into the physical-property libraries of all computer-aided process simulation and design programs. Constants for other

empirical vapor-pressure equations are tabulated for hundreds of compounds by Poling et al. [11]. At low pressures, the enthalpy of vaporization is given in terms of vapor pressure by classical thermodynamics:

$$\Delta H^{\text{vap}} = RT^2 \left(\frac{d \ln P^s}{dT} \right) \quad (2-40)$$

If (2-39) is used for the vapor pressure, (2-40) becomes

$$\Delta H^{\text{vap}} = RT^2 \left[-\frac{k_2}{(k_3 + T)^2} + k_4 + \frac{k_5}{T} + k_7k_6T^{k_7-1} \right] \quad (2-41)$$

The enthalpy of an ideal-liquid mixture is obtained by subtracting the molar enthalpy of vaporization from the ideal vapor molar enthalpy for each species, as given by (2-36), and summing these, as shown by (5) in Table 2.4. The entropy of the ideal-liquid mixture, given by (6), is obtained in a similar manner from the ideal-gas entropy by subtracting the molar entropy of vaporization, given by $\Delta H^{\text{vap}}/T$.

The final equation in Table 2.4 gives the expression for the ideal K -value, previously included in Table 2.3. Although it is usually referred to as the Raoult's law K -value, where Raoult's law is given by

$$p_i = x_i P_i^s \quad (2-42)$$

the assumption of Dalton's law is also required:

$$p_i = y_i P \quad (2-43)$$

Combination of (2-42) and (2-43) gives the Raoult's law K -value:

$$K_i \equiv \frac{y_i}{x_i} = \frac{P_i^s}{P} \quad (2-44)$$

The extended Antoine equation, (2-39) (or some other suitable expression), can be used to estimate vapor pressure. Note that the ideal K -value is independent of phase compositions, but is exponentially dependent on temperature, because of the vapor pressure, and inversely proportional to pressure. From (2-21), the relative volatility using (2-44) is independent of pressure.

EXAMPLE 2.3

Styrene is manufactured by catalytic dehydrogenation of ethylbenzene, followed by vacuum distillation to separate styrene from unreacted ethylbenzene [14]. Typical conditions for the feed to an industrial distillation unit are 77.5°C (350.6 K) and 100 torr (13.33 kPa) with the following vapor and liquid flows

at equilibrium:

Component	<i>n</i> , kmol/h	
	Vapor	Liquid
Ethylbenzene (EB)	76.51	27.31
Styrene (S)	61.12	29.03

Based on the property constants given below, and assuming that the ideal-gas, ideal-liquid-solution model of Table 2.4 is suitable at this low pressure, estimate values of v_V , ρ_V , h_V , s_V , v_L , ρ_L , h_L , and s_L in SI units, and the K -values and relative volatility.

Property Constants for (2-35), (2-38), (2-39) (In all cases, T is in K)		
	Ethylbenzene	Styrene
M , kg/kmol	106.168	104.152
$C_{P_V}^\circ$, J/kmol-K:		
$a_0 R$	-43,098.9	-28,248.3
$a_1 R$	707.151	615.878
$a_2 R$	-0.481063	-0.40231
$a_3 R$	1.30084×10^{-4}	9.93528×10^{-5}
$a_4 R$	0	0
P° , Pa:		
k_1	86.5008	130.542
k_2	-7,440.61	-9,141.07
k_3	0	0
k_4	0.00623121	0.0143369
k_5	-9.87052	-17.0918
k_6	4.13065×10^{-18}	1.8375×10^{-18}
k_7	6	6
ρ_L , kg/m ³ :		
A	289.8	299.2
B	0.268	0.264
T_c , K	617.9	617.1
$R = 8.314$ kJ/kmol-K or kPa-m ³ /kmol-K = 8,314 J/kmol-K		

SOLUTION

Phase mole-fraction compositions and average molecular weights: From $y_i = (n_{iV})/n_V$, $x_i = (n_{iL})/n_L$,

	Ethylbenzene	Styrene
y	0.5559	0.4441
x	0.4848	0.5152

From (1), Table 2.4,

$$M_V = (0.5559)(106.168) + (0.4441)(104.152) = 105.27$$

$$M_L = (0.4848)(106.168) + (0.5152)(104.152) = 105.13$$

Vapor molar volume and density: From (1), Table 2.4,

$$v_V = \frac{RT}{P} = \frac{(8.314)(350.65)}{(13.332)} = 219.2 \text{ m}^3/\text{kmol}$$

$$\rho_V = \frac{M_V}{v_V} = \frac{105.27}{219.2} = 0.4802 \text{ kg/m}^3$$

Vapor molar enthalpy (datum = ideal gas at 298.15 K and 0 kPa): From (2-36) for ethylbenzene,

$$\begin{aligned} h_{EBV}^\circ &= -43098.9(350.65 - 298.15) \\ &\quad + \left(\frac{707.151}{2}\right)(350.65^2 - 298.15^2) \\ &\quad - \left(\frac{0.481063}{3}\right)(350.65^3 - 298.15^3) \\ &\quad + \left(\frac{1.30084 \times 10^{-4}}{4}\right)(350.65^4 - 298.15^4) \\ &= 7,351,900 \text{ J/kmol} \end{aligned}$$

Similarly,

$$h_{SV}^\circ = 6,957,700 \text{ J/kmol}$$

From (2), Table 2.4, for the mixture,

$$\begin{aligned} h_V &= \sum y_i h_{iV}^\circ = (0.5559)(7,351,900) \\ &\quad + (0.4441)(6,957,100) = 7,176,800 \text{ J/kmol} \end{aligned}$$

Vapor molar entropy (datum = pure components as vapor at 298.15 K, 101.3 kPa): From (2-37), for each component,

$$\begin{aligned} \int_{T_0}^T \left(\frac{C_{P_V}^\circ}{T}\right) dT &= 22,662 \text{ J/kmol-K for ethylbenzene} \\ &\text{and } 21,450 \text{ J/kmol-K for styrene} \end{aligned}$$

From (3), Table 2.4, for the mixture,

$$\begin{aligned} s_V &= [(0.5559)(22,662.4) + (0.4441)(21,450.3)] \\ &\quad - 8,314 \ln \left(\frac{13.332}{101.3}\right) - 8,314[(0.5559) \ln(0.5559) \\ &\quad + (0.4441) \ln(0.4441)] = 44,695 \text{ J/kmol-K} \end{aligned}$$

Note that the terms for the pressure effect and the mixing effect are significant for this problem.

Liquid molar volume and density. From (2-38), for ethylbenzene,

$$\rho_{EBL} = (289.8)(0.268)^{-(1-350.65/617.9)^{2/7}} = 816.9 \text{ kg/m}^3$$

$$v_{EBL} = \frac{M_{EB}}{\rho_{EBL}} = 0.1300 \text{ m}^3/\text{kmol}$$

Similarly,

$$\rho_{SL} = 853.0 \text{ kg/m}^3$$

$$v_{SL} = 0.1221 \text{ m}^3/\text{kmol}$$

From (4), Table 2.4, for the mixture,

$$v_L = (0.4848)(0.1300) + (0.5152)(0.1221) = 0.1259 \text{ m}^3/\text{kmol}$$

$$\rho_L = \frac{M_L}{v_L} = \frac{105.13}{0.1259} = 835.0 \text{ kg/m}^3$$

Liquid molar enthalpy (datum = ideal gas at 298.15 K): Use (5) in Table 2.4 for the mixture. For the enthalpy of vaporization of

ethylbenzene, from (2-41),

$$\Delta H_{EB}^{\text{vap}} = 8,314(350.65)^2 \left[\frac{-(-7,440.61)}{(0 + 350.65)^2} + 0.00623121 \right. \\ \left. + \frac{-9.87052}{(350.65)} + 6(4.13065 \times 10^{-18})(350.65)^5 \right] \\ = 39,589,800 \text{ J/kmol}$$

Similarly,

$$\Delta H_S^{\text{vap}} = 40,886,700 \text{ J/kmol}$$

Then, applying (5), Table 2.4, using h_{EBv}^o and h_{Sv}^o from above,

$$h_L = [(0.4848)(7,351,900 - 39,589,800) \\ + (0.5152)(6,957,700 - 40,886,700)] \\ = -33,109,000 \text{ J/kmol}$$

Liquid molar entropy (datum = pure components as vapor at 298.15 K and 101.3 kPa): From (6), Table 2.4 for the mixture, using values for $\int_{T_0}^T (C_{Pv}^o/T) dT$ and ΔH^{vap} of EB and S from above,

$$s_L = (0.4848) \left(22,662 - \frac{39,589,800}{350.65} \right) \\ + (0.5152) \left(21,450 - \frac{40,886,700}{350.65} \right) \\ - 8,314 \ln \left(\frac{13.332}{101.3} \right) \\ - 8,314 [0.4848 \ln(0.4848) + 0.5152 \ln(0.5152)] \\ = -70,150 \text{ J/kmol-K}$$

K-values: Because (7), Table 2.4 will be used to compute the K-values, first estimate the vapor pressures using (2-39). For ethylbenzene,

$$\ln P_{EB}^s = 86.5008 + \left(\frac{-7,440.61}{(0 + 350.65)} \right) \\ + 0.00623121(350.65) + (-9.87052) \ln(350.65) \\ + 4.13065 \times 10^{-18}(350.65)^6 \\ = 9.63481 \\ P_{EB}^s = \exp(9.63481) = 15,288 \text{ Pa} = 15.288 \text{ kPa}$$

Similarly,

$$P_S^s = 11.492 \text{ kPa}$$

From (7), Table 2.4,

$$K_{EB} = \frac{15.288}{13.332} = 1.147$$

$$K_S = \frac{11.492}{13.332} = 0.862$$

Relative volatility: From (2-21),

$$\alpha_{EB,S} = \frac{K_{EB}}{K_S} = \frac{1.147}{0.862} = 1.331$$

2.4 GRAPHICAL CORRELATIONS OF THERMODYNAMIC PROPERTIES

Calculations of estimated thermodynamic and other physical properties for the design of separation operations are most commonly carried out with computer-aided, process design

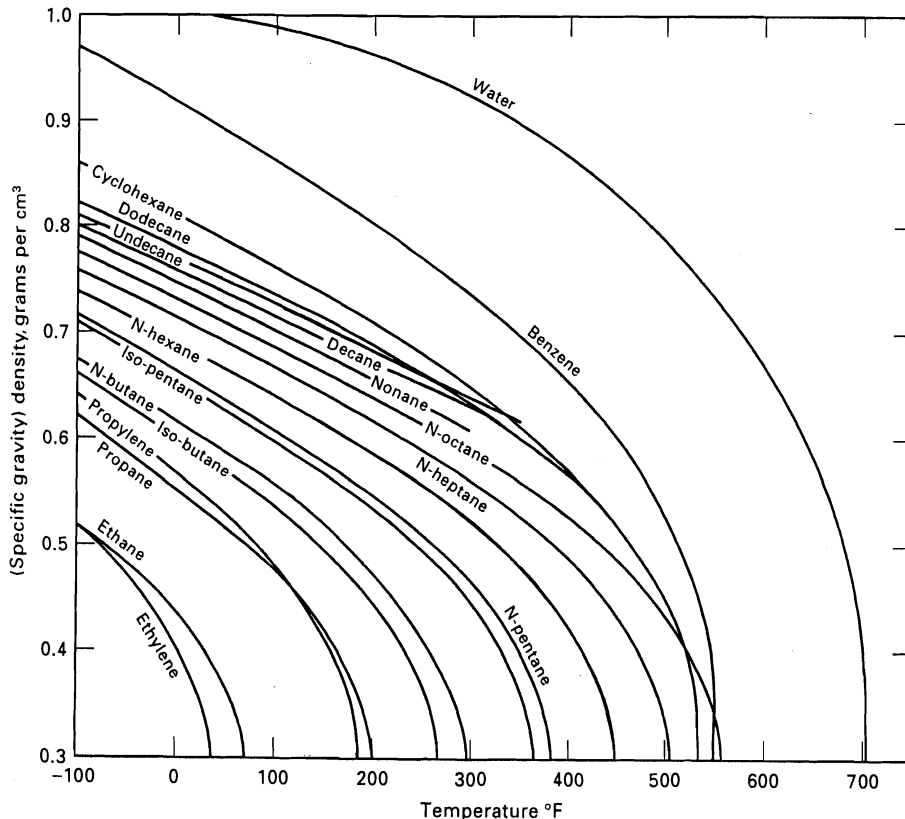


Figure 2.3 Hydrocarbon fluid densities.

[Adapted from G.G. Brown, D.L. Katz, G.G. Oberfell, and R.C. Alden, *Natural Gasoline and the Volatile Hydrocarbons*, Nat'l Gas Assoc. Amer., Tulsa, OK (1948).]

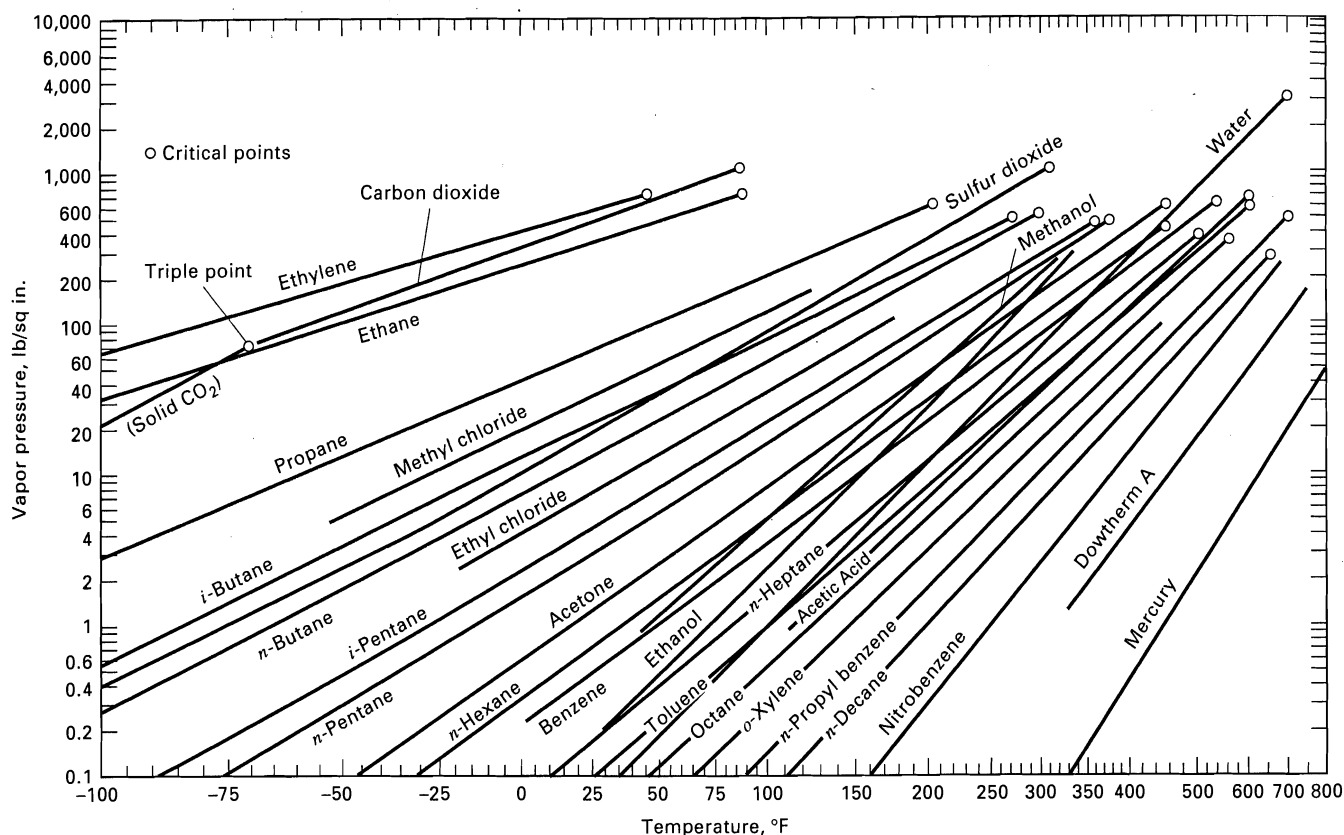


Figure 2.4 Vapor pressure as a function of temperature.

[Adapted from A.S. Faust, L.A. Wenzel, C.W. Clump, L. Maus, and L.B. Andersen, *Principles of Unit Operations*, John Wiley and Sons, New York (1960).]

and simulation programs, such as Aspen Plus, HYSYS, ChemCad, and Pro/II. However, plots of properties can best show effects of temperature and pressure. Some representative plots, which are readily generated by simulation programs, are shown in this section.

Saturated liquid densities as a function of temperature are plotted for some hydrocarbons in Figure 2.3. The density decreases rapidly as the critical temperature is approached until it becomes equal to the density of the vapor phase at the critical point. The liquid density curves are well correlated by the modified Rackett equation (2-38).

Figure 2.4 is a plot of liquid-state vapor pressures for some common chemicals, covering a wide range of temperature from below the normal boiling point to the critical temperature, where the vapor pressure terminates at the critical pressure. In general, the curves are found to fit the extended Antoine equation (2-39) reasonably well. This plot is useful for determining the phase state (liquid or vapor) of a pure substance and for estimating Raoult's law K -values from (2-44) [or (3) in Table 2.3].

Curves of ideal-gas, zero-pressure enthalpy over a wide range of temperature are given in Figure 2.5 for light-paraffin hydrocarbons. The datum is the liquid phase at 0°C, at which the enthalpy is zero. The derivatives of these curves fit the fourth-degree polynomial (2-35) for the ideal-gas heat

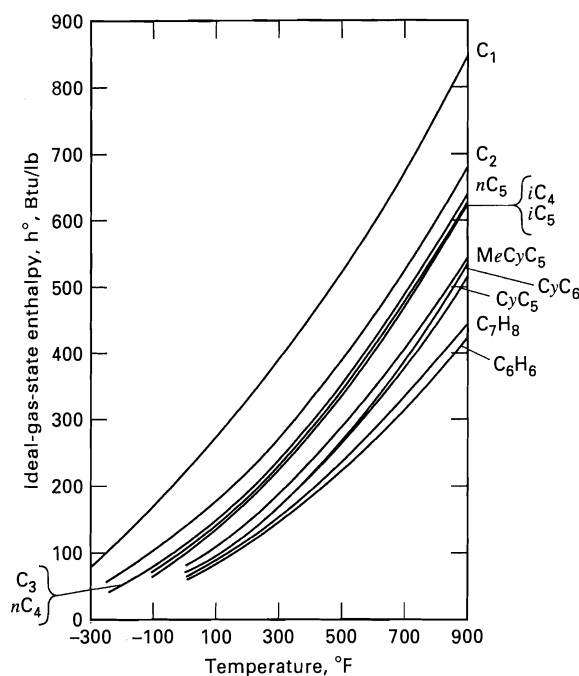


Figure 2.5 Ideal-gas-state enthalpy of pure components.

[Adapted from *Engineering Data Book*, 9th ed., Gas Processors Suppliers Association, Tulsa (1972).]

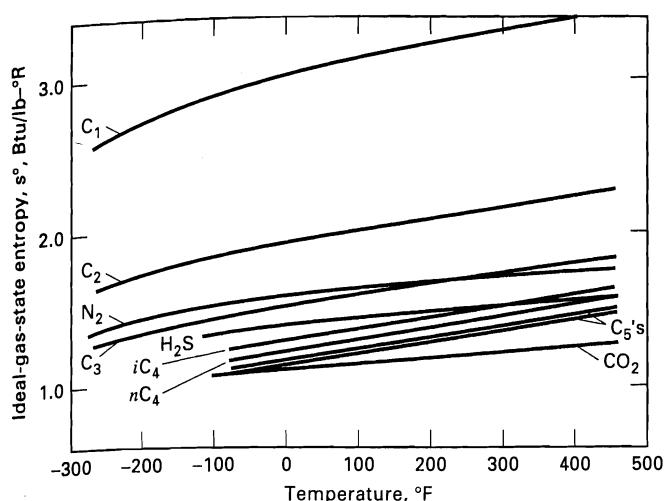


Figure 2.6 Ideal-gas-state entropy of pure components.

[Adapted from *Engineering Data Book*, 9th ed., Gas Processors Suppliers Association, Tulsa (1972).]

capacity reasonably well. Curves of ideal-gas entropy of several light gases, over a wide range of temperature, are given in Figure 2.6.

Enthalpies (heats) of vaporization are plotted as a function of saturation temperature in Figure 2.7 for light-paraffin hydrocarbons. These values are independent of pressure and decrease to zero at the critical point, where vapor and liquid phases become indistinguishable.

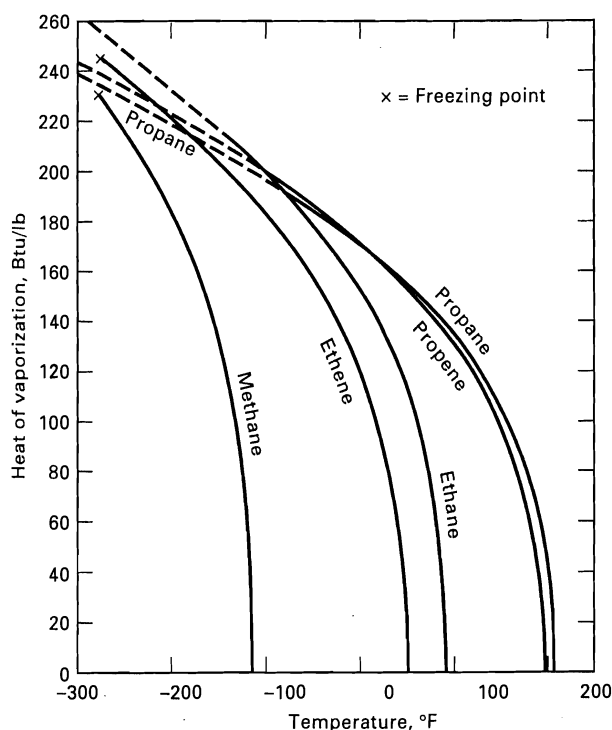


Figure 2.7 Heat of vaporization of light olefins and paraffins.

[Adapted from American Petroleum Institute, Technical Data Book, Washington, DC (Aug. 1963).]

Nomographs for determining the effects of temperature and pressure on the K -values of hydrocarbons and light gases are presented in Figures 2.8 and 2.9, which are taken from Hadden and Grayson [15]. In both charts, all K -values collapse to 1.0 at a pressure of 5,000 psia (34.5 MPa). This pressure, called the *convergence pressure*, depends on the boiling range of the components in the mixture. For example, in Figure 2.10 the components of the mixture (N_2 to nC_{10}) cover a very wide boiling-point range, resulting in a convergence pressure of close to 2,500 psia. For narrow-boiling mixtures, such as a mixture of ethane and propane, the convergence pressure is generally less than 1,000 psia. The K -value charts of Figures 2.8 and 2.9 apply strictly to a convergence pressure of 5,000 psia. A detailed procedure for correcting for the convergence pressure is given by Hadden and Grayson [15]. Use of the nomographs is illustrated below in Exercise 2.4.

No simple charts are available for estimating liquid-liquid equilibrium constants (distribution coefficients) because of the pronounced effect of composition. However, for ternary systems that are dilute in the solute and involve almost immiscible solvents, an extensive tabulation of distribution coefficients for the solute is given by Robbins [16].

EXAMPLE 2.4

Petroleum refining begins with the distillation, at near-atmospheric pressure, of crude oil into fractions of different boiling ranges. The fraction boiling from 0 to 100°C, the light naphtha, is a blending stock for gasoline. The fraction boiling from 100 to 200°C, the heavy naphtha, undergoes subsequent chemical processing into more useful products. One such process is steam cracking to produce a gas containing ethylene, propylene, and a number of other compounds, including benzene and toluene. This gas is then sent to a distillation train to separate the mixture into a dozen or more products. In the first column, hydrogen and methane are removed by cryogenic distillation at 3.2 MPa (464 psia). At a tray in the distillation column where the temperature is 40°F, use the appropriate K -value nomograph to estimate K -values for H_2 , CH_4 , C_2H_4 , and C_3H_6 .

SOLUTION

At 40°F, Figure 2.8 applies. The K -value of hydrogen depends on the other compounds in the mixture. Because appreciable amounts of benzene and toluene are present, locate a point (call it A) midway between the points for " H_2 in benzene" and " H_2 in toluene." Next, locate a point (call it B) at 40°F and 464 psia on the T-P grid. Connect points A and B with a straight line and read a value of $K = 100$ where the line intersects the K scale.

In a similar way, with the same location for point B, read $K = 11$ for methane. For ethylene (ethene) and propylene (propene), the point A is located on the normal boiling-point scale and the same point is used for B. Resulting K -values are 1.5 and 0.32, respectively.

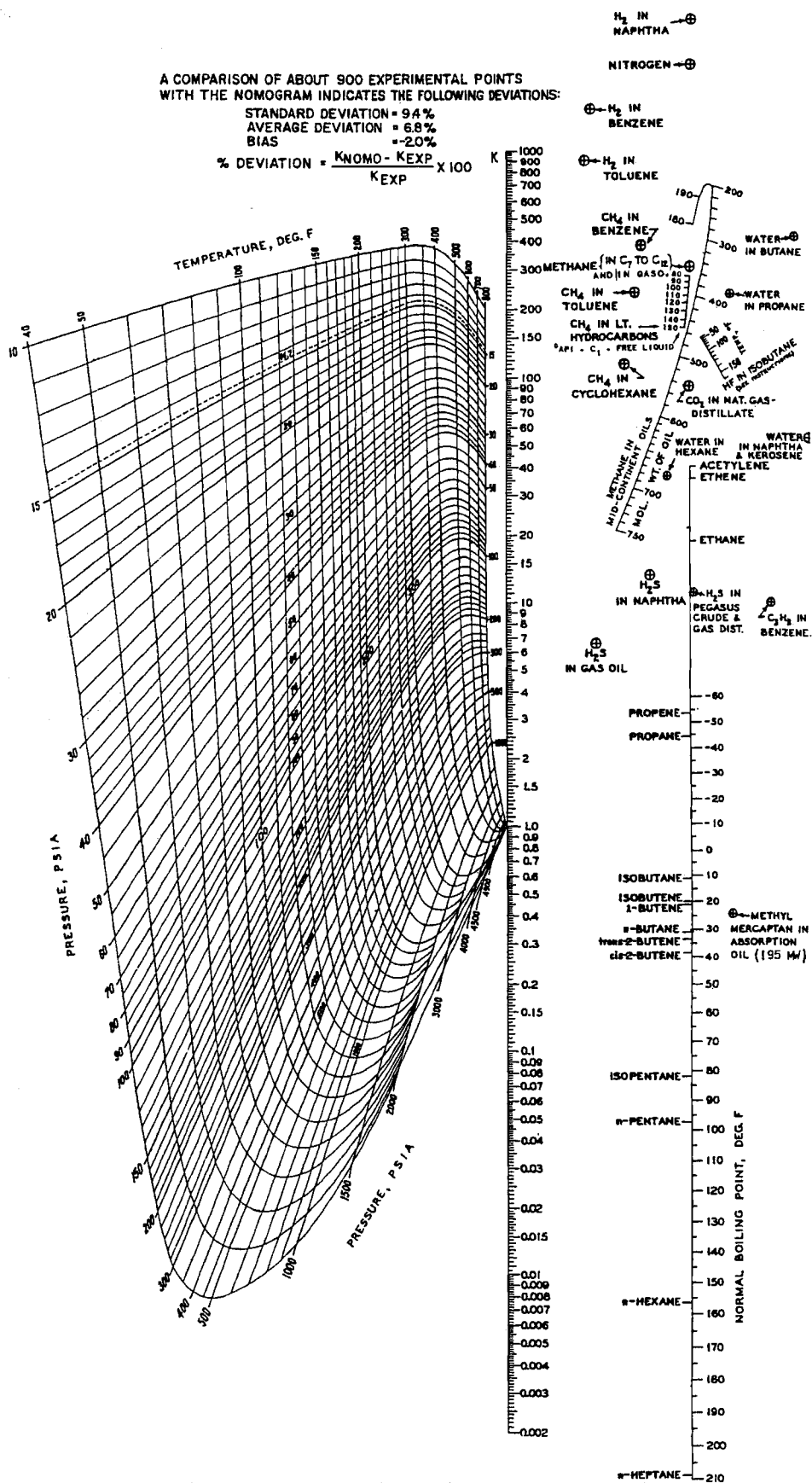
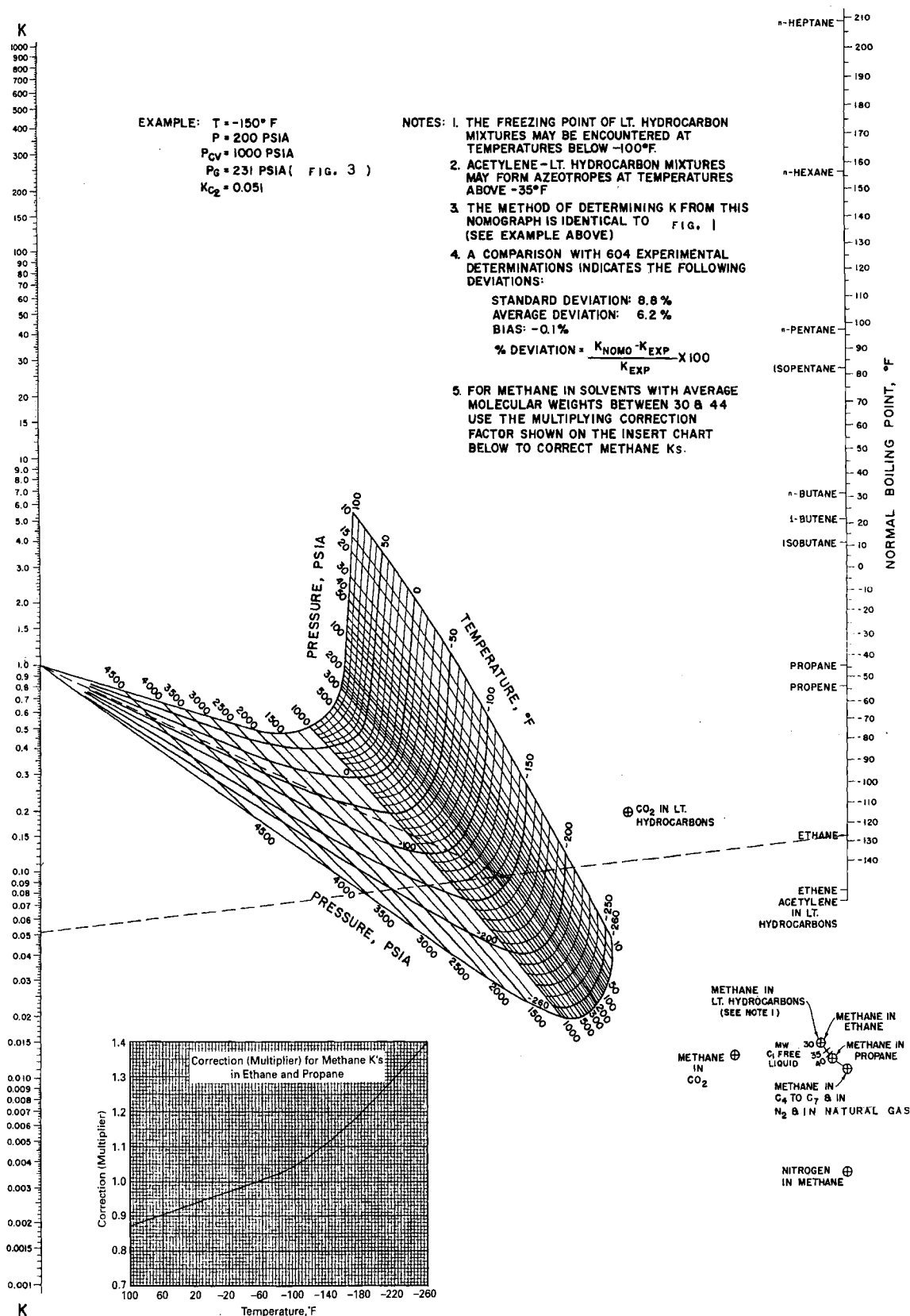


Figure 2.8 Vapor-liquid equilibria, 40 to 800°F.

[From S.T. Hadden and H.G. Grayson, *Hydrocarbon Proc. and Petrol. Refiner*, 40, 207 (Sept. 1961), with permission.]

Figure 2.9 Vapor-liquid equilibria, -260 to 100°F .[From S.T. Hadden and H.G. Grayson, *Hydrocarbon Proc. and Petrol. Refiner*, 40, 207 (Sept. 1961), with permission.]

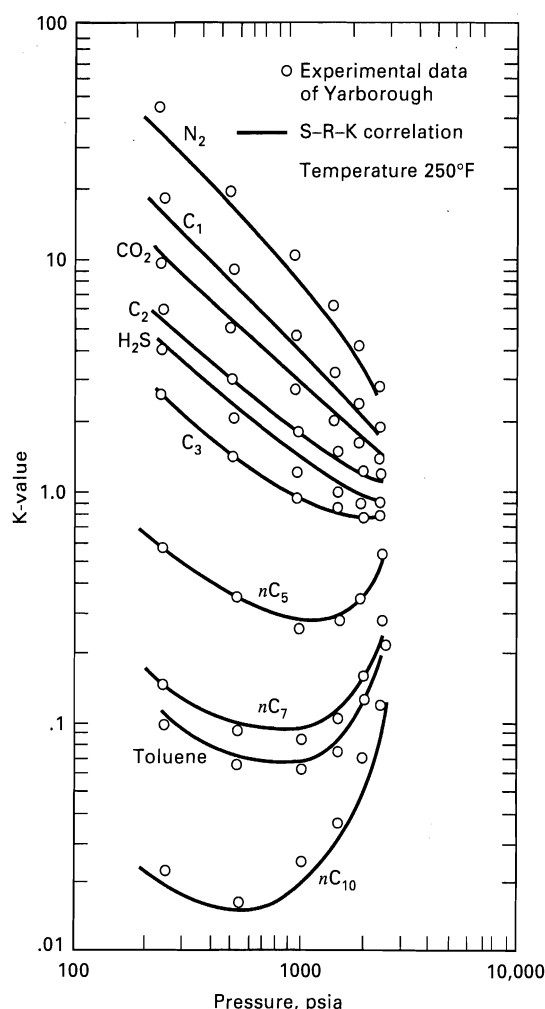


Figure 2.10 Comparison of experimental K -value data and S-R-K correlation.

2.5 NONIDEAL THERMODYNAMIC PROPERTY MODELS

Unlike the equations of Table 2.1, which are universally applicable to all pure substances and mixtures, whether ideal or nonideal, no universal equations are available for computing, for nonideal mixtures, values of thermodynamic properties such as density, enthalpy, entropy, fugacities, and activity coefficients as functions of temperature, pressure, and phase composition. Instead, two types of models are used: (1) P - v - T equation-of-state models and (2) activity coefficient or free-energy models. These are based on *constitutive equations* because they depend on the constitution or nature of the components in the mixture.

P - v - T Equation-of-State Models

The first type of model is a relationship between molar volume (or density), temperature, and pressure, usually referred to as a P - v - T equation of state. A large number of such equations have been proposed, mostly for the vapor phase. The simplest is the ideal-gas law, which applies only at low pressures or high temperatures because it neglects the volume occupied by the molecules and intermolecular forces among the molecules. All other equations of state attempt to correct for these two deficiencies. The equations of state that are most widely used by chemical engineers are listed in Table 2.5. These and other equations of state are discussed in some detail by Poling et al. [11].

Not included in Table 2.5 is the van der Waals equation, $P = RT/(v - b) - a/v^2$, where a and b are species-dependent constants that can be estimated from the critical temperature and pressure. The van der Waals equation was the first successful approach to the formulation of an equation of state for a nonideal gas. It is rarely used by chemical

Table 2.5 Useful Equations of State

Name	Equation	Equation Constants and Functions
(1) Ideal gas law	$P = \frac{RT}{v}$	None
(2) Generalized	$P = \frac{ZRT}{v}$	$Z = Z(P_r, T_r, Z_c \text{ or } \omega)$ as derived from data
(3) Redlich-Kwong (R-K)	$P = \frac{RT}{v - b} - \frac{a}{v^2 + bv}$	$b = 0.08664RT_c/P_c$ $a = 0.42748R^2T_c^{2.5}/P_cT^{0.5}$
(4) Soave-Redlich-Kwong (S-R-K or R-K-S)	$P = \frac{RT}{v - b} - \frac{a}{v^2 + bv}$	$b = 0.08664RT_c/P_c$ $a = 0.42748R^2T_c^2 [1 + f_\omega (1 - T_r^{0.5})]^2 / P_c$ $f_\omega = 0.48 + 1.574\omega - 0.176\omega^2$
(5) Peng-Robinson (P-R)	$P = \frac{RT}{v - b} - \frac{a}{v^2 + 2bv - b^2}$	$b = 0.07780RT_c/P_c$ $a = 0.45724R^2T_c^2 [1 + f_\omega (1 - T_r^{0.5})]^2 / P_c$ $f_\omega = 0.37464 + 1.54226\omega - 0.26992\omega^2$

engineers because its range of application is too narrow. However, its development did suggest that all species might have equal reduced molar volumes, $v_r = v/v_c$, at the same reduced temperature, $T_r = T/T_c$, and reduced pressure, $P_r = P/P_c$. This finding, referred to as the *law (principle or theorem) of corresponding states*, was utilized to develop the generalized equation of state given as (2) in Table 2.5. That equation defines the *compressibility factor*, Z , which is a function of P_r , T_r , and the critical compressibility factor, Z_c , or the *acentric factor*, ω , which is determined from experimental P - v - T data. The acentric factor, introduced by Pitzer et al. [17], accounts for differences in molecular shape and is determined from the vapor pressure curve:

$$\omega = \left[-\log \left(\frac{P^s}{P_c} \right)_{T_r=0.7} \right] - 1.000 \quad (2-45)$$

This definition results in a value for ω of zero for symmetric molecules. Some typical values of ω are 0.264, 0.490, and 0.649 for toluene, *n*-decane, and ethyl alcohol, respectively, as taken from the extensive tabulation of Poling et al. [11].

In 1949, Redlich and Kwong [18] published an equation of state that, like the van der Waals equation, contains only two constants, both of which can be determined from T_c and P_c , by applying the critical conditions

$$\left(\frac{\partial P}{\partial v} \right)_{T_c} = 0 \quad \text{and} \quad \left(\frac{\partial^2 P}{\partial v^2} \right)_{T_c} = 0$$

However, the R-K equation, given as (3) in Table 2.5, is a considerable improvement over the van der Waals equation. A study by Shah and Thodos [19] showed that the simple R-K equation, when applied to nonpolar compounds, has an accuracy that compares quite favorably with equations containing many more constants. Furthermore, the R-K equation can approximate the liquid-phase region.

If the R-K equation is expanded to obtain a common denominator, a cubic equation in v results. Alternatively, (2) and (3) in Table 2.5 can be combined to eliminate v to give the compressibility factor, Z , form of the R-K equation:

$$Z^3 - Z^2 + (A - B - B^2)Z - AB = 0 \quad (2-46)$$

where

$$A = \frac{aP}{R^2T^2} \quad (2-47)$$

$$B = \frac{bP}{RT} \quad (2-48)$$

Equation (2-46), which is cubic in Z , can be solved analytically for three roots (e.g., see *Perry's Handbook*, 7th ed., p. 4-20). In general, at supercritical temperatures, where only one phase can exist, one real root and a complex conjugate pair of roots are obtained. Below the critical temperature, where vapor and/or liquid phases can exist, three real roots are obtained, with the largest value of Z (largest v)

corresponding to the vapor phase—that is, Z_V —and the smallest Z (smallest v) corresponding to the liquid phase—that is, Z_L . The intermediate value of Z is of no practical use.

To apply the R-K equation to mixtures, *mixing rules* are used to average the constants a and b for each component in the mixture. The recommended rules for vapor mixtures of C components are

$$a = \sum_{i=1}^C \left[\sum_{j=1}^C y_i y_j (a_i a_j)^{0.5} \right] \quad (2-49)$$

$$b = \sum_{i=1}^C y_i b_i \quad (2-50)$$

EXAMPLE 2.5

Glanville, Sage, and Lacey [20] measured specific volumes of vapor and liquid mixtures of propane and benzene over wide ranges of temperature and pressure. Use the R-K equation to estimate specific volume of a vapor mixture containing 26.92 wt% propane at 400°F (477.6 K) and a saturation pressure of 410.3 psia (2,829 kPa). Compare the estimated and experimental values.

SOLUTION

Let propane be denoted by P and benzene by B. The mole fractions are

$$y_P = \frac{0.2692/44.097}{(0.2692/44.097) + (0.7308/78.114)} = 0.3949$$

$$y_B = 1 - 0.3949 = 0.6051$$

The critical constants for propane and benzene are given by Poling et al. [11]:

	Propane	Benzene
T_c , K	369.8	562.2
P_c , kPa	4,250	4,890

From the equations for the constants b and a in Table 2.5 for the R-K equation, using SI units,

$$b_P = \frac{0.08664(8.3144)(369.8)}{4,250} = 0.06268 \text{ m}^3/\text{kmol}$$

$$a_P = \frac{0.42748(8.3144)^2(369.8)^{2.5}}{(4,250)(477.59)^{0.5}} = 836.7 \text{ kPa}\cdot\text{m}^6/\text{kmol}^2$$

Similarly,

$$b_B = 0.08263 \text{ m}^3/\text{kmol}$$

$$a_B = 2,072 \text{ kPa}\cdot\text{m}^6/\text{kmol}^2$$

From (2-50),

$$b = (0.3949)(0.06268) + (0.6051)(0.08263) = 0.07475 \text{ m}^3/\text{kmol}$$

From (2-49),

$$\begin{aligned} a &= y_P^2 a_P + 2y_P y_B (a_P a_B)^{0.5} + y_B^2 a_B \\ &= (0.3949)^2 (836.7) + 2(0.3949)(0.6051)[(836.7)(2,072)]^{0.5} \\ &\quad + (0.6051)^2 (2,072) = 1,518 \text{ kPa}\cdot\text{m}^6/\text{kmol}^2 \end{aligned}$$

From (2-47) and (2-48) using SI units,

$$\begin{aligned} A &= \frac{(1,518)(2,829)}{(8.314)^2 (477.59)^2} = 0.2724 \\ B &= \frac{(0.07475)(2,829)}{(8.314)(477.59)} = 0.05326 \end{aligned}$$

From (2-46), we obtain the cubic Z form of the R-K equation:

$$Z^3 - Z^2 + 0.2163Z - 0.01451 = 0$$

Solving this equation gives one real root and a conjugate pair of complex roots:

$$Z = 0.7314, \quad 0.1314 + 0.04243i, \quad 0.1314 - 0.04243i$$

The one real root is assumed to be that for the vapor phase.

From (2) of Table 2.5, the molar volume is

$$v = \frac{ZRT}{P} = \frac{(0.7314)(8.314)(477.59)}{2,829} = 1.027 \text{ m}^3/\text{kmol}$$

The average molecular weight of the mixture is computed to 64.68 kg/kmol. The specific volume is

$$\frac{v}{M} = \frac{1.027}{64.68} = 0.01588 \text{ m}^3/\text{kg} = 0.2543 \text{ ft}^3/\text{lb}$$

Glanville et al. report experimental values of $Z = 0.7128$ and $v/M = 0.2478 \text{ ft}^3/\text{lb}$, which are within 3% of the above estimated values.

Following the success of earlier work by Wilson [21], Soave [6] added a third parameter, the acentric factor, ω , defined by (2-45), to the R-K equation. The resulting, so-called Soave-Redlich-Kwong (S-R-K) or Redlich-Kwong-Soave (R-K-S) equation of state, given as (4) in Table 2.5, was immediately accepted for application to mixtures containing hydrocarbons and/or light gases because of its simplicity and accuracy. The main improvement was to make the parameter a a function of the acentric factor and temperature so as to achieve a good fit to vapor pressure data of hydrocarbons and thereby greatly improve the ability of the equation to predict properties of the liquid phase.

Four years after the introduction of the S-R-K equation, Peng and Robinson [7] presented a further modification of the R-K and S-R-K equations in an attempt to achieve improved agreement with experimental data in the critical region and for liquid molar volume. The Peng-Robinson (P-R) equation of state is listed as (5) in Table 2.5. The S-R-K and P-R equations of state are widely applied in process calculations, particularly for saturated vapors and liquids. When applied to mixtures of hydrocarbons and/or light gases, the mixing rules are given by (2-49) and (2-50), except that (2-49) is often modified to include a binary interaction coefficient, k_{ij} :

$$a = \sum_{i=1}^C \left[\sum_{j=1}^C y_i y_j (a_i a_j)^{0.5} (1 - k_{ij}) \right] \quad (2-51)$$

Values of k_{ij} , back-calculated from experimental data, have been published for both the S-R-K and P-R equations. Knapp et al. [22] present an extensive tabulation. Generally, k_{ij} is taken as zero for hydrocarbons paired with hydrogen or other hydrocarbons.

Although the S-R-K and P-R equations were not intended to be applied to mixtures containing polar organic compounds, they are finding increasing use in such applications by employing large values of k_{ij} , in the vicinity of 0.5, as back-calculated from experimental data. However, a preferred procedure for mixtures containing polar organic compounds is to use a more theoretically based mixing rule such as that of Wong and Sandler, which is discussed in detail in Chapter 11 and which bridges the gap between a cubic equation of state and an activity-coefficient equation.

Another theoretical basis for polar and nonpolar substances is the virial equation of state due to Thiesen [23] and Onnes [24]. A common representation of the virial equation, which can be derived from the statistical mechanics of the forces between the molecules, is a power series in $1/v$ for Z :

$$Z = 1 + \frac{B}{v} + \frac{C}{v^2} + \cdots \quad (2-52)$$

An empirical modification of the virial equation is the Starling form [5] of the Benedict-Webb-Rubin (B-W-R) equation of state for hydrocarbons and light gases in both the gas and liquid phases. Walas [25] presents an extensive discussion of B-W-R-type equations, which because of the large number of terms and species constants (at least 8), is not widely used except for pure substances at cryogenic temperatures. A more useful modification of the B-W-R equation is a generalized corresponding-states form developed by Lee and Kesler [26] with an important extension to mixtures by Plöcker et al. [8]. All of the constants in the L-K-P equation are given in terms of the acentric factor and reduced temperature and pressure, as developed from P - v - T data for three simple fluids ($\omega = 0$), methane, argon, and krypton, and a reference fluid ($\omega = 0.398$), n -octane. The equations, constants, and mixing rules in terms of pseudo-critical properties are given by Walas [25]. The Lee-Kesler-Plöcker (L-K-P) equation of state describes vapor and liquid mixtures of hydrocarbons and/or light gases over wide ranges of temperature and pressure.

Derived Thermodynamic Properties from P - v - T Models

In the previous subsection, several useful P - v - T equations of state for the estimation of the molar volume (or density) or pure substances and mixtures in either the vapor or liquid phase were presented. If a temperature-dependent, ideal-gas heat capacity or enthalpy equation, such as (2-35) or (2-36), is also available, all other vapor- and liquid-phase properties can be derived in a consistent manner by applying the classical integral equations of thermodynamics given in Table 2.6. These equations, in the form of departure (from the ideal gas) equations of Table 2.4, and often referred to as residuals, are applicable to vapor or liquid phases.

Table 2.6 Classical Integral Departure Equations of Thermodynamics

At a given temperature and composition, the following equations give the effect of pressure above that for an ideal gas.

Mixture enthalpy:

$$(1) (h - h_V^o) = Pv - RT - \int_{\infty}^v \left[P - T \left(\frac{\partial P}{\partial T} \right)_v \right] dv$$

Mixture entropy:

$$(2) (s - s_V^o) = \int_{\infty}^v \left(\frac{\partial P}{\partial T} \right)_v dv - \int_{\infty}^v \frac{R}{v} dv$$

Pure-component fugacity coefficient:

$$(3) \phi_{iV} = \exp \left[\frac{1}{RT} \int_0^P \left(v - \frac{RT}{P} \right) dP \right]$$

$$= \exp \left[\frac{1}{RT} \int_v^{\infty} \left(P - \frac{RT}{v} \right) dv - \ln Z + (Z - 1) \right]$$

Partial fugacity coefficient:

$$(4) \bar{\phi}_{iV} = \exp \left\{ \frac{1}{RT} \int_V^{\infty} \left[\left(\frac{\partial P}{\partial N_i} \right)_{T,V,N_j} - \frac{RT}{V} \right] dV - \ln Z \right\}$$

$$\text{where } V = v \sum_{i=1}^C N_i$$

When the ideal-gas law, $P = RT/v$, is substituted into (1) to (4) of Table 2.6, the results for the vapor, as expected, are

$$(h - h_V^o) = 0 \quad \phi = 1$$

$$(s - s_V^o) = 0 \quad \bar{\phi} = 1$$

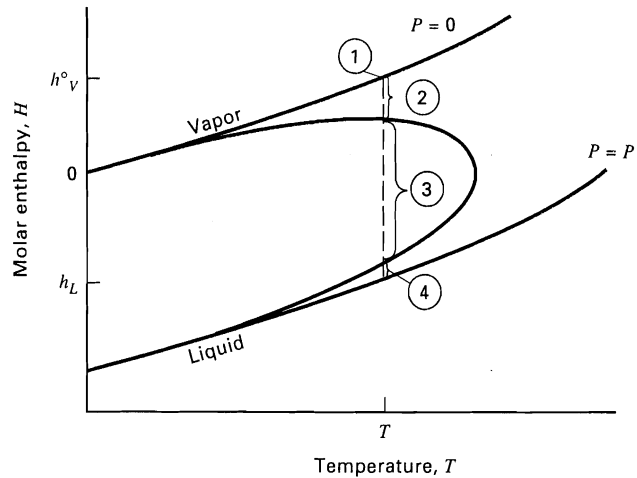
However, when the R-K equation is substituted into the equations of Table 2.6, the following results for the vapor phase are obtained after a rather tedious exercise in calculus:

$$h_V = \sum_{i=1}^C (y_i h_{iV}^o) + RT \left[Z_V - 1 - \frac{3A}{2B} \ln \left(1 + \frac{B}{Z_V} \right) \right] \quad (2-53)$$

$$s_V = \sum_{i=1}^C (y_i s_{iV}^o) - R \ln \left(\frac{P}{P^o} \right) - R \sum_{i=1}^C (y_i \ln y_i) + R \ln(Z_V - B) \quad (2-54)$$

$$\phi_V = \exp \left[Z_V - 1 - \ln(Z_V - B) - \frac{A}{B} \ln \left(1 + \frac{B}{Z_V} \right) \right] \quad (2-55)$$

$$\bar{\phi}_{iV} = \exp \left[(Z_V - 1) \frac{B_i}{B} - \ln(Z_V - B) - \frac{A}{B} \left(2\sqrt{\frac{A_i}{A}} - \frac{B_i}{B} \right) \ln \left(1 + \frac{B}{Z_V} \right) \right] \quad (2-56)$$

**Figure 2.11** Contributions to enthalpy.

The results for the liquid phase are identical if y_i and Z_V (but not h_{iV}^o) are replaced by x_i and Z_L , respectively. It may be surprising that the liquid-phase forms of (2-53) and (2-54) account for the enthalpy and entropy of vaporization, respectively. This is because the R-K equation of state, as well as the S-R-K and P-R equations, are continuous functions in passing between the vapor and liquid regions, as shown for enthalpy in Figure 2.11. Thus, the liquid enthalpy is determined by accounting for the following four effects for a pure species at a temperature below the critical. From (1), Table 2.6, the four contributions to enthalpy in Figure 2.11 are as follows:

$$h_L = h_V^o + Pv - RT - \int_{\infty}^v \left[P - T \left(\frac{\partial P}{\partial T} \right)_v \right] dv$$

$$= \underbrace{h_V^o}_{(1) \text{ Vapor at zero pressure}} + \underbrace{(Pv)_{V_s} - RT - \int_{\infty}^{v_{V_s}} \left[P - T \left(\frac{\partial P}{\partial T} \right)_v \right] dv}_{(2) \text{ Pressure correction for vapor to saturation pressure}} - \underbrace{T \left(\frac{\partial P}{\partial T} \right)_s (v_{V_s} - v_{L_s})}_{(3) \text{ Latent heat of vaporization}} + \underbrace{[(Pv)_L - (Pv)_{L_s}] - \int_{v_{L_s}}^{v_L} \left[P - T \left(\frac{\partial P}{\partial T} \right)_v \right] dv}_{(4) \text{ Correction to liquid for pressure in excess of saturation pressure}} \quad (2-57)$$

where the subscript s refers to the saturation pressure.

The fugacity coefficient, ϕ , of a pure species at temperature T and pressure P from the R-K equation, as given by (2-55), applies to the vapor for $P < P_i^s$. For $P > P_i^s$, ϕ is the fugacity coefficient of the liquid. Saturation pressure corresponds to the condition of $\phi_V = \phi_L$. Thus, at a temperature $T < T_c$, the saturation pressure (vapor pressure), P^s , can be estimated from the R-K equation of state by setting (2-55) for the vapor equal to (2-55) for the liquid and solving, by an iterative procedure, for P , which then equals P^s .

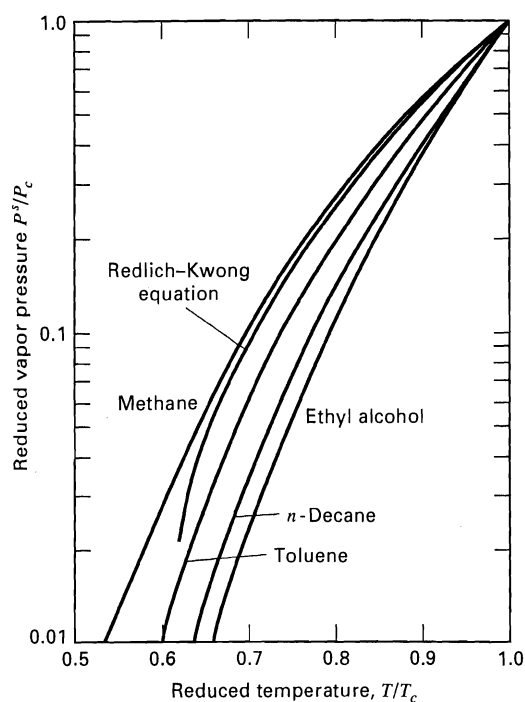
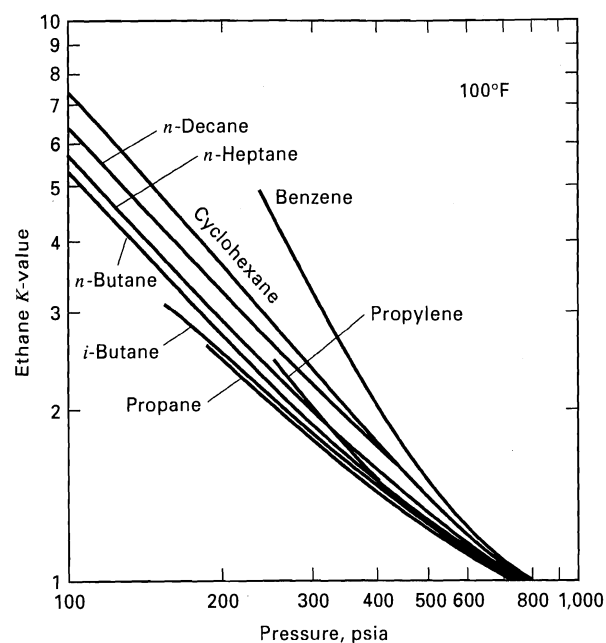


Figure 2.12 Reduced vapor pressure.

The results, as given by Edmister [27], are plotted in reduced form in Figure 2.12. The R-K vapor-pressure curve does not satisfactorily represent data for a wide range of molecular shapes, as witnessed by the experimental curves for methane, toluene, *n*-decane, and ethyl alcohol on the same plot. This failure represents one of the major shortcomings of the R-K equation and is the main reason why Soave [6] modified the R-K equation by introducing the acentric factor in such a way as to greatly improve agreement with experimental vapor-pressure data. Thus, while the critical constants, T_c and P_c alone are insufficient to generalize thermodynamic behavior, a substantial improvement is made by incorporating into the P - v - T equation a third parameter that represents the generic differences in the reduced-vapor-pressure curves.

As seen in (2-56), partial fugacity coefficients depend on pure-species properties, A_i and B_i , and mixture properties, A and B . Once ϕ_{iV} and ϕ_{iL} are computed from (2-56), a K -value can be estimated from (2-26).

The most widely used P - v - T equations of state for separation calculations involving vapor and liquid phases are the S-R-K, P-R, and L-K-P relations. These equations are combined with the integral departure equations of Table 2.6 to obtain useful equations for estimating the enthalpy, entropy, fugacity coefficients, partial fugacity coefficients of vapor and liquid phases, and K -values. The results of the integrations are even more complex than (2-53) to (2-56) and are unsuitable for manual calculations. However, computer programs for making calculations with these equations are rapid, accurate, and readily available. Such programs are



Curves represent experimental data of:
 Kay et al. (Ohio State Univ.) Robinson et al. (Univ. Alberta)
 Sage et al. (Calif. Inst. Tech.) Thodos (Northwestern)

Figure 2.13 K -values of ethane in binary hydrocarbon mixtures at 100°F.

incorporated into widely used steady-state, computer-aided process design and simulation programs, such as Aspen Plus, HYSYS, ChemCad, and Pro/II.

Ideal K -values as determined from Eq. (7) in Table 2.4, depend only on temperature and pressure, and not on composition. Most frequently, ideal K -values are applied to mixtures of nonpolar compounds, particularly hydrocarbons such as paraffins and olefins. Figure 2.13 shows experimental K -value curves for a light hydrocarbon, ethane, in various binary mixtures with other, less volatile hydrocarbons at 100°F (310.93 K) for pressures from 100 psia (689.5 kPa) to convergence pressures between 720 and 780 psia (4.964 MPa to 5.378 MPa). At the convergence pressure, separation by operations involving vapor-liquid equilibrium becomes impossible because all K -values become 1.0. The temperature of 100°F is close to the critical temperature of 550°R (305.6 K) for ethane. Figure 2.13 shows that ethane does not form ideal solutions at 100°F with all the other components because the K -values depend on the other component, even for paraffin homologs. For example, at 300 psia, the K -value of ethane in benzene is 80% higher than in propane.

The ability of equations of state, such as S-R-K, P-R, and L-K-P equations, to predict the effect of composition as well as the effect of temperature and pressure on K -values of multicomponent mixtures of hydrocarbons and light gases is shown in Figure 2.10. The mixture contains 10 species ranging in volatility from nitrogen to *n*-decane. The experimental data points, covering almost a 10-fold range of pressure at 250°F, are those of Yarborough [28]. Agreement with the S-R-K equation is very good.

EXAMPLE 2.6

In the high-pressure, high-temperature, thermal hydrodealkylation of toluene to benzene ($C_7H_8 + H_2 \rightarrow C_6H_6 + CH_4$), excess hydrogen is used to minimize cracking of aromatics to light gases. In practice, conversion of toluene per pass through the reactor is only 70%. To separate and recycle hydrogen, hot reactor effluent vapor of 5,597 kmol/h at 500 psia (3,448 kPa) and 275°F (408.2 K) is partially condensed to 120°F (322 K), with product phases separated in a flash drum. If the composition of the reactor effluent is as follows, and the flash drum pressure is 485 psia (3,344 kPa), calculate equilibrium compositions and flow rates of vapor and liquid leaving the flash drum and the amount of heat that must be transferred using a computer-aided, steady-state, simulation program with each of the equation-of-state models discussed above. Compare the results, including flash-drum K -values and enthalpy and entropy changes.

Component	Mole Fraction
Hydrogen (H)	0.3177
Methane (M)	0.5894
Benzene (B)	0.0715
Toluene (T)	0.0214
	1.0000

SOLUTION

The computations were made with a computer-aided, process-simulation program, using the S-R-K, P-R, and L-K-P equations of state. The results at 120°F and 485 psia are as follows:

	Equation of State		
	S-R-K	P-R	L-K-P
Vapor flows, kmol/h:			
Hydrogen	1,777.1	1,774.9	1,777.8
Methane	3,271.0	3,278.5	3,281.4
Benzene	55.1	61.9	56.0
Toluene	6.4	7.4	7.0
Total	5,109.6	5,122.7	5,122.2
Liquid flows, kmol/h:			
Hydrogen	1.0	3.3	0.4
Methane	27.9	20.4	17.5
Benzene	345.1	338.2	344.1
Toluene	113.4	112.4	112.8
Total	487.4	474.3	474.8
K -values:			
Hydrogen	164.95	50.50	466.45
Methane	11.19	14.88	17.40
Benzene	0.01524	0.01695	0.01507
Toluene	0.00537	0.00610	0.00575
Enthalpy change, GJ/h	35.267	34.592	35.173
Entropy change, MJ/h-K	-95.2559	-93.4262	-95.0287
Percent of benzene and toluene condensed	88.2	86.7	87.9

Because the reactor effluent is mostly hydrogen and methane, the effluent at 275°F and 500 psia, and the equilibrium vapor at 120°F and 485 psia are nearly ideal gases ($0.98 < Z < 1.00$), despite the moderately high pressures. Thus, the enthalpy and entropy changes are dominated by vapor heat capacity and latent heat effects, which are largely independent of which equation of state is used. Consequently, the enthalpy and entropy changes among the three equations of state differ by less than 2%.

Significant differences exist for the K -values of H_2 and CH_4 . However, because the values are in all cases large, the effect on the amount of equilibrium vapor is very small. Reasonable K -values for H_2 and CH_4 , based on experimental data, are 100 and 13, respectively. K -values for benzene and toluene differ among the three equations of state by as much as 11% and 14%, respectively, which, however, causes less than a 2% difference in the percentage of benzene and toluene condensed. Raoult's law K -values for benzene and toluene, based on vapor-pressure data, are 0.01032 and 0.00350, which are considerably lower than the values computed from each of the three equations of state because deviations to fugacities due to pressure are important in the liquid phase and, particularly, in the vapor phase.

Note that the material balances are precisely satisfied for each equation of state. However, the user of a computer-aided design and simulation program should never take this as an indication that the results are correct.

2.6 ACTIVITY-COEFFICIENT MODELS FOR THE LIQUID PHASE

In Sections 2.3 and 2.5, methods based on equations of state are presented for predicting thermodynamic properties of vapor and liquid mixtures. In this section, predictions of liquid properties based on *Gibbs free-energy models* for predicting liquid-phase activity coefficients and other excess functions such as volume and enthalpy of mixing are developed. Regular-solution theory, which can be applied to mixtures of nonpolar compounds using only constants for the pure components, is the first model presented. This is followed by a discussion of several models that can be applied to mixtures containing polar compounds, provided that experimental data are available to determine the *binary interaction parameters* in these models. If not, group-contribution methods, which have been extensively developed, can be used to make estimates. All models discussed can be applied to predict vapor-liquid phase equilibria; and some can estimate liquid-liquid equilibria, and even solid-liquid and polymer-liquid equilibria.

Except at high pressures, dependency of K -values on composition is due primarily to nonideal solution behavior in the liquid phase. Prausnitz, Edmister, and Chao [29] showed that the relatively simple *regular-solution theory* of Scatchard and Hildebrand [30] can be used to estimate deviations due to nonideal behavior of hydrocarbon-liquid mixtures. They expressed K -values in terms of (2-27), $K_i = \gamma_{iL} \phi_{iL} / \bar{\phi}_{iV}$. Chao and Seader [9] simplified and extended application of this equation to a general correlation for hydrocarbons and some light gases in the form of a compact set of equations

especially suitable for use with a digital computer, which was widely used before the availability of the S-R-K and P-R equations.

Simple models for the liquid-phase activity coefficient, γ_{iL} , based only on properties of pure species, are not generally accurate. However, for hydrocarbon mixtures, regular-solution theory is convenient and widely applied. The theory is based on the premise that nonideality is due to differences in van der Waals forces of attraction among the different molecules present. Regular solutions have an endothermic heat of mixing, and all activity coefficients are greater than one. These solutions are regular in the sense that molecules are assumed to be randomly dispersed. Unequal attractive forces between like and unlike molecule pairs tend to cause segregation of molecules. However, for regular solutions the species concentrations on a molecular level are identical to overall solution concentrations. Therefore, excess entropy due to segregation is zero and entropy of regular solutions is identical to that of ideal solutions, in which the molecules are randomly dispersed.

Activity Coefficients from Gibbs Free Energy

Activity-coefficient equations often have their basis in Gibbs free-energy models. For a nonideal solution, the molar Gibbs free energy, g , is the sum of the molar free energy of an ideal solution and an excess molar free energy g^E for non-ideal effects. For a liquid solution,

$$\begin{aligned} g &= \sum_{i=1}^C x_i g_i + RT \sum_{i=1}^C x_i \ln x_i + g^E \\ &= \sum_{i=1}^C x_i (g_i + RT \ln x_i + \bar{g}_i^E) \end{aligned} \quad (2-58)$$

where $g \equiv h - Ts$ and excess molar free energy is the sum of the partial excess molar free energies. The partial excess molar free energy is related by classical thermodynamics to the liquid-phase activity coefficient by

$$\begin{aligned} \frac{\bar{g}_i^E}{RT} &= \ln \gamma_i = \left[\frac{\partial (N_i g^E / RT)}{\partial N_i} \right]_{P,T,N_j} \\ &= \frac{g^E}{RT} - \sum_k x_k \left[\frac{\partial (g^E / RT)}{\partial x_k} \right]_{P,T,x_r} \end{aligned} \quad (2-59)$$

where $j \neq i$, $r \neq k$, $k \neq i$, and $r \neq i$.

The relationship between excess molar free energy and excess molar enthalpy and entropy is

$$g^E = h^E - Ts^E = \sum_{i=1}^C x_i (\bar{h}_i^E - T\bar{s}_i^E) \quad (2-60)$$

Regular-Solution Model

For a multicomponent, regular liquid solution, the excess molar free energy is based on nonideality due to differences in molecular size and intermolecular forces. The former are

expressed in terms of liquid molar volume and the latter in terms of the enthalpy of vaporization. The resulting model is

$$g^E = \sum_{i=1}^C (x_i v_{iL}) \left[\frac{1}{2} \sum_{i=1}^C \sum_{j=1}^C \Phi_i \Phi_j (\delta_i - \delta_j)^2 \right] \quad (2-61)$$

where Φ is the volume fraction assuming additive molar volumes, as given by

$$\Phi_i = \frac{x_i v_{iL}}{\sum_{j=1}^C x_j v_{jL}} = \frac{x_i v_{iL}}{v_L} \quad (2-62)$$

and δ is the solubility parameter, which is defined in terms of the volumetric internal energy of vaporization as

$$\delta_i = \left(\frac{\Delta E_i^{\text{vap}}}{v_{iL}} \right)^{1/2} \quad (2-63)$$

Values of the solubility parameter for many components can be obtained from process simulation programs.

Applying (2-59) to (2-61) gives an expression for the activity coefficient in a regular solution:

$$\ln \gamma_{iL} = \frac{v_{iL} \left(\delta_i - \sum_{j=1}^C \Phi_j \delta_j \right)^2}{RT} \quad (2-64)$$

Because $\ln \gamma_{iL}$ varies almost inversely with absolute temperature, v_{iL} and δ_j are frequently taken as constants at some convenient reference temperature, such as 25°C. Thus, the estimation of γ_L by regular-solution theory requires only the pure-species constants v_L and δ . The latter parameter is often treated as an empirical constant determined by back-calculation from experimental data. For species with a critical temperature below 25°C, v_L and δ at 25°C are hypothetical. However, they can be evaluated by back-calculation from phase-equilibria data.

When molecular-size differences, as reflected by liquid molar volumes, are appreciable, the following Flory-Huggins size correction can be added to the regular-solution free-energy contribution:

$$g^E = RT \sum_{i=1}^C x_i \ln \left(\frac{\Phi_i}{x_i} \right) \quad (2-65)$$

Substitution of (2-65) into (2-59) gives

$$\ln \gamma_{iL} = \ln \left(\frac{v_{iL}}{v_L} \right) + 1 - \left(\frac{v_{iL}}{v_L} \right) \quad (2-66)$$

The complete expression for the activity coefficient of a species in a regular solution, including the Flory-Huggins correction, is

$$\gamma_{iL} = \exp \left[\frac{v_{iL} \left(\delta_i - \sum_{j=1}^C \Phi_j \delta_j \right)^2}{RT} + \ln \left(\frac{v_{iL}}{v_L} \right) + 1 - \frac{v_{iL}}{v_L} \right] \quad (2-67)$$

EXAMPLE 2.7

Yerazunis, Plowright, and Smola [31] measured liquid-phase activity coefficients for the *n*-heptane/toluene system over the entire concentration range at 1 atm (101.3 kPa). Estimate activity coefficients for the range of conditions using regular-solution theory both with and without the Flory–Huggins correction. Compare estimated values with experimental data.

SOLUTION

Experimental liquid-phase compositions and temperatures for 7 of 19 points are as follows, where H denotes heptane and T denotes toluene:

$T, ^\circ\text{C}$	x_H	x_T
98.41	1.0000	0.0000
98.70	0.9154	0.0846
99.58	0.7479	0.2521
101.47	0.5096	0.4904
104.52	0.2681	0.7319
107.57	0.1087	0.8913
110.60	0.0000	1.0000

At 25°C, liquid molar volumes are $v_{HL} = 147.5 \text{ cm}^3/\text{mol}$ and $v_{TL} = 106.8 \text{ cm}^3/\text{mol}$. Solubility parameters are 7.43 and 8.914 $(\text{cal}/\text{cm}^3)^{1/2}$, respectively, for H and T. As an example, consider mole fractions in the above table for 104.52°C. From (2-62), volume fractions are

$$\Phi_H = \frac{0.2681(147.5)}{0.2681(147.5) + 0.7319(106.8)} = 0.3359$$

$$\Phi_T = 1 - \Phi_H = 1 - 0.3359 = 0.6641$$

Substitution of these values, together with the solubility parameters, into (2-64) gives

$$\gamma_H = \exp \left\{ \frac{147.5[7.430 - 0.3359(7.430) - 0.6641(8.914)]^2}{1.987(377.67)} \right\} = 1.212$$

Values of γ_H and γ_T computed in this manner for all seven liquid-phase conditions are plotted in Figure 2.14.

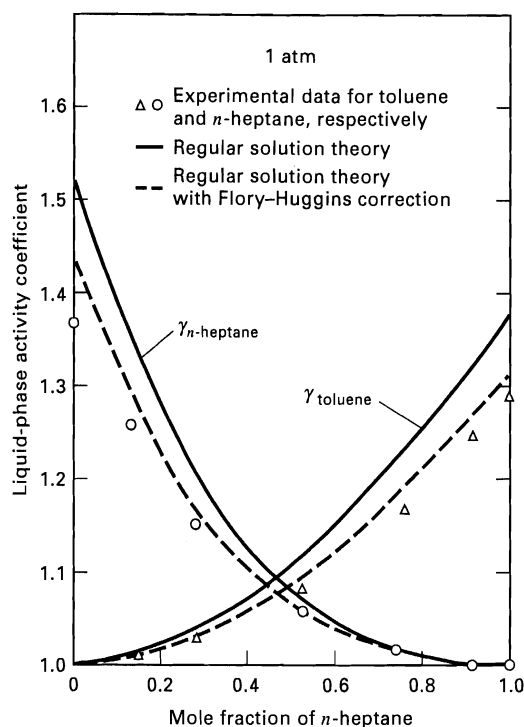


Figure 2.14 Liquid-phase activity coefficients for *n*-heptane/toluene system at 1 atm.

Applying (2-67), with the Flory–Huggins correction, to the same data point gives

$$\gamma_H = \exp \left[0.1923 + \ln \left(\frac{147.5}{117.73} \right) + 1 - \left(\frac{147.5}{117.73} \right) \right] = 1.179$$

Values of γ_H and γ_T computed in this manner are included in Figure 2.14. Deviations from experiment are not greater than 12% for regular-solution theory and not greater than 6% when the Flory–Huggins correction is included. Unfortunately, such good agreement is not always obtained with nonpolar hydrocarbon solutions, as shown, for example, by Hermesen and Prausnitz [32], who studied the cyclopentane/benzene system.

Nonideal Liquid Solutions

When liquids contain dissimilar polar species, particularly those that can form or break hydrogen bonds, the ideal-liquid solution assumption is almost always invalid and the regular-solution theory is not applicable. Ewell, Harrison, and Berg [33] provide a very useful classification of molecules based on the potential for association or solvation due to hydrogen-bond formation. If a molecule contains a hydrogen atom attached to a donor atom (O, N, F, and in certain cases C), the active hydrogen atom can form a bond with another molecule containing a donor atom. The classification in Table 2.7 permits qualitative estimates of deviations from Raoult's law for binary pairs when used in conjunction with Table 2.8. Positive deviations correspond to values of $\gamma_{iL} > 1$. Nonideality results in a variety of variations of γ_{iL} with composition, as shown in Figure 2.15 for several binary systems, where the

Roman numerals refer to classification groups in Tables 2.7 and 2.8. Starting with Figure 2.15a and taking the other plots in order, we offer the following explanations for the non-idealities. Normal heptane (V) breaks ethanol (II) hydrogen bonds, causing strong positive deviations. In Figure 2.15b, similar but less positive deviations occur when acetone (III) is added to formamide (I). Hydrogen bonds are broken and formed with chloroform (IV) and methanol (II) in Figure 2.15c, resulting in an unusual positive deviation curve for chloroform that passes through a maximum. In Figure 2.15d, chloroform (IV) provides active hydrogen atoms that can form hydrogen bonds with oxygen atoms of acetone (III), thus causing negative deviations. For water (I) and *n*-butanol (II) in Figure 2.15e, hydrogen bonds of both molecules are broken, and nonideality is sufficiently strong to cause formation of two immiscible liquid phases (*phase splitting*) over a wide region of overall composition.

Table 2.7 Classification of Molecules Based on Potential for Forming Hydrogen Bonds

Class	Description	Example
I	Molecules capable of forming three-dimensional networks of strong H-bonds	Water, glycols, glycerol, amino alcohols, hydroxylamines, hydroxyacids, polyphenols, and amides
II	Other molecules containing both active hydrogen atoms and donor atoms (O, N, and F)	Alcohols, acids, phenols, primary and secondary amines, oximes, nitro and nitrile compounds with α -hydrogen atoms, ammonia, hydrazine, hydrogen fluoride, and hydrogen cyanide
III	Molecules containing donor atoms but no active hydrogen atoms	Ethers, ketones, aldehydes, esters, tertiary amines (including pyridine type), and nitro and nitrile compounds without α -hydrogen atoms
IV	Molecules containing active hydrogen atoms but no donor atoms that have two or three chlorine atoms on the same carbon atom as a hydrogen or one chlorine on the carbon atom and one or more chlorine atoms on adjacent carbon atoms	CHCl_3 , CH_2Cl_2 , CH_3CHCl_2 , $\text{CH}_2\text{ClCH}_2\text{Cl}$, $\text{CH}_2\text{ClCHClCH}_2\text{Cl}$, and $\text{CH}_2\text{ClCHCl}_2$
V	All other molecules having neither active hydrogen atoms nor donor atoms	Hydrocarbons, carbon disulfide, sulfides, mercaptans, and haloalkanes not in class IV

Nonideal-solution effects can be incorporated into K -value formulations in two different ways. We have already described the use of $\bar{\phi}_i$, the partial fugacity coefficient, in conjunction with an equation of state and adequate mixing rules. This is the method most frequently used for handling nonidealities in the vapor phase. However, $\bar{\phi}_{iV}$ reflects the combined effects of a nonideal gas and a nonideal-gas solution. At low pressures, both effects are negligible. At moderate pressures, a vapor solution may still be ideal even though the gas mixture does not follow the ideal-gas law. Nonidealities in the liquid phase, however, can be severe even at low pressures. Earlier in this section, $\bar{\phi}_{iL}$ was used to express liquid-phase nonidealities for nonpolar species.

When polar species are present, mixing rules can be modified to include binary interaction parameters, k_{ij} , as in (2-51).

The other technique for handling solution nonidealities is to retain $\bar{\phi}_{iV}$ in the K -value formulation, but replace $\bar{\phi}_{iL}$ by the product of γ_{iL} and ϕ_{iL} , where the former quantity accounts for deviations from nonideal solutions. Equation (2-26) then becomes

$$K_i = \frac{\gamma_{iL} \phi_{iL}}{\bar{\phi}_{iV}} \quad (2-68)$$

which was derived previously as (2-27). At low pressures, from Table 2.2, $\phi_{iL} = P_i^s/P$ and $\bar{\phi}_{iV} = 1.0$, so (2-68)

Table 2.8 Molecule Interactions Causing Deviations from Raoult's Law

Type of Deviation	Classes	Effect on Hydrogen Bonding
Always negative	III + IV	H-bonds formed only
Quasi-ideal; always positive or ideal	III + III III + V IV + IV IV + V V + V	No H-bonds involved
Usually positive, but some negative	I + I I + II I + III II + II II + III	H-bonds broken and formed
Always positive	I + IV (frequently limited solubility) II + IV	H-bonds broken and formed, but dissociation of Class I or II is more important effect
Always positive	I + V II + V	H-bonds broken only

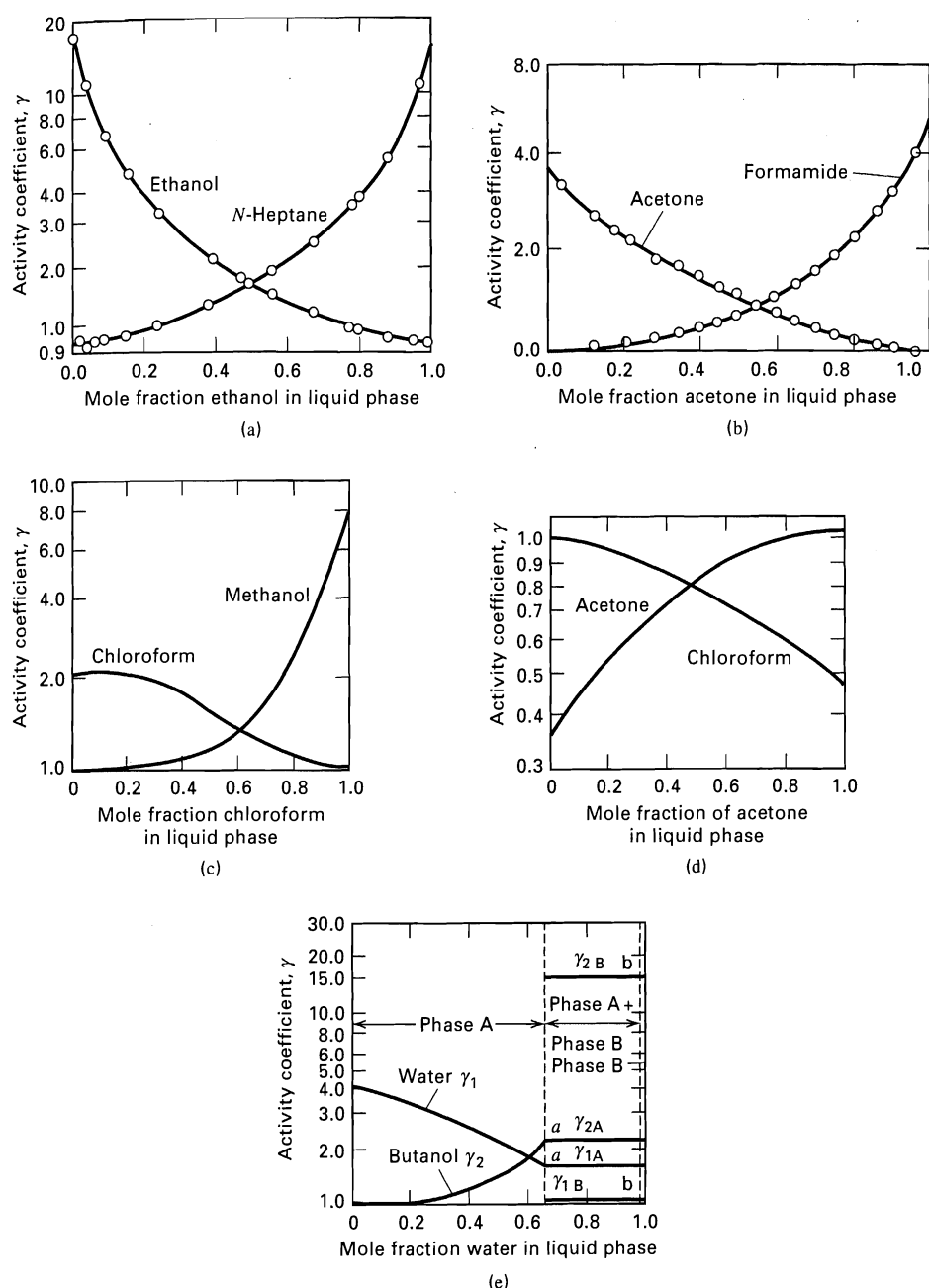


Figure 2.15 Typical variations of activity coefficients with composition in binary liquid systems: (a) ethanol(II)/*n*-heptane(V); (b) acetone(III)/formamide(I); (c) chloroform(IV)/methanol(II); (d) acetone(III)/chloroform(IV); (e) water(I)/*n*-butanol(II).

reduces to a modified Raoult's law K -value, which differs from (2-44) only in the γ_{iL} term:

$$K_i = \frac{\gamma_{iL} P_i^s}{P} \quad (2-69)$$

At moderate pressures, (5) of Table 2.3 is preferred over (2-69).

Regular-solution theory is useful only for estimating values of γ_{iL} for mixtures of nonpolar species. However, many empirical and semitheoretical equations exist for estimating activity coefficients of binary mixtures containing polar and/or nonpolar species. These equations contain binary interaction parameters, which are back-calculated from experimental data. Some of the more useful equations are listed in Table 2.9 in binary-pair form. For a given activity-

coefficient correlation, the equations of Table 2.10 can be used to determine excess volume, excess enthalpy, and excess entropy. However, unless the dependency on pressure of the parameters and properties used in the equations for activity coefficient is known, excess liquid volumes cannot be determined directly from (1) of Table 2.10. Fortunately, the contribution of excess volume to total mixture volume is generally small for solutions of nonelectrolytes. For example, consider a 50 mol% solution of ethanol in *n*-heptane at 25°C. From Figure 2.15a, this is a highly nonideal, but miscible, liquid mixture. From the data of Van Ness, Soczek, and Kochar [34], excess volume is only 0.465 cm³/mol, compared to an estimated ideal-solution molar volume of 106.3 cm³/mol. Once the partial molar excess functions are estimated for each species, the excess functions are computed from the mole fraction sums.

Table 2.9 Empirical and Semitheoretical Equations for Correlating Liquid-Phase Activity Coefficients of Binary Pairs

Name	Equation for Species 1	Equation for Species 2
(1) Margules	$\log \gamma_1 = Ax_2^2$	$\log \gamma_2 = Ax_1^2$
(2) Margules (two-constant)	$\log \gamma_1 = x_2^2[\bar{A}_{12} + 2x_1(\bar{A}_{21} - \bar{A}_{12})]$	$\log \gamma_2 = x_1^2[\bar{A}_{21} + 2x_2(\bar{A}_{12} - \bar{A}_{21})]$
(3) van Laar (two-constant)	$\ln \gamma_1 = \frac{A_{12}}{[1 + (x_1 A_{12})/(x_2 A_{21})]^2}$	$\ln \gamma_2 = \frac{A_{21}}{[1 + (x_2 A_{21})/(x_1 A_{12})]^2}$
(4) Wilson (two-constant)	$\ln \gamma_1 = -\ln(x_1 + \Lambda_{12}x_2) + x_2 \left(\frac{\Lambda_{12}}{x_1 + \Lambda_{12}x_2} - \frac{\Lambda_{21}}{x_2 + \Lambda_{21}x_1} \right)$	$\ln \gamma_2 = -\ln(x_2 + \Lambda_{21}x_1) - x_1 \left(\frac{\Lambda_{12}}{x_1 + \Lambda_{12}x_2} - \frac{\Lambda_{21}}{x_2 + \Lambda_{21}x_1} \right)$
(5) NRTL (three-constant)	$\ln \gamma_1 = \frac{x_2^2 \tau_{21} G_{21}^2}{(x_1 + x_2 G_{21})^2} + \frac{x_1^2 \tau_{12} G_{12}}{(x_2 + x_1 G_{12})^2}$ $G_{ij} = \exp(-\alpha_{ij} \tau_{ij})$	$\ln \gamma_2 = \frac{x_1^2 \tau_{12} G_{12}^2}{(x_2 + x_1 G_{12})^2} + \frac{x_2^2 \tau_{21} G_{21}}{(x_1 + x_2 G_{21})^2}$ $G_{ij} = \exp(-\alpha_{ij} \tau_{ij})$
(6) UNIQUAC (two-constant)	$\ln \gamma_1 = \ln \frac{\Psi_1}{x_1} + \frac{\bar{Z}}{2} q_1 \ln \frac{\theta_1}{\Psi_1} + \Psi_2 \left(l_1 - \frac{r_1}{r_2} l_2 \right) - q_1 \ln(\theta_1 + \theta_2 T_{21}) + \theta_2 q_1 \left(\frac{T_{21}}{\theta_1 + \theta_2 T_{21}} - \frac{T_{12}}{\theta_2 + \theta_1 T_{12}} \right)$	$\ln \gamma_2 = \ln \frac{\Psi_2}{x_2} + \frac{\bar{Z}}{2} q_2 \ln \frac{\theta_2}{\Psi_2} + \Psi_1 \left(l_2 - \frac{r_2}{r_1} l_1 \right) - q_2 \ln(\theta_2 + \theta_1 T_{12}) + \theta_1 q_2 \left(\frac{T_{12}}{\theta_2 + \theta_1 T_{12}} - \frac{T_{21}}{\theta_1 + \theta_2 T_{21}} \right)$

Margules Equations

The Margules equations (1) and (2) in Table 2.9 date back to 1895, and the two-constant form is still in common use because of its simplicity. These equations result from power-series expansions in mole fractions for $\bar{g}_i^E$ and conversion to activity coefficients by means of (2-59). The one-constant form is equivalent to symmetrical activity-coefficient curves, which are rarely observed experimentally.

van Laar Equation

Because of its flexibility, simplicity, and ability to fit many systems well, the van Laar equation is widely used. It was derived from the van der Waals equation of state, but the

constants, shown as A_{12} and A_{21} in (3) of Table 2.9, are best back-calculated from experimental data. These constants are, in theory, constant only for a particular binary pair at a given temperature. In practice, they are frequently computed from isobaric data covering a range of temperature. The van Laar theory expresses the temperature dependence of A_{ij} as

$$A_{ij} = \frac{A'_{ij}}{RT} \quad (2-70)$$

Regular-solution theory and the van Laar equation are equivalent for a binary solution if

$$A_{ij} = \frac{v_{iL}}{RT} (\delta_i - \delta_j)^2 \quad (2-71)$$

The van Laar equation can fit activity coefficient-composition curves corresponding to both positive and negative deviations from Raoult's law, but cannot fit curves that exhibit minima or maxima such as those in Figure 2.15c.

When data are isothermal, or isobaric over only a narrow range of temperature, determination of van Laar constants is conducted in a straightforward manner. The most accurate procedure is a nonlinear regression to obtain the best fit to the data over the entire range of binary composition, subject to minimization of some objective function. A less accurate, but extremely rapid, manual-calculation procedure can be used when experimental data can be extrapolated to infinite-dilution conditions. Modern experimental techniques are available for accurately and rapidly determining activity

Table 2.10 Classical Partial Molar Excess Functions of Thermodynamics

Excess volume:

$$(1) (\bar{v}_{iL} - \bar{v}_{iL}^{\text{ID}}) \equiv \bar{v}_{iL}^E = RT \left(\frac{\partial \ln \gamma_{iL}}{\partial P} \right)_{T,x}$$

Excess enthalpy:

$$(2) (\bar{h}_{iL} - \bar{h}_{iL}^{\text{ID}}) \equiv \bar{h}_{iL}^E = -RT^2 \left(\frac{\partial \ln \gamma_{iL}}{\partial T} \right)_{P,x}$$

Excess entropy:

$$(3) (\bar{s}_{iL} - \bar{s}_{iL}^{\text{ID}}) \equiv \bar{s}_{iL}^E = -R \left[T \left(\frac{\partial \ln \gamma_{iL}}{\partial T} \right)_{P,x} + \ln \gamma_{iL} \right]$$

ID = ideal mixture; E = excess because of nonideality.

coefficients at infinite dilution. Applying (3) of Table 2.9 to the conditions $x_i = 0$ and then $x_j = 0$, we have

$$A_{ij} = \ln \gamma_i^\infty, \quad x_i = 0$$

and

$$A_{ji} = \ln \gamma_j^\infty, \quad x_j = 0 \quad (2-72)$$

For practical applications, it is important that the van Laar equation predicts azeotrope formation correctly, where $x_i = y_i$ and $K_i = 1.0$. If activity coefficients are known or can be computed at the azeotropic composition—say, from (2-69), ($\gamma_{iL} = P/P_i^s$, since $K_i = 1.0$)—these coefficients can be used to determine the van Laar constants directly from the following equations obtained by solving simultaneously for A_{12} and A_{21} :

$$A_{12} = \ln \gamma_1 \left(1 + \frac{x_2 \ln \gamma_2}{x_1 \ln \gamma_1} \right)^2 \quad (2-73)$$

$$A_{21} = \ln \gamma_2 \left(1 + \frac{x_1 \ln \gamma_1}{x_2 \ln \gamma_2} \right)^2 \quad (2-74)$$

These equations are applicable to activity-coefficient data obtained at any single composition.

Mixtures of self-associated polar molecules (class II in Table 2.7) with nonpolar molecules such as hydrocarbons (class V) can exhibit the strong nonideality of the positive-deviation type shown in Figure 2.15a. Figure 2.16 shows experimental data of Sinor and Weber [35] for ethanol

(1)/*n*-hexane (2), a system of this type, at 101.3 kPa. These data were correlated with the van Laar equation by Orye and Prausnitz [36] to give $A_{12} = 2.409$ and $A_{21} = 1.970$. From $x_1 = 0.1$ to 0.9, the fit of the data to the van Laar equation is reasonably good; in the dilute regions, however, deviations are quite severe and the predicted activity coefficients for ethanol are low. An even more serious problem with these highly nonideal mixtures is that the van Laar equation may erroneously predict formation of two liquid phases (phase splitting) when values of activity coefficients exceed approximately 7.

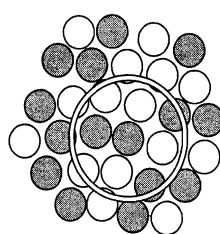
Local-Composition Concept and the Wilson Model

Since its introduction in 1964, the Wilson equation [37], shown in binary form in Table 2.9 as (4), has received wide attention because of its ability to fit strongly nonideal, but miscible, systems. As shown in Figure 2.16, the Wilson equation, with binary interaction parameters of $\Lambda_{12} = 0.0952$ and $\Lambda_{21} = 0.2713$ determined by Orye and Prausnitz [36], fits experimental data well even in dilute regions where the variation of γ_1 becomes exponential. Corresponding infinite-dilution activity coefficients computed from the Wilson equation are $\gamma_1^\infty = 21.72$ and $\gamma_2^\infty = 9.104$.

The Wilson equation accounts for effects of differences both in molecular size and intermolecular forces, consistent with a semitheoretical interpretation based on the Flory-Huggins relation (2-65). Overall solution-volume fractions ($\Phi_i = x_i v_{iL}/v_L$) are replaced by local-volume fractions, $\bar{\Phi}_i$, which are related to local-molecule segregations caused by differing energies of interaction between pairs of molecules. The concept of local compositions that differ from overall compositions is shown schematically for an overall, equimolar, binary solution in Figure 2.17, which is taken from Cukor and Prausnitz [38]. About a central molecule of type 1, the local mole fraction of molecules of type 2 is shown as $\frac{5}{8}$, while the overall composition is $\frac{1}{2}$.

For local-volume fraction, Wilson proposed

$$\bar{\Phi}_i = \frac{v_{iL} x_i \exp(-\lambda_{ii}/RT)}{\sum_{j=1}^C v_{jL} x_j \exp(-\lambda_{ij}/RT)} \quad (2-75)$$



15 of type 1
15 of type 2

Overall mole fractions: $x_1 = x_2 = 1/2$
Local mole fractions:
Molecules of 2 about a central molecule 1
 $x_{21} = \frac{\text{Total molecules about a central molecule 1}}{\text{Total molecules about a central molecule 1}}$
 $x_{21} + x_{11} = 1$, as shown
 $x_{12} + x_{22} = 1$
 $x_{11} \sim 3/8$
 $x_{21} \sim 5/8$

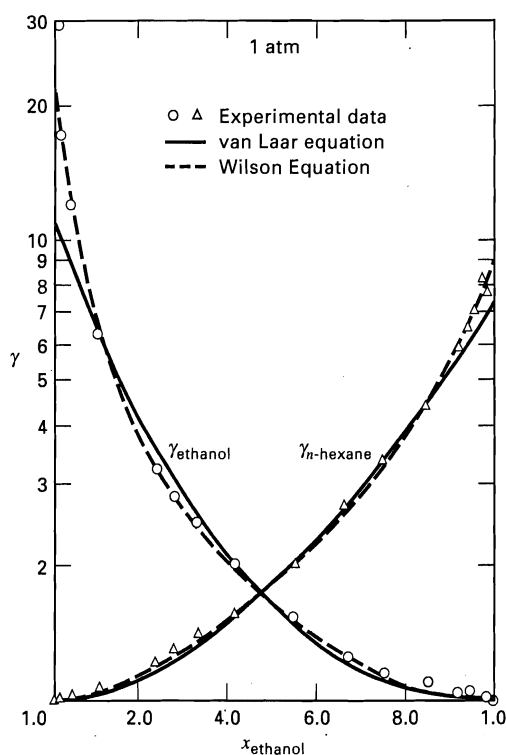


Figure 2.16 Liquid-phase activity coefficients for ethanol/*n*-hexane system.

[Data from J.E. Sinor and J.H. Weber, *J. Chem. Eng. Data*, 5, 243–247 (1960).]

Figure 2.17 The concept of local compositions.

[From P.M. Cukor and J.M. Prausnitz, *Int. Chem. Eng. Symp. Ser. No. 32*, 3, 88 (1969).]

where energies of interaction $\lambda_{ij} = \lambda_{ji}$, but $\lambda_{ii} \neq \lambda_{jj}$. Following the treatment by Orye and Prausnitz [36], substitution of the binary form of (2-75) into (2-65) and defining the binary interaction parameters as

$$\Lambda_{12} = \frac{v_{2L}}{v_{1L}} \exp \left[-\frac{(\lambda_{12} - \lambda_{11})}{RT} \right] \quad (2-76)$$

$$\Lambda_{21} = \frac{v_{1L}}{v_{2L}} \exp \left[-\frac{(\lambda_{12} - \lambda_{22})}{RT} \right] \quad (2-77)$$

leads to the following equation for a binary system:

$$\frac{g^E}{RT} = -x_1 \ln(x_1 + \Lambda_{12}x_2) - x_2 \ln(x_2 + \Lambda_{21}x_1) \quad (2-78)$$

The Wilson equation is very effective for dilute compositions where entropy effects dominate over enthalpy effects. The Orye–Prausnitz form of the Wilson equation for the activity coefficient, as given in Table 2.9, follows from combining (2-59) with (2-78). Values of $\Lambda_{ij} < 1$ correspond to positive deviations from Raoult's law, while values > 1 correspond to negative deviations. Ideal solutions result from $\Lambda_{ij} = 1$. Studies indicate that λ_{ii} and λ_{ij} are temperature-dependent. Values of v_{iL}/v_{jL} depend on temperature also, but the variation may be small compared to temperature effects on the exponential terms in (2-76) and (2-77).

The Wilson equation is readily extended to multicomponent mixtures by neglecting ternary and higher molecular interactions and assuming a pseudo-binary mixture. The following multicomponent Wilson equation involves only binary interaction constants:

$$\ln \gamma_k = 1 - \ln \left(\sum_{j=1}^C x_j \Lambda_{kj} \right) - \sum_{i=1}^C \left(\frac{x_i \Lambda_{ik}}{\sum_{j=1}^C x_j \Lambda_{ij}} \right) \quad (2-79)$$

where $\Lambda_{ii} = \Lambda_{jj} = \Lambda_{kk} = 1$.

As mixtures become highly nonideal, but still miscible, the Wilson equation becomes markedly superior to the Margules and van Laar equations. The Wilson equation is consistently superior for multicomponent solutions. Values of the constants in the Wilson equation for many binary systems are tabulated in the DECHEMA collection of Gmehling and Onken [39]. Two limitations of the Wilson equation are its inability to predict immiscibility, as in Figure 2.15e, and maxima and minima in the activity coefficient–mole fraction relationships, as shown in Figure 2.15c.

When insufficient experimental data are available to determine binary Wilson parameters from a best fit of activity coefficients over the entire range of composition, infinite-dilution or single-point values can be used. At infinite dilution, the Wilson equation in Table 2.9 becomes

$$\ln \gamma_1^\infty = 1 - \ln \Lambda_{12} - \Lambda_{21} \quad (2-80)$$

$$\ln \gamma_2^\infty = 1 - \ln \Lambda_{21} - \Lambda_{12} \quad (2-81)$$

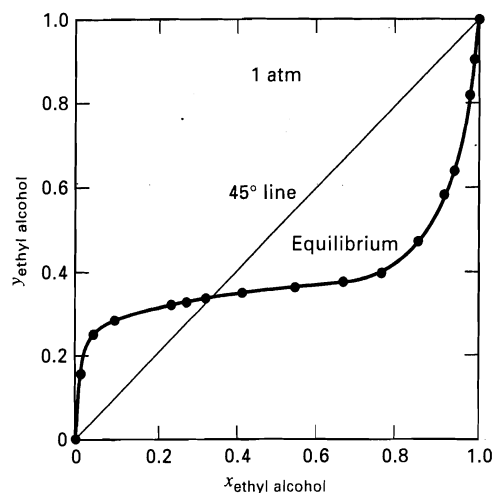


Figure 2.18 Equilibrium curve for *n*-hexane/ethanol system.

An iterative procedure is required to obtain Λ_{12} and Λ_{21} from these nonlinear equations. If temperatures corresponding to γ_1^∞ and γ_2^∞ are not close or equal, (2-76) and (2-77) should be substituted into (2-80) and (2-81) with values of $(\lambda_{12} - \lambda_{11})$ and $(\lambda_{12} - \lambda_{22})$ determined from estimates of pure-component liquid molar volumes.

When the experimental data of Sinor and Weber [35] for *n*-hexane/ethanol, shown in Figure 2.16, are plotted as a *y*–*x* diagram in ethanol (Figure 2.18), the equilibrium curve crosses the 45° line at an ethanol mole fraction of $x = 0.332$. The measured temperature corresponding to this composition is 58°C. Ethanol has a normal boiling point of 78.33°C, which is higher than the normal boiling point of 68.75°C for *n*-hexane. Nevertheless, ethanol is more volatile than *n*-hexane up to an ethanol mole fraction of $x = 0.322$, the minimum-boiling azeotrope. This occurs because of the relatively close boiling points of the two species and the high activity coefficients for ethanol at low concentrations. At the azeotropic composition, $y_i = x_i$; therefore, $K_i = 1.0$. Applying (2-69) to both species,

$$\gamma_1 P_1^s = \gamma_2 P_2^s \quad (2-82)$$

If species 2 is more volatile in the pure state ($P_2^s > P_1^s$), the criteria for formation of a minimum-boiling azeotrope are

$$\gamma_1 \geq 1 \quad (2-83)$$

$$\gamma_2 \geq 1 \quad (2-84)$$

and

$$\frac{\gamma_1}{\gamma_2} < \frac{P_2^s}{P_1^s} \quad (2-85)$$

for x_1 less than the azeotropic composition. These criteria are most readily applied at $x_1 = 0$. For example, for the *n*-hexane (2)/ethanol (1) system at 1 atm (101.3 kPa), when the liquid-phase mole fraction of ethanol approaches zero, temperature approaches 68.75°C (155.75°F), the boiling point of pure *n*-hexane. At this temperature, $P_1^s = 10$ psia

(68.9 kPa) and $P_2^s = 14.7$ psia (101.3 kPa). Also from Figure 2.16, $\gamma_1^\infty = 21.72$ when $\gamma_2 = 1.0$. Thus, $\gamma_1^\infty/\gamma_2 = 21.72$, but $P_2^s/P_1^s = 1.47$. Therefore, a minimum-boiling azeotrope will occur.

Maximum-boiling azeotropes are less common. They occur for relatively close-boiling mixtures when negative deviations from Raoult's law arise such that $\gamma_i < 1.0$. Criteria for their formation are derived in a manner similar to that for minimum-boiling azeotropes. At $x_1 = 1$, where species 2 is more volatile,

$$\gamma_1 = 1.0 \quad (2-86)$$

$$\gamma_2^\infty < 1.0 \quad (2-87)$$

and

$$\frac{\gamma_2^\infty}{\gamma_1} < \frac{P_1^s}{P_2^s} \quad (2-88)$$

For an azeotropic binary system, the two binary interaction parameters Λ_{12} and Λ_{21} can be determined by solving (4) of Table 2.9 at the azeotropic composition, as shown in the following example.

EXAMPLE 2.8

From measurements by Sinor and Weber [35] of the azeotropic condition for the ethanol/*n*-hexane system at 1 atm (101.3 kPa, 14.696 psia), calculate Λ_{12} and Λ_{21} .

SOLUTION

Let E denote ethanol and H denote *n*-hexane. The azeotrope occurs at $x_E = 0.332$, $x_H = 0.668$, and $T = 58^\circ\text{C}$ (331.15 K). At 1 atm, (2-69) can be used to approximate K -values. Thus, at azeotropic conditions, $\gamma_i = P/P_i^s$. The vapor pressures at 58°C are $P_E^s = 6.26$ psia and $P_H^s = 10.28$ psia. Therefore,

$$\gamma_E = \frac{14.696}{6.26} = 2.348$$

$$\gamma_H = \frac{14.696}{10.28} = 1.430$$

Substituting these values together with the above corresponding values of x_i into the binary form of the Wilson equation in Table 2.9 gives

$$\ln 2.348 = -\ln(0.332 + 0.668\Lambda_{EH}) + 0.668 \left(\frac{\Lambda_{EH}}{0.332 + 0.668\Lambda_{EH}} - \frac{\Lambda_{HE}}{0.332\Lambda_{HE} + 0.668} \right)$$

$$\ln 1.430 = -\ln(0.668 + 0.332\Lambda_{HE}) - 0.332 \left(\frac{\Lambda_{EH}}{0.332 + 0.668\Lambda_{EH}} - \frac{\Lambda_{HE}}{0.332\Lambda_{HE} + 0.668} \right)$$

Solving these two nonlinear equations simultaneously by an iterative procedure, we obtain $\Lambda_{EH} = 0.041$ and $\Lambda_{HE} = 0.281$. From these constants, the activity-coefficient curves can be predicted if the temperature variations of Λ_{EH} and Λ_{HE} are ignored. The results are plotted in Figure 2.19. The fit of experimental data is good except, perhaps, for near-infinite-dilution conditions, where $\gamma_E^\infty = 49.82$ and $\gamma_H^\infty = 9.28$. The former value is considerably

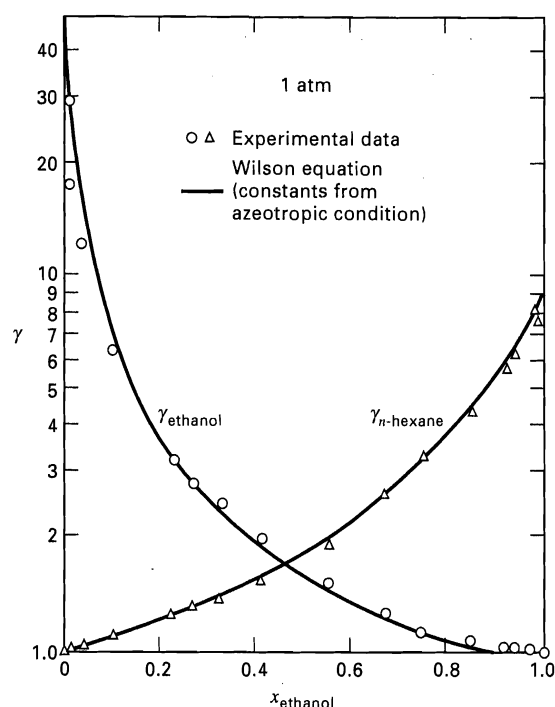


Figure 2.19 Liquid-phase activity coefficients for ethanol/*n*-hexane system.

greater than the value of 21.72 obtained by Orye and Prausnitz [36] from a fit of all experimental data points. However, if Figures 2.16 and 2.19 are compared, it is seen that widely differing γ_E^∞ values have little effect on γ in the composition region $x_E = 0.15$ to 1.00, where the two sets of Wilson curves are almost identical. For accuracy over the entire composition range, commensurate with the ability of the Wilson equation, data for at least three well-spaced liquid compositions per binary are preferred.

The Wilson equation can be extended to liquid-liquid or vapor-liquid-liquid systems by multiplying the right-hand side of (2-78) by a third binary-pair constant evaluated from experimental data [37]. However, for multicomponent systems of three or more species, the third binary-pair constants must be the same for all constituent binary pairs. Furthermore, as shown by Hiranuma [40], representation of ternary systems involving only one partially miscible binary pair can be extremely sensitive to the third binary-pair Wilson constant. For these reasons, application of the Wilson equation to liquid-liquid systems has not been widespread. Rather, the success of the Wilson equation for prediction of activity coefficients for miscible liquid systems greatly stimulated further development of the local-composition concept of Wilson in an effort to obtain more universal expressions for liquid-phase activity coefficients.

NRTL Model

The nonrandom, two-liquid (NRTL) equation developed by Renon and Prausnitz [41,42] as listed in Table 2.9, represents an accepted extension of Wilson's concept. The NRTL equation is applicable to multicomponent vapor-liquid,

liquid–liquid, and vapor–liquid–liquid systems. For multi-component vapor–liquid systems, only binary-pair constants from the corresponding binary-pair experimental data are required. For a multicomponent system, the NRTL expression for the activity coefficient is

$$\ln \gamma_i = \frac{\sum_{j=1}^C \tau_{ji} G_{ji} x_j}{\sum_{k=1}^C G_{ki} x_k} + \sum_{j=1}^C \left[\frac{x_j G_{ij}}{\sum_{k=1}^C G_{kj} x_k} \left(\tau_{ij} - \frac{\sum_{k=1}^C x_k \tau_{kj} G_{kj}}{\sum_{k=1}^C G_{kj} x_k} \right) \right] \quad (2-89)$$

where

$$G_{ji} = \exp(-\alpha_{ji} \tau_{ji}) \quad (2-90)$$

The coefficients τ are given by

$$\tau_{ij} = \frac{g_{ij} - g_{jj}}{RT} \quad (2-91)$$

$$\tau_{ji} = \frac{g_{ji} - g_{ii}}{RT} \quad (2-92)$$

where g_{ij} , g_{jj} , and so on are energies of interaction between molecule pairs. In the above equations, $G_{ji} \neq G_{ij}$, $\tau_{ij} \neq \tau_{ji}$, $G_{ii} = G_{jj} = 1$, and $\tau_{ii} = \tau_{jj} = 0$. Often $(g_{ij} - g_{jj})$ and other constants are linear in temperature. For ideal solutions, $\tau_{ji} = 0$.

The parameter α_{ji} characterizes the tendency of species j and species i to be distributed in a nonrandom fashion. When $\alpha_{ji} = 0$, local mole fractions are equal to overall solution mole fractions. Generally α_{ji} is independent of temperature and depends on molecule properties in a manner similar to the classifications in Tables 2.7 and 2.8. Values of α_{ji} usually lie between 0.2 and 0.47. When $\alpha_{ji} < 0.426$, phase immiscibility is predicted. Although α_{ji} can be treated as an adjustable parameter, to be determined from experimental binary-pair data, more commonly α_{ji} is set according to the following rules, which are occasionally ambiguous:

1. $\alpha_{ji} = 0.20$ for mixtures of saturated hydrocarbons and polar, nonassociated species (e.g., *n*-heptane/acetone).
2. $\alpha_{ji} = 0.30$ for mixtures of nonpolar compounds (e.g., benzene/*n*-heptane), except fluorocarbons and paraffins; mixtures of nonpolar and polar, nonassociated species (e.g., benzene/acetone); mixtures of polar species that exhibit negative deviations from Raoult's law (e.g., acetone/chloroform) and moderate positive deviations (e.g., ethanol/water); mixtures of water and polar nonassociated species (e.g., water/acetone).
3. $\alpha_{ji} = 0.40$ for mixtures of saturated hydrocarbons and homolog perfluorocarbons (e.g., *n*-hexane/perfluoro-*n*-hexane).
4. $\alpha_{ji} = 0.47$ for mixtures of an alcohol or other strongly self-associated species with nonpolar species (e.g., ethanol/benzene); mixtures of carbon tetrachloride with either acetonitrile or nitromethane; mixtures of water with either butyl glycol or pyridine.

UNIQUAC Model

In an attempt to place calculations of liquid-phase activity coefficients on a simple, yet more theoretical basis, Abrams and Prausnitz [43] used statistical mechanics to derive an expression for excess free energy. Their model, called UNIQUAC (universal quasichemical), generalizes a previous analysis by Guggenheim and extends it to mixtures of molecules that differ appreciably in size and shape. As in the Wilson and NRTL equations, local concentrations are used. However, rather than local volume fractions or local mole fractions, UNIQUAC uses the local area fraction θ_{ij} as the primary concentration variable.

The local area fraction is determined by representing a molecule by a set of bonded segments. Each molecule is characterized by two structural parameters that are determined relative to a standard segment taken as an equivalent sphere of a unit of a linear, infinite-length, polymethylene molecule. The two structural parameters are the relative number of segments per molecule, r (volume parameter), and the relative surface area of the molecule, q (surface parameter). Values of these parameters computed from bond angles and bond distances are given by Abrams and Prausnitz [43] and Gmehling and Onken [39] for a number of species. For other compounds, values can be estimated by the group-contribution method of Fredenslund et al. [46].

For a multicomponent liquid mixture, the UNIQUAC model gives the excess free energy as

$$\frac{g^E}{RT} = \sum_{i=1}^C x_i \ln \left(\frac{\Psi_i}{x_i} \right) + \frac{\bar{Z}}{2} \sum_{i=1}^C q_i x_i \ln \left(\frac{\theta_i}{\Psi_i} \right) - \sum_{i=1}^C q_i x_i \ln \left(\sum_{j=1}^C \theta_j T_{ji} \right) \quad (2-93)$$

The first two terms on the right-hand side account for *combinatorial* effects due to differences in molecule size and shape; the last term provides a *residual* contribution due to differences in intermolecular forces, where

$$\Psi_i = \frac{x_i r_i}{\sum_{i=1}^C x_i r_i} = \text{segment fraction} \quad (2-94)$$

$$\theta = \frac{x_i q_i}{\sum_{i=1}^C x_i q_i} = \text{area fraction} \quad (2-95)$$

where $\bar{Z}$ = lattice coordination number set equal to 10, and

$$T_{ji} = \exp \left(\frac{u_{ji} - u_{ii}}{RT} \right) \quad (2-96)$$

Equation (2-93) contains only two adjustable parameters for each binary pair, $(u_{ji} - u_{ii})$ and $(u_{ij} - u_{jj})$. Abrams and Prausnitz show that $u_{ji} = u_{ij}$ and $T_{ii} = T_{jj} = 1$. In general, $(u_{ji} - u_{ii})$ and $(u_{ij} - u_{jj})$ are linear functions of temperature.

If (2-59) is combined with (2-93), an equation for the liquid-phase activity coefficient for a species in a multicomponent mixture is obtained:

$$\ln \gamma_i = \ln \gamma_i^C + \ln \gamma_i^R$$

$$= \underbrace{\ln(\Psi_i/x_i) + (\bar{Z}/2) q_i \ln(\theta_i/\Psi_i) + l_i - (\Psi_i/x_i) \sum_{j=1}^C x_j l_j}_{C, \text{ combinatorial}} + \underbrace{q_i \left[1 - \ln \left(\sum_{j=1}^C \theta_j T_{ji} \right) - \sum_{j=1}^C \left(\frac{\theta_j T_{ij}}{\sum_{k=1}^C \theta_k T_{kj}} \right) \right]}_{R, \text{ residual}} \quad (2-97)$$

where

$$l_j = \left(\frac{\bar{Z}}{2} \right) (r_j - q_j) - (r_j - 1) \quad (2-98)$$

For a binary mixture of species 1 and 2, (2-97) reduces to (6) in Table 2.9 for $\bar{Z} = 10$.

UNIFAC Model

Liquid-phase activity coefficients must be estimated for nonideal mixtures even when experimental phase equilibria data are not available and when the assumption of regular solutions is not valid because polar compounds are present. For such predictions, Wilson and Deal [47] and then Derr and Deal [48], in the 1960s, presented methods based on treating a solution as a mixture of functional groups instead of molecules. For example, in a solution of toluene and acetone, the contributions might be 5 aromatic CH groups, 1 aromatic C group, and 1 CH₃ group from toluene; and 2 CH₃ groups plus 1 CO carbonyl group from acetone. Alternatively, larger groups might be employed to give 5 aromatic CH groups and 1 CCH₃ group from toluene; and 1 CH₃ group and 1 CH₃CO group from acetone. As larger and larger functional groups are used, the accuracy of molecular representation increases, but the advantage of the group-contribution method decreases because a larger number of groups is required. In practice, about 50 functional groups are used to represent literally thousands of multicomponent liquid mixtures.

To estimate the partial molar excess free energies, $\bar{g}_i^E$, and then the activity coefficients, size parameters for each functional group and binary interaction parameters for each pair of functional groups are required. Size parameters can be calculated from theory. Interaction parameters are back-calculated from existing phase-equilibria data and then used with the size parameters to predict phase-equilibria properties of mixtures for which no data are available.

The UNIFAC (UNIQUAC Functional-group Activity Coefficients) group-contribution method, first presented by Fredenslund, Jones, and Prausnitz [49] and further developed for use in practice by Fredenslund, Gmehling, and

Rasmussen [50], Gmehling, Rasmussen, and Fredenslund [51], and Larsen, Rasmussen, and Fredenslund [52], has several advantages over other group-contribution methods: (1) It is theoretically based on the UNIQUAC method; (2) the parameters are essentially independent of temperature; (3) size and binary interaction parameters are available for a wide range of types of functional groups; (4) predictions can be made over a temperature range of 275–425 K and for pressures up to a few atmospheres; and (5) extensive comparisons with experimental data are available. All components in the mixture must be condensable.

The UNIFAC method for predicting liquid-phase activity coefficients is based on the UNIQUAC equation (2-97), wherein the molecular volume and area parameters in the combinatorial terms are replaced by

$$r_i = \sum_k v_k^{(i)} R_k \quad (2-99)$$

$$q_i = \sum_k v_k^{(i)} Q_k \quad (2-100)$$

where $v_k^{(i)}$ is the number of functional groups of type k in molecule i , and R_k and Q_k are the volume and area parameters, respectively, for the type- k functional group.

The residual term in (2-97), which is represented by $\ln \gamma_i^R$, is replaced by the expression

$$\ln \gamma_i^R = \underbrace{\sum_k v_k^{(i)} (\ln \Gamma_k - \ln \Gamma_k^{(i)})}_{\text{all functional groups in mixture}} \quad (2-101)$$

where Γ_k is the residual activity coefficient of the functional group k in the actual mixture, and $\Gamma_k^{(i)}$ is the same quantity but in a reference mixture that contains only molecules of type i . The latter quantity is required so that $\gamma_i^R \rightarrow 1.0$ as $x_i \rightarrow 1.0$. Both Γ_k and $\Gamma_k^{(i)}$ have the same form as the residual term in (2-97). Thus,

$$\ln \Gamma_k = Q_k \left[1 - \ln \left(\sum_m \theta_m T_{mk} \right) - \sum_m \frac{\theta_m T_{mk}}{\sum_n \theta_n T_{nm}} \right] \quad (2-102)$$

where θ_m is the area fraction of group m , given by an equation similar to (2-95),

$$\theta_m = \frac{X_m Q_m}{\sum_n X_n Q_n} \quad (2-103)$$

where X_m is the mole fraction of group m in the solution,

$$X_m = \frac{\sum_j v_m^{(j)} x_j}{\sum_j \sum_n (v_n^{(j)} x_j)} \quad (2-104)$$

and T_{mk} is a group interaction parameter given by an equation similar to (2-96),

$$T_{mk} = \exp \left(-\frac{a_{mk}}{T} \right) \quad (2-105)$$

where $a_{mk} \neq a_{km}$. When $m = k$, then $a_{mk} = 0$ and $T_{mk} = 1.0$. For $\Gamma_k^{(i)}$, (2-102) also applies, where θ terms correspond to the pure component i . Although values of R_k and Q_k are different for each functional group, values of a_{mk} are equal for all subgroups within a main group. For example, main group CH_2 consists of subgroups CH_3 , CH_2 , CH , and C . Accordingly,

$$a_{\text{CH}_3, \text{CHO}} = a_{\text{CH}_2, \text{CHO}} = a_{\text{CH}, \text{CHO}} = a_{\text{C}, \text{CHO}}$$

Thus, the amount of experimental data required to obtain values of a_{mk} and a_{km} and the size of the corresponding bank of data for these parameters is not as large as might be expected.

The ability of a group-contribution method to predict liquid-phase activity coefficients has been further improved by introduction of a modified UNIFAC method by Gmehling [51], referred to as UNIFAC (Dortmund). To correlate data for mixtures having a wide range of molecular size, they modified the combinatorial part of (2-97). To handle temperature dependence more accurately, they replaced (2-105) with a three-coefficient equation. The resulting modification permits reasonably reliable predictions of liquid-phase activity coefficients (including applications to dilute solutions and multiple liquid phases), heats of mixing, and azeotropic compositions. Values of the UNIFAC (Dortmund) parameters for 51 groups are available in a series of publications starting in 1993 with Gmehling, Li, and Schiller [53] and more recently with Wittig, Lohmann, and Gmehling [54].

Liquid-Liquid Equilibria

When species are notably dissimilar and activity coefficients are large, two and even more liquid phases may coexist at equilibrium. For example, consider the binary system of methanol (1) and cyclohexane (2) at 25°C. From measurements of Takeuchi, Nitta, and Katayama [55], van Laar constants are $A_{12} = 2.61$ and $A_{21} = 2.34$, corresponding, respectively, to infinite-dilution activity coefficients of 13.6 and 10.4 obtained using (2-72). These values of A_{12} and A_{21} can be used to construct an equilibrium plot of y_1 against x_1 assuming an isothermal condition. By combining (2-69), where $K_i = y_i/x_i$, with

$$P = \sum_{i=1}^C x_i \gamma_{iL} P_i^s \quad (2-106)$$

one obtains the following relation for computing y_i from x_i :

$$y_1 = \frac{x_1 \gamma_1 P_1^s}{x_1 \gamma_1 P_1^s + x_2 \gamma_2 P_2^s} \quad (2-107)$$

Vapor pressures at 25°C are $P_1^s = 2.452$ psia (16.9 kPa) and $P_2^s = 1.886$ psia (13.0 kPa). Activity coefficients can be computed from the van Laar equation in Table 2.9. The resulting equilibrium plot is shown in Figure 2.20, where it is observed that over much of the liquid-phase region, three values of y_1 exist. This indicates phase instability. Experimentally, single liquid phases can exist only for cyclohexane-rich mixtures of

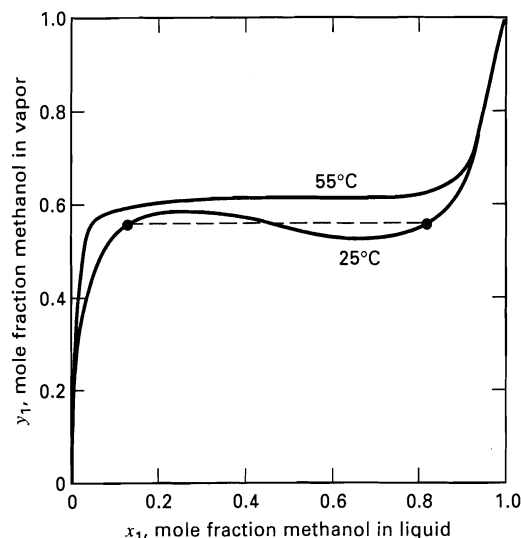


Figure 2.20 Equilibrium curves for methanol/cyclohexane systems.

[Data from K. Strubl, V. Svoboda, R. Holub, and J. Pick, *Collect. Czech. Chem. Commun.*, **35**, 3004–3019 (1970).]

$x_1 = 0.8248$ to 1.0 and for methanol-rich mixtures of $x_1 = 0.0$ to 0.1291. Because a coexisting vapor phase exhibits only a single composition, two coexisting liquid phases prevail at opposite ends of the dashed line in Figure 2.20. The liquid phases represent solubility limits of methanol in cyclohexane and cyclohexane in methanol.

For two coexisting equilibrium liquid phases, the relation $\gamma_{iL}^{(1)} x_i^{(1)} = \gamma_{iL}^{(2)} x_i^{(2)}$ must hold. This permits determination of the two-phase region in Figure 2.20 from the van Laar or other suitable activity-coefficient equation for which the constants are known. Also shown in Figure 2.20 is an equilibrium curve for the same binary system at 55°C based on data of Strubl et al. [56]. At this higher temperature, methanol and cyclohexane are completely miscible. The data of Kiser, Johnson, and Shetlar [57] show that phase instability ceases to exist at 45.75°C, the critical solution temperature. Rigorous thermodynamic methods for determining phase instability and, thus, existence of two equilibrium liquid phases are generally based on free-energy calculations, as discussed by Prausnitz et al. [4]. Most of the empirical and semitheoretical equations for the liquid-phase activity coefficient listed in Table 2.9 apply to liquid-liquid systems. The Wilson equation is a notable exception.

2.7 DIFFICULT MIXTURES

The equation-of-state and activity-coefficient models presented in Sections 2.5 and 2.6, respectively, are inadequate for estimating K -values of mixtures containing: (1) both polar and supercritical (light-gas) components, (2) electrolytes, and (3) both polymers and solvents. For these difficult mixtures, special models have been developed, some of which are briefly described in the following subsections. More detailed discussions of the following three topics are given by Prausnitz, Lichtenthaler, and de Azevedo [4].

Predictive Soave-Redlich-Kwong (PSRK) Model

Equation-of-state models, such as S-R-K and P-R, describe mixtures of nonpolar and slightly polar compounds. Gibbs free-energy activity-coefficient models are formulated for subcritical nonpolar and polar compounds. When a mixture contains both polar compounds and supercritical (light-gas) components (e.g., a mixture of hydrogen, carbon monoxide, methane, methyl acetate, and ethanol), neither method applies. To estimate vapor-liquid phase equilibria for such mixtures, a number of more theoretically based mixing rules for use with the S-R-K and P-R equations of state have been developed. In a different approach, Holderbaum and Gmehling [58] formulated a group-contribution equation of state referred to as the predictive Soave-Redlich-Kwong (PSRK) model, which combines a modified S-R-K equation of state with the UNIFAC model. To improve the ability of the S-R-K equation to predict vapor pressure of polar compounds, they provide an improved temperature dependence for the pure-component parameter, a , in Table 2.5. To handle mixtures of nonpolar, polar, and supercritical components, they use a mixing rule for a , which includes the UNIFAC model for handling nonideal effects more accurately. Additional and revised pure-component and group interaction parameters for use in the PSRK model are provided by Fischer and Gmehling [59]. In particular, [58] and [59] provide parameters for nine light gases (Ar, CO, CO₂, CH₄, H₂, H₂S, N₂, NH₃, and O₂) in addition to UNIFAC parameters for 50 groups.

Electrolyte Solution Models

Solutions of weak and/or strong electrolytes are common in chemical processes. For example, sour water, found in many petroleum plants, may consist of solvent (water) and five dissolved gases: CO, CO₂, CH₄, H₂S, and NH₃. The apparent composition of the solution is based on these six molecules. However, because of dissociation, which in this case is weak, the true composition of the aqueous solution includes ionic as well as molecular species. For sour water, the ionic species present at chemical equilibrium include H⁺, OH⁻, HCO₃⁻, CO₃⁼, HS⁻, S⁼, NH₄⁺, and NH₂COO⁻, with the total numbers of positive and negative ions subject to electroneutrality. For example, while the apparent concentration of NH₄ in the solution might be 2.46 moles per kg of water, when dissociation is taken into account, the molality is only 0.97, with NH₄⁺ having a molality of 1.49. All eight ionic species are nonvolatile, while all six molecular species are volatile to some extent. Accurate calculations of vapor-liquid equilibrium for multicomponent electrolyte solutions must consider both chemical and physical equilibrium, both of which involve liquid-phase activity coefficients.

A number of models have been developed for predicting activity coefficients in multicomponent systems of electrolytes. Of particular note are the models of Pitzer [60] and Chen and associates [61, 62, and 63], both of which are included in simulation programs. Both models can handle

dilute to concentrated solutions, but only the model of Chen and associates, which is a substantial modification of the NRTL model (see Section 2.6), can handle mixed-solvent systems, such as those containing water and alcohols.

Polymer Solution Models

Polymer processing often involves solutions of solvent, monomer, and an amorphous (noncrystalline) polymer, requiring vapor-liquid and, sometimes, liquid-liquid phase-equilibria calculations, for which estimation of activity coefficients of all components in the mixture is needed. In general, the polymer is nonvolatile, but the solvent and monomer are volatile. When the solution is dilute in the polymer, activity-coefficient methods of Section 2.6, such as the NRTL method, can be used. Of more interest are solutions with appreciable concentrations of polymer, for which the methods of Sections 2.5 and 2.6 are inadequate. Consequently, special-purpose empirical and theoretical models have been developed. One method, which is available in simulation programs, is the modified NRTL model of Chen [64], which combines a modification of the Flory-Huggins equation (12-65) for widely differing molecular size with the NRTL concept of local composition. Chen represents the polymer with segments. Thus, solvent-solvent, solvent-segment, and segment-segment binary interaction parameters are required, which are often available from the literature and may be assumed independent of temperature, polymer chain length, and polymer concentration, making the model quite flexible.

2.8 SELECTING AN APPROPRIATE MODEL

The three previous sections of this chapter have discussed the more widely used models for estimating fugacities, activity coefficients, and K -values for components in mixtures. These models and others are included in computer-aided, process-simulation programs. To solve a particular separations problem, it is necessary to select an appropriate model. This section presents recommendations for making at least a preliminary selection.

The selection procedure includes a few models not covered in this chapter, but for which a literature reference is given. The procedure begins by characterizing the mixture by chemical types present: Light gases (LG), Hydrocarbons (HC), Polar organic compounds (PC), and Aqueous solutions (A), with or without Electrolytes (E).

If the mixture is (A) with no (PC), then if electrolytes are present, select the modified NRTL equation. Otherwise, select a special model, such as one for sour water (containing NH₃, H₂S, CO₂, etc.) or aqueous amine solutions.

If the mixture contains (HC), with or without (LG), covering a wide boiling range, choose the corresponding-states method of Lee-Kesler-Plöcker [8, 65]. If the boiling range of a mixture of (HC) is not wide boiling, the selection depends on the pressure and temperature. For all temperatures and pressures, the Peng-Robinson equation is suitable. For

all pressures and noncryogenic temperatures, the Soave–Redlich–Kwong equation is suitable. For all temperatures, but not pressures in the critical region, the Benedict–Webb–Rubin–Starling [5, 66, 67] method is suitable.

If the mixture contains (PC), the selection depends on whether (LG) are present. If they are, the PSRK method is recommended. If not, then a suitable liquid-phase activity-

coefficient method is selected as follows. If the binary interaction coefficients are not available, select the UNIFAC method, which should be considered as only a first approximation. If the binary interaction coefficients are available and splitting in two liquid phases will not occur, select the Wilson or NRTL equation. Otherwise, if phase splitting is probable, select the NRTL or UNIQUAC equation.

SUMMARY

1. Separation processes are often energy-intensive. Energy requirements are determined by applying the first law of thermodynamics. Estimates of minimum energy needs can be made by applying the second law of thermodynamics with an entropy balance or an availability balance.
2. Phase equilibrium is expressed in terms of vapor–liquid and liquid–liquid K -values, which are formulated in terms of fugacity and activity coefficients.
3. For separation systems involving an ideal-gas mixture and an ideal-liquid solution, all necessary thermodynamic properties can be estimated from the ideal-gas law, a vapor heat-capacity equation, a vapor-pressure equation, and an equation for the liquid density as a function of temperature.
4. Graphical correlations of pure-component thermodynamic properties are widely available and useful for making rapid, manual calculations at near-ambient pressure for an ideal solution.

REFERENCES

1. MIX, T.W., J.S. DWECK, M. WEINBERG, and R.C. ARMSTRONG, *AIChE Symp. Ser.*, No. 192, Vol. 76, 15–23 (1980).
2. FELDER, R.M., and R.W. ROUSSEAU, *Elementary Principles of Chemical Processes*, 3rd ed., John Wiley & Sons, New York (2000).
3. DE NEVERS, N., and J.D. SEADER, *Latin Am. J. Heat and Mass Transfer*, **8**, 77–105 (1984).
4. PRASUNITZ, J.M., R.N. LICHTENTHALER, and E.G. DE AZEVEDO, *Molecular Thermodynamics of Fluid-Phase Equilibria*, 3rd ed., Prentice-Hall, Upper Saddle River, NJ (1999).
5. STARLING, K.E., *Fluid Thermodynamic Properties for Light Petroleum Systems*, Gulf Publishing, Houston, TX (1973).
6. SOAVE, G., *Chem. Eng. Sci.*, **27**, 1197–1203 (1972).
7. PENG, D.Y., and D.B. ROBINSON, *Ind. Eng. Chem. Fundam.*, **15**, 59–64 (1976).
8. PLÖCKER, U., H. KNAPP, and J.M. PRASUNITZ, *Ind. Eng. Chem. Process Des. Dev.*, **17**, 324–332 (1978).
9. CHAO, K.C., and J.D. SEADER, *AIChE J.*, **7**, 598–605 (1961).
10. GRAYSON, H.G., and C.W. STREED, Paper 20-P07, Sixth World Petroleum Conference, Frankfurt, June 1963.
11. POLING, B.E., J.M. PRASUNITZ, and J.P. O'CONNELL, *The Properties of Gases and Liquids*, 5th ed., McGraw-Hill, New York (2001).
12. RACKETT, H.G., *J. Chem. Eng. Data*, **15**, 514–517 (1970).
13. YAWS, C.L., H.-C. YANG, J.R. HOPPER, and W.A. CAWLEY, *Hydrocarbon Processing*, **71** (1), 103–106 (1991).
14. FRANK, J.C., G.R. GEYER, and H. KEHDE, *Chem. Eng. Prog.*, **65** (2), 79–86 (1969).
15. HADDEN, S.T., and H.G. GRAYSON, *Hydrocarbon Process., Petrol. Refiner*, **40** (9), 207–218 (1961).
16. ROBBINS, L.A., Section 15, "Liquid-Liquid Extraction Operations and Equipment," in R.H. Perry, D. Green, and J.O. Maloney, Eds., *Perry's Chemical Engineers' Handbook*, 7th ed., McGraw-Hill, New York (1997).
17. PITZER, K.S., D.Z. LIPPMAN, R.F. CURL, Jr., C.M. HUGGINS, and D.E. PETERSEN, *J. Am. Chem. Soc.*, **77**, 3433–3440 (1955).
18. REDLICH, O., and J.N.S. KWONG, *Chem. Rev.*, **44**, 233–244 (1949).
19. SHAH, K.K., and G. THODOS, *Ind. Eng. Chem.*, **57** (3), 30–37 (1965).
20. GLANVILLE, J.W., B.H. SAGE, and W.N. LACEY, *Ind. Eng. Chem.*, **42**, 508–513 (1950).
21. WILSON, G.M., *Adv. Cryogenic Eng.*, **11**, 392–400 (1966).
22. KNAPP, H., R. DORING, L. OELLRICH, U. PLÖCKER, and J.M. PRASUNITZ, *Vapor-Liquid Equilibria for Mixtures of Low Boiling Substances*, Chem. Data. Ser., Vol. VI, DECHEMA (1982).
23. THIESEN, M., *Ann. Phys.*, **24**, 467–492 (1885).
24. ONNES, K., *Konink. Akad. Wetens.*, p. 633 (1912).
25. WALAS, S.M., *Phase Equilibria in Chemical Engineering*, Butterworth, Boston (1985).
26. LEE, B.I., and M.G. KESSLER, *AIChE J.*, **21**, 510–527 (1975).
27. EDMISTER, W.C., *Hydrocarbon Processing*, **47** (9), 239–244 (1968).
28. YARBOROUGH, L., *J. Chem. Eng. Data*, **17**, 129–133 (1972).
29. PRASUNITZ, J.M., W.C. EDMISTER, and K.C. CHAO, *AIChE J.*, **6**, 214–219 (1960).
30. HILDEBRAND, J.H., J.M. PRASUNITZ, and R.L. SCOTT, *Regular and Related Solutions*, Van Nostrand Reinhold, New York (1970).
31. YERAZUNIS, S., J.D. PLOWRIGHT, and F.M. SMOLA, *AIChE J.*, **10**, 660–665 (1964).
32. HERMSEN, R.W., and J.M. PRASUNITZ, *Chem. Eng. Sci.*, **18**, 485–494 (1963).

33. EWELL, R.H., J.M. HARRISON, and L. BERG, *Ind. Eng. Chem.*, **36**, 871–875 (1944).
34. VAN NESS, H.C., C.A. SOCZEK, and N.K. KOCHAR, *J. Chem. Eng. Data*, **12**, 346–351 (1967).
35. SINOR, J.E., and J.H. WEBER, *J. Chem. Eng. Data*, **5**, 243–247 (1960).
36. ORYE, R.V., and J.M. PRAUSNITZ, *Ind. Eng. Chem.*, **57** (5), 18–26 (1965).
37. WILSON, G.M., *J. Am. Chem. Soc.*, **86**, 127–130 (1964).
38. CUKOR, P.M., and J.M. PRAUSNITZ, *Inst. Chem. Eng. Symp. Ser. No. 32*, **3**, 88 (1969).
39. GMEHLING, J., and U. ONKEN, *Vapor-Liquid Equilibrium Data Collection*, DECHEMA Chem. Data Ser., 1–8, (1977–1984).
40. HIRANUMA, M., *J. Chem. Eng. Japan*, **8**, 162–163 (1975).
41. RENON, H., and J.M. PRAUSNITZ, *AIChE J.*, **14**, 135–144 (1968).
42. RENON, H., and J.M. PRAUSNITZ, *Ind. Eng. Chem. Process Des. Dev.*, **8**, 413–419 (1969).
43. ABRAMS, D.S., and J.M. PRAUSNITZ, *AIChE J.*, **21**, 116–128 (1975).
44. ABRAMS, D.S., Ph.D. thesis in chemical engineering, University of California, Berkeley, 1974.
45. PRAUSNITZ, J.M., T.F. ANDERSON, E.A. GRENS, C.A. ECKERT, R. HSIEH, and J.P. O'CONNELL, *Computer Calculations for Multicomponent Vapor-Liquid and Liquid-Liquid Equilibria*, Prentice-Hall, Englewood Cliffs, NJ (1980).
46. FREDENSLUND, A., J. GMEHLING, M.L. MICHELSEN, P. RASMUSSEN, and J.M. PRAUSNITZ, *Ind. Eng. Chem. Process Des. Dev.*, **16**, 450–462 (1977).
47. WILSON, G.M., and C.H. DEAL, *Ind. Eng. Chem. Fundam.*, **1**, 20–23 (1962).
48. DERR, E.L., and C.H. DEAL, *Inst. Chem. Eng. Symp. Ser. No. 32*, **3**, 40–51 (1969).
49. FREDENSLUND, A., R.L. JONES, and J.M. PRAUSNITZ, *AIChE J.*, **21**, 1086–1099 (1975).
50. FREDENSLUND, A., J. GMEHLING, and P. RASMUSSEN, *Vapor-Liquid Equilibria Using UNIFAC, A Group Contribution Method*, Elsevier, Amsterdam (1977).
51. GMEHLING, J., P. RASMUSSEN, and A. FREDENSLUND, *Ind. Eng. Chem. Process Des. Dev.*, **21**, 118–127 (1982).
52. LARSEN, B.L., P. RASMUSSEN, and A. FREDENSLUND, *Ind. Eng. Chem. Res.*, **26**, 2274–2286 (1987).
53. GMEHLING, J., J. LI, and M. SCHILLER, *Ind. Eng. Chem. Res.*, **32**, 178–193 (1993).
54. WITTIG, R., J. LOHMANN, and J. GMEHLING, *Ind. Eng. Chem. Res.*, **42**, 183–188 (2003).
55. TAKEUCHI, S., T. NITTA, and T. KATAYAMA, *J. Chem. Eng. Japan*, **8**, 248–250 (1975).
56. STRUBL, K., V. SVOBODA, R. HOLUB, and J. PICK, *Collect. Czech. Chem. Commun.*, **35**, 3004–3019 (1970).
57. KISER, R.W., G.D. JOHNSON, and M.D. SHETLAR, *J. Chem. Eng. Data*, **6**, 338–341 (1961).
58. HOLDERBAUM, T., and J. GMEHLING, *Fluid Phase Equilibria*, **70**, 251–265 (1991).
59. FISCHER, K., and J. GMEHLING, *Fluid Phase Equilibria*, **121**, 185–206 (1996).
60. PITZER, K.S., *J. Phys. Chem.*, **77**, No. 2, 268–277 (1973).
61. CHEN, C.-C., H.I. BRITT, J.F. BOSTON, and L.B. EVANS, *AIChE Journal*, **28**, 588–596 (1982).
62. CHEN, C.-C., and L.B. EVANS, *AIChE Journal*, **32**, 444–459 (1986).
63. MOCK, B., L.B. EVANS, and C.-C. CHEN, *AIChE Journal*, **28**, 1655–1664 (1986).
64. CHEN, C.-C., *Fluid Phase Equilibria*, **83**, 301–312 (1993).
65. LEE, B.I., and M.G. KESLER, *AIChE Journal*, **21**, 510–527 (1975).
66. BENEDICT, M., G.B. WEBB, and L.C. RUBIN, *Chem. Eng. Progress*, **47** (8), 419 (1951).
67. BENEDICT, M., G.B. WEBB, and L.C. RUBIN, *Chem. Eng. Progress*, **47** (9), 449 (1951).

EXERCISES

Section 2.1

2.1 A hydrocarbon stream in a petroleum refinery is to be separated at 1,500 kPa into two products under the conditions shown below. Using the data given, compute the minimum work of separation, $W_{\min}$, in kJ/h for $T_0 = 298.15$ K.

Component	kmol/h	
	Feed	Product 1
Ethane	30	30
Propane	200	192
n-Butane	370	4
n-Pentane	350	0
n-Hexane	50	0

	Feed	Product 1	Product 2
Phase condition	Liquid	Vapor	Liquid
Temperature, K	364	313	394
Enthalpy, kJ/kmol	19,480	25,040	25,640
Entropy, kJ/kmol-K	36.64	33.13	54.84

2.2 In petroleum refineries, a mixture of paraffins and cycloparaffins is commonly reformed in a fixed-bed catalytic reactor to produce blending stocks for gasoline and aromatic precursors for making petrochemicals. A typical multicomponent product from catalytic reforming is a mixture of ethylbenzene with the three xylene isomers. If this mixture is separated, these four chemicals can then be subsequently processed to make styrene, phthalic anhydride, isophthalic acid, and terephthalic acid. Compute, using the following data, the minimum work of separation in Btu/h for $T_0 = 560^\circ\text{R}$ if the mixture below is separated at 20 psia into three products.

Component	Feed, lbmol/h	Split Fraction (SF)		
		Product 1	Product 2	Product 3
Ethylbenzene	150	0.96	0.04	0.000
p-Xylene	190	0.005	0.99	0.005
m-Xylene	430	0.004	0.99	0.006
o-Xylene	230	0.00	0.015	0.985

	Feed	Product 1	Product 2	Product 3
Phase condition	Liquid	Liquid	Liquid	Liquid
Temperature, °F	305	299	304	314
Enthalpy, Btu/lbmol	29,290	29,750	29,550	28,320
Entropy, Btu/lbmol-°R	15.32	12.47	13.60	14.68

2.3 Distillation column C3 in Figure 1.9 separates stream 5 into streams 6 and 7, according to the material balance in Table 1.5. A suitable column for the separation, if carried out at 700 kPa, contains 70 plates with a condenser duty of 27,300,000 kJ/h. Using the following data and an infinite surroundings temperature, T_0 , of 298.15 K, compute:

- The duty of the reboiler in kJ/h
- The irreversible production of entropy in kJ/h-K, assuming the use of cooling water at a nominal temperature of 25°C for the condenser and saturated steam at 100°C for the reboiler
- The lost work in kJ/h
- The minimum work of separation in kJ/h
- The second-law efficiency

Assume the shaft work of the reflux pump is negligible.

	Feed (Stream 5)	Distillate (Stream 6)	Bottoms (Stream 7)
Phase condition	Liquid	Liquid	Liquid
Temperature, K	348	323	343
Pressure, kPa	1,950	700	730
Enthalpy, kJ/kmol	17,000	13,420	15,840
Entropy, kJ/kmol-K	25.05	5.87	21.22

2.4 A spiral-wound, nonporous cellulose acetate membrane separator is to be used to separate a gas containing H_2 , CH_4 , and C_2H_6 . The permeate will be 95 mol% pure H_2 and will contain no ethane. The relative split ratio (separation power), SP, for H_2 relative to methane will be 47. Using the following data and an infinite surroundings temperature of 80°F, compute:

- The irreversible production of entropy in Btu/h-R
- The lost work in Btu/h
- The minimum work of separation in Btu/h. Why is it negative? What other method(s) might be used to make the separation?

Feed flow rates, lbmol/h

H_2	3,000
CH_4	884
C_2H_6	120

Stream properties:

	Feed	Permeate	Retentate
Phase condition	Vapor	Vapor	Vapor
Temperature, °F	80	80	80
Pressure, psia	365	50	365
Enthalpy, Btu/lbmol	8,550	8,380	8,890
Entropy, Btu/lbmol-R	1.520	4.222	2.742

Section 2.2

2.5 Which of the following K -value expressions, if any, is (are) rigorous? For those expressions that are not rigorous, cite the assumptions involved.

- $K_i = \bar{\phi}_{iL}/\bar{\phi}_{iV}$
- $K_i = \phi_{iL}/\phi_{iL}$
- $K_i = \phi_{iL}$
- $K_i = \gamma_{iL}\phi_{iL}/\bar{\phi}_{iV}$
- $K_i = P_i^s/P$
- $K_i = \gamma_{iL}\phi_{iL}/\gamma_{iV}\phi_{iV}$
- $K_i = \gamma_{iL}P_i^s/P$

2.6 Experimental measurements of Vaughan and Collins [*Ind. Eng. Chem.*, **34**, 885 (1942)] for the propane-isopentane system at 167°F and 147 psia show for propane a liquid-phase mole fraction of 0.2900 in equilibrium with a vapor-phase mole fraction of 0.6650. Calculate:

- The K -values for C_3 and iC_5 from the experimental data.
- Estimates of the K -values of C_3 and iC_5 from Raoult's law assuming vapor pressures at 167°F of 409.6 and 58.6 psia, respectively.

Compare the results of (a) and (b). Assuming the experimental values are correct, how could better estimates of the K -values be achieved? To respond to this question, compare the rigorous expression $K_i = \gamma_{iL}\phi_{iL}/\bar{\phi}_{iV}$ to the Raoult's law expression $K_i = P_i^s/P$.

2.7 Mutual solubility data for the isooctane (1)/furfural (2) system at 25°C are [*Chem. Eng. Sci.*, **6**, 116 (1957)]

	Liquid Phase I	Liquid Phase II
x_1	0.0431	0.9461

Compute:

- The distribution coefficients for isooctane and furfural
- The relative selectivity for isooctane relative to furfural
- The activity coefficient of isooctane in liquid phase 1 and the activity coefficient of furfural in liquid phase 2 assuming $\gamma_2^{(1)} = 1.0$ and $\gamma_1^{(2)} = 1.0$.

2.8 In petroleum refineries, streams rich in alkylbenzenes and alkylnaphthalenes result from catalytic cracking operations. Such streams can be hydrodealkylated to more valuable products such as benzene and naphthalene. At 25°C, solid naphthalene (normal melting point = 80.3°C) has the following solubilities in various liquid solvents [*Naphthalene*, API Publication 707, Washington, DC (Oct. 1978)], including benzene:

Solvent	Mole Fraction Naphthalene
Benzene	0.2946
Cyclohexane	0.1487
Carbon tetrachloride	0.2591
<i>n</i> -Hexane	0.1168
Water	0.18×10^{-5}

For each solvent, compute the activity coefficient of naphthalene in the liquid solvent phase using the following equations for the vapor pressure in torr of solid and liquid naphthalene:

$$\ln P_{\text{solid}}^s = 26.708 - 8,712/T$$

$$\ln P_{\text{liquid}}^s = 16.1426 - 3992.01/(T - 71.29)$$

where T is in K.

Section 2.3

2.9 A binary ideal-gas mixture of A and B undergoes an isothermal, isobaric separation at T_0 , the infinite surroundings temperature. Starting with Eq. (4), Table 2.1, derive an equation for the minimum work of separation, $W_{\min}$, in terms of mole fractions of the feed and the two products. Use your equation to prepare a plot of the dimensionless group, $W_{\min}/RT_0n_F$, as a function of mole fraction of A in the feed for:

- A perfect separation
- A separation with $SF_A = 0.98$, $SF_B = 0.02$
- A separation with $SR_A = 9.0$ and $SR_B = \frac{1}{9}$
- A separation with $SF = 0.95$ for A and $SP_{A,B} = 361$

How sensitive is $W_{\min}$ to product purities? Does $W_{\min}$ depend on the particular separation operation used?

Prove, by calculus, that the largest value of $W_{\min}$ occurs for a feed with equimolar quantities of A and B.

2.10 The separation of isopentane from *n*-pentane by distillation is difficult (approximately 100 trays are required), but is commonly practiced in industry. Using the extended Antoine vapor pressure equation, (2-39), with the constants below and in conjunction with Raoult's law, calculate relative volatilities for the isopentane/*n*-pentane system and compare the values on a plot with the following smoothed experimental values [*J. Chem. Eng. Data*, 8, 504 (1963)]:

Temperature, °F	α_{iC_5, nC_5}
125	1.26
150	1.23
175	1.21
200	1.18
225	1.16
250	1.14

What do you conclude about the applicability of Raoult's law in this temperature range for this binary system?

Vapor pressure constants for (2-39) with vapor pressure in kPa and T in K are

	iC_5	nC_5
k_1	13.6106	13.9778
k_2	-2,345.09	-2,554.60
k_3	-40.2128	-36.2529
k_4, k_5, k_6	0	0

2.11 Operating conditions at the top of a vacuum distillation column for the separation of ethylbenzene from styrene are given below, where the overhead vapor is condensed in an air-cooled condenser to give subcooled reflux and distillate. Using the property constants in Example 2.3, estimate the heat transfer rate (duty) for the condenser in kJ/h, assuming an ideal gas and ideal gas and liquid solutions.

	Overhead Vapor	Reflux	Distillate
Phase condition	Vapor	Liquid	Liquid
Temperature, K	331	325	325
Pressure, kPa	6.69	6.40	6.40
Component flow rates, kg/h:			
Ethylbenzene	77,500	66,960	10,540
Styrene	2,500	2,160	340

2.12 Toluene can be hydrodealkylated to benzene, but the conversion per pass through the reactor is only about 70%. Consequently, the toluene must be recovered and recycled. Typical conditions for the feed to a commercial distillation unit are 100°F, 20 psia, 415 lbmol/h of benzene, and 131 lbmol/h of toluene. Based on the property constants below, and assuming that the ideal gas, ideal liquid solution model of Table 2.4 applies at this low pressure, prove that the mixture is a liquid and estimate v_L and ρ_L in American engineering units.

Property constants for (2-38) and (2-39), where in all cases, T is in K, are

	Benzene	Toluene
M , kg/kmol	78.114	92.141
P^s , torr:		
k_1	15.9008	16.0137
k_2	-2,788.51	-3,096.52
k_3	-52.36	-53.67
k_4, k_5, k_6	0	0
ρ_L , kg/m ³ :		
A	304.1	290.6
B	0.269	0.265
T_c	562.0	593.1

Section 2.4

2.13 Measured conditions for the bottoms from a depropanizer distillation unit in a small refinery are given below. Using the data in Figure 2.3 and assuming an ideal liquid solution (volume of mixing = 0), compute the liquid density in lb/ft³, lb/gal, lb/bbl (42 gal), and kg/m³.

Phase Condition	Liquid
Temperature, °F	229
Pressure, psia	282
Flow rates, lbmol/h:	
C_3	2.2
iC_4	171.1
nC_4	226.6
iC_5	28.1
nC_5	17.5

2.14 Isopropanol, containing 13 wt% water, can be dehydrated to obtain almost pure isopropanol at a 90% recovery by azeotropic distillation with benzene. When condensed, the overhead vapor from the column splits into two immiscible liquid phases. Use the relations in Table 2.4 with data in Perry's Handbook and the operating conditions below to compute the rate of heat transfer in Btu/h and kJ/h for the condenser.

	Overhead	Water-Rich Phase	Organic-Rich Phase
Phase	Vapor	Liquid	Liquid
Temperature, °C	76	40	40
Pressure, bar	1.4	1.4	1.4
Flow rate, kg/h:			
Isopropanol	6,800	5,870	930
Water	2,350	1,790	560
Benzene	24,600	30	24,570

2.15 A hydrocarbon vapor-liquid mixture at 250°F and 500 psia contains N₂, H₂S, CO₂, and all the normal paraffins from methane to heptane. Use Figure 2.8 to estimate the K -value of each

component in the mixture. Which components will have a tendency to be present to a greater extent in the equilibrium vapor?

2.16 Acetone, a valuable solvent, can be recovered from air by absorption in water or by adsorption on activated carbon. If absorption is used, the conditions for the streams entering and leaving are as listed below. If the absorber operates adiabatically, estimate the temperature of the exiting liquid phase using a simulation program.

	Feed Gas	Absorbent	Gas Out	Liquid Out
Flow rate, lbmol/h:				
Air	687	0	687	0
Acetone	15	0	0.1	14.9
Water	0	1,733	22	1,711
Temperature, °F	78	90	80	—
Pressure, psia	15	15	14	15
Phase	Vapor	Liquid	Vapor	Liquid

Some concern has been expressed about the possible explosion hazard associated with the feed gas. The lower and upper flammability limits for acetone in air are 2.5 and 13 mol%, respectively. Is the mixture within the explosive range? If so, what can be done to remedy the situation?

Section 2.5

2.17 Subquality natural gas contains an intolerable amount of nitrogen impurity. Separation processes that can be used to remove nitrogen include cryogenic distillation, membrane separation, and pressure-swing adsorption. For the latter process, a set of typical feed and product conditions is given below. Assume a 90% removal of N₂ and a 97% methane natural-gas product. Using the R-K equation of state with the constants listed below, compute the flow rate in thousands of actual cubic feet per hour for each of the three streams.

	N ₂	CH ₄
Feed flow rate, lbmol/h:	176	704
T _c , K	126.2	190.4
P _c , bar	33.9	46.0

Stream conditions are

	Feed (Subquality Natural Gas)	Product (Natural Gas)	Waste Gas
Temperature, °F	70	100	70
Pressure, psia	800	790	280

2.18 Use the R-K equation of state to estimate the partial fugacity coefficients of propane and benzene in the vapor mixture of Example 2.5.

2.19 Use a computer-aided, steady-state simulation program to estimate the *K*-values, using the P-R and S-R-K equations of state, of an equimolar mixture of the two butane isomers and the four butene isomers at 220°F and 276.5 psia. Compare these values with the following experimental results [*J. Chem. Eng. Data*, 7, 331 (1962)]:

Component	<i>K</i> -value
Isobutane	1.067
Isobutene	1.024
<i>n</i> -Butane	0.922
1-Butene	1.024
<i>trans</i> -2-Butene	0.952
<i>cis</i> -2-Butene	0.876

2.20 The disproportionation of toluene to benzene and xylenes is carried out in a catalytic reactor at 500 psia and 950°F. The reactor effluent is cooled in a series of heat exchangers for heat recovery until a temperature of 235°F is reached at a pressure of 490 psia. The effluent is then further cooled and partially condensed by the transfer of heat to cooling water in a final exchanger. The resulting two-phase equilibrium mixture at 100°F and 485 psia is then separated in a flash drum. For the reactor effluent composition given below, use a computer-aided, steady-state simulation program with the S-R-K and P-R equations of state to compute the component flow rates in lbmol/h in both the resulting vapor and liquid streams, the component *K*-values for the equilibrium mixture, and the rate of heat transfer to the cooling water. Compare the results.

Component	Reactor Effluent, lbmol/h
H ₂	1,900
CH ₄	215
C ₂ H ₆	17
Benzene	577
Toluene	1,349
<i>p</i> -Xylene	508

Section 2.6

2.21 For an ambient separation process where the feed and products are all nonideal liquid solutions at the infinite surroundings temperature, *T*₀, (4) of Table 2.1 for the minimum work of separation reduces to

$$\frac{W_{\min}}{RT_0} = \sum_{\text{out}} n \left[\sum_i x_i \ln(\gamma_i x_i) \right] - \sum_{\text{in}} n \left[\sum_i x_i \ln(\gamma_i x_i) \right]$$

For the liquid-phase separation at ambient conditions (298 K, 101.3 kPa) of a 35 mol% mixture of acetone (1) in water (2) into 99 mol% acetone and 98 mol% water products, calculate the minimum work in kJ/kmol of feed. Liquid-phase activity coefficients at ambient conditions are correlated reasonably well by the van Laar equations with *A*₁₂ = 2.0 and *A*₂₁ = 1.7. What would the minimum rate of work be if acetone and water formed an ideal liquid solution?

2.22 The sharp separation of benzene and cyclohexane by distillation at ambient pressure is impossible because of the formation of an azeotrope at 77.6°C. K.C. Chao [Ph.D. thesis, University of Wisconsin (1956)] obtained the following vapor-liquid equilibrium data for the benzene (B)/cyclohexane (CH) system at 1 atm:

T, °C	<i>x</i> _B	<i>y</i> _B	<i>γ</i> _B	<i>γ</i> _{CH}
79.7	0.088	0.113	1.300	1.003
79.1	0.156	0.190	1.256	1.008
78.5	0.231	0.268	1.219	1.019
78.0	0.308	0.343	1.189	1.032
77.7	0.400	0.422	1.136	1.056
77.6	0.470	0.482	1.108	1.075
77.6	0.545	0.544	1.079	1.102
77.6	0.625	0.612	1.058	1.138
77.8	0.701	0.678	1.039	1.178
78.0	0.757	0.727	1.025	1.221
78.3	0.822	0.791	1.018	1.263
78.9	0.891	0.863	1.005	1.328
79.5	0.953	0.938	1.003	1.369

Vapor pressure is given by (2-39), where constants for benzene are in Exercise 2.12 and constants for cyclohexane are $k_1 = 15.7527$, $k_2 = -2766.63$, and $k_3 = -50.50$.

(a) Use the data to calculate and plot the relative volatility of benzene with respect to cyclohexane versus benzene composition in the liquid phase. What happens to the relative volatility in the vicinity of the azeotrope?

(b) From the azeotropic composition for the benzene/cyclohexane system, calculate the constants in the van Laar equation. With these constants, use the van Laar equation to compute the activity coefficients over the entire range of composition and compare them, in a plot like Figure 2.16, with the above experimental data. How well does the van Laar equation predict the activity coefficients?

2.23 Benzene can be used to break the ethanol/water azeotrope so as to produce nearly pure ethanol. The Wilson constants for the ethanol(1)/benzene(2) system at 45°C are $\Lambda_{12} = 0.124$ and $\Lambda_{21} = 0.523$. Use these constants with the Wilson equation to predict the liquid-phase activity coefficients for this system over the entire range of composition and compare them, in a plot like Figure 2.16, with the following experimental results [*Austral. J. Chem.*, 7, 264 (1954)]:

x_1	$\ln \gamma_1$	$\ln \gamma_2$
0.0374	2.0937	0.0220
0.0972	1.6153	0.0519

x_1	$\ln \gamma_1$	$\ln \gamma_2$
0.3141	0.7090	0.2599
0.5199	0.3136	0.5392
0.7087	0.1079	0.8645
0.9193	0.0002	1.3177
0.9591	-0.0077	1.3999

2.24 For the binary system ethanol(1)/isooctane(2) at 50°C, the infinite-dilution, liquid-phase activity coefficients are $\gamma_1^\infty = 21.17$ and $\gamma_2^\infty = 9.84$.

(a) Calculate the constants A_{12} and A_{21} in the van Laar equations.

(b) Calculate the constants Λ_{12} and Λ_{21} in the Wilson equations.

(c) Using the constants from (a) and (b), calculate γ_1 and γ_2 over the entire composition range and plot the calculated points as $\log \gamma$ versus x_1 .

(d) How well do the van Laar and Wilson predictions agree with the azeotropic point where $x_1 = 0.5941$, $\gamma_1 = 1.44$, and $\gamma_2 = 2.18$?

(e) Show that the van Laar equation erroneously predicts separation into two liquid phases over a portion of the composition range by calculating and plotting a y - x diagram like Figure 2.20.

Chapter 3

Mass Transfer and Diffusion

Mass transfer is the net movement of a component in a mixture from one location to another where the component exists at a different concentration. In many separation operations, the transfer takes place between two phases across an interface. Thus, the absorption by a solvent liquid of a solute from a carrier gas involves mass transfer of the solute through the gas to the gas-liquid interface, across the interface, and into the liquid. Mass-transfer models describe this and other processes such as passage of a species through a gas to the outer surface of a porous, adsorbent particle and into the adsorbent pores, where the species is adsorbed on the porous surface. Mass transfer also governs selective permeation through a nonporous, polymeric material of a component of a gas mixture. Mass transfer, as used here, does not refer to the flow of a fluid through a pipe. However, mass transfer might be superimposed on that flow. Mass transfer is not the flow of solids on a conveyor belt.

Mass transfer occurs by two basic mechanisms: (1) *molecular diffusion* by random and spontaneous microscopic movement of individual molecules in a gas, liquid, or solid as a result of thermal motion; and (2) *eddy (turbulent) diffusion* by random, macroscopic fluid motion. Both molecular and/or eddy diffusion frequently involve the movement of different species in opposing directions. When a net flow occurs in one of these directions, the total rate of mass transfer of individual species is increased or decreased by this bulk flow or *convection effect*, which may be considered a third mechanism of mass transfer. Molecular diffusion is extremely slow, whereas eddy diffusion is orders of magnitude more rapid. Therefore, if industrial separation processes are to be conducted in equipment of reasonable size, fluids must be agitated and interfacial areas maximized. If mass transfer in solids is involved, using small particles to decrease the distance in the direction of diffusion will increase the rate.

When separations involve two or more phases, the extent of the separation is limited by phase equilibrium, because, with time, the phases in contact tend to equilibrate by mass transfer between phases. When mass transfer is rapid, equilibration is approached in seconds or minutes, and design of separation equipment may be based on phase equilibrium, not mass transfer. For separations involving

barriers, such as membranes, differing species mass-transfer rates through the membrane govern equipment design.

In a binary mixture, molecular diffusion of component A with respect to B occurs because of different potentials or driving forces, which include differences (gradients) of concentration (ordinary diffusion), pressure (pressure diffusion), temperature (thermal diffusion), and external force fields (forced diffusion) that act unequally on the different chemical species present. Pressure diffusion requires a large pressure gradient, which is achieved for gas mixtures with a centrifuge. Thermal diffusion columns or cascades can be employed to separate liquid and gas mixtures by establishing a temperature gradient. More widely applied is forced diffusion in an electrical field, to cause ions of different charges to move in different directions at different speeds.

In this chapter, only molecular diffusion caused by concentration gradients is considered, because this is the most common type of molecular diffusion in separation processes. Furthermore, emphasis is on binary systems, for which molecular-diffusion theory is relatively simple and applications are relatively straightforward. Multicomponent molecular diffusion, which is important in many applications, is considered briefly in Chapter 12. Diffusion in multicomponent systems is much more complex than diffusion in binary systems, and is a more appropriate topic for advanced study using a text such as Taylor and Krishna [1].

Molecular diffusion occurs in solids and in fluids that are stagnant or in laminar or turbulent motion. Eddy diffusion occurs in fluids in turbulent motion. When both molecular diffusion and eddy diffusion occur, they take place in parallel and are additive. Furthermore, they take place because of the same concentration difference (gradient). When mass transfer occurs under turbulent-flow conditions, but across an interface or to a solid surface, conditions may be laminar or nearly stagnant near the interface or solid surface. Thus, even though eddy diffusion may be the dominant mechanism in the bulk of the fluid, the overall rate of mass transfer may be controlled by molecular diffusion because the eddy-diffusion mechanism is damped or even eliminated as the interface or solid surface is approached.

Mass transfer of one or more species results in a total net rate of bulk flow or flux in one direction relative to a fixed

plane or stationary coordinate system. When a net flux occurs, it carries all species present. Thus, the molar flux of an individual species is the sum of all three mechanisms. If N_i is the molar flux of species i with mole fraction x_i , and N is the total molar flux, with both fluxes in moles per unit time per unit area in a direction perpendicular to a stationary plane across which mass transfer occurs, then

$$N_i = x_i N + \text{molecular diffusion flux of } i \\ + \text{eddy diffusion flux of } i \quad (3-1)$$

where $x_i N$ is the bulk-flow flux. Each term in (3-1) is positive or negative depending on the direction of the flux relative to

the direction selected as positive. When the molecular and eddy-diffusion fluxes are in one direction and N is in the opposite direction, even though a concentration difference or gradient of i exists, the net mass-transfer flux, N_i , of i can be zero.

In this chapter, the subject of mass transfer and diffusion is divided into seven areas: (1) steady-state diffusion in stagnant media, (2) estimation of diffusion coefficients, (3) unsteady-state diffusion in stagnant media, (4) mass transfer in laminar flow, (5) mass transfer in turbulent flow, (6) mass transfer at fluid-fluid interfaces, and (7) mass transfer across fluid-fluid interfaces.

3.0 INSTRUCTIONAL OBJECTIVES

After completing this chapter, you should be able to:

- Explain the relationship between mass transfer and phase equilibrium.
- Explain why separation models for mass transfer and phase equilibrium are useful.
- Discuss mechanisms of mass transfer, including the effect of bulk flow.
- State, in detail, Fick's law of diffusion for a binary mixture and discuss its analogy to Fourier's law of heat conduction in one dimension.
- Modify Fick's law of diffusion to include the bulk flow effect.
- Calculate mass-transfer rates and composition gradients under conditions of equimolar, countercurrent diffusion and unimolecular diffusion.
- Estimate, in the absence of data, diffusivities (diffusion coefficients) in gas and liquid mixtures, and know of some sources of data for diffusion in solids.
- Calculate multidimensional, unsteady-state, molecular diffusion by analogy to heat conduction.
- Calculate rates of mass transfer by molecular diffusion in laminar flow for three common cases: (1) falling liquid film, (2) boundary-layer flow past a flat plate, and (3) fully developed flow in a straight, circular tube.
- Define a mass-transfer coefficient and explain its analogy to the heat-transfer coefficient and its usefulness, as an alternative to Fick's law, in solving mass-transfer problems.
- Understand the common dimensionless groups (Reynolds, Sherwood, Schmidt, and Peclet number for mass transfer) used in correlations of mass-transfer coefficients.
- Use analogies, particularly that of Chilton and Colburn, and more theoretically based equations, such as those of Churchill et al., to calculate rates of mass transfer in turbulent flow.
- Calculate rates of mass transfer across fluid-fluid interfaces using the two-film theory and the penetration theory.

3.1 STEADY-STATE, ORDINARY MOLECULAR DIFFUSION

Suppose a cylindrical glass vessel is partly filled with water containing a soluble red dye. Clear water is carefully added on top so that the dyed solution on the bottom is undisturbed. At first, a sharp boundary exists between the two layers, but after a time the upper layer becomes colored, while the layer below becomes less colored. The upper layer is more colored near the original interface between the two layers and less colored in the region near the top of the upper layer. During this color change, the motion of each dye molecule is random, undergoing collisions mainly with water molecules and sometimes with other dye molecules, moving first in one

direction and then in another, with no one direction preferred. This type of motion is sometimes referred to as a *random-walk process*, which yields a mean-square distance of travel for a given interval of time, but not a direction of travel. Thus, at a given horizontal plane through the solution in the cylinder, it is not possible to determine whether, in a given time interval, a given molecule will cross the plane or not. However, on the average, a fraction of all molecules in the solution below the plane will cross over into the region above and the same fraction will cross over in the opposite direction. Therefore, if the concentration of dye molecules in the lower region is greater than in the upper region, a net rate of mass transfer of dye molecules will take place from the

lower to the upper region. After a long time, a dynamic equilibrium will be achieved and the concentration of dye will be uniform throughout the solution. Based on these observations, it is clear that:

1. Mass transfer by ordinary molecular diffusion occurs because of a concentration, difference or gradient; that is, a species diffuses in the direction of decreasing concentration.
2. The mass-transfer rate is proportional to the area normal to the direction of mass transfer and not to the volume of the mixture. Thus, the rate can be expressed as a flux.
3. Net mass transfer stops when concentrations are uniform.

Fick's Law of Diffusion

The above observations were quantified by Fick in 1855, who proposed an extension of Fourier's 1822 heat-conduction theory. Fourier's first law of heat conduction is

$$q_z = -k \frac{dT}{dz} \quad (3-2)$$

where q_z is the heat flux by conduction in the positive z -direction, k is the thermal conductivity of the medium, and dT/dz is the temperature gradient, which is negative in the direction of heat conduction. Fick's first law of molecular diffusion also features a proportionality between a flux and a gradient. For a binary mixture of A and B,

$$J_{A_z} = -D_{AB} \frac{dc_A}{dz} \quad (3-3a)$$

and

$$J_{B_z} = -D_{BA} \frac{dc_B}{dz} \quad (3-3b)$$

where, in (3-3a), J_{A_z} is the molar flux of A by ordinary molecular diffusion relative to the molar-average velocity of the mixture in the positive z direction, D_{AB} is the *mutual diffusion coefficient* of A in B, discussed in the next section, c_A is the molar concentration of A, and dc_A/dz is the concentration gradient of A, which is negative in the direction of ordinary molecular diffusion. Similar definitions apply to (3-3b). The molar fluxes of A and B are in opposite directions. If the gas, liquid, or solid mixture through which diffusion occurs is isotropic, then values of k and D_{AB} are independent of direction. Nonisotropic (anisotropic) materials include fibrous and laminated solids as well as single, noncubic crystals. The diffusion coefficient is also referred to as the *diffusivity* and the mass diffusivity (to distinguish it from thermal and momentum diffusivities).

Many alternative forms of (3-3a) and (3-3b) are used, depending on the choice of driving force or potential in the gradient. For example, we can express (3-3a) as

$$J_A = -cD_{AB} \frac{dx_A}{dz} \quad (3-4)$$

where, for convenience, the z subscript on J has been dropped, c = total molar concentration or molar density ($c = 1/v = \rho/M$), and x_A = mole fraction of species A.

Equation (3-4) can also be written in the following equivalent mass form, where j_A is the mass flux of A by ordinary molecular diffusion relative to the mass-average velocity of the mixture in the positive z -direction, ρ is the mass density, and w_A is the mass fraction of A:

$$j_A = -\rho D_{AB} \frac{dw_A}{dz} \quad (3-5)$$

Velocities in Mass Transfer

It is useful to formulate expressions for velocities of chemical species in the mixture. If these velocities are based on the molar flux, N , and the molar diffusion flux, J , the molar average velocity of the mixture, v_M , relative to stationary coordinates is given for a binary mixture as

$$v_M = \frac{N}{c} = \frac{N_A + N_B}{c} \quad (3-6)$$

Similarly, the velocity of species i , defined in terms of N_i , is relative to stationary coordinates:

$$v_i = \frac{N_i}{c_i} \quad (3-7)$$

Combining (3-6) and (3-7) with $x_i = c_i/c$ gives

$$v_M = x_A v_A + x_B v_B \quad (3-8)$$

Alternatively, species diffusion velocities, v_{iD} , defined in terms of J_i , are relative to the molar-average velocity and are defined as the difference between the species velocity and the molar-average velocity for the mixture:

$$v_{iD} = \frac{J_i}{c_i} = v_i - v_M \quad (3-9)$$

When solving mass-transfer problems involving net movement of the mixture, it is not convenient to use fluxes and flow rates based on v_M as the frame of reference. Rather, it is preferred to use mass-transfer fluxes referred to stationary coordinates with the observer fixed in space. Thus, from (3-9), the total species velocity is

$$v_i = v_M + v_{iD} \quad (3-10)$$

Combining (3-7) and (3-10),

$$N_i = c_i v_M + c_i v_{iD} \quad (3-11)$$

Combining (3-11) with (3-4), (3-6), and (3-7),

$$N_A = \frac{n_A}{A} = x_A N - c D_{AB} \left(\frac{dx_A}{dz} \right) \quad (3-12)$$

and

$$N_B = \frac{n_B}{A} = x_B N - c D_{BA} \left(\frac{dx_B}{dz} \right) \quad (3-13)$$

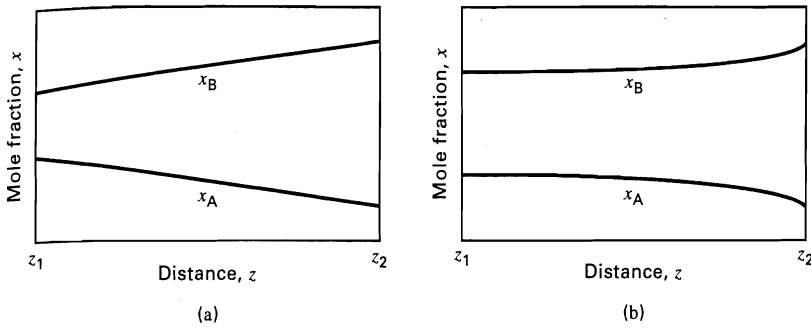


Figure 3.1 Concentration profiles for limiting cases of ordinary molecular diffusion in binary mixtures across a stagnant film: (a) equimolar counterdiffusion (EMD); (b) unimolecular diffusion (UMD).

where in (3-12) and (3-13), n_i is the molar flow rate in moles per unit time, A is the mass-transfer area, the first terms on the right-hand sides are the fluxes resulting from bulk flow, and the second terms on the right-hand sides are the ordinary molecular diffusion fluxes. Two limiting cases are important:

1. Equimolar counterdiffusion (EMD)
2. Unimolecular diffusion (UMD)

Equimolar Counterdiffusion

In equimolar counterdiffusion (EMD), the molar fluxes of A and B in (3-12) and (3-13) are equal but opposite in direction; thus,

$$N = N_A + N_B = 0 \quad (3-14)$$

Thus, from (3-12) and (3-13), the diffusion fluxes are also equal but opposite in direction:

$$J_A = -J_B \quad (3-15)$$

This idealization is closely approached in distillation. From (3-12) and (3-13), we see that in the absence of fluxes other than molecular diffusion,

$$N_A = J_A = -cD_{AB} \left(\frac{dx_A}{dz} \right) \quad (3-16)$$

and

$$N_B = J_B = -cD_{BA} \left(\frac{dx_B}{dz} \right) \quad (3-17)$$

If the total concentration, pressure, and temperature are constant and the mole fractions are maintained constant (but different) at two sides of a stagnant film between z_1 and z_2 , then (3-16) and (3-17) can be integrated from z_1 to any z between z_1 and z_2 to give

$$J_A = \frac{cD_{AB}}{z - z_1} (x_{A1} - x_A) \quad (3-18)$$

and

$$J_B = \frac{cD_{BA}}{z - z_1} (x_{B1} - x_B) \quad (3-19)$$

Thus, in the steady state, the mole fractions are linear in distance, as shown in Figure 3.1a. Furthermore, because c is constant through the film, where

$$c = c_A + c_B \quad (3-20)$$

by differentiation,

$$dc = 0 = dc_A + dc_B \quad (3-21)$$

Thus,

$$dc_A = -dc_B \quad (3-22)$$

From (3-3a), (3-3b), (3-15), and (3-22),

$$\frac{D_{AB}}{dz} = \frac{D_{BA}}{dz} \quad (3-23)$$

Therefore, $D_{AB} = D_{BA}$.

This equality of diffusion coefficients is always true in a binary system of constant molar density.

EXAMPLE 3.1

Two bulbs are connected by a straight tube, 0.001 m in diameter and 0.15 m in length. Initially the bulb at end 1 contains N_2 and the bulb at end 2 contains H_2 . The pressure and temperature are maintained constant at 25°C and 1 atm. At a certain time after allowing diffusion to occur between the two bulbs, the nitrogen content of the gas at end 1 of the tube is 80 mol% and at end 2 is 25 mol%. If the binary diffusion coefficient is 0.784 cm²/s, determine:

- (a) The rates and directions of mass transfer of hydrogen and nitrogen in mol/s
- (b) The species velocities relative to stationary coordinates, in cm/s

SOLUTION

(a) Because the gas system is closed and at constant pressure and temperature, mass transfer in the connecting tube is equimolar counterdiffusion by molecular diffusion.

The area for mass transfer through the tube, in cm², is $A = 3.14(0.1)^2/4 = 7.85 \times 10^{-3}$ cm². The total gas concentration (molar density) is $c = \frac{P}{RT} = \frac{1}{(82.06)(298)} = 4.09 \times 10^{-5}$ mol/cm³. Take the reference plane at end 1 of the connecting tube. Applying (3-18) to

N_2 over the length of the tube,

$$\begin{aligned} n_{N_2} &= \frac{cD_{N_2, H_2}}{z_2 - z_1} [(x_{N_2})_1 - (x_{N_2})_2] A \\ &= \frac{(4.09 \times 10^{-5})(0.784)(0.80 - 0.25)}{15} (7.85 \times 10^{-3}) \\ &= 9.23 \times 10^{-9} \text{ mol/s} \quad \text{in the positive } z\text{-direction} \\ n_{H_2} &= 9.23 \times 10^{-9} \text{ mol/s} \quad \text{in the negative } z\text{-direction} \end{aligned}$$

(b) For equimolar counterdiffusion, the molar-average velocity of the mixture, v_M , is 0. Therefore, from (3-9), species velocities are equal to species diffusion velocities. Thus,

$$\begin{aligned} v_{N_2} = (v_{N_2})_D &= \frac{J_{N_2}}{c_{N_2}} = \frac{n_{N_2}}{Acx_{N_2}} \\ &= \frac{9.23 \times 10^{-9}}{[(7.85 \times 10^{-3})(4.09 \times 10^{-5})x_{N_2}]} \\ &= \frac{0.0287}{x_{N_2}} \quad \text{in the positive } z\text{-direction} \end{aligned}$$

Similarly,

$$v_{H_2} = \frac{0.0287}{x_{H_2}} \quad \text{in the negative } z\text{-direction}$$

Thus, species velocities depend on species mole fractions, as follows:

z , cm	x_{N_2}	x_{H_2}	v_{N_2} , cm/s	v_{H_2} , cm/s
0 (end 1)	0.800	0.200	0.0351	-0.1435
5	0.617	0.383	0.0465	-0.0749
10	0.433	0.567	0.0663	-0.0506
15 (end 2)	0.250	0.750	0.1148	-0.0383

Note that species velocities vary across the length of the connecting tube, but at any location, z , $v_M = 0$. For example, at $z = 10$ cm, from (3-8),

$$v_M = (0.433)(0.0663) + (0.567)(-0.0506) = 0$$

Unimolecular Diffusion

In unimolecular diffusion (UMD), mass transfer of component A occurs through stagnant (nonmoving) component B. Thus,

$$N_B = 0 \quad (3-24)$$

and

$$N = N_A \quad (3-25)$$

Therefore, from (3-12),

$$N_A = x_A N_A - cD_{AB} \frac{dx_A}{dz} \quad (3-26)$$

which can be rearranged to a Fick's-law form,

$$N_A = -\frac{cD_{AB}}{(1-x_A)} \frac{dx_A}{dz} = -\frac{cD_{AB}}{x_B} \frac{dx_A}{dz} \quad (3-27)$$

The factor $(1-x_A)$ accounts for the bulk-flow effect. For a mixture dilute in A, the bulk-flow effect is negligible or small. In mixtures more concentrated in A, the bulk-flow effect can be appreciable. For example, in an equimolar mixture of A and B, $(1-x_A) = 0.5$ and the molar mass-transfer flux of A is twice the ordinary molecular-diffusion flux.

For the stagnant component, B, (3-13) becomes

$$0 = x_B N_A - cD_{BA} \frac{dx_B}{dz} \quad (3-28)$$

or

$$x_B N_A = cD_{BA} \frac{dx_B}{dz} \quad (3-29)$$

Thus, the bulk-flow flux of B is equal but opposite to its diffusion flux.

At quasi-steady-state conditions, that is, with no accumulation, and with constant molar density, (3-27) becomes in integral form:

$$\int_{z_1}^z dz = -\frac{cD_{AB}}{N_A} \int_{x_{A1}}^{x_A} \frac{dx_A}{1-x_A} \quad (3-30)$$

which upon integration yields

$$N_A = \frac{cD_{AB}}{z - z_1} \ln \left(\frac{1-x_A}{1-x_{A1}} \right) \quad (3-31)$$

Rearrangement to give the mole-fraction variation as a function of z yields

$$x_A = 1 - (1-x_{A1}) \exp \left[\frac{N_A(z-z_1)}{cD_{AB}} \right] \quad (3-32)$$

Thus, as shown in Figure 3.1b, the mole fractions are nonlinear in distance.

An alternative and more useful form of (3-31) can be derived from the definition of the log mean. When $z = z_2$, (3-31) becomes

$$N_A = \frac{cD_{AB}}{z_2 - z_1} \ln \left(\frac{1-x_{A2}}{1-x_{A1}} \right) \quad (3-33)$$

The log mean (LM) of $(1-x_A)$ at the two ends of the stagnant layer is

$$\begin{aligned} (1-x_A)_{LM} &= \frac{(1-x_{A2}) - (1-x_{A1})}{\ln[(1-x_{A2})/(1-x_{A1})]} \\ &= \frac{x_{A1} - x_{A2}}{\ln[(1-x_{A2})/(1-x_{A1})]} \end{aligned} \quad (3-34)$$

Combining (3-33) with (3-34) gives

$$\begin{aligned} N_A &= \frac{cD_{AB}}{z_2 - z_1} \frac{(x_{A1} - x_{A2})}{(1-x_A)_{LM}} = \frac{cD_{AB}}{(1-x_A)_{LM}} \frac{(-\Delta x_A)}{\Delta z} \\ &= \frac{cD_{AB}}{(x_B)_{LM}} \frac{(-\Delta x_A)}{\Delta z} \end{aligned} \quad (3-35)$$

EXAMPLE 3.2

As shown in Figure 3.2, an open beaker, 6 cm in height, is filled with liquid benzene at 25°C to within 0.5 cm of the top. A gentle breeze of dry air at 25°C and 1 atm is blown by a fan across the mouth of the beaker so that evaporated benzene is carried away by convection after it transfers through a stagnant air layer in the beaker. The vapor pressure of benzene at 25°C is 0.131 atm. The mutual diffusion coefficient for benzene in air at 25°C and 1 atm is 0.0905 cm²/s. Compute:

- The initial rate of evaporation of benzene as a molar flux in mol/cm²-s
- The initial mole-fraction profiles in the stagnant air layer
- The initial fractions of the mass-transfer fluxes due to molecular diffusion
- The initial diffusion velocities, and the species velocities (relative to stationary coordinates) in the stagnant layer
- The time in hours for the benzene level in the beaker to drop 2 cm from the initial level, if the specific gravity of liquid benzene is 0.874. Neglect the accumulation of benzene and air in the stagnant layer as it increases in height

SOLUTION

Let A = benzene, B = air.

$$c = \frac{P}{RT} = \frac{1}{(82.06)(298)} = 4.09 \times 10^{-5} \text{ mol/cm}^3$$

- (a) Take $z_1 = 0$. Then $z_2 - z_1 = \Delta z = 0.5$ cm. From Dalton's law, assuming equilibrium at the liquid benzene-air interface,

$$x_{A1} = \frac{p_{A1}}{P} = \frac{0.131}{1} = 0.131 \quad x_{A2} = 0$$

$$(1 - x_A)_{LM} = \frac{0.131}{\ln[(1 - 0)/(1 - 0.131)]} = 0.933 = (x_B)_{LM}$$

From (3-35),

$$N_A = \frac{(4.09 \times 10^{-5})(0.0905)}{0.5} \left(\frac{0.131}{0.933} \right) = 1.04 \times 10^{-6} \text{ mol/cm}^2\text{-s}$$

$$(b) \frac{N_A(z - z_1)}{cD_{AB}} = \frac{(1.04 \times 10^{-6})(z - 0)}{(4.09 \times 10^{-5})(0.0905)} = 0.281 z$$

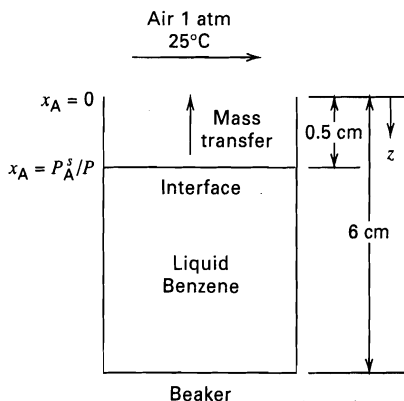


Figure 3.2 Evaporation of benzene from a beaker—Example 3.2.

From (3-32),

$$x_A = 1 - 0.869 \exp(0.281 z) \quad (1)$$

Using (1), the following results are obtained:

z, cm	x_A	x_B
0.0	0.1310	0.8690
0.1	0.1060	0.8940
0.2	0.0808	0.9192
0.3	0.0546	0.9454
0.4	0.0276	0.9724
0.5	0.0000	1.0000

These profiles are only slightly curved.

- (c) From (3-27) and (3-29), we can compute the bulk flow terms, $x_A N_A$ and $x_B N_A$, from which the molecular diffusion terms are obtained.

z, cm	$x_i N$ Bulk-Flow Flux, mol/cm ² -s $\times 10^6$		J_i Molecular-Diffusion Flux, mol/cm ² -s $\times 10^6$	
	A	B	A	B
0.0	0.1360	0.9040	0.9040	-0.9040
0.1	0.1100	0.9300	0.9300	-0.9300
0.2	0.0840	0.9560	0.9560	-0.9560
0.3	0.0568	0.9832	0.9832	-0.9832
0.4	0.0287	1.0113	1.0113	-1.0113
0.5	0.0000	1.0400	1.0400	-1.0400

Note that the molecular-diffusion fluxes are equal but opposite, and the bulk-flow flux of B is equal but opposite to its molecular-diffusion flux, so that its molar flux, N_B , is zero, making B (air) stagnant.

- (d) From (3-6),

$$v_M = \frac{N}{c} = \frac{N_A}{c} = \frac{1.04 \times 10^{-6}}{4.09 \times 10^{-5}} = 0.0254 \text{ cm/s} \quad (2)$$

From (3-9), the diffusion velocities are given by

$$v_{id} = \frac{J_i}{c_i} = \frac{J_i}{x_i c} \quad (3)$$

From (3-10), the species velocities relative to stationary coordinates are

$$v_i = v_{id} + v_M \quad (4)$$

Using (2) to (4), we obtain

z, cm	v_{id} Molecular-Diffusion Velocity, cm/s		J_i Species Velocity, cm/s	
	A	B	A	B
0.0	0.1687	-0.0254	0.1941	0
0.1	0.2145	-0.0254	0.2171	0
0.2	0.2893	-0.0254	0.3147	0
0.3	0.4403	-0.0254	0.4657	0
0.4	0.8959	-0.0254	0.9213	0
0.5	∞	-0.0254	∞	0

Note that v_B is zero everywhere, because its molecular-diffusion velocity is negated by the molar-mean velocity.

(e) The mass-transfer flux for benzene evaporation can be equated to the rate of decrease in the moles of liquid benzene per unit cross section of the beaker. Letting z = distance down from the mouth of the beaker and using (3-35) with $\Delta z = z$,

$$N_A = \frac{cD_{AB}}{z} \frac{(-\Delta x_A)}{(1-x_A)_{LM}} = \frac{\rho_L}{M_L} \frac{dz}{dt} \quad (5)$$

Separating variables and integrating,

$$\int_0^t dt = t = \frac{\rho_L(1-x_A)_{LM}}{M_L c D_{AB}(-\Delta x_A)} \int_{z_1}^{z_2} z dz \quad (6)$$

The coefficient of the integral on the right-hand side of (6) is constant at

$$\frac{0.874(0.933)}{78.11(4.09 \times 10^{-5})(0.0905)(0.131)} = 21,530 \text{ s/cm}^2$$

$$\int_{z_1}^{z_2} z dz = \int_{0.5}^{2.5} z dz = 3 \text{ cm}^2$$

From (6), $t = 21,530(3) = 64,590 \text{ s}$ or 17.94 h , which is a long time because of the absence of turbulence.

3.2 DIFFUSION COEFFICIENTS

Diffusivities or diffusion coefficients are defined for a binary mixture by (3-3) to (3-5). Measurement of diffusion coefficients must involve a correction for any bulk flow using (3-12) and (3-13) with the reference plane being such that there is no net molar bulk flow.

The binary diffusivities, D_{AB} and D_{BA} , are mutual or binary diffusion coefficients. Other coefficients include D_{iM} , the diffusivity of i in a multicomponent mixture; D_{ii} , the self-diffusion coefficient; and the tracer or interdiffusion coefficient. In this chapter, and throughout this book, the focus is on the mutual diffusion coefficient, which will be referred to as the diffusivity or diffusion coefficient.

Diffusivity in Gas Mixtures

As discussed by Poling, Prausnitz, and O'Connell [2], a number of theoretical and empirical equations are available for estimating the value of $D_{AB} = D_{BA}$ in gases at low to moderate pressures. The theoretical equations, based on Boltzmann's kinetic theory of gases, the theorem of corresponding states, and a suitable intermolecular energy-potential function, as developed by Chapman and Enskog, predict D_{AB} to be inversely proportional to pressure and almost independent of composition, with a significant increase for increasing temperature. Of greater accuracy and ease of use is the following empirical equation of Fuller, Schettler, and Giddings [3], which retains the form of the Chapman-Enskog theory but utilizes empirical constants

Table 3.1 Diffusion Volumes from Fuller, Ensley, and Giddings [*J. Phys. Chem.*, **73**, 3679-3685 (1969)] for Estimating Binary Gas Diffusivity by the Method of Fuller et al. [3]

Atomic Diffusion Volumes Atomic and Structural Diffusion-Volume Increments			
C	15.9	F	14.7
H	2.31	Cl	21.0
O	6.11	Br	21.9
N	4.54	I	29.8
Aromatic ring	-18.3	S	22.9
Heterocyclic ring	-18.3		
Diffusion Volumes of Simple Molecules			
He	2.67	CO	18.0
Ne	5.98	CO ₂	26.7
Ar	16.2	N ₂ O	35.9
Kr	24.5	NH ₃	20.7
Xe	32.7	H ₂ O	13.1
H ₂	6.12	SF ₆	71.3
D ₂	6.84	Cl ₂	38.4
N ₂	18.5	Br ₂	69.0
O ₂	16.3	SO ₂	41.8
Air	19.7		

derived from experimental data:

$$D_{AB} = D_{BA} = \frac{0.00143T^{1.75}}{PM_{AB}^{1/2}[(\sum_V)_A^{1/3} + (\sum_V)_B^{1/3}]^2} \quad (3-36)$$

where D_{AB} is in cm^2/s , P is in atm, T is in K,

$$M_{AB} = \frac{2}{(1/M_A) + (1/M_B)} \quad (3-37)$$

and $\sum_V$ = summation of atomic and structural diffusion volumes from Table 3.1, which includes diffusion volumes of some simple molecules.

Experimental values of binary gas diffusivity at 1 atm and near-ambient temperature range from about 0.10 to $10.0 \text{ cm}^2/\text{s}$. Poling, et al. [2] compared (3-36) to experimental data for 51 different binary gas mixtures at low pressures over a temperature range of 195 – $1,068 \text{ K}$. The average deviation was only 5.4% , with a maximum deviation of 25% . Only 9 of 69 estimated values deviated from experimental values by more than 10% . When an experimental diffusivity is available at values of T and P that are different from the desired conditions, (3-36) indicates that D_{AB} is proportional to $T^{1.75}/P$, which can be used to obtain the desired value. Some representative experimental values of binary gas diffusivity are given in Table 3.2.

Table 3.2 Experimental Binary Diffusivities of Some Gas Pairs at 1 atm

Gas pair, A-B	Temperature, K	D_{AB} , cm ² /s
Air—carbon dioxide	317.2	0.177
Air—ethanol	313	0.145
Air—helium	317.2	0.765
Air— <i>n</i> -hexane	328	0.093
Air—water	313	0.288
Argon—ammonia	333	0.253
Argon—hydrogen	242.2	0.562
Argon—hydrogen	806	4.86
Argon—methane	298	0.202
Carbon dioxide—nitrogen	298	0.167
Carbon dioxide—oxygen	293.2	0.153
Carbon dioxide—water	307.2	0.198
Carbon monoxide—nitrogen	373	0.318
Helium—benzene	423	0.610
Helium—methane	298	0.675
Helium—methanol	423	1.032
Helium—water	307.1	0.902
Hydrogen—ammonia	298	0.783
Hydrogen—ammonia	533	2.149
Hydrogen—cyclohexane	288.6	0.319
Hydrogen—methane	288	0.694
Hydrogen—nitrogen	298	0.784
Nitrogen—benzene	311.3	0.102
Nitrogen—cyclohexane	288.6	0.0731
Nitrogen—sulfur dioxide	263	0.104
Nitrogen—water	352.1	0.256
Oxygen—benzene	311.3	0.101
Oxygen—carbon tetrachloride	296	0.0749
Oxygen—cyclohexane	288.6	0.0746
Oxygen—water	352.3	0.352

From Marrero, T. R., and E. A. Mason, *J. Phys. Chem. Ref. Data*, **1**, 3–118 (1972).

EXAMPLE 3.3

Estimate the diffusion coefficient for the system oxygen (A)/benzene (B) at 38°C and 2 atm using the method of Fuller et al.

SOLUTION

From (3-37),

$$M_{AB} = \frac{2}{(1/32) + (1/78.11)} = 45.4$$

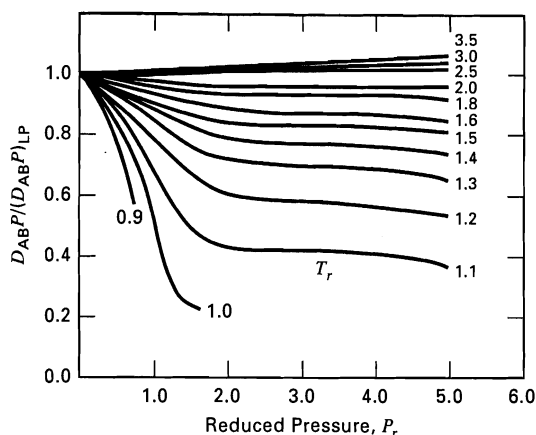
From Table 3.1, $(\sum V)_A = 16.3$ and $(\sum V)_B = 6(15.9) + 6(2.31) = 18.3 = 90.96$

From (3-36), at 2 atm and 311.2 K,

$$D_{AB} = D_{BA} = \frac{0.00143(311.2)^{1.75}}{(2)(45.4)^{1/2}[16.3^{1/3} + 90.96^{1/3}]^2} = 0.0495 \text{ cm}^2/\text{s}$$

At 1 atm, the predicted diffusivity is 0.0990 cm²/s, which is about 2% below the experimental value of 0.101 cm²/s in Table 3.2. The experimental value for 38°C can be extrapolated by the temperature dependency of (3-36) to give the following prediction at 200°C:

$$D_{AB} \text{ at } 200^\circ\text{C and 1 atm} = 0.102 \left(\frac{200 + 273.2}{38 + 273.2} \right)^{1.75} = 0.212 \text{ cm}^2/\text{s}$$

**Figure 3.3** Takahashi [4] correlation for effect of high pressure on binary gas diffusivity.

For binary mixtures of light gases, at pressures to about 10 atm, the pressure dependence on diffusivity is adequately predicted by the simple inverse relation (3-36), that is, PD_{AB} = a constant for a given temperature and gas mixture. At higher pressures, deviations from this relation are handled in a manner somewhat similar to the modification of the ideal-gas law by the compressibility factor based on the theorem of corresponding states. Although few reliable experimental data are available at high pressure, Takahashi [4] has published a tentative corresponding-states correlation, shown in Figure 3.3, patterned after an earlier correlation for self-diffusivities by Slattery [5]. In the Takahashi plot, $D_{AB}P/(D_{AB}P)_{LP}$ is given as a function of reduced temperature and pressure, where $(D_{AB}P)_{LP}$ is at low pressure where (3-36) applies. Mixture-critical temperature and pressure are molar-average values. Thus, a finite effect of composition is predicted at high pressure. The effect of high pressure on diffusivity is important in supercritical extraction, discussed in Chapter 11.

EXAMPLE 3.4

Estimate the diffusion coefficient for a 25/75 molar mixture of argon and xenon at 200 atm and 378 K. At this temperature and 1 atm, the diffusion coefficient is 0.180 cm²/s. Critical constants are

	T_c , K	P_c , atm
Argon	151.0	48.0
Xenon	289.8	58.0

SOLUTION

Calculate reduced conditions:

$$T_c = 0.25(151) + 0.75(289.8) = 255.1 \text{ K};$$

$$T_r = T/T_c = 378/255.1 = 1.48$$

$$P_c = 0.25(48) + 0.75(58) = 55.5;$$

$$P_r = P/P_c = 200/55.5 = 3.6$$

From Figure 3.3, $\frac{D_{AB}P}{(D_{AB}P)_{LP}} = 0.82$

$$D_{AB} = \frac{(D_{AB}P)_{LP}}{P} \left[\frac{D_{AB}P}{(D_{AB}P)_{LP}} \right] = \frac{(0.180)(1)}{200} (0.82) = 7.38 \times 10^{-4} \text{ cm}^2/\text{s}$$

Diffusivity in Liquid Mixtures

Diffusion coefficients in binary liquid mixtures are difficult to estimate because of the lack of a rigorous model for the liquid state. An exception is the case of a dilute solute (A) of very large, rigid, spherical molecules diffusing through a stationary solvent (B) of small molecules with no slip of the solvent at the surface of the solute molecules. The resulting relation, based on the hydrodynamics of creeping flow to describe drag, is the Stokes–Einstein equation:

$$(D_{AB})_{\infty} = \frac{RT}{6\pi\mu_B R_A N_A} \quad (3-38)$$

where R_A is the radius of the solute molecule and N_A is Avagadro's number. Although (3-38) is very limited in its application to liquid mixtures, it has long served as a starting point for more widely applicable empirical correlations for the diffusivity of solute (A) in solvent (B), where both A and B are of the same approximate molecular size. Unfortunately, unlike the situation in binary gas mixtures, $D_{AB} = D_{BA}$ in binary liquid mixtures can vary greatly with composition as shown in Example 3.7. Because the Stokes–Einstein equation does not provide a basis for extending

dilute conditions to more concentrated conditions, extensions of (3-38) have been restricted to binary liquid mixtures dilute in A, up to 5 and perhaps 10 mol%.

One such extension, which gives reasonably good predictions for small solute molecules, is the empirical Wilke–Chang [6] equation:

$$(D_{AB})_{\infty} = \frac{7.4 \times 10^{-8} (\phi_B M_B)^{1/2} T}{\mu_B v_A^{0.6}} \quad (3-39)$$

where the units are cm^2/s for D_{AB} ; cP (centipoises) for the solvent viscosity, μ_B ; K for T ; and cm^3/mol for v_A , the liquid molar volume of the solute at its normal boiling point. The parameter ϕ_B is an association factor for the solvent, which is 2.6 for water, 1.9 for methanol, 1.5 for ethanol, and 1.0 for unassociated solvents such as hydrocarbons. Note that the effects of temperature and viscosity are identical to the prediction of the Stokes–Einstein equation, while the effect of the radius of the solute molecule is replaced by v_A , which can be estimated by summing the atomic contributions in Table 3.3, which also lists values of v_A for dissolved light gases. Some representative experimental values of diffusivity in dilute binary liquid solutions are given in Table 3.4.

Table 3.3 Molecular Volumes of Dissolved Light Gases and Atomic Contributions for Other Molecules at the Normal Boiling Point

	Atomic Volume (m^3/kmol) $\times 10^3$		Atomic Volume (m^3/kmol) $\times 10^3$
C	14.8	Ring	
H	3.7	Three-membered, as in ethylene oxide	−6
O (except as below)	7.4	Four-membered	−8.5
Doubly bonded as carbonyl	7.4	Five-membered	−11.5
Coupled to two other elements:		Six-membered	−15
In aldehydes, ketones	7.4	Naphthalene ring	−30
In methyl esters	9.1	Anthracene ring	−47.5
In methyl ethers	9.9		
In ethyl esters	9.9		
In ethyl ethers	9.9		
In higher esters	11.0		
In higher ethers	11.0		
In acids (—OH)	12.0		
Joined to S, P, N	8.3		
N			
Doubly bonded	15.6	Air	29.9
In primary amines	10.5	O ₂	25.6
In secondary amines	12.0	N ₂	31.2
Br	27.0	Br ₂	53.2
Cl in RCHClR'	24.6	Cl ₂	48.4
Cl in RCl (terminal)	21.6	CO	30.7
F	8.7	CO ₂	34.0
I	37.0	H ₂	14.3
S	25.6	H ₂ O	18.8
P	27.0	H ₂ S	32.9
		NH ₃	25.8
		NO	23.6
		N ₂ O	36.4
		SO ₂	44.8

Source: G. Le Bas, *The Molecular Volumes of Liquid Chemical Compounds*, David McKay, New York (1915).

Table 3.4 Experimental Binary Liquid Diffusivities for Solutes, A, at Low Concentrations in Solvents, B

Solvent, B	Solute, A	Temperature, K	Diffusivity, D_{AB} , $\text{cm}^2/\text{s} \times 10^5$
Water	Acetic acid	293	1.19
	Aniline	293	0.92
	Carbon dioxide	298	2.00
	Ethanol	288	1.00
	Methanol	288	1.26
Ethanol	Allyl alcohol	293	0.98
	Benzene	298	1.81
	Oxygen	303	2.64
	Pyridine	293	1.10
	Water	298	1.24
Benzene	Acetic acid	298	2.09
	Cyclohexane	298	2.09
	Ethanol	288	2.25
	<i>n</i> -Heptane	298	2.10
	Toluene	298	1.85
<i>n</i> -Hexane	Carbon tetrachloride	298	3.70
	Methyl ethyl ketone	303	3.74
	Propane	298	4.87
	Toluene	298	4.21
Acetone	Acetic acid	288	2.92
	Formic acid	298	3.77
	Nitrobenzene	293	2.94
	Water	298	4.56

From Poling et al. [2].

EXAMPLE 3.5

Use the Wilke–Chang equation to estimate the diffusivity of aniline (A) in a 0.5 mol% aqueous solution at 20°C. At this temperature, the solubility of aniline in water is about 4 g/100 g of water or 0.77 mol% aniline. The experimental diffusivity value for an infinitely dilute mixture is $0.92 \times 10^{-5} \text{ cm}^2/\text{s}$.

SOLUTION

$$\mu_B = \mu_{\text{H}_2\text{O}} = 1.01 \text{ cP at } 20^\circ\text{C}$$

$$v_A = \text{liquid molar volume of aniline at its normal boiling point of } 457.6 \text{ K} = 107 \text{ cm}^3/\text{mol}$$

$$\phi_B = 2.6 \text{ for water} \quad M_B = 18 \text{ for water} \quad T = 293 \text{ K}$$

From (3-39),

$$D_{AB} = \frac{(7.4 \times 10^{-8})[2.6(18)]^{0.5}(293)}{1.01(107)^{0.6}} = 0.89 \times 10^{-5} \text{ cm}^2/\text{s}$$

This value is about 3% less than the experimental value for an infinitely dilute solution of aniline in water.

More recent liquid diffusivity correlations due to Hayduk and Minhas [7] give better agreement than the Wilke–Chang

equation with experimental values for nonaqueous solutions. For a dilute solution of one normal paraffin (C_5 to C_{32}) in another (C_5 to C_{16}),

$$(D_{AB})_\infty = 13.3 \times 10^{-8} \frac{T^{1.47} \mu_B^\epsilon}{v_A^{0.71}} \quad (3-40)$$

where

$$\epsilon = \frac{10.2}{v_A} - 0.791 \quad (3-41)$$

and the other variables have the same units as in (3-39).

For general nonaqueous solutions,

$$(D_{AB})_\infty = 1.55 \times 10^{-8} \frac{T^{1.29} (\mathcal{P}_B^{0.5} / \mathcal{P}_A^{0.42})}{\mu_B^{0.92} v_B^{0.23}} \quad (3-42)$$

where $\mathcal{P}$ is the parachor, which is defined as

$$\mathcal{P} = v\sigma^{1/4} \quad (3-43)$$

When the units of the liquid molar volume, v , are cm^3/mol and the surface tension, σ , are g/s^2 (dynes/cm), then the units of the parachor are $\text{cm}^3\text{-g}^{1/4}/\text{s}^{1/2}\text{-mol}$. Normally, at near-ambient conditions, $\mathcal{P}$ is treated as a constant, for which an extensive tabulation is available from Quayle [8], who also provides a group-contribution method for estimating parachors for compounds not listed. Table 3.5 gives values of parachors for a number of compounds, while Table 3.6 contains structural contributions for predicting the parachor in the absence of data.

The following restrictions apply to (3-42):

1. Solvent viscosity should not exceed 30 cP.
2. For organic acid solutes and solvents other than water, methanol, and butanols, the acid should be treated as a dimer by doubling the values of $\mathcal{P}_A$ and v_A .
3. For a nonpolar solute in monohydroxy alcohols, values of v_B and $\mathcal{P}_B$ should be multiplied by $8\mu_B$, where the viscosity is in centipoise.

Liquid diffusion coefficients for a solute in a dilute binary system range from about 10^{-6} to $10^{-4} \text{ cm}^2/\text{s}$ for solutes of molecular weight up to about 200 and solvents with viscosity up to about 10 cP. Thus, liquid diffusivities are five orders of magnitude less than diffusivities for binary gas mixtures at 1 atm. However, diffusion rates in liquids are not necessarily five orders of magnitude lower than in gases because, as seen in (3-5), the product of the concentration (molar density) and the diffusivity determines the rate of diffusion for a given concentration gradient in mole fraction. At 1 atm, the molar density of a liquid is three times that of a gas and, thus, the diffusion rate in liquids is only two orders of magnitude lower than in gases at 1 atm.

Table 3.5 Parachors for Representative Compounds

	Parachor, $\text{cm}^3\text{-g}^{1/4}/\text{s}^{1/2}\text{-mol}$		Parachor, $\text{cm}^3\text{-g}^{1/4}/\text{s}^{1/2}\text{-mol}$		Parachor, $\text{cm}^3\text{-g}^{1/4}/\text{s}^{1/2}\text{-mol}$
Acetic acid	131.2	Chlorobenzene	244.5	Methyl amine	95.9
Acetone	161.5	Diphenyl	380.0	Methyl formate	138.6
Acetonitrile	122	Ethane	110.8	Naphthalene	312.5
Acetylene	88.6	Ethylene	99.5	<i>n</i> -Octane	350.3
Aniline	234.4	Ethyl butyrate	295.1	1-Pentene	218.2
Benzene	205.3	Ethyl ether	211.7	1-Pentyne	207.0
Benzonitrile	258	Ethyl mercaptan	162.9	Phenol	221.3
<i>n</i> -Butyric acid	209.1	Formic acid	93.7	<i>n</i> -Propanol	165.4
Carbon disulfide	143.6	Isobutyl benzene	365.4	Toluene	245.5
Cyclohexane	239.3	Methanol	88.8	Triethyl amine	297.8

Source: Meissner, *Chem. Eng. Prog.*, **45**, 149–153 (1949).

Table 3.6 Structural Contributions for Estimating the Parachor

Carbon–hydrogen:		R—[—CO—]—R' (ketone)	
C	9.0	R + R' = 2	51.3
H	15.5	R + R' = 3	49.0
CH ₃	55.5	R + R' = 4	47.5
CH ₂ in —(CH ₂) _n		R + R' = 5	46.3
<i>n</i> < 12	40.0	R + R' = 6	45.3
<i>n</i> > 12	40.3	R + R' = 7	44.1
		—CHO	66
Alkyl groups			
1-Methylethyl	133.3	O (not noted above)	20
1-Methylpropyl	171.9	N (not noted above)	17.5
1-Methylbutyl	211.7	S	49.1
2-Methylpropyl	173.3	P	40.5
1-Ethylpropyl	209.5	F	26.1
1,1-Dimethylethyl	170.4	Cl	55.2
1,1-Dimethylpropyl	207.5	Br	68.0
1,2-Dimethylpropyl	207.9	I	90.3
1,1,2-Trimethylpropyl	243.5	Ethylenic bonds:	
C ₆ H ₅	189.6	Terminal	19.1
		2,3-position	17.7
		3,4-position	16.3
Special groups:			
—COO—	63.8	Triple bond	40.6
—COOH	73.8		
—OH	29.8		
—NH ₂	42.5	Ring closure:	
—O—	20.0	Three-membered	12
—NO ₂	74	Four-membered	6.0
—NO ₃ (nitrate)	93	Five-membered	3.0
—CO(NH ₂)	91.7	Six-membered	0.8

Source: Quale [8].

EXAMPLE 3.6

Estimate the diffusivity of formic acid (A) in benzene (B) at 25°C and infinite dilution, using the appropriate correlation of Hayduk and Minhas [7]. The experimental value is $2.28 \times 10^{-5} \text{ cm}^2/\text{s}$.

SOLUTION

Equation (3-42) applies, with $T = 298 \text{ K}$

$$\mathcal{P}_A = 93.7 \text{ cm}^3\text{-g}^{1/4}/\text{s}^{1/2}\text{-mol} \quad \mathcal{P}_B = 205.3 \text{ cm}^3\text{-g}^{1/4}/\text{s}^{1/2}\text{-mol}$$

$$\mu_B = 0.6 \text{ cP at } 25^\circ\text{C} \quad \nu_B = 96 \text{ cm}^3/\text{mol at } 80^\circ\text{C}$$

However, because formic acid is an organic acid, $\mathcal{P}_A$ is doubled to 187.4.

From (3-42),

$$(D_{AB})_\infty = 1.55 \times 10^{-8} \left[\frac{298^{1.29}(205.3^{0.5}/187.4^{0.42})}{0.6^{0.92}96^{0.23}} \right]$$

$$= 2.15 \times 10^{-5} \text{ cm}^2/\text{s}$$

which is within 6% of the experimental value.

The Stokes–Einstein and Wilke–Chang equations predict an inverse dependence of liquid diffusivity with viscosity. The Hayduk–Minhas equations predict a somewhat smaller dependence on viscosity. From data covering several orders of magnitude variation of viscosity, the liquid diffusivity is found to vary inversely with the viscosity raised to an exponent closer to 0.5 than to 1.0. The Stokes–Einstein and Wilke–Chang equations also predict that $D_{AB}\mu_B/T$ is a constant over a narrow temperature range. Because μ_B decreases exponentially with temperature, D_{AB} is predicted to increase exponentially with temperature. For example, for a dilute solution of water in ethanol, the diffusivity of water increases by a factor of almost 20 when the absolute temperature is increased 50%. Over a wide temperature range, it is preferable to express the effect of temperature on D_{AB} by an Arrhenius-type expression,

$$(D_{AB})_{\infty} = A \exp\left(\frac{-E}{RT}\right) \quad (3-44)$$

where, typically the activation energy for liquid diffusion, E , is no greater than 6,000 cal/mol.

Equations (3-39), (3-40), and (3-42) for estimating diffusivity in binary liquid mixtures only apply to the solute, A, in a dilute solution of the solvent, B. Unlike binary gas mixtures in which the diffusivity is almost independent of composition, the effect of composition on liquid diffusivity is complex, sometimes showing strong positive or negative deviations from linearity with mole fraction.

Based on a nonideal form of Fick's law, Vignes [9] has shown that, except for strongly associated binary mixtures such as chloroform/acetone, which exhibit a rare negative deviation from Raoult's law, infinite-dilution binary diffusivities, $(D)_{\infty}$, can be combined with mixture activity-coefficient data or correlations thereof to predict liquid binary diffusion coefficients D_{AB} and D_{BA} over the entire composition range. The Vignes equations are:

$$D_{AB} = (D_{AB})_{\infty}^{x_B} (D_{BA})_{\infty}^{x_A} \left(1 + \frac{\partial \ln \gamma_A}{\partial \ln x_A}\right)_{T,P} \quad (3-45)$$

$$D_{BA} = (D_{BA})_{\infty}^{x_A} (D_{AB})_{\infty}^{x_B} \left(1 + \frac{\partial \ln \gamma_B}{\partial \ln x_B}\right)_{T,P} \quad (3-46)$$

EXAMPLE 3.7

At 298 K and 1 atm, infinite-dilution diffusion coefficients for the methanol (A)/water (B) system are 1.5×10^{-5} cm²/s and 1.75×10^{-5} cm²/s for AB and BA, respectively.

Activity-coefficient data for the same conditions as estimated from the UNIFAC method are as follows:

x_A	γ_A	x_B	γ_B
0.0	2.245	1.0	1.000
0.1	1.748	0.9	1.013
0.2	1.470	0.8	1.044
0.3	1.300	0.7	1.087
0.4	1.189	0.6	1.140

x_A	γ_A	x_B	γ_B
0.5	1.116	0.5	1.201
0.6	1.066	0.4	1.269
0.7	1.034	0.3	1.343
0.8	1.014	0.2	1.424
0.9	1.003	0.1	1.511
1.0	1.000	0.0	1.605

Use the Vignes equations to estimate diffusion coefficients over the entire composition range.

SOLUTION

Using a spreadsheet to compute the derivatives in (3-45) and (3-46), which are found to be essentially equal at any composition, and the diffusivities from the same equations, the following results are obtained with $D_{AB} = D_{BA}$ at each composition. The calculations show a minimum diffusivity at a methanol mole fraction of 0.30.

x_A	$D_{AB}, \text{cm}^2/\text{s}$	$D_{BA}, \text{cm}^2/\text{s}$
0.20	1.10×10^{-5}	1.10×10^{-5}
0.30	1.08×10^{-5}	1.08×10^{-5}
0.40	1.12×10^{-5}	1.12×10^{-5}
0.50	1.18×10^{-5}	1.18×10^{-5}
0.60	1.28×10^{-5}	1.28×10^{-5}
0.70	1.38×10^{-5}	1.38×10^{-5}
0.80	1.50×10^{-5}	1.50×10^{-5}

If the diffusivity is assumed linear with mole fraction, the value at $x_A = 0.50$ is 1.625×10^{-5} , which is almost 40% higher than the predicted value of 1.18×10^{-5} .

Diffusivities of Electrolytes

In an electrolyte solute, the diffusion coefficient of the dissolved salt, acid, or base depends on the ions, since they are the diffusing entities. However, in the absence of an electric potential, only the molecular diffusion of the electrolyte molecule is of interest. The infinite-dilution diffusivity of a single salt in an aqueous solution in cm²/s can be estimated from the Nernst–Haskell equation:

$$(D_{AB})_{\infty} = \frac{RT[(1/n_+) + (1/n_-)]}{F^2[(1/\lambda_+) + (1/\lambda_-)]} \quad (3-47)$$

where

n_+ and n_- = valences of the cation and anion, respectively

λ_+ and λ_- = limiting ionic conductances in (A/cm²) (V/cm)(g-equiv/cm³), where A = amps and V = volts

F = Faraday's constant
= 96,500 coulombs/g-equiv

T = temperature, K

R = gas constant = 8.314 J/mol-K

Values of λ_+ and λ_- at 25°C are listed in Table 3.7. At other temperatures, these values are multiplied by $T/334\mu_B$,

Table 3.7 Limiting Ionic Conductances in Water at 25°C, in (A/cm²)(V/cm)(g-equiv/cm³)

Anion	λ_-	Cation	λ_+
OH ⁻	197.6	H ⁺	349.8
Cl ⁻	76.3	Li ⁺	38.7
Br ⁻	78.3	Na ⁺	50.1
I ⁻	76.8	K ⁺	73.5
NO ₃ ⁻	71.4	NH ₄ ⁺	73.4
ClO ₄ ⁻	68.0	Ag ⁺	61.9
HCO ₃ ⁻	44.5	Tl ⁺	74.7
HCO ₂ ⁻	54.6	($\frac{1}{2}$)Mg ²⁺	53.1
CH ₃ CO ₂ ⁻	40.9	($\frac{1}{2}$)Ca ²⁺	59.5
ClCH ₂ CO ₂ ⁻	39.8	($\frac{1}{2}$)Sr ²⁺	50.5
CNCH ₂ CO ₂ ⁻	41.8	($\frac{1}{2}$)Ba ²⁺	63.6
CH ₃ CH ₂ CO ₂ ⁻	35.8	($\frac{1}{2}$)Cu ²⁺	54
CH ₃ (CH ₂) ₂ CO ₂ ⁻	32.6	($\frac{1}{2}$)Zn ²⁺	53
C ₆ H ₅ CO ₂ ⁻	32.3	($\frac{1}{3}$)La ³⁺	69.5
HC ₂ O ₄ ⁻	40.2	($\frac{1}{3}$)Co(NH ₃) ₆ ³⁺	102
($\frac{1}{2}$)C ₂ O ₄ ²⁻	74.2		
($\frac{1}{2}$)SO ₄ ²⁻	80		
($\frac{1}{3}$)Fe(CN) ₆ ³⁻	101		
($\frac{1}{4}$)Fe(CN) ₆ ⁴⁻	111		

Source: Poling, Prausnitz, and O'Connell [2].

where T and μ_B are in kelvins and centipoise, respectively. As the concentration of the electrolyte increases, the diffusivity at first decreases rapidly by about 10% to 20% and then rises to values at a concentration of 2 N (normal) that approximate the infinite-dilution value. Some representative experimental values from Volume V of the International Critical Tables are given in Table 3.8.

Table 3.8 Experimental Diffusivities of Electrolytes in Aqueous Solutions

Solute	Concentration, Mol/L	Temperature, °C	Diffusivity, D_{AB} , cm ² /s $\times 10^5$
HCl	0.1	12	2.29
HNO ₃	0.05	20	2.62
	0.25	20	2.59
H ₂ SO ₄	0.25	20	1.63
KOH	0.01	18	2.20
	0.1	18	2.15
	1.8	18	2.19
NaOH	0.05	15	1.49
NaCl	0.4	18	1.17
	0.8	18	1.19
	2.0	18	1.23
KCl	0.4	18	1.46
	0.8	18	1.49
	2.0	18	1.58
MgSO ₄	0.4	10	0.39
Ca(NO ₃) ₂	0.14	14	0.85

EXAMPLE 3.8

Estimate the diffusivity of KCl in a dilute solution of water at 18.5°C. The experimental value is 1.7×10^{-5} cm²/s. At concentrations up to 2N, this value varies only from 1.5×10^{-5} to 1.75×10^{-5} cm²/s.

SOLUTION

At 18.5°C, $T/334\mu = 291.7/[(334)(1.05)] = 0.832$. Using Table 3.7, at 25°C, the corrected limiting ionic conductances are

$$\lambda_+ = 73.5(0.832) = 61.2 \quad \text{and} \quad \lambda_- = 76.3(0.832) = 63.5$$

From (3-47),

$$D_\infty = \frac{(8.314)(291.7)[(1/1) + (1/1)]}{96,500^2[(1/61.2) + (1/63.5)]} = 1.62 \times 10^{-5} \text{ cm}^2/\text{s}$$

which is close to the experimental value.

Diffusivity of Biological Solutes in Liquids

For dilute, aqueous, nonelectrolyte solutions, the Wilke–Chang equation (3-39) can be used for small solute molecules of liquid molar volumes up to 500 cm³/mol, which corresponds to molecular weights to almost 600. In biological applications, diffusivities of water-soluble protein macromolecules having molecular weights greater than 1,000 are of interest. In general, molecules with molecular weights to 500,000 have diffusivities at 25°C that range from 1×10^{-7} to 8×10^{-7} cm²/s, which is two orders of magnitude smaller than values of diffusivity for molecules with molecular weights less than 1,000. Data for many globular and fibrous protein macromolecules are tabulated by Sorber [10] with a few diffusivities given in Table 3.9. In the absence of data, the following semiempirical equation given by Geankoplis [11] and patterned after the Stokes–Einstein equation can be used:

$$D_{AB} = \frac{9.4 \times 10^{-15} T}{\mu_B (M_A)^{1/3}} \quad (3-48)$$

where the units are those of (3-39).

Also of interest in biological applications are diffusivities of small, nonelectrolyte molecules in aqueous gels containing up to 10 wt% of molecules such as certain polysaccharides (agar), which have a great tendency to swell. Diffusivities are given by Friedman and Kraemer [12]. In general, the diffusivities of small solute molecules in gels are not less than 50% of the values for the diffusivity of the solute in water, with values decreasing with increasing weight percent of gel.

Diffusivity in Solids

Diffusion in solids takes place by different mechanisms depending on the diffusing atom, molecule, or ion; the nature of the solid structure, whether it be porous or nonporous,

Table 3.9 Experimental Diffusivities of Large Biological Proteins in Aqueous Solutions

Protein	MW	Configuration	Temperature, °C	Diffusivity, D_{AB} , $\text{cm}^2/\text{s} \times 10^5$
Bovine serum albumin	67,500	globular	25	0.0681
γ -Globulin, human	153,100	globular	20	0.0400
Soybean protein	361,800	globular	20	0.0291
Urease	482,700	globular	25	0.0401
Fibrinogen, human	339,700	fibrous	20	0.0198
Lipoxidase	97,440	fibrous	20	0.0559

crystalline, or amorphous; and the type of solid material, whether it be metallic, ceramic, polymeric, biological, or cellular. Crystalline materials may be further classified according to the type of bonding, as molecular, covalent, ionic, or metallic, with most inorganic solids being ionic. However, ceramic materials can be ionic, covalent, or most often a combination of the two. Molecular solids have relatively weak forces of attraction among the atoms or molecules. In covalent solids, such as quartz silica, two atoms share two or more electrons equally. In ionic solids, such as inorganic salts, one atom loses one or more of its electrons by transfer to one or more other atoms, thus forming ions. In metals, positively charged ions are bonded through a field of electrons that are free to move. Unlike diffusion coefficients in gases and low-molecular-weight liquids, which each cover a range of only one or two orders of magnitude, diffusion coefficients in solids cover a range of many orders of magnitude. Despite the great complexity of diffusion in solids, Fick's first law can be used to describe diffusion if a measured diffusivity is available. However, when the diffusing solute is a gas, its solubility in the solid must also be known. If the gas dissociates upon dissolution in the solid, the concentration of the dissociated species must be used in Fick's law. In this section, many of the mechanisms of diffusion in solids are mentioned, but because they are exceedingly complex to quantify, the mechanisms are considered only qualitatively. Examples of diffusion in solids are considered, together with measured diffusion coefficients that can be used with Fick's first law. Emphasis is on diffusion of gas and liquid solutes through or into the solid, but movement of the atoms, molecules, or ions of the solid through itself is also considered.

Porous Solids

When solids are porous, predictions of the diffusivity of gaseous and liquid solute species in the pores can be made. These methods are considered only briefly here, with details deferred to Chapters 14, 15, and 16, where applications are made to membrane separations, adsorption, and leaching. This type of diffusion is also of great importance in the analysis and design of reactors using porous solid catalysts. It is sufficient to mention here that any of the following four mass-transfer

mechanisms or combinations thereof may take place:

1. Ordinary molecular diffusion through pores, which present tortuous paths and hinder the movement of large molecules when their diameter is more than 10% of the pore diameter
2. Knudsen diffusion, which involves collisions of diffusing gaseous molecules with the pore walls when the pore diameter and pressure are such that the molecular mean free path is large compared to the pore diameter
3. Surface diffusion involving the jumping of molecules, adsorbed on the pore walls, from one adsorption site to another based on a surface concentration-driving force
4. Bulk flow through or into the pores

When treating diffusion of solutes in porous materials where diffusion is considered to occur only in the fluid in the pores, it is common to refer to an effective diffusivity, D_{eff} , which is based on (1) the total cross-sectional area of the porous solid rather than the cross-sectional area of the pore and (2) on a straight path, rather than the pore path, which may be tortuous. If pore diffusion occurs only by ordinary molecular diffusion, Fick's law (3-3) can be used with an effective diffusivity. The effective diffusivity for a binary mixture can be expressed in terms of the ordinary diffusion coefficient, D_{AB} , by

$$D_{\text{eff}} = \frac{D_{AB}\epsilon}{\tau} \quad (3-49)$$

where ϵ is the fractional porosity (typically 0.5) of the solid and τ is the pore-path tortuosity (typically 2 to 3), which is the ratio of the pore length to the length if the pore were straight in the direction of diffusion. The effective diffusivity is either determined experimentally, without knowledge of the porosity or tortuosity, or predicted from (3-49) based on measurement of the porosity and tortuosity and use of the predictive methods for ordinary molecular diffusivity. As an example of the former, Boucher, Brier, and Osburn [13] measured effective diffusivities for the leaching of processed soybean oil (viscosity = 20.1 cP at 120°F) from 1/16-in.-thick porous clay plates with liquid tetrachloroethylene solvent. The rate of extraction was controlled by the rate of diffusion of the soybean oil in the clay plates. The measured value of

D_{eff} was $1.0 \times 10^{-6} \text{ cm}^2/\text{s}$. As might be expected from the effects of porosity and tortuosity, the effective value is about one order of magnitude less than the expected ordinary molecular diffusivity, D , of oil in the solvent.

Crystalline Solids

Diffusion through nonporous crystalline solids depends markedly on the crystal lattice structure and the diffusing entity. As discussed in Chapter 17 on crystallization, only seven different lattice structures are possible. For the cubic lattice (simple, body-centered, and face-centered), the diffusivity is the same in all directions (isotropic). In the six other lattice structures (including hexagonal and tetragonal), the diffusivity can be different in different directions (anisotropic). Many metals, including Ag, Al, Au, Cu, Ni, Pb, and Pt, crystallize into the face-centered cubic lattice structure. Others, including Be, Mg, Ti, and Zn, form anisotropic, hexagonal structures. The mechanisms of diffusion in crystalline solids include:

1. Direct exchange of lattice position by two atoms or ions, probably by a ring rotation involving three or more atoms or ions
2. Migration by small solutes through interlattice spaces called interstitial sites
3. Migration to a vacant site in the lattice
4. Migration along lattice imperfections (dislocations), or grain boundaries (crystal interfaces)

Diffusion coefficients associated with the first three mechanisms can vary widely and are almost always at least one order of magnitude smaller than diffusion coefficients in low-viscosity liquids. As might be expected, diffusion by the fourth mechanism can be faster than by the other three mechanisms. Typical experimental diffusivity values, taken mainly from Barrer [14], are given in Table 3.10. The diffusivities cover gaseous, ionic, and metallic solutes. The values cover an enormous 26-fold range. Temperature effects can be extremely large.

Metals

Important practical applications exist for diffusion of light gases through metals. To diffuse through a metal, a gas must first dissolve in the metal. As discussed by Barrer [14], all light gases do not dissolve in all metals. For example, hydrogen dissolves in such metals as Cu, Al, Ti, Ta, Cr, W, Fe, Ni, Pt, and Pd, but not in Au, Zn, Sb, and Rh. Nitrogen dissolves in Zr, but not in Cu, Ag, or Au. The noble gases do not dissolve in any of the common metals. When H_2 , N_2 , and O_2 dissolve in metals, they dissociate and may react to form hydrides, nitrides, and oxides, respectively. More complex molecules such as ammonia, carbon dioxide, carbon monoxide, and sulfur dioxide also dissociate. The following example illustrates how pressurized hydrogen gas can slowly leak through the wall of a small, thin pressure vessel.

Table 3.10 Diffusivities of Solute in Crystalline Metals and Salts

Metal/Salt	Solute	$T, ^\circ\text{C}$	$D, \text{cm}^2/\text{s}$
Ag	Au	760	3.6×10^{-10}
	Sb	20	3.5×10^{-21}
	Sb	760	1.4×10^{-9}
Al	Fe	359	6.2×10^{-14}
	Zn	500	2×10^{-9}
	Ag	50	1.2×10^{-9}
Cu	Al	20	1.3×10^{-30}
	Al	850	2.2×10^{-9}
	Au	750	2.1×10^{-11}
Fe	H_2	10	1.66×10^{-9}
	H_2	100	1.24×10^{-7}
	C	800	1.5×10^{-8}
Ni	H_2	85	1.16×10^{-8}
	H_2	165	1.05×10^{-7}
	CO	950	4×10^{-8}
W	U	1727	1.3×10^{-11}
AgCl	Ag^+	150	2.5×10^{-14}
	Ag^+	350	7.1×10^{-8}
	Cl^-	350	3.2×10^{-16}
KBr	H_2	600	5.5×10^{-4}
	Br_2	600	2.64×10^{-4}

EXAMPLE 3.9

Gaseous hydrogen at 200 psia and 300°C is stored in a small, 10-cm-diameter, steel pressure vessel having a wall thickness of 0.125 in. The solubility of hydrogen in steel, which is proportional to the square root of the hydrogen partial pressure in the gas, is equal to $3.8 \times 10^{-6} \text{ mol/cm}^3$ at 14.7 psia and 300°C . The diffusivity of hydrogen in steel at 300°C is $5 \times 10^{-6} \text{ cm}^2/\text{s}$. If the inner surface of the vessel wall remains saturated at the existing hydrogen partial pressure and the hydrogen partial pressure at the outer surface is zero, estimate the time, in hours, for the pressure in the vessel to decrease to 100 psia because of hydrogen loss by dissolving in and diffusing through the metal wall.

SOLUTION

Integrating Fick's first law, (3-3), where A is H_2 and B is the metal, assuming a linear concentration gradient, and equating the flux to the loss of hydrogen in the vessel,

$$-\frac{dn_A}{dt} = \frac{D_{AB} A \Delta c_A}{\Delta z} \quad (1)$$

Because $p_A = 0$ outside the vessel, $\Delta c_A = c_A = \text{solubility of A at the inside wall surface in mol/cm}^3$ and $c_A = 3.8 \times 10^{-6} \left(\frac{p_A}{14.7}\right)^{0.5}$, where p_A is the pressure of A in psia inside the vessel. Let p_{A_0} and n_{A_0} be the initial pressure and moles of A, respectively, in the vessel. Assuming the ideal-gas law and isothermal conditions,

$$n_A = n_{A_0} p_A / p_{A_0} \quad (2)$$

Differentiating (2) with respect to time,

$$\frac{dn_A}{dt} = \frac{n_{A_0}}{p_{A_0}} \frac{dp_A}{dt} \quad (3)$$

Combining (1) and (3),

$$\frac{dp_A}{dt} = - \frac{D_A A (3.8 \times 10^{-6}) p_A^{0.5} p_{A_0}}{n_{A_0} \Delta z (14.7)^{0.5}} \quad (4)$$

Integrating and solving for t ,

$$t = \frac{2n_{A_0} \Delta z (14.7)^{0.5}}{3.8 \times 10^{-6} D_A A p_{A_0}} (p_{A_0}^{0.5} - p_A^{0.5})$$

Assuming the ideal-gas law,

$$n_{A_0} = \frac{(200/14.7)[(3.14 \times 10^3)/6]}{82.05(300 + 273)} = 0.1515 \text{ mol}$$

The mean-spherical shell area for mass transfer, A , is

$$A = \frac{3.14}{2} [(10)^2 + (10.635)^2] = 336 \text{ cm}^2$$

The time for the pressure to drop to 100 psia is

$$t = \frac{2(0.1515)(0.125 \times 2.54)(14.7)^{0.5}}{3.8 \times 10^{-6}(5 \times 10^{-6})(336)(200)} (200^{0.5} - 100^{0.5}) \\ = 1.2 \times 10^6 \text{ s} = 332 \text{ h}$$

Silica and Glass

Another area of great interest is the diffusion of light gases through various forms of silica, whose two elements, Si and O, make up about 60% of the earth's crust. Solid silica can exist in three principal crystalline forms (quartz, tridymite, and cristobalite) and in various stable amorphous forms, including vitreous silica (a noncrystalline silicate glass or fused quartz). Table 3.11 includes diffusivities, D , and solubilities as Henry's law constants, H , at 1 atm for helium and hydrogen in fused quartz as calculated from correlations of experimental data by Swets, Lee, and Frank [15] and Lee [16], respectively. The product of the diffusivity and the solubility is called the permeability, P_M . Thus,

$$P_M = DH \quad (3-50)$$

Unlike metals, where hydrogen usually diffuses as the atom, hydrogen apparently diffuses as a molecule in glass.

Table 3.11 Diffusivities and Solubilities of Gases in Amorphous Silica at 1 atm

Gas	Temp, C	Diffusivity, cm ² /s	Solubility mol/cm ³ -atm
He	24	2.39×10^{-8}	1.04×10^{-7}
	300	2.26×10^{-6}	1.82×10^{-7}
	500	9.99×10^{-6}	9.9×10^{-8}
	1,000	5.42×10^{-5}	1.34×10^{-7}
H ₂	300	6.11×10^{-8}	3.2×10^{-14}
	500	6.49×10^{-7}	2.48×10^{-13}
	1,000	9.26×10^{-6}	2.49×10^{-12}
O ₂	1,000	6.25×10^{-9} (molecular)	
	1,000	9.43×10^{-15} (network)	

For both hydrogen and helium, diffusivities increase rapidly with increasing temperature. At ambient temperature the diffusivities are three orders of magnitude lower than in liquids. At elevated temperatures the diffusivities approach those observed in liquids. Solubilities vary only slowly with temperature. Hydrogen is orders of magnitude less soluble in glass than helium. For hydrogen, the diffusivity is somewhat lower than in metals. Diffusivities for oxygen are also included in Table 3.11 from studies by Williams [17] and Sucov [18]. At 1000°C, the two values differ widely because, as discussed by Kingery, Bowen, and Uhlmann [19], in the former case, transport occurs by molecular diffusion; while in the latter case, transport is by slower network diffusion as oxygen jumps from one position in the silicate network to another. The activation energy for the latter is much larger than for the former (71,000 cal/mol versus 27,000 cal/mol). The choice of glass can be very critical in high-vacuum operations because of the wide range of diffusivity.

Ceramics

Diffusion rates of light gases and elements in crystalline ceramics are very important because diffusion must precede chemical reactions and causes changes in the microstructure. Therefore, diffusion in ceramics has been the subject of numerous studies, many of which are summarized in Figure 3.4, taken from Kingery et al. [19], where diffusivity is plotted as a function of the inverse of temperature in the high-temperature range. In this form, the slopes of the curves are proportional to the activation energy for diffusion, E , where

$$D = D_0 \exp\left(-\frac{E}{RT}\right) \quad (3-51)$$

An insert at the middle-right region of Figure 3.4 relates the slopes of the curves to activation energy. The diffusivity curves cover a ninefold range from 10^{-6} to 10^{-15} cm²/s, with the largest values corresponding to the diffusion of potassium in β -Al₂O₃ and one of the smallest values for carbon in graphite. In general, the lower the diffusivity, the higher is the activation energy. As discussed in detail by Kingery et al. [19], diffusion in crystalline oxides depends not only on temperature but also on whether the oxide is stoichiometric or not (e.g., FeO and Fe_{0.95}O) and on impurities. Diffusion through vacant sites of nonstoichiometric oxides is often classified as metal-deficient or oxygen-deficient. Impurities can hinder diffusion by filling vacant lattice or interstitial sites.

Polymers

Thin, dense, nonporous polymer membranes are widely used to separate gas and liquid mixtures. As discussed in detail in Chapter 14, diffusion of gas and liquid species through polymers is highly dependent on the type of polymer, whether it be crystalline or amorphous and, if the latter, glassy or rubbery. Commercial crystalline polymers are

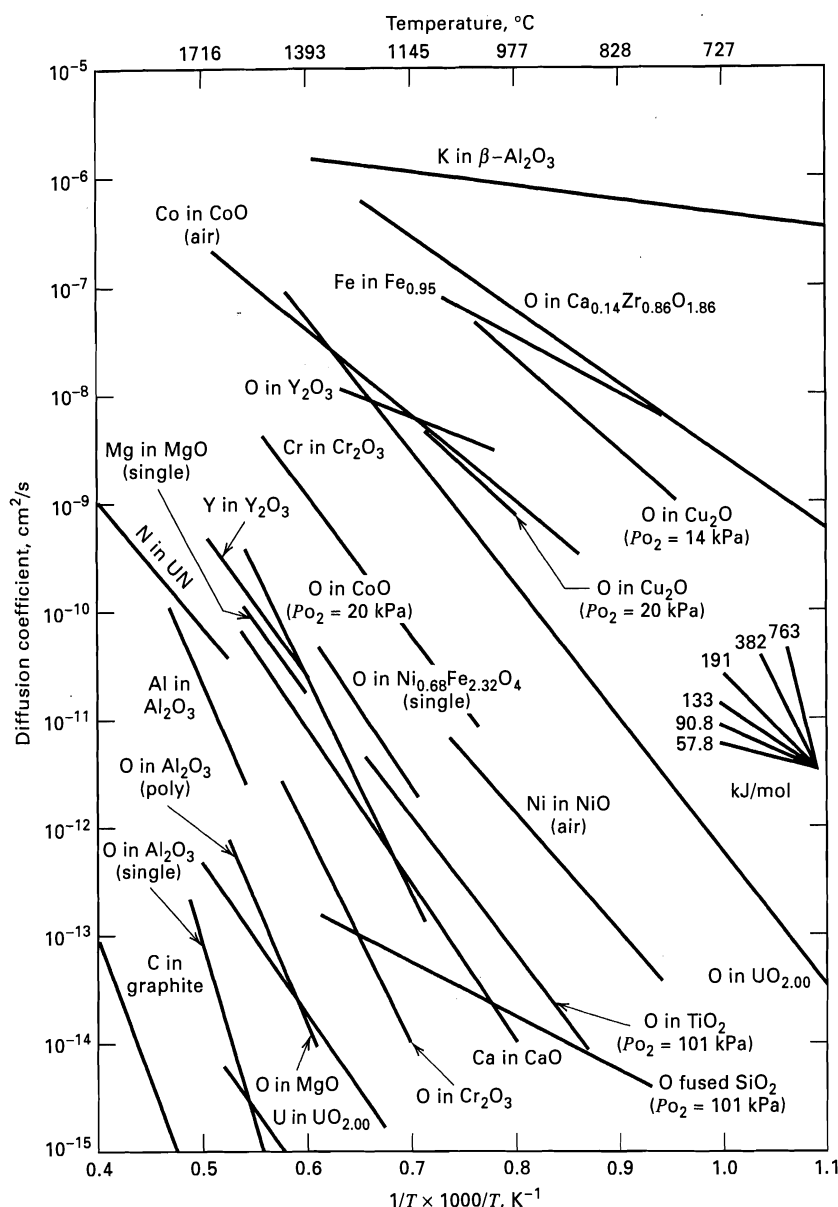


Figure 3.4 Diffusion coefficients for single- and polycrystalline ceramics.

[From W.D. Kingery, H.K. Bowen, and D.R. Uhlmann, *Introduction to Ceramics*, 2nd ed., Wiley Interscience, New York (1976) with permission.]

about 20% amorphous. It is mainly through the amorphous regions that diffusion occurs. As with the transport of gases through metals, transport of gaseous species through polymer membranes is usually characterized by the solution-diffusion mechanism of (3-50). Fick's first law, in the following integrated forms, is then applied to compute the mass transfer flux.

Gas species:

$$N_i = \frac{H_i D_i}{z_2 - z_1} (p_{i1} - p_{i2}) = \frac{P_{M_i}}{z_2 - z_1} (p_{i1} - p_{i2}) \quad (3-52)$$

where p_i is the partial pressure of the gas species at a polymer surface.

Liquid species:

$$N_i = \frac{K_i D_i}{z_2 - z_1} (c_{i1} - c_{i2}) \quad (3-53)$$

where K_i , the equilibrium partition coefficient, is equal to the ratio of the concentration in the polymer to the concentration, c_i , in the liquid adjacent to the polymer surface. The product $K_i D_i$ is the liquid permeability.

Values of diffusivity for light gases in four polymers, given in Table 14.6, range from 1.3×10^{-9} to 1.6×10^{-6} cm²/s, which is orders of magnitude less than for diffusion of the same species in a gas.

Diffusivities of liquids in rubbery polymers have been studied extensively as a means of determining viscoelastic parameters. In Table 3.12, taken from Ferry [20], diffusivities are given for different solutes in seven different rubber polymers at near-ambient conditions. The values cover a sixfold range, with the largest diffusivity being that for *n*-hexadecane in polydimethylsiloxane. The smallest diffusivities correspond to the case where the temperature is approaching the glass-transition temperature, where the polymer becomes glassy in structure. This more rigid structure hinders diffusion. In general, as would be expected,

Table 3.12 Diffusivities of Solutes in Rubbery Polymers

Polymer	Solute	Temperature, K	Diffusivity, cm ² /s
Polyisobutylene	<i>n</i> -Butane	298	1.19×10^{-9}
	<i>i</i> -Butane	298	5.3×10^{-10}
	<i>n</i> -Pentane	298	1.08×10^{-9}
	<i>n</i> -Hexadecane	298	6.08×10^{-10}
Hevea rubber	<i>n</i> -Butane	303	2.3×10^{-7}
	<i>i</i> -Butane	303	1.52×10^{-7}
	<i>n</i> -Pentane	303	2.3×10^{-7}
	<i>n</i> -Hexadecane	298	7.66×10^{-8}
Polymethylacrylate	Ethyl alcohol	323	2.18×10^{-10}
Polyvinylacetate	<i>n</i> -Propyl alcohol	313	1.11×10^{-12}
	<i>n</i> -Propyl chloride	313	1.34×10^{-12}
	Ethyl chloride	343	2.01×10^{-9}
	Ethyl bromide	343	1.11×10^{-9}
Polydimethylsiloxane	<i>n</i> -Hexadecane	298	1.6×10^{-6}
1,4-Polybutadiene	<i>n</i> -Hexadecane	298	2.21×10^{-7}
Styrene-butadiene rubber	<i>n</i> -Hexadecane	298	2.66×10^{-8}

smaller molecules have higher diffusivities. A more detailed study of the diffusivity of *n*-hexadecane in random styrene/butadiene copolymers at 25°C by Rhee and Ferry [21] shows a large effect on diffusivity of fractional free volume in the polymer.

Diffusion and permeability in crystalline polymers depend on the degree of crystallinity. Polymers that are 100% crystalline permit little or no diffusion of gases and liquids. For example, the diffusivity of methane at 25°C in polyoxyethylene oxyisophthaloyl decreases from 0.30×10^{-9} to 0.13×10^{-9} cm²/s when the degree of crystallinity increases from 0 (totally amorphous) to 40% [22]. A measure of crystallinity is the polymer density. The diffusivity of methane at 25°C in polyethylene decreases from 0.193×10^{-6} to 0.057×10^{-6} cm²/s when the specific gravity increases from 0.914 (low density) to 0.964 (high density) [22]. A plasticizer can cause the diffusivity to increase. For example, when polyvinylchloride is plasticized with 40% tricresyl triphosphate, the diffusivity of CO at 27°C increases from 0.23×10^{-8} to 2.9×10^{-8} cm²/s [22].

EXAMPLE 3.10

Hydrogen diffuses through a nonporous polyvinyltrimethylsilane membrane at 25°C. The pressures on the sides of the membrane are 3.5 MPa and 200 kPa. Diffusivity and solubility data are given in Table 14.9. If the hydrogen flux is to be 0.64 kmol/m²-h, how thick in micrometers should the membrane be?

SOLUTION

Equation (3-52) applies. From Table 14.9,

$$D = 160 \times 10^{-11} \text{ m}^2/\text{s} \quad H = S = 0.54 \times 10^{-4} \text{ mol/m}^3\text{-Pa}$$

From (3-50),

$$\begin{aligned} P_M &= DH = (160 \times 10^{-11})(0.54 \times 10^{-4}) \\ &= 86.4 \times 10^{-15} \text{ mol/m-s-Pa} \\ p_1 &= 3.5 \times 10^6 \text{ Pa} \quad p_2 = 0.2 \times 10^6 \text{ Pa} \end{aligned}$$

$$\text{Membrane thickness} = z_2 - z_1 = \Delta z = P_M(p_1 - p_2)/N$$

$$\begin{aligned} \Delta z &= \frac{86.4 \times 10^{-15}(3.5 \times 10^6 - 0.2 \times 10^6)}{[0.64(1000)/3600]} \\ &= 1.6 \times 10^{-6} \text{ m} = 1.6 \mu\text{m} \end{aligned}$$

As discussed in Chapter 14, polymer membranes must be very thin to achieve reasonable gas permeation rates.

Cellular Solids and Wood

As discussed by Gibson and Ashby [23], cellular solids consist of solid struts or plates that form edges and faces of cells, which are compartments or enclosed spaces. Cellular solids such as wood, cork, sponge, and coral exist in nature. Synthetic cellular structures include honeycombs, and foams (some with open cells) made from polymers, metals, ceramics, and glass. The word *cellulose* means “full of little cells.”

A widely used cellular solid is wood, whose annual world production of the order of 10^{12} kg is comparable to the production of iron and steel. Chemically, wood consists of lignin, cellulose, hemicellulose, and minor amounts of organic chemicals and elements. The latter are extractable, and the former three, which are all polymers, give wood its structure. Green wood also contains up to 25 wt% moisture in the cell walls and cell cavities. Adsorption or desorption of moisture in wood causes anisotropic swelling and shrinkage.

The structure of wood, which often consists of (1) highly elongated hexagonal or rectangular cells, called tracheids in softwood (coniferous species, e.g., spruce, pine, and fir) and fibers in hardwood (deciduous or broad-leaf species, e.g., oak, birch, and walnut); (2) radial arrays of rectangular-like cells, called rays, which are narrow and short in softwoods but wide and long in hardwoods; and (3) enlarged cells with large pore spaces and thin walls, called sap channels because they conduct fluids up the tree. The sap channels are less than 3 vol% of softwood, but as much as 55 vol% of hardwood.

Because the structure of wood is directional, many of its properties are anisotropic. For example, stiffness and strength are 2 to 20 times greater in the axial direction of the tracheids or fibers than in the radial and tangential directions of the trunk from which the wood is cut. This anisotropy extends to permeability and diffusivity of wood penetrants, such as moisture and preservatives. According to Stamm [24], the permeability of wood to liquids in the axial direction can be up to 10 times greater than in the transverse direction.

Movement of liquids and gases through wood and wood products takes time during drying and treatment with preservatives, fire retardants, and other chemicals. This movement takes place by capillarity, pressure permeability, and diffusion. Nevertheless, wood is not highly permeable because the cell voids are largely discrete and lack direct interconnections. Instead, communication among cells is through circular openings spanned by thin membranes with submicrometer-sized pores, called pits, and to a smaller extent, across the cell walls. Rays give wood some permeability in the radial direction. Sap channels do not contribute to permeability. All three mechanisms of movement of gases and liquids in wood are considered by Stamm [24]. Only diffusion is discussed here.

The simplest form of diffusion is that of a water-soluble solute through wood saturated with water, such that no dimensional changes occur. For the diffusion of urea, glycerine, and lactic acid into hardwood, Stamm [24] lists diffusivities in the axial direction that are about 50% of ordinary liquid diffusivities. In the radial direction, diffusivities are about 10% of the values in the axial direction. For example, at 26.7°C the diffusivity of zinc sulfate in water is $5 \times 10^{-6} \text{ cm}^2/\text{s}$. If loblolly pine sapwood is impregnated with zinc sulfate in the radial direction, the diffusivity is found to be $0.18 \times 10^{-6} \text{ cm}^2/\text{s}$ [24].

The diffusion of water in wood is more complex. Moisture content determines the degree of swelling or shrinkage. Water is held in the wood in different ways: It may be physically adsorbed on cell walls in monomolecular layers, condensed in preexisting or transient cell capillaries, or absorbed in cell walls to form a solid solution.

Because of the practical importance of lumber drying rates, most diffusion coefficients are measured under drying conditions in the radial direction across the fibers. Results depend on temperature and swollen-volume specific gravity.

Typical results are given by Sherwood [25] and Stamm [24]. For example, for beech with a swollen specific gravity of 0.4, the diffusivity increases from a value of about $1 \times 10^{-6} \text{ cm}^2/\text{s}$ at 10°C to $10 \times 10^{-6} \text{ cm}^2/\text{s}$ at 60°C.

3.3 ONE-DIMENSIONAL, STEADY-STATE AND UNSTEADY-STATE, MOLECULAR DIFFUSION THROUGH STATIONARY MEDIA

For conductive heat transfer in stationary media, Fourier's law is applied to derive equations for the rate of heat transfer for steady-state and unsteady-state conditions in shapes such as slabs, cylinders, and spheres. Many of the results are plotted in generalized charts. Analogous equations can be derived for mass transfer, using simplifying assumptions.

In one dimension, the molar rate of mass transfer of A in a binary mixture with B is given by a modification of (3-12), which includes bulk flow and diffusion:

$$n_A = x_A(n_A + n_B) - cD_{AB}A\left(\frac{dx_A}{dz}\right) \quad (3-54)$$

If A is a dissolved solute undergoing mass transfer, but B is stationary, $n_B = 0$. It is common to assume that c is a constant and x_A is small. The bulk-flow term is then eliminated and (3-54) accounts for diffusion only, becoming Fick's first law:

$$n_A = -cD_{AB}A\left(\frac{dx_A}{dz}\right) \quad (3-55)$$

Alternatively, (3-55) can be written in terms of concentration gradient:

$$n_A = -D_{AB}A\left(\frac{dc_A}{dz}\right) \quad (3-56)$$

This equation is analogous to Fourier's law for the rate of heat conduction, Q :

$$Q = -kA\left(\frac{dT}{dz}\right) \quad (3-57)$$

Steady State

For steady-state, one-dimensional diffusion, with constant D_{AB} , (3-56) can be integrated for various geometries, the most common results being analogous to heat conduction.

1. Plane wall with a thickness, $z_2 - z_1$:

$$n_A = D_{AB}A\left(\frac{c_{A1} - c_{A2}}{z_2 - z_1}\right) \quad (3-58)$$

2. Hollow cylinder of inner radius r_1 and outer radius r_2 , with diffusion in the radial direction outward:

$$n_A = 2\pi L \frac{D_{AB}(c_{A1} - c_{A2})}{\ln(r_2/r_1)} \quad (3-59)$$

or

$$n_A = D_{AB}A_{LM}\left(\frac{c_{A1} - c_{A2}}{r_2 - r_1}\right) \quad (3-60)$$

where

A_{LM} = log mean of the areas $2\pi rL$ at r_1 and r_2

L = length of the hollow cylinder

3. Spherical shell of inner radius r_1 and outer radius r_2 , with diffusion in the radial direction outward:

$$n_A = \frac{4\pi r_1 r_2 D_{AB}(c_{A1} - c_{A2})}{r_2 - r_1} \quad (3-61)$$

or

$$n_A = D_{AB} A_{GM} \left(\frac{c_{A1} - c_{A2}}{r_2 - r_1} \right) \quad (3-62)$$

where A_{GM} = geometric mean of the areas $4\pi r^2$ at r_1 and r_2 .

When $r_1/r_2 < 2$, the arithmetic mean area is no more than 4% greater than the log mean area. When $r_1/r_2 < 1.33$, the arithmetic mean area is no more than 4% greater than the geometric mean area.

Unsteady State

Equation (3-56) is applied to unsteady-state molecular diffusion by considering the accumulation or depletion of a species with time in a unit volume through which the species is diffusing. Consider the one-dimensional diffusion of species A in B through a differential control volume with diffusion in the z -direction only, as shown in Figure 3.5. Assume constant total concentration, $c = c_A + c_B$, constant diffusivity, and negligible bulk flow. The molar flow rate of species A by diffusion at the plane $z = z$ is given by (3-56):

$$n_{Az} = -D_{AB} A \left(\frac{\partial c_A}{\partial z} \right)_z \quad (3-63)$$

At the plane, $z = z + \Delta z$, the diffusion rate is

$$n_{A_{z+\Delta z}} = -D_{AB} A \left(\frac{\partial c_A}{\partial z} \right)_{z+\Delta z} \quad (3-64)$$

The accumulation of species A in the control volume is

$$A \frac{\partial c_A}{\partial t} \Delta z \quad (3-65)$$

Since rate in – rate out = accumulation,

$$-D_{AB} A \left(\frac{\partial c_A}{\partial z} \right)_z + D_{AB} A \left(\frac{\partial c_A}{\partial z} \right)_{z+\Delta z} = A \left(\frac{\partial c_A}{\partial t} \right) \Delta z \quad (3-66)$$

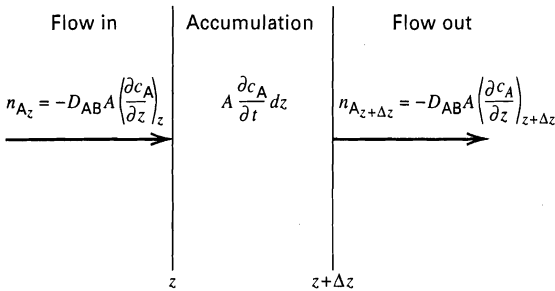


Figure 3.5 Unsteady-state diffusion through a differential volume $A \, dz$.

Rearranging and simplifying,

$$D_{AB} \left[\frac{(\partial c_A / \partial z)_{z+\Delta z} - (\partial c_A / \partial z)_z}{\Delta z} \right] = \frac{\partial c_A}{\partial t} \quad (3-67)$$

In the limit, as $\Delta z \rightarrow 0$,

$$\frac{\partial c_A}{\partial t} = D_{AB} \frac{\partial^2 c_A}{\partial z^2} \quad (3-68)$$

Equation (3-68) is Fick's second law for one-dimensional diffusion. The more general form, for three-dimensional rectangular coordinates, is

$$\frac{\partial c_A}{\partial t} = D_{AB} \left(\frac{\partial^2 c_A}{\partial x^2} + \frac{\partial^2 c_A}{\partial y^2} + \frac{\partial^2 c_A}{\partial z^2} \right) \quad (3-69)$$

For one-dimensional diffusion in the radial direction only, for cylindrical and spherical coordinates, Fick's second law becomes, respectively,

$$\frac{\partial c_A}{\partial t} = \frac{D_{AB}}{r} \frac{\partial}{\partial r} \left(r \frac{\partial c_A}{\partial r} \right) \quad (3-70)$$

and

$$\frac{\partial c_A}{\partial t} = \frac{D_{AB}}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial c_A}{\partial r} \right) \quad (3-71)$$

Equations (3-68) to (3-71) are analogous to Fourier's second law of heat conduction where c_A is replaced by temperature, T , and diffusivity, D_{AB} , is replaced by thermal diffusivity, $\alpha = k/\rho C_P$. The analogous three equations for heat conduction for constant, isotropic properties are, respectively:

$$\frac{\partial T}{\partial t} = \alpha \left(\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} + \frac{\partial^2 T}{\partial z^2} \right) \quad (3-72)$$

$$\frac{\partial T}{\partial t} = \frac{\alpha}{r} \frac{\partial}{\partial r} \left(r \frac{\partial T}{\partial r} \right) \quad (3-73)$$

$$\frac{\partial T}{\partial t} = \frac{\alpha}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial T}{\partial r} \right) \quad (3-74)$$

Analytical solutions to these partial differential equations in either Fick's law or Fourier's law form are available for a variety of boundary conditions. Many of these solutions are derived and discussed by Carslaw and Jaeger [26] and Crank [27]. Only a few of the more useful solutions are presented here.

Semi-infinite Medium

Consider the semi-infinite medium shown in Figure 3.6, which extends in the z -direction from $z = 0$ to $z = \infty$. The x and y coordinates extend from $-\infty$ to $+\infty$, but are not of interest because diffusion takes place only in the z -direction. Thus, (3-68) applies to the region $z \geq 0$. At time $t \leq 0$, the concentration is c_{A_0} for $z \geq 0$. At $t = 0$, the surface of the semi-infinite medium at $z = 0$ is instantaneously brought to the concentration $c_{A_s} > c_{A_0}$ and held there for $t > 0$. Therefore, diffusion into the medium occurs. However, because

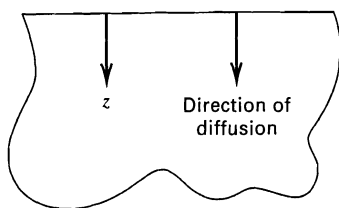


Figure 3.6 One-dimensional diffusion into a semi-infinite medium.

the medium is infinite in the z -direction, diffusion cannot extend to $z = \infty$ and, therefore, as $z \rightarrow \infty$, $c_A = c_{A_0}$ for all $t \geq 0$. Because the partial differential equation (3-68) and its one boundary (initial) condition in time and two boundary conditions in distance are linear in the dependent variable, c_A , an exact solution can be obtained. Either the method of combination of variables [28] or the Laplace transform method [29] is applicable. The result, in terms of the fractional accomplished concentration change, is

$$\theta = \frac{c_A - c_{A_0}}{c_{A_s} - c_{A_0}} = \text{erfc}\left(\frac{z}{2\sqrt{D_{AB}t}}\right) \quad (3-75)$$

where the complementary error function, erfc , is related to the error function, erf , by

$$\text{erfc}(x) = 1 - \text{erf}(x) = 1 - \frac{2}{\sqrt{\pi}} \int_0^x e^{-\eta^2} d\eta \quad (3-76)$$

The error function is included in most spreadsheet programs and handbooks, such as *Handbook of Mathematical Functions* [30]. The variation of $\text{erf}(x)$ and $\text{erfc}(x)$ is as follows:

x	$\text{erf}(x)$	$\text{erfc}(x)$
0	0.0000	1.0000
0.5	0.5205	0.4795
1.0	0.8427	0.1573
1.5	0.9661	0.0339
2.0	0.9953	0.0047
∞	1.0000	0.0000

Equation (3-75) is used to compute the concentration in the semi-infinite medium, as a function of time and distance from the surface, assuming no bulk flow. Thus, it applies most rigorously to diffusion in solids, and also to stagnant liquids and gases when the medium is dilute in the diffusing solute.

In (3-75), when $(z/2\sqrt{D_{AB}t}) = 2$, the complementary error function is only 0.0047, which represents less than a 1% change in the ratio of the concentration change at $z = z$ to the change at $z = 0$. Thus, it is common to refer to $z = 4\sqrt{D_{AB}t}$ as the penetration depth and to apply (3-75) to media of finite thickness as long as the thickness is greater than the penetration depth.

The instantaneous rate of mass transfer across the surface of the medium at $z = 0$ can be obtained by taking the derivative of (3-75) with respect to distance and substituting it into Fick's first law applied at the surface of the medium.

Thus, using the Leibnitz rule for differentiating the integral of (3-76), with $x = z/2\sqrt{D_{AB}t}$,

$$\begin{aligned} n_A &= -D_{AB}A \left(\frac{\partial c_A}{\partial z} \right)_{z=0} \\ &= D_{AB}A \left(\frac{c_{A_s} - c_{A_0}}{\sqrt{\pi D_{AB}t}} \right) \exp\left(-\frac{z^2}{4D_{AB}t}\right) \Big|_{z=0} \end{aligned} \quad (3-77)$$

Thus,

$$n_A|_{z=0} = \sqrt{\frac{D_{AB}}{\pi t}} A(c_{A_s} - c_{A_0}) \quad (3-78)$$

We can also determine the total number of moles of solute, N_A , transferred into the semi-infinite medium by integrating (3-78) with respect to time:

$$\begin{aligned} N_A &= \int_0^t n_A|_{z=0} dt = \sqrt{\frac{D_{AB}}{\pi}} A(c_{A_s} - c_{A_0}) \int_0^t \frac{dt}{\sqrt{t}} \\ &= 2A(c_{A_s} - c_{A_0}) \sqrt{\frac{D_{AB}t}{\pi}} \end{aligned} \quad (3-79)$$

EXAMPLE 3.11

Determine how long it will take for the dimensionless concentration change, $\theta = (c_A - c_{A_0})/(c_{A_s} - c_{A_0})$, to reach 0.01 at a depth $z = 1$ m in a semi-infinite medium, which is initially at a solute concentration c_{A_0} , after the surface concentration at $z = 0$ increases to c_{A_s} , for diffusivities representative of a solute diffusing through a stagnant gas, a stagnant liquid, and a solid.

SOLUTION

For a gas, assume $D_{AB} = 0.1 \text{ cm}^2/\text{s}$. We know that $z = 1 \text{ m} = 100 \text{ cm}$. From (3-75) and (3-76),

$$\theta = 0.01 = 1 - \text{erf}\left(\frac{z}{2\sqrt{D_{AB}t}}\right)$$

Therefore,

$$\text{erf}\left(\frac{z}{2\sqrt{D_{AB}t}}\right) = 0.99$$

From tables of the error function,

$$\left(\frac{z}{2\sqrt{D_{AB}t}}\right) = 1.8214$$

Solving,

$$t = \left[\frac{100}{1.8214(2)} \right]^2 \frac{1}{0.10} = 7,540 \text{ s} = 2.09 \text{ h}$$

In a similar manner, the times for typical gas, liquid, and solid media are:

Semi-infinite Medium	D_{AB} , cm^2/s	Time for $\theta = 0.01$ at 1 m
Gas	0.10	2.09 h
Liquid	1×10^{-5}	2.39 year
Solid	1×10^{-9}	239 centuries

These results show that molecular diffusion is very slow, especially in liquids and solids. In liquids and gases, the rate of mass

transfer can be greatly increased by agitation to induce turbulent motion. For solids, it is best to reduce the diffusion path to as small a dimension as possible by reducing the size of the solid.

Medium of Finite Thickness with Sealed Edges

Consider a rectangular, parallelepiped medium of finite thickness $2a$ in the z -direction, and either infinitely long dimensions in the y - and x -directions or finite lengths of $2b$ and $2c$, respectively, in those directions. Assume that in Figure 3.7a the edges parallel to the z -direction are sealed, so diffusion occurs only in the z -direction and initially the concentration of the solute in the medium is uniform at c_{A_0} . At time $t = 0$, the two unsealed surfaces of the medium at $z = \pm a$ are brought to and held at concentration $c_{A_s} > c_{A_0}$. Because of symmetry, $\partial c_A / \partial z = 0$ at $z = 0$. Assume constant D_{AB} . Again (3-68) applies, and an exact solution can be obtained because both (3-68) and the boundary conditions are linear in c_A . By the method of separation of variables [28] or the Laplace transform method [29], the result from Carslaw and Jaeger [26], in terms of the fractional, unaccomplished concentration change, E , is

$$E = 1 - \theta = \frac{c_{A_s} - c_A}{c_{A_s} - c_{A_0}} = \frac{4}{\pi} \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)} \times \exp[-D_{AB}(2n+1)^2 \pi^2 t / 4a^2] \cos \frac{(2n+1)\pi z}{2a} \quad (3-80)$$

or, in terms of the complementary error function,

$$E = 1 - \theta = \frac{c_{A_s} - c_A}{c_{A_s} - c_{A_0}} = \sum_{n=0}^{\infty} (-1)^n \times \left[\operatorname{erfc} \frac{(2n+1)a - z}{2\sqrt{D_{AB}t}} + \operatorname{erfc} \frac{(2n+1)a + z}{2\sqrt{D_{AB}t}} \right] \quad (3-81)$$

For large values of $D_{AB}t/a^2$, which is referred to as the Fourier number for mass transfer, the infinite series solutions of (3-80) and (3-81) converge rapidly, but for small values

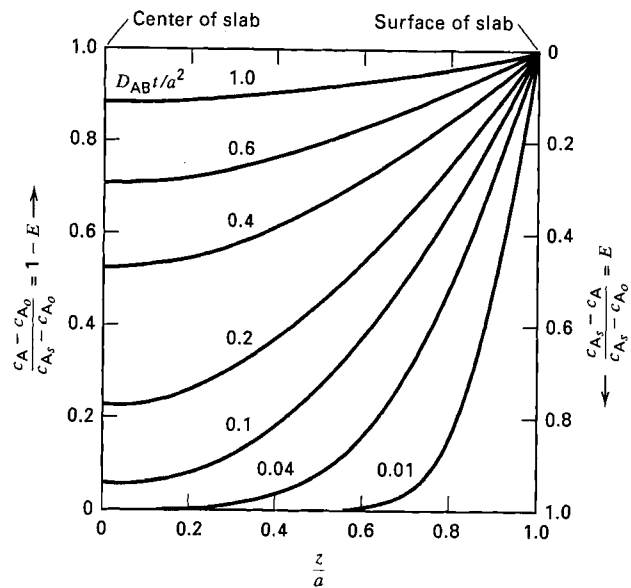


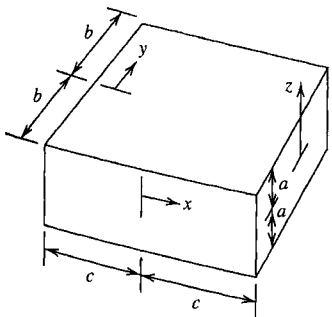
Figure 3.8 Concentration profiles for unsteady-state diffusion in a slab.

[Adapted from H.S. Carslaw and J.C. Jaeger, *Conduction of Heat in Solids*, 2nd ed., Oxford University Press, London (1959).]

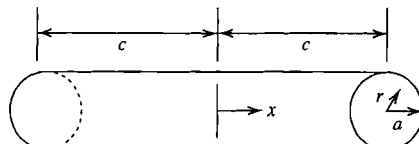
(e.g., short times), they do not. However, in the latter case, the solution for the semi-infinite medium applies for $D_{AB}t/a^2 < \frac{1}{16}$. A convenient plot of the exact solution is given in Figure 3.8.

The instantaneous rate of mass transfer across the surface of either unsealed face of the medium (i.e., at $z = \pm a$) is obtained by differentiating (3-80) with respect to z , evaluating the result at $z = a$, followed by substitution into Fick's first law to give

$$n_A|_{z=a} = \frac{2D_{AB}(c_{A_s} - c_{A_0})A}{a} \times \sum_{n=0}^{\infty} \exp \left[-\frac{D_{AB}(2n+1)^2 \pi^2 t}{4a^2} \right] \quad (3-82)$$



(a) Slab. Edges at $x = +c$ and $-c$ and at $y = +b$ and $-b$ are sealed.



(b) Cylinder. Two circular ends at $x = +c$ and $-c$ are sealed.



(c) Sphere

Figure 3.7 Unsteady-state diffusion in media of finite dimensions.

We can also determine the total number of moles transferred across either unsealed face by integrating (3-82) with respect to time. Thus,

$$\mathcal{N}_A = \int_0^t n_A|_{z=a} dt = \frac{8(c_{A_s} - c_{A_o})Aa}{\pi^2} \times \sum_{n=0}^{\infty} \frac{1}{(2n+1)^2} \left\{ 1 - \exp \left[-\frac{D_{AB}(2n+1)^2\pi^2 t}{4a^2} \right] \right\} \quad (3-83)$$

In addition, the average concentration of the solute in the medium, $c_{A_{avg}}$, as a function of time, can be obtained in the case of a slab from:

$$\frac{c_{A_s} - c_{A_{avg}}}{c_{A_s} - c_{A_o}} = \frac{\int_0^a (1 - \theta) dz}{a} \quad (3-84)$$

Substitution of (3-80) into (3-84) followed by integration gives

$$\begin{aligned} E_{avg,slab} &= (1 - \theta_{ave})_{slab} = \frac{c_{A_s} - c_{A_{avg}}}{c_{A_s} - c_{A_o}} \\ &= \frac{8}{\pi^2} \sum_{n=0}^{\infty} \frac{1}{(2n+1)^2} \exp \left[-\frac{D_{AB}(2n+1)^2\pi^2 t}{4a^2} \right] \end{aligned} \quad (3-85)$$

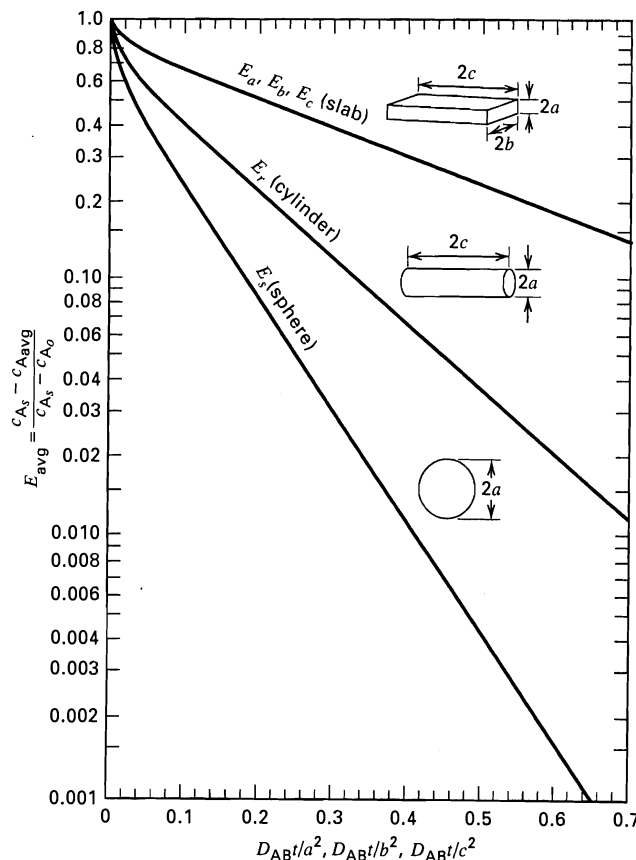


Figure 3.9 Average concentrations for unsteady-state diffusion.

[Adapted from R.E. Treybal, *Mass-Transfer Operations*, 3rd ed., McGraw-Hill, New York (1980).]

This equation is plotted in Figure 3.9. It is important to note that concentrations are in mass of solute per mass of dry solid or mass of solute/volume. This assumes that during diffusion the solid does not shrink or expand so that the mass of dry solid per unit volume of wet solid will remain constant. Then, we can substitute a concentration in terms of mass or moles of solute per mass of dry solid, i.e., the moisture content on the dry basis.

When the edges of the slab in Figure 3.7a are not sealed, the method of Newman [31] can be used with (3-69) to determine concentration changes within the slab. In this method, E or E_{avg} is given in terms of the E values from the solution of (3-68) for each of the coordinate directions by

$$E = E_x E_y E_z \quad (3-86)$$

Corresponding solutions for infinitely long, circular cylinders and spheres are available in Carslaw and Jaeger [26] and are plotted in Figures 3.9, 3.10, and 3.11, respectively. For a short cylinder, where the ends are not sealed, E or E_{ave} is given by the method of Newman as

$$E = E_r E_x \quad (3-87)$$

Some materials such as crystals and wood, have thermal conductivities and diffusivities that vary markedly with direction. For these anisotropic materials, Fick's second law in the form of (3-69) does not hold. Although the general anisotropic case is exceedingly complex, as shown in the following example, the mathematical treatment is relatively simple when the principal axes of diffusivity coincide with the coordinate system.

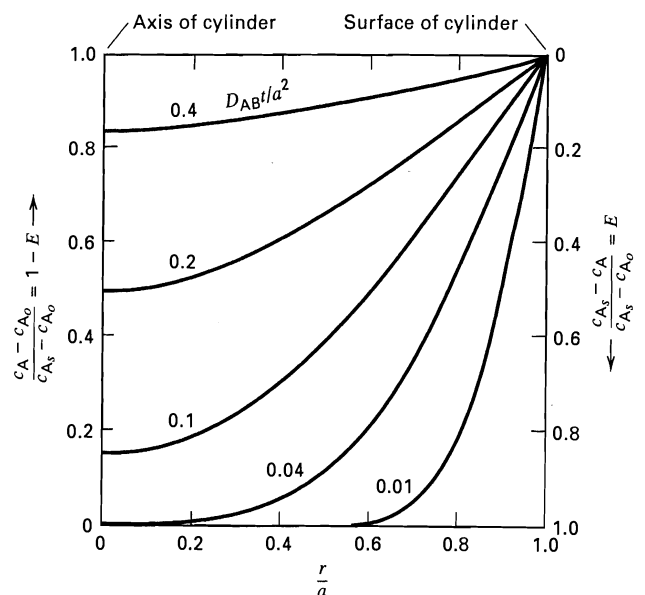


Figure 3.10 Concentration profiles for unsteady-state diffusion in a cylinder.

[Adapted from H.S. Carslaw and J.C. Jaeger, *Conduction of Heat in Solids*, 2nd ed., Oxford University Press, London (1959).]

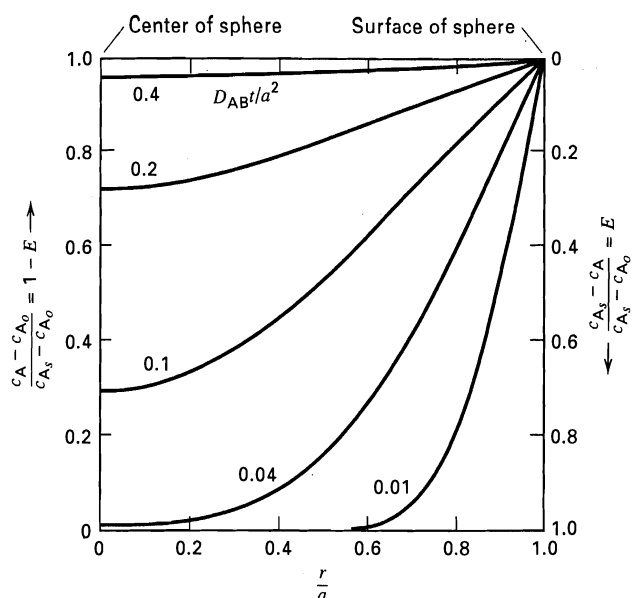


Figure 3.11 Concentration profiles for unsteady-state diffusion in a sphere.

[Adapted from H.S. Carslaw and J.C. Jaeger, *Conduction of Heat in Solids*, 2nd ed., Oxford University Press, London (1959).]

EXAMPLE 3.12

A piece of lumber, measuring $5 \times 10 \times 20$ cm, initially contains 20 wt% moisture. At time 0, all six faces are brought to an equilibrium moisture content of 2 wt%. Diffusivities for moisture at 25°C are $2 \times 10^{-5} \text{ cm}^2/\text{s}$ in the axial (z) direction along the fibers and $4 \times 10^{-6} \text{ cm}^2/\text{s}$ in the two directions perpendicular to the fibers. Calculate the time in hours for the average moisture content to drop to 5 wt% at 25°C . At that time, determine the moisture content at the center of the piece of lumber. All moisture contents are on a dry basis.

SOLUTION

In this case, the solid is anisotropic, with $D_x = D_y = 4 \times 10^{-6} \text{ cm}^2/\text{s}$ and $D_z = 2 \times 10^{-5} \text{ cm}^2/\text{s}$, where dimensions $2c$, $2b$, and $2a$ in the x , y , and z directions are 5, 10, and 20 cm, respectively. Fick's second law for an isotropic medium, (3-69), must be rewritten for this anisotropic material as

$$\frac{\partial c_A}{\partial t} = D_x \left[\frac{\partial^2 c_A}{\partial x^2} + \frac{\partial^2 c_A}{\partial y^2} \right] + D_z \frac{\partial^2 c_A}{\partial z^2} \quad (1)$$

as discussed by Carslaw and Jaeger [26].

To transform (1) into the form of (3-69), let

$$x_1 = x \sqrt{\frac{D}{D_x}} \quad y_1 = y \sqrt{\frac{D}{D_y}} \quad z_1 = z \sqrt{\frac{D}{D_z}} \quad (2)$$

where D is chosen arbitrarily. With these changes in variables, (1) becomes

$$\frac{\partial c_A}{\partial t} = D \left(\frac{\partial^2 c_A}{\partial x_1^2} + \frac{\partial^2 c_A}{\partial y_1^2} + \frac{\partial^2 c_A}{\partial z_1^2} \right) \quad (3)$$

Since this is the same form as (3-69) and since the boundary conditions do not involve diffusivities, we can apply Newman's method, using Figure 3.9, where concentration, c_A , is replaced by weight-percent moisture on a dry basis.

From (3-86) and (3-85),

$$E_{\text{ave,slab}} = E_{\text{avg}_x} E_{\text{avg}_y} E_{\text{avg}_z} = \frac{c_{A_{\text{ave}}} - c_{A_s}}{c_{A_0} - c_{A_s}} = \frac{5 - 2}{20 - 2} = 0.167$$

Let $D = 1 \times 10^{-5} \text{ cm}^2/\text{s}$.

z_1 Direction (axial):

$$a_1 = a \left(\frac{D}{D_z} \right)^{1/2} = \frac{20}{2} \left(\frac{1 \times 10^{-5}}{2 \times 10^{-5}} \right)^{1/2} = 7.07 \text{ cm}$$

$$\frac{Dt}{a_1^2} = \frac{1 \times 10^{-5} t}{7.07^2} = 2.0 \times 10^{-7} t, \text{ s}$$

y_1 Direction:

$$b_1 = b \left(\frac{D}{D_y} \right)^{1/2} = \frac{10}{2} \left(\frac{1 \times 10^{-5}}{4 \times 10^{-6}} \right)^{1/2} = 7.906 \text{ cm}$$

$$\frac{Dt}{b_1^2} = \frac{1 \times 10^{-5} t}{7.906^2} = 1.6 \times 10^{-7} t, \text{ s}$$

x_1 -Direction:

$$c_1 = c \left(\frac{D}{D_x} \right)^{1/2} = \frac{5}{2} \left(\frac{1 \times 10^{-5}}{4 \times 10^{-6}} \right)^{1/2} = 3.953 \text{ cm}$$

$$\frac{Dt}{c_1^2} = \frac{1 \times 10^{-5} t}{3.953^2} = 6.4 \times 10^{-7} t, \text{ s}$$

Use Figure 3.9 iteratively with assumed values of time in seconds to obtain values of E_{avg} for each of the three coordinates until (3-86) equals 0.167.

$t, \text{ h}$	$t, \text{ s}$	$E_{\text{avg}_{z_1}}$	$E_{\text{avg}_{y_1}}$	$E_{\text{avg}_{x_1}}$	E_{avg}
100	360,000	0.70	0.73	0.46	0.235
120	432,000	0.67	0.70	0.41	0.193
135	486,000	0.65	0.68	0.37	0.164

Therefore, it takes approximately 136 h.

For 136 h = 490,000 s, the Fourier numbers for mass transfer are

$$\frac{Dt}{a_1^2} = \frac{(1 \times 10^{-5})(490,000)}{7.07^2} = 0.0980$$

$$\frac{Dt}{b_1^2} = \frac{(1 \times 10^{-5})(490,000)}{7.906^2} = 0.0784$$

$$\frac{Dt}{c_1^2} = \frac{(1 \times 10^{-5})(490,000)}{3.953^2} = 0.3136$$

From Figure 3.8, at the center of the slab,

$$E_{\text{center}} = E_{z_1} E_{y_1} E_{x_1} = (0.945)(0.956)(0.605) = 0.547$$

$$= \frac{c_{A_s} - c_{A_{\text{center}}}}{c_{A_s} - c_{A_0}} = \frac{2 - c_{A_{\text{center}}}}{20 - 2} = 0.547$$

Solving,

$$c_A \text{ at the center} = 11.8 \text{ wt\% moisture}$$

3.4 MOLECULAR DIFFUSION IN LAMINAR FLOW

Many mass-transfer operations involve diffusion in fluids in laminar flow. The fluid may be a film flowing slowly down a vertical or inclined surface, a laminar boundary layer that forms as the fluid flows slowly past a thin plate, or the fluid may flow through a small tube or slowly through a large pipe or duct. Mass transfer may occur between a gas and a liquid film, between a solid surface and a fluid, or between a fluid and a membrane surface.

Falling Liquid Film

Consider a thin liquid film, of a mixture of volatile A and nonvolatile B, falling in laminar flow at steady state down one side of a vertical surface and exposed to pure gas, A, on the other side of the film, as shown in Figure 3.12. The surface is infinitely wide in the x -direction (normal to the page). In the absence of mass transfer of A into the liquid film, the liquid velocity in the z -direction, u_z , is zero. In the absence of end effects, the equation of motion for the liquid film in fully developed laminar flow in the downward y -direction is

$$\mu \frac{d^2 u_y}{dz^2} + \rho g = 0 \quad (3-88)$$

Usually, fully developed flow, where u_y is independent of the distance y , is established quickly. If δ is the thickness of the film and the boundary conditions are $u_y = 0$ at $z = \delta$ (no-slip condition at the solid surface) and $du_y/dz = 0$ at $z = 0$ (no drag at the liquid–gas interface), (3-88) is readily integrated to give a parabolic velocity profile:

$$u_y = \frac{\rho g \delta^2}{2\mu} \left[1 - \left(\frac{z}{\delta} \right)^2 \right] \quad (3-89)$$

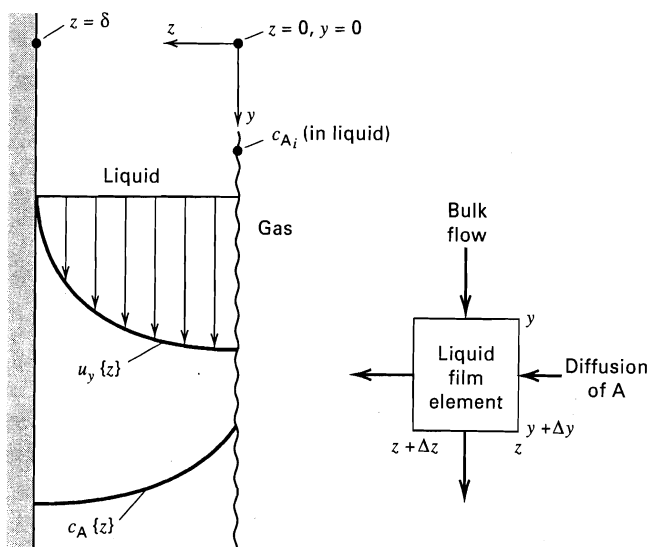


Figure 3.12 Mass transfer from a gas into a falling, laminar liquid film.

Thus, the maximum liquid velocity, which occurs at $z = 0$, is

$$(u_y)_{\max} = \frac{\rho g \delta^2}{2\mu} \quad (3-90)$$

The bulk-average velocity in the liquid film is

$$\bar{u}_y = \frac{\int_0^\delta u_y dz}{\delta} = \frac{\rho g \delta^2}{3\mu} \quad (3-91)$$

Thus, the film thickness for fully developed flow is independent of location y and is

$$\delta = \left(\frac{3\bar{u}_y \mu}{\rho g} \right)^{1/2} = \left(\frac{3\mu \Gamma}{\rho^2 g} \right)^{1/3} \quad (3-92)$$

where Γ = liquid film flow rate per unit width of film, W .

For film flow, the Reynolds number, which is the ratio of the inertial force to the viscous force, is

$$N_{\text{Re}} = \frac{4r_H \bar{u}_y \rho}{\mu} = \frac{4\delta \bar{u}_y \rho}{\mu} = \frac{4\Gamma}{\mu} \quad (3-93)$$

where r_H = hydraulic radius = (flow cross section)/(wetted perimeter) = $(W\delta)/W = \delta$ and, by the equation of continuity, $\Gamma = \bar{u}_y \rho \delta$.

As reported by Grimley [32], for $N_{\text{Re}} < 8$ to 25, depending on the surface tension and viscosity, the flow in the film is laminar and the interface between the liquid film and the gas is flat. The value of 25 is obtained with water. For $8 < 25 < N_{\text{Re}} < 1,200$, the flow is still laminar, but ripples and waves may appear at the interface unless suppressed by the addition of wetting agents to the liquid.

For a flat liquid–gas interface and a small rate of mass transfer of A into the liquid film, (3-88) to (3-93) hold and the film velocity profile is given by (3-89). Now consider a mole balance on A for an incremental volume of liquid film of constant density, as shown in Figure 3.12. Neglect bulk flow in the z -direction and axial diffusion in the y -direction. Then, at steady state, neglecting accumulation or depletion of A in the incremental volume,

$$\begin{aligned} -D_{AB}(\Delta y)(\Delta x) \left(\frac{\partial c_A}{\partial z} \right)_z + u_y c_A|_y (\Delta z)(\Delta x) \\ = -D_{AB}(\Delta y)(\Delta x) \left(\frac{\partial c_A}{\partial z} \right)_{z+\Delta z} + u_y c_A|_{y+\Delta y} (\Delta z)(\Delta x) \end{aligned} \quad (3-94)$$

Rearranging and simplifying (3-94),

$$\left[\frac{u_y c_A|_{y+\Delta y} - u_y c_A|_y}{\Delta y} \right] = D_{AB} \left[\frac{(\partial c_A / \partial z)_{z+\Delta z} - (\partial c_A / \partial z)_z}{\Delta z} \right] \quad (3-95)$$

In the limit, as $\Delta z \rightarrow 0$ and $\Delta y \rightarrow 0$,

$$u_y \frac{\partial c_A}{\partial y} = D_{AB} \frac{\partial^2 c_A}{\partial z^2} \quad (3-96)$$

Substituting (3-89) into (3-96),

$$\frac{\rho g \delta^2}{2\mu} \left[1 - \left(\frac{z}{\delta} \right)^2 \right] \frac{\partial c_A}{\partial y} = D_{AB} \frac{\partial^2 c_A}{\partial z^2} \quad (3-97)$$

This equation was solved by Johnstone and Pigford [33] and later by Olbrich and Wild [34], for the following boundary conditions:

$$\begin{aligned} c_A &= c_{A_i} \quad \text{at } z = 0 \quad \text{for } y > 0 \\ c_A &= c_{A_0} \quad \text{at } y = 0 \quad \text{for } 0 < z < \delta \\ \partial c_A / \partial z &= 0 \quad \text{at } z = \delta \quad \text{for } 0 < y < L \end{aligned}$$

where L = height of the vertical surface. The solution of Olbrich and Wild is in the form of an infinite series, giving c_A as a function of z and y . However, of more interest is the average concentration at $y = L$, which, by integration, is

$$\bar{c}_{A_y} = \frac{1}{\bar{u}_y \delta} \int_0^\delta u_y c_{A_y} dz \quad (3-98)$$

For the condition $y = L$, the result is

$$\frac{c_{A_i} - \bar{c}_{A_L}}{c_{A_i} - c_{A_0}} = 0.7857e^{-5.1213\eta} + 0.09726e^{-39.661\eta} + 0.036093e^{-106.25\eta} + \dots \quad (3-99)$$

where

$$\eta = \frac{2D_{AB}L}{3\delta^2\bar{u}_y} = \frac{8/3}{N_{Re}N_{Sc}(\delta/L)} = \frac{8/3}{(\delta/L)N_{Pe_M}} \quad (3-100)$$

$$\begin{aligned} N_{Sc} &= \text{Schmidt number} = \frac{\mu}{\rho D_{AB}} \\ &= \frac{\text{momentum diffusivity, } \mu/\rho}{\text{mass diffusivity, } D_{AB}} \end{aligned} \quad (3-101)$$

$$\begin{aligned} N_{Pe_M} &= N_{Re}N_{Sc} = \text{Peclet number for mass transfer} \\ &= \frac{4\delta\bar{u}_y}{D_{AB}} \end{aligned} \quad (3-102)$$

The Schmidt number is analogous to the Prandtl number, used in heat transfer:

$$N_{Pr} = \frac{C_P\mu}{k} = \frac{(\mu/\rho)}{(k/\rho C_P)} = \frac{\text{momentum diffusivity}}{\text{thermal diffusivity}}$$

The Peclet number for mass transfer is analogous to the Peclet number for heat transfer:

$$N_{Pe_H} = N_{Re}N_{Pr} = \frac{4\delta\bar{u}_y C_P \rho}{k}$$

Both Peclet numbers are ratios of convective transport to molecular transport.

The total rate of absorption of A from the gas into the liquid film for height L and width W is

$$n_A = \bar{u}_y \delta W (\bar{c}_{A_L} - c_{A_0}) \quad (3-103)$$

Mass-Transfer Coefficients

Mass-transfer problems involving fluids are most often solved using mass-transfer coefficients, analogous to heat-transfer coefficients. For the latter, *Newton's law of cooling* defines a heat-transfer coefficient, h :

$$Q = hA \Delta T \quad (3-104)$$

where

Q = rate of heat transfer

A = area for heat transfer (normal to the direction of heat transfer)

ΔT = temperature-driving force for heat transfer

For mass transfer, a composition driving force replaces ΔT . As discussed later in this chapter, because composition can be expressed in a number of ways, different mass-transfer coefficients are defined. If we select Δc_A as the driving force for mass transfer, we can write

$$n_A = k_c A \Delta c_A \quad (3-105)$$

which defines a mass-transfer coefficient, k_c , in mol/time-area-driving force, for a concentration driving force. Unfortunately, no name is in general use for (3-105).

For the falling laminar film, we take $\Delta c_A = c_{A_i} - \bar{c}_A$, which varies with vertical location, y , because even though c_{A_i} is independent of y , the average film concentration, $\bar{c}_A$, increases with y . To derive an expression for k_c , we equate (3-105) to Fick's first law at the gas-liquid interface:

$$k_c A (c_{A_i} - \bar{c}_A) = -D_{AB} A \left(\frac{\partial c_A}{\partial z} \right)_{z=0} \quad (3-106)$$

Although this is the most widely used approach for defining a mass-transfer coefficient, in this case of a falling film it fails because $(\partial c_A / \partial z)$ at $z = 0$ is not defined. Therefore, for this case we use another approach as follows. For an incremental height, we can write for film width W ,

$$n_A = \bar{u}_y \delta W d\bar{c}_A = k_c (c_{A_i} - \bar{c}_A) W dy \quad (3-107)$$

This defines a local value of k_c , which varies with distance y because $\bar{c}_A$ varies with y . An average value of k_c , over a height L , can be defined by separating variables and integrating (3-107):

$$\begin{aligned} k_{c_{avg}} &= \frac{\int_0^L k_c dy}{L} = \frac{\bar{u}_y \delta \int_{c_{A_0}}^{c_{A_L}} [d\bar{c}_A / (c_{A_i} - \bar{c}_A)]}{L} \\ &= \frac{\bar{u}_y \delta}{L} \ln \frac{c_{A_i} - c_{A_0}}{c_{A_i} - \bar{c}_{A_L}} \end{aligned} \quad (3-108)$$

In general, the argument of the natural logarithm in (3-108) is obtained from the reciprocal of (3-99). For values of η in (3-100) greater than 0.1, only the first term in (3-99) is significant (error is less than 0.5%). In that case,

$$k_{c_{avg}} = \frac{\bar{u}_y \delta}{L} \ln \frac{e^{5.1213\eta}}{0.7857} \quad (3-109)$$

Since $\ln e^x = x$,

$$k_{c_{avg}} = \frac{\bar{u}_y \delta}{L} (0.241 + 5.1213\eta) \quad (3-110)$$

In the limit, for large η , using (3-100) and (3-102), (3-110) becomes

$$k_{c_{avg}} = 3.414 \frac{D_{AB}}{\delta} \quad (3-111)$$

In a manner suggested by the Nusselt number, $N_{Nu} = h\delta/k$ for heat transfer, where δ = a characteristic length, we define a Sherwood number for mass transfer, which for a falling film of characteristic length δ is

$$N_{Sh_{avg}} = \frac{k_{c_{avg}} \delta}{D_{AB}} \quad (3-112)$$

From (3-111), $N_{Sh_{avg}} = 3.414$, which is the smallest value that the Sherwood number can have for a falling liquid film.

The average mass-transfer flux of A is given by

$$N_{A_{avg}} = \frac{n_{A_{avg}}}{A} = k_{c_{avg}}(c_{A_i} - \bar{c}_A)_{mean} \quad (3-113)$$

For values $\eta < 0.001$ in (3-100), when the liquid-film flow regime is still laminar without ripples, the time of contact of the gas with the liquid is short and mass transfer is confined to the vicinity of the gas-liquid interface. Thus, the film acts as if it were infinite in thickness. In this limiting case, the downward velocity of the liquid film in the region of mass transfer is just $u_{y_{max}}$, and (3-96) becomes

$$u_{y_{max}} \frac{\partial c_A}{\partial y} = D_{AB} \frac{\partial^2 c_A}{\partial z^2} \quad (3-114)$$

Since from (3-90) and (3-91), $u_{y_{max}} = 3\bar{u}_y/2$, (3-114) can be rewritten as

$$\frac{\partial c_A}{\partial y} = \left(\frac{2D_{AB}}{3\bar{u}_y} \right) \frac{\partial^2 c_A}{\partial z^2} \quad (3-115)$$

where the boundary conditions are

$$c_A = c_{A_0} \quad \text{for } z > 0 \quad \text{and } y > 0$$

$$c_A = c_{A_i} \quad \text{for } z = 0 \quad \text{and } y > 0$$

$$c_A = c_{A_0} \quad \text{for large } z \quad \text{and } y > 0$$

Equation (3-115) and the boundary conditions are equivalent to the case of the semi-infinite medium, as developed above. Thus, by analogy to (3-68), (3-75), and (3-76) the solution is

$$E = 1 - \theta = \frac{c_{A_i} - c_A}{c_{A_i} - c_{A_0}} = \text{erf} \left(\frac{z}{2\sqrt{2D_{AB}y/3\bar{u}_y}} \right) \quad (3-116)$$

Assuming that the driving force for mass transfer in the film is $c_{A_i} - c_{A_0}$, we can use Fick's first law at the gas-liquid interface to define a mass-transfer coefficient:

$$N_A = -D_{AB} \left. \frac{\partial c_A}{\partial z} \right|_{z=0} = k_c(c_{A_i} - c_{A_0}) \quad (3-117)$$

The error function is defined as

$$\text{erf } z = \frac{2}{\sqrt{\pi}} \int_0^z e^{-t^2} dt \quad (3-118)$$

Using the Leibnitz rule with (3-116) to differentiate this integral function,

$$\left. \frac{\partial c_A}{\partial z} \right|_{z=0} = -(c_{A_i} - c_{A_0}) \sqrt{\frac{3\bar{u}_y}{2\pi D_{AB}y}} \quad (3-119)$$

Substituting (3-119) into (3-117) and introducing the Peclet number for mass transfer from (3-102), we obtain an expression for the local mass-transfer coefficient as a function of distance down from the top of the wall:

$$k_c = \sqrt{\frac{3D_{AB}^2 N_{Pe_M}}{8\pi y \delta}} = \sqrt{\frac{3D_{AB}\Gamma}{2\pi y \delta \rho}} \quad (3-120)$$

The average value of k_c over the height of the film, L , is obtained by integrating (3-120) with respect to y , giving

$$k_{c_{avg}} = \sqrt{\frac{6D_{AB}\Gamma}{\pi \delta \rho L}} = \sqrt{\frac{3D_{AB}^2 N_{Pe_M}}{2\pi \delta L}} \quad (3-121)$$

Combining (3-121) with (3-112) and (3-102),

$$N_{Sh_{avg}} = \sqrt{\frac{3\delta}{2\pi L}} N_{Pe_M} = \sqrt{\frac{4}{\pi \eta}} \quad (3-122)$$

where, by (3-108), the proper mean to use with $k_{c_{avg}}$ is the log mean. Thus,

$$\begin{aligned} (c_{A_i} - \bar{c}_A)_{mean} &= (c_{A_i} - \bar{c}_A)_{LM} \\ &= \frac{(c_{A_i} - c_{A_0}) - (c_{A_i} - c_{A_L})}{\ln[(c_{A_i} - c_{A_0})/(c_{A_i} - \bar{c}_A)]} \end{aligned} \quad (3-123)$$

When ripples are present, values of $k_{c_{avg}}$ and $N_{Sh_{avg}}$ can be considerably larger than predicted by these equations.

In the above development, asymptotic, closed-form solutions are obtained with relative ease for large and small values of η , defined by (3-100). These limits, in terms of the average Sherwood number, are shown in Figure 3.13. The

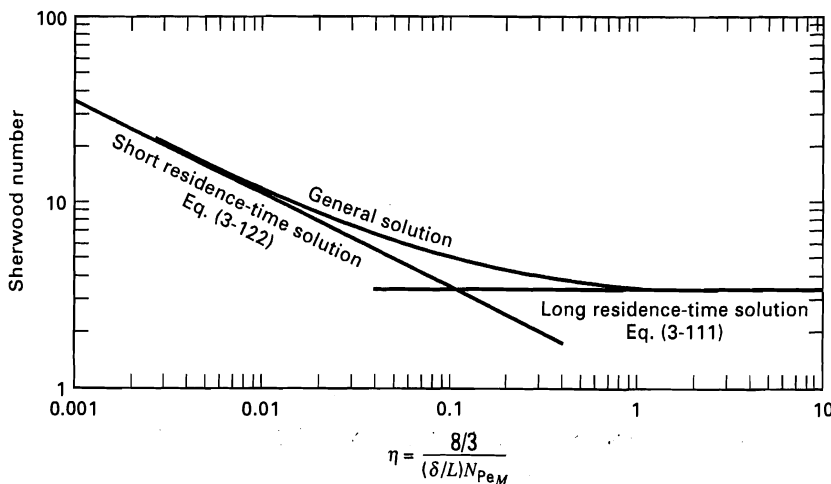


Figure 3.13 Limiting and general solutions for mass transfer to a falling, laminar liquid film.

general solution for intermediate values of η is not available in closed form. Similar limiting solutions for large and small values of appropriate parameters, usually dimensionless groups, have been obtained for a large variety of transport and kinetic phenomena, as discussed by Churchill [35]. Often the two limiting cases can be patched together to provide a reasonable estimate of the intermediate solution, if a single intermediate value is available from experiment or the general numerical solution. The procedure is discussed by Churchill and Usagi [36]. The general solution of Emmert and Pigford [37] to the falling, laminar liquid film problem is included in Figure 3.13.

EXAMPLE 3.13

Water (B) at 25°C, in contact with pure CO₂ (A) at 1 atm, flows as a film down a vertical wall 1 m wide and 3 m high at a Reynolds number of 25. Using the following properties, estimate the rate of absorption of CO₂ into water in kmol/s:

$$D_{AB} = 1.96 \times 10^{-5} \text{ cm}^2/\text{s}; \quad \rho = 1.0 \text{ g/cm}^3; \\ \mu_L = 0.89 \text{ cP} = 0.00089 \text{ kg/m-s}$$

Solubility of CO₂ in water at 1 atm and 25°C = $3.4 \times 10^{-5} \text{ mol/cm}^3$.

SOLUTION

From (3-93),

$$\Gamma = \frac{N_{\text{Re}} \mu}{4} = \frac{25(0.89)(0.001)}{4} = 0.00556 \frac{\text{kg}}{\text{m-s}}$$

From (3-101),

$$N_{\text{Sc}} = \frac{\mu}{\rho D_{AB}} = \frac{(0.89)(0.001)}{(1.0)(1,000)(1.96 \times 10^{-5})(10^{-4})} = 454$$

From (3-92),

$$\delta = \left[\frac{3(0.89)(0.001)(0.00556)}{1.0^2(1,000)^2(9.807)} \right]^{1/3} = 1.15 \times 10^{-4} \text{ m}$$

From (3-90) and (3-91), $\bar{u}_y = (2/3)u_{y,\text{max}}$. Therefore,

$$\bar{u}_y = \frac{2}{3} \left[\frac{(1.0)(1,000)(9.807)(1.15 \times 10^{-4})^2}{2(0.89)(0.001)} \right] = 0.0486 \text{ m/s}$$

From (3-100),

$$\eta = \frac{8/3}{(25)(454)[(1.15 \times 10^{-4})/3]} = 6.13$$

Therefore, (3-111) applies, giving

$$k_{c,\text{avg}} = \frac{3.41(1.96 \times 10^{-5})(10^{-4})}{1.15 \times 10^{-4}} = 5.81 \times 10^{-5} \text{ m/s}$$

To determine the rate of absorption, $\bar{c}_{A_L}$ must be determined. From (3-103) and (3-113),

$$n_A = \bar{u}_y \delta W (\bar{c}_{A_L} - c_{A_0}) = k_{c,\text{avg}} A \frac{(\bar{c}_{A_L} - c_{A_0})}{\ln[(c_{A_i} - c_{A_0})/(c_{A_i} - c_{A_L})]}$$

Thus,

$$\ln \left(\frac{c_{A_i} - c_{A_0}}{c_{A_i} - \bar{c}_{A_L}} \right) = \frac{k_{c,\text{avg}} A}{\bar{u}_y \delta W}$$

Solving for $\bar{c}_{A_L}$,

$$\bar{c}_{A_L} = c_{A_i} - (c_{A_i} - c_{A_0}) \exp \left(- \frac{k_{c,\text{avg}} A}{\bar{u}_y \delta W} \right)$$

$$L = 3 \text{ m}, \quad W = 1 \text{ m}, \quad A = WL = (1)(3) = 3 \text{ m}^2 \\ c_{A_0} = 0, \quad c_{A_i} = 3.4 \times 10^{-5} \text{ mol/cm}^3 = 3.4 \times 10^{-2} \text{ kmol/m}^3 \\ \bar{c}_{A_L} = 3.4 \times 10^{-2} \left\{ 1 - \exp \left[- \frac{(5.81 \times 10^{-5})(3)}{(0.0486)(1.15 \times 10^{-4})(1)} \right] \right\} \\ = 3.4 \times 10^{-2} \text{ kmol/m}^3$$

Thus, the exiting liquid film is saturated with CO₂, which implies equilibrium at the gas-liquid interface. From (3-103),

$$n_A = 0.0486(1.15 \times 10^{-4})(3.4 \times 10^{-2}) = 1.9 \times 10^{-7} \text{ kmol/s}$$

Boundary-Layer Flow on a Flat Plate

Consider the flow of a fluid (B) over a thin, flat plate parallel with the direction of flow of the fluid upstream of the plate, as shown in Figure 3.14. A number of possibilities for mass transfer of another species, A, into B exist: (1) The plate might consist of material A, which is slightly soluble in B. (2) Component A might be held in the pores of an inert solid plate, from which it evaporates or dissolves into B. (3) The plate might be an inert, dense polymeric membrane, through which species A can pass into fluid B. Let the fluid velocity profile upstream of the plate be uniform at a free-system velocity of u_0 . As the fluid passes over the plate, the velocity u_x in the direction of flow is reduced to zero at the wall, which establishes a velocity profile due to drag. At a certain distance z , normal to and out from the solid surface, the fluid velocity is 99% of u_0 . This distance, which increases with increasing distance x from the leading edge of the plate, is arbitrarily defined as the velocity boundary-layer thickness, δ . Essentially all flow retardation occurs in the boundary layer, as first suggested by Prandtl [38]. The buildup of this layer, the velocity profile in the layer, and the drag force can be determined for laminar flow by solving the equations of continuity and motion (Navier-Stokes equations) for the x -direction. For a Newtonian fluid of constant density and viscosity, in the absence of pressure gradients in the x - and

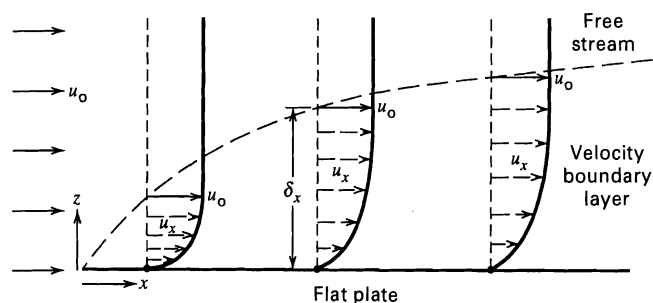


Figure 3.14 Laminar boundary-layer development for flow across a flat plate.

y - (normal to the x - z plane) directions, these equations for the region of the boundary layer are

$$\frac{\partial u_x}{\partial x} + \frac{\partial u_z}{\partial z} = 0 \quad (3-124)$$

$$u_x \frac{\partial u_x}{\partial x} + u_z \frac{\partial u_x}{\partial z} = \frac{\mu}{\rho} \left(\frac{\partial^2 u_x}{\partial z^2} \right) \quad (3-125)$$

The boundary conditions are

$$\begin{aligned} u_x &= u_o \text{ at } x = 0 \text{ for } z > 0 & u_x &= 0 \text{ at } z = 0 \text{ for } x > 0 \\ u_z &= u_o \text{ at } z = \infty \text{ for } x > 0 & u_z &= 0 \text{ at } z = 0 \text{ for } x > 0 \end{aligned}$$

The solution of (3-124) and (3-125) in the absence of heat and mass transfer, subject to these boundary conditions, was first obtained by Blasius [39] and is described in detail by Schlichting [40]. The result in terms of a local friction factor, f_x , a local shear stress at the wall, τ_{wx} , and a local drag coefficient at the wall, C_{D_x} , is

$$\frac{C_{D_x}}{2} = \frac{f_x}{2} = \frac{\tau_{wx}}{\rho u_o^2} = \frac{0.322}{N_{Re_x}^{0.5}} \quad (3-126)$$

where

$$N_{Re_x} = \frac{x u_o \rho}{\mu} \quad (3-127)$$

Thus, the drag is greatest at the leading edge of the plate, where the Reynolds number is smallest. Average values of the drag coefficient are obtained by integrating (3-126) from $x = 0$ to L , giving

$$\frac{C_{D_{avg}}}{2} = \frac{f_{avg}}{2} = \frac{0.664}{(N_{Re_L})^{0.5}} \quad (3-128)$$

The thickness of the velocity boundary layer increases with distance along the plate:

$$\frac{\delta}{x} = \frac{4.96}{N_{Re_x}^{0.5}} \quad (3-129)$$

A reasonably accurate expression for the velocity profile was obtained by Pohlhausen [41], who assumed the empirical form $u_x = C_1 z + C_2 z^3$.

If the boundary conditions,

$$u_x = 0 \text{ at } z = 0 \quad u_x = u_o \text{ at } z = \delta \quad \partial u_x / \partial z = 0 \text{ at } z = \delta$$

are applied to evaluate C_1 and C_2 , the result is

$$\frac{u_x}{u_o} = 1.5 \left(\frac{z}{\delta} \right) - 0.5 \left(\frac{z}{\delta} \right)^3 \quad (3-130)$$

This solution is valid only for a laminar boundary layer, which by experiment persists to $N_{Re_x} = 5 \times 10^5$.

When mass transfer of A into the boundary layer occurs, the following species continuity equation applies at constant diffusivity:

$$u_x \frac{\partial c_A}{\partial x} + u_z \frac{\partial c_A}{\partial z} = D_{AB} \left(\frac{\partial^2 c_A}{\partial z^2} \right) \quad (3-131)$$

If mass transfer begins at the leading edge of the plate and if the concentration in the fluid at the solid-fluid interface is constant, the additional boundary conditions are

$$\begin{aligned} c_A &= c_{A_o} \text{ at } x = 0 \text{ for } z > 0, \\ c_A &= c_{A_i} \text{ at } z = 0 \text{ for } x > 0, \\ \text{and } c_A &= c_{A_o} \text{ at } z = \infty \text{ for } x > 0 \end{aligned}$$

If the rate of mass transfer is low, the velocity profiles are undisturbed. The solution to the analogous problem in heat transfer was first obtained by Pohlhausen [42] for $N_{Pr} > 0.5$, as described in detail by Schlichting [40]. The results for mass transfer are

$$\frac{N_{Sh_x}}{N_{Re_x} N_{Sc}^{1/3}} = \frac{0.332}{N_{Re_x}^{0.5}} \quad (3-132)$$

where

$$N_{Sh_x} = \frac{x k_{c_x}}{D_{AB}} \quad (3-133)$$

and the driving force for mass transfer is $c_{A_i} - c_{A_o}$.

The concentration boundary layer, where essentially all of the resistance to mass transfer resides, is defined by

$$\frac{c_{A_i} - c_A}{c_{A_i} - c_{A_o}} = 0.99 \quad (3-134)$$

and the ratio of the concentration boundary-layer thickness, δ_c , to the velocity boundary thickness, δ , is

$$\delta_c / \delta = 1 / N_{Sc}^{1/3} \quad (3-135)$$

Thus, for a liquid boundary layer, where $N_{Sc} > 1$, the concentration boundary layer builds up more slowly than the velocity boundary layer. For a gas boundary layer, where $N_{Sc} \approx 1$, the two boundary layers build up at about the same rate. By analogy to (3-130), the concentration profile is given by

$$\frac{c_{A_i} - c_A}{c_{A_i} - c_{A_o}} = 1.5 \left(\frac{z}{\delta_c} \right) - 0.5 \left(\frac{z}{\delta_c} \right)^3 \quad (3-136)$$

Equation (3-132) gives the local Sherwood number. If this expression is integrated over the length of the plate, L , the average Sherwood number is found to be

$$N_{Sh_{avg}} = 0.664 N_{Re_L}^{1/2} N_{Sc}^{1/3} \quad (3-137)$$

where

$$N_{Sh_{avg}} = \frac{L k_{c_{avg}}}{D_{AB}} \quad (3-138)$$

EXAMPLE 3.14

Air at 100°C, 1 atm, and a free-stream velocity of 5 m/s flows over a 3-m-long, thin, flat plate of naphthalene, causing it to sublime.

- Determine the length over which a laminar boundary layer persists.
- For that length, determine the rate of mass transfer of naphthalene into air.
- At the point of transition of the boundary layer to turbulent flow, determine the thicknesses of the velocity and concentration boundary layers.

Assume the following values for physical properties:

Vapor pressure of naphthalene = 10 torr

Viscosity of air = 0.0215 cP

Molar density of air = 0.0327 kmol/m³

Diffusivity of naphthalene in air = 0.94×10^{-5} m²/s

SOLUTION

(a) $N_{Re_x} = 5 \times 10^5$ for transition. From (3-127),

$$x = L = \frac{\mu N_{Re_x}}{u_{op}} = \frac{[(0.0215)(0.001)](5 \times 10^5)}{(5)[(0.0327)(29)]} = 2.27 \text{ m}$$

at which transition to turbulent flow begins.

$$(b) \ c_{A_0} = 0 \quad c_{A_i} = \frac{10(0.0327)}{760} = 4.3 \times 10^{-4} \text{ kmol/m}^3$$

From (3-101),

$$N_{Sc} = \frac{\mu}{\rho D_{AB}} = \frac{[(0.0215)(0.001)]}{[(0.0327)(29)](0.94 \times 10^{-5})} = 2.41$$

From (3-137),

$$N_{Sh_{avg}} = 0.664(5 \times 10^5)^{1/2}(2.41)^{1/3} = 630$$

From (3-138),

$$k_{c_{avg}} = \frac{630(0.94 \times 10^{-5})}{2.27} = 2.61 \times 10^{-3} \text{ m/s}$$

For a width of 1 m,

$$A = 2.27 \text{ m}^2$$

$$\begin{aligned} n_A &= k_{c_{avg}} A (c_{A_i} - c_{A_0}) = 2.61 \times 10^{-3} (2.27) (4.3 \times 10^{-4}) \\ &= 2.55 \times 10^{-6} \text{ kmol/s} \end{aligned}$$

(c) From (3-129), at $x = L = 2.27$ m,

$$\delta = \frac{3.46(2.27)}{(5 \times 10^5)^{0.5}} = 0.0111 \text{ m}$$

From (3-135),

$$\delta_c = \frac{0.0111}{(2.41)^{1/3}} = 0.0083 \text{ m}$$

build up as shown at planes *b*, *c*, and *d*. In this region, the central core outside the boundary layer has a flat velocity profile where the flow is accelerated over the entrance velocity. Finally, at plane *e*, the boundary layer fills the tube. From here the velocity profile is fixed and the flow is said to be fully developed. The distance from the plane *a* to plane *e* is the entry region.

For fully developed laminar flow in a straight, circular tube, by experiment, the Reynolds number, $N_{Re} = D\bar{u}_x\rho/\mu$, where $\bar{u}_x$ is the flow-average velocity in the axial direction, x , and D is the inside diameter of the tube, must be less than 2,100. For this condition, the equation of motion in the axial direction for horizontal flow and constant properties is

$$\frac{\mu}{r} \frac{\partial}{\partial r} \left(r \frac{\partial u_x}{\partial r} \right) - \frac{dP}{dx} = 0 \quad (3-139)$$

where the boundary conditions are

$$r = 0 \text{ (axis of the tube), } \partial u_x / \partial r = 0$$

$$\text{and } r = r_w \text{ (tube wall), } u_x = 0$$

Equation (3-139) was integrated by Hagen in 1839 and Poiseuille in 1841. The resulting equation for the velocity profile, expressed in terms of the flow-average velocity, is

$$u_x = 2\bar{u}_x \left[1 - \left(\frac{r}{r_w} \right)^2 \right] \quad (3-140)$$

or, in terms of the maximum velocity at the tube axis,

$$u_x = u_{x_{max}} \left[1 - \left(\frac{r}{r_w} \right)^2 \right] \quad (3-141)$$

From the form of (3-141), the velocity profile is parabolic in nature.

The shear stress, pressure drop, and Fanning friction factor are obtained from solutions to (3-139):

$$\tau_w = -\mu \left(\frac{\partial u_x}{\partial r} \right) \bigg|_{r=r_w} = \frac{4\mu\bar{u}_x}{r_w} \quad (3-142)$$

$$-\frac{dP}{dx} = \frac{32\mu\bar{u}_x}{D^2} = \frac{2f\rho\bar{u}_x^2}{D} \quad (3-143)$$

with

$$f = \frac{16}{N_{Re}} \quad (3-144)$$

Fully Developed Flow in a Straight, Circular Tube

Figure 3.15 shows the formation and buildup of a laminar velocity boundary layer when a fluid flows from a vessel into a straight, circular tube. At the entrance, plane *a*, the velocity profile is flat. A velocity boundary layer then begins to

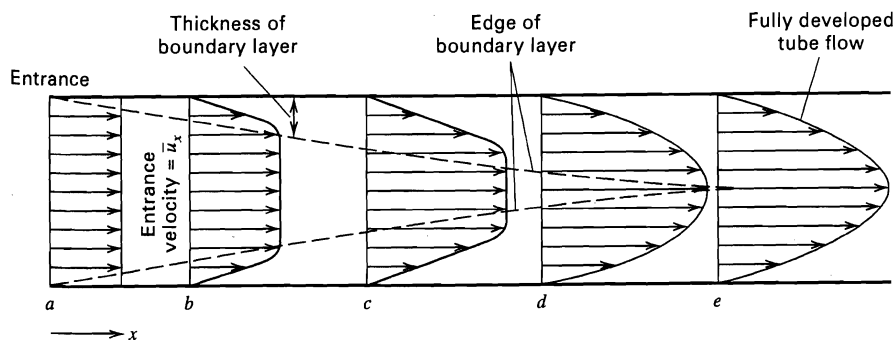


Figure 3.15 Buildup of a laminar velocity boundary layer for flow in a straight, circular tube.

The entry length to achieve fully developed flow is defined as the axial distance, L_e , from the entrance to the point at which the centerline velocity is 99% of the fully developed flow value. From the analysis of Langhaar [43] for the entry region,

$$\frac{L_e}{D} = 0.0575 N_{Re} \quad (3-145)$$

Thus, at the upper limit of laminar flow, $N_{Re} = 2,100$, $L_e/D = 121$, a rather large ratio. For $N_{Re} = 100$, the ratio is only 5.75. In the entry region, Langhaar's analysis shows the friction factor is considerably higher than the fully developed flow value given by (3-144). At $x = 0$, f is infinity, but then decreases exponentially with x , approaching the fully developed flow value at L_e . For example, for $N_{Re} = 1,000$, (3-144) gives $f = 0.016$, with $L_e/D = 57.5$. In the region from $x = 0$ to $x/D = 5.35$, the average friction factor from Langhaar is 0.0487, which is about three times higher than the fully developed value.

In 1885, Graetz [44] obtained a theoretical solution to the problem of convective heat transfer between the wall of a circular tube, held at a constant temperature, and a fluid flowing through the tube in fully developed laminar flow. Assuming constant properties and negligible conduction in the axial direction, the energy equation, after substituting (3-140) for u_x , is

$$2\bar{u}_x \left[1 - \left(\frac{r}{r_w} \right)^2 \right] \frac{\partial T}{\partial x} = \frac{k}{\rho C_p} \left[\frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial T}{\partial r} \right) \right] \quad (3-146)$$

The boundary conditions are

$$x = 0 \text{ (where heat transfer begins), } T = T_0, \text{ for all } r$$

$$x > 0, \quad r = r_w, \quad T = T_i \quad x > 0, \quad r = 0, \quad \partial T / \partial r = 0$$

The analogous species continuity equation for mass transfer, neglecting bulk flow in the radial direction and diffusion in the axial direction, is

$$2\bar{u}_x \left[1 - \left(\frac{r}{r_w} \right)^2 \right] \frac{\partial c_A}{\partial x} = D_{AB} \left[\frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial c_A}{\partial r} \right) \right] \quad (3-147)$$

with analogous boundary conditions.

The Graetz solution of (3-147) for the temperature profile or the concentration profile is in the form of an infinite series, and can be obtained from (3-146) by the method of separation of variables using the method of Frobenius. A detailed solution is given by Sellars, Tribus, and Klein [45]. From the concentration profile, expressions for the mass-transfer coefficient and the Sherwood number are obtained. When x is large, the concentration profile is fully developed and the local Sherwood number, N_{Sh_x} , approaches a limiting value of 3.656. At the other extreme, when x is small such that the concentration boundary layer is very thin and confined to a region where the fully developed velocity profile is linear, the local Sherwood number is obtained from the classic Leveque [46] solution, presented by Knudsen and Katz [47]:

$$N_{Sh_x} = \frac{k_{c_x} D}{D_{AB}} = 1.077 \left[\frac{N_{PeM}}{(x/D)} \right]^{1/3} \quad (3-148)$$

where

$$N_{PeM} = \frac{D \bar{u}_x}{D_{AB}} \quad (3-149)$$

The limiting solutions, together with the general Graetz solution, are shown in Figure 3.16, where it is seen that $N_{Sh_x} = 3.656$ is valid for $N_{PeM}/(x/D) < 4$ and (3-148) is valid for $N_{PeM}/(x/D) > 100$. The two limiting solutions can be patched together if one point of the general solution is available where the two solutions intersect.

Over a length of tube where mass transfer occurs, an average Sherwood number can be derived by integrating the general expression for the local Sherwood number. An empirical representation for that average, proposed by Hausen [48], is

$$N_{Sh_{avg}} = 3.66 + \frac{0.0668 [N_{PeM}/(x/D)]}{1 + 0.04 [N_{PeM}/(x/D)]^{2/3}} \quad (3-150)$$

which is based on a log-mean concentration driving force.

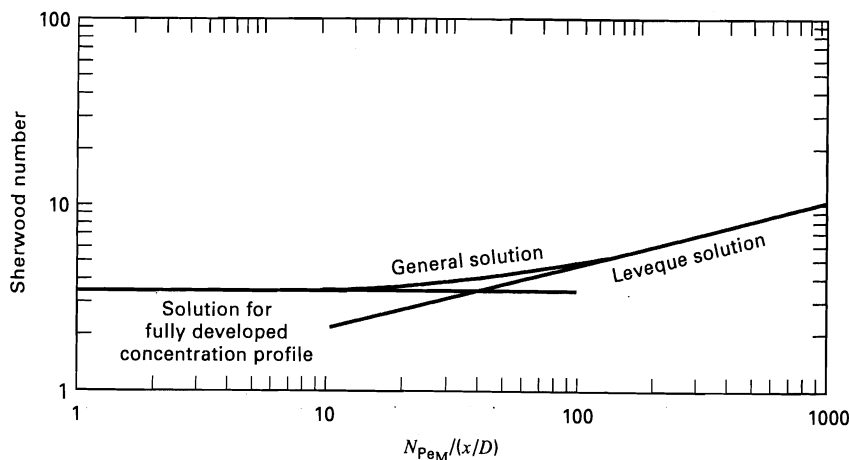


Figure 3.16 Limiting and general solutions for mass transfer to a fluid in laminar flow in a straight, circular tube.

EXAMPLE 3.15

Linton and Sherwood [49] conducted experiments on the dissolution of cast tubes of benzoic acid (A) into water (B) flowing through the tubes in laminar flow. They obtained good agreement with predictions based on the Graetz and Leveque equations. Consider a 5.23-cm-inside-diameter by 32-cm-long tube of benzoic acid, preceded by 400 cm of straight metal pipe of the same inside diameter where a fully developed velocity profile is established. Pure water enters the system at 25°C at a velocity corresponding to a Reynolds number of 100. Based on the following property data at 25°C, estimate the average concentration of benzoic acid in the water leaving the cast tube before a significant increase in the inside diameter of the benzoic acid tube occurs because of dissolution.

Solubility of benzoic acid in water = 0.0034 g/cm³

Viscosity of water = 0.89 cP = 0.0089 g/cm-s

Diffusivity of benzoic acid in water at infinite dilution
= 9.18×10^{-6} cm²/s

SOLUTION

$$N_{Sc} = \frac{0.0089}{(1.0)(9.18 \times 10^{-6})} = 970$$

$$N_{Re} = \frac{D\bar{u}_x\rho}{\mu} = 100$$

from which

$$\bar{u}_x = \frac{(100)(0.0089)}{(5.23)(1.0)} = 0.170 \text{ cm/s}$$

From (3-149),

$$N_{PeM} = \frac{(5.23)(0.170)}{9.18 \times 10^{-6}} = 9.69 \times 10^4$$

$$\frac{x}{D} = \frac{32}{5.23} = 6.12$$

$$\frac{N_{PeM}}{(x/D)} = \frac{9.69 \times 10^4}{6.12} = 1.58 \times 10^4$$

From (3-150),

$$N_{Shav} = 3.66 + \frac{0.0668(1.58 \times 10^4)}{1 + 0.04(1.58 \times 10^4)^{2/3}} = 44$$

$$k_{cav} = N_{Shav} \left(\frac{D_{AB}}{D} \right) = 44 \frac{(9.18 \times 10^{-6})}{5.23} = 7.7 \times 10^{-5} \text{ cm/s}$$

Using a log-mean driving force,

$$n_A = \bar{u}_x S (\bar{c}_{A_x} - c_{A_0}) = k_{cav} A \frac{[(c_{A_i} - c_{A_0}) - (c_{A_i} - \bar{c}_{A_x})]}{\ln[(c_{A_i} - c_{A_0})/(c_{A_i} - \bar{c}_{A_x})]}$$

where S is the cross-sectional area for flow. Simplifying,

$$\ln \left(\frac{c_{A_i} - c_{A_0}}{c_{A_i} - \bar{c}_{A_x}} \right) = \frac{k_{cav} A}{\bar{u}_x S}$$

$$c_{A_0} = 0 \quad \text{and} \quad c_{A_i} = 0.0034 \text{ g/cm}^3$$

$$S = \frac{\pi D^2}{4} = \frac{(3.14)(5.23)^2}{4} = 21.5 \text{ cm}^2 \quad \text{and}$$

$$A = \pi D x = (3.14)(5.23)(32) = 526 \text{ cm}^2$$

$$\ln \left(\frac{0.0034}{0.0034 - \bar{c}_{A_x}} \right) = \frac{(7.7 \times 10^{-5})(526)}{(0.170)(21.5)}$$

$$= 0.0111$$

$$\bar{c}_{A_x} = 0.0034 - \frac{0.0034}{e^{0.0111}} = 0.000038 \text{ g/cm}^3$$

Thus, the concentration of benzoic acid in the water leaving the cast tube is far from saturation.

3.5 MASS TRANSFER IN TURBULENT FLOW

In the two previous sections, diffusion in stagnant media and in laminar flow were considered. For both cases, Fick's law can be applied to obtain rates of mass transfer. A more common occurrence in engineering is turbulent flow, which is accompanied by much higher transport rates, but for which theory is still under development and the estimation of mass-transfer rates relies more on empirical correlations of experimental data and analogies with heat and momentum transfer. A summary of the dimensionless groups used in these correlations and the analogies is given in Table 3.13.

As shown by the famous dye experiment of Osborne Reynolds [50] in 1883, a fluid in laminar flow moves parallel to the solid boundaries in streamline patterns. Every particle of fluid moves with the same velocity along a streamline and there are no fluid velocity components normal to these streamlines. For a Newtonian fluid in laminar flow, the momentum transfer, heat transfer, and mass transfer are by molecular transport, governed by Newton's law of viscosity, Fourier's law of heat conduction, and Fick's law of molecular diffusion, respectively.

In turbulent flow, the rates of momentum, heat, and mass transfer are orders of magnitude greater than for molecular transport. This occurs because streamlines no longer exist and particles or eddies of fluid, which are large compared to the mean free path of the molecules in the fluid, mix with each other by moving from one region to another in fluctuating motion. This eddy mixing by velocity fluctuations occurs not only in the direction of flow but also in directions normal to flow, with the latter being of more interest. Momentum, heat, and mass transfer now occur by two parallel mechanisms: (1) molecular motion, which is slow; and (2) turbulent or eddy motion, which is rapid except near a solid surface, where the flow velocity accompanying turbulence decreases to zero. Mass transfer by bulk flow may also occur as given by (3-1).

In 1877, Boussinesq [51] modified Newton's law of viscosity to account for eddy motion. Analogous expressions were subsequently developed for turbulent-flow heat and mass transfer. For flow in the x -direction and transport in the z -direction normal to flow, these expressions are written in the following forms in the absence of bulk flow in the z -direction:

$$\tau_{zx} = -(\mu + \mu_t) \frac{du_x}{dz} \quad (3-151)$$

$$q_z = -(k + k_t) \frac{dT}{dz} \quad (3-152)$$

$$N_{A_z} = -(D_{AB} + D_t) \frac{dc_A}{dz} \quad (3-153)$$

where the double subscript, zx , on the shear stress, τ , stands for x -momentum in the z -direction. The molecular contributions, μ , k , and D_{AB} , are molecular properties of the fluid and depend on chemical composition, temperature, and pressure; the turbulent contributions, μ_t , k_t , and D_t , depend on the

Table 3.13 Some Useful Dimensionless Groups

Name	Formula	Meaning	Analogy
Fluid Mechanics			
Drag Coefficient	$C_D = \frac{2F_D}{Au^2\rho}$	$\frac{\text{Drag force}}{\text{Projected area} \times \text{Velocity head}}$	
Fanning Friction Factor	$f = \frac{\Delta P}{L} \frac{D}{2\bar{u}^2\rho}$	$\frac{\text{Pipe wall shear stress}}{\text{Velocity head}}$	
Froude Number	$N_{Fr} = \frac{\bar{u}^2}{gL}$	$\frac{\text{Inertial force}}{\text{Gravitational force}}$	
Reynolds Number	$N_{Re} = \frac{L\bar{u}\rho}{\mu} = \frac{L\bar{u}}{\nu} = \frac{LG}{\mu}$	$\frac{\text{Inertial force}}{\text{Viscous force}}$	
Weber Number	$N_{We} = \frac{\bar{u}^2\rho L}{\sigma}$	$\frac{\text{Inertial force}}{\text{Surface-tension force}}$	
Heat Transfer			
j -Factor for Heat Transfer	$j_H = N_{StH}(N_{Pr})^{2/3}$		j_M
Nusselt Number	$N_{Nu} = \frac{hL}{k}$	$\frac{\text{Convective heat transfer}}{\text{Conductive heat transfer}}$	N_{Sh}
Peclet Number for Heat Transfer	$N_{PeH} = N_{Re}N_{Pr} = \frac{L\bar{u}\rho C_P}{k}$	$\frac{\text{Bulk transfer of heat}}{\text{Conductive heat transfer}}$	N_{PeM}
Prandtl Number	$N_{Pr} = \frac{C_P\mu}{k} = \frac{\nu}{\alpha}$	$\frac{\text{Momentum diffusivity}}{\text{Thermal diffusivity}}$	N_{Sc}
Stanton Number for Heat Transfer	$N_{StH} = \frac{N_{Nu}}{N_{Re}N_{Pr}} = \frac{h}{C_P G}$	$\frac{\text{Heat transfer}}{\text{Thermal capacity}}$	N_{StM}
Mass Transfer			
j -Factor for Mass Transfer	$j_M = N_{StM}(N_{Sc})^{2/3}$		j_H
Lewis Number	$N_{Le} = \frac{N_{Sc}}{N_{Pr}} = \frac{k}{\rho C_P D_{AB}} = \frac{\alpha}{D_{AB}}$	$\frac{\text{Thermal diffusivity}}{\text{Mass diffusivity}}$	
Peclet Number for Mass Transfer	$N_{PeM} = N_{Re}N_{Sc} = \frac{L\bar{u}}{D_{AB}}$	$\frac{\text{Bulk transfer of mass}}{\text{Molecular diffusion}}$	N_{PeH}
Schmidt Number	$N_{Sc} = \frac{\mu}{\rho D_{AB}} = \frac{\nu}{D_{AB}}$	$\frac{\text{Momentum diffusivity}}{\text{Mass diffusivity}}$	N_{Pr}
Sherwood Number	$N_{Sh} = \frac{k_c L}{D_{AB}}$	$\frac{\text{Convective mass transfer}}{\text{Molecular diffusion}}$	N_{Nu}
Stanton Number for Mass Transfer	$N_{StM} = \frac{N_{Sh}}{N_{Re}N_{Sc}} = \frac{k_c}{\bar{u}\rho}$	$\frac{\text{Mass transfer}}{\text{Mass capacity}}$	N_{StH}

L = characteristic length G = mass velocity = $\bar{u}\rho$

Subscripts: M = mass transfer H = heat transfer

mean fluid velocity in the direction of flow and on position in the fluid with respect to the solid boundaries.

In 1925, in an attempt to quantify turbulent transport, Prandtl [52] developed an expression for μ_t in terms of an eddy mixing length, l , which is a function of position. The eddy mixing length is a measure of the average distance that an eddy travels before it loses its identity and mingles with other eddies. The mixing length is analogous to the mean free path of gas molecules, which is the average distance a molecule travels before it collides with another molecule.

By analogy, the same mixing length is valid for turbulent-flow heat transfer and mass transfer. To use this analogy, (3-151) to (3-153) are rewritten in diffusivity form:

$$\frac{\tau_{zx}}{\rho} = -(\nu + \epsilon_M) \frac{du_x}{dz} \quad (3-154)$$

$$\frac{q_z}{C_P \rho} = -(\alpha + \epsilon_H) \frac{dT}{dz} \quad (3-155)$$

$$N_{A_x} = -(D_{AB} + \epsilon_D) \frac{dc_A}{dz} \quad (3-156)$$

where ϵ_M , ϵ_H , are ϵ_D are momentum, heat, and mass eddy diffusivities, respectively; ν is the momentum diffusivity (kinematic viscosity), μ/ρ ; and α is the thermal diffusivity, $k/\rho C_P$. As a first approximation, the three eddy diffusivities may be assumed equal. This assumption is reasonably valid for ϵ_H and ϵ_D , but experimental data indicate that $\epsilon_M/\epsilon_H = \epsilon_M/\epsilon_D$ is sometimes less than 1.0 and as low as 0.5 for turbulence in a free jet.

Reynolds Analogy

If (3-154) to (3-156) are applied at a solid boundary, they can be used to determine transport fluxes based on transport coefficients, with driving forces from the wall, i , at $z = 0$, to the bulk fluid, designated with an overbar, $\bar{\cdot}$:

$$\frac{\tau_{zx}}{\bar{u}_x} = -(\nu + \epsilon_M) \left. \frac{d(\rho u_x / \bar{u}_x)}{dz} \right|_{z=0} = \frac{f\rho}{2} \bar{u}_x \quad (3-157)$$

$$q_z = -(\alpha + \epsilon_H) \left. \frac{d(\rho C_P T)}{dz} \right|_{z=0} = h(T_i - \bar{T}) \quad (3-158)$$

$$N_{A_i} = -(D_{AB} + \epsilon_D) \left. \frac{dc_A}{dz} \right|_{z=0} = k_c(c_{A_i} - \bar{c}_A) \quad (3-159)$$

We define dimensionless velocity, temperature, and solute concentration by

$$\theta = \frac{u_x}{\bar{u}_x} = \frac{T_i - T}{T_i - \bar{T}} = \frac{c_{A_i} - c_A}{c_{A_i} - \bar{c}_A} \quad (3-160)$$

If (3-160) is substituted into (3-157) to (3-159),

$$\begin{aligned} \left. \frac{\partial \theta}{\partial z} \right|_{z=0} &= \frac{f \bar{u}_x}{2(\nu + \epsilon_M)} = \frac{h}{\rho C_P (\alpha + \epsilon_H)} \\ &= \frac{k_c}{(D_{AB} + \epsilon_D)} \end{aligned} \quad (3-161)$$

This equation defines the analogies among momentum, heat, and mass transfer. Assuming that the three eddy diffusivities are equal and that the molecular diffusivities are either everywhere negligible or equal,

$$\frac{f}{2} = \frac{h}{\rho C_P \bar{u}_x} = \frac{k_c}{\bar{u}_x} \quad (3-162)$$

Equation (3-162) defines the Stanton number for heat transfer,

$$N_{St_h} = \frac{h}{\rho C_P \bar{u}_x} = \frac{h}{G C_P} \quad (3-163)$$

where $G = \text{mass velocity} = \bar{u}_x \rho$, and the Stanton number for mass transfer,

$$N_{St_m} = \frac{k_c}{\bar{u}_x} = \frac{k_c \rho}{G} \quad (3-164)$$

both of which are included in Table 3.13.

Equation (3-162) is referred to as the Reynolds analogy. It can be used to estimate values of heat and mass transfer coefficients from experimental measurements of the Fanning friction factor for turbulent flow, but only when

$N_{Pr} = N_{Sc} = 1$. Thus, the Reynolds analogy has limited practical value and is rarely applied in practice. Reynolds postulated the existence of the analogy in 1874 [53] and derived it in 1883 [50].

Chilton-Colburn Analogy

A widely used extension of the Reynolds analogy to Prandtl and Schmidt numbers other than 1 was presented by Colburn [54] for heat transfer and by Chilton and Colburn [55] for mass transfer. They showed that the Reynolds analogy for turbulent flow could be corrected for differences in velocity, temperature, and concentration distributions by incorporating N_{Pr} and N_{Sc} into (3-162) to define the following three Chilton-Colburn j -factors, included in Table 3.13.

$$\begin{aligned} j_M \equiv \frac{f}{2} &= j_H \equiv \frac{h}{G C_P} (N_{Pr})^{2/3} \\ &= j_D \equiv \frac{k_c \rho}{G} (N_{Sc})^{2/3} \end{aligned} \quad (3-165)$$

Equation (3-165) is the Chilton-Colburn analogy or the Colburn analogy for estimating average transport coefficients for turbulent flow. When $N_{Pr} = N_{Sc} = 1$, (3-165) reduces to (3-162).

In general, j -factors are uniquely determined by the geometric configuration and the Reynolds number. Based on the analysis, over many years, of experimental data on momentum, heat, and mass transfer, the following representative correlations have been developed for turbulent transport to or from smooth surfaces. Other correlations are presented in other chapters. In general, these correlations are reasonably accurate for N_{Pr} and N_{Sc} in the range of 0.5 to 10, but should be used with caution outside this range.

1. Flow through a straight, circular tube of inside diameter D :

$$j_M = j_H = j_D = 0.023 (N_{Re})^{-0.2} \quad (3-166)$$

for $10,000 < N_{Re} = DG/\mu < 1,000,000$

2. Average transport coefficients for flow across a flat plate of length L :

$$j_M = j_H = j_D = 0.037 (N_{Re})^{-0.2} \quad (3-167)$$

for $5 \times 10^5 < N_{Re} = Lu_o \rho / \mu < 5 \times 10^8$

3. Average transport coefficients for flow normal to a long, circular cylinder of diameter D , where the drag coefficient includes both form drag and skin friction, but only the skin friction contribution applies to the analogy:

$$(j_M)_{\text{skin friction}} = j_H = j_D = 0.193 (N_{Re})^{-0.382} \quad (3-168)$$

for $4,000 < N_{Re} < 40,000$

$$(j_M)_{\text{skin friction}} = j_H = j_D = 0.0266 (N_{Re})^{-0.195} \quad (3-169)$$

for $40,000 < N_{Re} < 250,000$

with $N_{Re} = \frac{DG}{\mu}$

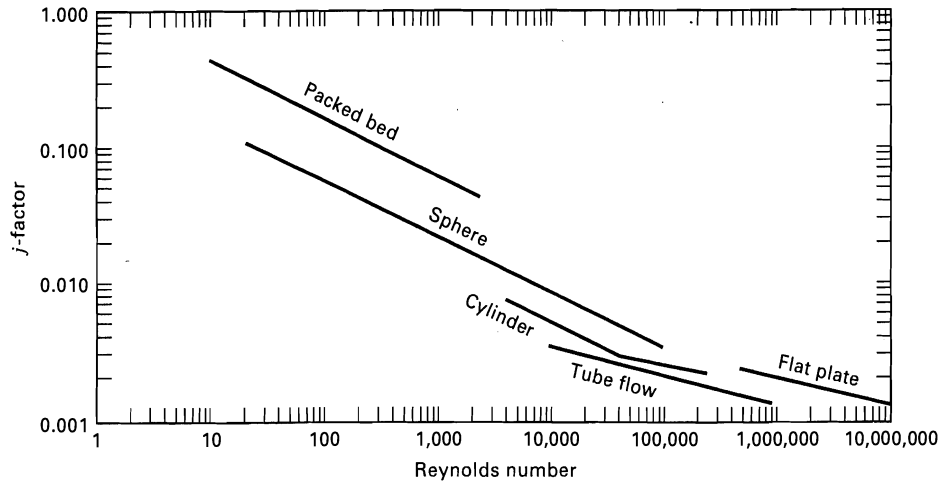


Figure 3.17 Chilton-Colburn j -factor correlations.

4. Average transport coefficients for flow past a single sphere of diameter D :

$$(j_M)_{\text{skin friction}} = j_H = j_D = 0.37(N_{\text{Re}})^{-0.4} \quad (3-170)$$

for $20 < N_{\text{Re}} = \frac{DG}{\mu} < 100,000$

5. Average transport coefficients for flow through beds packed with spherical particles of uniform size D_P :

$$j_H = j_D = 1.17(N_{\text{Re}})^{-0.415} \quad (3-171)$$

for $10 < N_{\text{Re}} = \frac{D_P G}{\mu} < 2,500$

The above correlations are plotted in Figure 3.17, where the curves do not coincide because of the differing definitions of the Reynolds number. However, the curves are not widely separated. When using the correlations in the presence of appreciable temperature and/or composition differences, Chilton and Colburn recommend that N_{Pr} and N_{Sc} be evaluated at the average conditions from the surface to the bulk stream.

Other Analogies

New turbulence theories have led to improvements and extensions of the Reynolds analogy, resulting in expressions for the Fanning friction factor and the Stanton numbers for heat and mass transfer that are less empirical than the Chilton-Colburn analogy. The first major improvement was by Prandtl [56] in 1910, who divided the flow into two regions: (1) a thin laminar-flow sublayer of thickness δ next to the wall boundary, where only molecular transport occurs; and (2) a turbulent region dominated by eddy transport, with $\epsilon_M = \epsilon_H = \epsilon_D$.

Further important theoretical improvements to the Reynolds analogy were made by von Karman, Martinelli, and Deissler, as discussed in detail by Knudsen and Katz [47]. The first two investigators inserted a buffer zone between the laminar sublayer and turbulent core. Deissler gradually reduced the eddy diffusivities as the wall was approached.

Other improvements were made by van Driest [64], who used a modified form of the Prandtl mixing length, Reichardt [65], who eliminated the zone concept by allowing the eddy diffusivities to decrease continuously from a maximum to zero at the wall, and Friend and Metzner [57], who modified the approach of Reichardt to obtain improved accuracy at very high Prandtl and Schmidt numbers (to 3,000). Their results for turbulent flow through a straight, circular tube are

$$N_{\text{StH}} = \frac{f/2}{1.20 + 11.8\sqrt{f/2}(N_{\text{Pr}} - 1)(N_{\text{Pr}})^{-1/3}} \quad (3-172)$$

$$N_{\text{StM}} = \frac{f/2}{1.20 + 11.8\sqrt{f/2}(N_{\text{Sc}} - 1)(N_{\text{Sc}})^{-1/3}} \quad (3-173)$$

Over a wide range of Reynolds number (10,000–10,000,000), the Fanning friction factor is estimated from the explicit empirical correlation of Drew, Koo, and McAdams [66],

$$f = 0.00140 + 0.125(N_{\text{Re}})^{-0.32} \quad (3-174)$$

which is in excellent agreement with the experimental data of Nikuradse [67] and is preferred over (3-165) with (3-166), which is valid only to $N_{\text{Re}} = 1,000,000$. For two- and three-dimensional turbulent-flow problems, some success has been achieved with the κ (kinetic energy of turbulence)– ϵ (rate of dissipation) model of Launder and Spalding [68], which is widely used in computational fluid dynamics (CFD) computer programs.

Theoretical Analogy of Churchill and Zajic

An alternative to (3-151) to (3-153) or the equivalent diffusivity forms of (3-154) to (3-156) for the development of transport equations for turbulent flow is to start with the time-averaged equations of Newton, Fourier, and Fick. For example, let us derive a form of Newton's law of viscosity for molecular and turbulent transport of momentum in parallel. In a turbulent-flow field in the axial x -direction, instantaneous velocity components, u_x and u_z , are

$$u_x = \bar{u}_x + u'_x$$

$$u_z = u'_z$$

where the “overbarred” component is the time-averaged (mean) local velocity and the primed component is the local fluctuating component that denotes instantaneous deviation from the local mean value. The mean velocity in the perpendicular z -direction is zero. The mean local velocity in the x -direction over a long period Θ of time θ is given by

$$\bar{u}_x = \frac{1}{\Theta} \int_0^\Theta u_x d\theta = \frac{1}{\Theta} \int_0^\Theta (\bar{u}_x + u'_x) d\theta \quad (3-175)$$

The time-averaged fluctuating components u'_x and u'_z equal zero.

The local instantaneous rate of momentum transfer by turbulence in the z -direction of x -direction turbulent momentum per unit area at constant density is

$$\rho u'_z(\bar{u}_x + u'_x) \quad (3-176)$$

The time-average of this turbulent momentum transfer is equal to the turbulent component of the shear stress, τ_{zx} ,

$$\begin{aligned} \tau_{zx} &= \frac{\rho}{\Theta} \int_0^\Theta u'_z(\bar{u}_x + u'_x) d\theta \\ &= \frac{\rho}{\Theta} \left[\int_0^\Theta u'_z(\bar{u}_x) d\theta + \int_0^\Theta u'_z(u'_x) d\theta \right] \end{aligned} \quad (3-177)$$

Because the time-average of the first term is zero, (3-177) reduces to

$$\tau_{zx} = \rho(\overline{u'_z u'_x}) \quad (3-178)$$

which is referred to as a Reynolds stress. Combining (3-178) with the molecular component of momentum transfer gives the turbulent-flow form of Newton's law of viscosity,

$$\tau_{zx} = -\mu \frac{du_x}{dz} + \rho(\overline{u'_z u'_x}) \quad (3-179)$$

If (3-179) is compared to (3-151), it is seen that an alternative approach to turbulence is to develop a correlating equation for the Reynolds stress, $(\overline{u'_z u'_x})$, first introduced by Churchill and Chan [73], rather than an expression for a turbulent viscosity μ_t . This stress, which is a complex function of position and rate of flow, has been correlated quite accurately for fully developed turbulent flow in a straight, circular tube by Heng, Chan, and Churchill [69]. In generalized form, with a the radius of the tube and $y = (a - z)$ the distance from the inside wall to the center of the tube, their equation is

$$\begin{aligned} (\overline{u'_z u'_x})^{++} &= \left(\left[0.7 \left(\frac{y^+}{10} \right)^3 \right]^{-8/7} + \left| \exp \left\{ \frac{-1}{0.436 y^+} \right\} \right. \right. \\ &\quad \left. \left. - \frac{1}{0.436 a^+} \left(1 + \frac{6.95 y^+}{a^+} \right) \right|^{-8/7} \right)^{-7/8} \end{aligned} \quad (3-180)$$

where

$$\begin{aligned} (\overline{u'_z u'_x})^{++} &= -\rho \overline{u'_z u'_x} / \tau \\ a^+ &= a(\tau_w \rho)^{1/2} / \mu \\ y^+ &= y(\tau_w \rho)^{1/2} / \mu \end{aligned}$$

Equation (3-180) is a highly accurate quantitative representation of turbulent flow because it is based on experimental data and numerical simulations described by Churchill and Zajic [70] and in considerable detail by Churchill [71]. From (3-142) and (3-143), the shear stress at the wall, τ_w , is related to the Fanning friction factor by

$$f = \frac{2\tau_w}{\rho \bar{u}_x^2} \quad (3-181)$$

where $\bar{u}_x$ is the flow-average velocity in the axial direction. Combining (3-179) with (3-181) and performing the required integrations, both numerically and analytically, lead to the following implicit equation for the Fanning friction factor as a function of the Reynolds number, $N_{Re} = 2a\bar{u}_x\rho/\mu$:

$$\begin{aligned} \left(\frac{2}{f} \right)^{1/2} &= 3.2 - 227 \frac{\left(\frac{2}{f} \right)^{1/2}}{\frac{N_{Re}}{2}} + 2500 \left[\frac{\left(\frac{2}{f} \right)^{1/2}}{\frac{N_{Re}}{2}} \right]^2 \\ &\quad + \frac{1}{0.436} \ln \left[\frac{\frac{N_{Re}}{2}}{\left(\frac{2}{f} \right)^{1/2}} \right] \end{aligned} \quad (3-182)$$

This equation is in excellent agreement with experimental data over a Reynolds number range of 4,000–3,000,000 and can probably be used to a Reynolds number of 100,000,000. Table 3.14 presents a comparison of the Churchill–Zajic equation, (3-182), with (3-174) of Drew et al. and (3-166) of Chilton and Colburn. Equation (3-174) gives satisfactory agreement for Reynolds numbers from 10,000 to 10,000,000, while (3-166) is useful only from 100,000 to 1,000,000.

Churchill and Zajic [70] show that if the equation for the conservation of energy is time averaged, a turbulent-flow form of Fourier's law of conduction can be obtained with the fluctuation term $(\overline{u'_z T'})$. Similar time averaging leads to a turbulent-flow form of Fick's law of diffusion with $(\overline{u'_z c'_A})$. To extend (3-180) and (3-182) to obtain an expression for the Nusselt number for turbulent-flow convective heat transfer in a straight, circular tube, Churchill and Zajic employ an analogy that is free of empiricism, but not exact. The result

Table 3.14 Comparison of Fanning Friction Factors for Fully Developed Turbulent Flow in a Smooth, Straight Circular Tube

N_{Re}	f , Drew et al. (3-174)	f , Chilton–Colburn (3-166)	f , Churchill–Zajic (3-182)
10,000	0.007960	0.007291	0.008087
100,000	0.004540	0.004600	0.004559
1,000,000	0.002903	0.002902	0.002998
10,000,000	0.002119	0.001831	0.002119
100,000,000	0.001744	0.001155	0.001573

for Prandtl numbers greater than 1 is

$$N_{Nu} = \frac{1}{\left(\frac{N_{Pr}}{N_{Pr_i}}\right) \frac{1}{N_{Nu_i}} + \left[1 - \left(\frac{N_{Pr}}{N_{Pr_i}}\right)^{2/3}\right] \frac{1}{N_{Nu_\infty}}} \quad (3-183)$$

where, from Yu, Ozoe, and Churchill [72],

$$N_{Pr_i} = \text{turbulent Prandtl number} = 0.85 + \frac{0.015}{N_{Pr}} \quad (3-184)$$

which replaces $(\overline{u_z' T'})$, as introduced by Churchill [74],

$$\begin{aligned} N_{Nu_i} &= \text{Nusselt number for } (N_{Pr} = N_{Pr_i}) \\ &= \frac{N_{Re} \left(\frac{f}{2}\right)}{1 + 145 \left(\frac{2}{f}\right)^{-5/4}} \end{aligned} \quad (3-185)$$

$$\begin{aligned} N_{Nu_\infty} &= \text{Nusselt number for } (N_{Pr} = \infty) \\ &= 0.07343 \left(\frac{N_{Pr}}{N_{Pr_i}}\right)^{1/3} N_{Re} \left(\frac{f}{2}\right)^{1/2} \end{aligned} \quad (3-186)$$

The accuracy of (3-183) is due to (3-185) and (3-186), which are known from theoretical considerations. Although (3-184) is somewhat uncertain, its effect is negligible.

A comparison of the Churchill et al. correlation of (3-183) with the Nusselt forms of (3-172) of Friend and Metzner and (3-166) of Chilton and Colburn, where from Table 3.13, $N_{Nu} = N_{St} N_{Re} N_{Pr}$, is given in Table 3.15 for a wide range of Reynolds number and Prandtl numbers of 1 and 1,000.

In Table 3.15, at a Prandtl number of 1, which is typical of low-viscosity liquids and close to that of most gases, the

Chilton–Colburn correlation, which is widely used, is within 10% of the more theoretically based Churchill–Zajic equation for Reynolds numbers up to 1,000,000. However, beyond that, serious deviations occur (25% at $N_{Re} = 10,000,000$ and almost 50% at $N_{Re} = 100,000,000$). Deviations of the Friend–Metzner correlation from the Churchill–Zajic equation vary from about 15% to 30% over the entire range of Reynolds number in Table 3.15. At all Reynolds numbers, the Churchill–Zajic equation predicts higher Nusselt numbers and, therefore, higher heat-transfer coefficients.

At a Prandtl number of 1,000, which is typical of high-viscosity liquids, the Friend–Metzner correlation is in fairly close agreement with the Churchill–Zajic equation, predicting values from 6 to 13% higher. The Chilton–Colburn correlation is seriously in error over the entire range of Reynolds number, predicting values ranging from 74 to 27% of those from the Churchill–Zajic equation as the Reynolds number increases. It is clear that the Chilton–Colburn correlation should not be used at high Prandtl numbers for heat transfer or (by analogy) at high Schmidt numbers for mass transfer.

The Churchill–Zajic equation for predicting the Nusselt number provides an effective power dependence on the Reynolds number as the Reynolds number increases. This is in contrast to the typically cited constant exponent of 0.8, as in the Chilton–Colburn correlation. For the Churchill–Zajic equation, at a Prandtl number of 1, the exponent increases with Reynolds number from 0.79 to 0.88; at a Prandtl number of 1,000, the exponent increases from 0.87 to 0.93.

Extension of the Churchill–Zajic equation to low Prandtl numbers, typical of molten metals, and to other geometries, such as parallel plates, is discussed by Churchill [71], who also considers the important effect of boundary conditions

Table 3.15 Comparison of Nusselt Numbers for Fully Developed Turbulent Flow in a Smooth, Straight Circular Tube

Prandtl number, $N_{Pr} = 1$			
N_{Re}	N_{Nu} , Friend–Metzner (3-172)	N_{Nu} , Chilton–Colburn (3-166)	N_{Nu} , Churchill–Zajic (3-183)
10,000	33.2	36.5	37.8
100,000	189	230	232
1,000,000	1210	1450	1580
10,000,000	8830	9160	11400
100,000,000	72700	57800	86000
Prandtl number, $N_{Pr} = 1000$			
N_{Re}	N_{Nu} , Friend–Metzner (3-172)	N_{Nu} , Chilton–Colburn (3-166)	N_{Nu} , Churchill–Zajic (3-183)
10,000	527	365	491
100,000	3960	2300	3680
1,000,000	31500	14500	29800
10,000,000	267800	91600	249000
100,000,000	2420000	578000	2140000

(e.g., constant wall temperature and uniform heat flux) at low-to-moderate Prandtl numbers.

For calculation of convective mass-transfer coefficients, k_c , for turbulent flow of gases and liquids in straight, smooth, circular tubes, it is recommended that the Churchill–Zajic equation be employed by applying the analogy between heat and mass transfer. Thus, as illustrated in the following example, in (3-183) to (3-186), from Table 3.13, the Sherwood number, N_{Sh} , is substituted for the Nusselt number, N_{Nu} ; and the Schmidt number, N_{Sc} , is substituted for the Prandtl number, N_{Pr} .

EXAMPLE 3.16

Linton and Sherwood [49] conducted experiments on the dissolving of cast tubes of cinnamic acid (A) into water (B) flowing through the tubes in turbulent flow. In one run, with a 5.23-cm-i.d. tube, $N_{Re} = 35,800$, and $N_{Sc} = 1,450$, they measured a Stanton number for mass transfer, N_{StM} , of 0.0000351. Compare this experimental value with predictions by the Reynolds, Chilton–Colburn, and Friend–Metzner analogies, and by the more theoretically-based Churchill–Zajic equation.

SOLUTION

From either (3-174) or (3-182), the Fanning friction factor is 0.00576.

Reynolds analogy:

From (3-162), $N_{StM} = \frac{f}{2} = 0.00576/2 = 0.00288$, which, as expected, is in poor agreement with the experimental value because the effect of Schmidt number is ignored.

Chilton–Colburn analogy:

From (3-165),

$$N_{StM} = \left(\frac{f}{2}\right) / (N_{Sc})^{2/3} = \left(\frac{0.00576}{2}\right) / (1450)^{2/3} = 0.0000225,$$

which is 64% of the experimental value.

Friend–Metzner analogy:

From (3-173), $N_{StM} = 0.0000350$, which is almost identical to the experimental value.

Churchill–Zajic equation:

Using mass-transfer analogs,

$$(3-184) \text{ gives } N_{Sc_f} = 0.850$$

$$(3-185) \text{ gives } N_{Sh_i} = 94$$

$$(3-186) \text{ gives } N_{Sh_{\infty}} = 1686$$

$$(3-183) \text{ gives } N_{Sh} = 1680$$

From Table 3.13,

$$N_{StM} = \frac{N_{Sh}}{N_{Re} N_{Sc}} = \frac{1680}{(35800)(1450)} = 0.0000324,$$

which is an acceptable 92% of the experimental value.

3.6 MODELS FOR MASS TRANSFER AT A FLUID–FLUID INTERFACE

In the three previous sections, diffusion and mass transfer within solids and fluids were considered, where the interface was a smooth solid surface. Of greater interest in separation processes is mass transfer across an interface between a gas and a liquid or between two liquid phases. Such interfaces exist in absorption, distillation, extraction, and stripping. At fluid–fluid interfaces, turbulence may persist to the interface. The following theoretical models have been developed to describe mass transfer between a fluid and such an interface.

Film Theory

A simple theoretical model for turbulent mass transfer to or from a fluid-phase boundary was suggested in 1904 by Nernst [58], who postulated that the entire resistance to mass transfer in a given turbulent phase is in a thin, stagnant region of that phase at the interface, called a film. This film is similar to the laminar sublayer that forms when a fluid flows in the turbulent regime parallel to a flat plate. This is shown schematically in Figure 3.18a for the case of a gas–liquid interface, where the gas is pure component A, which diffuses into nonvolatile liquid B. Thus, a process of absorption of A into liquid B takes place, without desorption of B into

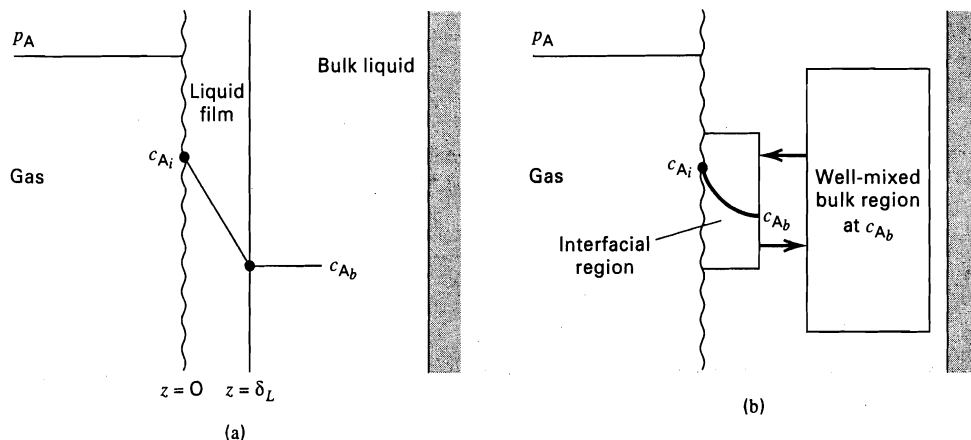


Figure 3.18 Theories for mass transfer from a fluid–fluid interface into a liquid: (a) film theory; (b) penetration and surface-renewal theories.

gaseous A. Because the gas is pure A at total pressure $P = p_A$, there is no resistance to mass transfer in the gas phase. At the gas-liquid interface, phase equilibrium is assumed so the concentration of A, c_{A_i} , is related to the partial pressure of A, p_A , by some form of Henry's law, for example, $c_{A_i} = H_A p_A$. In the thin, stagnant liquid film of thickness δ , molecular diffusion only occurs with a driving force of $c_{A_i} - c_{A_b}$. Since the film is assumed to be very thin, all of the diffusing A passes through the film and into the bulk liquid. If, in addition, bulk flow of A is neglected, the concentration gradient is linear as in Figure 3.18a. Accordingly, Fick's first law, (3-3a), for the diffusion flux integrates to

$$J_A = \frac{D_{AB}}{\delta}(c_{A_i} - c_{A_b}) = \frac{cD_{AB}}{\delta}(x_{A_i} - x_{A_b}) \quad (3-187)$$

If the liquid phase is dilute in A, the bulk-flow effect can be neglected and (3-187) applies to the total flux:

$$N_A = \frac{D_{AB}}{\delta}(c_{A_i} - c_{A_b}) = \frac{cD_{AB}}{\delta}(x_{A_i} - x_{A_b}) \quad (3-188)$$

If the bulk-flow effect is not negligible, then, from (3-31),

$$N_A = \frac{cD_{AB}}{\delta} \ln \left[\frac{1 - x_{A_b}}{1 - x_{A_i}} \right] = \frac{cD_{AB}}{\delta(1 - x_A)_{LM}} (x_{A_i} - x_{A_b}) \quad (3-189)$$

where

$$(1 - x_A)_{LM} = \frac{x_{A_i} - x_{A_b}}{\ln[(1 - x_{A_b})/(1 - x_{A_i})]} = (x_B)_{LM} \quad (3-190)$$

In practice, the ratios D_{AB}/δ in (3-188) and $D_{AB}/\delta(1 - x_A)_{LM}$ in (3-189) are replaced by mass transfer coefficients k_c and k'_c , respectively, because the film thickness, δ , which depends on the flow conditions, is not known and the subscript, c , refers to a concentration driving force.

The film theory, which is easy to understand and apply, is often criticized because it appears to predict that the rate of mass transfer is directly proportional to the molecular diffusivity. This dependency is at odds with experimental data, which indicate a dependency of D^n , where n ranges from about 0.5 to 0.75. However, if D_{AB}/δ is replaced with k_c , which is then estimated from the Chilton-Colburn analogy, Eq. (3-165), we obtain k_c proportional to $D_{AB}^{2/3}$, which is in better agreement with experimental data. In effect, δ depends on D_{AB} (or N_{Sc}). Regardless of whether the criticism of the film theory is valid, the theory has been and continues to be widely used in the design of mass-transfer separation equipment.

EXAMPLE 3.17

Sulfur dioxide is absorbed from air into water in a packed absorption tower. At a certain location in the tower, the mass-transfer flux is $0.0270 \text{ kmol SO}_2/\text{m}^2\text{-h}$ and the liquid-phase mole fractions are 0.0025 and 0.0003, respectively, at the two-phase interface and in

the bulk liquid. If the diffusivity of SO_2 in water is $1.7 \times 10^{-5} \text{ cm}^2/\text{s}$, determine the mass-transfer coefficient, k_c , and the film thickness, neglecting the bulk-flow effect.

SOLUTION

$$N_{\text{SO}_2} = \frac{0.027(1,000)}{(3,600)(100)^2} = 7.5 \times 10^{-7} \frac{\text{mol}}{\text{cm}^2\text{-s}}$$

For dilute conditions, the concentration of water is

$$c = \frac{1}{18.02} = 5.55 \times 10^{-2} \text{ mol/cm}^3$$

From (3-188),

$$k_c = \frac{D_{AB}}{\delta} = \frac{N_A}{c(x_{A_i} - x_{A_b})} = \frac{7.5 \times 10^{-7}}{5.55 \times 10^{-2}(0.0025 - 0.0003)} = 6.14 \times 10^{-3} \text{ cm/s}$$

Therefore,

$$\delta = \frac{D_{AB}}{k_c} = \frac{1.7 \times 10^{-5}}{6.14 \times 10^{-3}} = 0.0028 \text{ cm}$$

which is very small and typical of turbulent-flow mass-transfer processes.

Penetration Theory

A more realistic physical model of mass transfer from a fluid-fluid interface into a bulk liquid stream is provided by the penetration theory of Higbie [59], shown schematically in Figure 3.18b. The stagnant-film concept is replaced by Boussinesq eddies that, during a cycle, (1) move from the bulk to the interface; (2) stay at the interface for a short, fixed period of time during which they remain static so that molecular diffusion takes place in a direction normal to the interface; and (3) leave the interface to mix with the bulk stream. When an eddy moves to the interface, it replaces another static eddy. Thus, the eddies are alternately static and moving. Turbulence extends to the interface.

In the penetration theory, unsteady-state diffusion takes place at the interface during the time the eddy is static. This process is governed by Fick's second law, (3-68), with boundary conditions

$$\begin{aligned} c_A &= c_{A_b} & \text{at } t = 0 & \text{ for } 0 \leq z \leq \infty; \\ c_A &= c_{A_i} & \text{at } z = 0 & \text{ for } t > 0; \text{ and} \\ c_A &= c_{A_b} & \text{at } z = \infty & \text{ for } t > 0 \end{aligned}$$

These are the same boundary conditions as in unsteady-state diffusion in a semi-infinite medium. Thus, the solution can be written by a rearrangement of (3-75):

$$\frac{c_{A_i} - c_A}{c_{A_i} - c_{A_b}} = \text{erf} \left(\frac{z}{2\sqrt{D_{AB}t_c}} \right) \quad (3-191)$$

where t_c = "contact time" of the static eddy at the interface during one cycle. The corresponding average mass-transfer flux of A in the absence of bulk flow is given by the

following form of (3-79):

$$N_A = 2\sqrt{\frac{D_{AB}}{\pi t_c}}(c_{A_i} - c_{A_b}) \quad (3-192)$$

or

$$N_A = k_c(c_{A_i} - c_{A_b}) \quad (3-193)$$

Thus, the penetration theory gives

$$k_c = 2\sqrt{\frac{D_{AB}}{\pi t_c}} \quad (3-194)$$

which predicts that k_c is proportional to the square root of the molecular diffusivity, which is at the lower limit of experimental data.

The penetration theory is most useful when mass transfer involves bubbles or droplets, or flow over random packing. For bubbles, the contact time, t_c , of the liquid surrounding the bubble is taken as the ratio of bubble diameter to bubble-rise velocity. For example, an air bubble of 0.4-cm diameter rises through water at a velocity of about 20 cm/s. Thus, the estimated contact time, t_c , is $0.4/20 = 0.02$ s. For a liquid spray, where no circulation of liquid occurs inside the droplets, the contact time is the total time for the droplets to fall through the gas. For a packed tower, where the liquid flows as a film over particles of random packing, mixing can be assumed to occur each time the liquid film passes from one piece of packing to another. Resulting contact times are of the order of about 1 s. In the absence of any method of estimating the contact time, the liquid-phase mass-transfer coefficient is sometimes correlated by an empirical expression consistent with the 0.5 exponent on D_{AB} , given by (3-194) with the contact time replaced by a function of geometry and the liquid velocity, density, and viscosity.

EXAMPLE 3.18

For the conditions of Example 3.17, estimate the contact time for Higbie's penetration theory.

SOLUTION

From Example 3.17, $k_c = 6.14 \times 10^{-3}$ cm/s and $D_{AB} = 1.7 \times 10^{-5}$ cm²/s. From a rearrangement of (3-194),

$$t_c = \frac{4D_{AB}}{\pi k_c^2} = \frac{4(1.7 \times 10^{-5})}{3.14(6.14 \times 10^{-3})^2} = 0.57 \text{ s}$$

Surface-Renewal Theory

The penetration theory is not satisfying because the assumption of a constant contact time for all eddies that temporarily reside at the surface is not reasonable, especially for stirred tanks, contactors with random packings, and bubble and spray columns where the bubbles and droplets cover a wide range of sizes. In 1951, Danckwerts [60] suggested an improvement to the penetration theory that involves the replacement of the constant eddy contact time with the assumption of a residence-time distribution, wherein the probability of an eddy at the surface being replaced by a fresh eddy is independent of the age of the surface eddy.

Following the Levenspiel [61] treatment of residence-time distribution, let $F(t)$ be the fraction of eddies with a contact time of less than t . For $t = 0$, $F\{t\} = 0$, and $F\{t\}$ approaches 1 as t goes to infinity. A plot of $F\{t\}$ versus t , as shown in Figure 3.19, is referred to as a residence-time or age distribution. If $F\{t\}$ is differentiated with respect to t , we obtain another function:

$$\phi\{t\} = dF\{t\}/dt$$

where $\phi\{t\}dt$ = the probability that a given surface eddy will have a residence time t . The sum of probabilities is

$$\int_0^\infty \phi\{t\} dt = 1 \quad (3-195)$$

Typical plots of $F\{t\}$ and $\phi\{t\}$ are shown in Figure 3.19, where it is seen that $\phi\{t\}$ is similar to a normal probability curve.

For steady-state flow in and out of a well-mixed vessel, Levenspiel shows that

$$F\{t\} = 1 - e^{-t/\bar{t}} \quad (3-196)$$

where $\bar{t}$ is the average residence time. This function forms the basis, in reaction engineering, of the ideal model of a continuous, stirred-tank reactor (CSTR). Danckwerts selected the same model for his surface-renewal theory, using the corresponding $\phi\{t\}$ function:

$$\phi\{t\} = se^{-st} \quad (3-197)$$

where $s = 1/\bar{t}$ = fractional rate of surface renewal. As shown in Example 3.19 below, plots of (3-196) and (3-197) are much different from those in Figure 3.19.

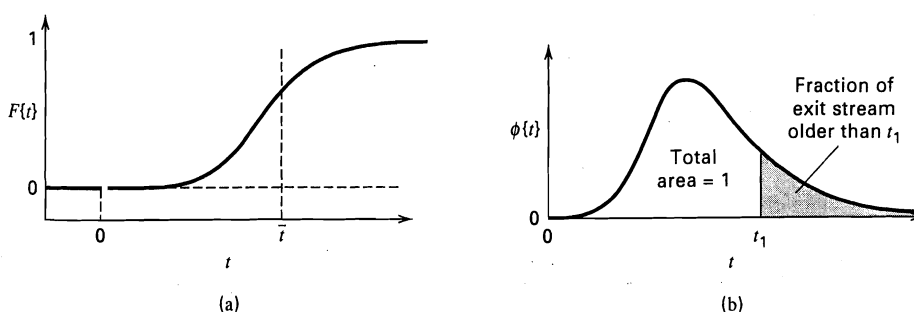


Figure 3.19 Residence-time distribution plots: (a) typical F curve; (b) typical age distribution.

[Adapted from O. Levenspiel, *Chemical Reaction Engineering*, 2nd ed., John Wiley and Sons, New York (1972).]

The instantaneous mass-transfer rate for an eddy with an age t is given by (3-192) for the penetration theory in flux form as

$$N_{A_t} = \sqrt{\frac{D_{AB}}{\pi t}} (c_{A_i} - c_{A_b}) \quad (3-198)$$

The integrated average rate is

$$(N_A)_{\text{avg}} = \int_0^\infty \phi\{t\} N_{A_t} dt \quad (3-199)$$

Combining (3-197), (3-198), and (3-199), and integrating:

$$(N_A)_{\text{avg}} = \sqrt{D_{AB}s} (c_{A_i} - c_{A_b}) \quad (3-200)$$

Thus,

$$k_c = \sqrt{D_{AB}s} \quad (3-201)$$

The more reasonable surface-renewal theory predicts the same dependency of the mass-transfer coefficient on molecular diffusivity as the penetration theory. Unfortunately, s , the fractional rate of surface renewal, is as elusive a parameter as the constant contact time, t_c .

EXAMPLE 3.19

For the conditions of Example 3.17, estimate the fractional rate of surface renewal, s , for Danckwert's theory and determine the residence time and probability distributions.

SOLUTION

From Example 3.17,

$$k_c = 6.14 \times 10^{-3} \text{ cm/s} \quad \text{and} \quad D_{AB} = 1.7 \times 10^{-5} \text{ cm}^2/\text{s}$$

From (3-201),

$$s = \frac{k_c^2}{D_{AB}} = \frac{(6.14 \times 10^{-3})^2}{1.7 \times 10^{-5}} = 2.22 \text{ s}^{-1}$$

Thus, the average residence time of an eddy at the surface is $1/2.22 = 0.45 \text{ s}$.

From (3-197),

$$\phi\{t\} = 2.22 e^{-2.22t} \quad (1)$$

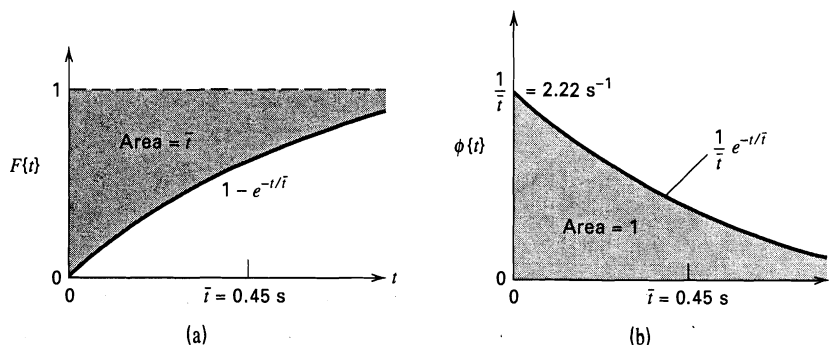


Figure 3.20 Age distribution curves for Example 3.19: (a) F curve; (b) $\phi\{t\}$ curve.

From (3-196), the residence-time distribution is given by

$$F\{t\} = 1 - e^{-t/0.45}, \quad (2)$$

where t is in seconds. Equations (1) and (2) are plotted in Figure 3.20. These curves are much different from the curves of Figure 3.19.

Film-Penetration Theory

Toor and Marchello [62], in 1958, combined features of the film, penetration, and surface-renewal theories to develop a film-penetration model, which predicts a dependency of the mass-transfer coefficient k_c , on the diffusivity, that varies from $\sqrt{D_{AB}}$ to D_{AB} . Their theory assumes that the entire resistance to mass transfer resides in a film of fixed thickness δ . Eddies move to and from the bulk fluid and this film. Age distributions for time spent in the film are of the Higbie or Danckwerts type.

Fick's second law, (3-68), still applies, but the boundary conditions are now

$$\begin{aligned} c_A &= c_{A_b} & \text{at } t = 0 & \text{for } 0 \leq z \leq \infty, \\ c_A &= c_{A_i} & \text{at } z = 0 & \text{for } t > 0; \quad \text{and} \\ c_A &= c_{A_b} & \text{at } z = \delta & \text{for } t > 0 \end{aligned}$$

Infinite-series solutions are obtained by the method of Laplace transforms. The rate of mass transfer is then obtained in the usual manner by applying Fick's first law (3-117) at the fluid-fluid interface. For small t , the solution, given as

$$N_{A_t} = (c_{A_i} - c_{A_b}) \left(\frac{D_{AB}}{\pi t} \right)^{1/2} \left[1 + 2 \sum_{n=1}^{\infty} \exp \left(-\frac{n^2 \delta^2}{D_{AB} t} \right) \right] \quad (3-202)$$

converges rapidly. For large t ,

$$\begin{aligned} N_{A_t} &= (c_{A_i} - c_{A_b}) \left(\frac{D_{AB}}{\delta} \right) \\ &\times \left[1 + 2 \sum_{n=1}^{\infty} \exp \left(-n^2 \pi^2 \frac{D_{AB} t}{\delta^2} \right) \right] \end{aligned} \quad (3-203)$$

Equation (3-199) with $\phi\{t\}$ from (3-197) can then be used to obtain average rates of mass transfer. Again, we can write two equivalent series solutions, which converge

at different rates. Equations (3-202) and (3-203) become, respectively,

$$N_{A_{avg}} = k_c(c_{A_i} - c_{A_b}) = (c_{A_i} - c_{A_b})(sD_{AB})^{1/2} \times \left[1 + 2 \sum_{n=1}^{\infty} \exp\left(-2n\delta\sqrt{\frac{s}{D_{AB}}}\right) \right] \quad (3-204)$$

$$N_{A_{avg}} = k_c(c_{A_i} - c_{A_b}) = (c_{A_i} - c_{A_b})\left(\frac{D_{AB}}{\delta}\right) \times \left[1 + 2 \sum_{n=1}^{\infty} \frac{1}{1 + n^2\pi^2\frac{D_{AB}}{s\delta^2}} \right] \quad (3-205)$$

In the limit, for a high rate of surface renewal, $s\delta^2/D_{AB}$, (3-204) reduces to the surface-renewal theory, (3-200). For low rates of renewal, (3-205) reduces to the film theory, (3-188). At conditions in between, k_c is proportional to D_{AB}^n , where n is in the range of 0.5–1.0. The application of the film-penetration theory is difficult because of lack of data for δ and s , but the predicted effect of molecular diffusivity brackets experimental data.

3.7 TWO-FILM THEORY AND OVERALL MASS-TRANSFER COEFFICIENTS

Separation processes that involve contacting two fluid phases require consideration of mass-transfer resistances in both phases. In 1923, Whitman [63] suggested an extension of the film theory to two fluid films in series. Each film presents a resistance to mass transfer, but concentrations in the two fluids at the interface are assumed to be in phase equilibrium. That is, there is no additional interfacial resistance to mass transfer. This concept has found extensive application in modeling of steady-state, gas–liquid, and liquid–liquid separation processes.

The assumption of phase equilibrium at the phase interface, while widely used, may not be valid when gradients of interfacial tension are established during mass transfer between two fluids. These gradients give rise to interfacial turbulence resulting, most often, in considerably increased mass-transfer coefficients. This phenomenon, referred to as

the *Marangoni effect*, is discussed in some detail by Bird, Stewart, and Lightfoot [28], who cite additional references. The effect can occur at both vapor–liquid and liquid–liquid interfaces, with the latter receiving the most attention. By adding surfactants, which tend to concentrate at the interface, the Marangoni effect may be reduced because of stabilization of the interface, even to the extent that an interfacial mass-transfer resistance may result, causing the overall mass-transfer coefficient to be reduced. In this book, unless otherwise indicated, the Marangoni effect will be ignored and phase equilibrium will always be assumed at the phase interface.

Gas–Liquid Case

Consider the steady-state mass transfer of A from a gas phase, across an interface, into a liquid phase. It could be postulated, as shown in Figure 3.21a, that a thin gas film exists on one side of the interface and a thin liquid film exists on the other side, with the controlling factors being molecular diffusion through each film. However, this postulation is not necessary, because instead of writing

$$N_A = \frac{(D_{AB})_G}{\delta_G}(c_{A_b} - c_{A_i})_G = \frac{(D_{AB})_L}{\delta_L}(c_{A_i} - c_{A_b})_L \quad (3-206)$$

we can express the rate of mass transfer in terms of mass-transfer coefficients determined from any suitable theory, with the concentration gradients visualized more realistically as in Figure 3.21b. In addition, we can use any number of different mass-transfer coefficients, depending on the selection of the driving force for mass transfer. For the gas phase, under dilute or equimolar counter diffusion (EMD) conditions, we write the mass-transfer rate in terms of partial pressures:

$$N_A = k_p(p_{A_b} - p_{A_i}) \quad (3-207)$$

where k_p is a gas-phase mass-transfer coefficient based on a partial-pressure driving force.

For the liquid phase, we use molar concentrations:

$$N_A = k_c(c_{A_i} - c_{A_b}) \quad (3-208)$$

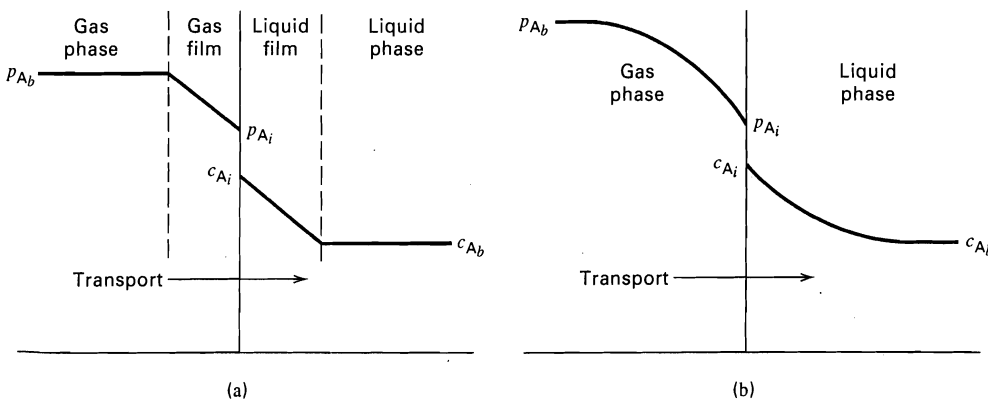


Figure 3.21 Concentration gradients for two-resistance theory: (a) film theory; (b) more realistic gradients.

At the phase interface, c_{A_i} and p_{A_i} are assumed to be in phase equilibrium. Applying a version of Henry's law different from that in Table 2.3,¹

$$c_{A_i} = H_A p_{A_i} \quad (3-209)$$

Equations (3-207) to (3-209) are a commonly used combination for vapor-liquid mass transfer. Computations of mass-transfer rates are generally made from a knowledge of bulk concentrations, which in this case are p_{A_b} and c_{A_b} . To obtain an expression for N_A in terms of an overall driving force for mass transfer, (3-207) to (3-209) are combined in the following manner to eliminate the interfacial concentrations, c_{A_i} and p_{A_i} . Solve (3-207) for p_{A_i} :

$$p_{A_i} = p_{A_b} - \frac{N_A}{k_p} \quad (3-210)$$

Solve (3-208) for c_{A_i} :

$$c_{A_i} = c_{A_b} + \frac{N_A}{k_c} \quad (3-211)$$

Combine (3-211) with (3-209) to eliminate c_{A_i} and combine the result with (3-210) to eliminate p_{A_i} to give

$$N_A = \frac{p_{A_b} H_A - c_{A_b}}{(H_A/k_p) + (1/k_c)} \quad (3-212)$$

It is customary to define: (1) a fictitious liquid-phase concentration $c_A^* = p_{A_b} H_A$, which is the concentration that would be in equilibrium with the partial pressure in the bulk gas; and (2) an overall mass-transfer coefficient, K_L . Thus, (3-212) is rewritten as

$$N_A = K_L (c_A^* - c_{A_b}) = \frac{(c_A^* - c_{A_b})}{(H_A/k_p) + (1/k_c)} \quad (3-213)$$

where

$$\frac{1}{K_L} = \frac{H_A}{k_p} + \frac{1}{k_c} \quad (3-214)$$

in which K_L is the overall mass-transfer coefficient based on the liquid phase. The quantities H_A/k_p and $1/k_c$ are measures of the mass-transfer resistances of the gas phase and the liquid phase, respectively. When $1/k_c \gg H_A/k_p$, (3-214) becomes

$$N_A = k_c (c_A^* - c_{A_b}) \quad (3-215)$$

Since resistance in the gas phase is then negligible, the gas-phase driving force is $p_{A_b} - p_{A_i} \approx 0$ and $p_{A_b} \approx p_{A_i}$.

¹Many different forms of Henry's law are found in the literature. They include

$$p_A = H_A x_A, \quad p_A = \frac{c_A}{H_A}, \quad \text{and} \quad y_A = H_A x_A$$

When a Henry's-law constant, H_A , is given without citing the equation that defines it, the defining equation can be determined from the units of the constant. For example, if the constant has the units of atm or atm/mole fraction, Henry's law is given by $p_A = H_A x_A$. If the units are mol/L-mmHg, Henry's law is $p_A = \frac{c_A}{H_A}$.

Alternatively, (3-207) to (3-209) can be combined to define an overall mass-transfer coefficient, K_G , based on the gas phase. The result is

$$N_A = \frac{p_{A_b} - c_{A_b}/H_A}{(1/k_p) + (1/H_A k_c)} \quad (3-216)$$

In this case, it is customary to define: (1) a fictitious gas-phase partial pressure $p_A^* = c_{A_b}/H_A$, which is the partial pressure that would be in equilibrium with the bulk liquid; and (2) an overall mass-transfer coefficient for the gas phase, K_G , based on a partial-pressure driving force. Thus, (3-216) can be rewritten as

$$N_A = K_G (p_{A_b} - p_A^*) = \frac{(p_{A_b} - p_A^*)}{(1/k_p) + (1/H_A k_c)} \quad (3-217)$$

where

$$\frac{1}{K_G} = \frac{1}{k_p} + \frac{1}{H_A k_c} \quad (3-218)$$

In this, the resistances are $1/k_p$ and $1/(H_A k_c)$. When $1/k_p \gg 1/H_A k_c$,

$$N_A = k_p (p_{A_b} - p_A^*) \quad (3-219)$$

Since the resistance in the liquid phase is then negligible, the liquid-phase driving force is $c_{A_i} - c_{A_b} \approx 0$ and $c_{A_i} \approx c_{A_b}$.

The choice between using (3-213) or (3-217) is arbitrary, but is usually made on the basis of which phase has the largest mass-transfer resistance; if the liquid, use (3-213); if the gas, use (3-217). Another common combination for vapor-liquid mass transfer uses mole fraction-driving forces, which define another set of mass-transfer coefficients:

$$N_A = k_y (y_{A_b} - y_{A_i}) = k_x (x_{A_i} - x_{A_b}) \quad (3-220)$$

In this case, phase equilibrium at the interface can be expressed in terms of the K -value for vapor-liquid equilibrium. Thus,

$$K_A = y_{A_i}/x_{A_i} \quad (3-221)$$

Combining (3-220) and (3-221) to eliminate y_{A_i} and x_{A_i} ,

$$N_A = \frac{y_{A_b} - x_{A_b}}{(1/K_A k_y) + (1/k_x)} \quad (3-222)$$

This time we define fictitious concentration quantities and overall mass-transfer coefficients for mole-fraction driving forces. Thus, $x_A^* = y_{A_b}/K_A$ and $y_A^* = K_A x_{A_b}$. If the two values of K_A are equal, we obtain

$$N_A = K_x (x_A^* - x_{A_b}) = \frac{x_A^* - x_{A_b}}{(1/K_A k_y) + (1/k_x)} \quad (3-223)$$

and

$$N_A = K_y (y_{A_b} - y_A^*) = \frac{y_{A_b} - y_A^*}{(1/k_y) + (K_A/k_x)} \quad (3-224)$$

where K_x and K_y are overall mass-transfer coefficients based on mole-fraction driving forces with

$$\frac{1}{K_x} = \frac{1}{K_A k_y} + \frac{1}{k_x} \quad (3-225)$$

and

$$\frac{1}{K_y} = \frac{1}{k_y} + \frac{K_A}{k_x} \quad (3-226)$$

When using correlations to estimate mass-transfer coefficients for use in the above equations, it is important to determine which coefficient (k_p , k_c , k_y , or k_x) is correlated. This can usually be done by checking the units or the form of the Sherwood or Stanton numbers. Coefficients correlated by the Chilton–Colburn analogy are k_c for either the liquid or gas phase. The different coefficients are related by the following expressions, which are summarized in Table 3.16.

Liquid phase:

$$k_x = k_c c = k_c \left(\frac{\rho L}{M} \right) \quad (3-227)$$

Ideal-gas phase:

$$k_y = k_p P = (k_c)_g \frac{P}{RT} = (k_c)_g c = (k_c)_g \left(\frac{\rho_G}{M} \right) \quad (3-228)$$

Typical units are

	SI	American Engineering
k_c	m/s	ft/h
k_p	kmol/s·m ² ·kPa	lbmol/h·ft ² ·atm
k_y, k_x	kmol/s·m ²	lbmol/h·ft ²

When unimolecular diffusion (UMD) occurs under non-dilute conditions, the effect of bulk flow must be included in the above equations. For binary mixtures, one method for doing this is to define modified mass-transfer coefficients, designated with a prime, as follows.

For the liquid phase, using k_c or k_x ,

$$k' = \frac{k}{(1 - x_A)_{LM}} = \frac{k}{(x_B)_{LM}} \quad (3-229)$$

For the gas phase, using k_p , k_y , or k_c ,

$$k' = \frac{k}{(1 - y_A)_{LM}} = \frac{k}{(y_B)_{LM}} \quad (3-230)$$

The expressions for k' are most readily used when the mass-transfer rate is controlled mainly by one of the two resistances. Experimental mass-transfer coefficient data reported in the literature are generally correlated in terms of k rather than k' . Mass-transfer coefficients estimated from the Chilton–Colburn analogy [e.g., equations (3-166) to (3-171)] are k_c , not k'_c .

Liquid–Liquid Case

For mass transfer across two liquid phases, equilibrium is again assumed at the interface. Denoting the two phases by $L^{(1)}$ and $L^{(2)}$, (3-223) and (3-224) can be rewritten as

$$N_A = K_x^{(2)} (x_A^{(2)*} - x_{A_b}^{(2)}) = \frac{x_A^{(2)*} - x_{A_b}^{(2)}}{(1/K_{DA} k_x^{(1)}) + (1/k_x^{(2)})} \quad (3-231)$$

and

$$N_A = K_x^{(1)} (x_{A_b}^{(1)} - x_A^{(1)*}) = \frac{x_{A_b}^{(1)} - x_A^{(1)*}}{(1/k_x^{(1)}) + (K_{DA}/k_x^{(2)})} \quad (3-232)$$

where

$$K_{DA} = \frac{x_{A_i}^{(1)}}{x_{A_i}^{(2)}} \quad (3-233)$$

Case of Large Driving Forces for Mass Transfer

When large driving forces exist for mass transfer, phase equilibria ratios such as H_A , K_A , and K_{DA} may not be constant across the two phases. This occurs particularly when one or both phases are not dilute with respect to the diffusing solute, A. In that case, expressions for the mass-transfer flux must be revised.

For example, if mole-fraction driving forces are used, we write, from (3-220) and (3-224),

$$N_A = k_y (y_{A_b} - y_{A_i}) = K_y (y_{A_b} - y_A^*) \quad (3-234)$$

Thus,

$$\frac{1}{K_y} = \frac{y_{A_b} - y_A^*}{k_y (y_{A_b} - y_{A_i})} \quad (3-235)$$

or

$$\frac{1}{K_y} = \frac{(y_{A_b} - y_{A_i}) + (y_{A_i} - y_A^*)}{k_y (y_{A_b} - y_{A_i})} = \frac{1}{k_y} + \frac{1}{k_y} \left(\frac{y_{A_i} - y_A^*}{y_{A_b} - y_{A_i}} \right) \quad (3-236)$$

From (3-220),

$$\frac{k_x}{k_y} = \frac{(y_{A_b} - y_{A_i})}{(x_{A_i} - x_{A_b})} \quad (3-237)$$

Table 3.16 Relationships among Mass-Transfer Coefficients

Equimolar Counterdiffusion (EMD):

Gases: $N_A = k_y \Delta y_A = k_c \Delta c_A = k_p \Delta p_A$
 $k_y = k_c \frac{P}{RT} = k_p P$ if ideal gas

Liquids: $N_A = k_x \Delta x_A = k_c \Delta c_A$
 $k_x = k_c c$, where c = total molar concentration (A + B)

Unimolecular Diffusion (UMD):

Gases: Same equations as for EMD with k replaced by $k' = \frac{k}{(y_B)_{LM}}$

Liquids: Same equations as for EMD with k replaced by $k' = \frac{k}{(x_B)_{LM}}$

When using concentration units for both phases, it is convenient to use:

$$k_G(\Delta c_G) = k_c(\Delta c) \text{ for the gas phase}$$

$$k_L(\Delta c_L) = k_c(\Delta c) \text{ for the liquid phase}$$

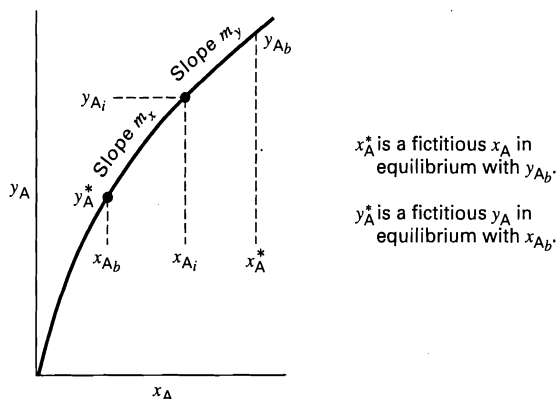


Figure 3.22 Curved equilibrium line.

Combining (3-234) and (3-237),

$$\frac{1}{K_y} = \frac{1}{k_y} + \frac{1}{k_x} \left(\frac{y_{A_i} - y_A^*}{x_{A_i} - x_{A_b}} \right) \quad (3-238)$$

In a similar manner,

$$\frac{1}{K_x} = \frac{1}{k_x} + \frac{1}{k_y} \left(\frac{x_A^* - x_{A_i}}{y_{A_b} - y_{A_i}} \right) \quad (3-239)$$

A typical curved equilibrium line is shown in Figure 3.22 with representative values of y_{A_b} , y_{A_i} , y_A^* , x_A^* , x_{A_i} , and x_{A_b} indicated. Because the line is curved, the vapor-liquid equilibrium ratio, $K_A = y_A/x_A$, is not constant across the two phases. As shown, the slope of the curve and thus, K_A , decreases with increasing concentration of A. Denote two slopes of the equilibrium line by

$$m_x = \left(\frac{y_{A_i} - y_A^*}{x_{A_i} - x_{A_b}} \right) \quad (3-240)$$

and

$$m_y = \left(\frac{y_{A_b} - y_{A_i}}{x_A^* - x_{A_i}} \right) \quad (3-241)$$

Substituting (3-240) and (3-241) into (3-238) and (3-239), respectively, gives

$$\frac{1}{K_y} = \frac{1}{k_y} + \frac{m_x}{k_x} \quad (3-242)$$

and

$$\frac{1}{K_x} = \frac{1}{k_x} + \frac{1}{m_y k_y} \quad (3-243)$$

EXAMPLE 3.20

Sulfur dioxide (A) is absorbed into water in a packed column. At a certain location, the bulk conditions are 50°C, 2 atm, $y_{A_b} = 0.085$, and $x_{A_b} = 0.001$. Equilibrium data for SO₂ between air and water at 50°C are

p_{SO_2} , atm	c_{SO_2} , lbmol/ft ³
0.0382	0.00193
0.0606	0.00290
0.1092	0.00483
0.1700	0.00676

Experimental values of the mass transfer coefficients are as follows.

Liquid phase: $k_c = 0.18$ m/h

Gas phase: $k_p = 0.040 \frac{\text{kmol}}{\text{h} \cdot \text{m}^2 \cdot \text{kPa}}$

Using mole-fraction driving forces, compute the mass-transfer flux by:

- Assuming an average Henry's-law constant and a negligible bulk-flow effect.
- Utilizing the actual curved equilibrium line and assuming a negligible bulk-flow effect.
- Utilizing the actual curved equilibrium line and taking into account the bulk-flow effect.

In addition,

- Determine the relative magnitude of the two resistances and the values of the mole fractions at the interface from the results of part (c).

SOLUTION

The equilibrium data are converted to mole fractions by assuming Dalton's law, $y_A = p_A/P$, for the gas and using $x_A = c_A/c$ for the liquid. The concentration of the liquid is close to that of pure water or 3.43 lbmol/ft³ or 55.0 kmol/m³. Thus, the mole fractions at equilibrium are:

y_{SO_2}	x_{SO_2}
0.0191	0.000563
0.0303	0.000846
0.0546	0.001408
0.0850	0.001971

These data are fitted with average and maximum absolute deviations of 0.91% and 1.16%, respectively, by the quadratic equation

$$y_{\text{SO}_2} = 29.74x_{\text{SO}_2} + 6,733x_{\text{SO}_2}^2 \quad (1)$$

Thus, differentiating, the slope of the equilibrium curve is given by

$$m = \frac{dy}{dx} = 29.74 + 13,466x_{\text{SO}_2} \quad (2)$$

The given mass-transfer coefficients can be converted to k_x and k_y by (3-227) and (3-228):

$$k_x = k_c c = 0.18(55.0) = 9.9 \frac{\text{kmol}}{\text{h} \cdot \text{m}^2}$$

$$k_y = k_p P = 0.040(2)(101.3) = 8.1 \frac{\text{kmol}}{\text{h} \cdot \text{m}^2}$$

- From (1) for $x_{A_b} = 0.001$, $y_A^* = 29.74(0.001) + 6,733(0.001)^2 = 0.0365$. From (1), for $y_{A_b} = 0.085$, we solve the quadratic equation to obtain $x_A^* = 0.001975$.

The average slope in this range is

$$m = \frac{0.085 - 0.0365}{0.001975 - 0.001} = 49.7.$$

From an examination of (3-242) and (3-243), the liquid-phase resistance is controlling because the term in k_x is much larger than the term in k_y . Therefore, from (3-243), using $m = m_x$,

$$\frac{1}{K_x} = \frac{1}{9.9} + \frac{1}{49.7(8.1)} = 0.1010 + 0.0025 = 0.1035$$

$$\text{or } K_x = 9.66 \frac{\text{kmol}}{\text{h-m}^2}$$

From (3-223),

$$N_A = 9.66(0.001975 - 0.001) = 0.00942 \frac{\text{kmol}}{\text{h-m}^2}.$$

(b) From part (a), the gas-phase resistance is almost negligible. Therefore, $y_{A_i} \approx y_{A_b}$ and $x_{A_i} \approx x_A^*$.

From (3-241), the slope m_y must, therefore, be taken at the point $y_{A_b} = 0.085$ and $x_A^* = 0.001975$ on the equilibrium line.

From (2), $m_y = 29.74 + 13,466(0.001975) = 56.3$. From (3-243),

$$K_x = \frac{1}{(1/9.9) + [1/(56.3)(8.1)]} = 9.69 \frac{\text{kmol}}{\text{h-m}^2},$$

giving $N_A = 0.00945 \text{ kmol/h-m}^2$. This is only a slight change from part (a).

(c) We now correct for bulk flow. From the results of parts (a) and (b), we have

$$y_{A_b} = 0.085, y_{A_i} = 0.085, x_{A_i} = 0.1975, x_{A_b} = 0.001$$

$$(y_B)_{LM} = 1.0 - 0.085 = 0.915 \text{ and } (x_B)_{LM} \approx 0.9986$$

From (3-229),

$$k'_x = \frac{9.9}{0.9986} = 9.9 \frac{\text{kmol}}{\text{h-m}^2} \text{ and } k'_y = \frac{8.1}{0.915} = 8.85 \frac{\text{kmol}}{\text{h-m}^2}$$

From (3-243),

$$K_x = \frac{1}{(1/9.9) + [1/56.3(8.85)]} = 9.71 \frac{\text{kmol}}{\text{h-m}^2}$$

From (3-223),

$$N_A = 9.71(0.001975 - 0.001) = 0.00947 \frac{\text{kmol}}{\text{h-m}^2}$$

which is only a very slight change from parts (a) and (b), where the bulk-flow effect was ignored. The effect is very small because here it is important only in the gas phase; but the liquid-phase resistance is controlling.

(d) The relative magnitude of the mass-transfer resistances can be written as

$$\frac{1/m_y k'_y}{1/k'_x} = \frac{1/(56.3)(8.85)}{1/9.9} = 0.02$$

Thus, the gas-phase resistance is only 2% of the liquid-phase resistance. The interface vapor mole fraction can be obtained from (3-223), after accounting for the bulk-flow effect:

$$y_{A_i} = y_{A_b} - \frac{N_A}{k'_y} = 0.085 - \frac{0.00947}{8.85} = 0.084$$

Similarly,

$$x_{A_i} = \frac{N_A}{k'_x} + x_{A_b} = \frac{0.00947}{9.9} + 0.001 = 0.00196$$

SUMMARY

1. Mass transfer is the net movement of a component in a mixture from one region to another region of different concentration, often between two phases across an interface. Mass transfer occurs by molecular diffusion, eddy diffusion, and bulk flow. Molecular diffusion occurs by a number of different driving forces, including concentration (the most important), pressure, temperature, and external force fields.
2. Fick's first law for steady-state conditions states that the mass-transfer flux by ordinary molecular diffusion is equal to the product of the diffusion coefficient (diffusivity) and the negative of the concentration gradient.
3. Two limiting cases of mass transfer are equimolar counterdiffusion (EMD) and unimolecular diffusion (UMD). The former is also a good approximation for dilute conditions. The latter must include the bulk-flow effect.
4. When experimental data are not available, diffusivities in gas and liquid mixtures can be estimated. Diffusivities in solids, including porous solids, crystalline solids, metals, glass, ceramics, polymers, and cellular solids, are best measured. For some solids—for example, wood—diffusivity is an anisotropic property.

5. Diffusivity values vary by orders of magnitude. Typical values are 0.10, 1×10^{-5} , and $1 \times 10^{-9} \text{ cm}^2/\text{s}$ for ordinary molecular diffusion of a solute in a gas, liquid, and solid, respectively.
6. Fick's second law for unsteady-state diffusion is readily applied to semi-infinite and finite stagnant media, including certain anisotropic materials.
7. Molecular diffusion under laminar-flow conditions can be determined from Fick's first and second laws, provided that velocity profiles are available. Common cases include falling liquid-film flow, boundary-layer flow on a flat plate, and fully developed flow in a straight, circular tube. Results are often expressed in terms of a mass-transfer coefficient embedded in a dimensionless group called the Sherwood number. The mass-transfer flux is given by the product of the mass-transfer coefficient and a concentration driving force.
8. Mass transfer in turbulent flow is often predicted by analogy to heat transfer. Of particular importance is the Chilton-Colburn analogy, which utilizes empirical j -factor correlations and the dimensionless Stanton number for mass transfer. A more accurate equation by Churchill and Zajic should be used for flow in tubes, particularly at high Schmidt and Reynolds numbers.

9. A number of models have been developed for mass transfer across a two-fluid interface and into a liquid. These include the film theory, penetration theory, surface-renewal theory, and the film-penetration theory. These theories predict mass-transfer coefficients that are proportional to the diffusivity raised to an exponent that varies from 0.5 to 1.0. Most experimental data provide exponents ranging from 0.5 to 0.75.

10. The two-film theory of Whitman (more properly referred to as a two-resistance theory) is widely used to predict the mass-transfer flux from one fluid phase, across an interface, and into another fluid phase, assuming equilibrium at the interface. One resistance is often controlling. The theory defines an overall mass-transfer coefficient that is determined from the separate coefficients for each of the two phases and the equilibrium relationship at the interface.

REFERENCES

1. TAYLOR, R., and R. KRISHNA, *Multicomponent Mass Transfer*, John Wiley and Sons, New York (1993).
2. POLING, B.E., J.M. PRASNITZ, and J.P. O'CONNELL, *The Properties of Liquids and Gases*, 5th ed., McGraw-Hill, New York (2001).
3. FULLER, E.N., P.D. SCHETTLER, and J.C. GIDDINGS, *Ind. Eng. Chem.*, **58** (5), 18–27 (1966).
4. TAKAHASHI, S., *J. Chem. Eng. Jpn.*, **7**, 417–420 (1974).
5. SLATTERY, J.C., M.S. thesis, University of Wisconsin, Madison (1955).
6. WILKE, C.R., and P. CHANG, *AIChE J.*, **1**, 264–270 (1955).
7. HAYDUK, W., and B.S. MINHAS, *Can. J. Chem. Eng.*, **60**, 295–299 (1982).
8. QUAYLE, O.R., *Chem. Rev.*, **53**, 439–589 (1953).
9. VIGNES, A., *Ind. Eng. Chem. Fundam.*, **5**, 189–199 (1966).
10. SORBER, H.A., *Handbook of Biochemistry, Selected Data for Molecular Biology*, 2nd ed., Chemical Rubber Co., Cleveland, OH (1970).
11. GEANKOPLIS, C.J., *Transport Processes and Separation Process Principles*, 4th ed., Prentice-Hall, Upper Saddle River, NJ (2003).
12. FRIEDMAN, L., and E.O. KRAEMER, *J. Am. Chem. Soc.*, **52**, 1298–1304, 1305–1310, 1311–1314 (1930).
13. BOUCHER, D.F., J.C. BRIER, and J.O. OSBURN, *Trans. AIChE*, **38**, 967–993 (1942).
14. BARRER, R.M., *Diffusion in and through Solids*, Oxford University Press, London (1951).
15. SWETS, D.E., R.W. LEE, and R.C. FRANK, *J. Chem. Phys.*, **34**, 17–22 (1961).
16. LEE, R. W., *J. Chem. Phys.*, **38**, 44–455 (1963).
17. WILLIAMS, E.L., *J. Am. Ceram. Soc.*, **48**, 190–194 (1965).
18. SUCOV, E.W., *J. Am. Ceram. Soc.*, **46**, 14–20 (1963).
19. KINGERY, W.D., H.K. BOWEN, and D.R. UHLMANN, *Introduction to Ceramics*, 2nd ed., John Wiley and Sons, New York (1976).
20. FERRY, J.D., *Viscoelastic Properties of Polymers*, John Wiley and Sons, New York (1980).
21. RHEE, C.K., and J.D. FERRY, *J. Appl. Polym. Sci.*, **21**, 467–476 (1977).
22. BRANDRUP, J., and E.H. IMMERGUT, Eds., *Polymer Handbook*, 3rd ed., John Wiley and Sons, New York (1989).
23. GIBSON, L.J., and M.F. ASHBY, *Cellular Solids, Structure and Properties*, Pergamon Press, Elmsford, NY (1988).
24. STAMM, A.J., *Wood and Cellulose Science*, Ronald Press, New York (1964).
25. SHERWOOD, T.K., *Ind. Eng. Chem.*, **21**, 12–16 (1929).
26. CARSLAW, H.S., and J.C. JAEGER, *Heat Conduction in Solids*, 2nd ed., Oxford University Press, London (1959).
27. CRANK, J., *The Mathematics of Diffusion*, Oxford University Press, London (1956).
28. BIRD, R.B., W.E. STEWART, and E.N. LIGHTFOOT, *Transport Phenomena*, 2nd ed., John Wiley and Sons, New York (2002).
29. CHURCHILL, R.V., *Operational Mathematics*, 2nd ed., McGraw-Hill, New York (1958).
30. ABRAMOWITZ, M., and I.A. STEGUN, Eds., *Handbook of Mathematical Functions*, National Bureau of Standards, Applied Mathematics Series 55, Washington, DC (1964).
31. NEWMAN, A.B., *Trans. AIChE*, **27**, 310–333 (1931).
32. GRIMLEY, S.S., *Trans. Inst. Chem. Eng.* (London), **23**, 228–235 (1948).
33. JOHNSTONE, H.F., and R.L. PIGFORD, *Trans. AIChE*, **38**, 25–51 (1942).
34. OLBRICH, W.E., and J.D. WILD, *Chem. Eng. Sci.*, **24**, 25–32 (1969).
35. CHURCHILL, S.W., *The Interpretation and Use of Rate Data: The Rate Concept*, McGraw-Hill, New York (1974).
36. CHURCHILL, S.W., and R. USAGI, *AIChE J.*, **18**, 1121–1128 (1972).
37. EMMERT, R.E., and R.L. PIGFORD, *Chem. Eng. Prog.*, **50**, 87–93 (1954).
38. PRANDTL, L., *Proc. 3rd Int. Math. Congress*, Heidelberg (1904); reprinted in *NACA Tech. Memo 452* (1928).
39. BLASIUS, H., *Z. Math. Phys.*, **56**, 1–37 (1908); reprinted in *NACA Tech. Memo 1256*.
40. SCHLICHTING, H., *Boundary Layer Theory*, 4th ed., McGraw-Hill, New York (1960).
41. POHLHAUSEN, E., *Z. Angew. Math. Mech.*, **1**, 252 (1921).
42. POHLHAUSEN, E., *Z. Angew. Math. Mech.*, **1**, 115–121 (1921).
43. LANGHAAR, H.L., *Trans. ASME*, **64**, A-55 (1942).
44. GRAETZ, L., *Ann. d. Physik*, **25**, 337–357 (1885).
45. SELLARS, J.R., M. TRIBUS, and J.S. KLEIN, *Trans. ASME*, **78**, 441–448 (1956).
46. LEVEQUE, J., *Ann. Mines*, [12], **13**, 201, 305, 381 (1928).
47. KNUDSEN, J.G., and D.L. KATZ, *Fluid Dynamics and Heat Transfer*, McGraw-Hill, New York (1958).
48. HAUSEN, H., *Verfahrenstechnik Beih. z. Ver. Deut. Ing.*, **4**, 91 (1943).
49. LINTON, W.H. Jr., and T.K. SHERWOOD, *Chem. Eng. Prog.*, **46**, 258–264 (1950).
50. REYNOLDS, O., *Trans. Roy. Soc. (London)*, **174A**, 935–982 (1883).
51. BOUSSINESQ, J., *Mem. Pre. Par. Div. Sav.*, XXIII, Paris (1877).
52. PRANDTL, L., *Z. Angew. Math. Mech.*, **5**, 136 (1925); reprinted in *NACA Tech. Memo 1231* (1949).
53. REYNOLDS, O., *Proc. Manchester Lit. Phil. Soc.*, **14**, 7 (1874).
54. COLBURN, A.P., *Trans. AIChE*, **29**, 174–210 (1933).
55. CHILTON, T.H., and A.P. COLBURN, *Ind. Eng. Chem.*, **26**, 1183–1187 (1934).
56. PRANDTL, L., *Physik. Z.*, **11**, 1072 (1910).

57. FRIEND, W.L., and A.B. METZNER, *AIChE J.*, **4**, 393–402 (1958).
58. NERNST, W., *Z. Phys. Chem.*, **47**, 52 (1904).
59. HIGBIE, R., *Trans. AIChE*, **31**, 365–389 (1935).
60. DANCKWERTS, P.V., *Ind. Eng. Chem.*, **43**, 1460–1467 (1951).
61. LEVENSPIEL, O., *Chemical Reaction Engineering*, 3rd ed., John Wiley and Sons, New York (1999).
62. TOOR, H.L., and J.M. MARCHELLO, *AIChE J.*, **4**, 97–101 (1958).
63. WHITMAN, W.G., *Chem. Met. Eng.*, **29**, 146–148 (1923).
64. VAN DRIEST, E.R., *J. Aero Sci.*, 1007–1011 and 1036 (1956).
65. REICHARDT, H., *Fundamentals of Turbulent Heat Transfer*, NACA Report TM-1408 (1957).
66. DREW, T.B., E.C. KOO, and W.H. MCADAMS, *Trans. Am. Inst. Chem. Engrs.*, **28**, 56 (1933).
67. NIKURADSE, J., *VDI-Forschungsheft*, p. 361 (1933).
68. LAUNDER, B.E., and D.B. SPALDING, *Lectures in Mathematical Models of Turbulence*, Academic Press, New York (1972).
69. HENG, L., C. CHAN, and S.W. CHURCHILL, *Chem. Eng. J.*, **71**, 163 (1998).
70. CHURCHILL, S.W., and S.C. ZAJIC, *AIChE J.*, **48**, 927–940 (2002).
71. CHURCHILL, S.W., “Turbulent Flow and Convection: The Prediction of Turbulent Flow and Convection in a Round Tube,” in *Advances in Heat Transfer*, J.P. Hartnett, and T.F. Irvine, Jr., Ser. Eds., Academic Press, New York, **34**, 255–361 (2001).
72. YU, B., H. OZOE, and S.W. CHURCHILL, *Chem. Eng. Sci.*, **56**, 1781 (2001).
73. CHURCHILL, S.W., and C. CHAN, *Ind. Eng. Chem. Res.*, **34**, 1332 (1995).
74. CHURCHILL, S.W., *AIChE J.*, **43**, 1125 (1997).

EXERCISES

Section 3.1

3.1 A beaker filled with an equimolar liquid mixture of ethyl alcohol and ethyl acetate evaporates at 0°C into still air at 101 kPa (1 atm) total pressure. Assuming Raoult's law applies, what will be the composition of the liquid remaining when half the original ethyl alcohol has evaporated, assuming that each component evaporates independently of the other? Also assume that the liquid is always well mixed. The following data are available:

	Vapor Pressure, kPa at 0°C	Diffusivity in Air m ² /s
Ethyl acetate (AC)	3.23	6.45×10^{-6}
Ethyl alcohol (AL)	1.62	9.29×10^{-6}

3.2 An open tank, 10 ft in diameter and containing benzene at 25°C, is exposed to air in such a manner that the surface of the liquid is covered with a stagnant air film estimated to be 0.2 in. thick. If the total pressure is 1 atm and the air temperature is 25°C, what loss of material in pounds per day occurs from this tank? The specific gravity of benzene at 60°F is 0.877. The concentration of benzene at the outside of the film is so low that it may be neglected. For benzene, the vapor pressure at 25°C is 100 torr, and the diffusivity in air is 0.08 cm²/s.

3.3 An insulated glass tube and condenser are mounted on a reboiler containing benzene and toluene. The condenser returns liquid reflux so that it runs down the wall of the tube. At one point in the tube the temperature is 170°F, the vapor contains 30 mol% toluene, and the liquid reflux contains 40 mol% toluene. The effective thickness of the stagnant vapor film is estimated to be 0.1 in. The molar latent heats of benzene and toluene are equal. Calculate the rate at which toluene and benzene are being interchanged by equimolar countercurrent diffusion at this point in the tube in lbmol/h-ft².

Diffusivity of toluene in benzene = 0.2 ft²/h.

Pressure = 1 atm total pressure (in the tube).

Vapor pressure of toluene at 170°F = 400 torr.

3.4 Air at 25°C with a dew-point temperature of 0°C flows past the open end of a vertical tube filled with liquid water maintained at 25°C. The tube has an inside diameter of 0.83 in., and the liquid

level was originally 0.5 in. below the top of the tube. The diffusivity of water in air at 25°C is 0.256 cm²/s.

- (a) How long will it take for the liquid level in the tube to drop 3 in.?
- (b) Make a plot of the liquid level in the tube as a function of time for this period.

3.5 Two bulbs are connected by a tube, 0.002 m in diameter and 0.20 m in length. Initially bulb 1 contains argon, and bulb 2 contains xenon. The pressure and temperature are maintained at 1 atm and 105°C, at which the diffusivity is 0.180 cm²/s. At time $t = 0$, diffusion is allowed to occur between the two bulbs. At a later time, the argon mole fraction in the gas at end 1 of the tube is 0.75, and 0.20 at the other end. Determine at the later time:

- (a) The rates and directions of mass transfer of argon and xenon
- (b) The transport velocity of each species
- (c) The molar average velocity of the mixture

Section 3.2

3.6 The diffusivity of toluene in air was determined experimentally by allowing liquid toluene to vaporize isothermally into air from a partially filled vertical tube 3 mm in diameter. At a temperature of 39.4°C, it took 96×10^4 s for the level of the toluene to drop from 1.9 cm below the top of the open tube to a level of 7.9 cm below the top. The density of toluene is 0.852 g/cm³, and the vapor pressure is 57.3 torr at 39.4°C. The barometer reading was 1 atm. Calculate the diffusivity and compare it with the value predicted from (3-36). Neglect the counterdiffusion of air.

3.7 An open tube, 1 mm in diameter and 6 in. long, has pure hydrogen blowing across one end and pure nitrogen blowing across the other. The temperature is 75°C.

- (a) For equimolar counterdiffusion, what will be the rate of transfer of hydrogen into the nitrogen stream (mol/s)? Estimate the diffusivity from (3-36).

- (b) For part (a), plot the mole fraction of hydrogen against distance from the end of the tube past which nitrogen is blown.

3.8 Some HCl gas diffuses across a film of air 0.1 in. thick at 20°C. The partial pressure of HCl on one side of the film is 0.08 atm and it is zero on the other. Estimate the rate of diffusion, as mol HCl/s-cm², if the total pressure is (a) 10 atm, (b) 1 atm, (c) 0.1 atm. The diffusivity of HCl in air at 20°C and 1 atm is 0.145 cm²/s.

3.9 Estimate the diffusion coefficient for the gaseous binary system nitrogen (A)/toluene (B) at 25°C and 3 atm using the method of Fuller et al.

3.10 For the mixture of Example 3.3, estimate the diffusion coefficient if the pressure is increased to 100 atm using the method of Takahashi.

3.11 Estimate the diffusivity of carbon tetrachloride at 25°C in a dilute solution of: (a) Methanol, (b) Ethanol, (c) Benzene, and (d) *n*-Hexane by the method of Wilke–Chang and Hayduk–Minhas. Compare the estimated values with the following experimental observations:

Solvent	Experimental D_{AB} , cm ² /s
Methanol	1.69×10^{-5} cm ² /s at 15°C
Ethanol	1.50×10^{-5} cm ² /s at 25°C
Benzene	1.92×10^{-5} cm ² /s at 25°C
<i>n</i> -Hexane	3.70×10^{-5} cm ² /s at 25°C

3.12 Estimate the liquid diffusivity of benzene (A) in formic acid (B) at 25°C and infinite dilution. Compare the estimated value to that of Example 3.6 for formic acid at infinite dilution in benzene.

3.13 Estimate the liquid diffusivity of acetic acid at 25°C in a dilute solution of: (a) Benzene, (b) Acetone, (c) Ethyl acetate, and (d) Water by an appropriate method. Compare the estimated values with the following experimental values:

Solvent	Experimental D_{AB} , cm ² /s
Benzene	2.09×10^{-5} cm ² /s at 25°C
Acetone	2.92×10^{-5} cm ² /s at 25°C
Ethyl acetate	2.18×10^{-5} cm ² /s at 25°C
Water	1.19×10^{-5} cm ² /s at 20°C

3.14 Water in an open dish exposed to dry air at 25°C is found to vaporize at a constant rate of 0.04 g/h·cm². Assuming the water surface to be at the wet-bulb temperature of 11.0°C, calculate the effective gas-film thickness (i.e., the thickness of a stagnant air film that would offer the same resistance to vapor diffusion as is actually encountered at the water surface).

3.15 Isopropyl alcohol is undergoing mass transfer at 35°C and 2 atm under dilute conditions through water, across a phase boundary, and then through nitrogen. Based on the data given below, estimate for isopropyl alcohol:

- The diffusivity in water using the Wilke–Chang equation
- The diffusivity in nitrogen using the Fuller et al. equation
- The product, $D_{AB}\rho_M$, in water
- The product, $D_{AB}\rho_M$, in air

where ρ_M is the molar density of the mixture.

Using the above results, compare:

- The diffusivities in parts (a) and (b)
- The diffusivity-molar density products in Parts (c) and (d)

Lastly:

- What conclusions can you come to about molecular diffusion in the liquid phase versus the gaseous phase?

Data:

Component	T_c , °R	P_c , psia	Z_c	v_L , cm ³ /mol
Nitrogen	227.3	492.9	0.289	—
Isopropyl alcohol	915	691	0.249	76.5

3.16 Experimental liquid-phase activity-coefficient data are given in Exercise 2.23 for the ethanol/benzene system at 45°C. Estimate and plot diffusion coefficients for both ethanol and benzene over the entire composition range.

3.17 Estimate the diffusion coefficient of NaOH in a 1-M aqueous solution at 25°C.

3.18 Estimate the diffusion coefficient of NaCl in a 2-M aqueous solution at 18°C. Compare your estimate with the experimental value of 1.28×10^{-5} cm²/s.

3.19 Estimate the diffusivity of N₂ in H₂ in the pores of a catalyst at 300°C and 20 atm if the porosity is 0.45 and the tortuosity is 2.5. Assume ordinary molecular diffusion in the pores.

3.20 Gaseous hydrogen at 150 psia and 80°F is stored in a small, spherical, steel pressure vessel having an inside diameter of 4 in. and a wall thickness of 0.125 in. At these conditions, the solubility of hydrogen in steel is 0.094 lbmol/ft³ and the diffusivity of hydrogen in steel is 3.0×10^{-9} cm²/s. If the inner surface of the vessel remains saturated at the existing hydrogen pressure and the hydrogen partial pressure at the outer surface is assumed to be zero, estimate:

- The initial rate of mass transfer of hydrogen through the metal wall
- The initial rate of pressure decrease inside the vessel
- The time in hours for the pressure to decrease to 50 psia, assuming the temperature stays constant at 80°F

3.21 A polyisoprene membrane of 0.8-μm thickness is to be used to separate a mixture of methane and H₂. Using the data in Table 14.9 and the following compositions, estimate the mass-transfer flux of each of the two species.

	Partial Pressures, MPa	
	Membrane Side 1	Membrane Side 2
Methane	2.5	0.05
Hydrogen	2.0	0.20

Section 3.3

3.22 A 3-ft depth of stagnant water at 25°C lies on top of a 0.10-in. thickness of NaCl. At time $t < 0$, the water is pure. At time $t = 0$, the salt begins to dissolve and diffuse into the water. If the concentration of salt in the water at the solid–liquid interface is maintained at saturation (36 g NaCl/100 g H₂O) and the diffusivity of NaCl in water is 1.2×10^{-5} cm²/s, independent of concentration, estimate, by assuming the water to act as a semi-infinite medium, the time and the concentration profile of salt in the water when

- 10% of the salt has dissolved
- 50% of the salt has dissolved
- 90% of the salt has dissolved

3.23 A slab of dry wood of 4-in. thickness and sealed edges is exposed to air of 40% relative humidity. Assuming that the two unsealed faces of the wood immediately jump to an equilibrium moisture content of 10 lb H₂O per 100 lb of dry wood, determine the time for the moisture to penetrate to the center of the slab (2 in. from either face). Assume a diffusivity of water in the wood as 8.3×10^{-6} cm²/s.

3.24 A wet, clay brick measuring 2 × 4 × 6 in. has an initial uniform moisture content of 12 wt%. At time $t = 0$, the brick is exposed on all sides to air such that the surface moisture content is

maintained at 2 wt%. After 5 h, the average moisture content is 8 wt%. Estimate:

- The diffusivity of water in the clay in cm^2/s .
- The additional time for the average moisture content to reach 4 wt%. All moisture contents are on a dry basis.

3.25 A spherical ball of clay, 2 in. in diameter, has an initial moisture content of 10 wt%. The diffusivity of water in the clay is $5 \times 10^{-6} \text{ cm}^2/\text{s}$. At time $t = 0$, the surface of the clay is brought into contact with air such that the moisture content at the surface is maintained at 3 wt%. Estimate the time for the average moisture content in the sphere to drop to 5 wt%. All moisture contents are on a dry basis.

Section 3.4

3.26 Estimate the rate of absorption of pure oxygen at 10 atm and 25°C into water flowing as a film down a vertical wall 1 m high and 6 cm in width at a Reynolds number of 50 without surface ripples. Assume the diffusivity of oxygen in water is $2.5 \times 10^{-5} \text{ cm}^2/\text{s}$ and that the mole fraction of oxygen in water at saturation for the above temperature and pressure is 2.3×10^{-4} .

3.27 For the conditions of Example 3.13, determine at what height from the top the average concentration of CO_2 would correspond to 50% of saturation.

3.28 Air at 1 atm flows at 2 m/s across the surface of a 2-in.-long surface that is covered with a thin film of water. If the air and water are maintained at 25°C , and the diffusivity of water in air at these conditions is $0.25 \text{ cm}^2/\text{s}$, estimate the mass flux for the evaporation of water at the middle of the surface assuming laminar boundary-layer flow. Is this assumption reasonable?

3.29 Air at 1 atm and 100°C flows across a thin, flat plate of naphthalene that is 1 m long, causing the plate to sublime. The Reynolds number at the trailing edge of the plate is at the upper limit for a laminar boundary layer. Estimate:

- The average rate of sublimation in kmol/s-m^2
- The local rate of sublimation at a distance of 0.5 m from the leading edge of the plate

Physical properties are given in Example 3.14.

3.30 Air at 1 atm and 100°C flows through a straight, 5-cm-diameter circular tube, cast from naphthalene, at a Reynolds number of 1,500. Air entering the tube has an established laminar-flow velocity profile. Properties are given in Example 3.14. If pressure drop through the tube is negligible, calculate the length of tube needed for the average mole fraction of naphthalene in the exiting air to be 0.005.

3.31 A spherical water drop is suspended from a fine thread in still, dry air. Show:

- That the Sherwood number for mass transfer from the surface of the drop into the surroundings has a value of 2 if the characteristic length is the diameter of the drop.

If the initial drop diameter is 1 mm, the air temperature is 38°C , the drop temperature is 14.4°C , and the pressure is 1 atm, calculate:

- The initial mass of the drop in grams.
- The initial rate of evaporation in grams per second.
- The time in seconds for the drop diameter to be reduced to 0.2 mm.
- The initial rate of heat transfer to the drop. If the Nusselt number is also 2, is the rate of heat transfer sufficient to supply the heat of vaporization and sensible heat of the evaporated water? If not, what will happen?

Section 3.5

3.32 Water at 25°C flows at 5 ft/s through a straight, cylindrical tube cast from benzoic acid, of 2-in. inside diameter. If the tube is 10 ft long, and fully developed, turbulent flow is assumed, estimate the average concentration of benzoic acid in the water leaving the tube. Physical properties are given in Example 3.15.

3.33 Air at 1 atm flows at a Reynolds number of 50,000 normal to a long, circular, 1-in.-diameter cylinder made of naphthalene. Using the physical properties of Example 3.14 for a temperature of 100°C , calculate the average sublimation flux in kmol/s-m^2 .

3.34 For the conditions of Exercise 3.33, calculate the initial average rate of sublimation in kmol/s-m^2 for a spherical particle of 1-in. initial diameter. Compare this result to that for a bed packed with naphthalene spheres with a void fraction of 0.5.

Section 3.6

3.35 Carbon dioxide is stripped from water by air in a wetted-wall tube. At a certain location, where the pressure is 10 atm and the temperature is 25°C , the mass-transfer flux of CO_2 is $1.62 \text{ lbmol/h-ft}^2$. The partial pressures of CO_2 are 8.2 atm at the interface and 0.1 atm in the bulk gas. The diffusivity of CO_2 in air at these conditions is $1.6 \times 10^{-2} \text{ cm}^2/\text{s}$. Assuming turbulent flow of the gas, calculate by the film theory, the mass-transfer coefficient k_c for the gas phase and the film thickness.

3.36 Water is used to remove CO_2 from air by absorption in a column packed with Pall rings. At a certain region of the column where the partial pressure of CO_2 at the interface is 150 psia and the concentration in the bulk liquid is negligible, the absorption rate is $0.017 \text{ lbmol/h-ft}^2$. The diffusivity of CO_2 in water is $2.0 \times 10^{-5} \text{ cm}^2/\text{s}$. Henry's law for CO_2 is $p = Hx$, where $H = 9,000$ psia. Calculate:

- The liquid-phase mass-transfer coefficient and the film thickness
- Contact time for the penetration theory
- Average eddy residence time and the probability distribution for the surface-renewal theory

3.37 Determine the diffusivity of H_2S in water, using the penetration theory, from the following data for the absorption of H_2S into a laminar jet of water at 20°C .

Jet diameter = 1 cm, Jet length = 7 cm, and Solubility of H_2S in water = 100 mol/m^3

The average rate of absorption varies with the flow rate of the jet as follows:

Jet Flow Rate, cm^3/s	Rate of Absorption, $\text{mol/s} \times 10^6$
0.143	1.5
0.568	3.0
1.278	4.25
2.372	6.15
3.571	7.20
5.142	8.75

Section 3.7

3.38 In a test on the vaporization of H_2O into air in a wetted-wall column, the following data were obtained:

Tube diameter, 1.46 cm, Wetted-tube length, 82.7 cm
Air rate to tube at 24°C and 1 atm, $720 \text{ cm}^3/\text{s}$

Temperature of inlet water, 25.15°C, Temperature of outlet water, 25.35°C

Partial pressure of water in inlet air, 6.27 torr, and in outlet air, 20.1 torr

The value for the diffusivity of water vapor in air is 0.22 cm²/s at 0°C and 1 atm. The mass velocity of air is taken relative to the pipe wall. Calculate:

(a) Rate of mass transfer of water into the air

(b) K_G for the wetted-wall column

3.39 The following data were obtained by Chamber and Sherwood [*Ind. Eng. Chem.*, **29**, 1415 (1937)] on the absorption of ammonia from an ammonia-air system by a strong acid in a wetted-wall column 0.575 in. in diameter and 32.5 in. long:

Inlet acid (2-N H ₂ SO ₄) temperature, °F	76
Outlet acid temperature, °F	81
Inlet air temperature, °F	77
Outlet air temperature, °F	84
Total pressure, atm	1.00
Partial pressure NH ₃ in inlet gas, atm	0.0807
Partial pressure NH ₃ in outlet gas, atm	0.0205
Air rate, lbmol/h	0.260

The operation was countercurrent, with the gas entering at the bottom of the vertical tower and the acid passing down in a thin film on the inner wall. The change in acid strength was inappreciable, and the vapor pressure of ammonia over the liquid may be assumed to have been negligible because of the use of a strong acid for absorption. Calculate the mass-transfer coefficient, k_p , from the data.

3.40 A new type of cooling-tower packing is being tested in a laboratory column. At two points in the column, 0.7 ft apart, the following data have been taken. Calculate the overall volumetric mass-transfer coefficient $K_y a$ that can be used to design a large, packed-bed cooling tower, where a is the mass-transfer area, A , per unit volume, V , of tower.

	Bottom	Top
Water temperature, °F	120	126
Water vapor pressure, psia	1.69	1.995
Mole fraction H ₂ O in air	0.001609	0.0882
Total pressure, psia	14.1	14.3
Air rate, lbmol/h	0.401	0.401
Column area, ft ²	0.5	0.5
Water rate, lbmol/h (approximate)	20	20

Chapter 4

Single Equilibrium Stages and Flash Calculations

The simplest separation process is one in which two phases in contact are brought to physical equilibrium, followed by phase separation. If the separation factor between two species in the two phases is very large, a single contacting stage may be sufficient to achieve a desired separation between them; if not, multiple stages are required. For example, if a vapor phase is in equilibrium with a liquid phase, the separation factor is the relative volatility, α , of a volatile component called the light key, LK, with respect to a less-volatile component called the heavy key, HK, where $\alpha_{LK,HK} = K_{LK}/K_{HK}$. If the separation factor is 10,000, an

almost perfect separation is achieved in a single stage. If the separation factor is only 1.10, an almost perfect separation requires hundreds of stages. In this chapter, only a single equilibrium stage is considered, but a wide spectrum of separation operations is described. In all cases, the calculations are made by combining material balances with phase equilibria relations. When a phase change such as vaporization occurs, or when heat of mixing effects are large, an energy balance must be added. In the next chapter, arrangements of multiple stages, called cascades, are described.

4.0 INSTRUCTIONAL OBJECTIVES

After completing this chapter, you should be able to:

- Explain what an equilibrium stage is and why it may not be sufficient to achieve a desired separation.
- Use the Gibbs phase rule to determine the number of intensive variables that must be specified to fix the remaining intensive variables for a system at equilibrium.
- Extend Gibbs phase rule to include extensive variables so that the number of degrees of freedom (number of variables minus the number of independent relations among the variables) can be determined for a continuous separation process.
- Explain and utilize ways that binary vapor–liquid equilibrium data are presented.
- Define relative volatility between two components of a vapor–liquid mixture.
- Use T – y – x and y – x diagrams of binary mixtures, with the concept of the q -line, to determine equilibrium phase compositions.
- Understand the difference between minimum- and maximum-boiling azeotropes and how they form.
- Use component material-balance equations with K -values to calculate bubble-point, dew-point, and equilibrium-flash conditions for multicomponent mixtures.
- Use triangular phase diagrams for ternary systems with component material balances to determine equilibrium compositions of liquid–liquid mixtures.
- Use distribution coefficients, usually determined from activity coefficients, with component material-balance equations to calculate liquid–liquid phase equilibria for multicomponent systems.
- Use equilibrium diagrams with component material balances to determine equilibrium-phase amounts and compositions for solid–fluid systems (leaching, crystallization, sublimation, desublimation, and adsorption) and for light gas–liquid systems (absorption).
- Calculate phase amounts and compositions for multicomponent vapor–liquid–liquid systems.

4.1 THE GIBBS PHASE RULE AND DEGREES OF FREEDOM

The description of a single-stage system at physical equilibrium involves *intensive variables*, which are independent of the size of the system, and *extensive variables*, which do

depend on system size. Intensive variables are temperature, pressure, and phase compositions (mole fractions, mass fractions, concentrations, etc.). Extensive variables include mass or moles and energy for a batch system, and mass or molar flow rates and energy transfer rates for a flow system.

Regardless of whether only intensive variables or both intensive and extensive variables are considered, only a few of the variables are independent; when these are specified, all other variables become fixed. The number of independent variables is referred to as the *variance* or the number of *degrees of freedom*, $\mathcal{F}$, for the system.

The phase rule of J. Willard Gibbs, which applies only to the intensive variables at equilibrium, is used to determine $\mathcal{F}$. The rule states that

$$\mathcal{F} = C - \mathcal{P} + 2 \quad (4-1)$$

where C is the number of components and $\mathcal{P}$ is the number of phases at equilibrium. Equation (4-1) is derived by counting, at physical equilibrium, the number of intensive variables and the number of independent equations that relate these variables. The number of intensive variables, $\mathcal{V}$, is

$$\mathcal{V} = C\mathcal{P} + 2 \quad (4-2)$$

where the 2 refers to the equilibrium temperature and pressure, while the term $C\mathcal{P}$ is the total number of composition variables (e.g., mole fractions) for components distributed among $\mathcal{P}$ equilibrium phases. The number of independent equations, $\mathcal{E}$, relating the intensive variables is

$$\mathcal{E} = \mathcal{P} + C(\mathcal{P} - 1) \quad (4-3)$$

where the first term, $\mathcal{P}$, refers to the requirement that mole or mass fractions sum to one for each phase and the second term, $C(\mathcal{P} - 1)$, refers to the number of independent K -value equations of the general form

$$K_i = \frac{\text{mole fraction of } i \text{ in phase (1)}}{\text{mole fraction of } i \text{ in phase (2)}}$$

where (1) and (2) refer to equilibrium phases. For two phases, there are C independent expressions of this type; for three phases, $2C$; for four phases, $3C$; and so on. For example, for three phases ($V, L^{(1)}, L^{(2)}$), we can write $3C$ different K -value equations:

$$K_i^{(1)} = y_i/x_i^{(1)} \quad i = 1 \text{ to } C$$

$$K_i^{(2)} = y_i/x_i^{(2)} \quad i = 1 \text{ to } C$$

$$K_{Di} = x_i^{(1)}/x_i^{(2)} \quad i = 1 \text{ to } C$$

However, only $2C$ of these equations are independent, because

$$K_{Di} = K_i^{(2)}/K_i^{(1)}$$

Thus, the term for the number of independent K -value equations is $C(\mathcal{P} - 1)$, not $C\mathcal{P}$.

Degrees-of-Freedom Analysis

The degrees of freedom is the number of intensive variables, $\mathcal{V}$, less the number of equations, $\mathcal{E}$. Thus, from (4-2) and (4-3), $\mathcal{F} = \mathcal{V} - \mathcal{E} = (C\mathcal{P} + 2) - [\mathcal{P} + C(\mathcal{P} - 1)] = C - \mathcal{P} + 2$ which completes the derivation of (4-1). When the number, $\mathcal{F}$, of intensive variables is specified, the remaining $\mathcal{P} + C(\mathcal{P} - 1)$ intensive variables are determined from the $\mathcal{P} + C(\mathcal{P} - 1)$ equations.

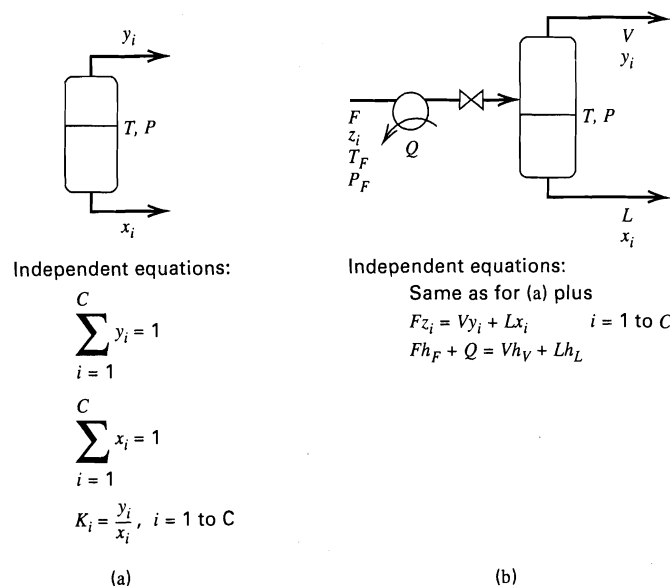


Figure 4.1 Different treatments of degrees of freedom for vapor-liquid phase equilibria: (a) Gibbs phase rule (considers equilibrium intensive variables only); (b) general analysis (considers all intensive and extensive variables).

As an example, consider the vapor-liquid equilibrium ($\mathcal{P} = 2$) shown in Figure 4.1a, where the equilibrium intensive variables are labels on the sketch located above the list of independent equations relating these variables. Suppose there are $C = 3$ components. From (4-1), $\mathcal{F} = 3 - 2 + 2 = 3$. The equilibrium intensive variables are $T, P, x_1, x_2, x_3, y_1, y_2$, and y_3 . If values are specified for T, P , and one of the mole fractions, the remaining five mole fractions are fixed and can be computed from the five independent equations listed in Figure 4.1a. Irrational specifications lead to infeasible results. For example, if the components are $\text{H}_2\text{O}, \text{N}_2$, and O_2 , and $T = 100^\circ\text{F}$ and $P = 15$ psia are specified, a specification of $x_{\text{N}_2} = 0.90$ is not feasible because nitrogen is not nearly this soluble in water.

In using the Gibbs phase rule, it should be noted that the K -values are not variables, but are thermodynamic functions that depend on the intensive variables discussed in Chapter 2.

The Gibbs phase rule is limited because it does not deal with feed streams sent to the equilibrium stage nor with extensive variables used when designing or analyzing separation operations. However, the phase rule can be extended for process applications, by adding the feed stream and extensive variables, and additional independent equations relating feed variables, extensive variables, and the intensive variables already considered by the rule.

Consider the single-stage, vapor-liquid ($\mathcal{P} = 2$) equilibrium separation process shown in Figure 4.1b. By comparison with Figure 4.1a, the additional variables are z_i, T_F, P_F, F, Q, V , and L , or $C + 6$ variables, all of which are indicated in the diagram. In general, for $\mathcal{P}$ phases, the additional variables number $C + \mathcal{P} + 4$. The additional independent equations, listed below the diagram, are the C component material balances and the energy balance, or $C + 1$ equations. Note that, like K -values, stream enthalpies are not

counted as variables because they are thermodynamic functions that depend on intensive variables.

For the general degrees-of-freedom analysis for phase equilibrium, with C components, $\mathcal{P}$ phases, and a single feed phase, (4-2) and (4-3) are extended by adding the number of additional variables and equations, respectively:

$$\begin{aligned}\mathcal{V} &= (C\mathcal{P} + 2) + (C + \mathcal{P} + 4) = \mathcal{P} + C\mathcal{P} + C + 6 \\ \mathcal{E} &= [\mathcal{P} + C(\mathcal{P} - 1)] + (C + 1) = \mathcal{P} + C\mathcal{P} + 1 \quad (4-4) \\ \mathcal{F} &= \mathcal{V} - \mathcal{E} = C + 5\end{aligned}$$

For example, if the $C + 5$ degrees of freedom are used to specify all z_i and the five variables F , T_F , P_F , T , and P , the remaining variables are computed from the equations shown in Figure 4.1.¹ To apply the Gibbs phase rule, (4-1), the number of phases must be known. When applying (4-4), the determination of the number of equilibrium phases, $\mathcal{P}$, is implicit in the computational procedure as illustrated in later sections of this chapter.

In the following sections, the Gibbs phase rule, (4-1), and the equation for the number of degrees of freedom of a flow system, (4-4), are applied to (1) tabular equilibrium data, (2) graphical equilibrium data, or (3) thermodynamic equations for K -values and enthalpies for vapor-liquid, liquid-liquid, solid-liquid, gas-liquid, gas-solid, vapor-liquid-solid, and vapor-liquid-liquid systems at equilibrium.

4.2 BINARY VAPOR-LIQUID SYSTEMS

Experimental vapor-liquid equilibrium data for systems containing two components, A and B, are widely available. Sources include *Perry's Handbook* [1] and Gmehling and Onken [2]. Because $y_B = 1 - y_A$ and $x_B = 1 - x_A$, the data are presented in terms of just four intensive variables: T , P , y_A , and x_A . Most commonly T , y_A , and x_A are tabulated for a fixed P for ranges of y_A and x_A from 0 to 1, where A is the more-volatile component ($y_A > x_A$). However, if an azeotrope (see Section 4.3) forms, B becomes the more volatile component on one side of the azeotropic point. By the Gibbs phase rule, (4-1), $\mathcal{F} = 2 - 2 + 2 = 2$. Thus, with pressure fixed, phase compositions are completely defined if temperature is also fixed, and the separation factor, that is, the relative volatility in the case of vapor-liquid equilibria,

$$\alpha_{A,B} = \frac{K_A}{K_B} = \frac{(y_A/x_A)}{(y_B/x_B)} = \frac{(y_A/x_A)}{(1 - y_A)/(1 - x_A)} \quad (4-5)$$

is also fixed.

Vapor-liquid equilibria data of the form T - y_A - x_A for 1 atm pressure of three binary systems of industrial importance are given in Table 4.1. Included are values of relative volatility computed from (4-5). As discussed in Chapter 2,

Table 4.1 Vapor-Liquid Equilibrium Data for Three Common Binary Systems at 1 atm Pressure

a. Water (A)-Glycerol (B) System

$P = 101.3$ kPa

Data of Chen and Thompson, *J. Chem. Eng. Data*, **15**, 471 (1970)

Temperature, °C	y_A	x_A	$\alpha_{A,B}$
100.0	1.0000	1.0000	
101.7	0.9998	0.9534	250
104.6	0.9996	0.8846	333
109.8	0.9991	0.7731	332
122.0	0.9982	0.5610	440
128.8	0.9980	0.4742	544
140.4	0.9974	0.3622	683
148.2	0.9964	0.3077	627
158.6	0.9940	0.2460	507
175.2	0.9898	0.1756	456
207.0	0.9804	0.0945	481
244.5	0.9341	0.0491	275
282.5	0.8308	0.0250	191
290.0	0.0000	0.0000	

b. Methanol (A)-Water (B) System

$P = 101.3$ kPa

Data of J.G. Dunlop, M.S. thesis, Brooklyn Polytechnic Institute (1948)

Temperature, °C	y_A	x_A	$\alpha_{A,B}$
64.5	1.000	1.000	
65.0	0.979	0.950	2.45
66.0	0.958	0.900	2.53
67.5	0.915	0.800	2.69
69.3	0.870	0.700	2.87
71.2	0.825	0.600	3.14
73.1	0.779	0.500	3.52
75.3	0.729	0.400	4.04
78.0	0.665	0.300	4.63
81.7	0.579	0.200	5.50
84.4	0.517	0.150	6.07
87.7	0.418	0.100	6.46
89.3	0.365	0.080	6.61
91.2	0.304	0.060	6.84
93.5	0.230	0.040	7.17
96.4	0.134	0.020	7.58
100.0	0.000	0.000	

c. *Para*-xylene (A)-*Meta*-xylene (B) System

$P = 101.3$ kPa

Data of Kato, Sato, and Hirata, *J. Chem. Eng. Jpn.*, **4**, 305 (1970)

Temperature, °C	y_A	x_A	$\alpha_{A,B}$
138.335	1.0000	1.0000	
138.414	0.9019	0.9000	1.0021
138.491	0.8033	0.8000	1.0041
138.568	0.7043	0.7000	1.0061
138.644	0.6049	0.6000	1.0082
138.720	0.5051	0.5000	1.0102
138.795	0.4049	0.4000	1.0123
138.869	0.3042	0.3000	1.0140
138.943	0.2032	0.2000	1.0160
139.016	0.1018	0.1000	1.0180
139.088	0.0000	0.0000	

¹The development of (4-4) assumes that the sum of the mole fractions in the feed will equal one. Alternatively, the equation $\sum_{i=1}^C z_i = 1$ can be added to the number of independent equations (thus forcing the feed mole fractions to sum to one). Then, the degrees of freedom becomes one less or $C + 4$.

$\alpha_{A,B}$ depends on T , P , and the compositions of the equilibrium vapor and liquid. At 1 atm, where $\alpha_{A,B}$ is approximated well by $\gamma_A P_A^s / \gamma_B P_B^s$, $\alpha_{A,B}$ depends only on T and x_A , since vapor-phase nonidealities are small. Because of the dependency on x_A , $\alpha_{A,B}$ is not a constant, but varies from point to point. For the three binary systems in Table 4.1, the vapor and liquid phases become richer in the less-volatile component, B, as temperature increases. For $x_A = 1$, the temperature is the normal boiling point of A; for $x_A = 0$, the temperature is the normal boiling point of B. For the three systems, all other data points are at temperatures between the two boiling points. Except for the pure components ($x_A = 1$ or 0), $y_A > x_A$ and $\alpha_{A,B} > 1$.

For the water–glycerol system, the difference in normal boiling points is 190°C. Therefore, relative volatility values are very high, making it possible to achieve a reasonably good separation in a single equilibrium stage. Industrially, the separation is often conducted in an evaporator, which produces a nearly pure water vapor and a glycerol-rich liquid. For example, from Table 4.1, at 207°C, a vapor of 98 mol% water is in equilibrium with a liquid phase containing more than 90 mol% glycerol.

For the methanol–water system, the difference in normal boiling points is 35.5°C. As a result, the relative volatility is an order of magnitude lower than for the water–glycerol system. A sharp separation cannot be made with a single stage. About 30 trays are required in a distillation operation to obtain a 99 mol% methanol distillate and a 98 mol% water bottoms, an acceptable industrial separation.

For the aromatic paraxylene–metaxylene isomer system, the normal boiling-point difference is only 0.8°C. Thus, the relative volatility is very close to 1.0, making the separation by distillation impractical because about 1,000 trays are required to produce nearly pure products. Instead, crystallization and adsorption, which have much higher separation factors, are used commercially to make the separation.

Experimental vapor–liquid equilibrium data for the methanol–water system are given in Table 4.2 in the form of P – y_A – x_A for fixed temperatures of 50, 150, and 250°C. The three sets of data cover a pressure range of 1.789 to 1,234 psia, with the higher pressures corresponding to the higher temperatures. At 50°C, relative volatilities are moderately high at an average value of 4.94 over the composition range. At 150°C, the average relative volatility is only 3.22; and at 250°C, it decreases to 1.75. Thus, as the temperature and pressure increase, the relative volatility decreases significantly. In Table 4.2, for the data set at 250°C, it is seen that as the compositions become richer in methanol, a point is reached in the neighborhood of 1,219 psia, at a methanol mole fraction of 0.772, where the relative volatility is 1.0 and no separation by distillation is possible because the compositions of the vapor and liquid are identical and the two phases become one phase. This is the critical point of a mixture of this composition. It is intermediate between the critical points of pure methanol

Table 4.2 Vapor–Liquid Equilibrium Data for the Methanol–Water System at Temperatures of 50, 150, and 250°C

a. Methanol (A)–Water (B) System

$T = 50^\circ\text{C}$

Data of McGlashan and Williamson,
J. Chem. Eng. Data, **21**, 196 (1976)

Pressure, psia	y_A	x_A	$\alpha_{A,B}$
1.789	0.0000	0.0000	
2.373	0.2661	0.0453	7.64
2.838	0.4057	0.0863	7.23
3.369	0.5227	0.1387	6.80
3.764	0.5898	0.1854	6.32
4.641	0.7087	0.3137	5.32
5.163	0.7684	0.4177	4.63
5.771	0.8212	0.5411	3.90
6.122	0.8520	0.6166	3.58
6.811	0.9090	0.7598	3.16
7.280	0.9455	0.8525	3.00
7.800	0.9817	0.9514	2.74
8.072	1.0000	0.0000	

b. Methanol (A)–Water (B) System

$T = 150^\circ\text{C}$

Data of Griswold and Wong, *Chem. Eng. Prog. Symp. Ser.*, **48** (3), 18 (1952)

Pressure, psia	y_A	x_A	$\alpha_{A,B}$
73.3	0.060	0.009	7.03
79.0	0.135	0.022	6.94
85.7	0.213	0.044	5.88
93.9	0.286	0.079	4.67
114.9	0.459	0.186	3.71
139.7	0.610	0.374	2.62
148.6	0.662	0.459	2.31
160.4	0.731	0.578	1.98
177.4	0.832	0.748	1.67
193.5	0.929	0.893	1.57
194.5	0.943	0.913	1.58
196.5	0.960	0.936	1.64
197.7	0.972	0.953	1.71
199.2	0.982	0.969	1.75

c. Methanol (A)–Water (B) System

$T = 250^\circ\text{C}$

Data of Griswold and Wong, *Chem. Eng. Prog. Symp. Ser.*, **48** (3), 18 (1952)

Pressure, psia	y_A	x_A	$\alpha_{A,B}$
681	0.163	0.066	2.76
764	0.280	0.132	2.56
818	0.344	0.180	2.39
889	0.423	0.254	2.15
949	0.487	0.331	1.92
994	0.542	0.404	1.75
1049	0.596	0.483	1.58
1099	0.643	0.553	1.46
1159	0.698	0.631	1.35
1204	0.756	0.732	1.13
1219	0.772	0.772	1.00
1234	0.797	0.797	1.00

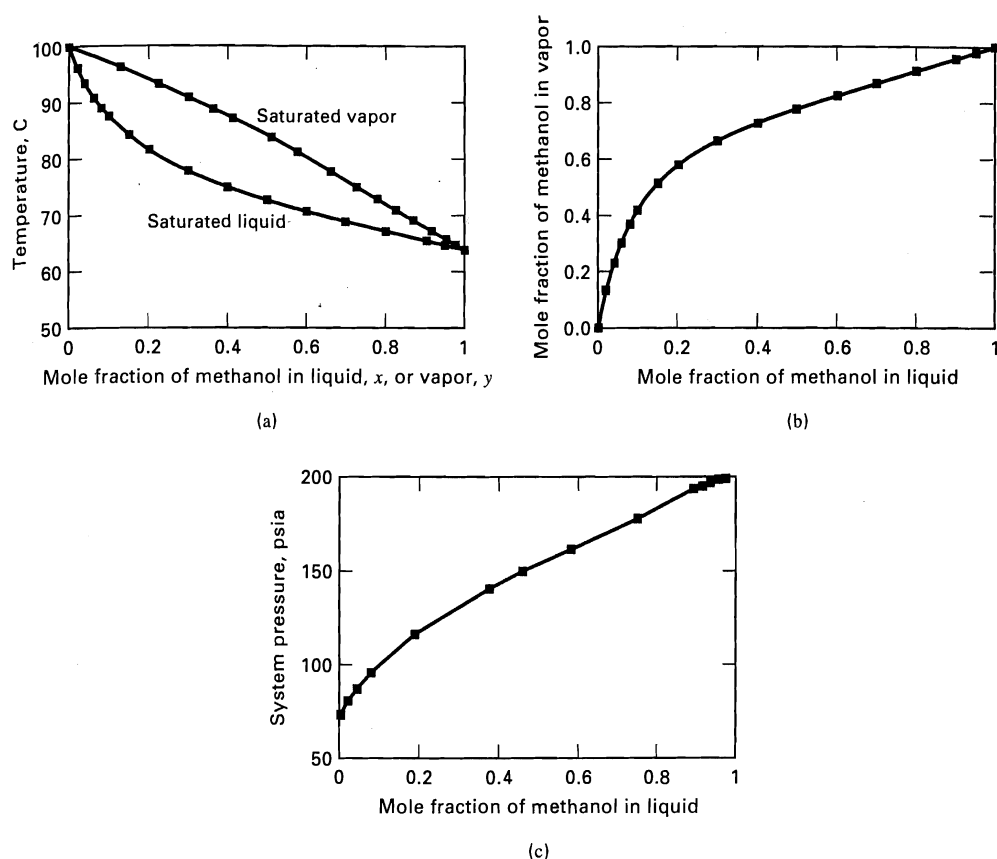


Figure 4.2 Vapor–liquid equilibrium conditions for the methanol–water system: (a) T – y – x diagram for 1 atm pressure; (b) y – x diagram for 1 atm pressure; (c) P – x diagram for 150°C.

and pure water:

$y_A = x_A$	$T_c, ^\circ\text{C}$	P_c, psia
0.000	374.1	3,208
0.772	250	1,219
1.000	240	1,154

A set of critical conditions exists for each binary-mixture composition. In industry, distillation columns operate at pressures well below the critical pressure of the mixture to be separated to avoid relative volatilities that approach a value of 1.

The data of Tables 4.1 and 4.2 for the methanol–water system are plotted in three different ways in Figure 4.2: (a) T versus y_A or x_A at $P = 1$ atm; (b) y_A versus x_A at $P = 1$ atm; and (c) P versus x_A at $T = 150^\circ\text{C}$. These three plots all satisfy the requirement of the Gibbs phase rule that when two intensive variables are fixed, all other variables are fixed by the governing equilibrium equations and mole-fraction-summation constraints. Of the three diagrams in Figure 4.2, only (a) contains the complete data; (b) does not contain temperatures; and (c) does not contain vapor-phase mole fractions. Although mass fractions could be used in place of mole fractions, the latter are preferred because theoretical phase-equilibrium relations are based on molar properties.

Plots like Figure 4.2a are useful for determining phase states, phase-transition temperatures, equilibrium-phase compositions, and equilibrium-phase amounts for a given feed of known composition. Consider the T – y – x plot in Figure 4.3 for the normal hexane (H)–normal octane (O)

system at 101.3 kPa. Because normal hexane is the more volatile species, the mole fractions are for that component. The upper curve, labeled “saturated vapor,” gives the dependency on the dew-point temperature of the vapor mole fraction y_H ; the lower curve, labeled “saturated liquid,” gives the dependency of the bubble-point temperature on the liquid-phase mole fraction, x_H . The two curves converge at $x_H = 0$, the normal boiling point of pure normal octane (258.2°F),

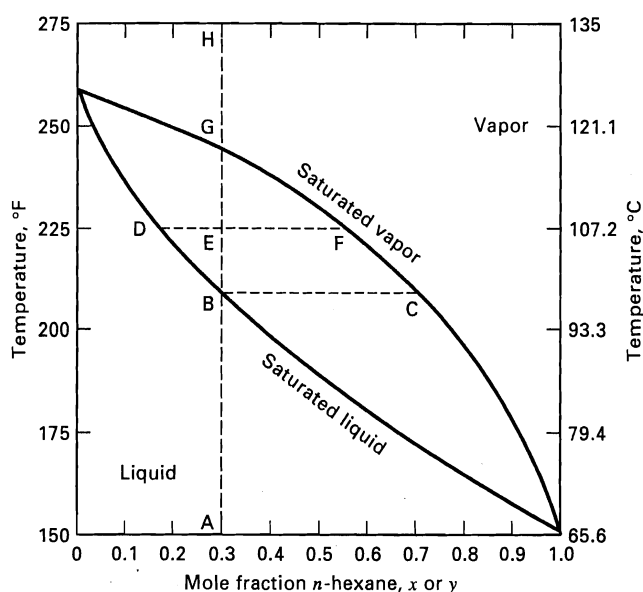


Figure 4.3 Use of the T – y – x phase equilibrium diagram for the normal hexane–normal octane system at 1 atm.

and at $x_H = 1$, the normal boiling point of normal hexane (155.7°F). In order for two phases to exist, a point representing the overall composition of the two-phase binary mixture at a given temperature must be located in the two-phase region between the two curves. If the point lies above the saturated-vapor curve, only a superheated vapor is present; if the point lies below the saturated-liquid curve, only a subcooled liquid exists.

Suppose we have a mixture of 30 mol% H at 150°F. From Figure 4.3, at point A we have a subcooled liquid with $x_H = 0.3$ ($x_O = 0.7$). When this mixture is heated at a constant pressure of 1 atm, the liquid state is maintained until a temperature of 210°F is reached, which corresponds to point B on the saturated-liquid curve. Point B is the *bubble point* because the first bubble of vapor appears. This bubble is a saturated vapor in equilibrium with the liquid at the same temperature. Thus, its composition is determined by following a *tie line*, BC from $x_H = 0.3$ to $y_H = 0.7$ ($y_O = 0.3$). The tie line is horizontal because the temperatures of the two equilibrium phases are the same. As the temperature of the two-phase mixture is increased to point E, on horizontal tie line DEF at 225°F, the mole fraction of H in the liquid phase decreases to $x_H = 0.17$ (because it is more volatile than O and preferentially vaporizes) and correspondingly the mole fraction of H in the vapor phase increases to $y_H = 0.55$. Throughout the two-phase region, the vapor is at its dew point, while the liquid is at its bubble point. The overall composition of the two phases remains at a mole fraction of 0.30 for hexane. At point E, the relative molar amounts of the two equilibrium phases is determined by the inverse lever-arm rule based on the lengths of the line segments DE and EF. Thus, referring to Figures 4.1b and 4.3, $V/L = DE/EF$ or $V/F = DE/DEF$. When the temperature is increased to 245°F, point G, the dew point for $y_H = 0.3$ is reached, where only one droplet of equilibrium liquid remains with a composition from the tie line FG at point F of $x_H = 0.06$. A further increase in temperature—say, to point H at 275°F—gives a superheated vapor with $y_H = 0.30$. The steps are reversible starting from point H and moving down to point A.

Constant-pressure x - y plots like Figure 4.2b are also useful because the equilibrium-vapor-and-liquid compositions are represented by points on the equilibrium curve. However, no phase-temperature information is included. Such plots usually include a 45° reference line, $y = x$. Consider the y - x plot in Figure 4.4 for H-O at 101.3 kPa. This plot is convenient for determining equilibrium-phase compositions for various values of mole-percent vaporization of a feed mixture of a given composition by geometric constructions.

Suppose we have a feed mixture, F , shown in Figure 4.1b, of overall composition $z_H = 0.6$. To determine the equilibrium-phase compositions if, say, 60 mol% of the feed is vaporized, we develop the dashed-line construction in Figure 4.4. Point A on the 45° line represents z_H . Point B on the equilibrium curve is reached by extending a line, called

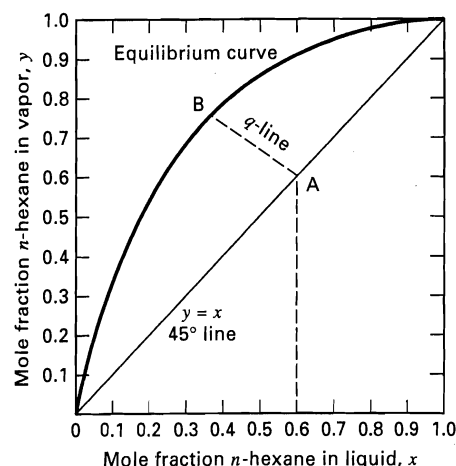


Figure 4.4 Use of the y - x phase equilibrium diagram for the normal hexane-normal octane system at 1 atm.

the q -line, upward and to the left toward the equilibrium curve at a slope equal to $[(V/F) - 1]/(V/F)$. Thus, for 60 mol% vaporization, the slope $= (0.6 - 1)/0.6 = -2/3$. Point B at the intersection of line AB with the equilibrium curve gives the equilibrium composition as $y_H = 0.76$ and $x_H = 0.37$. This computation requires a trial-and-error placement of a horizontal line if we use Figure 4.3. The derivation of the slope of the q -line in Figure 4.4 follows by combining the mole-balance equation of Figure 4.1a,

$$Fz_H = Vy_H + Lx_H$$

with the total mole balance,

$$F = V + L$$

to eliminate L , giving the equation for the q -line:

$$y_H = \left[\frac{(V/F) - 1}{(V/F)} \right] x_H + \left[\frac{1}{(V/F)} \right] z_H$$

Thus, the slope of the q -line passing through the equilibrium point (y_H, x_H) is $[(V/F) - 1]/(V/F)$.

Figure 4.2c is the least used of the three plots in Figure 4.2. However, such a plot does illustrate, for a fixed temperature, the extent to which the binary mixture deviates from an ideal solution. If Raoult's law applies, the total pressure above the liquid is

$$\begin{aligned} P &= P_A^s x_A + P_B^s x_B \\ &= P_A^s x_A + P_B^s (1 - x_A) \\ &= P_B^s + x_A (P_A^s - P_B^s) \end{aligned} \quad (4-6)$$

Thus, a plot of P versus x_A is a straight line with intersections at the vapor pressure of B for $x_A = 0$ and the vapor pressure of A for $x_B = 0$ ($x_A = 1$). The greater the departure from a straight line, the greater is the deviation from the assumptions of an ideal gas and/or an ideal-liquid solution.

If the pressures are sufficiently low that the equilibrium-vapor phase is ideal and the curve is convex, deviations from Raoult's law are positive, and species liquid-phase activity coefficients are greater than 1; if the curve is concave, deviations are negative and activity coefficients are less than 1. In either case, the total pressure is given by

$$P = \gamma_A P_A^s x_A + \gamma_B P_B^s x_B \quad (4-7)$$

If the vapor does not obey the ideal-gas law, (4-7) does not apply. In Figure 4.2c, system pressures are sufficiently high that some deviation from the ideal-gas law occurs. However, the convexity is due mainly to activity coefficients that are greater than 1.

For relatively close (narrow)-boiling binary mixtures that exhibit ideal or nearly ideal behavior, the relative volatility, $\alpha_{A,B}$, varies little with pressure. If $\alpha_{A,B}$ is assumed constant over the entire composition range, the y - x phase-equilibrium curve can be determined and plotted from a rearrangement of (4-5):

$$y_A = \frac{\alpha_{A,B} x_A}{1 + x_A(\alpha_{A,B} - 1)} \quad (4-8)$$

For an ideal solution, $\alpha_{A,B}$ can be approximated with Raoult's law to give

$$\alpha_{A,B} = \frac{K_A}{K_B} = \frac{P_A^s/P}{P_B^s/P} = \frac{P_A^s}{P_B^s} \quad (4-9)$$

Thus, from a knowledge of just the vapor pressures of the two components at a temperature, say, midway between the two boiling points at the given pressure, a y - x phase-equilibrium curve can be approximated using only one value of $\alpha_{A,B}$. Families of curves, as shown in Figure 4.5, can be used for preliminary calculations in the absence of detailed experimental data. The use of (4-8) and (4-9) is not recommended for wide-boiling or nonideal mixtures.

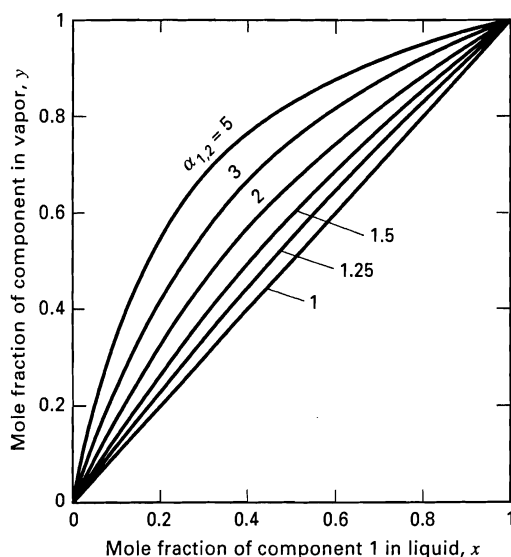


Figure 4.5 Vapor-liquid phase equilibrium curves for constant values of relative volatility.

4.3 AZEOTROPIC SYSTEMS

Departures from Raoult's law frequently manifest themselves in the formation of *azeotropes*, particularly for mixtures of close-boiling species of different chemical types whose liquid solutions are nonideal. Azeotropes are formed by liquid mixtures exhibiting maximum- or minimum-boiling points. These represent, respectively, negative or positive deviations from Raoult's law. Vapor and liquid compositions are identical at the azeotropic composition; thus, all K -values are 1 and no separation of species can take place.

If only one liquid phase exists, the mixture forms a *homogeneous* azeotrope; if more than one liquid phase is present, the azeotrope is *heterogeneous*. In accordance with the Gibbs phase rule, at constant pressure in a two-component system, the vapor can coexist with no more than two liquid phases, while in a ternary mixture up to three liquid phases can coexist with the vapor.

Figures 4.6, 4.7, and 4.8 show three types of azeotropes that are commonly encountered with binary mixtures. The most common type by far is the minimum-boiling homogeneous azeotrope, illustrated in Figure 4.6 for the isopropyl ether-isopropyl alcohol system. In Figure 4.6a, for a temperature of 70°C, the maximum total pressure is greater than the vapor pressure of either component because activity coefficients are greater than 1. The y - x diagram in Figure 4.6b shows that for a pressure of 1 atm the azeotropic mixture occurs at 78 mol% ether. Figure 4.6c is a T - x diagram for a pressure of 101 kPa, where the azeotrope is seen to boil at 66°C. In Figure 4.6a, for 70°C, the azeotrope, at 123 kPa (923 torr), is 72 mol% ether. Thus, the azeotropic composition shifts with pressure. In distillation, the minimum-boiling azeotropic mixture is the overhead product.

For the maximum-boiling homogeneous azeotropic acetone-chloroform system in Figure 4.7a, the minimum total pressure is below the vapor pressures of the pure components because activity coefficients are less than 1. The azeotrope concentrates in the bottoms in a distillation operation.

Heterogeneous azeotropes are always minimum-boiling mixtures because activity coefficients must be significantly greater than 1 to cause splitting into two liquid phases. The region a-b in Figure 4.8a for the water-normal butanol system is a two-phase region where total and partial pressures remain constant as the relative amounts of the two phases change, but the phase compositions do not. The y - x diagram in Figure 4.8b shows a horizontal line over the immiscible region, and the phase diagram of Figure 4.8c shows a minimum constant temperature.

Azeotropes limit the separation achievable by ordinary distillation. It is possible to shift the equilibrium by changing the pressure sufficiently to "break" the azeotrope, or move it away from the region where the required separation must be made. For example, ethyl alcohol and water form a homogeneous minimum-boiling azeotrope of 95.6 wt% alcohol at

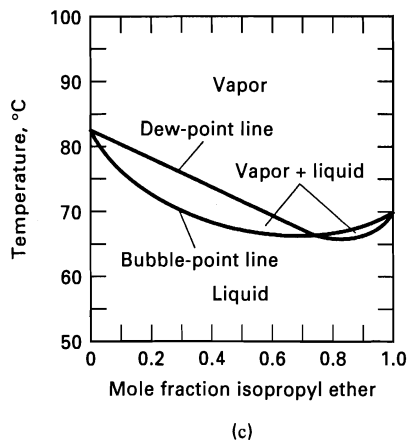
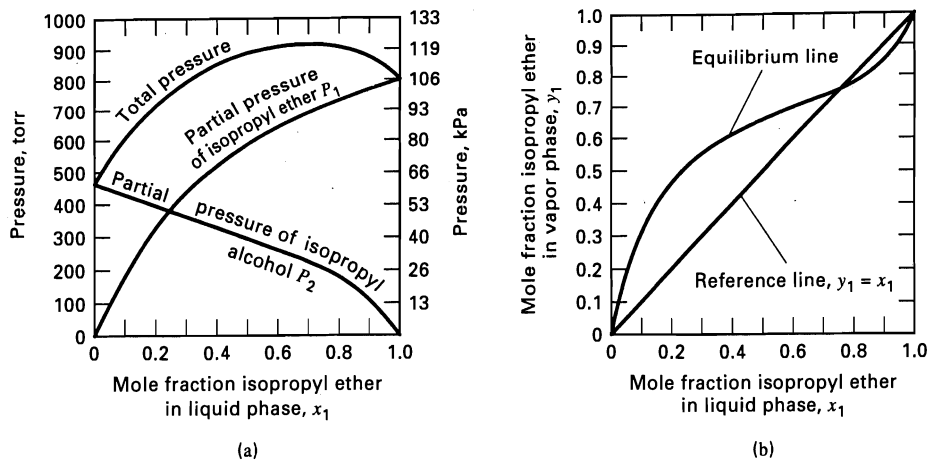


Figure 4.6 Minimum-boiling-point azeotrope, isopropyl ether–isopropyl alcohol system: (a) partial and total pressures at 70°C ; (b) vapor–liquid equilibria at 101 kPa; (c) phase diagram at 101 kPa.

[Adapted from O.A. Hougen, K.M. Watson, and R.A. Ragatz, *Chemical Process Principles. Part II*, 2nd ed., John Wiley and Sons, New York (1959).]

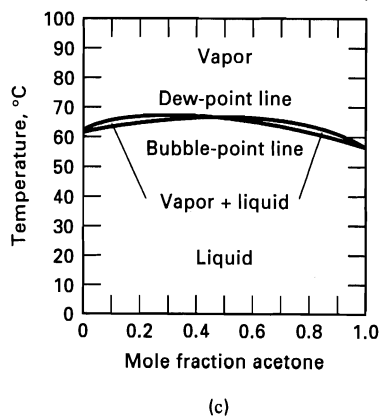
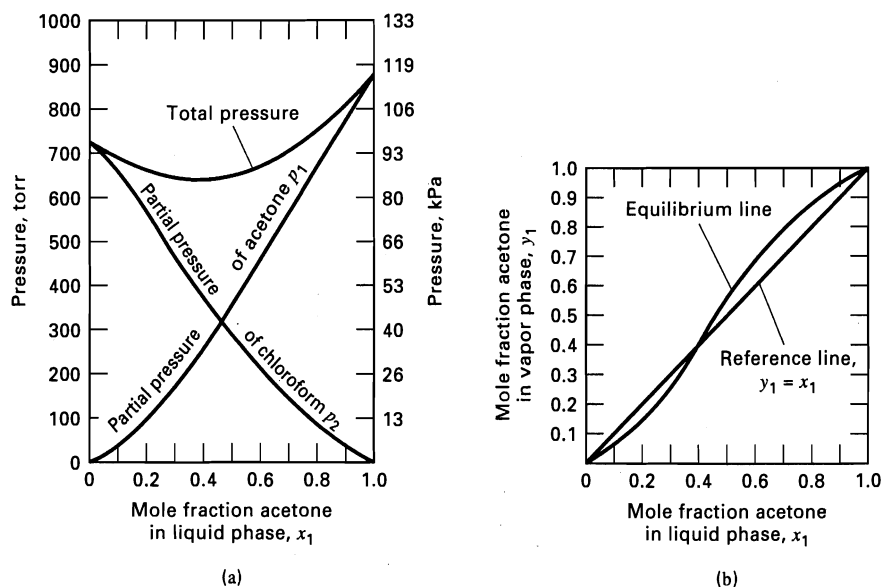


Figure 4.7 Maximum-boiling-point azeotrope, acetone–chloroform system: (a) partial and total pressures at 60°C ; (b) vapor–liquid equilibria at 101 kPa; (c) phase diagram at 101 kPa pressure.

[Adapted from O.A. Hougen, K.M. Watson, and R.A. Ragatz, *Chemical Process Principles. Part II*, 2nd ed., John Wiley and Sons, New York (1959).]

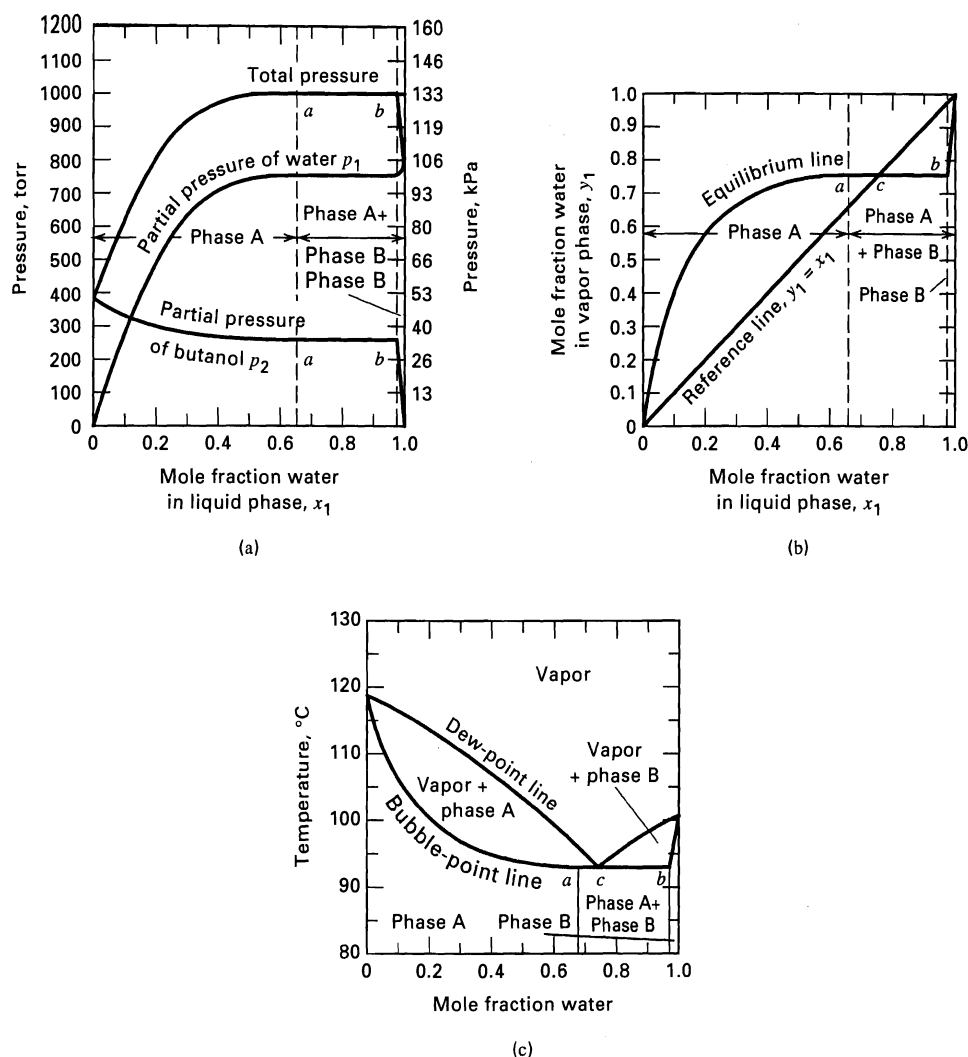


Figure 4.8 Minimum-boiling-point (two liquid phases) water/n-butanol system: (a) partial and total pressures at 100°C; (b) vapor-liquid equilibria at 101 kPa; (c) phase diagram at 101 kPa pressure.

[Adapted from O.A. Hougen, K.M. Watson, and R.A. Ragatz, *Chemical Process Principles. Part II*, 2nd ed., John Wiley and Sons, New York (1959).]

78.15°C and 101.3 kPa. However, at vacuums of less than 9.3 kPa, no azeotrope is formed. Ternary azeotropes also occur, and these offer the same barrier to complete separation as do binary azeotropes.

Azeotrope formation in general, and heterogeneous azeotropes in particular, can be employed to achieve difficult separations. As discussed in Chapter 1, an entrainer is added for the purpose of combining with one or more of the components in the feed to form a minimum-boiling azeotrope, which is then recovered as the distillate.

Figure 4.9 shows the Keyes process [3] for making pure ethyl alcohol by *heterogeneous azeotropic distillation*. Water and ethyl alcohol form a binary, minimum-boiling azeotrope containing 95.6 wt% alcohol and boiling at 78.15°C at 101.3 kPa. Thus, it is impossible to obtain pure alcohol (boiling point = 78.40°C) by ordinary distillation at 1 atm. The addition of benzene to an alcohol-water mixture results in the formation of a minimum-boiling, heterogeneous ternary, azeotrope containing, by weight, 18.5% alcohol, 74.1% benzene, and 7.4% water, boiling at 64.85°C. Upon condensation, the ternary azeotrope separates into two liquid layers: a top layer containing 14.5% alcohol, 84.5% benzene, and 1% water, and a bottoms layer of

53% alcohol, 11% benzene, and 36% water, all by weight. The benzene-rich layer is returned as reflux. The other layer is sent to a second distillation column for recovery and recycling of alcohol and benzene. Absolute alcohol, which has a boiling point above that of the ternary azeotrope, is removed at the bottom of the column.

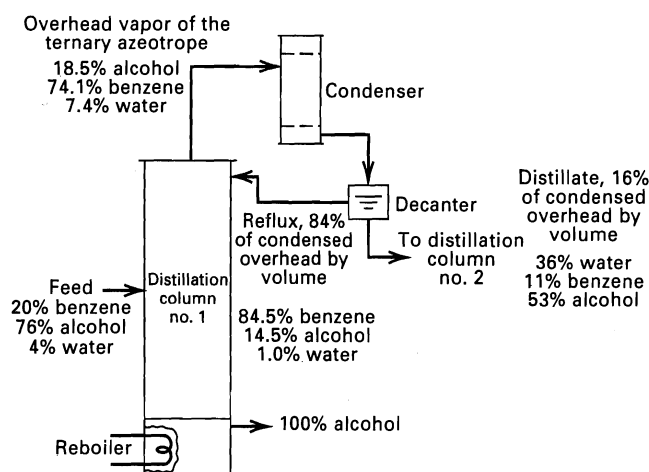


Figure 4.9 The Keyes process for absolute alcohol.

In *extractive distillation*, as discussed in Chapter 1, a solvent is added, usually near the top of the column, to selectively alter the activity coefficients in order to increase the relative volatility between the two species to be separated. The solvent is generally a relatively polar, high-boiling constituent, such as phenol, aniline, or furfural, which concentrates at the bottom of the column.

4.4 MULTICOMPONENT FLASH, BUBBLE-POINT, AND DEW-POINT CALCULATIONS

A *flash* is a single-equilibrium-stage distillation in which a feed is partially vaporized to give a vapor richer in the more-volatile components than the remaining liquid. In Figure 4.10a, a liquid feed is heated under pressure and flashed adiabatically across a valve to a lower pressure, resulting in the creation of a vapor phase that is separated from the remaining liquid in a flash drum. If the valve is omitted, a low-pressure liquid can be partially vaporized in the heater and then separated into two phases in the flash drum. Alternatively, a vapor feed can be cooled and partially condensed, with phase separation in a flash drum, as in Figure 4.10b, to give a liquid that is richer in the less-volatile components. In both cases, if the equipment is properly designed, the vapor and liquid leaving the drum are in equilibrium [4].

Unless the relative volatility is very large, the degree of separation achievable between two components in a single equilibrium stage is poor. Therefore, flashing (partial vaporization) or partial condensation are usually auxiliary operations used to prepare streams for further processing.

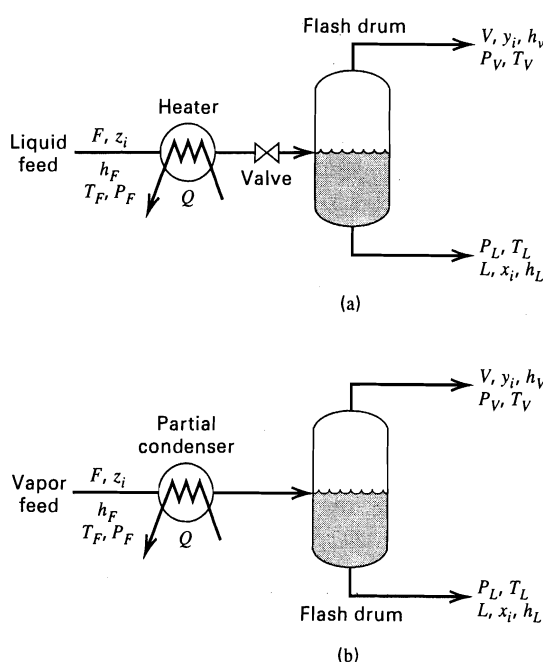


Figure 4.10 Continuous, single-stage equilibrium separation: (a) flash vaporization (adiabatic flash with valve, isothermal flash without valve when T_V is specified); (b) partial condensation (analogous to isothermal flash when T_V is specified).

Typically, the vapor phase is sent to a vapor separation system, while the liquid phase is sent to a liquid separation system. Computational methods for a single-stage flash calculation are of fundamental importance. Such calculations are used not only for the operations in Figure 4.10, but also to determine, anywhere in a process, the phase condition of a stream or batch of known composition, temperature, and pressure.

For the single-stage equilibrium operation with one feed stream and two product streams, shown in Figure 4.10, the $2C + 5$ equations listed in Table 4.3 apply. (In Figure 4.10, T and P are given separately for the vapor and liquid products to emphasize the subsequent need to assume mechanical and thermal equilibrium.) They relate the $3C + 10$ variables ($F, V, L, z_i, y_i, x_i, T_F, T_V, T_L, P_F, P_V, P_L, Q$) and leave $C + 5$ degrees of freedom. Assuming that $C + 3$ feed variables F, T_F, P_F , and C values of z_i are known, two additional variables can be specified. The most common sets of specifications are

T_V, P_V	Isothermal flash
$V/F = 0, P_L$	Bubble-point temperature
$V/F = 1, P_V$	Dew-point temperature
$T_L, V/F = 0$	Bubble-point pressure
$T_V, V/F = 1$	Dew-point pressure
$Q = 0, P_V$	Adiabatic flash
Q, P_V	Nonadiabatic flash
$V/F, P_V$	Percent vaporization flash

Calculation procedures, described in the following for all these cases, are well known and widely used. They all assume that specified values of feed mole fractions, z_i , sum to one.

Isothermal Flash

If the equilibrium temperature T_V (or T_L) and the equilibrium pressure P_V (or P_L) of a multicomponent mixture are specified, values of the remaining $2C + 5$ variables are determined from the same number of equations in Table 4.3.

Table 4.3 Equations for Single-Stage Flash Vaporization and Partial Condensation Operations

Equation	Number of Equations
(1) $P_V = P_L$ (mechanical equilibrium)	1
(2) $T_V = T_L$ (thermal equilibrium)	1
(3) $y_i = K_i x_i$ (phase equilibrium)	C
(4) $F z_i = V y_i + L x_i$ (component material balance)	C
(5) $F = V + L$ (total material balance)	1
(6) $h_F F + Q = h_V V + h_L L$ (energy balance)	1
(7) $\sum_i y_i - \sum_i x_i = 0$ (summations)	1
$\mathcal{E} = 2C + 5$	
$K_i = K_i\{T_V, P_V, y, x\}$	$h_F = h_F\{T_F, P_F, z\}$
$h_V = h_V\{T_V, P_V, y\}$	$h_L = h_L\{T_L, P_L, x\}$

Table 4.4 Rachford–Rice Procedure for Isothermal-Flash Calculations When K -Values Are Independent of CompositionSpecified variables: $F, T_F, P_F, z_1, z_2, \dots, z_C, T_V, P_V$ **Steps**

- (1) $T_L = T_V$
- (2) $P_L = P_V$
- (3) Solve

$$f\{\Psi\} = \sum_{i=1}^C \frac{z_i(1 - K_i)}{1 + \Psi(K_i - 1)} = 0$$

for $\Psi = V/F$, where $K_i = K_i\{T_V, P_V\}$.

- (4) $V = F\Psi$

$$(5) x_i = \frac{z_i}{1 + \Psi(K_i - 1)}$$

$$(6) y_i = \frac{z_i K_i}{1 + \Psi(K_i - 1)} = x_i K_i$$

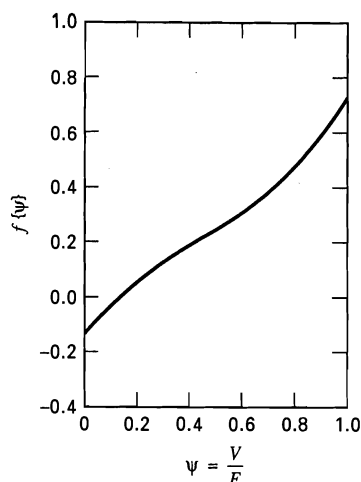
- (7) $L = F - V$

$$(8) Q = h_V V + h_L L - h_F F$$

The computational procedure, referred to as the *isothermal-flash calculation*, is not straightforward because Eq. (4) in Table 4.3 is a nonlinear equation in the unknowns V, L, y_i , and x_i . Many solution strategies have been developed, but the generally preferred procedure, as given in Table 4.4, is that of Rachford and Rice [5] when K -values are independent (or nearly independent) of equilibrium-phase compositions.

Equations containing only a single unknown are solved first. Thus, Eqs. (1) and (2) in Table 4.3 are solved, respectively, for P_L and T_L . The unknown Q appears only in Eq. (6), so Q is computed only after all other equations have been solved. This leaves Eqs. (3), (4), (5), and (7) in Table 4.3 to be solved for V, L , and all values of y and x . These equations can be partitioned so as to solve for the unknowns in a sequential manner by substituting Eq. (5) into Eq. (4) to eliminate L and combining the result with Eq. (3) to obtain Eqs. (5) and (6) in Table 4.4. Here (5) is in x_i , but not y_i , and (6) is in y_i but not x_i . Summing these two equations and combining them with $\sum y_i - \sum x_i = 0$ to eliminate y_i and x_i gives Eq. (3) in Table 4.4; a nonlinear equation in V (or $\Psi = V/F$) only. Upon solving this equation numerically in an iterative manner for Ψ and then V , from Eq. (4) of Table 4.4, one can obtain the remaining unknowns directly from Eqs. (5) through (8) in Table 4.4. When T_F and/or P_F are not specified, Eq. (6) of Table 4.3 is not solved for Q . By this isothermal-flash procedure, the equilibrium-phase condition of a mixture at a known temperature ($T_V = T_L$) and pressure ($P_V = P_L$) is determined.

Equation (3) of Table 4.4 can be solved iteratively by guessing values of Ψ between 0 and 1 until the function $f\{\Psi\} = 0$. A typical form of the function, as will be computed in Example 4.1, is shown in Figure 4.11. The most widely employed numerical method for solving Eq. (3) of Table 4.4 is Newton's method [6]. A predicted value of the Ψ

**Figure 4.11** Rachford–Rice function for Example 4.1.

root for iteration $k + 1$ is computed from the recursive relation

$$\Psi^{(k+1)} = \Psi^{(k)} - \frac{f\{\Psi^{(k)}\}}{f'\{\Psi^{(k)}\}} \quad (4-10)$$

where the superscript is the iteration index, and the derivative of $f\{\Psi\}$, from Eq. (3) in Table 4.4, with respect to Ψ is

$$f'\{\Psi^{(k)}\} = \sum_{i=1}^C \frac{z_i(1 - K_i)^2}{[1 + \Psi^{(k)}(K_i - 1)]^2} \quad (4-11)$$

The iteration can be initiated by assuming $\Psi^{(1)} = 0.5$. Sufficient accuracy will be achieved by terminating the iterations when $|\Psi^{(k+1)} - \Psi^{(k)}|/\Psi^{(k)} < 0.0001$.

One should check the existence of a valid root ($0 \leq \Psi \leq 1$), before employing the procedure of Table 4.4, by checking to see if the equilibrium condition corresponds to subcooled liquid or superheated vapor rather than partial vaporization or partial condensation. A first estimate of whether a multicomponent feed gives a two-phase equilibrium mixture when flashed at a given temperature and pressure can be made by inspecting the K -values. If all K -values are greater than 1, the exit phase is superheated vapor above the dew point. If all K -values are less than 1, the single exit phase is a subcooled liquid below the bubble point. If one or more K -values are greater than 1 and one or more K -values are less than 1, the check is made as follows. First, $f\{\Psi\}$ is computed from Eq. (3) for $\Psi = 0$. If the resulting $f\{0\} > 0$, the mixture is below its bubble point (subcooled liquid). Alternatively, if $f\{1\} < 0$, the mixture is above the dew point (superheated vapor).

EXAMPLE 4.1

A 100-kmol/h feed consisting of 10, 20, 30, and 40 mol% of propane (3), *n*-butane (4), *n*-pentane (5), and *n*-hexane (6), respectively, enters a distillation column at 100 psia (689.5 kPa) and

200°F (366.5°K). Assuming equilibrium, what mole fraction of the feed enters as liquid, and what are the liquid and vapor compositions?

SOLUTION

At flash conditions, from Figure 2.8, $K_3 = 4.2$, $K_4 = 1.75$, $K_5 = 0.74$, $K_6 = 0.34$, independent of compositions. Because some K -values are greater than 1 and some are less than 1, it is first necessary to compute values of $f\{0\}$ and $f\{1\}$ for Eq. (3) in Table 4.4 to determine if the mixture is between the bubble point and the dew point.

$$f\{0\} = \frac{0.1(1-4.2)}{1} + \frac{0.2(1-1.75)}{1} + \frac{0.3(1-0.74)}{1} + \frac{0.4(1-0.34)}{1} = -0.128$$

Since $f\{0\}$ is not greater than zero, the mixture is above the bubble point.

$$f\{1\} = \frac{0.1(1-4.2)}{1+(4.2-1)} + \frac{0.2(1-1.75)}{1+(1.75-1)} + \frac{0.3(1-0.74)}{1+(0.74-1)} + \frac{0.4(1-0.34)}{1+(0.34-1)} = 0.720$$

Since $f\{1\}$ is not less than zero, the mixture is below the dew point. Therefore, the mixture is part vapor and substitution of z_i and K_i values in Eq. (3) of Table 4.4 gives

$$0 = \frac{0.1(1-4.2)}{1+\Psi(4.2-1)} + \frac{0.2(1-1.75)}{1+\Psi(1.75-1)} + \frac{0.3(1-0.74)}{1+\Psi(0.74-1)} + \frac{0.4(1-0.34)}{1+\Psi(0.34-1)}$$

Solution of this equation by Newton's method using an initial guess for Ψ of 0.50 gives the following iteration history:

k	$\Psi^{(k)}$	$f\{\Psi^{(k)}\}$	$f'\{\Psi^{(k)}\}$	$\Psi^{(k+1)}$	$\frac{\Psi^{(k+1)} - \Psi^{(k)}}{\Psi^{(k)}}$
1	0.5000	0.2515	0.6259	0.0982	0.8037
2	0.0982	-0.0209	0.9111	0.1211	0.2335
3	0.1211	-0.0007	0.8539	0.1219	0.0065
4	0.1219	0.0000	0.8521	0.1219	0.0000

For this example, convergence is very rapid, giving $\Psi = V/F = 0.1219$. From Eq. (4) of Table 4.4, the equilibrium-vapor flow rate is $0.1219(100) = 12.19$ kmol/h, and the equilibrium-liquid flow rate from Eq. (7) is $(100 - 12.19) = 87.81$ kmol/h. The liquid and vapor compositions computed from Eqs. (5) and (6) are

	x	y
Propane	0.0719	0.3021
<i>n</i> -Butane	0.1833	0.3207
<i>n</i> -Pentane	0.3098	0.2293
<i>n</i> -Hexane	0.4350	0.1479
	1.0000	1.0000

A plot of $f\{\Psi\}$ as a function of Ψ is shown in Figure 4.11.

Bubble and Dew Points

Often, it is desirable to bring a mixture to the bubble point or the dew point. At the bubble point, $\Psi = 0$ and $f\{0\} = 0$. Therefore, from Eq. (3), Table 4.4,

$$f\{0\} = \sum_i z_i(1 - K_i) = \sum_i z_i - \sum_i z_i K_i = 0$$

However, $\sum_i z_i = 1$. Therefore, the bubble-point equation is

$$\sum_i z_i K_i = 1 \quad (4-12)$$

At the dew point, $\Psi = 1$ and $f\{1\} = 0$. Therefore, from Eq. (3), Table 4.4,

$$f\{1\} = \sum_i \frac{z_i(1 - K_i)}{K_i} = \sum_i \frac{z_i}{K_i} - \sum_i z_i = 0$$

Therefore, the dew-point equation is

$$\sum_i \frac{z_i}{K_i} = 1 \quad (4-13)$$

For a given feed composition, z_i , (4-12) or (4-13) can be used to find T for a specified P or to find P for a specified T .

Because of the K -values, the bubble- and dew-point equations are generally highly nonlinear in temperature, but only moderately nonlinear in pressure, except in the region of the *convergence pressure*, where K -values of very light or very heavy species change radically with pressure, as in Figure 2.10. Therefore, iterative procedures are required to solve for bubble- and dew-point conditions. One exception is where Raoult's law K -values are applicable. Substitution of $K_i = P_i^s/P$ into (4-12) leads to an equation for the direct calculation of bubble-point pressure:

$$P_{\text{bubble}} = \sum_{i=1}^C z_i P_i^s \quad (4-14)$$

where P_i^s is the temperature-dependent vapor pressure of species i . Similarly, from (4-13), the dew-point pressure is

$$P_{\text{dew}} = \left(\sum_{i=1}^C \frac{z_i}{P_i^s} \right)^{-1} \quad (4-15)$$

Another useful exception occurs for mixtures at the bubble point when K -values can be expressed by the modified Raoult's law, $K_i = \gamma_i P_i^s/P$. Substituting this equation into (4-12),

$$P_{\text{bubble}} = \sum_{i=1}^C \gamma_i z_i P_i^s \quad (4-16)$$

Liquid-phase activity coefficients can be computed for a known temperature and composition, since $x_i = z_i$ at the bubble point.

Bubble- and dew-point calculations are used to determine saturation conditions for liquid and vapor streams, respectively. It is important to note that when vapor-liquid equilibrium is established, the vapor is at its dew point and the liquid is at its bubble point.

EXAMPLE 4.2

In Figure 1.9, the nC_4 -rich bottoms product from column C3 has the composition given in Table 1.5. If the pressure at the bottom of the distillation column is 100 psia (689 kPa), estimate the temperature of the mixture.

SOLUTION

The bottoms product will be a liquid at its bubble point with the following composition:

Component	kmol/h	$z_i = x_i$
<i>i</i> -Butane	8.60	0.0319
<i>n</i> -Butane	215.80	0.7992
<i>i</i> -Pentane	28.10	0.1041
<i>n</i> -Pentane	17.50	0.0648
	270.00	1.0000

The bubble-point temperature can be estimated by finding the temperature that will satisfy (4-12), using K -values from Figure 2.8. Because the bottoms product is rich in nC_4 , assume that the K -value of nC_4 is 1. From Figure 2.8, for 100 psia, $T = 150^\circ\text{F}$. For this temperature, using Figure 2.8 to obtain the K -values of the other three hydrocarbons and substituting these values and the z -values into (4-12),

$$\begin{aligned}\sum z_i K_i &= 0.0319(1.3) + 0.7992(1.0) + 0.1041(0.47) \\ &\quad + 0.0648(0.38) \\ &= 0.042 + 0.799 + 0.049 + 0.025 = 0.915\end{aligned}$$

Because the sum is not 1.0, another temperature must be assumed and the summation repeated. To increase the sum, the K -values must be greater and, thus, the temperature must be higher. Because the sum is dominated by nC_4 , assume that its K -value must be $1.000(1.00/0.915) = 1.09$. This corresponds to a temperature of 160°F , which results in a summation of 1.01. By linear interpolation, $T = 159^\circ\text{F}$.

EXAMPLE 4.3

Cyclopentane is to be separated from cyclohexane by liquid-liquid extraction with methanol at 25°C . In extraction it is important that the liquid mixtures be maintained at pressures greater than the bubble-point pressure. Calculate the bubble-point pressure using the following equilibrium liquid-phase compositions, activity coefficients, and vapor pressures:

	Methanol	Cyclohexane	Cyclopentane
Vapor pressure, psia	2.45	1.89	6.14
Methanol-rich layer:			
x	0.7615	0.1499	0.0886
γ	1.118	4.773	3.467
Cyclohexane-rich layer:			
x	0.1737	0.5402	0.2861
γ	4.901	1.324	1.074

SOLUTION

Because the bubble-point pressure is likely to be below ambient pressure, the modified Raoult's law in the form of (4-16) applies for either liquid phase. If the methanol-rich layer data are used:

$$\begin{aligned}P_{\text{bubble}} &= 1.118(0.7615)(2.45) + 4.773(0.1499)(1.89) \\ &\quad + 3.467(0.0886)(6.14) \\ &= 5.32 \text{ psia (36.7 kPa)}\end{aligned}$$

A similar calculation based on the cyclohexane-rich layer gives an identical result because the data are consistent with phase equilibrium theory such that $\gamma_{iL}^{(1)} x_i^{(1)} = \gamma_{iL}^{(2)} x_i^{(2)}$. A pressure higher than 5.32 psia will prevent formation of vapor at this location in the extraction process. Thus, operation at atmospheric pressure is a good choice.

EXAMPLE 4.4

Propylene (P) is to be separated from 1-butene (B) by distillation into a vapor distillate containing 90 mol% propylene. Calculate the column operating pressure assuming the exit temperature from the partial condenser is 100°F (37.8°C), the minimum attainable temperature with cooling water. Determine the composition of the liquid reflux. In Figure 4.12, K -values estimated from Eq. (5), Table 2.3, using the Redlich-Kwong equation of state for the vapor fugacity, are plotted and compared to experimental data [7] and Raoult's law K -values.

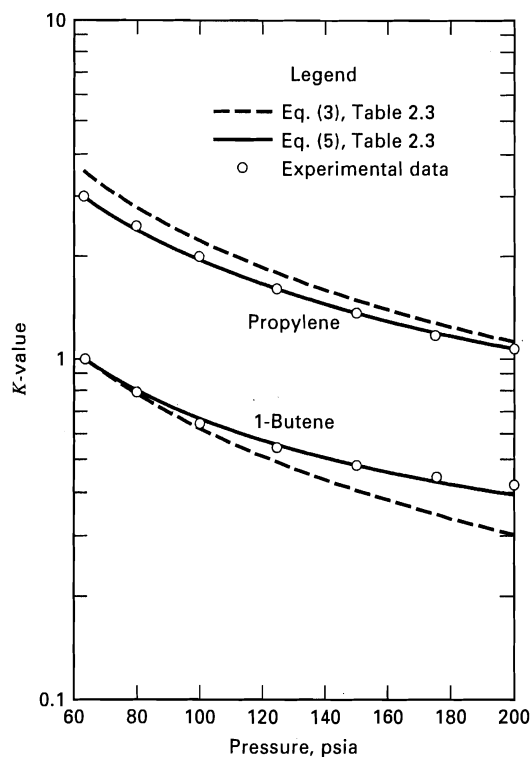


Figure 4.12 K -values for propylene/1-butene system at 100°F .

SOLUTION

The operating pressure corresponds to a dew-point condition for the vapor-distillate composition. The composition of the reflux corresponds to the liquid in equilibrium with the vapor distillate at its dew point. The method of false position [8] can be used to perform the iterative calculations by rewriting (4-13) in the form

$$f\{P\} = \sum_{i=1}^C \frac{z_i}{K_i} - 1 \quad (1)$$

The recursion relationship for the method of false position is based on the assumption that $f\{P\}$ is linear in P such that

$$P^{(k+2)} = P^{(k+1)} - f\{P^{(k+1)}\} \left[\frac{P^{(k+1)} - P^{(k)}}{f\{P^{(k+1)}\} - f\{P^{(k)}\}} \right] \quad (2)$$

This assumption is reasonable because, at low pressures, K -values in (2) are almost inversely proportional to pressure. Two values of P are required to initialize this formula. Choose 100 psia and 190 psia. At 100 psia, reading the K -values from the solid lines in Figure 4.12,

$$f\{P\} = \frac{0.90}{2.0} + \frac{0.10}{0.68} - 1.0 = -0.40$$

Subsequent iterations give

k	$P^{(k)}$, psia	K_P	K_B	$f\{P^{(k)}\}$
1	100	2.0	0.68	-0.40
2	190	1.15	0.42	+0.02
3	186	1.18	0.425	-0.0020

Iterations are terminated when $|P^{(k+2)} - P^{(k+1)}|/P^{(k+1)} < 0.005$.

An operating pressure of 186 psia (1,282 kPa) at the partial condenser outlet is indicated. The composition of the liquid reflux is obtained from $x_i = z_i/K_i$ with the result

Component	Equilibrium Mole Fraction	
	Vapor Distillate	Liquid Reflux
Propylene	0.90	0.76
1-Butene	0.10	0.24
	1.00	1.00

Adiabatic Flash

When the pressure of a liquid stream of known composition, flow rate, and temperature (or enthalpy) is reduced adiabatically across a valve as in Figure 4.10a, an *adiabatic-flash* calculation is made to determine the resulting temperature, compositions, and flow rates of the equilibrium liquid and vapor streams for a specified pressure downstream of the valve. For an adiabatic flash, the isothermal-flash calculation procedure can be applied in the following iterative manner. A guess is made of the flash temperature, T_V . Then Ψ , V , x , y , and L are determined, as for an isothermal flash, from steps 3 through 7 in Table 4.4. The guessed value of T_V (equal to T_L) is next checked by an energy balance obtained by combining Eqs. (7) and (8) of Table 4.4 with $Q = 0$ to give

$$f\{T_V\} = \frac{\Psi h_V + (1 - \Psi)h_L - h_F}{1,000} = 0 \quad (4-17)$$

where the division by 1,000 is to make the terms of the order of 1. If the computed value of $f\{T_V\}$ is not zero, the entire procedure is repeated for two or more guesses of T_V . A plot of $f\{T_V\}$ versus T_V is interpolated to determine the correct value of T_V . The procedure is tedious because it involves inner-loop iteration on Ψ and outer-loop iteration on T_V .

Outer-loop iteration on T_V is very successful when Eq. (3) of Table 4.4 is not sensitive to the guess of T_V . This is the case for wide-boiling mixtures. For close-boiling mixtures (e.g., isomers), the algorithm may fail because of extreme sensitivity to the value of T_V . In this case, it is preferable to do the outer-loop iteration on Ψ and solve Eq. (3) of Table 4.4 for T_V in the inner loop, using a guessed value for Ψ to initiate the process:

$$f\{T_V\} = \sum_{i=1}^C \frac{z_i(1 - K_i)}{1 + \Psi(K_i - 1)} = 0 \quad (4-18)$$

Then, Eqs. (5) and (6) of Table 4.4 are solved for x and y , respectively. Equation (4-17) is then solved directly for Ψ , since

$$f\{\Psi\} = \frac{\Psi h_V + (1 - \Psi)h_L - h_F}{1,000} = 0 \quad (4-19)$$

from which

$$\Psi = \frac{h_F - h_L}{h_V - h_L} \quad (4-20)$$

If Ψ from (4-20) is not equal to the value of Ψ guessed to solve (4-18), the new value of Ψ is used to repeat the outer loop starting with (4-18).

Multicomponent, isothermal-flash, bubble-point, dew-point, and adiabatic-flash calculations can be very tedious because of their iterative nature. They are unsuitable for manual calculations for nonideal vapor and liquid mixtures because of the complexity of the expressions for the thermodynamic properties, K , h_V , and h_L . However, robust algorithms for making such calculations are incorporated into widely used steady-state simulation computer programs such as ASPEN PLUS, CHEMCAD, HYSYS, and PRO/II.

EXAMPLE 4.5

The equilibrium liquid from the flash drum at 120°F and 485 psia in Example 2.6 is fed to a distillation tower to remove the remaining hydrogen and methane. A tower for this purpose is often referred to as a *stabilizer*. Pressure at the feed plate of the stabilizer is 165 psia (1,138 kPa). Calculate the percent vaporization of the feed if the pressure is decreased adiabatically from 485 to 165 psia by valve and pipeline pressure drop.

SOLUTION

This problem is most conveniently solved by using a steady-state simulation program. If the CHEMCAD program is used with

K -values and enthalpies estimated from the P-R equation of state, the following results are obtained:

Component	kmol/h		
	Feed 120°F 485 psia	Vapor 112°F 165 psia	Liquid 112°F 165 psia
Hydrogen	1.0	0.7	0.3
Methane	27.9	15.2	12.7
Benzene	345.1	0.4	344.7
Toluene	113.4	0.04	113.36
Total	487.4	16.34	471.06
Enthalpy, kJ/h	-1,089,000	362,000	-1,451,000

This case involves a wide-boiling feed, so the procedure involving (4-17) is the best choice. The above results show that only a small amount of vapor ($\Psi = 0.0035$), predominantly H_2 and CH_4 , is produced by the adiabatic flash. The computed flash temperature of 112°F is 8°F below the feed temperature. The enthalpy of the feed is equal to the sum of the vapor and liquid product enthalpies for this adiabatic operation.

4.5 TERNARY LIQUID-LIQUID SYSTEMS

Ternary mixtures that undergo phase splitting to form two separate liquid phases can differ as to the extent of solubility of the three components in each of the two liquid phases. The simplest case is shown in Figure 4.13a, where only the solute, component B, has any appreciable solubility in either the carrier, A, or the solvent, C, both of which have negligible (although never zero) solubility in each other. In this case, the equations can be derived for a single equilibrium stage, using the variables F , S , $L^{(1)}$, and $L^{(2)}$ to refer, respectively, to the flow rates (or amounts) of the feed, solvent, exiting extract, and exiting raffinate. By definition, the extract is the exiting liquid phase that contains the solvent and the extracted solute; the raffinate is the exiting liquid phase that contains the carrier, A, of the feed and the portion of the solute, B, that is not extracted. Although the extract is shown in Figure 4.13a as leaving from the top of the stage, this will only be so if the extract is the lighter (lower-density) exiting phase. Assuming that the entering solvent contains no solute, B, it is convenient to write material balance and phase-equilibrium equations for the solute, B. These two equations may be written in terms of molar or mass flow rates. To obtain the simplest result, it is preferable to express

compositions of the solute as mass or mole ratios instead of mass or mole fractions.

Let:

F_A = feed rate of carrier A

S = flow rate of solvent C

X_B = ratio of mass (or moles) of solute B, to mass (or moles) of the other component in the feed (F), raffinate (R), or extract (E)

Then, the solute material balance is

$$X_B^{(F)} F_A = X_B^{(E)} S + X_B^{(R)} F_A \quad (4-21)$$

and the distribution of solute at equilibrium is given by

$$X_B^{(E)} = K'_{D_B} X_B^{(R)} \quad (4-22)$$

where K'_{D_B} is the distribution coefficient defined in terms of mass or mole ratios. Substituting (4-22) into (4-21) to eliminate $X_B^{(E)}$ gives

$$X_B^{(R)} = \frac{X_B^{(F)} F_A}{F_A + K'_{D_B} S} \quad (4-23)$$

It is convenient to define an extraction factor, E_B , for the solute B:

$$E_B = K'_{D_B} S / F_A \quad (4-24)$$

The larger the value of E , the greater the extent to which the solute is extracted. Large values of E result from large values of the distribution coefficient, K'_{D_B} , or large ratios of solvent to carrier. Substituting (4-24) in (4-23) gives the fraction of B that is not extracted as

$$X_B^{(R)} / X_B^{(F)} = \frac{1}{1 + E_B} \quad (4-25)$$

where it is clear that the larger the extraction factor, the smaller the fraction of B not extracted.

Values of mass (mole) ratios, X , are related to mass (mole) fractions, x , by

$$X_i = x_i / (1 - x_i) \quad (4-26)$$

Values of the distribution coefficient, K'_{D_i} , in terms of ratios, are related to K_{D_i} in terms of fractions as given in (2-20) by

$$K'_{D_i} = \frac{x_i^{(1)} / (1 - x_i^{(1)})}{x_i^{(2)} / (1 - x_i^{(2)})} = K_{D_i} \left(\frac{1 - x_i^{(2)}}{1 - x_i^{(1)}} \right) \quad (4-27)$$

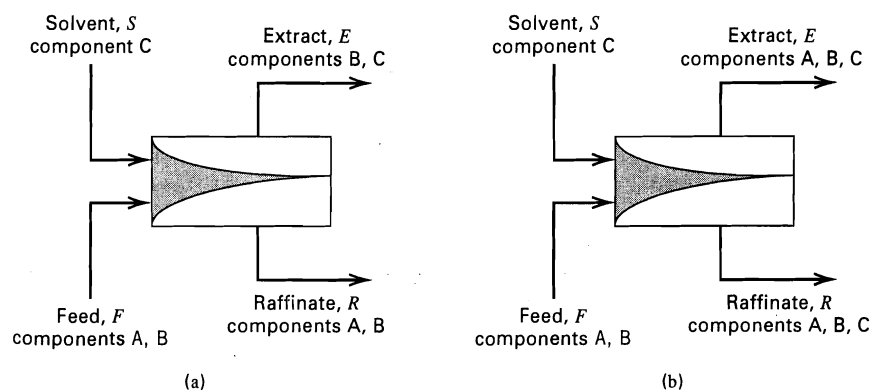


Figure 4.13 Phase splitting of ternary mixtures: (a) components A and C mutually insoluble; (b) components A and C partially soluble.

When values of x_i are small, K'_D approaches K_D . As discussed in Chapter 2, the distribution coefficient, K_D , which can be determined from activity coefficients using the expression $K_{D_B} = \gamma_B^{(2)} / \gamma_B^{(1)}$ when mole fractions are used, is a strong function of equilibrium-phase compositions and temperature. However, when the raffinate and extract are both dilute in the solute, activity coefficients of the solute can be approximated by the values at infinite dilution so that K_{D_B} can be taken as a constant at a given temperature. An extensive listing of such K_{D_B} values in mass fraction units for various ternary systems is given in *Perry's Handbook* [9]. If values for F_B , $X_B^{(F)}$, S , and K_{D_B} are given, (4-25) can be solved for $X_B^{(R)}$.

EXAMPLE 4.6

A feed of 13,500 kg/h consists of 8 wt% acetic acid (B) in water (A). The removal of the acetic acid is to be accomplished by liquid–liquid extraction at 25°C with methyl isobutyl ketone solvent (C), because distillation of the feed would require vaporization of large amounts of water. If the raffinate is to contain only 1 wt% acetic acid, estimate the kilograms per hour of solvent required if a single equilibrium stage is used.

SOLUTION

Assume that the carrier (water) and the solvent are immiscible. From *Perry's Handbook*, take $K_D = 0.657$ in mass-fraction units for this system. For the relatively low concentrations of acetic acid in this problem, assume that $K'_D = K_D$.

$$F_A = (0.92)(13,500) = 12,420 \text{ kg/h}$$

$$X_B^{(F)} = (13,500 - 12,420) / 12,420 = 0.087$$

The raffinate is to contain 1 wt% B. Therefore,

$$X_B^{(R)} = 0.01 / (1 - 0.01) = 0.0101$$

From (4-25), solving for E_B ,

$$E_B = \frac{X_B^{(F)}}{X_B^{(R)}} - 1 = (0.087 / 0.0101) - 1 = 7.61$$

From (4-24), the definition of the extraction factor,

$$S = \frac{E_B F_A}{K'_D} = 7.61(12,420 / 0.657) = 144,000 \text{ kg/h}$$

This is a very large solvent flow rate compared to the feed rate—more than a factor of 10! Multiple stages should be used to reduce the solvent rate or a solvent with a larger distribution coefficient should be sought. For 1-butanol as the solvent, $K_D = 1.613$.

In the ternary liquid–liquid system, shown in Figure 4.13b, components A and C are partially soluble in each other and component B again distributes between the extract and raffinate phases. Both of these exiting phases contain all components present in the feed and solvent. This case is by far the most commonly encountered, and a number of different phase diagrams and computational techniques have been

devised to determine the equilibrium compositions. Examples of phase diagrams are shown in Figure 4.14 for the water (A)–ethylene glycol (B)–furfural (C) system at 25°C and a pressure of 101 kPa, which is above the bubble-point pressure, so no vapor phase exists. Experimental data for this system were obtained by Conway and Norton [18]. The pairs water–ethylene glycol and furfural–ethylene glycol are each completely miscible. The only partially miscible pair is furfural–water. Furfural might be used as a solvent to remove the solute, ethylene glycol, from water; the furfural-rich phase is the extract, and the water-rich phase is the raffinate.

Figure 4.14a, an equilateral-triangular diagram, is the most common display of ternary liquid–liquid equilibrium data in the chemical literature. Any point located within or on an edge of the triangle represents a mixture composition. Such a diagram has the property that the sum of the lengths of the perpendicular lines drawn from any interior point to the sides equals the altitude of the triangle. Thus, if each of the three altitudes is scaled from 0 to 100, the percent of, say, furfural, at any point such as M, is simply the length of the line perpendicular to the base opposite the pure furfural apex, which represents 100% furfural. Figure 4.14a is constructed for compositions based on mass fractions (mole fractions and volume fractions are also sometimes used). Thus, the point M in Figure 4.14a represents a mixture of feed and solvent (before phase separation) containing 18.9 wt% water, 20 wt% ethylene glycol, and 61.1 wt% furfural.

The miscibility limits for the furfural–water binary system are at D and G. The miscibility boundary (saturation or binodal curve) DEPRG can be obtained experimentally by a *cloud-point titration*; water, for example, is added to a (clear) 50 wt% solution of furfural and glycol, and it is noted that the onset of cloudiness due to the formation of a second phase occurs when the mixture is 11% water, 44.5% furfural, and 44.5% glycol by weight. Other miscibility data are given in Table 4.5, from which the miscibility curve in Figure 4.14a was drawn.

Table 4.5 Equilibrium Miscibility Data in Weight Percent for the Furfural–Ethylene Glycol–Water System at 25°C and 101 kPa

Furfural	Ethylene Glycol	Water
95.0	0.0	5.0
90.3	5.2	4.5
86.1	10.0	3.9
75.1	20.0	4.9
66.7	27.5	5.8
49.0	41.5	9.5
34.3	50.5	15.2
27.5	52.5	20.0
13.9	47.5	38.6
11.0	40.0	49.0
9.7	30.0	60.3
8.4	15.0	76.6
7.7	0.0	92.3

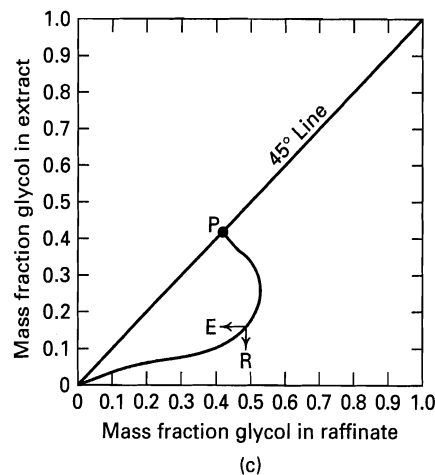
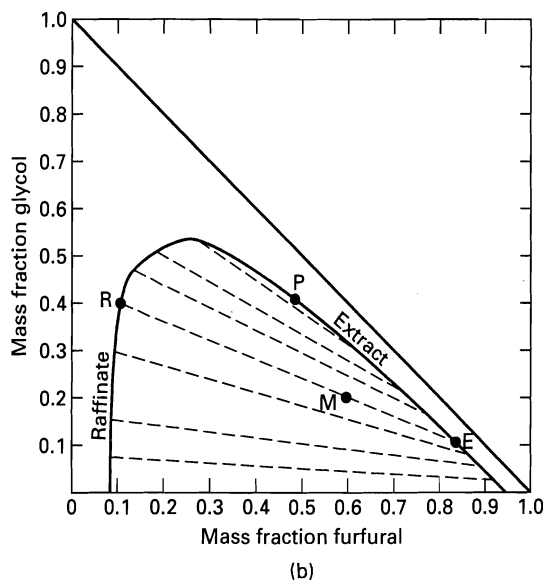
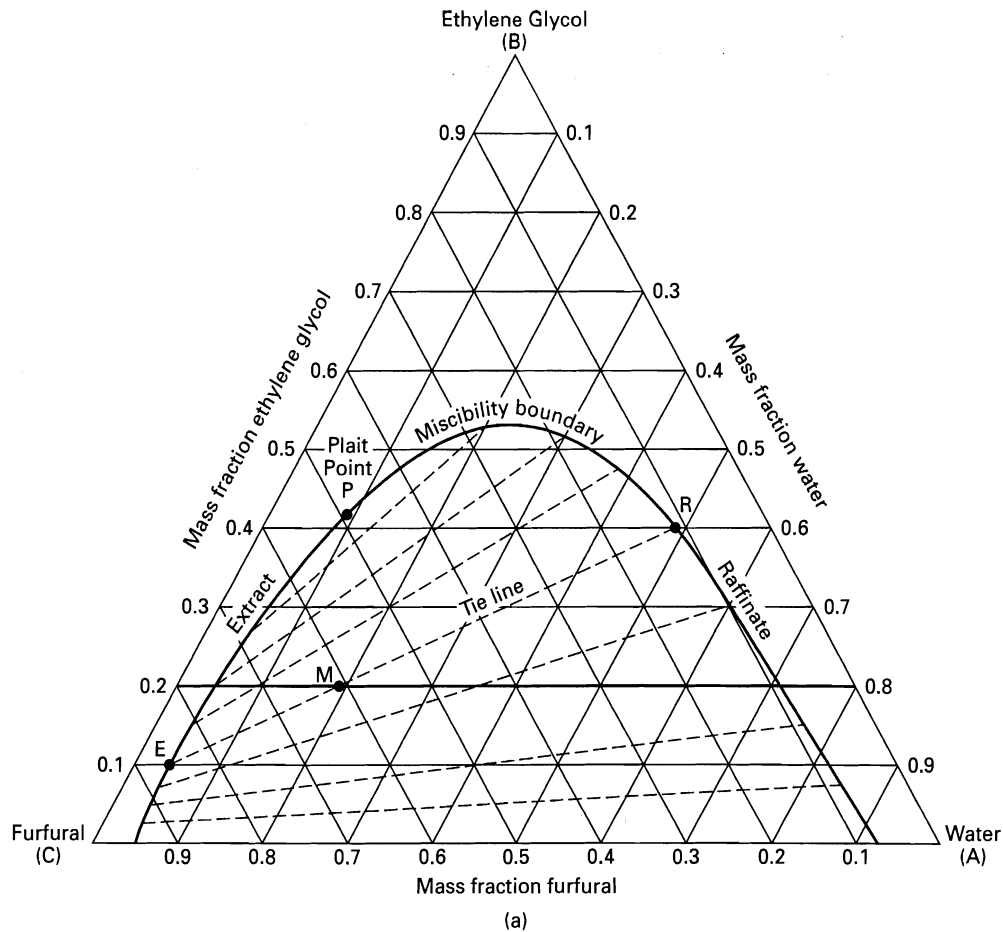


Figure 4.14 Liquid-liquid equilibrium, ethylene glycol-furfural-water, 25°C, 101 kPa: (a) equilateral triangular diagram; (b) right triangular diagram; (c) equilibrium solute diagram in mass fractions (*continues*).

Tie lines, shown as dashed lines below the miscibility boundary in Figure 4.14a, are used to connect points on the miscibility boundary, DEPRG, that represent equilibrium-phase compositions. To obtain data to construct tie lines, such as ER, it is necessary to make a mixture such as M (20% glycol, 18.9% water, and 61.1% furfural),

equilibrate it, and then chemically analyze the resulting equilibrium extract and raffinate phases E and R (in this case, 10% glycol, 3.9% water, and 86.1% furfural; and 40% glycol, 49% water, and 11% furfural, respectively). At point P, the *plait point*, the two liquid phases have identical compositions. Therefore, the tie lines converge

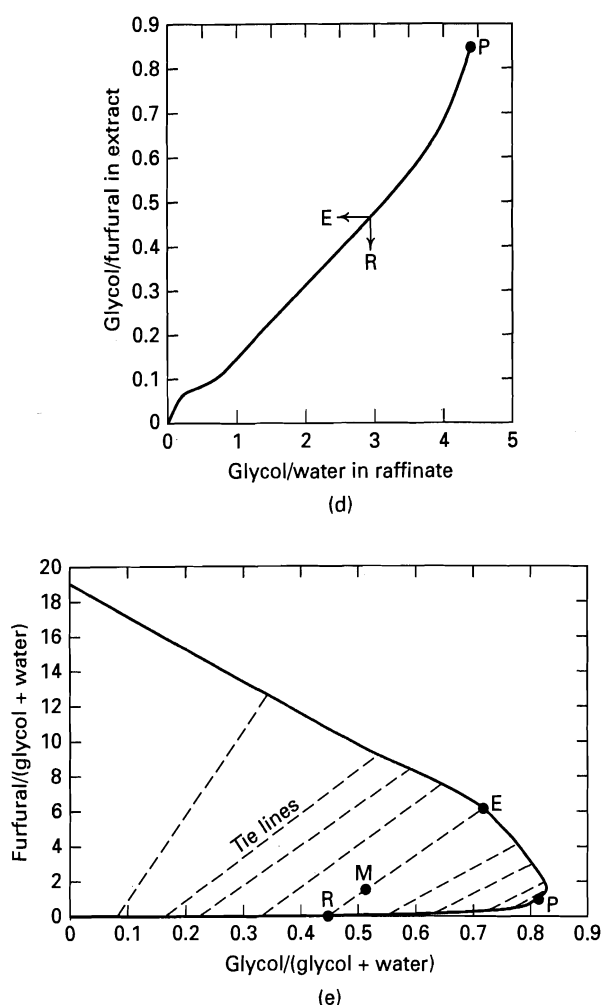


Figure 4.14 (Continued) (d) equilibrium solute diagram in mass ratios; (e) Janecke diagram.

Table 4.6 Mutual Equilibrium (Tie Line)
Data for the Furfural–Ethylene
Glycol–Water System at 25°C and 101 kPa

Glycol in Water Layer, wt%	Glycol in Furfural Layer, wt%
41.5	41.5
50.5	32.5
52.5	27.5
51.5	20.0
47.5	15.0
40.0	10.0
30.0	7.5
20.0	6.2
15.0	5.2
7.3	2.5

to a point and the two phases become one phase. Tie-line data for this system are given in Table 4.6, in terms of glycol composition.

When there is mutual solubility between two phases, the thermodynamic variables necessary to define the equilib-

rium system are temperature, pressure, and the concentrations of the components in each phase. According to the phase rule, (4-1), for a three-component, two-liquid-phase system, there are three degrees of freedom. At constant temperature and pressure, specification of the concentration of one component in either of the phases suffices to completely define the state of the system. Thus, as shown in Figure 4.14a, one value for glycol weight percent on the miscibility boundary curve fixes the composition of the corresponding phase and, by means of the tie line, the composition of the other equilibrium phase.

Figure 4.14b is a representation of the same system on a right-triangular diagram. Here the concentrations in weight percent of any two of the three components (normally the solute and solvent) are given, the concentration of the third being obtained by difference from 100 wt%. Diagrams like this are easier to construct and read than equilateral triangular diagrams.

However, equilateral triangular diagrams are conveniently constructed with the computer program, CSpace, which can be downloaded from the web site at www.ugr.es/~cspace.

Figures 4.14c and 4.14d are representations of the same ternary system in terms of weight fraction and weight ratios of the solute, respectively. Figure 4.14c is simply a plot of the equilibrium (tie-line) data of Table 4.6 in terms of solute mass fraction. In Figure 4.14d, mass ratios of solute (ethylene glycol) to furfural and water for the extract and raffinate phases, respectively, are used. Such curves can be used to interpolate tie lines, since only a limited number of tie lines are shown on triangular graphs. Because of this, such diagrams are often referred to as *distribution diagrams*. When mole (rather than mass) fractions are used in a diagram like Figure 4.14c, a nearly straight line is often evident near the origin, where the slope is the distribution coefficient, K_D , for the solute at infinite dilution.

In 1906, Janecke [10] suggested the equilibrium data display shown as Figure 4.14e. Here, the mass of solvent per unit mass of solvent-free material, furfural/(water + glycol), is plotted as the ordinate versus the mass ratio, on a solvent-free basis, of glycol/(water + glycol) as abscissa. The ordinate and abscissa apply to both phases. Equilibrium conditions are related by tie lines. Mole ratios can be used also to construct Janecke diagrams.

Any of the five diagrams in Figure 4.14 can be used for solving problems involving material balances subject to liquid–liquid equilibrium constraints, as is demonstrated in the following example.

EXAMPLE 4.7

Determine the composition of the equilibrium extract and raffinate phases produced when a 45% by weight glycol (B)–55% water (A) solution is contacted with twice its weight of pure furfural solvent (C) at 25°C and 101 kPa. Use each of the five diagrams in Figure 4.14, if possible.

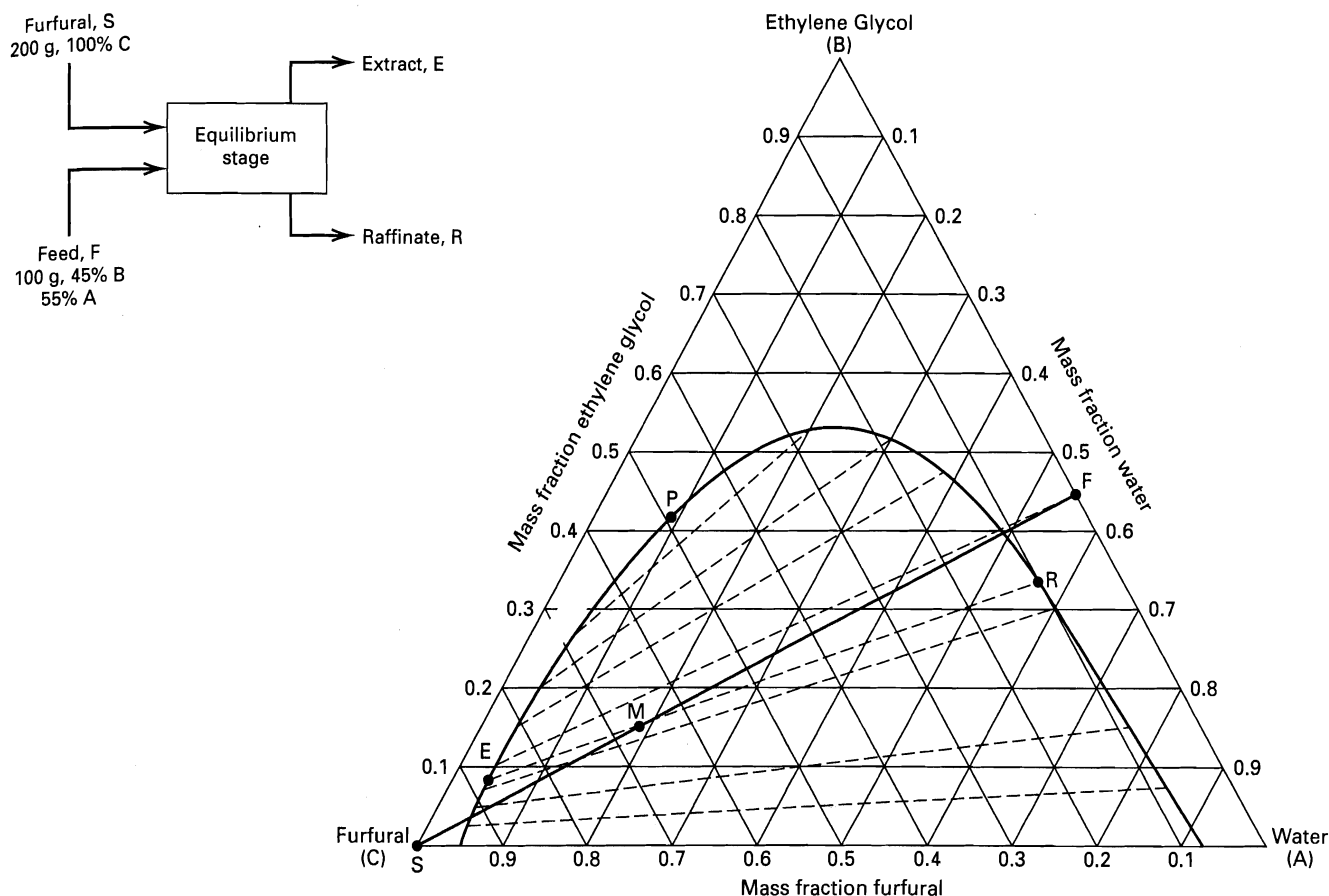


Figure 4.15 Solution to Example 4.7a.

SOLUTION

Assume a basis of 100 g of 45% glycol–water feed. Thus, in Figure 4.13b, the feed (F) is 55 g of A and 45 g of B. The solvent (S) is 200 g of C. Let E = the extract, and R = the raffinate.

(a) By an equilateral-triangular diagram, Figure 4.15:

Step 1. Locate the feed and solvent compositions at points F and S , respectively.

Step 2. Define M , the mixing point, as $M = F + S = E + R$

Step 3. Apply the inverse-lever-arm rule to the equilateral-triangular phase-equilibrium diagram. Let $w_i^{(1)}$ be the mass fraction of species i in the extract, $w_i^{(2)}$ be the mass fraction of species i in the raffinate, and $w_i^{(M)}$ be the mass fraction of species i in the combined feed and solvent phases.

From a balance on the solvent, C: $(F + S)w_C^{(M)} = Fw_C^{(F)} + Sw_C^{(S)}$.

Therefore,

$$\frac{F}{S} = \frac{w_C^{(S)} - w_C^{(M)}}{w_C^{(M)} - w_C^{(F)}} \quad (1)$$

Thus, points S , M , and F lie on a straight line, and, by the inverse lever arm rule,

$$\frac{F}{S} = \frac{SM}{MF} = \frac{1}{2}$$

The composition at point M is 18.3% A, 15.0% B, and 66.7% C.

Step 4. Since M lies in the two-phase region, the mixture must separate along an interpolated dash-dot tie line into the extract

phase at point E (8.5% B, 4.5% A, and 87.0% C) and the raffinate at point R (34.0% B, 56.0% A, and 10.0% C).

Step 5. The inverse-lever-arm rule applies to points E , M , and R , so $E = M(RM/ER)$. Because $M = 100 + 200 = 300$ g, and from measurements of the line segments, $E = 300(147/200) = 220$ g and $R = M - E = 300 - 220 = 80$ g.

(b) By a right-triangular diagram, Figure 4.16:

Step 1. Locate the points F and S for the two feed streams.

Step 2. Define the mixing point $M = F + S$.

Step 3. The inverse-lever-arm rule also applies to right-triangular diagrams, so $MF/MS = \frac{1}{2}$.

Step 4. Points R and E are on the ends of the interpolated dash-dot tie line passing through point M .

The numerical results of part (b) are identical to those of part (a).

(c) By an equilibrium solute diagram, Figure 4.14c. A material balance on glycol, B,

$$Fw_B^{(F)} + Sw_B^{(S)} = 45 = Ew_B^{(E)} + Rw_B^{(R)} \quad (2)$$

must be solved simultaneously with a phase-equilibrium relationship. It is not possible to do this graphically using Figure 4.14c in any straightforward manner unless the solvent (C) and carrier (A) are mutually insoluble. The outlet-stream composition can be found, however, by the following iterative procedure.

Step 1. Guess a value for $w_B^{(E)}$ and read the equilibrium value, $w_B^{(R)}$, from Figure 4.14c.

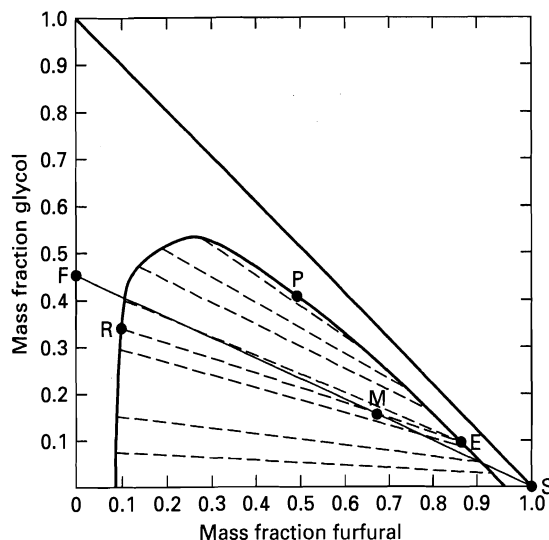


Figure 4.16 Solution to Example 4.7b.

Step 2. Substitute these two values into the equation obtained by combining (2) with the overall balance, $E + R = 300$, to eliminate R . Solve for E and then R .

Step 3. Check to see if the furfural (or water) balance is satisfied using the equilibrium data from Figures 4.14a, 4.14b, or 4.14c. If not, repeat steps 1 to 3 with a new guess for $w_B^{(E)}$. This procedure leads to the same results obtained in parts (a) and (b).

(d) By an equilibrium solute diagram in mass fractions, Figure 4.14d: This plot suffers from the same limitations as Figure 4.13c in that a solution must be achieved by an iterative procedure.

(e) By a Janecke diagram, Figure 4.17:

Step 1. The feed mixture is located at point F. With the addition of 200 g of pure furfural solvent, $M = F + S$ is located as shown, since the ratio of glycol to (glycol + water) remains the same.

Step 2. The mixture at point M separates into the two phases at points E and R, using the interpolated dash-dot tie line, with the coordinates (7.1, 0.67) at E and (0.10, 0.37) at R.

Step 3. Let Z^E and Z^R equal the total mass of components A and B in the extract and raffinate, respectively. Then, the following

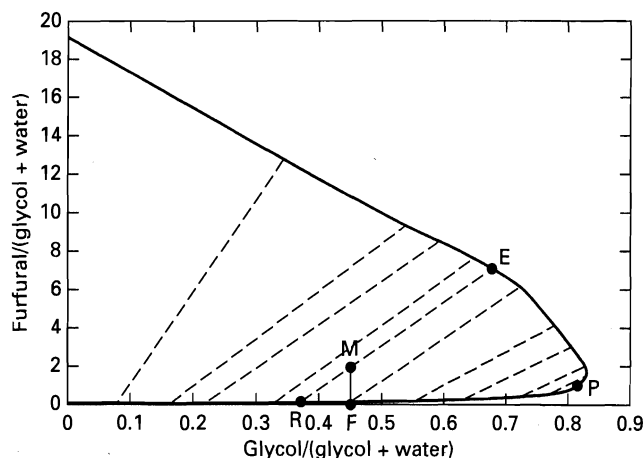


Figure 4.17 Solution to Example 4.7e.

balances apply:

$$\text{Furfural: } 7.1Z^E + 0.10Z^R = 200$$

$$\text{Glycol: } 0.67Z^E + 0.37Z^R = 45$$

Solving these two simultaneous equations, we obtain $Z^E = 27$ g, $Z^R = 73$ g.

Thus, the furfural in the extract = $(7.1)(27 \text{ g}) = 192$ g, the furfural in the raffinate = $200 - 192 = 8$ g, the glycol in the extract = $(0.67)(27 \text{ g}) = 18$ g, the glycol in the raffinate = $45 - 18 = 27$ g, the water in the raffinate = $73 - 27 = 46$ g, and the water in the extract = $27 - 18 = 9$ g. The total extract is $192 + 27 = 219$ g, which is close to the results obtained in part (a). The raffinate composition and amount can be obtained just as readily.

It should be noted on the Janecke diagram that $\overline{ME}/\overline{MR}$ does not equal R/E ; it equals the ratio of R/E on a solvent-free basis.

In Figure 4.14, two pairs of components are mutually soluble, while one pair is only partially soluble. Ternary systems where two pairs and even all three pairs are only partially soluble are also common. Figure 4.18 shows examples, taken from Francis [11] and Findlay [12], of four different cases where two pairs of components are only partially soluble.

In Figure 4.18a, two separate two-phase regions are formed, while in Figure 4.18c, in addition to the two-phase regions, a three-phase region, RST, is formed. In Figure 4.18b, the two separate two-phase regions merge. For a ternary mixture, as temperature is reduced, phase behavior may progress from Figure 4.18a to 4.18b to 4.18c. In Figures 4.18a, 4.18b, and 4.18c, all tie lines slope in the same direction. In some systems of importance, *solutropy*, a reversal of tie-line slopes, occurs.

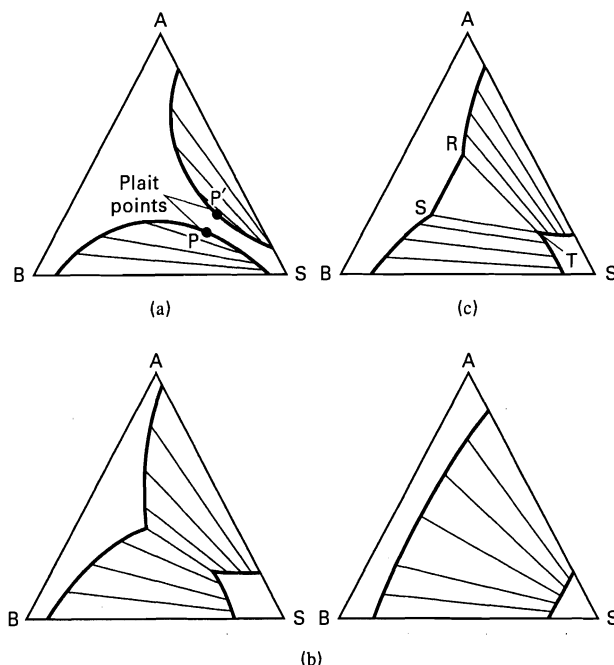


Figure 4.18 Equilibria for 3/2 systems: (a) miscibility boundaries are separate; (b) miscibility boundaries and tie-line equilibria merge; (c) tie lines do not merge and the three-phase region RST is formed.

4.6 MULTICOMPONENT LIQUID-LIQUID SYSTEMS

Quaternary and higher multicomponent mixtures are often encountered in liquid-liquid extraction processes, particularly when two solvents are used for liquid-liquid extraction. As discussed in Chapter 2, multicomponent liquid-liquid equilibria are very complex and there is no compact graphical way of representing experimental phase equilibria. Accordingly, the computation of the equilibrium-phase compositions is best made by a computer-assisted algorithm using activity coefficient equations from Chapter 2 that account for the effect of composition (e.g., NRTL, UNIQUAC, or UNIFAC). One such algorithm is a modification of the Rachford-Rice algorithm for vapor-liquid equilibrium, given in Tables 4.3 and 4.4. To apply these tables to multicomponent, liquid-liquid equilibria, the following symbol transformations are made, where all flow rates and compositions are in moles:

Vapor-Liquid Equilibria (Tables 4.3, 4.4)

Feed, F
Equilibrium vapor, V
Equilibrium liquid, L
Feed mole fractions, z_i
Vapor mole fractions, y_i
Liquid mole fractions, x_i
 K -value, K_i
 $\Psi = V/F$

Liquid-Liquid Equilibria

Feed, F , + solvent, S
Extract, E ($L^{(1)}$)
Raffinate, R ($L^{(2)}$)
Mole fractions of combined F and S
Extract mole fractions, $x_i^{(1)}$
Raffinate mole fractions, $x_i^{(2)}$
Distribution coefficient, K_{Di}
 $\Psi = E/F$

Most liquid-liquid equilibria are achieved under adiabatic conditions, thus necessitating consideration of an energy balance. However, if both feed and solvent enter the stage at identical temperatures, the only energy effect is the heat of mixing, which is often sufficiently small that only a very small temperature change occurs. Accordingly, the calculations are often made isothermally.

The modified Rachford-Rice algorithm is shown in the flow chart of Figure 4.19. This algorithm is applicable for either an isothermal vapor-liquid (V-L) or liquid-liquid (L-L) equilibrium-stage calculation when K -values depend strongly on phase compositions. For the L-L case, the algorithm assumes that the feed and solvent flow rates and compositions are fixed. The equilibrium pressure and temperature are also specified. An initial estimate is made of the phase compositions, $x_i^{(1)}$ and $x_i^{(2)}$, and corresponding estimates of the distribution coefficients are made from liquid-phase activity coefficients, using (2-30) with, for example, the NRTL or UNIQUAC equations discussed in Chapter 2. Equation 3 of Table 4.4 is then solved iteratively for $\Psi = E/(F + S)$, from which values of $x_i^{(2)}$ and $x_i^{(1)}$ are computed from Eqs. (5) and (6), respectively, of Table 4.4. The resulting values of $x_i^{(1)}$ and $x_i^{(2)}$ will not usually sum, respectively, to 1 for each liquid phase and are therefore normalized. The normalized values are obtained from equations of the form $x'_i = x_i / \sum x_j$, where x'_i are the normalized values that force $\sum x'_j$ to equal 1. The normalized values replace the values computed from Eqs. (5) and (6). The iterative

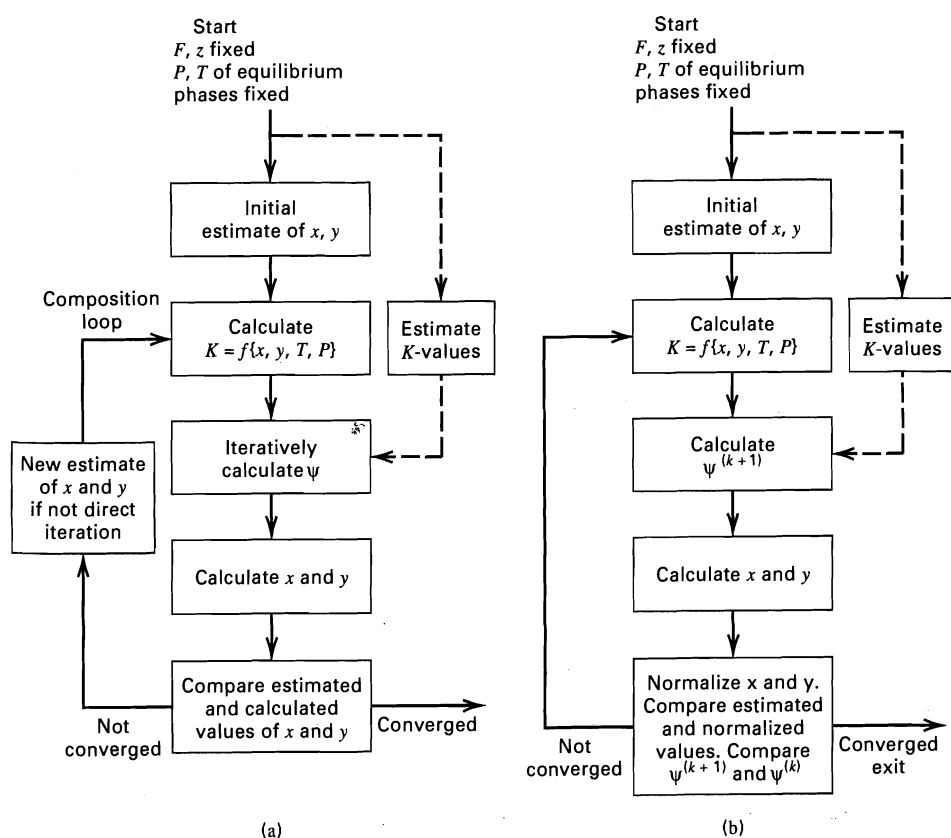


Figure 4.19 Algorithm for isothermal-flash calculation when K -values are composition-dependent: (a) separate nested iterations on Ψ and (x, y) ; (b) simultaneous iteration on Ψ and (x, y) .

procedure is repeated until the compositions $x_i^{(1)}$ and $x_i^{(2)}$, to say three or four significant digits, no longer change from one iteration to the next. Multicomponent liquid–liquid equilibrium calculations are best carried out with a steady-state simulation computer program.

EXAMPLE 4.8

An azeotropic mixture of isopropanol, acetone, and water is being dehydrated with ethyl acetate in a distillation system of two columns. Benzene was previously used as the dehydrating agent, but recent legislation has made the use of benzene undesirable because it is carcinogenic. Ethyl acetate is far less toxic. The overhead vapor from the first column, with the composition given below, at a pressure of 20 psia and a temperature of 80°C is condensed and cooled to 35°C, without significant pressure drop, causing the formation of two liquid phases in equilibrium. Estimate the amounts of the two phases in kilograms per hour and the phase compositions in weight percent.

Component	kg/h
Isopropanol	4,250
Acetone	850
Water	2,300
Ethyl acetate	43,700

Note that the specification of this problem satisfies the degrees of freedom from (4-4), which for $C = 4$ is 9.

SOLUTION

This example was solved with the ChemCAD program using the UNIFAC group contribution method to estimate liquid-phase activity coefficients. The results are as follows:

Component	Weight Fraction	
	Organic-Rich Phase	Water-Rich Phase
Isopropanol	0.0843	0.0615
Acetone	0.0169	0.0115
Water	0.0019	0.8888
Ethyl acetate	0.8969	0.0382
	1.0000	1.0000
Flow rate, kg/h	48,617	2,483

It is of interest to compare the distribution coefficients computed from the above results based on the UNIFAC method to experimental values given in *Perry's Handbook* [1]:

Component	Distribution Coefficient (wt % Basis)	
	UNIFAC	<i>Perry's Handbook</i>
Isopropanol	1.37	1.205 (20°C)
Acetone	1.47	1.50 (30°C)
Water	0.0021	—
Ethyl acetate	23.5	—

Results for isopropanol and acetone are in reasonably good agreement at these relatively dilute conditions, considering that no temperature corrections were made.

4.7 SOLID–LIQUID SYSTEMS

Solid–liquid separation operations include leaching, crystallization, and adsorption. In a leaching operation (solid–liquid extraction), a multicomponent solid mixture is separated by contacting the solid with a solvent that selectively dissolves some, but not all, components in the solid. Although this operation is quite similar to liquid–liquid extraction, two aspects of leaching make it a much more difficult separation operation in practice. Diffusion in solids is very slow compared to diffusion in liquids, thus making it difficult to achieve equilibrium. Also, it is virtually impossible to completely separate a solid phase from a liquid phase. A clear liquid phase can be obtained, but the solids will be accompanied by some liquid. In comparison, the separation of two liquid phases is fairly easy to accomplish.

A second solid–liquid system involves the crystallization of one or more, but not all, components from a liquid mixture. This operation is analogous to distillation. However, although equilibrium can be achieved, a sharp phase separation is again virtually impossible.

A third application of solid–liquid systems, adsorption, involves the use of a porous solid agent, which does not undergo phase change or composition change. The solid selectively adsorbs, on its exterior and interior surface, certain components of the liquid mixture. The adsorbed species are then desorbed and the solid adsorbing agent is regenerated for repeated use. Variations of adsorption include ion exchange and chromatography. A solid–liquid system is also utilized in membrane-separation operations, where the solid is a membrane that selectively absorbs and transports certain species, thus effecting a separation.

Solid–liquid separation processes, such as leaching and crystallization, almost always involve phase-separation operations such as gravity sedimentation, filtration, and centrifugation. These operations are not covered in this textbook, but are discussed in Section 18 of *Perry's Handbook* [1].

Leaching

A leaching stage for a ternary system is shown in Figure 4.20. The solid mixture to be separated consists of particles containing insoluble A and solute B. The solvent, C, selectively dissolves B. The overflow from the stage is a solids-free liquid of solvent C and dissolved B. The underflow is a wet solid or slurry of liquid and solid A. In an ideal, equilibrium leaching stage, all of the solute is dissolved by the solvent; none of the solid A is dissolved. In addition, the composition of the retained liquid phase in the underflow is identical to the composition of the liquid overflow, and the

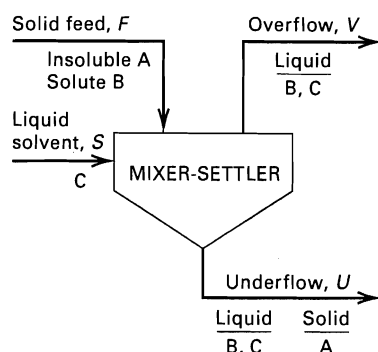


Figure 4.20 Leaching stage.

overflow is free of solids. The mass ratio of solid to liquid in the underflow depends on the properties of the two phases and the type of equipment used, and is best determined from experience or tests with prototype equipment. In general, if the viscosity of the liquid phases increases with increasing solute concentration, the mass ratio of solid to liquid in the underflow will decrease because the solid will retain more liquid.

Ideal leaching calculations can be carried out algebraically or graphically, with diagrams like those shown in Figure 4.21, using the following nomenclature in mass units:

F = total mass flow rate of feed to be leached

S = total mass flow rate of entering solvent

U = total mass flow rate of the underflow, including solids

V = total mass flow rate of the overflow

X_A = mass ratio of insoluble solid A to (solute B + solvent C) in the feed flow, F , or underflow, U

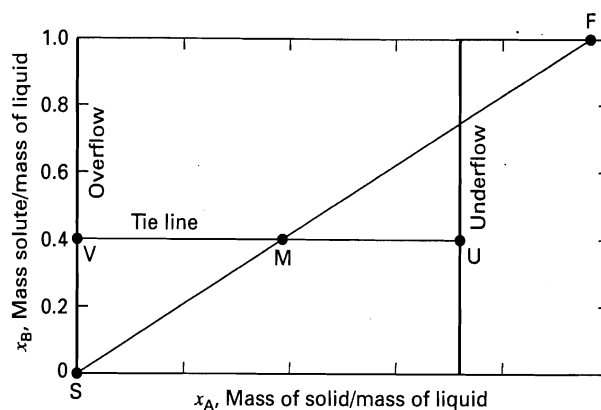
Y_A = mass ratio of insoluble solid A to (solute B + solvent C) in the entering solvent flow, S , or overflow, V

X_B = mass ratio of solute B to (solute B + solvent C) in the feed flow, F , or underflow, U

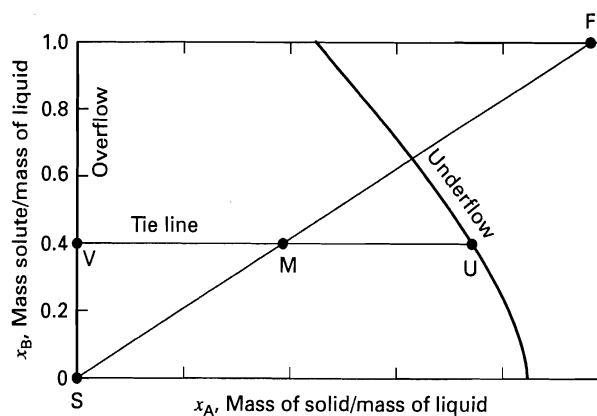
Y_B = mass ratio of solute B to (solute B + solvent C) in the solvent flow, S , or overflow, V

Figure 4.21a depicts ideal leaching conditions when, in the underflow, the mass ratio of insoluble solid to liquid, X_A , is a constant, independent of the concentration, X_B , of solute in the solids-free liquid. The resulting tie line is vertical. This case is referred to as *constant-solution underflow*. Figure 4.21b depicts ideal leaching conditions when X_A varies with X_B . This case is referred to as *variable-solution underflow*. In both ideal cases, we assume (1) an entering feed, F , free of solvent such that $X_B = 1$; (2) a solids-free and solute-free solvent, S , such that $Y_A = 0$ and $Y_B = 0$; (3) equilibrium between the exiting liquid solutions in the underflow, U , and the overflow, V , such that $X_B = Y_B$; and (4) a solids-free overflow, V , such that $Y_A = 0$.

As with ternary, liquid-liquid extraction calculations, discussed in Section 4.5, a mixing point, M , can be defined for



(a)



(b)

Figure 4.21 Underflow-overflow conditions for ideal leaching: (a) constant solution underflow; (b) variable solution underflow.

($F + S$), equal to that for the sum of the two products of the leaching stage, ($U + V$). Typical mixing points and inlet and outlet compositions are included in Figures 4.21a and b. In both cases, as shown in the next example, the inverse-lever-arm rule can be applied to the line UMV to obtain the flow rates of the underflow, U , and overflow, V .

EXAMPLE 4.9

Soybeans are the predominant oilseed crop in the world, followed by cottonseed, peanuts, and sunflower seed. While soybeans are not generally consumed directly by humans; they can be processed to produce valuable products. Large-scale production of soybeans in the United States began after World War II, increasing in recent years to more than 140 billion pounds per year. Most of the soybeans are processed to obtain soy oil and vitamins like niacin and lecithin for human consumption, and a defatted meal for livestock feed. Compared to other vegetable oils, soy oil is more economical, more stable, and healthier. Typically, 100 pounds of soybeans yields 18 pounds of soy oil and 79 pounds of defatted meal.

To recover oil, soybeans are first cleaned, cracked to loosen the seeds from the hulls, dehulled, and dried to 10–11% moisture. They are then leached with a hexane solution to remove the oil. However,

before leaching, the soybeans are flaked to reduce the time required for mass transfer of the oil out of the bean and into hexane. Following leaching, the hexane in the overflow is separated from the soy oil and recovered for recycle by evaporation, while the underflow is treated to remove residual hexane and toasted with high-temperature air to produce defatted meal. Modern soybean-extraction plants crush up to 3,000 tons of soybeans per day. This example is concerned with just the leaching step.

Oil is to be leached from 100,000 kg/h of soybean flakes, containing 19% by weight oil, in a single stage by 100,000 kg/h of a hexane solvent. Experimental data indicate that the oil content of the flakes will be reduced to 0.5 wt%. For the type of equipment to be used, the expected contents of the underflows is as follows:

β , Mass fraction of solids in underflow	0.68	0.67	0.65	0.62	0.58	0.53
Mass ratio of solute in underflow liquid, X_B	0.0	0.2	0.4	0.6	0.8	1.0

Calculate by both a graphical and an analytical method, the compositions and flow rates of the underflow and overflow, assuming an ideal leaching stage. What percentage of the oil in the feed is recovered in the overflow?

SOLUTION

The flakes contain $(0.19)(100,000) = 19,000$ kg/h of oil and $(100,000 - 19,000) = 81,000$ kg/h of insolubles. However, all of the oil is not leached. For convenience in the calculations, lump the unleached oil with the insolubles to give an effective A. The flow rate of unleached oil = $(81,000)(0.5/99.5) = 407$ kg/h. Therefore, take the flow rate of A as $(81,000 + 407) = 81,407$ kg/h and the oil in the feed as just the amount that is leached or $(19,000 - 407) = 18,593$ kg/h of B. Therefore, in the feed, F,

$$Y_A = (81,407/18,593) = 4.38$$

$$X_B = 1.0$$

The sum of the liquid solutions in the underflow and overflow includes 100,000 kg/h of hexane and 18,593 kg/h of leached oil. Therefore, for the underflow and overflow,

$$X_B = Y_B = [18,593/(100,000 + 18,593)] = 0.157$$

This is a case of variable-solution underflow. Using data in the above table, convert values of β , the mass fraction of solids in the underflow, to values of X_A , the mass ratio of insolubles to liquid in the underflow, which by material balance is

$$X_A = \frac{\text{kg/h A}}{\text{kg/h (B + C)}} = \frac{\beta U}{(1 - \beta)U} = \frac{\beta}{(1 - \beta)} \quad (1)$$

Using (1), the following values of X_A are computed from the previous table:

X_A	2.13	2.03	1.86	1.63	1.38	1.13
X_B	0.0	0.2	0.4	0.6	0.8	1.0

Graphical Method

Figure 4.22 is a plot of X_A as a function of X_B . Data for the underflow line are obtained from the preceding table. Because no solids

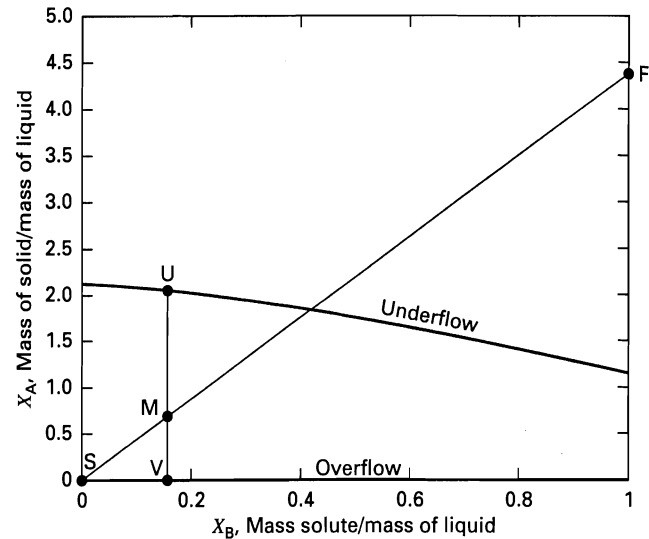


Figure 4.22 Constructions for Example 4.9.

leave in the overflow, that line is drawn horizontally at $X_A = 0$. On this plot are composition points for the feed flakes, F, and the entering solvent hexane, S, with a straight line drawn between them. A point for the overflow, V, is plotted at $X_A = 0$ and, from above, $X_B = 0.157$. Since $Y_B = X_B = 0.157$, the value of X_A in the underflow is obtained at the intersection of a vertical line drawn from the overflow point, V, to the underflow line. This value is $X_A = 2.05$. The two lines FS and UV intersect at the mixing point, M.

We now compute the compositions of the underflow and overflow. In the overflow, from $X_B = 0.157$, the mass fractions of solute B and solvent C are, respectively, 0.157 and $(1 - 0.157) = 0.843$. In the underflow, using $X_A = 2.05$ and $X_B = 0.157$, the mass fractions of solids, B, and C, are, respectively, $[2.05/(1 + 2.05)] = 0.672$, $0.157(1 - 0.672) = 0.0515$, and $(1 - 0.672 - 0.0515) = 0.2765$.

The inverse-lever-arm rule can be used to compute the amounts of underflow and overflow. Here, the rule applies only to the liquid phases in the two exiting streams because the coordinates of Figure 4.22 are compositions on a solids-free basis. The mass ratio of liquid flow rate in the underflow to liquid flow rate in the overflow is given by the ratio of line MV to line MU. With M located at $X_A = 0.69$, this ratio = $(0.69 - 0.0)/(2.05 - 0.69) = 0.51$. Thus, the liquid flow rate in the underflow = $(100,000 + 18,593)(0.51)/(1 + 0.51) = 40,054$ kg/h. Adding to this the flow rates of carrier and unextracted oil, computed above, gives $U = 40,054 + 81,407 = 121,461$ kg/h or say 121,000 kg/h. The overflow rate = $V = 200,000 - 121,000 = 79,000$ kg/h.

The oil flow rate in the feed is 19,000 kg/h. The oil flow rate in the overflow = $Y_B V = 0.157(79,000) = 12,400$ kg/h. Thus, the percentage of the oil in the feed that is recovered in the overflow = $12,400/19,000 = 0.653$ or 65.3%. Adding washing stages, as described in Section 5.2, can increase the oil recovery.

Algebraic Method

Instead of using the inverse-lever-arm rule with Figure 4.22, mass-balance equations can be applied. As with the graphical method, $X_B = 0.157$, giving a value from the previous table of $X_A = 2.05$. Then, since the flow rate of solids in the underflow = 81,407 kg/h, the flow rate of liquid in the underflow = $81,407/2.05 = 39,711$ kg/h.

The total flow rate of underflow is $U = 81,407 + 39,711 = 121,118$ kg/h. By mass balance, the flow rate of overflow = $200,000 - 121,118 = 78,882$ kg/h. These values are close to those obtained by the graphical method. The percentage recovery of oil, and compositions of the underflow and overflow, are computed in the same manner as in the graphical method.

Crystallization

Crystallization may take place from aqueous or nonaqueous solutions. The simplest case is for a binary mixture of two organic chemicals such as naphthalene and benzene, whose solubility or solid-liquid phase-equilibrium diagram for a pressure of 1 atm is shown in Figure 4.23. Points A and B are the melting (freezing) points of pure benzene (5.5°C) and pure naphthalene (80.2°C), respectively. When benzene is dissolved in liquid naphthalene or naphthalene is dissolved in liquid benzene, the freezing point of the solvent is depressed. Point E is the *eutectic point*, corresponding to a eutectic temperature (-3°C) and eutectic composition (80 wt% benzene). The word “eutectic” is derived from a Greek word that means “easily fused,” and in Figure 4.23 it represents the binary mixture of naphthalene and benzene, as separate solid phases, with the lowest freezing (melting) point.

Temperature-composition points located above the curve AEB correspond to a homogeneous liquid phase. Curve AE is the solubility curve for benzene in naphthalene. For example, at 0°C the solubility is very high, 87 wt% benzene or 6.7 kg benzene/kg naphthalene. Curve EB is the solubility curve for naphthalene. At 25°C the solubility is 41 wt% naphthalene or 0.7 kg naphthalene/kg benzene. At 50°C , the solubility of naphthalene is much higher, 1.9 kg naphthalene/kg benzene. For this mixture, as with most mixtures, solubility increases with increasing temperature.

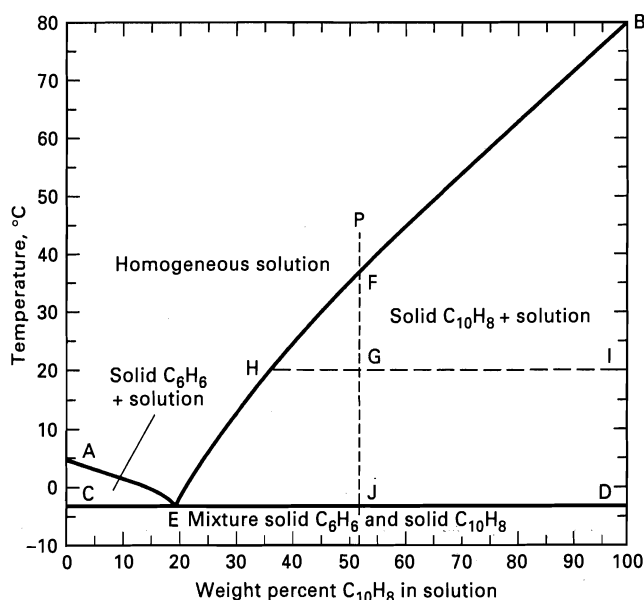


Figure 4.23 Solubility of naphthalene in benzene.

[Adapted from O.A. Hougen, K.M. Watson, and R.A. Ragatz, *Chemical Process Principles. Part I*, 2nd ed., John Wiley and Sons, New York (1954).]

If a liquid solution of composition and temperature represented by point P is cooled along the vertical, dashed line, it will remain a liquid until the line intersects the solubility curve at point F. If the temperature is lowered further, crystals of naphthalene form and the remaining liquid, called the *mother liquor*, becomes richer in benzene. For example, when point G is reached, pure naphthalene crystals and a mother liquor, given by point H on solubility curve EB, coexist at equilibrium, with the composition of the solution being 37 wt% naphthalene. This is in agreement with the Gibbs phase rule (4-1), because with $C = 2$ and $\mathcal{P} = 2$, $\mathcal{F} = 2$ and for fixed T and P , the phase compositions are fixed. The fraction of the solution crystallized can be determined by applying the inverse-lever-arm rule. Thus, in Figure 4.23, the fraction is kilograms naphthalene crystals/kilograms original solution = length of line GH/length of line HI = $(52 - 37)/(100 - 37) = 0.238$.

As the temperature is lowered further until line CED, corresponding to the eutectic temperature, is reached at point J, the two-phase system consists of naphthalene crystals and a mother liquor of the eutectic composition given by point E. Any further removal of heat causes the eutectic solution to solidify.

EXAMPLE 4.10

A total of 8,000 kg/h of a liquid solution of 80 wt% naphthalene and 20 wt% benzene at 70°C is cooled to 30°C to form naphthalene crystals. Assuming that equilibrium is achieved, determine the amount of crystals formed and the composition of the equilibrium mother liquor.

SOLUTION

From Figure 4.23, at 30°C , the solubility of naphthalene is 45 wt% naphthalene. By the inverse-lever-arm rule, for an original 80 wt% solution,

$$\frac{\text{kg naphthalene crystals}}{\text{kg original mixture}} = \frac{(80 - 45)}{(100 - 45)} = 0.636$$

The flow rate of crystals = $0.636 (8,000) = 5,090$ kg/h.

The composition of the remaining 2,910 kg/h of mother liquor is 55 wt% benzene and 45 wt% naphthalene.

Crystallization of a salt from an aqueous solution is frequently complicated by the formation of hydrates of the salt with water in certain definite molar proportions. These hydrates can be stable solid compounds within certain ranges of temperature as given in the solid-liquid phase equilibrium diagram. A rather extreme, but common, case is that of MgSO_4 , which can form the stable hydrates $\text{MgSO}_4 \cdot 12\text{H}_2\text{O}$, $\text{MgSO}_4 \cdot 7\text{H}_2\text{O}$, $\text{MgSO}_4 \cdot 6\text{H}_2\text{O}$, and $\text{MgSO}_4 \cdot \text{H}_2\text{O}$. The high hydrate is stable at low temperatures, while the low hydrate is the stable form at higher temperatures.

A simpler example is that of Na_2SO_4 in mixtures with water. As seen in the phase diagram of Figure 4.24, only one

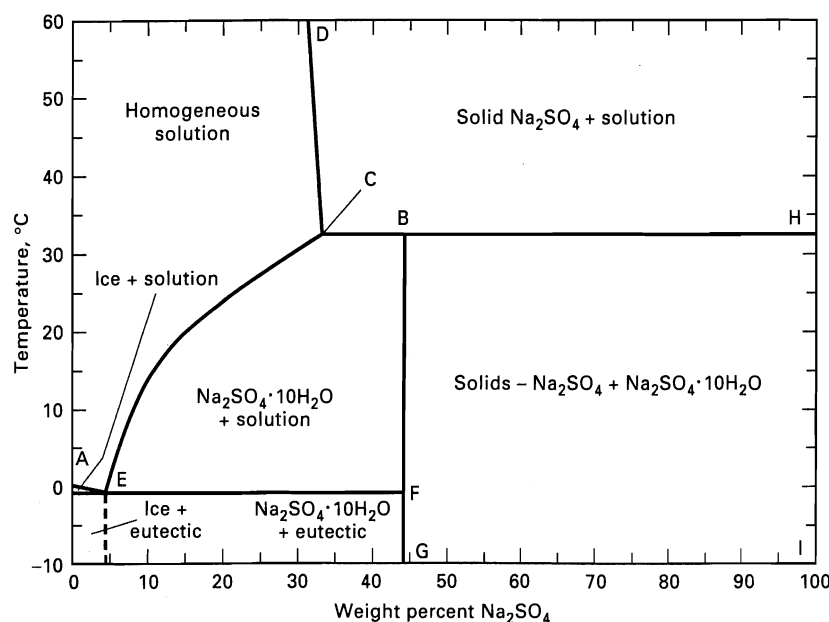


Figure 4.24 Solubility of sodium sulfate in water.

[Adapted from O.A. Hougen, K.M. Watson, and R.A. Ragatz, *Chemical Process Principles. Part I*, 2nd ed., John Wiley and Sons, New York (1954).]

stable hydrate is formed, $\text{Na}_2\text{SO}_4 \cdot 10\text{H}_2\text{O}$, commonly known as Glauber's salt. Not shown in Figure 4.24 is the metastable hydrate $\text{Na}_2\text{SO}_4 \cdot 7\text{H}_2\text{O}$. Since the molecular weights are 142.05 for Na_2SO_4 and 18.016 for H_2O , the weight percent Na_2SO_4 in the decahydrate is 44.1, which corresponds to the vertical line BFG.

The freezing point of water, 0°C , is at A in Figure 4.24, but the melting point of Na_2SO_4 , 884°C , is not shown because the temperature scale is terminated at 60°C . The decahydrate melts at 32.4°C , point B, to form solid Na_2SO_4 and a mother liquor, point C, of 32.5 wt% Na_2SO_4 . As Na_2SO_4 is dissolved in water, the freezing point is depressed slightly along curve AE until the eutectic, point E, is reached.

Curves EC and CD represent the solubilities of the decahydrate crystals and anhydrous sodium sulfate, respectively, in water. Note that the solubility of Na_2SO_4 decreases slightly with increasing temperature.

For each region, the coexisting phases are indicated. For example, in the region below GFBHI, a solid solution of the anhydrous and decahydrate forms exists. The amounts of the coexisting phases can be determined by the inverse-lever-arm rule.

EXAMPLE 4.11

A 30 wt% aqueous Na_2SO_4 solution of 5,000 lb/h enters a cooling-type crystallizer at 50°C . At what temperature will crystallization begin? Will the crystals be the decahydrate or anhydrous form? To what temperature will the mixture have to be cooled to crystallize 50% of the Na_2SO_4 ?

SOLUTION

From Figure 4.24, the original solution of 30 wt% Na_2SO_4 at 50°C corresponds to a point in the homogeneous liquid solution region. If a vertical line is dropped from that point, it intersects the solubility

curve EC at 31°C . Below this temperature, the crystals formed are the decahydrate.

The feed contains $(0.30)(5,000) = 1,500$ lb/h of Na_2SO_4 and $(5,000 - 1,500) = 3,500$ lb/h of H_2O . Thus, $(0.5)(1,500) = 750$ lb/h are to be crystallized. The decahydrate crystals include water of hydration in an amount given by ratioing molecular weights or

$$750 \left[\frac{(10)(18.016)}{(142.05)} \right] = 950 \text{ lb/h}$$

Thus, the total amount of decahydrate is $750 + 950 = 1,700$ lb/h. The water remaining in the mother liquor is $3,500 - 950 = 2,550$ lb/h. The composition of the mother liquor is $750/(2,550 + 750) (100\%) = 22.7$ wt% Na_2SO_4 . From Figure 4.24, the temperature corresponding to 22.7 wt% Na_2SO_4 on the solubility curve EC is 26°C .

The amount of crystals can be verified by applying the inverse-lever-arm rule, which gives $5,000 [(30 - 22.7)/(44.1 - 22.7)] = 1,700$ lb/h.

Liquid Adsorption

When a liquid mixture is brought into contact with a micro-porous solid, adsorption of certain components in the mixture takes place on the internal surface of the solid. The maximum extent of adsorption occurs when equilibrium is reached. The solid, which is essentially insoluble in the liquid, is the *adsorbent*. The component(s) being adsorbed are called *solutes* when in the liquid and constitute the *adsorbate* upon adsorption on the solid. In general, the higher the concentration of the solute, the higher is the equilibrium adsorbate concentration on the adsorbent. The component(s) of the liquid mixture other than the solute(s), that is, the solvent (carrier), are assumed not to adsorb.

No theory for predicting adsorption-equilibrium curves, based on molecular properties of the solute and solid, is universally embraced. Instead, laboratory experiments must

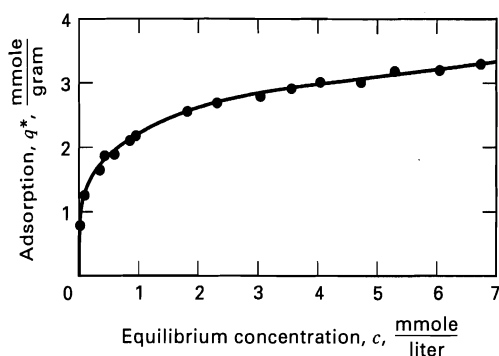


Figure 4.25 Adsorption isotherm for phenol from an aqueous solution in the presence of activated carbon at 20°C.

be performed at a fixed temperature for each liquid mixture and adsorbent to provide data for plotting curves, called *adsorption isotherms*. Figure 4.25, taken from the data of Fritz and Schluender [13], is an isotherm for the adsorption of phenol from an aqueous solution onto activated carbon at 20°C. Activated, powdered, or granular carbon is a microcrystalline, nongraphitic form of carbon that has a microporous structure to give it a very high internal surface area per unit mass of carbon, and therefore a high capacity for adsorption. Activated carbon preferentially adsorbs organic compounds rather than water when contacted with an aqueous phase containing dissolved organics. As shown in Figure 4.25, as the concentration of phenol in the aqueous phase is increased, the extent of adsorption increases very rapidly at first, followed by a much-slower increase. When the concentration of phenol is 1.0 mmol/L (0.001 mol/L of aqueous solution or 0.000001 mol/g of aqueous solution), the concentration of phenol on the activated carbon is somewhat more than 2.16 mmol/g (0.00216 mol/g of carbon or 0.203 g phenol/g of carbon). Thus, the affinity of this adsorbent for phenol is extremely high. The extent of adsorption depends markedly on the nature of the process used to produce the activated carbon. Adsorption isotherms like Figure 4.25 can be used to determine the amount of adsorbent required to selectively remove a given amount of solute from a liquid.

Consider the ideal, single-stage adsorption process of Figure 4.26, where A is the carrier liquid, B is the solute, and C is the solid adsorbent. Let

c_B = concentration of solute in the carrier liquid, mol/unit volume

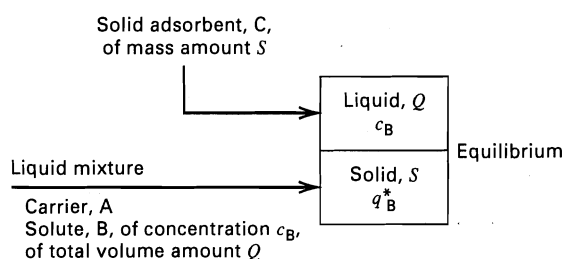


Figure 4.26 Equilibrium stage for liquid adsorption.

q_B = concentration of adsorbate, mol/unit mass of adsorbent

Q = volume of liquid (assumed to remain constant during adsorption)

S = mass of adsorbent (solute-free basis)

A material balance on the solute, assuming that the entering adsorbent is free of solute and that adsorption equilibrium is achieved, as designated by the asterisk superscript on q , gives

$$c_B^{(F)} Q = c_B Q = q_B^* S \quad (4-28)$$

This equation can be rearranged to the form of a straight line that can be plotted on the graph of an adsorption isotherm of the type in Figure 4.25, to obtain a graphical solution at equilibrium for c_B and q_B^* . Thus, solving (4-28) for q_B^* ,

$$q_B^* = -\frac{Q}{S} c_B + c_B^{(F)} \frac{Q}{S} \quad (4-29)$$

The intercept on the c_B axis is $c_B^{(F)} Q/S$, and slope is $-(Q/S)$. The intersection of (4-29) with the adsorption isotherm is the equilibrium condition, c_B and q_B^* .

Alternatively, an algebraic solution can be obtained. Adsorption isotherms for equilibrium-liquid adsorption of a species i can frequently be fitted with the empirical Freundlich equation, discussed in Chapter 15:

$$q_i^* = A c_i^{(1/n)} \quad (4-30)$$

where A and n depend on the solute, carrier, and particular adsorbent. The constant, n , is greater than 1, and A is a function of temperature. Freundlich developed his equation from experimental data on the adsorption on charcoal of organic solutes from aqueous solutions. Substitution of (4-30) into (4-29) gives

$$A c_B^{(1/n)} = -\frac{Q}{S} c_B + c_B^{(F)} \frac{Q}{S} \quad (4-31)$$

which is a nonlinear equation in c_B that can be solved numerically by an iterative method, as illustrated in the following example.

EXAMPLE 4.12

One liter of an aqueous solution containing 0.010 mol of phenol is brought to equilibrium at 20°C with 5 g of activated carbon having the adsorption isotherm shown in Figure 4.25. Determine the percent adsorption of the phenol and the equilibrium concentrations of phenol on carbon by:

(a) A graphical method

(b) A numerical algebraic method

For the latter case, the curve of Figure 4.25 is fitted quite well with the Freundlich equation (4-30), giving

$$q_B^* = 2.16 c_B^{(1/4.35)} \quad (1)$$

SOLUTION

From the data given, $c^{(F)} = 10$ mmol/L, $Q = 1$ L, and $S = 5$ g.

- (a) *Graphical method.* From (4-29), $q_B^* = -(\frac{1}{5})c_B + 10(\frac{1}{5}) = -0.2c_B + 2$

This equation, with a slope of -0.2 and an intercept of 2 , when plotted on Figure 4.25, yields an intersection with the equilibrium curve at $q_B^* = 1.9$ mmol/g and $c_B = 0.57$ mmol/liter. Thus, the percent adsorption of phenol is

$$\frac{c_B^{(F)} - c_B}{c_B^{(F)}} = \frac{10 - 0.57}{10} = 0.94 \quad \text{or} \quad 94\%$$

- (b) *Numerical algebraic method.* Applying Eq. (1) from the problem statement and (4-31),

$$2.16c_B^{0.23} = -0.2c_B + 2 \quad (2)$$

or

$$f\{c_B\} = 2.16c_B^{0.23} + 0.2c_B - 2 = 0 \quad (3)$$

This nonlinear equation for c_B can be solved by any of a number of iterative numerical techniques. For example, Newton's method [14] can be applied to Eq. (3) by using the iteration rule:

$$c_B^{(k+1)} = c_B^{(k)} - f^{(k)}\{c_B\}/f'^{(k)}\{c_B\} \quad (4)$$

where k is the iteration index. For this example, $f\{c_B\}$ is given by Eq. (3) and $f'\{c_B\}$ is obtained by differentiating Eq. (3) with respect to c_B to give

$$f'^{(k)}\{c_B\} = 0.497c_B^{-0.77} + 0.2$$

A convenient initial guess for c_B can be made by assuming almost 100% adsorption of phenol to give $q_B^* = 2$ mmol/g. Then, from (4-30),

$$c_B^{(0)} = (q_B^*/A)^n = (2/2.16)^{4.35} = 0.72 \text{ mmol/L}$$

where the (0) superscript designates the starting guess. The Newton iteration rule of Eq. (4) can now be applied, giving the following results:

k	$c_B^{(k)}$	$f^{(k)}\{c_B\}$	$f'^{(k)}\{c_B\}$	$c_B^{(k+1)}$
0	0.72	0.1468	0.8400	0.545
1	0.545	-0.0122	0.9928	0.558
2	0.558	-0.00009	0.9793	0.558

These results indicate convergence to $f\{c_B\} = 0$ for a value of $c_B = 0.558$ after only three iterations. From Eq. (1),

$$q_B^* = 2.16(0.558)^{(1/4.35)} = 1.89 \text{ mmol/g}$$

The result of the numerical method is within the accuracy of the graphical method.

4.8 GAS-LIQUID SYSTEMS

Vapor-liquid systems were covered in Sections 4.2, 4.3, and 4.4. There, the vapor was a mixture of species, most or all of which were condensable. Although the terms *vapor* and *gas* are often used interchangeably, the term *gas* is used to designate a mixture for which the temperature is above the critical temperatures of most or all of the species in the mixture. Thus, the components of a gas mixture are not easily

condensed to a liquid. In this section, the physical equilibrium of gas-liquid mixtures is considered.

Even though components of a gas mixture are at a temperature above critical, they can dissolve in an appropriate liquid solvent to an extent that depends on the temperature and their partial pressure in the gas mixture. With good mixing, equilibrium between the two phases can be achieved in a short time unless the liquid is very viscous.

Unlike equilibrium vapor-liquid mixtures, where, as discussed in Chapter 2, a number of theoretical relationships are in use for estimating K -values from molecular properties, no widely accepted theory exists for gas-liquid mixtures. Instead, experimental data, plots of experimental data, or empirical correlations are used.

Experimental solubility data for 13 common gases dissolved in water are plotted over a range of temperature from 0 to as high as 100°C in Figure 4.27. The ordinate is the

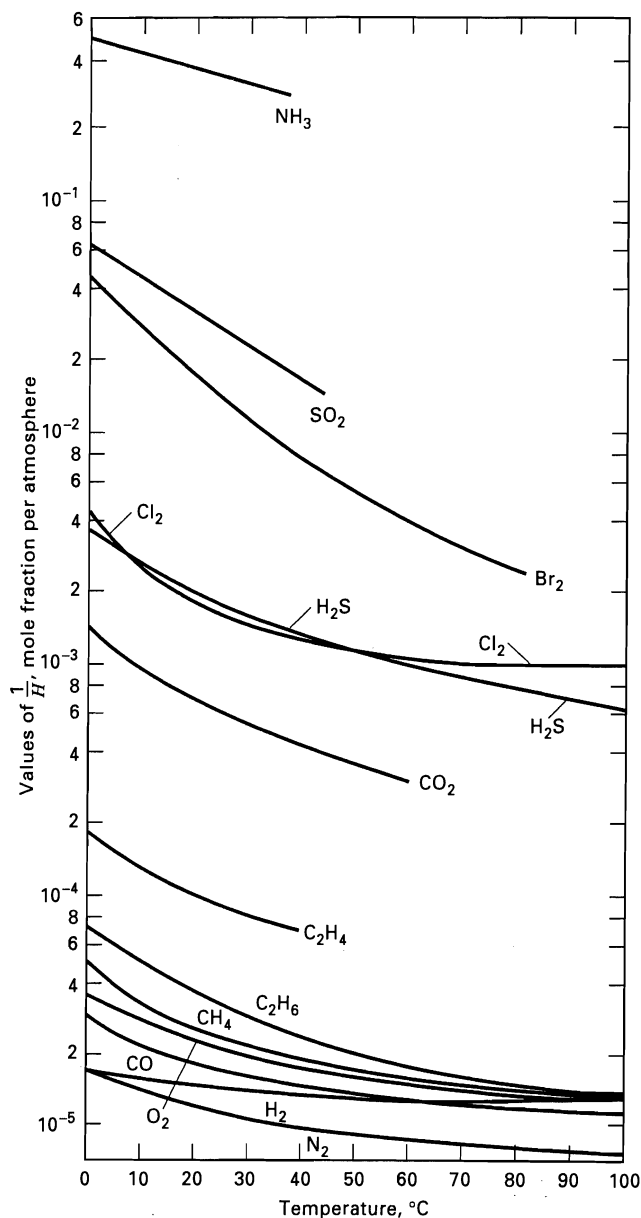


Figure 4.27 Henry's law constant for solubility of gases in water. [Adapted from O.A. Hougen, K.M. Watson, and R.A. Ragatz, *Chemical Process Principles. Part I*, 2nd ed., John Wiley and Sons, New York (1954).]

equilibrium mole fraction of the gas (solute) in the liquid when the pressure of the gas is 1 atm. The curves of Figure 4.27 can be used to estimate the solubility in water at other pressures and for mixtures of gases by applying Henry's law with the partial pressure of the solute, provided that mole-fraction solubilities are low and no chemical reactions occur among the gas species or with water. Henry's law, discussed briefly in Chapter 2 and given in Table 2.3, is rewritten for use with Figure 4.27 as

$$x_i = \left(\frac{1}{H_i} \right) y_i P \quad (4-32)$$

where H_i = Henry's law constant, atm.

For gases with a high solubility, such as ammonia, Henry's law may not be applicable, even at low partial pressures. In that case, experimental data for the actual conditions of pressure and temperature are necessary as in Example 4.14. In either case, calculations of equilibrium conditions are made in the manner illustrated in previous sections of this chapter by combining material balances with equilibrium relationships or data. The following two examples illustrate single-stage, gas-liquid equilibria calculation methods.

EXAMPLE 4.13

An ammonia plant, located at the base of a 300-ft (91.44-m)-high mountain, employs a unique absorption system for disposing of by-product CO_2 . The CO_2 is absorbed in water at a CO_2 partial pressure of 10 psi (68.8 kPa) above that required to lift water to the top of the mountain. The CO_2 is then vented to the atmosphere at the top of the mountain, the water being recirculated as shown in Figure 4.28. At 25°C, calculate the amount of water required to dispose of 1,000 ft³ (28.31 m³)(STP) of CO_2 .

SOLUTION

Basis: 1,000 ft³ (28.31 m³) of CO_2 at 0°C and 1 atm (STP). From Figure 4.27, the reciprocal of the Henry's law constant for CO_2 at 25°C is 6×10^{-4} mole fraction/atm. The CO_2 pressure in the absorber (at the foot of the mountain) is

$$p_{\text{CO}_2} = \frac{10}{14.7} + \frac{300 \text{ ft H}_2\text{O}}{34 \text{ ft H}_2\text{O/atm}} = 9.50 \text{ atm} = 960 \text{ kPa}$$

At this partial pressure, the equilibrium concentration of CO_2 in the water is

$$x_{\text{CO}_2} = 9.50(6 \times 10^{-4}) = 5.7 \times 10^{-3} \text{ mole fraction CO}_2 \text{ in water}$$

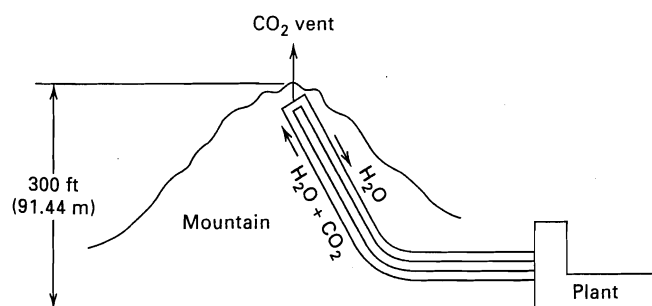


Figure 4.28 Flowsheet for Example 4.13.

The corresponding ratio of dissolved CO_2 to water is

$$\frac{5.7 \times 10^{-3}}{1 - 5.7 \times 10^{-3}} = 5.73 \times 10^{-3} \text{ mol CO}_2/\text{mol H}_2\text{O}$$

The total number of moles of CO_2 to be absorbed is

$$\frac{1,000 \text{ ft}^3}{359 \text{ ft}^3/\text{lbmol (at STP)}} = \frac{1,000}{359} = 2.79 \text{ lbmol}$$

or $(2.79)(44)(0.454) = 55.73 \text{ kg}$.

Assuming all the absorbed CO_2 is vented at the mountain top, the number of moles of water required is $2.79/(5.73 \times 10^{-3}) = 458 \text{ lbmol} = 8,730 \text{ lb} = 3,963 \text{ kg}$.

If one corrects for the fact that the pressure on top of the mountain is 101 kPa, so that not all of the CO_2 is vented, 4,446 kg (9,810 lb) of water are required.

EXAMPLE 4.14

The partial pressure of ammonia (A) in air-ammonia mixtures in equilibrium with their aqueous solutions at 20°C is given in Table 4.7. Using these data, and neglecting the vapor pressure of water and the solubility of air in water, construct an equilibrium diagram at 101 kPa using mole ratios $Y_A = \text{mol NH}_3/\text{mol air}$, and $X_A = \text{mol NH}_3/\text{mol H}_2\text{O}$ as coordinates. Henceforth, the subscript A is dropped. If 10 mol of gas, of composition $Y = 0.3$, are contacted with 10 mol of a solution of composition $X = 0.1$, what are the compositions of the resulting phases at equilibrium? The process is assumed to be isothermal and at atmospheric pressure.

SOLUTION

The equilibrium data given in Table 4.7 are recalculated in terms of mole ratios in Table 4.8 and plotted in Figure 4.29.

Mol NH_3 in entering gas = $10[Y/(1 + Y)] = 10(0.3/1.3) = 2.3$

Mol NH_3 in entering liquid = $10[X/(1 + X)] = 10(0.1/1.1) = 0.91$

Table 4.7 Partial Pressure of Ammonia over Ammonia-Water Solutions at 20°C

NH_3 Partial Pressure, kPa	g $\text{NH}_3/\text{g H}_2\text{O}$
4.23	0.05
9.28	0.10
15.2	0.15
22.1	0.20
30.3	0.25

Table 4.8 Y-X Data for Ammonia-Water, 20°C

Y , mol $\text{NH}_3/\text{mol Air}$	X , mol $\text{NH}_3/\text{mol H}_2\text{O}$
0.044	0.053
0.101	0.106
0.176	0.159
0.279	0.212
0.426	0.265

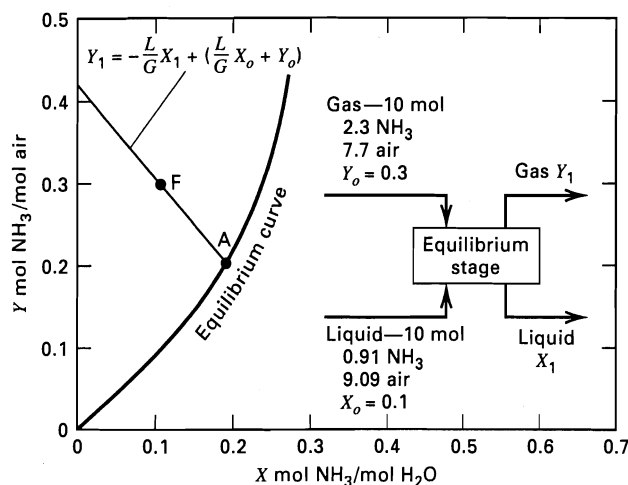


Figure 4.29 Equilibrium for air–NH₃–H₂O at 20°C, 1 atm, in Example 4.14.

A molar material balance for ammonia about the equilibrium stage is

$$GY_0 + LX_0 = GY_1 + LX_1 \quad (1)$$

where G = moles of air and L = moles of H₂O. Then $G = 10 - 2.3 = 7.7$ mol and $L = 10 - 0.91 = 9.09$ mol. Solving for Y_1 from Eq. (1),

$$Y_1 = -\frac{L}{G}X_1 + \left(\frac{L}{G}X_0 + Y_0\right) \quad (2)$$

This material-balance relationship is an equation of a straight line of slope $(L/G) = -9.09/7.7 = -1.19$, with an intercept of $(L/G)(X_0) + Y_0 = 0.42$.

The intersection of this material-balance line with the equilibrium curve, as shown in Figure 4.29, gives the ammonia composition of the gas and liquid phases leaving the stage as $Y_1 = 0.195$ and $X_1 = 0.19$. This result can be checked by an NH₃ balance, since the amount of NH₃ leaving is $(0.195)(7.70) + (0.19)(9.09) = 3.21$, which equals the total moles of NH₃ entering.

It is of importance to recognize that Eq. (2), the material balance line, called an *operating line* and discussed in great detail in Chapters 5 to 8, is the locus of all passing stream pairs; thus, X_0 , Y_0 (point F) also lies on this operating line.

4.9 GAS–SOLID SYSTEMS

Systems consisting of gas and solid phases that tend to equilibrium are involved in sublimation, desublimation, and adsorption separation operations.

Sublimation and Desublimation

In sublimation, a solid vaporizes into a gas phase without passing through a liquid state. In desublimation, one or more components (solutes) in the gas phase are condensed to a solid phase without passing through a liquid state. At low pressure both sublimation and desublimation are governed by the solid vapor pressure of the solute. Sublimation of the solid takes place when the partial pressure of the solute in the gas phase is less than the vapor pressure of the solid at

the system temperature. When the partial pressure of the solute in the gas phase exceeds the vapor pressure of the solid, desublimation occurs. At equilibrium, the vapor pressure of the species as a solid is equal to the partial pressure of the species as a solute in the gas phase. This is illustrated in the following example.

EXAMPLE 4.15

Ortho-xylene is partially oxidized in the vapor phase with air to produce phthalic anhydride, PA, in a catalytic reactor (fixed bed or fluidized bed) operating at about 370°C and 780 torr. However, a very large excess of air must be used to keep the xylene content of the reactor feed below 1 mol% to avoid an explosive mixture. In a typical plant, 8,000 lbmol/h of reactor effluent gas, containing 67 lbmol/h of PA and other amounts of N₂, O₂, CO, CO₂, and water vapor are cooled to separate the PA by desublimation to a solid at a total pressure of 770 torr. If the gas is cooled to 206°F, where the vapor pressure of solid PA is 1 torr, calculate the number of pounds of PA condensed per hour as a solid, and the percent recovery of PA from the gas if equilibrium is achieved. Assume that the xylene is converted completely to PA.

SOLUTION

At these conditions, only the PA condenses. At equilibrium, the partial pressure of PA is equal to the vapor pressure of solid PA, or 1 torr. Thus, the amount of PA in the cooled gas is given by Dalton's law of partial pressures:

$$(n_{PA})_G = \frac{P_{PA}}{P} n_G \quad (1)$$

where

$$n_G = (8,000 - 67) + (n_{PA})_G \quad (2)$$

and n = lbmol/h. Combining Eqs. (1) and (2),

$$\begin{aligned} (n_{PA})_G &= \frac{P_{PA}}{P} [(8,000 - 67) + (n_{PA})_G] \\ &= \frac{1}{770} [(8,000 - 67) + (n_{PA})_G] \end{aligned} \quad (3)$$

Solving this linear equation gives

$$(n_{PA})_G = 10.3 \text{ lbmol/h of PA}$$

The amount of PA desublimed is $67 - 10.3 = 56.7$ lbmol/h. The percent recovery of PA is $56.7/67 = 0.846$ or 84.6%. The amount of PA remaining in the gas is a very large quantity. In a typical plant, the gas is cooled to a much lower temperature, perhaps 140°F, where the vapor pressure of PA is less than 0.1 torr, bringing the recovery of PA to almost 99%.

Gas Adsorption

As with liquid mixtures, one or more components of a gas mixture can be adsorbed on the surface of a solid adsorbent. Data for a single solute can be represented by an adsorption isotherm of the type shown in Figure 4.25, or in similar diagrams, where the partial pressure of the solute in the gas is used in place of the concentration. However, when two components of a gas mixture are adsorbed and the purpose

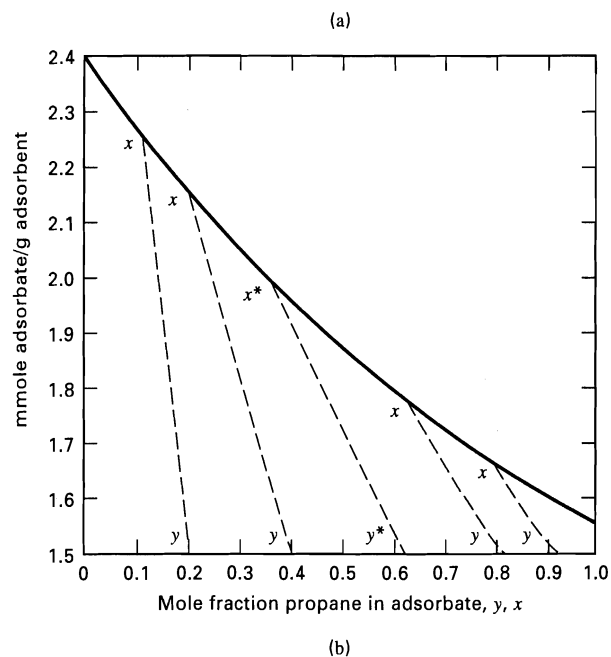
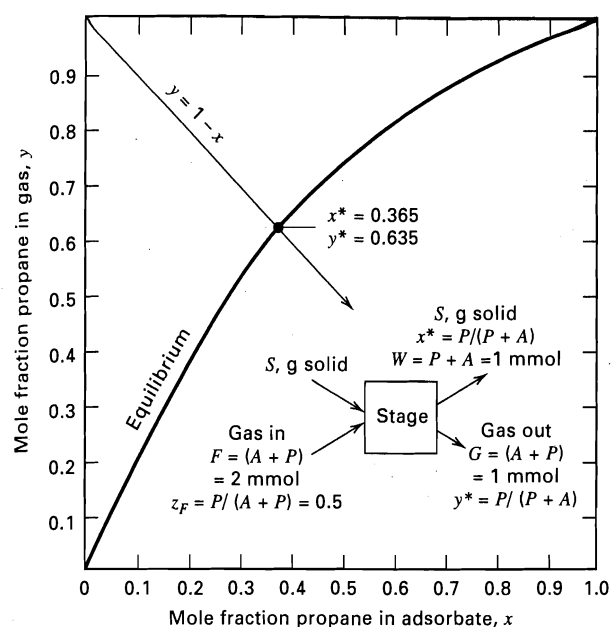


Figure 4.30 Adsorption equilibrium at 25°C and 101 kPa of propane and propylene on silica gel.

[Adapted from W.K. Lewis, E.R. Gilliland, B. Chertow, and W. H. Hoffman, *J. Am. Chem. Soc.*, **72**, 1153 (1950).]

of adsorption is to separate these two components, other methods of representing the experimental data may be preferred. One such representation is shown in Figure 4.30, from the data of Lewis et al. [15], for the adsorption of a propane (P)–propylene (A) gas mixture with silica gel at 25°C and 101 kPa. At 25°C, a pressure of at least 1,000 kPa is required to initiate condensation (dew point) of a mixture of propylene and propane. However, in the presence of silica gel, significant amounts of the gas are adsorbed at 101 kPa.

Figure 4.30a is similar to a binary vapor–liquid equilibrium plot of the type discussed in Section 4.2. For adsorption equilibria, the liquid-phase mole fraction is replaced by the

mole fraction in the adsorbate. For the propylene–propane mixture, propylene is adsorbed more strongly. For example, for an equimolar mixture in the gas phase, the adsorbate contains only 27 mol% propane. Figure 4.30b combines the data for the equilibrium mole fractions in the gas and adsorbate with the amount of adsorbate per unit of adsorbent. The mole fractions are obtained by reading the abscissa at the two ends of a tie line. For example, for equilibrium with $y_P = y^* = 0.50$, Figure 4.30b gives $x_P = x^* = 0.27$ and 2.08 mmol of adsorbate/g adsorbent. Therefore, $y_A = 0.50$, and $x_A = 0.73$. The separation factor analogous to the relative volatility for distillation is

$$(0.50/0.27)/(0.50/0.73) = 2.7$$

This value is much higher than the α -value for distillation, which, from Figure 2.8, at 25°C and 1,100 kPa is only 1.13. Accordingly, the separation of propylene and propane by adsorption has received some attention. Equilibrium calculations using data such as that shown in Figure 4.30 are made in the usual manner by combining such data with material-balance equations, as illustrated in the following example.

EXAMPLE 4.16

Propylene (A) and propane (P), are to be separated by preferential adsorption on silica gel (S) at 25°C and 101 kPa.

Two millimoles of a gas containing 50 mol% P and 50 mol% A is equilibrated with silica gel at 25°C and 101 kPa. Manometric measurements show that 1 mmol of gas is adsorbed. If the data of Figure 4.30 apply, what is the mole fraction of propane in the equilibrium gas and adsorbate, and how many grams of silica gel are used?

SOLUTION

A pictorial representation of the process is included in Figure 4.30a, where W = millimoles of adsorbate, G = millimoles of gas leaving, and z_F = mole fraction of propane in the feed.

The propane mole balance is

$$Fz_F = Wx^* + Gy^* \quad (1)$$

With $F = 2$, $z_F = 0.5$, $W = 1$, and $G = F - W = 1$, Eq. (1) becomes $1 = x^* + y^*$.

The operating (material-balance) line $y^* = 1 - x^*$ is the locus of all solutions of the material-balance equation, and is shown in Figure 4.30a. It intersects the equilibrium curve at $x^* = 0.365$, $y^* = 0.635$. From Figure 4.30b, at the point x^* , there must be 2.0 mmol adsorbate/g adsorbent; therefore there are $1.0/2 = 0.50$ g of silica gel in the system.

4.10 MULTIPHASE SYSTEMS

In previous sections of this chapter, only two phases were considered to be in equilibrium. In some applications of multiphase systems, three or more phases coexist. Figure 4.31

Air
<i>n</i> -hexane-rich liquid
Aniline-rich liquid
Water-rich liquid
Phosphorous liquid
Gallium liquid
Mercury liquid

Figure 4.31 Seven phases in equilibrium.

is a schematic diagram of a photograph of a laboratory curiosity taken from Hildebrand [16], which shows seven phases in equilibrium at near-ambient temperature. The phase on top is air, followed by six liquid phases in order of increasing density: hexane-rich, aniline-rich, water-rich, phosphorous, gallium, and mercury. Each phase contains all components in the seven-phase mixture, but the mole fractions in many cases are extremely small. For example, the aniline-rich phase contains on the order of 10 mol% *n*-hexane, 20 mol% water, but much less than 1 mol% each of dissolved air, phosphorous, gallium, and mercury. Note that even though the hexane-rich phase is not in direct contact with the water-rich phase, an equilibrium amount of water (approximately 0.06 mol%) is present in the hexane-rich phase because each phase is in equilibrium with each of the other phases, as attested by the equality of component fugacities:

$$f_i^{(1)} = f_i^{(2)} = f_i^{(3)} = f_i^{(4)} = f_i^{(5)} = f_i^{(6)} = f_i^{(7)}$$

More practical multiphase systems include the vapor-liquid-solid systems present in evaporative crystallization and pervaporation, and the vapor-liquid-liquid systems that occur when distilling certain mixtures of water and hydrocarbons or other organic chemicals having a limited solubility in water. Actually, all of the two-phase systems considered in the previous sections of this chapter involve a third phase, the containing vessel. However, the material of the container is selected on the basis of its inertness to and lack of solubility in the phases it contains, and therefore the material of the container does not normally enter into phase-equilibria calculations.

Although calculations of multiphase equilibrium are based on the same principles as for two-phase systems (material balances, energy balances, and phase-equilibria criteria such as equality of fugacity), the computations can be quite complex unless simplifying assumptions are made, in which case approximate results are obtained. Rigorous calculations are best made with a computer algorithm. In this section both types of calculations are illustrated.

Approximate Method for a Vapor-Liquid-Solid System

The simplest case of multiphase equilibrium is that encountered in an evaporative crystallizer involving crystallization of an inorganic compound, B, from its aqueous solution at its bubble point in the presence of its vapor. Assume that only two components are present, B and water. In that case, it is common to assume that B has no vapor pressure and water is not present in the solid phase. Thus, the vapor is pure water (steam), the liquid is a mixture of water and B, and the solid phase is pure B. Then, the solubility of B in the liquid phase is not influenced by the presence of the vapor, and the system pressure at a given temperature can be approximated by applying Raoult's law to the water in the liquid phase:

$$P = P_{\text{H}_2\text{O}}^s x_{\text{H}_2\text{O}} \quad (4-33)$$

where $x_{\text{H}_2\text{O}}$ can be obtained from the solubility of B.

EXAMPLE 4.17

A 5,000-lb batch of 20 wt% aqueous MgSO_4 solution is fed to a vacuum, evaporative crystallizer operating at 160°F. At this temperature, the stable solid phase is the monohydrate, with a MgSO_4 solubility of 36 wt%. If 75% of the water is evaporated, calculate:

- Pounds of water evaporated
- Pounds of monohydrate crystals, $\text{MgSO}_4 \cdot \text{H}_2\text{O}$
- Crystallizer pressure

SOLUTION

- The feed solution is $0.20(5,000) = 1,000$ lb MgSO_4 , and $5,000 - 1,000 = 4,000$ lb H_2O . The amount of water evaporated is $0.75(4,000) = 3,000$ lb H_2O .
- Let W = amount of MgSO_4 remaining in solution. Then MgSO_4 in the crystals = $1,000 - W$.
 MW of H_2O = 18 and MW of MgSO_4 = 120.4.
 Water of crystallization for the monohydrate
 $= (1,000 - W)(18/120.4) = 0.15(1,000 - W)$.
 Water remaining in solution
 $= 4,000 - 3,000 - 0.15(1,000 - W) = 850 + 0.15W$.
 Total amount of solution remaining
 $= 850 + 0.15W + W = 850 + 1.15W$.
 From the solubility of MgSO_4 ,

$$0.36 = \frac{W}{850 + 1.15W}$$

Solving: $W = 522$ pounds of dissolved MgSO_4 .
 MgSO_4 crystallized = $1,000 - 522 = 478$ lb.
 Water of crystallization = $0.15(1,000 - W)$
 $= 0.15(1,000 - 522) = 72$ lb.
 Total monohydrate crystals = $478 + 72 = 550$ lb.

- Crystallizer pressure is given by (4-33). At 160°F, the vapor pressure of H_2O is 4.74 psia. Then water remaining in solution = $(850 + 0.15W)/18 = 51.6$ lbmol.

MgSO_4 remaining in solution = $522/120.4 = 4.3$ lbmol.

Hence

$$x_{\text{H}_2\text{O}} = 51.6/(51.6 + 4.3) = 0.923$$

By Raoult's law, $p_{\text{H}_2\text{O}} = P = 4.74(0.923) = 4.38$ psia

Approximate Method for a Vapor-Liquid-Liquid System

Another case suitable for an approximate method is that of a mixture containing water and hydrocarbons (HCs), at conditions such that a vapor phase and two liquid phases, HC-rich (1) and water-rich (2) coexist. Often the solubilities of water in the liquid HC phase and HCs in the water phase are less than 0.1 mol% and may be neglected. In that case, if the liquid HC phase obeys Raoult's law, the total pressure of the system is given by the sum of the pressures exhibited by the separate phases:

$$P = P_{\text{H}_2\text{O}}^s + \sum_{\text{HCs}} P_i^s x_i^{(1)} \quad (4-34)$$

For more general cases, at low pressures where the vapor phase is ideal but the liquid HC phase may be nonideal,

$$P = P_{\text{H}_2\text{O}}^s + P \sum_{\text{HCs}} K_i x_i^{(1)} \quad (4-35)$$

which can be rearranged to

$$P = \frac{P_{\text{H}_2\text{O}}^s}{1 - \sum_{\text{HCs}} K_i x_i^{(1)}} \quad (4-36)$$

Equations (4-34) and (4-36) can be used directly to estimate the pressure for a given temperature and liquid-phase composition or iteratively to estimate the temperature for a given pressure. An important aspect of the calculation is the determination of the particular phases present from all six possible cases, namely, V , $V-L^{(1)}$, $V-L^{(1)}-L^{(2)}$, $V-L^{(2)}$, $L^{(1)}-L^{(2)}$, and L . It is not always obvious how many and which phases may be present. Indeed, if a $V-L^{(1)}-L^{(2)}$ solution to a problem exists, almost always $V-L^{(1)}$ and $V-L^{(2)}$ solutions also exist. In that case, the three-phase solution is the correct one. It is important, therefore, to seek the three-phase solution first.

EXAMPLE 4.18

A mixture of 1,000 kmol of 75 mol% water and 25 mol% n -octane is cooled under equilibrium conditions at a constant pressure 133.3 kPa (1,000 torr) from an initial temperature of 136°C to a final temperature of 25°C. Determine:

- The initial phase condition
- The temperature, phase amounts, and compositions when each phase change occurs

Assume that water and n -octane are immiscible liquids. The vapor pressure of octane is included in Figure 2.4.

SOLUTION

- (a) Initial phase conditions are $T = 136^\circ\text{C} = 276.8^\circ\text{F}$ and $P = 133.3$ kPa = 19.34 psia. Vapor pressures at 276.8°F and $P_{\text{H}_2\text{O}}^s = 46.7$ psia and $P_{n\text{C}_8}^s = 19.5$ psia. Because the initial pressure is less than the vapor pressure of each component, the initial phase condition is all vapor, with partial pressures

$$p_{\text{H}_2\text{O}} = y_{\text{H}_2\text{O}} P = 0.75(19.34) = 14.5 \text{ psia}$$

$$p_{n\text{C}_8} = y_{n\text{C}_8} P = 0.25(19.34) = 4.8 \text{ psia}$$

- (b) As the temperature is decreased, the first phase change occurs when a temperature is reached where either $P_{\text{H}_2\text{O}}^s = p_{\text{H}_2\text{O}} = 14.5$ psia or $P_{n\text{C}_8}^s = p_{n\text{C}_8} = 4.8$ psia. The corresponding temperatures where these vapor pressures occur are 211°F for H_2O and 194°F for $n\text{C}_8$. The highest temperature applies. Therefore, water condenses first when the temperature reaches 211°F . This is the dew-point temperature of the initial mixture at the system pressure. As the temperature is further reduced, the number of moles of water in the vapor decreases, causing the partial pressure of water to decrease below 14.5 psia and the partial pressure of $n\text{C}_8$ to increase above 4.8 psia. Thus, $n\text{C}_8$ begins to condense, forming a second liquid phase, at a temperature higher than 194°F but lower than 211°F . This temperature, referred to as the *secondary dew point*, must be determined iteratively. The calculation is simplified if the bubble point of the mixture is computed first.

From (4-34),

$$P = 19.34 \text{ psi} = P_{\text{H}_2\text{O}}^s + P_{n\text{C}_8}^s \quad (1)$$

Thus, a temperature is sought, as follows, to cause Eq. (1) to be satisfied:

$T, ^\circ\text{F}$	$P_{\text{H}_2\text{O}}^s, \text{psia}$	$P_{n\text{C}_8}^s, \text{psia}$	P, psia
194	10.17	4.8	14.97
202	12.01	5.6	17.61
206	13.03	6.1	19.13
207	13.30	6.2	19.50

By linear interpolation, $T = 206.7^\circ\text{F}$ for $P = 19.34$ psia. Below this temperature, the vapor phase disappears and only two immiscible liquid phases are present.

To determine the temperature at which one of the liquid phases disappears, which is the same condition as when the second liquid phase begins to appear (secondary dew point), it is noted for this case, with only pure water and a pure HC present, that vaporization starting from the bubble point is at a constant temperature until one of the two liquid phases is completely vaporized. Thus, the secondary dew-point temperature is the same as the bubble-point temperature or 206.7°F . At the secondary dew point, the partial pressures are $p_{\text{H}_2\text{O}} = 13.20$ psia and $p_{n\text{C}_8} = 6.14$ psia, with all of the $n\text{C}_8$ in the vapor phase. Therefore, the phase amounts and compositions are

Component	Vapor		H ₂ O-Rich Liquid
	kmol	y	kmol
H ₂ O	53.9	0.683	21.1
$n\text{C}_8$	25.0	0.317	0.0
	78.9	1.000	21.1

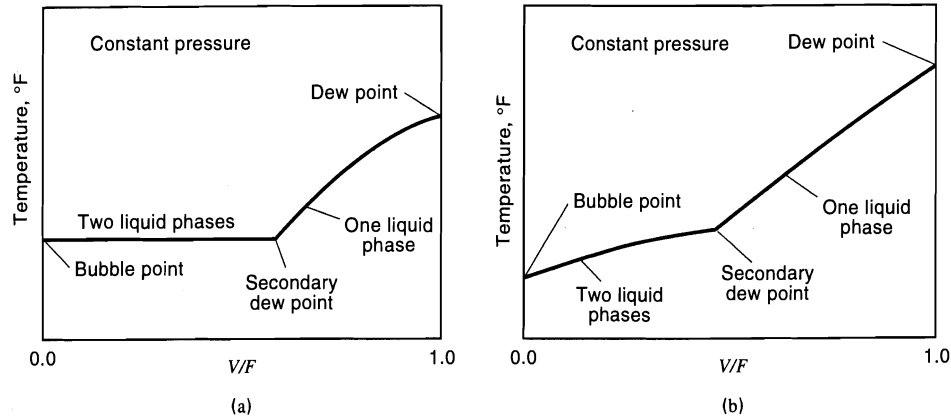


Figure 4.32 Typical flash curves for immiscible liquid mixtures of water and hydrocarbons at constant pressure: (a) only one hydrocarbon species present; (b) more than one hydrocarbon species present.

If desired, additional flash calculations can be made for conditions between the dew point and secondary dew point. The resulting flash curve is Figure 4.32a. If more than one HC species is present, the liq-

uid HC phase does not evaporate at a constant composition and the secondary dew-point temperature is higher than the bubble-point temperature. In that case, the flash is described by Figure 4.32b.

Rigorous Method for a Vapor-Liquid-Liquid System

The rigorous method for treating a vapor-liquid-liquid system at a given temperature and pressure is called a *three-phase isothermal flash*. As first presented by Henley and Rosen [17], it is analogous to the isothermal two-phase flash algorithm developed in Section 4.4. The system is shown schematically in Figure 4.33. The usual material balances and phase-equilibrium relations apply for each component:

$$Fz_i = Vy_i + L^{(1)}x_i^{(1)} + L^{(2)}x_i^{(2)} \quad (4-37)$$

$$K_i^{(1)} = y_i/x_i^{(1)} \quad (4-38)$$

$$K_i^{(2)} = y_i/x_i^{(2)} \quad (4-39)$$

Alternatively, the following relation can be substituted for (4-38) and (4-39):

$$K_{Di} = x_i^{(1)}/x_i^{(2)} \quad (4-40)$$

These equations can be solved by a modification of the Rachford-Rice procedure if we let $\Psi = V/F$ and $\xi = L^{(1)}/(L^{(1)} + L^{(2)})$, where $0 \leq \Psi \leq 1$ and $0 \leq \xi \leq 1$. By combining

(4-37), (4-38), and (4-39) with

$$\sum x_i^{(1)} - \sum y_i = 0 \quad (4-41)$$

and

$$\sum x_i^{(1)} - \sum x_i^{(2)} = 0 \quad (4-42)$$

to eliminate y_i , $x_i^{(1)}$, and $x_i^{(2)}$, two simultaneous equations in Ψ and ξ are obtained:

$$\sum_i \frac{z_i(1 - K_i^{(1)})}{\xi(1 - \Psi) + (1 - \Psi)(1 - \xi)K_i^{(1)}/K_i^{(2)} + \Psi K_i^{(1)}} = 0 \quad (4-43)$$

and

$$\sum_i \frac{z_i(1 - K_i^{(1)}/K_i^{(2)})}{\xi(1 - \Psi) + (1 - \Psi)(1 - \xi)K_i^{(1)}/K_i^{(2)} + \Psi K_i^{(1)}} = 0 \quad (4-44)$$

Values of Ψ and ξ are computed by solving the nonlinear equations (4-43) and (4-44) simultaneously by an appropriate numerical method such as that of Newton. Then the amounts and compositions of the three phases are determined from

$$V = \Psi F \quad (4-45)$$

$$L^{(1)} = \xi(F - V) \quad (4-46)$$

$$L^{(2)} = F - V - L^{(1)} \quad (4-47)$$

$$y_i = \frac{z_i}{\xi(1 - \Psi)/K_i^{(1)} + (1 - \Psi)(1 - \xi)/K_i^{(2)} + \Psi} \quad (4-48)$$

$$x_i^{(1)} = \frac{z_i}{\xi(1 - \Psi) + (1 - \Psi)(1 - \xi)(K_i^{(1)}/K_i^{(2)}) + \Psi K_i^{(1)}} \quad (4-49)$$

$$x_i^{(2)} = \frac{z_i}{\xi(1 - \Psi)(K_i^{(2)}/K_i^{(1)}) + (1 - \Psi)(1 - \xi) + \Psi K_i^{(2)}} \quad (4-50)$$

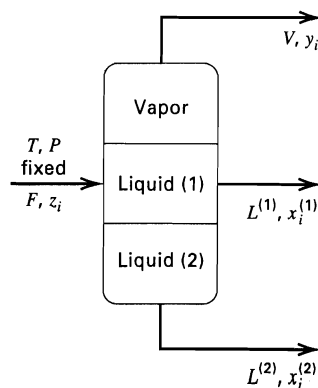


Figure 4.33 Conditions for a three-phase isothermal flash.

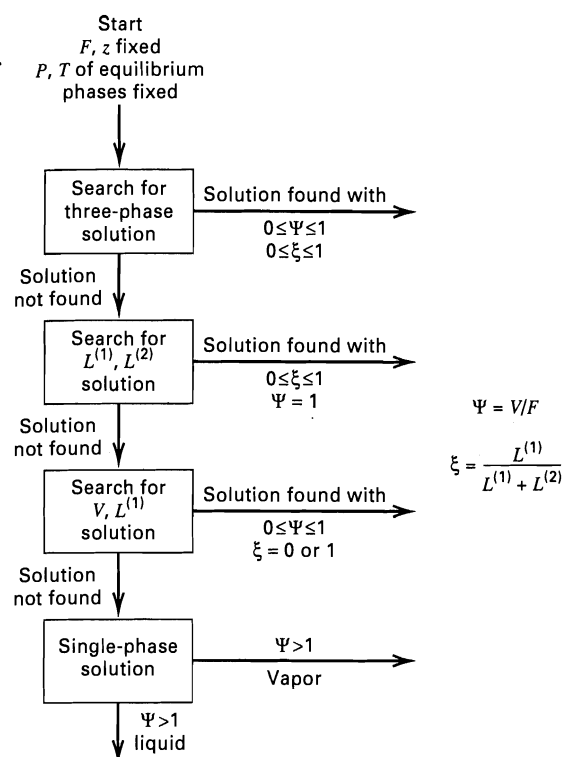


Figure 4.34 Algorithm for an isothermal three-phase flash.

Calculations for an isothermal three-phase flash are difficult and tedious because of the strong dependency of K -values on liquid-phase compositions when two immiscible liquid phases are present. In addition, it is usually not obvious that three phases will be present, and calculations may be necessary for other combinations of phases. A typical algorithm for determining the phase conditions is shown in Figure 4.34. Because of the complexity of the isothermal three-phase flash algorithm, calculations are best made with a steady-state, process-simulation computer program. Such programs can also perform adiabatic or nonadiabatic three-phase flashes by iterating on temperature until the enthalpy balance,

$$h_F F + Q = h_V V + h_{L(1)} L^{(1)} + h_{L(2)} L^{(2)} = 0 \quad (4-51)$$

is satisfied.

SUMMARY

1. The phase rule of Gibbs, which applies to intensive variables at equilibrium, determines the number of independent variables that can be specified. This rule can be extended to the more general determination of the degrees of freedom (number of allowable specifications) for a flow system, including consideration of extensive variables. The intensive and extensive variables are related by material and energy balance equations together with phase equilibrium data in the form of equations, tables, and/or graphs.
2. Vapor-liquid equilibrium conditions for binary systems are conveniently represented and determined with T - y - x , y - x , and P - x diagrams. The relative volatility for a binary system tends to 1.0 as the critical point is approached.

EXAMPLE 4.19

In a process for producing styrene from toluene and methanol, the gaseous reactor effluent is as follows:

Component	kmol/h
Hydrogen	350
Methanol	107
Water	491
Toluene	107
Ethylbenzene	141
Styrene	350

If this stream is brought to equilibrium at 38°C and 300 kPa, compute the amounts and compositions of the phases present.

SOLUTION

Because water, hydrocarbons, and a light gas are present in the mixture, the possibility exists that a vapor and two liquid phases may be present, with the methanol being distributed among all three phases. The isothermal three-phase flash module of the ChemCAD simulation program was used with Henry's law for hydrogen and the UNIFAC method for estimating liquid-phase activity coefficients for the other components, to obtain the following results:

Component	kmol/h		
	V	$L^{(1)}$	$L^{(2)}$
Hydrogen	349.96	0.02	0.02
Methanol	9.54	14.28	83.18
Water	7.25	8.12	475.63
Toluene	1.50	105.44	0.06
Ethylbenzene	0.76	140.20	0.04
Styrene	1.22	348.64	0.14
Totals	370.23	616.70	559.07

As would be expected, little of the hydrogen is dissolved in either of the two liquid phases. Little of the other components is left uncondensed. The water-rich liquid phase contains little of the hydrocarbons, but much of the methanol. The organic-rich phase contains most of the hydrocarbons and small amounts of water and methanol.

Additional calculations at 300 kPa indicate that the organic phase condenses first with dew point = 143°C and secondary dew point = 106°C.

3. Minimum- or maximum-boiling azeotropes, which are formed by close-boiling, nonideal liquid mixtures, are conveniently represented by the same types of diagrams used for nonazeotropic (zeotropic) binary mixtures. Highly nonideal liquid mixtures can form heterogeneous azeotropes involving two liquid phases.
4. For multicomponent mixtures, vapor-liquid equilibrium-phase compositions and amounts can be determined by isothermal-flash, adiabatic-flash, and bubble- and dew-point calculations. When the mixtures are nonideal, the computations are best done with process-simulation computer programs.
5. Liquid-liquid equilibrium conditions for ternary mixtures are best determined graphically from triangular and other equilibrium

diagrams, unless only one of the three components (called the solute) is soluble in the two liquid phases and the system is dilute in the solute. In that case, the conditions can be readily determined algebraically using phase-distribution ratios for the solute.

6. Liquid-liquid equilibrium conditions for multicomponent mixtures of four or more components are best determined with process-simulation computer programs, particularly when the system is not dilute with respect to the solute(s).

7. Solid-liquid equilibrium commonly occurs in leaching, crystallization, and adsorption. Leaching calculations commonly assume that the solute is completely dissolved in the solvent and that the remaining solid leaving in the underflow is accompanied by a known fraction of liquid. Crystallization calculations are best made with a solid-liquid phase equilibrium diagram. For crystallization of inorganic salts from an aqueous solution, formation of hydrates must be considered. Equilibrium adsorption

can be represented algebraically or graphically by adsorption isotherms.

8. Solubility of gases that are only sparingly soluble in a liquid are well represented by a Henry's law constant that depends on temperature.

9. Solid vapor pressure can be used to determine equilibrium sublimation and desublimation conditions for gas-solid systems. Adsorption isotherms and y - x diagrams are useful in determining adsorption-equilibrium conditions for gas mixtures in the presence of a solid adsorbent.

10. Calculations of equilibrium when more than two phases are present are best made with computer simulation programs. However, approximate manual procedures are readily applied to vapor-liquid-solid systems when no component is found in all three phases and for vapor-liquid-liquid systems when only one component distributes in all three phases.

REFERENCES

1. PERRY, R.H., D.W. GREEN, and J.O. MALONEY, Eds., *Perry's Chemical Engineers' Handbook*, 7th ed., McGraw-Hill, New York, Section 13 (1997).
2. GMEHLING, J., and U. ONKEN, *Vapor-Liquid Equilibrium Data Collection*, DECHEMA Chemistry Data Series, 1-8 (1977-1984).
3. KEYES, D.B., *Ind. Eng. Chem.*, **21**, 998-1001 (1929).
4. HUGHES, R.R., H.D. EVANS, and C.V. STERNLING, *Chem. Eng. Progr.*, **49**, 78-87 (1953).
5. RACHFORD, H.H., JR., and J.D. RICE, *J. Pet. Tech.*, **4** (10), Section 1, p. 19, and Section 2, p. 3 (Oct. 1952).
6. PRESS, W.H., S.A. TEUKOLSKY, W.T. VETTERLING, and B.P. FLANNERY, *Numerical Recipes in FORTRAN*, 2nd ed., Cambridge University Press, Cambridge, chap. 9 (1992).
7. GOFF, G.H., P.S. FARRINGTON, and B.H. SAGE, *Ind. Eng. Chem.*, **42**, 735-743 (1950).
8. CONSTANTINIDES, A., and N. MOSTOUFI, *Numerical Methods for Chemical Engineers with MATLAB Applications*, Prentice Hall PTR, Upper Saddle River, NJ (1999).
9. ROBBINS, L.A., in R.H. PERRY, D.H. GREEN, and J.O. MALONEY, Eds., *Perry's Chemical Engineers' Handbook*, 7th ed., McGraw-Hill, New York, pp. 15-10 to 15-15 (1997).
10. JANECKE, E., *Z. Anorg. Allg. Chem.*, **51**, 132-157 (1906).
11. FRANCIS, A.W., *Liquid-Liquid Equilibria*, Interscience, New York (1963).
12. FINDLAY, A., *Phase Rule*, Dover, New York (1951).
13. FRITZ, W., and E.-U. SCHULUENDER, *Chem. Eng. Sci.*, **29**, 1279-1282 (1974).
14. FELDER, R.M., and R.W. ROUSSEAU, *Elementary Principles of Chemical Processes*, 3rd ed., John Wiley and Sons, New York, pp. 613-616 (1986).
15. LEWIS, W.K., E.R. GILLILAND, B. CHERTON, and W.H. HOFFMAN, *J. Am. Chem. Soc.*, **72**, 1153-1157 (1950).
16. HILDEBRAND, J.H., *Principles of Chemistry*, 4th ed., Macmillan, New York (1940).
17. HENLEY, E.J., and E.M. ROSEN, *Material and Energy Balance Computations*, John Wiley and Sons, New York, pp. 351-353 (1969).
18. CONWAY, J.B., and J.J. NORTON, *Ind. Eng. Chem.*, **43**, 1433-1435 (1951).

EXERCISES

Section 4.1

4.1 Consider the equilibrium stage shown in Figure 4.35. Conduct a degrees-of-freedom analysis by performing the following steps:

- (a) List and count the variables.
- (b) Write and count the equations relating the variables.
- (c) Calculate the degrees of freedom.
- (d) List a reasonable set of design variables.

4.2 Can the following problems be solved uniquely?

- (a) The feed streams to an adiabatic equilibrium stage consist of liquid and vapor streams of known composition, flow rate, temperature, and pressure. Given the stage (outlet) temperature and pressure, calculate the composition and amounts of equilibrium vapor and liquid leaving the stage.

- (b) The same as part (a), except that the stage is not adiabatic.

- (c) A multicomponent vapor of known temperature, pressure, and composition is to be partially condensed in a condenser. The outlet pressure of the condenser and the inlet cooling water temperature are fixed. Calculate the cooling water required.

4.3 Consider an adiabatic equilibrium flash. The variables are all as indicated in Figure 4.36.

- (a) Determine the number of variables.
- (b) Write all the independent equations that relate the variables.
- (c) Determine the number of equations.
- (d) Determine the number of degrees of freedom.
- (e) What variables would you prefer to specify in order to solve a typical adiabatic flash problem?

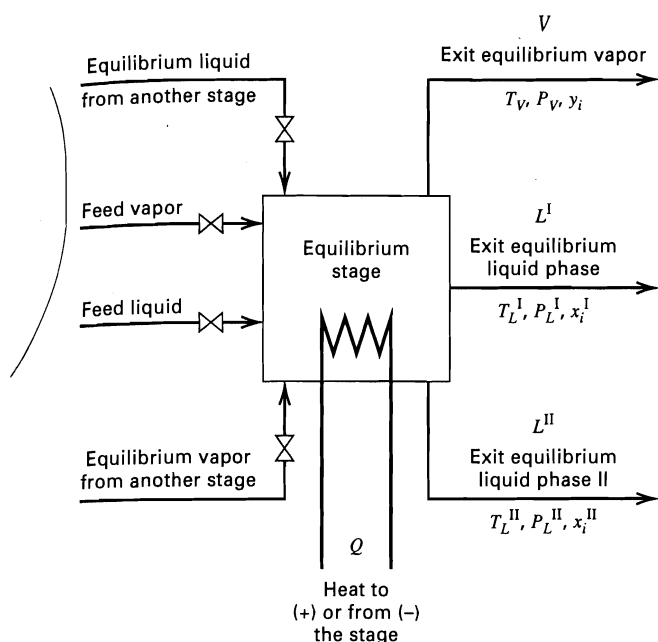


Figure 4.35 Conditions for Exercise 4.1.

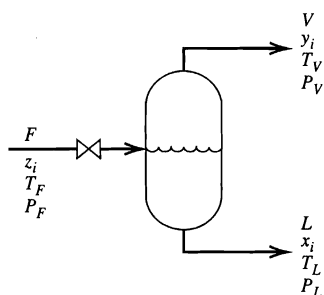


Figure 4.36 Conditions for Exercise 4.3.

4.4 Determine the number of degrees of freedom for a non-adiabatic equilibrium flash for one liquid feed, one vapor stream product, and two immiscible liquid stream products as shown in Figure 4.33.

4.5 Consider the seven-phase equilibrium system shown in Figure 4.31. Assume that air consists of N_2 , O_2 , and argon. How many degrees of freedom are computed by the Gibbs phase rule? What variables might be specified to fix the system?

Section 4.2

4.6 A liquid mixture containing 25 mol% benzene and 75 mol% ethyl alcohol, in which components are miscible in all proportions, is heated at a constant pressure of 1 atm (101.3 kPa, 760 torr) from a temperature of 60°C to 90°C. Using the following T - x - y experimental data, perform calculations to determine the answers to parts (a) through (f).

EXPERIMENTAL T - x - y DATA FOR BENZENE-ETHYL ALCOHOL AT 1 ATM

Temperature, °C:	78.4	77.5	75	72.5	70	68.5	67.7	68.5	72.5	75	77.5	80.1
Mole percent benzene in vapor:	0	7.5	28	42	54	60	68	73	82	88	95	100
Mole percent benzene in liquid:	0	1.5	5	12	22	31	68	81	91	95	98	100

- At what temperature does vaporization begin?
- What is the composition of the first bubble of equilibrium vapor formed?
- What is the composition of the residual liquid when 25 mol% has evaporated? Assume that all vapor formed is retained within the apparatus and that it is completely mixed and in equilibrium with the residual liquid.
- Repeat part (c) for 90 mol% vaporized.
- Repeat part (d) if, after 25 mol% is vaporized as in part (c), the vapor formed is removed and an additional 35 mol% is vaporized by the same technique used in part (c).
- Plot the temperature versus the percent vaporized for parts (c) and (e).
- Use the following vapor pressure data in conjunction with Raoult's and Dalton's laws to construct a T - x - y diagram, and compare it for the answers obtained in parts (a) and (f) with those obtained using the experimental T - x - y data. What do you conclude about the applicability of Raoult's law to this binary system?

VAPOR PRESSURE DATA

Vapor pressure, torr:	20	40	60	100	200	400	760
Ethanol, °C:	8	19.0	26.0	34.9	48.4	63.5	78.4
Benzene, °C:	-2.6	7.6	15.4	26.1	42.2	60.6	80.1

4.7 Stearic acid is to be steam distilled at 200°C in a direct-fired still, heat-jacketed to prevent condensation. Steam is introduced into the molten acid in small bubbles, and the acid in the vapor leaving the still has a partial pressure equal to 70% of the vapor pressure of pure stearic acid at 200°C. Plot the kilograms of acid distilled per kilogram of steam added as a function of total pressure from 101.3 kPa down to 3.3 kPa at 200°C. The vapor pressure of stearic acid at 200°C is 0.40 kPa.

4.8 The relative volatility, α , of benzene to toluene at 1 atm is 2.5. Construct an x - y diagram for this system at 1 atm. Repeat the construction using vapor pressure data for benzene from Exercise 4.6 and for toluene from the following table in conjunction with Raoult's and Dalton's laws. Also construct a T - x - y diagram.

- A liquid containing 70 mol% benzene and 30 mol% toluene is heated in a container at 1 atm until 25 mol% of the original liquid is evaporated. Determine the temperature. The phases are then separated mechanically, and the vapors condensed. Determine the composition of the condensed vapor and the liquid residue.
- Calculate and plot the K -values as a function of temperature at 1 atm.

VAPOR PRESSURE OF TOLUENE

Vapor pressure, torr:	20	40	60	100	200	400	760	1,520
Temperature, °C:	18.4	31.8	40.3	51.9	69.5	89.5	110.6	136

4.9 The vapor pressure of toluene is given in Exercise 4.8, and that of n -heptane is given in the accompanying table.

VAPOR PRESSURE OF *n*-HEPTANE

Vapor pressure, torr:							
20	40	60	100	200	400	760	1,520
Temperature, °C:							
9.5	22.3	30.6	41.8	58.7	78.0	98.4	124

- Plot an x - y equilibrium diagram for this system at 1 atm by using Raoult's and Dalton's laws.
- Plot the T - x bubble-point curve at 1 atm.
- Plot α and K -values versus temperature.
- Repeat part (a) using the arithmetic average value of α , calculated from the two extreme values.
- Compare your x - y and T - x - y diagrams with the following experimental data of Steinhauser and White [*Ind. Eng. Chem.*, **41**, 2912 (1949)].

VAPOR-LIQUID EQUILIBRIUM DATA FOR *n*-HEPTANE/TOLUENE AT 1 ATM

$x_{n\text{-heptane}}$	$y_{n\text{-heptane}}$	$T, ^\circ\text{C}$
0.025	0.048	110.75
0.129	0.205	106.80
0.250	0.349	104.50
0.354	0.454	102.95
0.497	0.577	101.35
0.692	0.742	99.73
0.843	0.864	98.90
0.940	0.948	98.50
0.994	0.993	98.35

4.10 Saturated-liquid feed, of $F = 40$ mol/h, containing 50 mol% A and B is supplied continuously to the apparatus shown in Figure 4.37. The condensate from the condenser is split so that half of it is returned to the still pot.

- If heat is supplied at such a rate that $W = 30$ mol/h and $\alpha = 2$, as subsequently defined, what will be the composition of the overhead and the bottoms product?
- If the operation is changed so that no condensate is returned to the still pot and $W = 3D$ as before, what will be the composition of the products?

$$\alpha = \frac{P_A^s}{P_B^s} = \frac{y_A x_B}{y_B x_A}$$

4.11 It is required to design a fractionation tower to operate at 101.3 kPa to obtain a distillate consisting of 95 mol% acetone (A) and 5 mol% water, and a residue containing 1 mol% A. The feed

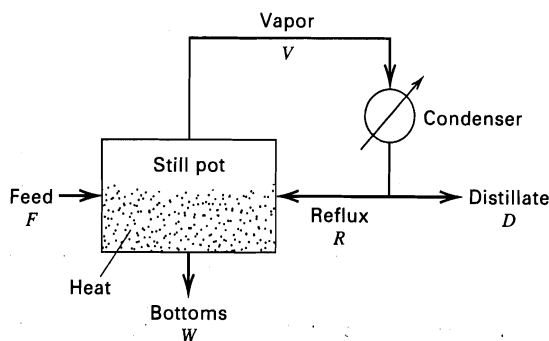


Figure 4.37 Conditions for Exercise 4.10.

liquid is at 125°C and 687 kPa and contains 57 mol% A. The feed is introduced to the column through an expansion valve so that it enters the column partially vaporized at 60°C. From the data below, determine the molar ratio of liquid to vapor in the partially vaporized feed. Enthalpy and equilibrium data are as follows:

Molar latent heat of A = 29,750 kJ/kmol (constant)

Molar latent heat of H₂O = 42,430 kJ/kmol (constant)

Molar specific heat of A = 134 kJ/kmol-K (constant)

Molar specific heat of H₂O = 75.3 kJ/kmol-K (constant)

Enthalpy of high-pressure, hot feed before adiabatic expansion = 0

Enthalpies of feed phases after expansion: $h_V = 27,200$ kJ/kmol, $h_L = -5,270$ kJ/kmol

VAPOR-LIQUID EQUILIBRIUM DATA FOR ACETONE-H₂O AT 101.3 kPa

	$T, ^\circ\text{C}$						
	56.7	57.1	60.0	61.0	63.0	71.7	100
Mol% A in liquid:	100	92.0	50.0	33.0	17.6	6.8	0
Mol% A in vapor:	100	94.4	85.0	83.7	80.5	69.2	0

4.12 Using vapor pressure data from Exercises 4.6 and 4.8 and the enthalpy data provided below:

- Construct an h - x - y diagram for the benzene-toluene system at 1 atm (101.3 kPa) based on the use of Raoult's and Dalton's laws.

$T, ^\circ\text{C}$	Saturated Enthalpy, kJ/kg		h_L	h_V
	Benzene	Toluene		
60	79	487	77	471
80	116	511	114	495
100	153	537	151	521

- Calculate the energy required for 50 mol% vaporization of a 30 mol% liquid solution of benzene in toluene, initially at saturation temperature. If the vapor is then condensed, what is the heat load on the condenser in kJ/kg of solution if the condensate is saturated and if it is subcooled by 10°C?

Section 4.3

4.13 Vapor-liquid equilibrium data at 101.3 kPa are given for the chloroform-methanol system on p. 13-11 of *Perry's Chemical Engineers' Handbook*, 6th ed. From these data, prepare plots like Figures 4.6b and 4.6c. From the plots, determine the azeotropic composition and temperature at 101.3 kPa. Is the azeotrope of the minimum- or maximum-boiling type?

4.14 Vapor-liquid equilibrium data at 101.3 kPa are given for the water-formic acid system on p. 13-14 of *Perry's Chemical Engineers' Handbook*, 6th ed. From these data, prepare plots like Figures 4.7b and 4.7c. From the plots, determine the azeotropic composition and temperature at 101.3 kPa. Is the azeotrope of the minimum- or maximum-boiling type?

4.15 Vapor-liquid equilibrium data for mixtures of water and isopropanol at 1 atm (101.3 kPa, 760 torr) are given below.

- Prepare T - x - y and x - y diagrams.

(b) When a solution containing 40 mol% isopropanol is slowly vaporized, what will be the composition of the initial vapor formed?

(c) If this same 40% mixture is heated under equilibrium conditions until 75 mol% has been vaporized, what will be the compositions of the vapor and liquid produced?

VAPOR-LIQUID EQUILIBRIUM FOR ISOPROPANOL AND WATER AT 1 ATM

T, °C	Mol% Isopropanol	
	Liquid	Vapor
93.00	1.18	21.95
89.75	3.22	32.41
84.02	8.41	46.20
83.85	9.10	47.06
82.12	19.78	52.42
81.64	28.68	53.44
81.25	34.96	55.16
80.62	45.25	59.26
80.32	60.30	64.22
80.16	67.94	68.21
80.21	68.10	68.26
80.28	76.93	74.21
80.66	85.67	82.70
81.51	94.42	91.60

Notes:

Composition of the azeotrope: $x = y = 68.54\%$.

Boiling point of azeotrope: 80.22°C.

Boiling point of pure isopropanol: 82.5°C.

(d) Calculate K -values and values of α at 80°C and 89°C.

(e) Compare your answers in parts (a), (b), and (c) to those obtained from T - x - y and x - y diagrams based on the following vapor pressure data and Raoult's and Dalton's laws.

What do you conclude about the applicability of Raoult's law to this system?

Vapor Pressures of Isopropanol and Water

Vapor pressure, torr	200	400	760
Isopropanol, °C	53.0	67.8	82.5
Water, °C	66.5	83	100

Section 4.4

4.16 Using the y - x and T - y - x diagrams in Figures 4.3 and 4.4, determine the temperature, amounts, and compositions of the equilibrium vapor and liquid phases at 101 kPa for the following conditions with a 100-kmol mixture of nC_6 (H) and nC_8 (C).

(a) $z_H = 0.5$, $\Psi = V/F = 0.2$

(b) $z_H = 0.4$, $y_H = 0.6$

(c) $z_H = 0.6$, $x_C = 0.7$

(d) $z_H = 0.5$, $\Psi = 0$

(e) $z_H = 0.5$, $\Psi = 1.0$

(f) $z_H = 0.5$, $T = 200^\circ F$

4.17 For a binary mixture of components 1 and 2, show that the equilibrium phase compositions and amounts can be computed directly from the following reduced forms of Eqs. (5), (6), and (3)

of Table 4.4.

$$x_1 = (1 - K_2)/(K_1 - K_2)$$

$$x_2 = 1 - x_1$$

$$y_1 = (K_1 K_2 - K_1)/(K_2 - K_1)$$

$$y_2 = 1 - y_1$$

$$\Psi = \frac{V}{F} = \frac{z_1[(K_1 - K_2)/(1 - K_2)] - 1}{K_1 - 1}$$

4.18 Consider the Rachford-Rice form of the flash equation,

$$\sum_{i=1}^C \frac{z_i(1 - K_i)}{1 + (V/F)(K_i - 1)} = 0$$

Under what conditions can this equation be satisfied?

4.19 A liquid containing 60 mol% toluene and 40 mol% benzene is continuously distilled in a single-equilibrium-stage unit at atmospheric pressure. What percent of benzene in the feed leaves in the vapor if 90% of the toluene entering in the feed leaves in the liquid? Assume a relative volatility of 2.3 and obtain the solution graphically.

4.20 Solve Exercise 4.19 by assuming an ideal solution and using vapor pressure data from Figure 2.4. Also determine the temperature.

4.21 A seven-component mixture is flashed at a specified temperature and pressure.

(a) Using the K -values and feed composition given below, make a plot of the Rachford-Rice flash function

$$f(\Psi) = \sum_{i=1}^C \frac{z_i(1 - K_i)}{1 + \Psi(K_i - 1)}$$

at intervals of Ψ of 0.1, and from the plot estimate the correct root of Ψ .

(b) An alternative form of the flash function is

$$f(\Psi) = \sum_{i=1}^C \frac{z_i K_i}{1 + \Psi(K_i - 1)} - 1$$

Make a plot of this equation also at intervals of Ψ of 0.1 and explain why the Rachford-Rice function is preferred.

Component	z_i	K_i
1	0.0079	16.2
2	0.1321	5.2
3	0.0849	2.6
4	0.2690	1.98
5	0.0589	0.91
6	0.1321	0.72
7	0.3151	0.28

4.22 One hundred kilomoles of a feed composed of 25 mol% n -butane, 40 mol% n -pentane, and 35 mol% n -hexane are flashed at steady-state conditions. If 80% of the hexane is to be recovered in the liquid at 240°F, what pressure is required, and what are the liquid and vapor compositions? Obtain K -values from Figure 2.8.

4.23 An equimolar mixture of ethane, propane, n -butane, and n -pentane is subjected to a flash vaporization at 150°F and 205 psia. What are the expected amounts and compositions of the liquid and vapor products? Is it possible to recover 70% of the ethane in the

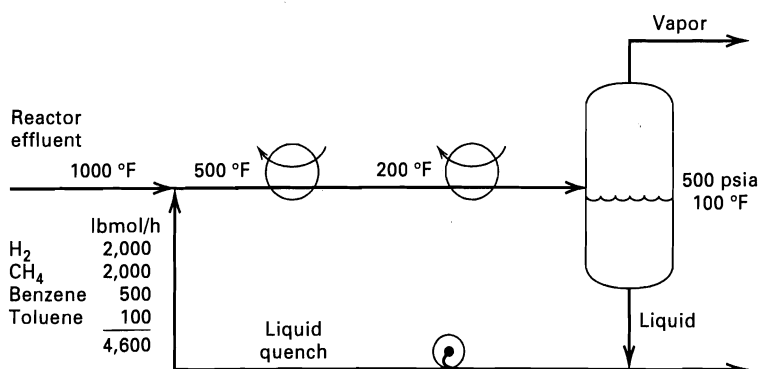


Figure 4.38 Conditions for Exercise 4.24.

vapor by a single-stage flash at other conditions without losing more than 5% of nC_4 to the vapor? Obtain K -values from Figure 2.8.

4.24 The system shown in Figure 4.38 is used to cool the reactor effluent and separate the light gases from the heavier hydrocarbons. K -values for the components at 500 psia and 100°F are

Component	K_i
H ₂	80
CH ₄	10
Benzene	0.010
Toluene	0.004

(a) Calculate the composition and flow rate of the vapor leaving the flash drum.

(b) Does the flow rate of liquid quench influence the result? Prove your answer analytically.

4.25 The mixture shown in Figure 4.39 is partially condensed and separated into two phases. Calculate the amounts and compositions of the equilibrium phases, V and L .

4.26 The following stream is at 200 psia and 200°F. Determine whether it is a subcooled liquid or a superheated vapor, or whether it is partially vaporized, without making a flash calculation.

Component	lbmol/h	K -value
C ₃	125	2.056
nC_4	200	0.925
nC_5	175	0.520

4.27 The overhead system for a distillation column is shown in Figure 4.40. The composition of the total distillates is indicated, with 10 mol% of it being taken as vapor. Determine the pressure in the reflux drum, if the temperature is 100°F. Use the following K -values by assuming that K is inversely proportional to pressure.

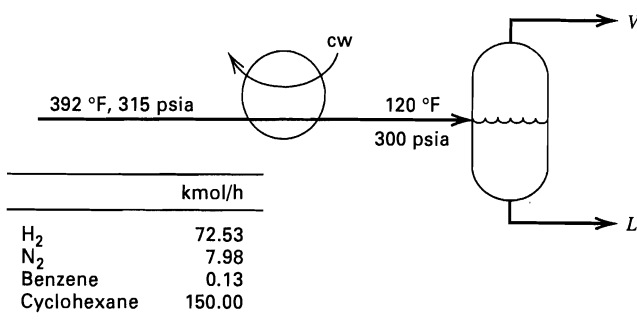


Figure 4.39 Conditions for Exercise 4.25.

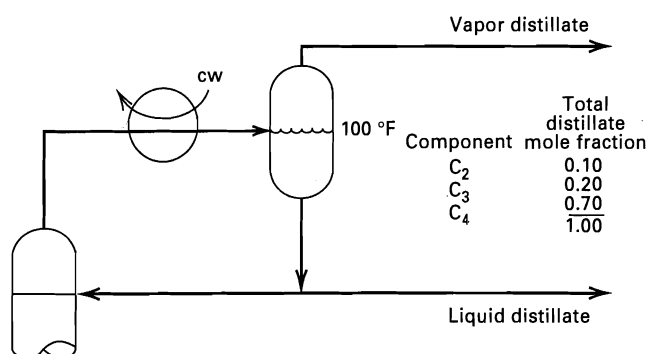


Figure 4.40 Conditions for Exercise 4.27.

Component K at 100°F, 200 psia

C ₂	2.7
C ₃	0.95
C ₄	0.34

4.28 Determine the phase condition of a stream having the following composition at 7.2°C and 2,620 kPa.

Component	kmol/h
N ₂	1.0
C ₁	124.0
C ₂	87.6
C ₃	161.6
nC_4	176.2
nC_5	58.5
nC_6	33.7

Perform the calculations with a computer simulation program using at least three different options for K -values. Does the choice of K -value method influence the results?

4.29 A liquid mixture consisting of 100 kmol of 60 mol% benzene, 25 mol% toluene, and 15 mol% *o*-xylene is flashed at 1 atm and 100°C.

(a) Compute the amounts of liquid and vapor products and their composition.

(b) Repeat the calculation at 100°C and 2 atm.

(c) Repeat the calculation at 105°C and 0.1 atm.

(d) Repeat the calculation at 150°C and 1 atm.

Assume ideal solutions and use the vapor pressure curves of Figure 2.4 for benzene and toluene. For *o*-xylene, draw a vapor pressure line that goes through the points (100.2°C, 200 torr) and (144°C, 760 torr).

4.30 Prove that the vapor leaving an equilibrium flash is at its dew point and that the liquid leaving an equilibrium flash is at its bubble point.

4.31 The following mixture is introduced into a distillation column as saturated liquid at 1.72 MPa. Calculate the bubble-point temperature using the K -values of Figure 2.8.

Compound	kmol/h
Ethane	1.5
Propane	10.0
<i>n</i> -Butane	18.5
<i>n</i> -Pentane	17.5
<i>n</i> -Hexane	3.5

4.32 An equimolar solution of benzene and toluene is totally evaporated at a constant temperature of 90°C. What are the pressures at the beginning and end of the vaporization process? Assume an ideal solution and use the vapor pressure curves of Figure 2.4.

4.33 The following equations are given by Sebastiani and Lacquaniti [*Chem. Eng. Sci.*, **22**, 1155 (1967)] for the liquid-phase activity coefficients of the water (W)–acetic acid (A) system.

$$\log \gamma_W = x_A^2 [A + B(4x_W - 1) + C(x_W - x_A)(6x_W - 1)]$$

$$\log \gamma_A = x_W^2 [A + B(4x_W - 3) + C(x_W - x_A)(6x_W - 5)]$$

$$A = 0.1182 + \frac{64.24}{T(K)}$$

$$B = 0.1735 - \frac{43.27}{T(K)}$$

$$C = 0.1081$$

Find the dew point and bubble point of a mixture of composition $x_W = 0.5$, $x_A = 0.5$ at 1 atm. Flash the mixture at a temperature halfway between the dew point and the bubble point.

4.34 Find the bubble-point and dew-point temperatures of a mixture of 0.4 mole fraction toluene (1) and 0.6 mole fraction *n*-butanol (2) at 101.3 kPa. The K -values can be calculated from (2-72), the modified Raoult's law, using vapor-pressure data, and γ_1 and γ_2 from the van Laar equation of Table 2.9 with $A_{12} = 0.855$ and $A_{21} = 1.306$. If the same mixture is flashed at a temperature midway between the bubble point and dew point, and 101.3 kPa, what fraction is vaporized, and what are the compositions of the two phases?

4.35 (a) For a liquid solution having a molar composition of ethyl acetate (A) of 80% and ethyl alcohol (E) of 20%, calculate the bubble-point temperature at 101.3 kPa and the composition of the corresponding vapor using (2-72) with vapor pressure data and the van Laar equation of Table 2.9 with $A_{AE} = 0.855$, $A_{EA} = 0.753$.

(b) Find the dew point of the mixture.

(c) Does the mixture form an azeotrope? If so, predict the temperature and composition.

4.36 A binary solution at 107°C contains 50 mol% water (W) and 50 mol% formic acid (F). Using (2-72) with vapor pressure data and the van Laar equation of Table 2.9 with $A_{WF} = -0.2935$ and $A_{FW} = -0.2757$, compute:

(a) The bubble-point pressure.

(b) The dew-point pressure.

Also determine whether the mixture forms a maximum- or minimum-boiling azeotrope. If so, predict the azeotropic pressure at 107°C and the azeotropic composition.

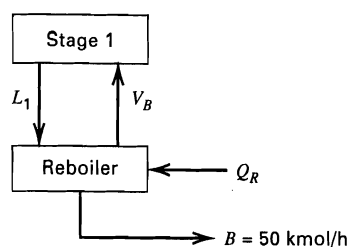


Figure 4.41 Conditions for Exercise 4.38.

4.37 For a mixture consisting of 45 mol% *n*-hexane, 25 mol% *n*-heptane, and 30 mol% *n*-octane at 1 atm, use a simulation computer program to:

(a) Find the bubble- and dew-point temperatures.

(b) Find the flash temperature, and the compositions and relative amounts of the liquid and vapor products if the mixture is subjected to a flash distillation at 1 atm so that 50 mol% of the feed is vaporized.

(c) Find how much of the octane is taken off as vapor if 90% of the hexane is taken off as vapor.

Repeat parts (a) and (b) at 5 atm and 0.5 atm.

4.38 In Figure 4.41, 150 kmol/h of a saturated liquid, L_1 , at 758 kPa, of molar composition, propane 10%, *n*-butane 40%, and *n*-pentane 50%, enters the reboiler from stage 1. What are the compositions and amounts of V_B and B ? What is Q_R , the reboiler duty? Use a simulation computer program to find the answers.

4.39 (a) Find the bubble-point temperature of the following mixture at 50 psia, using K -values from Figure 2.8 or Figure 2.9.

Component	z_i
Methane	0.005
Ethane	0.595
<i>n</i> -Butane	0.400

(b) Find the temperature that results in 25% vaporization at this pressure. Determine the corresponding liquid and vapor compositions.

4.40 As shown in Figure 4.42, a hydrocarbon mixture is heated and expanded before entering a distillation column. Calculate, using a simulation computer program, the mole percent vapor phase and vapor and liquid phase mole fractions at each of the three locations indicated by a pressure specification.

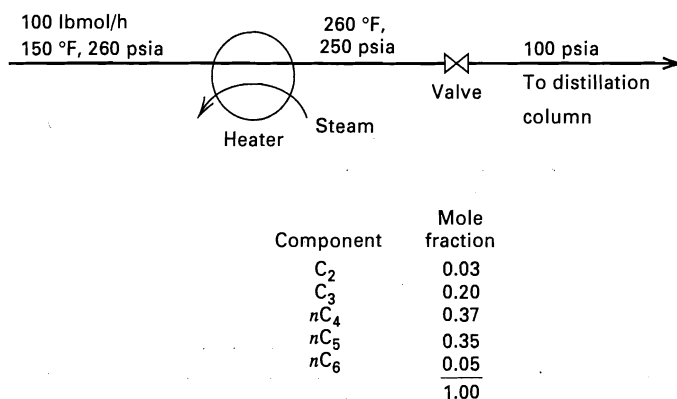


Figure 4.42 Conditions for Exercise 4.40.

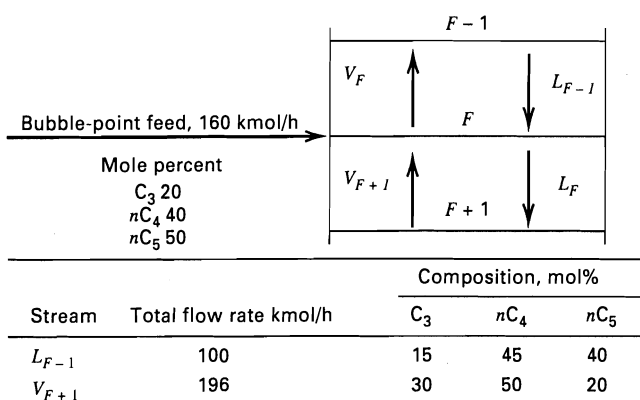


Figure 4.43 Conditions for Exercise 4.41.

4.41 Streams entering stage F of a distillation column are shown in Figure 4.43. What is the temperature of stage F and the compositions and amounts of streams V_F and L_F if the pressure is 785 kPa for all streams? Use a simulation computer program to obtain the answers.

4.42 Flash adiabatically, across a valve, a stream composed of the six hydrocarbons given below. The feed upstream of the valve is at 250°F and 500 psia. The pressure downstream of the valve is 300 psia.

Component	z_i
C ₂ H ₄	0.02
C ₂ H ₆	0.03
C ₃ H ₆	0.05
C ₃ H ₈	0.10
iC ₄	0.20
nC ₄	0.60

Compute using a simulation computer program:

- The phase condition upstream of the valve.
- The temperature downstream of the valve.
- The molar fraction vaporized downstream of the valve.
- The mole fraction compositions of the vapor and liquid phases downstream of the valve.

4.43 Propose a detailed algorithm like Figure 4.19a and Table 4.4 for a flash where the percent vaporized and the flash pressure are to be specified.

4.44 Determine algorithms for carrying out the following flash calculations, assuming that expressions for K -values and enthalpies are available.

Given	Find
h_F, P	Ψ, T
h_F, T	Ψ, P
h_F, Ψ	T, P
Ψ, T	h_F, P
Ψ, P	h_F, T
T, P	h_F, Ψ

Section 4.5

4.45 A feed of 13,500 kg/h consists of 8 wt% acetic acid (B) in water (A). The removal of the acetic acid is to be accomplished by liquid-liquid extraction at 25°C. The raffinate is to contain only

1 wt% acetic acid. The following four solvents, with accompanying distribution coefficients in mass-fraction units, are being considered. Water and each solvent (C) can be considered immiscible. For each solvent, estimate the kilograms required per hour if a single equilibrium stage is used.

Solvent	K_D
Methyl acetate	1.273
Isopropyl ether	0.429
Heptadecanol	0.312
Chloroform	0.178

4.46 Forty-five kilograms of a solution containing 30 wt% ethylene glycol in water is to be extracted with furfural. Using Figures 4.14a and 4.14e, calculate:

- The minimum quantity of solvent.
- The maximum quantity of solvent.
- The weights of solvent-free extract and raffinate for 45 kg solvent, and the percent glycol extracted.
- The maximum possible purity of glycol in the finished extract and the maximum purity of water in the raffinate for one equilibrium stage.

4.47 Prove that, in a triangular diagram, where each vertex represents a pure component, the composition of the system at any point inside the triangle is proportional to the length of the respective perpendicular drawn from the point to the side of the triangle opposite the vertex in question. It is not necessary to assume a special case (i.e., a right or equilateral triangle).

4.48 A mixture of chloroform (CHCl_3) and acetic acid at 18°C and 1 atm (101.3 kPa) is to be extracted with water to recover the acid.

- Forty-five kilograms of a mixture containing 35 wt% CHCl_3 and 65 wt% acid is treated with 22.75 kg of water at 18°C in a simple one-stage batch extraction. What are the compositions and weights of the raffinate and extract layers produced?
- If the raffinate layer from the above treatment is extracted again with one-half its weight of water, what will be the compositions and weights of the new layers?
- If all the water is removed from this final raffinate layer, what will its composition be?

Solve this exercise using the following equilibrium data to construct one or more of the types of diagrams in Figure 4.14.

LIQUID-LIQUID EQUILIBRIUM DATA FOR $\text{CHCl}_3\text{-H}_2\text{O-CH}_3\text{COOH}$ AT 18°C AND 1 ATM

Heavy Phase (wt%)			Light Phase (wt%)		
CHCl_3	H_2O	CH_3COOH	CHCl_3	H_2O	CH_3COOH
99.01	0.99	0.00	0.84	99.16	0.00
91.85	1.38	6.77	1.21	73.69	25.10
80.00	2.28	17.72	7.30	48.58	44.12
70.13	4.12	25.75	15.11	34.71	50.18
67.15	5.20	27.65	18.33	31.11	50.56
59.99	7.93	32.08	25.20	25.39	49.41
55.81	9.58	34.61	28.85	23.28	47.87

4.49 Isopropyl ether (E) is used to separate acetic acid (A) from water (W). The liquid-liquid equilibrium data at 25°C and 1 atm

(101.3 kPa) are presented below.

(a) One hundred kilograms of a 30 wt% A–W solution is contacted with 120 kg of ether in an equilibrium stage. What are the compositions and weights of the resulting extract and raffinate? What would be the concentration of acid in the (ether-rich) extract if all the ether were removed?

(b) A mixture containing 52 kg A and 48 kg W is contacted with 40 kg of E. What are the extract and raffinate compositions and quantities?

LIQUID-LIQUID EQUILIBRIUM DATA FOR ACETIC ACID (A), WATER (W), AND ISOPROPANOL ETHER (E) AT 25°C AND 1 ATM

Water-Rich Layer			Ether-Rich Layer		
Wt% A	Wt% W	Wt% E	Wt% A	Wt% W	Wt% E
1.41	97.1	1.49	0.37	0.73	98.9
2.89	95.5	1.61	0.79	0.81	98.4
6.42	91.7	1.88	1.93	0.97	97.1
13.30	84.4	2.3	4.82	1.88	93.3
25.50	71.1	3.4	11.4	3.9	84.7
36.70	58.9	4.4	21.6	6.9	71.5
45.30	45.1	9.6	31.1	10.8	58.1
46.40	37.1	16.5	36.2	15.1	48.7

Section 4.6

4.50 Diethylene glycol (DEG) is used as a solvent in the UDEX liquid–liquid extraction process [H.W. Grote, *Chem Eng. Progr.*, **54** (8), 43 (1958)] to separate paraffins from aromatics. If 280 lbmol/h of 42.86 mol% *n*-hexane, 28.57 mol% *n*-heptane, 17.86 mol% benzene, and 10.71 mol% toluene is contacted with 500 lbmol/h of 90 mol% aqueous DEG at 325°F and 300 psia, calculate, using a simulation computer program and the UNIFAC L/L method for estimating liquid-phase activity coefficients, the flow rates and molar compositions of the resulting two liquid phases. Is DEG more selective for the paraffins or the aromatics?

4.51 A feed of 110 lbmol/h includes 5, 3, and 2 lbmol/h, respectively, of formic acid, acetic acid, and propionic acid in water. If the acids are extracted in a single equilibrium stage with 100 lbmol/h of ethyl acetate (EA), calculate with a simulation computer program using the UNIFAC method, flow rates and molar compositions of the resulting two liquid phases. What is the order of selectivity of EA for the three organic acids?

Section 4.7

4.52 Repeat Example 4.9 for 200,000 kg/h of hexane.

4.53 Water is to be used in a single equilibrium stage to dissolve 1,350 kg/h of Na_2CO_3 from 3,750 kg/h of a solid, where the balance is an insoluble oxide. If 4,000 kg/h of water is used and the underflow from the stage is 40 wt% solvent on a solute-free basis, compute the flow rates and compositions of the overflow and the underflow.

4.54 Repeat Exercise 4.53 if the residence time is only sufficient to leach 80% of the carbonate.

4.55 A total of 6,000 lb/h of a liquid solution of 40 wt% benzene in naphthalene at 50°C is cooled to 15°C. Assuming that equilibrium is achieved, use Figure 4.23 to determine the amount of crystals formed, and the flow rate and composition of the mother liquor. Are the crystals benzene or naphthalene?

4.56 Repeat Example 4.10, except determine the temperature necessary to crystallize 80% of the naphthalene.

4.57 A total of 10,000 kg/h of a 10 wt% liquid solution of naphthalene in benzene is cooled from 30°C to 0°C. Assuming that equilibrium is achieved, determine the amount of crystals formed and the composition and flow rate of the mother liquid. Are the crystals benzene or naphthalene? Use Figure 4.23.

4.58 Repeat Example 4.11, except let the original solution be 20 wt% Na_2SO_4 .

4.59 At 20°C, 1,000 kg of a mixture of 50 wt% $\text{Na}_2\text{SO}_4 \cdot 10\text{H}_2\text{O}$ and 50 wt% Na_2SO_4 crystals exists. How many kilograms of water must be added to just completely dissolve the crystals if the temperature is kept at 20°C and equilibrium is maintained? Use Figure 4.24.

4.60 Repeat Example 4.12, except determine the grams of activated carbon to achieve:

(a) 75% adsorption of phenol.

(b) 90% adsorption of phenol.

(c) 98% adsorption of phenol.

4.61 A colored substance (B) is to be removed from a mineral oil by adsorption with clay particles at 25°C. The original oil has a color index of 200 units/100 kg oil, while the decolorized oil must have an index of only 20 units/100 kg oil. The following experimental adsorption equilibrium data have been measured in a laboratory:

c_B , color units/ 100 kg oil	200	100	60	40	10
q_B , color units/ 100 kg clay	10	7.0	5.4	4.4	2.2

(a) Fit the data to the Freundlich equation.

(b) Compute the kilograms of clay needed to treat 500 kg of oil if one equilibrium contact is used.

Section 4.8

4.62 Vapor–liquid equilibrium data in mole fractions for the system acetone–air–water at 1 atm (101.3 kPa) are as follows:

y , acetone in air:	0.004	0.008	0.014	0.017	0.019	0.020
x , acetone in water:	0.002	0.004	0.006	0.008	0.010	0.012

(a) Plot the data as (1) a graph of moles acetone per mole air versus moles acetone per mole water, (2) partial pressure of acetone versus g acetone per g water, and (3) y versus x .

(b) If 20 moles of gas containing 0.015 mole fraction acetone is brought into contact with 15 moles of water in an equilibrium stage, what would be the composition of the discharge streams? Solve graphically. For both parts, neglect partitioning of water and air.

4.63 It has been proposed that oxygen be separated from nitrogen by absorbing and desorbing air in water. Pressures from 101.3 to 10,130 kPa and temperatures between 0 and 100°C are to be used.

(a) Devise a workable scheme for doing the separation assuming the air is 79 mol% N_2 and 21 mol% O_2 .

(b) Henry's law constants for O_2 and N_2 are given in Figure 4.27. How many batch absorption steps would be necessary to make 90 mol% pure oxygen? What yield of oxygen (based on total amount of oxygen feed) would be obtained?

4.64 A vapor mixture having equal volumes of NH_3 and N_2 is to be contacted at 20°C and 1 atm (760 torr) with water to absorb a portion of the NH_3 . If 14 m^3 of this mixture is brought into contact with 10 m^3 of water and if equilibrium is attained, calculate the percent of the ammonia originally in the gas that will be absorbed. Both temperature and total pressure will be maintained constant during the absorption. The partial pressure of NH_3 over water at 20°C is as follows:

Partial Pressure of NH_3 in Air, torr	Grams of Dissolved $\text{NH}_3/100\text{ g of H}_2\text{O}$
470	40
298	30
227	25
166	20
114	15
69.6	10
50.0	7.5
31.7	5.0
24.9	4.0
18.2	3.0
15.0	2.5
12.0	2.0

Section 4.9

4.65 Repeat Example 4.15 for temperatures corresponding to the following vapor pressures for solid PA:

- 0.7 torr
- 0.4 torr
- 0.1 torr

Plot the percent recovery of PA versus the solid vapor pressure for the range from 0.1 torr to 1.0 torr.

4.66 Nitrogen at 760 torr and 300°C contains 10 mol% anthraquinone (A). If this gas is cooled to 200°C , calculate the percent desublimation of A. Vapor pressure data for solid A are as follows:

T, $^\circ\text{C}$:	190.0	234.2	264.3	285.0
Vapor pressure, torr:	1	10	40	100

These data can be fitted to the Antoine equation (2-39) using the first three constants.

4.67 At 25°C and 101 kPa, 2 mol of a gas containing 35 mol% propylene in propane is equilibrated with 0.1 kg of silica gel adsorbent. Using the equilibrium data of Figure 4.30, calculate the moles and composition of the gas adsorbed and the equilibrium composition of the gas not adsorbed.

4.68 A gas containing 50 mol% propylene in propane is to be separated with silica gel having the equilibrium properties shown in Figure 4.30. The final products are to be 90 mol% propylene and 75 mol% propane. If 1,000 lb of silica gel/lbmol of feed gas or less is used, can the desired separation be made in one equilibrium stage? If not, what separation can be achieved?

Section 4.10

4.69 Repeat Example 4.17 for 90% evaporation of the water.

4.70 A 5,000-kg/h aqueous solution of 20 wt% Na_2SO_4 is fed to an evaporative crystallizer operating at 60°C . Equilibrium data are given in Figure 4.24. If 80% of the Na_2SO_4 is to be crystallized, calculate:

- The kilograms of water that must be evaporated per hour
- The crystallizer pressure in torr

4.71 Calculate the dew-point pressure, secondary dew-point pressure, and bubble-point pressure of the following mixtures at 50°C , assuming that the liquid aromatics and water are mutually insoluble:

- 50 mol% benzene and 50 mol% water.
- 50 mol% toluene and 50 mol% water.
- 40 mol% benzene, 40 mol% toluene, and 20 mol% water.

4.72 Repeat Exercise 4.71, except compute temperatures for a pressure of 2 atm.

4.73 A liquid containing 30 mol% toluene, 40 mol% ethylbenzene, and 30 mol% water is subjected to a continuous, flash distillation at a total pressure of 0.5 atm. Assuming that mixtures of ethylbenzene and toluene obey Raoult's law and that the hydrocarbons are completely immiscible in water and vice versa, calculate the temperature and composition of the vapor phase at the bubble-point temperature.

4.74 As shown in Figure 4.8, water (W) and *n*-butanol (B) can form a three-phase system at 101 kPa. For a mixture of overall composition of 60 mol% W and 40 mol% B, use a simulation computer program and the UNIFAC method to estimate:

- Dew-point temperature and composition of the first drop of liquid.
- Bubble-point temperature and composition of the first bubble of vapor.
- Compositions and relative amounts of all three phases for 50 mol% vaporization.

4.75 Repeat Example 4.19 for a temperature of 25°C . Are the changes significant?

Cascades and Hybrid Systems

In the previous chapter, a single equilibrium stage was utilized to separate a mixture. In practice, a single stage is rarely sufficient to perform the desired separation. This chapter introduces separation *cascades*, which are collections of contacting stages. Cascades are used in industrial processes to (1) accomplish separations that cannot be achieved in a single stage, and/or (2) reduce the required amount of the mass- or energy-separating agent.

A typical cascade is shown in Figure 5.1, where, in each stage, an attempt is made to bring two or more process streams of different phase state and composition into intimate contact to promote rapid mass and heat transfer, so as to approach physical equilibrium. The resulting phases, whose compositions and temperatures are now closer to, or at, equilibrium, are then separated and each is sent to another stage in the cascade, or withdrawn as a product. Although equilibrium conditions may not be achieved in each stage, it is common to design and analyze cascades using equilibrium-stage models. Alternatively, in the case of membrane separations, where

phase equilibrium is not a consideration and mass-transfer rates through the membrane determine the separation, cascades of membranes can enable separations that cannot be achieved by contact of the feed mixture with a single-membrane separator.

Cascades are prevalent in *unit operations*, such as distillation, absorption, stripping, and liquid–liquid extraction. In cases where the extent of separation by a single-unit operation is limited or the energy required is excessive, it is worthwhile to consider a *hybrid system* of two different unit operations, such as the combination of distillation and pervaporation, which is used to separate azeotropic mixtures. In the last decade, with increased awareness of the need for conserving energy, much attention is being given to hybrid systems. This chapter introduces both cascades and hybrid systems. To illustrate the benefits of cascades, the calculations are based on simple models. Rigorous models, best implemented by computer calculations, are deferred to Chapters 10–12.

5.0 INSTRUCTIONAL OBJECTIVES

After completing this chapter, you should be able to:

- Explain how multi-equilibrium-stage cascades with countercurrent flow can achieve a significantly better separation than a single equilibrium stage.
- Explain the difference between a single-section cascade and a two-section cascade and the limits of what each type can achieve.
- Estimate the recovery of a key component in countercurrent leaching and washing cascades.
- Estimate recovery of a key component in each of three types of liquid–liquid extraction cascades.
- Define and explain the significance of absorption and stripping factors.
- Estimate the recoveries of all components in a single-section, countercurrent cascade using the Kremser method.
- Estimate recoveries of all components in a two-section, countercurrent cascade using the Edmister extension of the Kremser method.
- Configure a membrane cascade to improve a membrane separation.
- Explain the merits and give examples of hybrid separation systems.
- Determine degrees of freedom and a set of specifications for a separation process or any element included in the process.

5.1 CASCADE CONFIGURATIONS

Cascades can be configured in many ways, as shown by the examples in Figure 5.2, where stages are represented by either boxes, as in Figure 5.1, or as horizontal lines in Figure 5.2d,e.

Depending on the mechanical design of the stages, cascades may be arranged vertically or horizontally. The feed to be separated is designated by F ; the mass-separating agent, if used, is designated by S ; and products are designated by P_i .

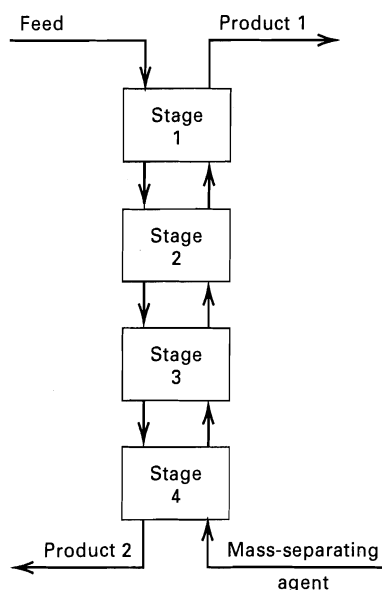


Figure 5.1 Cascade of contacting stages.

In the *countercurrent cascade*, shown in Figures 5.1 and 5.2a, the two phases flow countercurrently to each other between stages. As will be shown in examples, this configuration is very efficient and is widely used for absorption, stripping, liquid-liquid extraction, leaching, and washing. The *crosscurrent cascade*, shown in Figure 5.2b, is, in most cases, not as efficient as the countercurrent cascade, but it is

easier to apply in a batchwise manner. It differs from the countercurrent cascade in that the solvent is divided into portions fed individually to each stage.

A complex diamond variation of the crosscurrent cascade is shown in Figure 5.2c. Unlike the two former cascades, which are linear or one-dimensional, the diamond configuration is two-dimensional. One application is to batch crystallization. Feed F is separated in stage 1 into crystals, which pass to stage 2, and mother liquor, which passes to stage 4. In each of the other stages, partial crystallization or recrystallization occurs by processing crystals, mother liquor, or combinations of the two. Final products are purified crystals and impurity-bearing mother liquors.

The first three cascades in Figure 5.2 consist of *single sections* with streams entering and leaving only from the ends. Such cascades are used to recover components from a feed stream and are not generally useful for making a sharp separation between two selected feed components, called *key components*. To do this, it is best to provide a cascade consisting of *two sections*. The countercurrent cascade of Figure 5.2d is often used. It consists of one section above the feed and one below. If two solvents are used, where S_1 selectively dissolves certain components of the feed, while S_2 is more selective for the other components, the process, referred to as *fractional liquid-liquid extraction*, achieves a sharp separation. If S_1 is a liquid absorbent and S_2 is a vapor stripping agent, added to the cascade, as shown, or produced internally by condensation heat transfer at the top to give liquid reflux, and boiling heat transfer at the bottom to give vapor boilup, the process is simple *distillation*, for which a sharp split between two key components can be achieved if a reasonably high relative volatility exists between the two key components and if reflux, boilup, and the number of stages are sufficient.

Figure 5.2e shows an interlinked system of two distillation columns containing six countercurrent cascade sections. Reflux and boilup for the first column are provided by the second column. This system is capable of taking a ternary (three-component) feed, F , and producing three relatively pure products, P_1 , P_2 , and P_3 .

In this chapter, algebraic equations are developed for modeling idealized cascades to illustrate, quantitatively, their capabilities and advantages. First, a simple countercurrent, single-section cascade for a solid-liquid leaching and/or washing process is considered. Then, cocurrent, crosscurrent, and countercurrent single-section cascades, based on simplified component distribution coefficients, are compared for a liquid-liquid extraction process. A two-section, countercurrent cascade is subsequently developed for a vapor-liquid distillation operation. Finally, membrane cascades are described. In the first three cases, a set of linear algebraic equations is reduced to a single relation for estimating the extent of separation as a function of the number of stages in the cascade, the separation factor, and the flow ratio of the mass- or energy-separating agent to the feed. More rigorous models for design and analysis purposes are

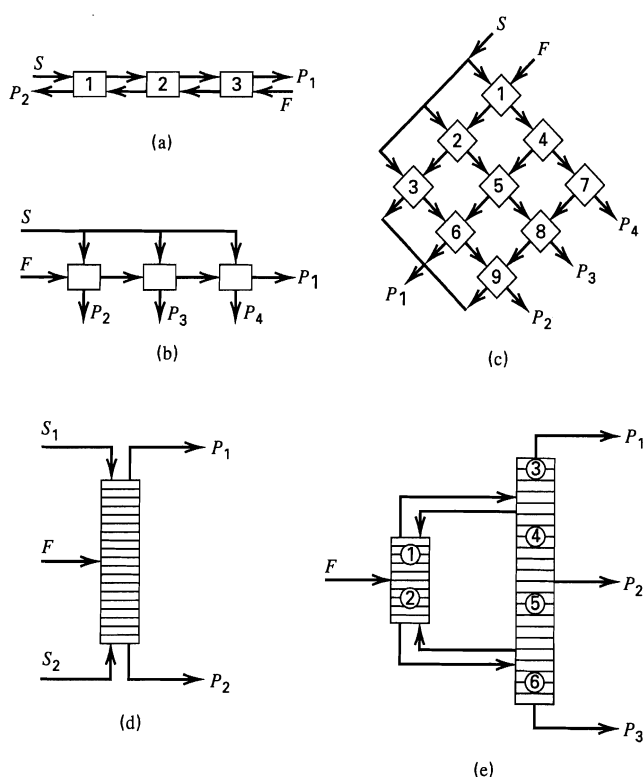


Figure 5.2 Examples of cascade configurations: (a) countercurrent cascade; (b) crosscurrent cascade; (c) two-dimensional, diamond cascade; (d) two-section, countercurrent cascade; (e) interlinked system of countercurrent cascades.

developed in subsequent chapters. As will be seen, single relations cannot be obtained from rigorous models because of the nonlinear nature of rigorous models, making computer calculations a necessity.

5.2 SOLID-LIQUID CASCADES

Consider the N -stage, countercurrent leaching-washing process shown in Figure 5.3. This cascade is an extension of the single-stage systems discussed in Section 4.7. The solid feed, entering stage 1, consists of two components A and B, of mass flow rates F_A and F_B . Pure liquid solvent, C, which can dissolve B completely, but not A at all, enters stage N at a mass flow rate S . Thus, A passes through the cascade as an insoluble solid. It is convenient to express liquid-phase concentrations of B, the solute, in terms of mass ratios of solute to solvent. The liquid *overflow* from each stage, j , contains Y_j mass of soluble material per mass of solute-free solvent, and no insoluble material. The *underflow* from each stage is a slurry consisting of a mass flow F_A of insoluble solids, a constant ratio of mass of solvent/mass of insoluble solids equal to R , and X_j mass of soluble material/mass of solute-free solvent. For a given solid feed, a relationship between the exiting underflow concentration of the soluble component, X_N , the solvent feed rate, and the number of stages is derived as follows.

If equilibrium is achieved at each stage, the overflow solute concentration, Y_j , equals the underflow solute concentration in the liquid, X_j , which refers to liquid held by the solid in the underflow. Assume that all soluble material, A_j is dissolved or leached in stage 1 and all other stages are then washing stages for reducing the amount of soluble material lost in the underflow leaving the last stage, N , and thereby increasing the amount of soluble material leaving in the overflow from stage 1. By solvent material balances, it is readily shown that for constant R , the flow rate of solvent leaving in the overflow from stages 2 to N is S . The flow rate of solvent leaving in the underflow from stages 1 to N is RF_A . Therefore, the flow rate of solvent leaving in the overflow from stage 1 is $S - RF_A$.

A material balance for the soluble material around any interior stage n from $n = 2$ to $N - 1$ is given by

$$Y_{n+1}S + X_{n-1}RF_A = Y_nS + X_nRF_A \quad (5-1)$$

For terminal stages 1 and N , the material balances on the soluble material are, respectively,

$$Y_2S + F_B = Y_1(S - RF_A) + X_1RF_A \quad (5-2)$$

$$X_{N-1}RF_A = Y_NS + X_NRF_A \quad (5-3)$$

Assuming equilibrium, the concentration of soluble material in the overflow from each stage is equal to the concentration of soluble material in the liquid part of the underflow from the same stage. Thus,

$$X_n = Y_n \quad (5-4)$$

In addition, it is convenient to define a *washing factor*, W , as

$$W = \frac{S}{RF_A} \quad (5-5)$$

If (5-1), (5-2), and (5-3) are each combined with (5-4) to eliminate Y , and the resulting equations are rearranged to allow the substitution of (5-5), the following equations result:

$$X_1 - X_2 = \left(\frac{F_B}{S} \right) \quad (5-6)$$

$$\left(\frac{1}{W} \right) X_{n-1} - \left(\frac{1+W}{W} \right) X_n + X_{n+1} = 0, \quad n = 2 \text{ to } N - 1 \quad (5-7)$$

$$\left(\frac{1}{W} \right) X_{N-1} - \left(\frac{1+W}{W} \right) X_N = 0 \quad (5-8)$$

Equations (5-6) to (5-8) constitute a set of N linear algebraic equations in N unknowns, $X_n (n = 1 \text{ to } N)$. The equations are of a tridiagonal, sparse-matrix form, which—for example, with $N = 5$ —is given by

$$\begin{bmatrix} 1 & -1 & 0 & 0 & 0 \\ \left(\frac{1}{W}\right) & -\left(\frac{1+W}{W}\right) & 1 & 0 & 0 \\ 0 & \left(\frac{1}{W}\right) & -\left(\frac{1+W}{W}\right) & 1 & 0 \\ 0 & 0 & \left(\frac{1}{W}\right) & -\left(\frac{1+W}{W}\right) & 1 \\ 0 & 0 & 0 & \left(\frac{1}{W}\right) & -\left(\frac{1+W}{W}\right) \end{bmatrix} \times \begin{bmatrix} X_1 \\ X_2 \\ X_3 \\ X_4 \\ X_5 \end{bmatrix} = \begin{bmatrix} \left(\frac{F_B}{S}\right) \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} \quad (5-9)$$

Equations of type (5-9) can be solved by Gaussian elimination by starting from the top and eliminating unknowns X_1 , X_2 , etc. in order to obtain

$$X_N = \left(\frac{F_B}{S} \right) \left(\frac{1}{W^{N-1}} \right) \quad (5-10)$$

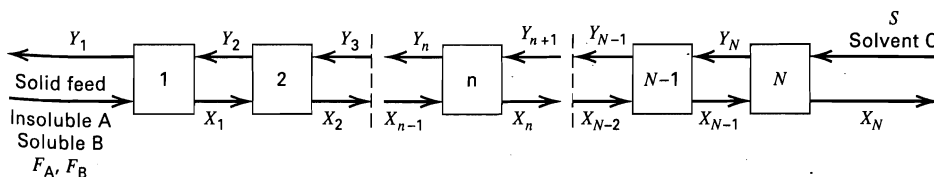


Figure 5.3 Countercurrent leaching or washing system.

By back-substitution, interstage values of X are given by

$$X_n = \left(\frac{F_B}{S} \right) \left(\frac{\sum_{k=0}^{N-n} W^k}{W^{N-1}} \right) \quad (5-11)$$

For example, with $N = 5$,

$$X_1 = Y_1 = \left(\frac{F_B}{S} \right) \left(\frac{1 + W + W^2 + W^3 + W^4}{W^4} \right)$$

The purpose of the cascade, for any given S , is to maximize Y_1 , the amount of soluble solids dissolved in the solvent leaving in the overflow from stage 1, and to minimize X_N , the amount of soluble solid dissolved in the solvent leaving the underflow with the insoluble material from stage N . Equation (5-10) indicates that this can be achieved for a given soluble-solids feed rate, F_B , by specifying a large

solvent feed rate, S , a large number of stages, N , and/or by employing a large washing factor, which can be achieved by minimizing the amount of liquid underflow compared to overflow. It should be noted that the minimum amount of solvent required corresponds to zero overflow from stage 1, or

$$S_{\min} = R F_A \quad (5-12)$$

For this minimum value, $W = 1$ from (5-5) and all soluble solids leave in the underflow from the last stage, N , regardless of the number of stages. Therefore, it is best to specify a value of S significantly greater than $S_{\min}$. Equations (5-10) and (5-5) show that the value of X_N is reduced exponentially by increasing the number of stages, N . Thus, the countercurrent cascade can be very effective. For two or more stages, X_N is also reduced exponentially by increasing the solvent rate, S . For three or more stages, the value of X_N is reduced exponentially by decreasing the underflow ratio R .

EXAMPLE 5.1

Pure water is to be used to dissolve 1,350 kg/h of Na_2CO_3 from 3,750 kg/h of a solid, where the balance is an insoluble oxide. If 4,000 kg/h of water is used as the solvent for the carbonate and the total underflow from each stage is 40 wt% solvent on a solute-free basis, compute and plot the percent recovery of the carbonate in the overflow product for one stage and for two to five countercurrent stages, as in Figure 5.3.

SOLUTION

Soluble solids feed rate = $F_B = 1,350$ kg/h

Insoluble solids feed rate = $F_A = 3,750 - 1,350 = 2,400$ kg/h

Solvent feed rate = $S = 4,000$ kg/h

Underflow ratio $R = 40/60 = 2/3$

Washing factor $W = S/R F_A = 4,000/[(2/3)(2,400)] = 2.50$

Overall fractional recovery of soluble solids = $Y_1(S - R F_A)/F_B$

By overall material balance on soluble solids for N stages,

$$F_B = Y_1(S - R F_A) + X_N R F_A$$

Solving for Y_1 and using (5-5) to introduce the washing factor,

$$Y_1 = \frac{(F_B/S) - (1/W)X_N}{(1 - 1/W)}$$

From the given data,

$$Y_1 = \frac{(1,350/4,000) - (1/2.50)X_N}{(1 - 1/2.50)} \quad \text{or} \quad (1)$$

$$Y_1 = 0.5625 - 0.6667 X_N$$

where, from (5-10),

$$X_N = \left(\frac{1,350}{4,000} \right) \frac{1}{2.50^{N-1}} = \frac{0.3375}{2.50^{N-1}} \quad (2)$$

The percent recovery of soluble material is

$$\begin{aligned} Y_1(S - R F_A)/F_B &= Y_1[4,000 - (2/3)(2,400)]/1,350 \times 100\% \\ &= 177.8 Y_1 \end{aligned} \quad (3)$$

Results for one to five stages, as computed from (1) to (3), are

No. of Stages in Cascade, N	X_N	Y_1	Percent Recovery of Soluble Solids
1	0.3375	0.3375	60.0
2	0.1350	0.4725	84.0
3	0.0540	0.5265	93.6
4	0.0216	0.5481	97.4
5	0.00864	0.5567	99.0

A plot of the percent recovery of soluble solids as a function of the number of stages is shown in Figure 5.4. Although only a 60% recovery is obtained with one stage, 99% recovery is achieved for five stages. To achieve 99% recovery with one stage, a water rate of 160,000 kg/h is required, which is 40 times that required for five stages. Thus, the use of multiple stages in a countercurrent cascade to increase recovery of soluble material can be much more effective than increased use of a mass-separating agent with a single stage.

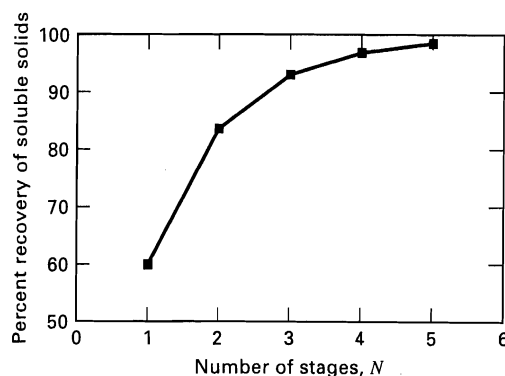


Figure 5.4 Effect of number of stages on percent recovery in Example 5.1.

5.3 SINGLE-SECTION, LIQUID-LIQUID EXTRACTION CASCADES

Three possible two-stage, single-section, liquid-liquid extraction cascades are the cocurrent, crosscurrent, and countercurrent arrangements in Figure 5.5. The countercurrent arrangement is generally preferred because, as will be shown in this section, that arrangement results in a higher degree of extraction for a given amount of solvent and number of equilibrium stages.

In Section 4.5, (4-25), for the fraction of solute, B, that is not extracted, was derived for a single liquid-liquid equilibrium extraction stage, assuming the use of pure solvent, and a constant value for the distribution coefficient, K'_{D_B} , for the solute, B, dissolved in components A and C, which are mutually insoluble. That equation is now extended to multiple stages for each type of cascade shown in Figure 5.5.

Cocurrent Cascade

If additional stages are added in the cocurrent arrangement in Figure 5.5a, the equation for the first stage is that of a single stage. That is, from (4-25) in mass ratio units,

$$X_B^{(1)} / X_B^{(F)} = \frac{1}{1 + E} \quad (5-13)$$

where E is the extraction factor, given by

$$E = K'_{D_B} S / F_A \quad (5-14)$$

Since $Y_B^{(1)}$ is in equilibrium with $X_B^{(1)}$ according to

$$K'_{D_B} = Y_B^{(1)} / X_B^{(1)} \quad (5-15)$$

the combination of (5-15) with (5-13) gives

$$Y_B^{(1)} / X_B^{(F)} = K'_{D_B} / (1 + E) \quad (5-16)$$

For the second stage, a material balance for B gives

$$X_B^{(1)} F_A + Y_B^{(1)} S = X_B^{(2)} F_A + Y_B^{(2)} S \quad (5-17)$$

with

$$K'_{D_B} = Y_B^{(2)} / X_B^{(2)} \quad (5-18)$$

Combining (5-17) with (5-18), (5-13), and (5-16) to eliminate $X_B^{(1)}$, $Y_B^{(1)}$, and $Y_B^{(2)}$ gives

$$\frac{X_B^{(2)}}{X_B^{(F)}} = \frac{1}{1 + E} = \frac{X_B^{(N)}}{X_B^{(F)}} \quad (5-19)$$

Comparison of (5-19) with (5-13) shows that $X_B^{(2)} = X_B^{(1)}$. Thus, no additional extraction takes place in the second stage. This is as expected because the two streams leaving the first stage are at equilibrium and when they are recontacted in stage 2, no additional net mass transfer of B occurs. Accordingly, a cocurrent cascade of equilibrium stages has no merit other than to provide increased residence time.

Crosscurrent Cascade

For the crosscurrent cascade shown in Figure 5.5b, the feed progresses through each stage, starting with stage 1 and finishing with stage N . The solvent flow rate, S , is divided into portions that are sent to each stage. If the portions are equal, the following mass ratios are obtained by application of (5-13), where S is replaced by S/N , so that E is replaced by E/N :

$$\begin{aligned} X_B^{(1)} / X_B^{(F)} &= 1 / (1 + E/N) \\ X_B^{(2)} / X_B^{(1)} &= 1 / (1 + E/N) \\ &\vdots \\ X_B^{(N)} / X_B^{(N-1)} &= 1 / (1 + E/N) \end{aligned} \quad (5-20)$$

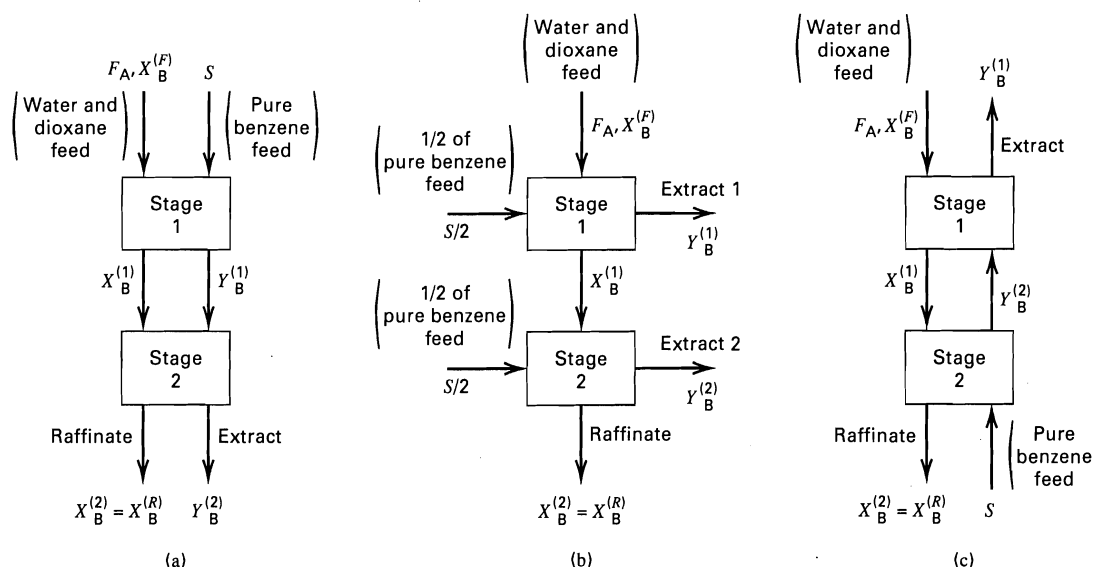


Figure 5.5 Two-stage arrangements: (a) cocurrent cascade; (b) crosscurrent cascade; (c) countercurrent cascade.

Combining the equations in (5-20) to eliminate all intermediate interstage variables, $X_B^{(n)}$, the final raffinate mass ratio is given by

$$X_B^{(N)}/X_B^{(F)} = X_B^{(R)}/X_B^{(F)} = \frac{1}{(1 + E/N)^N} \quad (5-21)$$

Interstage values of $X_B^{(n)}$ are obtained similarly from

$$X_B^{(n)}/X_B^{(F)} = \frac{1}{(1 + E/N)^n} \quad (5-22)$$

Thus, unlike the cocurrent cascade, the value of X_B decreases in each successive stage. For an infinite number of equilibrium stages, (5-21) becomes

$$X_B^{(\infty)}/X_B^{(F)} = 1/\exp(E) \quad (5-23)$$

Thus, even for an infinite number of stages, $X_B^{(R)} = X_B^{(\infty)}$ cannot be reduced to zero.

Countercurrent Cascade

In the countercurrent arrangement for two stages in Figure 5.5c, the feed liquid passes through the cascade counter-currently to the solvent. For a two-stage system, the material balance and equilibrium equations for solute, B, for each stage are as follows.

Stage 1:

$$X_B^{(F)} F_A + Y_B^{(2)} S = X_B^{(1)} F_A + Y_B^{(1)} S \quad (5-24)$$

$$K'_{D_B} = \frac{Y_B^{(1)}}{X_B^{(1)}} \quad (5-25)$$

Stage 2:

$$X_B^{(1)} F_A = X_B^{(2)} F_A + Y_B^{(2)} S \quad (5-26)$$

$$K'_{D_B} = \frac{Y_B^{(2)}}{X_B^{(2)}} \quad (5-27)$$

Combining (5-24) to (5-27) with (5-14) to eliminate $Y_B^{(1)}$, $Y_B^{(2)}$, and $X_B^{(1)}$ gives

$$X_B^{(2)}/X_B^{(F)} = X_B^{(R)}/X_B^{(F)} = \frac{1}{1 + E + E^2} \quad (5-28)$$

If the number of countercurrent stages is extended to N stages, the result is

$$X_B^{(N)}/X_B^{(F)} = X_B^{(R)}/X_B^{(F)} = 1 / \sum_{n=0}^N E^n \quad (5-29)$$

Interstage values of $X_B^{(n)}$ are obtained in a similar fashion, giving

$$X_B^{(n)}/X_B^{(F)} = \sum_{k=0}^{N-n} E^k / \sum_{k=0}^N E^k \quad (5-30)$$

As with the crosscurrent arrangement, the value of X_B decreases in each successive stage. The amount of decrease for

the countercurrent arrangement is greater than for the cross-current arrangement, and the difference increases exponentially with increasing extraction factor, E . Therefore, the countercurrent cascade is the most efficient of the three linear cascades.

For an infinite number of equilibrium stages, the limit of (5-28) gives two results:

$$X_B^{(\infty)}/X_B^{(F)} = 0, \quad 1 \leq E \leq \infty \quad (5-31)$$

$$X_B^{(\infty)}/X_B^{(F)} = (1 - E), \quad E \leq 1 \quad (5-32)$$

Thus, complete extraction can be achieved in a countercurrent cascade, but only for an extraction factor, E , greater than 1.

EXAMPLE 5.2

Ethylene glycol can be catalytically dehydrated completely to *p*-dioxane (a cyclic diether) by the reaction $2\text{HOCH}_2\text{CH}_2\text{HO} \rightarrow \text{H}_2\text{CCH}_2\text{OCH}_2\text{CH}_2\text{O} + 2\text{H}_2\text{O}$. Water and *p*-dioxane have normal boiling points of 100°C and 101.1°C, respectively, and cannot be separated economically by distillation. However, liquid-liquid extraction at 25°C (298.15 K), using benzene as a solvent, is reasonably effective. Assume that 4,536 kg/h (10,000 lb/h) of a 25 wt% solution of *p*-dioxane in water is to be separated continuously by using 6,804 kg/h (15,000 lb/h) of pure benzene. Assuming that benzene and water are mutually insoluble, determine the effect of the number and arrangement of stages on the percent extraction of *p*-dioxane. The flowsheet is shown in Figure 5.6.

SOLUTION

Three different arrangements of stages will be examined: (a) cocurrent cascade, (b) crosscurrent cascade, and (c) countercurrent cascade. Because water and benzene are almost mutually insoluble, (5-13), (5-21), and (5-29) can be used, respectively, to estimate $X_B^{(R)}/X_B^{(F)}$, the fraction of *p*-dioxane not extracted, as a function of the number of stages. From the equilibrium data of Berdt and Lynch [1], the distribution coefficient for *p*-dioxane, $K'_{D_B} = Y_B/X_B$, where Y refers to the benzene phase and X refers to the water phase, varies from 1.0 to 1.4 over the concentration range of interest. For this example, assume a constant value of 1.2. From the given data, $S = 6,804$ kg/h of benzene, $F_A = 4,536(0.75) = 3,402$ kg/h of water, and $X_B^{(F)} = 0.25/0.75 = 1/3$. From (5-14),

$$E = 1.2(6,804)/3,402 = 2.4$$

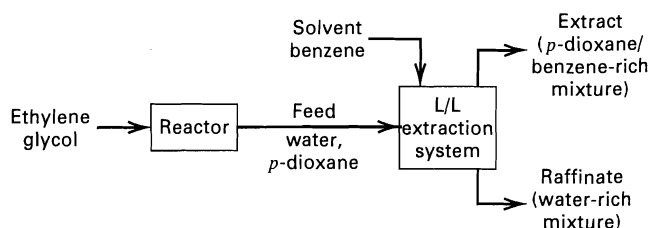


Figure 5.6 Flowsheet for Example 5.2.

Single equilibrium stage

All three arrangements give identical results for a single stage. From (5-13),

$$X_B^{(1)} / X_B^{(F)} = 1 / (1 + 2.4) = 0.294$$

The corresponding fractional extraction is

$$1 - X_B^{(1)} / X_B^{(F)} = 1 - 0.294 = 0.706 \quad \text{or} \quad 70.6\%$$

More than one equilibrium stage

(a) Cocurrent cascade. For any number of stages, the percent extraction is the same as for one stage, 70.6%.

(b) Crosscurrent cascade. For any number of stages, (5-21) applies. For example, for two stages, assuming equal flow rates of solvent to each stage,

$$X_B^{(2)} / X_B^{(F)} = 1 / (1 + E/2)^2 = 1 / (1 + 2.4/2)^2 = 0.207$$

and the percent extraction is 79.3%. Results for other numbers of stages are obtained in the same manner.

(c) Countercurrent cascade. For any number of stages, (5-29) applies. For example, for two stages,

$$X_B^{(2)} / X_B^{(F)} = 1 / (1 + E + E^2) = 1 / (1 + 2.4 + 2.4^2) = 0.109$$

and the percent extraction is 89.1%. Results for other numbers of stages are obtained in the same manner.

A plot of percent extraction as a function of the number of equilibrium stages for up to five stages is shown in Figure 5.7 for each of the three arrangements. The probability-scale ordinate is convenient because for the countercurrent arrangement, with $E > 1$, 100% extraction is approached as the number of stages approaches infinity. For the crosscurrent arrangement, a maximum percent extraction of 90.9% is computed from (5-23). For five stages, Figure 5.7 shows that the countercurrent cascade has already achieved 99% extraction.

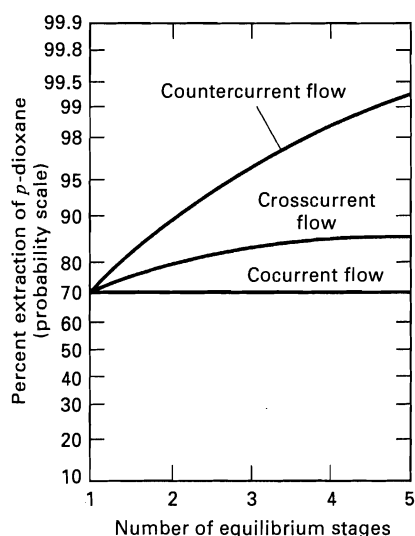


Figure 5.7 Effect of multiple-stage cascade arrangement on extraction efficiency.

5.4 MULTICOMPONENT VAPOR-LIQUID CASCADES

Countercurrent cascades are used extensively for vapor-liquid separation operations, including absorption, stripping, and distillation. For absorption and stripping, a *single-section cascade* is used to recover one selected component from the feed. For distillation, a *two-section cascade* is effective in achieving a separation between two selected components referred to as the key components. For both cases, approximate calculation procedures relate compositions of multicomponent vapor and liquid streams entering and exiting the cascade to the number of equilibrium stages required. These approximate procedures are called *group methods* because they provide only an overall treatment of the group of stages in the cascade, without considering detailed changes in temperature, phase compositions, and flows from stage to stage.

Single-Section Cascades by Group Methods

Kremser [2] originated the group method by deriving an equation for the fractional absorption of a species from a gas into a liquid absorbent for a multistage countercurrent absorber. His method also applies to strippers. The treatment presented here is similar to that of Edmister [3] for general application to vapor-liquid separation operations. An alternative treatment is given by Smith and Brinkley [4].

Consider first the countercurrent cascade of N adiabatic, equilibrium stages used, as shown in Figure 5.8a, to absorb species present in the entering vapor. Assume that these species are absent in the entering liquid. Stages are numbered from top to bottom. It is convenient to express stream compositions in terms of component molar flow rates, v_i and l_i , in the vapor and liquid phases, respectively. However, in the following derivation, the subscript i is dropped. A material balance around the top of the absorber, including stages 1 through $N - 1$, for any absorbed species gives

$$v_N = v_1 + l_{N-1} \quad (5-33)$$

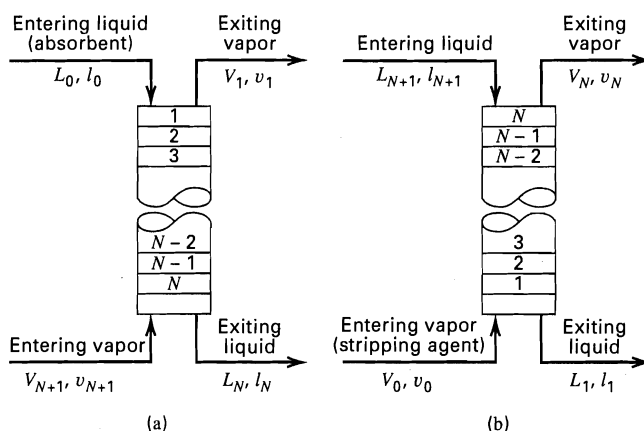


Figure 5.8 Countercurrent cascades of N adiabatic stages: (a) absorber; (b) stripper.

where

$$v = yV \quad (5-34)$$

$$l = xL \quad (5-35)$$

and $l_0 = 0$. From equilibrium considerations for stage N , the definition of the vapor-liquid equilibrium ratio or K -value can be employed to give

$$y_N = K_N x_N \quad (5-36)$$

Combining (5-34), (5-35), and (5-36), v_N becomes

$$v_N = \frac{l_N}{L_N/(K_N V_N)} \quad (5-37)$$

An absorption factor A , analogous to the extraction factor, E , for a given stage and component is defined by

$$A = \frac{L}{KV} \quad (5-38)$$

Combining (5-37) and (5-38),

$$v_N = \frac{l_N}{A_N} \quad (5-39)$$

Substituting (5-39) into (5-33),

$$l_N = (l_{N-1} + v_1)A_N \quad (5-40)$$

The internal flow rate, l_{N-1} , is eliminated by successive substitution using material balances around successively smaller sections of the top of the cascade. For stages 1 through $N - 2$,

$$l_{N-1} = (l_{N-2} + v_1)A_{N-1} \quad (5-41)$$

Substituting (5-41) into (5-40),

$$l_N = l_{N-2}A_{N-1}A_N + v_1(A_N + A_{N-1}A_N) \quad (5-42)$$

Continuing this process to the top stage, where $l_1 = v_1A_1$, ultimately converts (5-42) into

$$l_N = v_1(A_1A_2A_3 \dots A_N + A_2A_3 \dots A_N + A_3 \dots A_N + \dots + A_N) \quad (5-43)$$

A more useful form is obtained by combining (5-43) with the overall component balance

$$l_N = v_{N+1} - v_1 \quad (5-44)$$

to give an equation for the exiting vapor in terms of the entering vapor and a recovery fraction:

$$v_1 = v_{N+1}\phi_A \quad (5-45)$$

where, by definition, the recovery fraction is

$$\phi_A = \frac{1}{A_1A_2A_3 \dots A_N + A_2A_3 \dots A_N + A_3 \dots A_N + \dots + A_N + 1}$$

= fraction of species in entering vapor that is not absorbed (5-46)

In the group method, an average effective absorption factor, A_e , replaces the separate absorption factors for each

stage. Equation (5-46) now becomes

$$\phi_A = \frac{1}{A_e^N + A_e^{N-1} + A_e^{N-2} + \dots + A_e + 1} \quad (5-47)$$

When multiplied and divided by $(A_e - 1)$, (5-47) reduces to

$$\phi_A = \frac{A_e - 1}{A_e^{N+1} - 1} \quad (5-48)$$

Note that each component has a different A_e and, therefore, a different value of ϕ_A . Figure 5.9 from Edmister [3] is a plot of (5-48) with a probability scale for ϕ_A , a logarithmic scale for A_e , and N as a parameter. This plot, in linear coordinates, was first developed by Kremser [2].

Consider next the countercurrent stripper shown in Figure 5.8b. Assume that the components stripped from the liquid are absent in the entering vapor, and ignore condensation or absorption of the stripping agent. In this case, stages are numbered from bottom to top to facilitate the derivation. The pertinent stripping equations follow in a manner analogous to the absorber equations. The results are

$$l_1 = l_{N+1}\phi_S \quad (5-49)$$

where

$$\phi_S = \frac{S_e - 1}{S_e^{N+1} - 1} = \text{fraction of species in entering liquid that is not stripped} \quad (5-50)$$

$$S = \frac{KV}{L} = \frac{1}{A} = \text{stripping factor} \quad (5-51)$$

Figure 5.9 also applies to (5-50). As shown in Figure 5.10, absorbers are frequently coupled with strippers or distillation columns to permit regeneration and recycle of absorbent. Since stripping action is not perfect, recycled absorbent entering the absorber contains species present in the vapor entering the absorber. Vapor passing up through the absorber can strip these as well as the absorbed species introduced in the makeup absorbent. A general absorber equation is obtained by combining (5-45) for absorption of species from the entering vapor with a modified form of (5-49) for stripping of the same species from the entering liquid. For stages numbered from top to bottom, as in Figure 5.8a, (5-49) becomes

$$l_N = l_0\phi_S \quad (5-52)$$

or, since

$$l_0 = v_1 + l_N$$

$$v_1 = l_0(1 - \phi_S) \quad (5-53)$$

The total balance in the absorber for a component appearing in both entering vapor and entering liquid is obtained by adding (5-45) and (5-53) to give

$$v_1 = v_{N+1}\phi_A + l_0(1 - \phi_S) \quad (5-54)$$